

Noncommutative Algebraic Topology I

K-Theory & Homotopical Algebra

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1 Introduction

In this introduction, we will provide information relevant to this thesis in the following order:

1. Our overarching goals.
2. Prerequisites and reading recommendations.
3. Conventions & general structure.
4. Acknowledgements.

We strongly recommend reading these sections as they contain information relevant for pleasant reading of this thesis.

1.1 Background, Overarching Goals & Structure

The intersection between operator algebras and homotopy theory is not a new one; it has existed since the days when the Brown-Douglas-Fillmore $\text{Ext}(-)$ groups were invented to study the classification theory of essentially normal operators and when Atiyah's topological K-theory was extended to the realm of C^* -algebras & Banach algebras. Since then, there has been a wave of interest in the use of methods from algebraic topology to study operator algebraic structures. The advent of bivariant homology theories on categories of operator algebras and the crucial role they play in the formulation of important conjectures has become a large part of modern research in the theory of operator algebras.

On the other side of the bridge, algebraic topology has also undergone rapid change. Indeed, the notion of “homotopy”, once thought to be something useful only for studying spaces, has now gained a life of its own and is the main notion studied in *homotopy theory*. Over the years, homotopy theory has found itself becoming a branch of mathematics which is distinct from topology. In a rather famous letter entitled *The Future of Homotopy Theory*, Clark Barwick actually makes a much stronger claim about homotopy theory in relation to topology, among other important topics.

Some of us call ourselves *algebraic topologists*, but this has the unhelpful effect of making the subject appear to be an area of topology, which I think is profoundly inaccurate. It so happens that one way (and historically the first way) to model homotopical thinking is to employ a very particular class of topological spaces. Today, the praxis of homotopy theory interacts with topology no more often than it does with arithmetic geometry and category theory, and the interactions with areas like representation theory are growing rapidly. *Homotopy theory is not a branch of topology.* [12, p. 1]

Indeed, it is certainly true that the notion of “homotopy” is pervasive across different parts of mathematics, even when there are no spaces in sight, and that it plays a crucial role in constructing invariants which are useful, for example, in solving classification problems. In that sense, it would not be correct to say that it is a purely topological notion. Nevertheless, homotopy theory *did* arise from algebraic topology, so one might expect to see many interactions between homotopy theory and operator algebras. Indeed, there have been *some* developments over the years but it has only recently become the case that we are deliberately using homotopy-theoretic methods for the purpose of understanding (and potentially solving) conjectures in operator algebra theory. So, this requires some rationalization.

One possible reason for the lack of such interactions until recent years has to do with the structures that are studied in homotopy theory. The classical structures of interest in homotopy theory, such as the stable homotopy category, did not have obvious analogues in the setting of C^* -algebras. Moreover, the methods stable homotopy theory, which used a lot of homological algebra and category theory, were only barely being introduced into operator algebra theory by the work of (Joachim) Cuntz ([31]), (Nigel) Higson ([51]), (Jonathan) Rosenberg ([88]) and others¹, though it took a number of years before their work was pushed forward *using ideas from homotopy theory*. Of course, one should also be aware that the current status of homotopy theory also took a lot of time to properly develop. For instance, it may surprise the reader to know that model categories were not viewed favorably for a long time by many homotopy theorists. Indeed, Andrew Blumberg, another famous homotopy theorist, mentions as much in a set of lecture notes on equivariant homotopy theory.

Elmendorf's original proof of his theorem was in the 1980s and did not use model categories, even though Quillen had already introduced them at the time. Until the mid-1990s (30 years after Quillen introduced model categories in [78]), many homotopy theorists avoided them, thinking of them as formal gobbledygook. However, around the time [37] introduced a symmetric monoidal category of spectra, people began realizing they were unavoidable. [34, p. 9]

So, it is not so surprising that model categories have only “recently” become structures of interest to operator algebraists. In fact, some operator algebraists might find it somewhat surprising to learn that the possible use of model categories in operator algebra theory was already considered as early as 1984 in the work of (Claude) Schochet, which we quote below.

We have not yet formally defined “admissible category of C^* -algebras”. Roughly, a category of C^* -algebras is admissible if it is closed under the various homotopy constructions required to develop elementary homotopy theory. Here are some admissible categories:

1. all C^* -algebras,
2. separable C^* -algebras,
3. separable nuclear C^* -algebras,

In each case, we allow all C^* -maps as morphisms. This is a provisional definition. It should be replaced by the closed model categories of Quillen. However, there is as yet no good notion of loop space or fibration. Thus, we proceed in an ad hoc manner... [89, p. 402]

¹The references we provide are some examples of older work by these mathematicians which have a strong homological/categorical flavor.

It suffices to say that there was much potential for early interactions between homotopy theory and operator algebras. Of course, we must emphasize that recent years have brought many more interactions. In fact, we should mention that newer developments in homotopy theory have brought the so-called $(\infty, 1)$ -categories to the frontier. These are, in a sense, supposed to “replace” model categories as the “correct” setting in which one should do homotopy theory. Nevertheless, we emphasize that model categories are often used in tandem with $(\infty, 1)$ -categories and, in fact, some results about the latter actually rely on the existence of certain model structures on concrete categories of interest.

Of course, none of this also accounts for various other structures that were of interest in homotopy theory and still continue to be studied today. For instance, triangulated categories, which arose from algebraic geometry and classical stable homotopy theory, could also be viewed as “precursors” to $(\infty, 1)$ -categories and, in fact, are the main structure used in the study of bivariant homology theories on C^* -algebras. Indeed, it was realized very early on, due to the work of (Ralf) Meyer and (Ryszard) Nest in [68], that Kasparov’s KK-category actually has the structure of a triangulated category and this structure has been concretely helpful in recent K-theory computations, as well as in presenting a homological picture of the Baum-Connes Conjecture. The close relationship between triangulated categories and (stable) $(\infty, 1)$ -categories has, through the work of (Ulrich) Bunke, (Alexander) Engel and (Markus) Land in [22], [23], [21] and [24], resulted in certain bivariant homology theories attaining an $(\infty, 1)$ -categorical enhancement. This seems to have potential for applications towards, say, K-theory computations of crossed products and the UCT problem.

We say a few words about conventions in Section 1.2. We discuss the prerequisites for the thesis and some reading recommendations in Section 1.3. Finally, we discuss some aspects of foundations in Section 1.4.

1.2 Conventions

We introduce notation, acronyms and specific categories that may be used throughout the thesis.

General Mathematical Notation

$\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

$\mathbb{F}$ denotes any field.

$\mathbb{N}$ denotes the set of natural numbers (excluding 0)

$\mathbb{N}_0$ is the set of natural numbers including 0

$\mathbb{R}_0^+$ denotes the set of all non-negative real numbers.

$\mathbb{R}^+$ denotes the set of all positive real numbers.

If A, B are mathematical objects, then $A \stackrel{\text{def}}{=} B$ means that A is defined to be equal to B .

If A, B are statements, then $A \stackrel{\text{def}}{\Leftrightarrow} B$ means that A is defined to be equivalent to B .

Common Acronyms

WLOG - Without loss of generality.

TFAE - The following are equivalent.

TFSH - The following statements hold.

nbhd - neighborhood.

Some Categories

Set is the category of sets (with functions as morphisms).

Grp is the category of groups (with group homomorphisms as morphisms).

AbGrp is the full subcategory of Grp consisting of abelian groups.

Top is the category of topological spaces (with continuous maps as morphisms).

CHaus is the full subcategory of Top consisting of compact Hausdorff spaces.

Grp(Top) is the category of topological groups (with continuous group homomorphisms as morphisms).

Ab(Top) is the full subcategory of Grp(Top) consisting of topological abelian groups.

Grp(LCH) is the full subcategory of Grp(Top) consisting of locally compact Hausdorff groups.

Grp(CHaus) is the full subcategory of Grp(Top) consisting of compact Hausdorff groups.

Ab(LCH) is the full subcategory of Grp(Top) consisting of locally compact Hausdorff groups.

Ab(CHaus) is the full subcategory of Grp(Top) consisting of compact Hausdorff abelian groups.

Ban $_{\mathbb{C}}$ is the category of $\mathbb{C}$ -Banach spaces (with continuous linear maps as morphisms).

$C^*\text{Alg}$ is the category of C^* -algebras (with $\star$ -homomorphisms as morphisms).

${}^1C^*\text{Alg}$ is the subcategory of $C^*\text{Alg}$ consisting of unital C^* -algebras & unital $\star$ -homomorphisms.

${}_{\omega}C^*\text{Alg}$ is the category of separable C^* -algebras.

BanAlg is the category of $\mathbb{C}$ -Banach algebras (with continuous algebra homomorphisms as morphisms).

${}^1\text{BanAlg}$ is the subcategory of BanAlg consisting of unital Banach algebras and unital morphisms.

Categorical Notation

If $\mathcal{C}$ is a category and x is an object in $\mathcal{C}$, then we denote that by writing $x \in \mathcal{C}$.

If $\mathcal{C}$ is a category, then $\text{Mor}(\mathcal{C})$ refers to the set^a of all morphisms of $\mathcal{C}$.

If $\mathcal{C}$ is a category and $x, y \in \mathcal{C}$, then $\mathcal{C}(x, y)$ is the set of all morphisms from x to y .

If $\mathcal{C}$ is a category and $x \in \mathcal{C}$, then $\text{Aut}(x)$ is the set of all isomorphisms with (co)domain x .

If $\mathcal{C}$ is a pointed category, then the zero object will typically be denoted by 0 unless otherwise stated.

^aThis will make sense through the notion of a Grothendieck universe, as discussed in Section 1.4.

We also mention that there are several places where we have labeled certain definitions/results with the name of the author. This has only been done just for the sake of clarity about our contributions. In other places where we reprove results or simply extend what is in the literature, we do state that very clearly. Let us also explain ?? a little more. Throughout the thesis, we will have boxes of the following form which appear in the main text but also in proofs:

| **Claim 1.1.** THIS IS A TEST BOX.

These are supposed to encode claims whose proofs we do want to include but doing so in the main thesis breaks the flow of the exposition. So, instead, we simply include their statements in the main thesis so that the reader will certainly come across them but we place the proofs in ??. For electronic usage, the box is hyperlinked, so clicking on “Claim [Number]” will lead you to proof of the claim and similarly, clicking on “Proof [Number]” will lead you back to the claim. For the reader who prefers to use a printed copy, we will state the page number where the proof can be found.

1.3 Prerequisites & Reading Recommendations

Due to the fact that some parts of this thesis involve new results, it will not always be possible to write in a self-contained way. For full appreciation of everything written here, we will need basic background in functional analysis, operator

algebras and category theory. Here are some reading recommendations for these topics:

Functional Analysis: One can certainly use [75] because it is extremely well-written.

Operator Algebras: [71] is a more modern reference but [56] was a more pleasant read, even with its clearly analytic tilt. It also contains a nice introduction to the necessary concepts from functional analysis.

Category Theory: [83] is accessible but lacks examples from functional analysis and these might be very interesting for readers who would like an early glimpse into the relevance of category theory towards analytic topics. For that, it is a good idea to look into the rather accessible resource [59]; this also doubles up as an accessible treatment of functional analysis.

We do use homotopy theory in Sections ?? and ?? but we introduce most of the structures we use, with the sole exception of $(\infty, 1)$ -category theory. Nevertheless, we do try to make our exposition intuitive and readable.

1.4 A Word About Foundations

This thesis does make extensive use of category theory and, to a much lesser extent, some $(\infty, 1)$ -category theory. Therefore, it is necessary to say a few words about the foundations we will use. This will effectively play no major role for us, so we mainly just include this as a formality. Nevertheless, there may be instances where use phrases like “consider a set of morphisms in a category” and since we are not using the word “set” or “collection” in just the sense of ZFC set theory, it is best to clarify what exactly we mean. Indeed, we need slightly more than the axioms of ZFC set theory to formalize category theory. This is typically done with the notion of a *Grothendieck universe*.

Definition 1.1

A Grothendieck universe is a set U such that:

1. If $S \in U$ and $x \in S$, then $x \in U$.
2. If $S \in U$, then $\mathcal{P}(S) \in U$.
3. $\mathbb{N} \in U$.
4. If $I \in U$ and $f : I \rightarrow U$ is a map, then $\bigcup_{i \in I} f(i) \in U$.

If U is a Grothendieck universe, then its elements are known as U -small sets.

One may, now, state the additional axiom of interest.

Axiom 1.2. The Existence of Grothendieck Universes

Let x be a set. Then, there exists a Grothendieck universe that contains x .

The typical way in which this axiom is used is that one will typically fix finitely many universes $U_1, \dots, U_n$ such that U_i is contained in U_{i+1} for each i and simply keep track of constructions which may result in one needing to go to a “higher universe”. Such constructions include localizations of categories (this applies both for usual categories and $(\infty, 1)$ -categories). Again, we do not wish to make a big deal out of this because it will not matter for our purposes. Indeed, we will not even explicitly use this material later in the thesis because it simply hampers readability.

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2 Motivating Projective/Injective Modules

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

Motivation.

2.1 The Structure of Projectives

In all cases above, the idea is to narrow down our focus on projective modules and to look at the **formal difference** of two projective modules. The formal difference bit is handled in the second step below, while the focus on projective modules is handled over here. Pick your poison from the above list and realize that we're somehow interested in constructing K_0 by looking at the structure of projective modules. So, let's say something about them over here. For the sake of readers who couldn't care less, we include their definition in the next bit of terminology.

Terminology 2.1. Let $R \in \text{Rng}^1$ and let $M \in {}_R\text{Mod}$. M is projective iff ${}_R\text{Mod}(M, -)$ is exact.

At this point, we are not going to carry out a full review of the basic theory of projective modules and we're not going to introduce facts about free modules & such from scratch. There are two reasons for this:

1. This is basic algebra, so we expect you to either know it or to have the capacity to quickly learn it.
2. We will not even really need any of these facts unless we're specifically looking at proving facts about modules.

To be more precise about the second point, it turns out that there is a concrete description of K_0 in terms of idempotents and this is far more amenable to calculations. In fact, a large part of our exposition on the K-theory of C^* -algebras is dedicated towards showing how one actually computes with these idempotents. In fact, we will significantly simplify this concrete description of K_0 in the case of C^* -algebras so that we can work with projections instead. Before we get to any of that stuff, let's first state a standard characterization of projectivity for modules.

Proposition 2.2

Let $R \in \text{Rng}^1$ and let $M \in {}_R\text{Mod}$. TFAE:

1. M is projective.
2. For all R -modules and all solid morphisms in the following diagram:

$$\begin{array}{ccc} & & P \\ & \nearrow \psi & \downarrow g \\ M & \xrightarrow{f} & N \end{array}$$

there exists a morphism $\psi : M \rightarrow P$ such that the diagram commutes when g is surjective^a.

3. Every surjective R -linear map $\alpha : N \rightarrow M$ is split epi in ${}_R\text{Mod}$.
4. M is a direct summand in every module of which it is a quotient.
5. M is a direct summand in a free module.

^aIn this case, the morphism ψ is known as a lift and this is known as the lifting property.

Proof. For a proof in the literature, see [53, Theorem 4.7, p.25]. I have used this resource as inspiration for the proof in [REF]. The reason for studying projectives in there was in light of their direct application towards the UCT. ■

The crucial corollary of this result is a characterization of finitely-generated projective modules. We did prove this in [REF] but we recommend doing it as an exercise here since it's not complicated.

Claim 2.1. Let $R \in \text{Rng}^1$ and let $M \in {}_R\text{Mod}$. TFAE:

1. M is a finitely-generated projective module.
2. M is a direct summand of R^n for some $n \in \mathbb{N}$.
3. There is an idempotent $e \in {}_R\text{Mod}(R^n)$ such that $M \cong_{{}_R\text{Mod}} \text{ran}(e)$.

We have a remark about this result.

Remark 2.3. For the moment, the important piece of this corollary is the second statement, as this is what we'll use to construct K_0 . However, as we alluded to earlier, the concrete description of K_0 via idempotents will be inspired essentially by the third statement of this corollary.

Here's some notation that will shorten our sentences drastically.

Notation 2.4. Let $R \in \text{Rng}^1$ and let $M \in {}_R\text{Mod}$. We will use the acronym:

f.g.p.,

as an abbreviation for finitely-generated projective. We denote by:

${}_R\text{Mod}_{\text{f.g.p.}}$,

the full subcategory of ${}_R\text{Mod}$ consisting of finitely-generated projective R -modules.

So, before we construct the functor associated to the first step, we need a small result.

Claim 2.2. Let $R \in \text{Rng}^1$ and let $M, N \in {}_R\text{Mod}_{\text{f.g.p.}}$. Then, $M \oplus N \in {}_R\text{Mod}_{\text{f.g.p.}}$.

2.2 The Geometric View: Classifying Vector Bundles & The Serre-Swan Theorem

Let's start with the basic theory of vector bundles. Since vector bundles will also come up again in our work on noncommutative geometry much later, we will treat the theory over here in some level of generality. At the same time, we don't want to dive too deep into it, so we will only present that which is needed for the proof of the so-called Serre-Swan Theorem. Indeed, this theorem provides an example of a geometry-algebra correspondence which serves as an alternative way of thinking about K-theory. Much of our approach to (vector) bundles owes itself to [3, Chapter I], [55, Chapter 1-4, p.1-54], [99] and [49, Chapter I.5, p.21-37]. Let's start with the definition of a vector bundle on a topological space.

Definition 2.5

Let $X \in \text{Top}$. A $\mathbb{K}$ -bundle over X is pair (E, p) consisting of the following data:

1. $E \in \text{Top}$.
2. A continuous map $p : E \rightarrow X$.
3. For each $x \in X$, we have that:

$E_x \stackrel{\text{def}}{=} p^{\leftarrow}(\{x\}),$
is a $\mathbb{K}$ -normed space such that $E = \bigsqcup_{x \in X} E_x$.

We have names for these constituents of the data of a $\mathbb{K}$ -bundle:

1. p is the projection map.
2. E is the total space of the bundle.
3. X is the base space of the bundle.
4. For each $x \in X$, the space E_x is the fiber over x .

If $(E, p), (F, q)$ are $\mathbb{K}$ -bundles over X , then a morphism:

$$\phi : (E, p) \rightarrow (F, q),$$

of $\mathbb{K}$ -bundles is a continuous map $\phi : E \rightarrow F$ such that the following diagram:

$$\begin{array}{ccc} E & \xrightarrow{p} & X \\ \phi \downarrow & \nearrow q & \\ F & & \end{array}$$

commutes and for each $x \in X$, the map $\phi|_{E_x} \in \mathcal{L}(E_x, F_x)$.

Let's extract a category from the given data. We need two exercises for that purpose.

| **Claim 2.3.** Let $X \in \text{Top}$ and let (E, p) be a $\mathbb{K}$ -bundle over X . Then, Id_E is a $\mathbb{K}$ -bundle morphism.

| **Claim 2.4.** The composition of two $\mathbb{K}$ -bundle morphisms of $\mathbb{K}$ -bundles is a $\mathbb{K}$ -bundle morphism.

These two exercises allow us build our category of $\mathbb{K}$ -bundles over X . As always, we do this as an example.

Example 2.6. Let $X \in \text{Top}$. We define a structure $\text{Bund}_{\mathbb{K}}(X)$ as follows:

1. The objects are $\mathbb{K}$ -bundles over X .
2. The morphisms are just $\mathbb{K}$ -bundle morphisms, with the obvious composition law.

This data forms a category; this is known as the category of $\mathbb{K}$ -bundles over X .

Okay, we have two remarks about Definition 2.5; please read through these carefully.

Remark 2.7. Let's highlight some differences between our definition and the one presented in some books.

1. The name “quasi-bundle” is typically used.
2. The bundles in books typically have finite-dimensional fibers.

Obviously, the underlying reason for (2) is that finite-dimensional things are easier to deal with and one can just use standard techniques from linear algebra. However, we will eventually want to deal with bundles that are not bundles that do not have finite-dimensional fibers so we thought it would be a good idea to get rid of the finite-dimensionality, especially since it's not needed for the definition to work^a.

^aThis is also a slight subtlety about Atiyah's treatment of K-theory in [3, Chapter I]; he often uses the fact that certain operations are continuous but he doesn't specify any topology on the fibers. Obviously, most people just assume that the topology is the Euclidean one but, a priori, it's not clear that this is the only choice of reasonable topology. Indeed, standard results from functional analysis ensure that we can narrow down our choice of topology, though these are obviously not trivial.

Remark 2.8. As we mentioned in our motivation above, one often doesn't just consider infinite-dimensional bundles over a topological space. Usually, the topological space will be equipped with some measure-theoretic structure for various purposes. For instance, one may reasonably require consideration of $\mathbb{K}$ -bundles over a positive Radon measure space & this can be used for the applications we've alluded to in the discussion above.

Let's introduce some terminology.

Terminology 2.9. Let $X \in \text{Top}$ and let $(E, p) \in \text{Bund}_{\mathbb{K}}(X)$.

1. A $\mathbb{K}$ -bundle in which all fibers are finite-dimensional is known as a finite-dimensional $\mathbb{K}$ -bundle.

2. A $\mathbb{K}$ -bundle in which all fibers are $\mathbb{K}$ -Banach spaces is known as a $\mathbb{K}$ -Banach bundle.
3. A $\mathbb{K}$ -bundle in which all fibers are $\mathbb{K}$ -Hilbert spaces is known as a $\mathbb{K}$ -Hilbert bundle.

For the moment, our real interest is in *finite-dimensional* (vector) bundles; note that in this case, we require all fibers to be of the same dimension and we will just view the fibers as $\mathbb{K}^n$ when it is convenient. Indeed, the category of finite-dimensional vector bundles is one of the two main actors in the Serre-Swan Theorem, as we will demonstrate.

Let us introduce a bit of notation regarding some interesting subcategories/categories of bundles.

Notation 2.10. Let $X \in \text{Top}$.

1. The category of bundles over $\mathbb{K}$ with finite-dimensional fibers is denoted by $\text{f.d. Bund}_{\mathbb{K}}$.
2. The category of all vector bundles over $\mathbb{K}$ is denoted by $\text{VecBund}_{\mathbb{K}}$.
3. The category of all vector bundles over $\mathbb{K}$ is denoted by $\text{f.d. VecBund}_{\mathbb{K}}$.

We have to fuss about these distinctions because all of these notions will be useful for us to keep around.

Here's the canonical example of a $\mathbb{K}$ -bundle over a topological space; it will be very important.

Example 2.11. Let $X \in \text{Top}$ and let V be a $\mathbb{K}$ -normed space. Define:

1. $E \stackrel{\text{def}}{=} X \times V$.
2. We define:

$$p : E \rightarrow X, (x, v) \mapsto p(x, v) \stackrel{\text{def}}{=} x,$$

as the projection map.

Now, we have the following claim.

| **Claim.** (E, p) forms a $\mathbb{K}$ -bundle over X .

You'll verify this in **Claim 2.5**; this is known as the product bundle.

Here's the associated exercise.

| **Claim 2.5.** (E, p) forms a $\mathbb{K}$ -bundle over X .

Here's another reason why we care about the classification of (vector) bundles (which we introduce in a bit).

Example 2.12. A $\mathbb{K}$ -bundle over a one-point space is literally just a $\mathbb{K}$ -normed space. In other words, the classification of $\mathbb{K}$ -vector spaces is generalized by the classification of $\mathbb{K}$ -bundles. In fact, we will see that bundles over a one-point space are just vector bundles. Of course, this leads us naturally to the interpretation that the study of vector bundles is a kind of "parametrized linear algebra". We will see more evidence of this viewpoint later.

Now, what exactly is a vector bundle? Well, a bundle is just something that locally looks like a product bundle. This means that it's easy to "do algebra" locally on some vector bundle and that's encapsulated by the next definition.

Definition 2.13

Let $X \in \text{Top}$ and let $(E, p) \in \text{Bund}_{\mathbb{K}}(X)$.

1. (E, p) is trivial iff it is isomorphic to some product bundle.
2. (E, p) is locally trivial iff for every $x \in X$, there is a nbhd $U \in \mathcal{N}_x$ such that $E|_U$ is trivial over U .

A locally trivial $\mathbb{K}$ -bundle is also known as a vector $\mathbb{K}$ -bundle.

There are a number of interesting constructions involving bundles and we will certainly discuss these whenever we use bundle theory in our work (or in other works). For now, we discuss the ones necessary for the Serre-Swan Theorem.

Construction 2.14. Let $X \in \text{Top}$, let $Y \subseteq X$ be a subspace and let $(E, p) \in \text{Bund}_{\mathbb{K}}(X)$. Then, we define:

1. $E|_Y \stackrel{\text{def}}{=} \bigsqcup_{x \in Y} p^{\leftarrow}(\{x\})$.
2. $q \stackrel{\text{def}}{=} p|_{E|_Y}$.

Now, we have the following claim

| **Claim.** $(E|_Y, q)$ is a $\mathbb{K}$ -bundle over Y .

This holds quite literally by construction; the only thing we need to check is that the restriction of q to each fiber is a linear map but this is true because the restriction of p to each fiber is a linear map.

Let's generalize this construction; let $Y \in \text{Top}$ and let $f : Y \rightarrow X$ be continuous. Define:

1. $f^*(E) \stackrel{\text{def}}{=} \bigsqcup_{y \in Y} p^{\leftarrow}(\{f(y)\})$.
2. We give $p^{\leftarrow}(\{f(y)\})$ the vector space structure of $E_{f(y)}$ for each $y \in Y$.
3. We know that $f^*(E)$ is a subspace of E , so we just define $q = p|_{f^*(E)}$.

Now, we have the following claim.

| **Claim.** $(f^*(E), q)$ is a $\mathbb{K}$ -bundle over Y .

This is clear by construction. We refer to this as the induced bundle of (E, p) over f . In the case where f is the

inclusion of Y as a subspace of X into X , we obtain the restriction construction which we went through earlier^a.

^aThey're not technically the same but they are isomorphic, as you can show easily.

Construction 2.15. Let $X \in \mathbf{Top}$ and suppose that we have $\mathbb{K}$ -bundles:

$$\begin{aligned} E &\xrightarrow{p} X, \\ F &\xrightarrow{q} X. \end{aligned}$$

These fit into the following pullback diagram^a:

$$\begin{array}{ccc} E \oplus F & \longrightarrow & F \\ \downarrow & & \downarrow q \\ E & \xrightarrow{p} & X \end{array}$$

where we see that:

$$E \oplus F = \{(x, x') \in E \times F : p(x) = q(x')\}.$$

Now, we define:

$$\phi : E \oplus F \rightarrow X, (x, x') \mapsto \phi(x, x') \stackrel{\text{def}}{=} p(x),$$

and this is well-defined by our explicit expression for $E \oplus F$. Here's a simple result.

Claim. The pair $(E \oplus F, \phi)$ forms a $\mathbb{K}$ -bundle.

This is clear. If both bundles are vector bundles, then this should also be a vector bundle.

Claim. If both bundles are vector bundles, then the pair $(E \oplus F, \phi)$ is a vector $\mathbb{K}$ -bundle.

You'll prove this in **Claim 2.6**. We refer to this as the Whitney sum of two vector $\mathbb{K}$ -bundles.

^aWe are computing the pullback in $\mathbf{Top}$ and this is a complete category, so it makes sense.

Here's the associated exercise.

Claim 2.6. Prove the second claim of **Construction 2.15**.

Of course, we have notation regarding categories of vector bundles.

Notation 2.16. Let $X \in \mathbf{Top}$.

1. $\mathbf{VecBund}_{\mathbb{K}}(X)$ is the full subcategory of $\mathbf{Bund}_{\mathbb{K}}(X)$ consisting of vector $\mathbb{K}$ -bundles.
 2. $\mathbf{f.d. VecBund}_{\mathbb{K}}(X)$ is the full subcategory of $\mathbf{f.d. Bund}_{\mathbb{K}}(X)$ consisting of finite-dimensional vector $\mathbb{K}$ -bundles.
- Of course, a finite-dimensional vector $\mathbb{K}$ -bundle is just a vector $\mathbb{K}$ -bundle which is a finite-dimensional bundle^a.

^aObviously, all of this is happening over the base space X .

To move towards the statement of the Serre-Swan Theorem, the main result of this lecture, we will need some terminology and an interesting construction. This is where we bring in module theory into the mix.

Terminology 2.17. Let $X \in \mathbf{Top}$ and let $(E, p) \in \mathbf{Bund}_{\mathbb{K}}(X)$. A section of p is an $s \in \mathcal{C}(X, E)$ such that:

$$\forall x \in X : (p \circ s)(x) = x.$$

We also say that s is a section of the bundle; the set of all sections is denoted by $\Gamma(X, E)$.

Obviously, what's happening here is that s is some kind of “generalized function” which is picking out an $s(x)$ from each E_x for all $x \in X$. You might have seen concrete instances in which doing this can be useful, both in geometry & in other parts of mathematics². Now, we proceed with our construction; the goal is to turn the set of all sections of a bundle into a module. Under relatively mild assumptions, we will prove that this module is finitely-generated and projective.

Construction 2.18. Let $X \in \mathbf{Top}$ and let $(E, p) \in \mathbf{Bund}_{\mathbb{K}}(X)$.

$$\forall s_1, s_2 \in \Gamma(X, E) : \forall x \in X : (s_1 + s_2)(x) \stackrel{\text{def}}{=} s_1(x) + s_2(x),$$

$$\forall f \in \mathcal{C}(X, \mathbb{K}) : \forall s \in \Gamma(X, E) : \forall x \in X : (fs)(x) \stackrel{\text{def}}{=} f(x)s(x).$$

So, this is how we define the structure of a left $\mathcal{C}(X, \mathbb{K})$ -module on $\Gamma(X, E)$. Now, does this make sense?

1. Well, if $s_1, s_2 \in \Gamma(X, E)$, then $s_1(x), s_2(x) \in E_x$ because:

$$p(s_1(x)) = p(s_2(x)) = x.$$

So, we are just adding two vectors in the vector space E_x and this is perfectly legitimate.

2. The multiplication is well-defined as well, because $s(x) \in E_x$ and $f(x)$ just scales it.

Now, one needs to actually check that $\Gamma(X, E) \in \mathcal{C}(X, \mathbb{K})\mathbf{Mod}$ but this is a very easy computation. The more important point is that this construction does seem to be functorial. Indeed, we construct a map of categories:

$$\Gamma(X, -) : \mathbf{Bund}_{\mathbb{K}}(X) \rightarrow \mathcal{C}(X, \mathbb{K})\mathbf{Mod},$$

as follows:

²For instance, recall any situation where you have a sequence $(A_n)_{n \in \mathbb{N}}$ of sets and you're asked to construct a sequence of elements by selecting an $x_n \in A_n$ for each $n \in \mathbb{N}$. Formally, this will just be a function from $\mathbb{N}$ to $\bigcup_{n \in \mathbb{N}} A_n$.

1. If $(E, p) \in \text{Bund}_{\mathbb{K}}(X)$, then $\Gamma(X, -)(E, p) \stackrel{\text{def}}{=} \Gamma(X, E)$.
2. If $(E, p), (F, q) \in \text{Bund}_{\mathbb{K}}(X)$ and $\phi : E \rightarrow F$ is a morphism of bundles, then we define:

$$\Gamma(X, \phi) : \Gamma(X, E) \rightarrow \Gamma(X, F), \quad s \mapsto \Gamma(X, \phi)(s) \stackrel{\text{def}}{=} \phi \circ s.$$

It's easily shown that this is well-defined^a.

It is easily shown that this is a functor. In fact, we can show even more.

| **Claim.** If $(E, p), (F, q) \in \text{Bund}_{\mathbb{K}}$, then $\Gamma(X, E \oplus F) \simeq \Gamma(X, E) \oplus \Gamma(X, F)$.

You'll show this in **Claim 2.7**. So, there is a plausible connection between finitely-generated projective modules over $\mathcal{C}(X, \mathbb{K})$ and finite-dimensional vector bundles over X . The trick is to make this connection far more precise.

^aRecall that you need to show that $\phi \circ s$ is a section.

Here's the exercise associated to the construction.

| **Claim 2.7.** Prove the claim in [Construction 2.18](#).

So, here's the statement of the Serre-Swan Theorem.

Theorem 2.19. The Serre-Swan Theorem

Let $X \in \text{CHaus}$. The functor:

$$\Gamma(X, -) : \text{f.d. VecBund}_{\mathbb{K}}(X) \rightarrow \mathcal{C}(X, \mathbb{K})\text{Mod}_{\text{f.g.p.}},$$

is an equivalence of categories.

Here are two remarks about this result.

| **Remark 2.20.** In my mind, the best way to view what this result does is that it's a Gelfand Duality-type result. Indeed, it tells us which

| **Remark 2.21.** Some algebraic geometry thing about Riemann-Roch and Serre's theorem about varieties.

Let us establish the results needed to prove the theorem above. This will constitute the rest of the content of this section and the important thing for you to pick up here is the geometric mode of thinking about the structure of this result.

Proposition 2.22

Let V, W be finite-dimensional $\mathbb{K}$ -normed spaces. TFSH:

1. If $X \in \text{Top}$, then $\mathcal{C}(X, V) \cong_{\mathcal{C}(X, \mathbb{K})\text{Mod}} \Gamma(X, X \times V)$.
2. If $X \in \text{Top}$, then $\mathcal{C}(X, \mathcal{L}(V, W))$ is in bijection with $\text{VecBund}_{\mathbb{K}}(X) (X \times V, X \times W)$.
3. If $X \in \text{CHaus}$ and $(E, p) \in \text{f.d. VecBund}_{\mathbb{K}}(X)$, then $\Gamma(X, E)$ is finitely-generated.

You'll certainly prove this with us.

Proof of Proposition 2.22. We'll prove each statement one-by-one.

| **Claim.** Statement (1) holds.

Of course, one can easily turn $\mathcal{C}(X, V)$ into a $\mathcal{C}(X, \mathbb{K})$ -module simply by defining the operations pointwise; this is standard so we do not spell it out here. On the other hand, let $f \in \mathcal{C}(X, V)$ and define:

$$\phi(f) : X \rightarrow X \times V, \quad x \mapsto \phi(f)(x) \stackrel{\text{def}}{=} (x, f(x)).$$

This gives us a map $\phi : \mathcal{C}(X, V) \rightarrow \Gamma(X, X \times V)$. Actually, does it?

| **Claim 2.8.** $\text{ran}(\phi) \subseteq \Gamma(X, X \times V)$.

Finally, we need to show that ϕ is an isomorphism.

| **Claim 2.9.** ϕ is bijective.

This proves statement (1).

| **Claim.** Statement (2) holds.

In this situation, let $f \in \mathcal{C}(X, \mathcal{L}(V, W))$. Now, we define:

$$\hat{f} : X \times V \rightarrow X \times W, \quad (x, v) \mapsto \hat{f}(x, v) \stackrel{\text{def}}{=} (x, f(x)(v)).$$

At this stage, we leave it as an exercise to show the following statements:

1. $\hat{f}$ is a morphism of the given product bundles.
2. $f \mapsto \hat{f}$ is a bijection.

Showing them is a matter of carrying out completely easy computations as we have done above.

| **Claim.** Statement (3) holds.

For each $x \in X$, there is an open nbhd U_x of x such that $E|_{U_x}$ is trivial over U_x . So, we have an open cover $(U_x)_{x \in X}$ of

X and we can extract a finite sequence $(x_i)_{i=1}^m$ such that:

$$X = \bigcup_{i=1}^m U_{x_i}.$$

Now, we define two classes of functions:

1. We define the constant functions $e_j^{x_i} : U_{x_i} \rightarrow \mathbb{K}^n$ which map to the standard basis of $\mathbb{K}^n$.
2. We define a partition of unity $(f_i)_{i=1}^m$ subordinate to the finite subcover $(U_{x_i})_{i=1}^m$.

Now, we define:

$$e_{ij} \stackrel{\text{def}}{=} f_i e_j^{x_i},$$

where this is defined on U_{x_i} point-wise and it can be extended to X via continuation by 0 because each f_i is supported in U_{x_i} . By construction, we see that the functions (e_{ij}) generate $\Gamma(X, E)$ as a $\mathcal{C}(X, \mathbb{K})$ -module and since this is a finite generating set, we've shown that $\Gamma(X, E)$ is finitely-generated. This proves the desired result.

In fact, we can extend statement (3) above and even say that $\Gamma(X, E)$ is projective.

Proposition 2.23

Let $X \in \text{CHaus}$ and let $(E, p) \in {}_{\text{f.d.}} \text{VecBund}_{\mathbb{K}}(X)$. Then, $\Gamma(X, E) \in {}_{\mathcal{C}(X, \mathbb{K})} \text{Mod}_{\text{f.g.p.}}$.

| **Proof.** This is a calculation. ■

At this stage, we are ready for the proof of the Serre-Swan Theorem.

| **Proof of Theorem 2.19.** Blah Blah Blah. ■

2.3 The Analytic View: The von Neumann Dimension

3 Constructing Unital Algebraic K_0

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| Motivation.

3.1 Some Basics on Projective Modules & The Grothendieck Construction

So, as we mentioned above, we are going to take the algebraic approach to K-theory and we'll start by constructing K_0 . This is a functor $K_0 : C^*Alg \rightarrow AbGrp$. Since our approach is algebraic, we will instead construct a functor:

$$K_0 : Rng \rightarrow AbGrp.$$

In this lecture, we will focus on the unital situation so what you can expect, by the end of this section, is that we would have constructed K_0 on Rng^1 . Once we have that, the extension to all rings is a matter of working with the unitization of a ring. But let's not get ahead of ourselves. Let's describe the process for constructing K_0 . There are just two steps:

1. We construct a functor:

$$\mathbb{V} : Rng^1 \rightarrow AbSemiGrp,$$

which takes a unital ring R and produces an abelian semigroup $\mathbb{V}(R)$ from it.

2. We compose $\mathbb{V}$ with the so-called Grothendieck construction, a functor $\mathcal{G} : AbSemiGrp \rightarrow AbGrp$.

So, the first half of this section is dedicated towards explaining the first step and the second half explains the second step.

3.1.1 The First Step

Let's start by discussing the technicalities behind the first step. Here's some notation.

Notation 3.1. Let $R \in Rng^1$ and let $M \in {}_RMod_{f.g.p.}$.

1. We let:

$$[M],$$

denote the isomorphism class of M ; this is the set of all f.g.p. R -modules which are isomorphic to M .

2. We let $\mathbb{V}(R)$ denote the set of all isomorphism classes of f.g.p. R -modules.

Note that in [REF], we use $\mathbb{V}(R)$ to denote simply all projective R -modules. So, there is a slight disparity in notation but it will not cause any confusion here because we will, soon, refrain from working so much with R -modules.

Now, we can proceed with the construction.

Construction 3.2. Let $R \in Rng^1$ and let $M, N \in {}_RMod_{f.g.p.}$. We define:

$$[M] + [N] \stackrel{\text{def}}{=} [M \oplus N].$$

This gives $\mathbb{V}(R)$ a binary operation. In fact, we can say much more.

| **Claim.** $\mathbb{V}(R)$ becomes an abelian semigroup.

You'll prove this in **Claim 3.1**. Now, how do we cook up a functor from this construction? Let $R, S \in Rng^1$ and let $f \in Rng^1(R, S)$. We want to define $\mathbb{V}(f) : \mathbb{V}(R) \rightarrow \mathbb{V}(S)$ as follows; let $M \in {}_RMod_{f.g.p.}$. Let's consider^a:

$$S \otimes_f M.$$

So, one first needs to check that this is f.g.p.

| **Claim.** $S \otimes_f M$ is f.g.p over S .

You'll prove this in **Claim 3.2**. So, we define:

$$\mathbb{V}(f)([M]) \stackrel{\text{def}}{=} [S \otimes_f M].$$

Since tensor products respect isomorphisms, it follows that this is well-defined. In fact, we can say more.

| **Claim.** $\mathbb{V}(f)$ is a morphism of abelian semigroups.

You'll prove this in **Claim 3.3**. So, we have successfully constructed a map:

$$\mathbb{V} : Rng^1 \rightarrow AbSemiGrp,$$

of categories. The last step is to demonstrate functoriality.

| **Claim.** $\mathbb{V}$ is a functor.

You'll prove this in **Claim 3.4**. This completes step 1 of the process of constructing K_0 .

^aOne should think about why this makes sense.

Here are the promised exercises.

| **Claim 3.1.** Prove claim (1) of [Construction 3.2](#).

| **Claim 3.2.** Prove claim (2) of [Construction 3.2](#).

| **Claim 3.3.** Prove claim (3) of [Construction 3.2](#).

| **Claim 3.4.** Prove claim (4) of [Construction 3.2](#).

This construction demonstrates that we can cook up an abelian semigroup from f.g.p modules over a unital ring. Our next goal is to provide a way for us to “turn” an abelian semigroup into an abelian group.

3.1.2 The Second Step

More specifically, we want to construct a functor:

$$\mathcal{G} : \text{AbSemiGrp} \rightarrow \text{AbGrp},$$

which allows us to obtain abelian groups from abelian semigroups. This is the so-called Grothendieck construction and composing this with the functor $\mathbb{V}$ as constructed above will give us K_0 on Rng^1 . Developing this material is an exercise in gaining control over the construction. For the sake of completeness, we will actually provide two constructions which do this for us, though it will turn out that the second of these is very intuitive and far more useful for computational purposes. Let's start by defining the Grothendieck group of an abelian semigroup.

Definition 3.3

Let $S \in \text{AbSemiGrp}$. A Grothendieck group of S is a pair (G, γ_S) consisting of:

1. $G \in \text{AbGrp}$.
2. An additive map $\gamma_S : S \rightarrow G$.

such that if $H \in \text{AbGrp}$ and $f : S \rightarrow H$ is an additive map, then:

$$\exists! g \in \text{AbGrp}(G, H) : f = g \circ \gamma_S.$$

That is, there exists a unique group homomorphism $g : G \rightarrow H$ such that the following diagram:

$$\begin{array}{ccc} S & \xrightarrow{\gamma_S} & G \\ & \searrow f & \downarrow g \\ & & H \end{array}$$

commutes.

Here's an easy result.

| **Claim 3.5.** The Grothendieck group of an abelian semigroup is unique up to unique isomorphism.

This result motivates the next bit of notation.

| **Notation 3.4.** Let $S \in \text{AbSemiGrp}$. We denote by:

$$(G(S), \gamma_S),$$

the Grothendieck group of S if it exists.

We will give some examples of abelian semigroups with Grothendieck groups via exercises.

| **Claim 3.6.** If $S \in \text{AbGrp}$, then $G(S) = S$ and $\gamma_S = \text{Id}_S$.

| **Claim 3.7.** If $S = \emptyset$, then $G(S) = 0$ and γ_S is the unique map from $\emptyset$ to 0 .

| **Claim 3.8.** Let $S = \mathbb{N}$. Then:

$$G(S) = \mathbb{Z},$$

and γ_S is just the inclusion of $\mathbb{N}$ into $\mathbb{Z}$.

| **Claim 3.9.** Let $S = \mathbb{N}_0$. Then:

$$G(S) = \mathbb{Z},$$

and γ_S is the inclusion of $\mathbb{N}_0$ into $\mathbb{Z}$.

| **Claim 3.10.** Let $S = \mathbb{Z} \setminus \{0\}$ with its standard multiplicative semigroup structure. Then:

$$G(S) = \mathbb{Q} \setminus \{0\},$$

with its multiplicative structure and the map γ_S is the inclusion of S into $G(S)$.

| **Claim 3.11.** Let $S \in \text{AbSemiGrp}$. We define:

$$S(\infty) \stackrel{\text{def}}{=} S \cup \{\infty\},$$

and we require that:

$$\forall s \in S(\infty) : s + \infty \stackrel{\text{def}}{=} \infty.$$

Then, we have that:

$$G(S(\infty)) = 0,$$

and γ_S is the zero map from $S \rightarrow 0$.

So, it remains to prove that the Grothendieck group of an abelian semigroup exists. Now, there are two constructions of the Grothendieck group that we would like to present. The first of these is absolutely straightforward.

| **Construction 3.5.** Let $S \in \text{AbSemiGrp}$ and define:

$$F \stackrel{\text{def}}{=} \mathbb{Z}^{(S)} \stackrel{\text{def}}{=} \bigoplus_{s \in S} \mathbb{Z}.$$

That is, this is the free abelian group with basis S . Now, we consider the inclusion:

$$i : S \rightarrow F.$$

This is not an additive map but we turn it into one by taking a suitable quotient. Define:

$$F_0 \stackrel{\text{def}}{=} \{i(x) + i(y) - i(x+y) : x, y \in S\}.$$

This is a subgroup of F . Now, we define:

$$G(S) \stackrel{\text{def}}{=} F/F_0,$$

$$\gamma_S \stackrel{\text{def}}{=} \pi \circ i,$$

where $\pi : F \rightarrow F/F_0$ is the canonical projection. We have the following claim.

| Claim. $(G(S), \gamma_S)$ is the Grothendieck group of S .

In fact, this will follow essentially by the universal property of the free abelian group. So, let $H \in \text{AbGrp}$ and let $f : S \rightarrow H$ be an additive map. By the universal property of the free abelian group, we obtain a group homomorphism $\hat{f} : F \rightarrow H$ such that $\hat{f} \circ i = f$. Now, we simply descend $\hat{f}$ to the quotient $G(S)$ and since every group homomorphism obtained at each step of the process above is unique, the claim is proved.

The second construction is much less short and sweet but it is more useful for computational purposes. Actually, it is a generalization of something we've already done at a much earlier point; we previously constructed the integers from the natural numbers simply by taking formal differences of natural numbers & considering their equivalence classes under a reasonable equivalence relation. To be more precise, here are the concrete steps we took:

1. We construct the cartesian square $\mathbb{N} \times \mathbb{N}$.
2. We define a relation on $\mathbb{N} \times \mathbb{N}$ as follows:

$$\forall a, a', b, b' \in \mathbb{N} : (a, b) \sim (a', b') \Leftrightarrow a + b' = a' + b.$$

One can show that this is an equivalence relation.

3. Define $\mathbb{Z}$ as $(\mathbb{N} \times \mathbb{N}) / \sim$.

Now, $\mathbb{N}$ has some nice properties that allowed us to carry out the construction above with no problems but it's more complicated with abelian semigroups. The key bit that needs to be changed is conditions (2), as we'll explain now.

Construction 3.6. Let $S \in \text{AbSemiGrp}$. We define:

$$\forall x, x', y, y' \in S : (x, y) \sim (x', y') \stackrel{\text{def}}{\Leftrightarrow} (\exists z \in S : (x + y') + z = (x' + y) + z).$$

Let's start with the following claim.

| Claim. This is an equivalence relation on $S \times S$.

You'll prove this in **Claim 3.12**. For $x, y \in S$, we further write:

$$\langle x, y \rangle \stackrel{\text{def}}{=} \{(x', y') \in S \times S : (x, y) \sim (x', y')\},$$

for the equivalence class of (x, y) under the equivalence relation. Now, we define:

$$G(S) \stackrel{\text{def}}{=} (S \times S) / \sim.$$

Now, we need to equip $G(S)$ with the structure of a group. Define:

$$\forall x, x', y, y' \in S : \langle x, y \rangle + \langle x', y' \rangle \stackrel{\text{def}}{=} \langle x + x', y + y' \rangle.$$

We will leave it to you to establish that this turns $G(S)$ into an abelian group.

| Claim. With the addition above, $G(S)$ becomes an abelian group.

You'll prove this in **Claim 3.13**. We still need to define the map from S into $G(S)$. Well, this is easy:

$$\gamma_S : S \rightarrow G(S), \quad x \mapsto \gamma_S(x) \stackrel{\text{def}}{=} \langle x + y, y \rangle.$$

In the next result, we establish some key properties of this map.

Claim. TFSH:

1. Let $y, y' \in S$. Then:

$$\forall x \in S : \langle x + y, y \rangle = \langle x + y', y' \rangle.$$

This shows that γ_S does not depend on y .

2. $\forall x, y \in S : \langle x, y \rangle = \gamma_S(x) - \gamma_S(y)$.
3. γ_S is additive.

You'll prove this in **Claim 3.14**. At this stage, we have everything we need to prove that this construction works.

| Claim. $(G(S), \gamma_S)$ is the Grothendieck group of S .

We'll need to verify that this satisfies the universal property. Let's start with uniqueness.

Subclaim. Let ψ, ψ' be group homomorphisms such that:

$$\phi = \psi \circ \gamma_S = \psi' \circ \gamma_S.$$

Thn, $\psi = \psi'$.

Let $x, y \in S$. Now, we have the following computation^a:

$$\begin{aligned}\psi(\langle x, y \rangle) &= \psi(\gamma_S(x) - \gamma_S(y)) \\ &= \psi(\gamma_S(x)) - \psi(\gamma_S(y)) \\ &= \psi'(\gamma_S(x)) - \psi'(\gamma_S(y)) \\ &= \psi'(\gamma_S(x) - \gamma_S(y)) \\ &= \psi'(\langle x, y \rangle),\end{aligned}$$

as was desired. Now, the proof of uniqueness tells us how we should define ψ .

Subclaim. The pair above satisfies the universal property.

We define:

$$\forall x, y \in S : \psi(\langle x, y \rangle) \stackrel{\text{def}}{=} \gamma_S(x) - \gamma_S(y).$$

Let's check that this relation is a function; let $x, x', y, y' \in S$ with $(x, y) \sim (x', y')$. This means that:

$$\exists z \in S : x + y' + z = x' + y + z.$$

Applying γ_S on both sides, we find that:

$$\gamma_S(x) + \gamma_S(y') + \gamma_S(z) = \gamma_S(x') + \gamma_S(y) + \gamma_S(z).$$

Since $G(S)$ is a group, we can cancel $\gamma_S(z)$ on both sides. Therefore, we get that:

$$\gamma_S(x) - \gamma_S(y) = \gamma_S(x') - \gamma_S(y'),$$

which implies that ψ is a function. Now, let's check that this preserves products; let $x, x', y, y' \in S$. Then:

$$\begin{aligned}\psi(\langle x, y \rangle + \langle x', y' \rangle) &= \psi(\langle x + x', y + y' \rangle) \\ &= \gamma_S(x + x') - \gamma_S(y + y') \\ &= \gamma_S(x) + \gamma_S(x') - \gamma_S(y) - \gamma_S(y') \\ &= (\gamma_S(x) - \gamma_S(y)) + (\gamma_S(x') - \gamma_S(y')) \\ &= \psi(\langle x, y \rangle) + \psi(\langle x', y' \rangle),\end{aligned}$$

which proves that ψ is a group homomorphism. Finally, it's clear^b that ψ causes the given diagram to commute.

This completes our second construction of the Grothendieck group of S .

^aOver here, we use the properties of γ_S specified above.

^bIt follows immediately by the definition of γ_S and a two-step calculation.

Here are the exercises associated to the construction.

Claim 3.12. Prove claim (1) of [Construction 3.6](#).

Claim 3.13. Prove claim (2) of [Construction 3.6](#).

Claim 3.14. Prove claim (3) of [Construction 3.6](#).

We have a remark about the conceptual nature of the construction.

Remark 3.7. The construction above is quite similar to what one does with the localization of a unital commutative ring. Indeed, let $R \in \text{AbRng}^1$ and let $S \subseteq R$ be multiplicative. We define a relation on $R \times S$ as follows:

$$\forall (x, y), (x', y') \in R \times S : (x, y) \sim (x', y') \stackrel{\text{def}}{\iff} (\exists t \in S : tx'y' = tx'y).$$

The specific use of the element t is meant to turn this relation into an equivalence relation. Of course, this is not necessary when we're constructing $\mathbb{Q}$ from $\mathbb{Z}$ because $\mathbb{Z}$ is an integral domain^a. These two constructions seem to be analogous in the sense that there is a simpler "version" of them that needs to be "fixed" when generalized.

^aIn general, it's not needed for the fraction field construction.

The next two results highlight specific properties of the map γ_S obtained in [Construction 3.6](#).

Claim 3.15. Let $S \in \text{AbSemiGrp}$ and let $x, y \in S$. TFAE:

1. $\gamma_S(x) = \gamma_S(y)$.
2. $\exists z \in S : x + z = y + z$.

Claim 3.16. Let $S \in \text{AbSemiGrp}$. TFAE:

1. γ_S is injective.
2. S has the cancellation property.

With all of this work, we are lead to an essential result which will let us cook up a functor.

Claim 3.17. Let $S, S' \in \text{AbSemiGrp}$ and let $\phi : S \rightarrow S'$ be additive. Then:

$$\exists! \psi \in \text{AbGrp}(G(S), G(S')) : \psi \circ \gamma_S = \gamma_{S'} \circ \phi.$$

That is, there is a unique $\psi \in \text{AbGrp}(G(S), G(S'))$ such that the following diagram:

$$\begin{array}{ccc} S & \xrightarrow{\phi} & S' \\ \gamma_S \downarrow & & \downarrow \gamma_{S'} \\ G(S) & \xrightarrow{\psi} & G(S') \end{array}$$

commutes. We will write ψ as $G(\phi)$.

Okay, we're ready to construct a functor associated to our work above.

Example 3.8. We define a map of categories:

$$\mathcal{G} : \text{AbSemiGrp} \rightarrow \text{AbGrp},$$

as follows:

1. If S is an abelian semigroup, then $\mathcal{G}(S) \stackrel{\text{def}}{=} G(S)$.
2. If S, S' are abelian semigroups and $\phi : S \rightarrow S'$ is an additive map, then $\mathcal{G}(\phi)$ simply denotes the unique group homomorphism from $G(S)$ to $G(S')$ whose existence is guaranteed by **Claim 3.17**.

Let's show that this data actually defines a functor.

| **Claim.** $\mathcal{G}$ is a functor.

You'll handle this in **Claim 3.18**; this is known as the Grothendieck construction on AbSemiGrp .

Here's the exercise associated to the example.

| **Claim 3.18.** Prove the claim of **Example 3.8**.

Here's a simple result.

| **Claim 3.19.** The functor $\mathcal{G}$ is left-adjoint to the forgetful functor $\iota : \text{AbGrp} \rightarrow \text{AbSemiGrp}$.

We have a general remark about this second construction.

Remark 3.9. One may note that this second construction of the Grothendieck embodies the idea of forming a group out of an abelian semigroup a lot better. Indeed, we really DO take formal differences of elements in the abelian semigroup to obtain a group, modulo an appropriate equivalence relation so that everything works.

With this preparation at hand, we are finally ready to define K_0 for unital rings.

Definition 3.10

We define:

$$K_0 \stackrel{\text{def}}{=} \mathcal{G} \circ \mathbb{V},$$

where $\mathbb{V}$ was defined in **Construction 3.2**. This is a functor $\text{Rng}^1 \rightarrow \text{AbGrp}$.

Here's a simple result which highlights easy properties of K_0 .

Claim 3.20. Let $R, S \in \text{Rng}^1$. TFSH:

1. $\mathbb{V}(R^{\text{op}}) \simeq \mathbb{V}(R)$.
2. $\mathbb{V}(R \oplus S) \simeq \mathbb{V}(R) \oplus \mathbb{V}(S)$.
3. If $A, B \in \text{AbSemiGrp}$, then $\mathcal{G}(A \oplus B) \simeq \mathcal{G}(A) \oplus \mathcal{G}(B)$.
4. $K_0(R^{\text{op}}) \simeq K_0(R)$.
5. $K_0(R \times S) \simeq K_0(R) \oplus K_0(S)$.

Okay, so this picture of algebraic K_0 is quite nice and you would, hopefully, agree that it is rather clean. Our final goal, before we move on to some concrete calculations, is to present a standard portrait of algebraic K_0 , something which we'll improve on in the next lecture. Here's some questionable notation.

Notation 3.11. Let $R \in \text{Rng}^1$ and let $M \in {}_R\text{Mod}_{\text{f.g.p.}}$. Then, we use the notation:

$$[M],$$

to denote an element of $K_0(R)$.

So, the reason this notation is questionable is because $[M]$ is technically an element of $\mathbb{V}(R)$. Therefore, we really should be using $\gamma([M])$ for an element of $K_0(R)$, where $\gamma : \mathbb{V}(R) \rightarrow K_0(R)$ is the Grothendieck map associated to the Grothendieck group. Nevertheless, we are not typically going to be interested in the functor $\mathbb{V}$, which is why the notation above is acceptable. In any event, here's the standard portrait as promised.

Proposition 3.12. The Standard Portrait of Unital Algebraic K_0 (Modules)

Let $R \in \text{Rng}^1$. TFSH:

1. $K_0(R) = \{[P] - [Q] : P, Q \in {}_R\text{Mod}_{\text{f.g.p.}}\} = \{[P] - [R^n] : n \in \mathbb{N}_0 \wedge P \in {}_R\text{Mod}_{\text{f.g.p.}}\}.$

2. If $P, Q \in {}_R\text{Mod}_{\text{f.g.p.}}$, then TFAE:

(a) $[P] = [Q]$.

(b) There exists a $L \in {}_R\text{Mod}_{\text{f.g.p.}}$ such that $P \oplus L = Q \oplus L$.

Moreover, we can replace L above by some R^n for $n \in \mathbb{N}_0$.

Proof. We'll prove each statement one-by-one.

1. The first equality follows from statement (2) of ?? and our choice of notation above. So, we just need to explain the second equality. Let $P, Q \in {}_R\text{Mod}_{\text{f.g.p.}}$ and let $Q' \in {}_R\text{Mod}$ be a direct summand in a finitely-generated free module alongside Q . That is:

$$\exists n \in \mathbb{N}_0 : Q \oplus Q' \simeq R^n.$$

Then, we see that:

$$\begin{aligned} [P] - [Q] &= [P] + [Q'] - [R^n] \\ &= [P \oplus Q'] - [R^n], \end{aligned}$$

which is exactly what we wanted so we're done.

2. The equivalence is just a special case of **Claim 3.15** so it remains to prove the last statement. Let $L' \in {}_R\text{Mod}$ so that $L \oplus L' \simeq R^n$ for some $n \in \mathbb{N}_0$. Then, we have the following computation:

$$\begin{aligned} P \oplus R^n &\simeq P \oplus (L \oplus L') \\ &\simeq (P \oplus L) \oplus L' \\ &\simeq (Q \oplus L) \oplus L' \\ &\simeq Q \oplus (L \oplus L') \\ &\simeq Q \oplus R^n, \end{aligned}$$

as was desired. ■

Here's a remark about this result.

Remark 3.13. The standard portrait above, while easy to establish, contains a lot of useful information. For example, it opens up the possibility of placing an order structure on K_0 -groups, at least for certain types of rings. We will certainly do this in the case of unital AF algebras and this extra structure will build up towards a K-theoretic classification theorem involving such algebras.

This result prompts the following bit of terminology that will be useful for later work.

Terminology 3.14. Let $R \in \text{Rng}^1$ and let $M, N \in {}_R\text{Mod}$. We say that M and N are stably isomorphic iff:

$$\exists P \in {}_R\text{Mod} : M \oplus P \simeq N \oplus P.$$

In particular, if M, N are f.g.p, then $[M] = [N]$ iff M and N are stably isomorphic.

So, while the description of algebraic K_0 is nice and everything, you will see in Section 3.2 that calculating with this standard portrait is rather annoying, especially if we wish to do calculations where we can't rely on convenient structure theorems. It would be nice if we could have a concrete picture and this is what we'll develop in the next lecture.

3.2 Elementary Computations of Algebraic K_0

For now, let us put the material presented above to good use by providing a number of interesting calculations. For the moment, many of our examples will be algebraic or algebraic-geometric.

Example 3.15. Let k be a field. We have the claim

Claim. $K_0(k) \simeq \mathbb{Z}$.

The goal is to show that:

$$\mathbb{V}(k) \simeq \mathbb{N}_0.$$

If this is true, then we have that:

$$K_0(k) = \mathcal{G}(\mathbb{V}(k)) \simeq \mathcal{G}(\mathbb{N}_0) \simeq \mathbb{Z}.$$

Firstly, every f.g.p module over k is just a finite-dimensional k -vector space. Now, we define:

$$\Phi : \text{Proj}(k) \rightarrow \mathbb{N}_0, [M] \mapsto \Phi([M]) \stackrel{\text{def}}{=} \dim(M),$$

where this is well-defined because the dimension is an isomorphism invariant of finite-dimensional k -vector spaces. In fact, Φ is an additive bijection; the bijectivity follows from the fact that $\dim$ is an isomorphism invariant and the additivity follows from standard results about the dimension. So, we see that $\mathbb{V}(k) \simeq \mathbb{N}_0$ and we're done.

We would like to generalize the above for PIDs. That is, for $R \in \text{Rng}^1$ being a PID, we want to show that:

$$K_0(R) \simeq \mathbb{Z}.$$

For the reader who is familiar with the structure theorem for finitely-generated modules over PIDs, one may expect that this is true. For those who aren't familiar with that result, we don't expect you to start learning its proof now as it is non-trivial. So, we present a self-contained classification of f.g.p modules over PIDs.

Proposition 3.16. Classification of Finitely-Generated Projective Modules Over PIDs

Let $R \in \text{Rng}^1$ be a PID and let $M \in {}_R\text{Mod}$ be f.g.p. Then:

$$\exists! n \in \mathbb{N}_0 : M \simeq R^n.$$

We refer to n as the rank of M and denote it by $\text{rank}(M)$.

Proof. See [87, Theorem 1.3.1, p.11]. ■

Now, we can proceed with the computation.

Example 3.17. Let R be a PID. We have the following claim.

Claim. $K_0(R) \simeq \mathbb{Z}$.

In fact, the argument is exactly as in Example 3.15. We define:

$$\Phi : \text{Proj}(R) \rightarrow \mathbb{N}_0, [M] \mapsto \Phi([M]) \stackrel{\text{def}}{=} \text{rank}(M).$$

This is an isomorphism in AbSemiGrp so we obtain an isomorphism in K -theory and we're done.

So, we cannot distinguish between fields and PIDs using K_0 . Here's another algebraic calculation.

Example 3.18. Let $k \in \text{Rng}^1$ and let $R = \text{End}_k(k^{(\mathbb{N})})$. We have a claim.

Claim. $K_0(R) = 0$.

We will show this in two steps:

1. Let's show that:

$$R \oplus R \simeq R.$$

We define a map:

$$\Phi : R \oplus R \rightarrow R, (T, S) \mapsto \Phi(T, S) \stackrel{\text{def}}{=} T \oplus S.$$

In fact, this is clearly a bijection and one can just construct an explicit inverse precisely because $\mathbb{N}$ contains countable proper subsets. Of course, it's clear that this is an R -linear map.

2. Let M be a f.g.p R -module. Let's show that:

$$M \oplus M \simeq M.$$

The cleanest argument for this is given as follows:

$$\begin{aligned} M &\simeq M \otimes_R R \\ &\simeq M \otimes_R (R \oplus R) \\ &\simeq (M \otimes_R R) \oplus (M \otimes_R R) \\ &\simeq M \oplus M, \end{aligned}$$

as was desired.

These two steps give us the claim because if M is a f.g.p R -module, then:

$$[M] = [M \oplus M] = [M] + [M],$$

which means that $[M] = 0$. That is, $K_0(R) = 0$.

Okay, these examples are nice and if you're unfamiliar with some of the algebraic results that went into doing these computations, then we suggest familiarizing yourself with them. The next natural step is to restrict one's focus to commutative unital rings and try to use structure theorems from commutative algebra to compute the K_0 -groups of certain special rings. Actually, one reason for doing this is that we wish to also be able to use geometric arguments and these are sometimes very useful in elucidating the *reasons* behind a certain computation going through.

Of course, the dream is to obtain geometric interpretations of these computations. In particular, we'd like to have a geometric picture of the rings that we start with and have that be closely related to the K_0 -group that we get at the end of the computation. We will try to be as articulate as we can about such connections, though one should realize that we can't be extremely detailed since this is not a book in algebraic geometry. For the first computation below, we will state a result from commutative algebra which is similar to Proposition 3.16.

Proposition 3.19

Let $R \in \text{AbRng}^1$ be local and let $M \in {}_R\text{Mod}$ be f.g.p. TFSH:

1. M is free.
2. M has a uniquely defined rank.

Proof. See [REF]. ■

Okay, here's the first example.

Example 3.20. Let $R \in \text{AbRng}^1$ be local. Then, we have the following claim.

Claim. $K_0(R) \simeq \mathbb{Z}$, with the isomorphism class of a free module of rank 1 being the generator.

| In light of Proposition 3.19, the argument is literally the same as in the two previous examples.

It is also desirable to compute the K_0 -groups of unital commutative Noetherian rings of Krull dimension 0. Of course, these are just the Artinian rings by a theorem of Hopkins as proved in [REF].

Example 3.21. Let $R \in \mathbf{AbRng}^1$ be Artinian. Here's our claim.

| **Claim.** $K_0(R) \simeq \mathbb{Z}^{|\mathrm{Spec}(R)|}$.

Actually, this follows from the structure theorem for Artinian rings^a. Indeed, we have that:

$$\begin{aligned} K_0(R) &\simeq K_0\left(\prod_{m \in \mathrm{Spec}(R)} R_{(m)}\right) \\ &\simeq \prod_{m \in \mathrm{Spec}(R)} K_0(R_{(m)}) \\ &\simeq \prod_{m \in \mathrm{Spec}(R)} \mathbb{Z}, \end{aligned}$$

where we used the result of Example 3.20 for this computation.

^aThis was covered in [REF].

For Example 3.21, there is a nice geometric argument available which effectively encodes the language of the structure theorem. The rest of this particular section of calculations will be spent explaining this geometric argument and this will require some background in algebraic geometry. In fact, it would be useful for someone to learn some of this material even if they're not interested in this particular direction for two major reasons:

1. It gives an algebro-geometric description of projective modules which might be more compelling than any other description you may have seen from other contexts. It may not be your preferred way of thinking about such modules but it's still nice to know that such an approach exists.
2. It will get you to learn something else for a change.

So, this concludes the totality of that which we wanted to discuss for this lecture. We've successfully constructed algebraic K_0 on the category of unital rings, we've established a standard picture for it and we've done a few calculations. In the next lecture, we will want to extend K_0 to the category of all rings and give an alternative description of K_0 in terms of idempotents. This will be crucial for computations of K-theory and it will be relevant even in our discussion of the K-theory of continuous functions, as we will cover in ??.

4 The Idempotent Picture of Algebraic K_0

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

Motivation.

4.1 Idempotents Are Important

The first piece of evidence that K_0 can be described alternatively by idempotents comes from statement (3) of **Claim 2.1**, where we characterized f.g.p modules as the ranges of idempotent matrices. Since the elements of K_0 -groups are isomorphism classes of f.g.p modules, it seems reasonable for us to have a characterization of when the ranges of idempotent matrices are isomorphic. Before we talk about that result, here's some useful notation.

Notation 4.1. Let $R \in \text{Rng}$. The set of all idempotents in R will be denoted by $\text{Idem}(R)$.

Now, we are lead to the following result.

Proposition 4.2

Let $R \in \text{Rng}^1$ and let $e, f \in \text{Idem}(M_{n \times n}(R))$. TFAE:

1. $\text{ran}(e)$ and $\text{ran}(f)$ are isomorphic.
2. $\exists x, y \in M_{n \times n}(R) : xy = e \wedge yx = f$.

In (2), we can always find x and y such that $xyx = x$ and $yxy = y$.

Proof. Let's start with the equivalence.

(1) $\Rightarrow$ (2): Let $\alpha : \text{ran}(e) \rightarrow \text{ran}(f)$ be an isomorphism. We define x and y by the following respective compositions:

$$R^n \xrightarrow{e} \text{ran}(e) \xrightarrow{\alpha} \text{ran}(f) \hookrightarrow R^n$$

$$R^n \xrightarrow{f} \text{ran}(f) \xrightarrow{\alpha^{-1}} \text{ran}(e) \hookrightarrow R^n$$

Observe that y has range $\text{ran}(e)$. So, we see that:

$$xy = \alpha \circ e \circ \alpha^{-1} \circ f = \alpha \circ \alpha^{-1} \circ f = f.$$

Similarly, x has range $\text{ran}(f)$. So, we see that:

$$yx = \alpha^{-1} \circ f \circ \alpha \circ e = \alpha^{-1} \circ \alpha \circ e = e,$$

and we're done.

(2) $\Rightarrow$ (1): Observe that:

$$xf = xyx = ex,$$

$$ye = yxy = fy.$$

So, x maps $\text{ran}(f)$ to $\text{ran}(e)$ while y maps $\text{ran}(e)$ to $\text{ran}(f)$. By the relations given in (2), we see that x and y are mutually inverse isomorphisms. This proves that $\text{ran}(f)$ and $\text{ran}(e)$ are isomorphic.

As for the last part, we leave it as an instructive exercise as follows.

Claim 4.1. Let x, y be as given in (2) and define:

$$x' \stackrel{\text{def}}{=} xyx,$$

$$y' \stackrel{\text{def}}{=} yxy.$$

Then, this satisfies (2). Also, $x'y'x' = x'$ and $y'x'y' = y'$.

This completes the proof. ■

The result prompts us to introduce the following piece of terminology.

Terminology 4.3. Let $R \in \text{Rng}$ and let $e, f \in R$.

1. e and f are algebraically equivalent iff:

$$\exists x, y \in R : e = xy \wedge f = yx.$$

We say that x and y implements the algebraic equivalence between e and f .

2. The pair $(x, y) \in R \times R$ is reduced^a iff they implement the algebraic equivalence between e and f while satisfying:

$$ex = x = xf,$$

$$fy = y = ye.$$

These are the reduced identities for the pair (x, y) .

3. Assume that R is unital. Then, e and f are similar iff:

$$\exists u \in R^\times : ue = fu.$$

That is, e and f are the “same” up to a “change-of-basis”.

Typically, we require that e and f are idempotents.

^aThis is inspired directly by the last part of Proposition 4.2.

We have a remark about this piece of terminology.

Remark 4.4. It doesn't take a genius to figure out that the notion of algebraic equivalence above looks remarkably similar to the notion of Murray-von Neumann equivalence of projections that we previously introduced in the theory of von Neumann algebras. I find it incredible that, in fact, the notion of MvN equivalence becomes absolutely crucial in understanding the K_0 -groups of C^* -algebras, as we will see in Section 7.

The point is that Proposition 4.2 is saying that whenever we have an algebraic equivalence between matrix idempotents, then we can always find a reduced pair which implements that algebraic equivalence. Here are other basic results.

Claim 4.2. Let $R \in \text{Rng}^1$ and let $e, f \in \text{Idem}(R)$ be similar. Then, they are algebraically equivalent.

Claim 4.3. Both algebraic equivalence and similarity form equivalence relations over idempotents.

Now, it's not true that algebraic equivalence implies similarity; here's a simple counterexample.

Example 4.5. Let $V \in \text{Vec}_{\mathbb{F}}$ have countably infinite dimension. Then, every idempotent $p \in \text{Idem}(\text{Vec}_{\mathbb{F}}(V))$ of infinite rank is algebraically equivalent to Id_V . However, Id_V can only be similar to itself.

Let us introduce new notation for algebraic equivalence and similarity.

Notation 4.6. Let $R \in \text{Rng}^1$ and let $p, q \in R$. If they are similar, then we denote this by writing:

$$p \sim_{\text{similar}} q.$$

If they are algebraically equivalent, then we denote this by writing:

$$p \sim_{\text{AlgEquiv}} q,$$

which is quite intuitive.

We can show that algebraic equivalence does imply similarity in 2×2 -matrices. Before we prove this, let us introduce a useful bit of notation that will be given its own meaning later in this section as part of our goal to describe K_0 concretely.

Notation 4.7. Let $R \in \text{Rng}$ and let $x, y \in R$. We define:

$$x \oplus y \stackrel{\text{def}}{=} \begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}.$$

If x and y are idempotents, then this will be known as the idempotent sum^a.

^aIn that specific case, it's a rather notable operation since this addition produces an idempotent element of the ring $M_{2 \times 2}(R)$.

Here's a simple fact that we feel comfortable enough leaving as an exercise.

Claim 4.4. Let $R \in \text{Rng}^1$ and let $p \in R$. Then:

$$p \oplus 0 \sim_{\text{similar}} 0 \oplus p,$$

within these two 2×2 -matrices over R .

The next result has a rather interesting proof.

Proposition 4.8

Let $R \in \text{Rng}^1$ and let $p, q \in \text{Idem}(R) \setminus \{0\}$. TFSH:

1. $p \sim_{\text{similar}} q$ iff $p \sim_{\text{AlgEquiv}} q$ and $1 - p \sim_{\text{AlgEquiv}} 1 - q$.
2. If $p \sim_{\text{AlgEquiv}} q$, then $p \oplus 0 \sim_{\text{similar}} q \oplus 0$ in $M_{2 \times 2}(R)$.

You'll participate in giving this proof with us.

Proof of Proposition 4.8. We'll prove several claims.

Claim. $\Rightarrow$ holds in statement (1).

Let $u \in R^\times$ so that $p = uqu^{-1}$. Then, we see that:

$$\begin{aligned} 1 - p &= uu^{-1} - p \\ &= uu^{-1} - uqu^{-1} \\ &= u(1 - q)u^{-1}, \end{aligned}$$

which means that $1 - p \sim_{\text{similar}} 1 - q$. So, **Claim 4.2** implies that $p \sim_{\text{AlgEquiv}} q$ and $1 - p \sim_{\text{AlgEquiv}} 1 - q$.

Claim. $\Leftarrow$ holds in statement (1).

Let $(x, y), (x', y') \in R \times R$ be reduced pairs such that:

$$\begin{aligned} p &= xy, \\ q &= yx, \\ 1 - p &= x'y', \\ 1 - q &= y'x'. \end{aligned}$$

Then, we define³:

$$u \stackrel{\text{def}}{=} y + y'.$$

Let's check that this works.

Claim 4.5. u is invertible with inverse^a $x + x'$.

^aJust show this as usual. The only issue is that when you multiply everything out, you won't immediately get a 1 on the other side. To fix that, you will need to use the reduced identities.

It's not hard to show that u actually works.

Claim 4.6. $q = upu^{-1}$.

So, it follows that p and q are similar.

Claim. Statement (2) holds.

Let $x, y \in R$ be a reduced pair such that:

$$p = xy$$

$$q = yx.$$

We define:

$$u \stackrel{\text{def}}{=} \begin{pmatrix} 1 - p & x \\ y & 1 - q \end{pmatrix}.$$

Now, the following result follows by a calculation.

Claim 4.7. $u^2 = 1$. That is, u is invertible.

The next result, which is also a calculation, nearly finishes off the proof.

Claim 4.8. $u(p \oplus 0)u = 0 \oplus q$.

So, we've shown that $p \oplus 0$ is similar to $0 \oplus q$ but we know that $0 \oplus q$ is similar to $q \oplus 0$ so it follows that:

$$p \oplus 0 \sim_{\text{similar}} q \oplus 0,$$

and that completes the proof.

So, Proposition 4.2 suggests that there is a way to construct algebraic K_0 for unital rings by studying the structure of idempotent matrices. However, note that for $R \in \text{Rng}^1$ and for $M, N \in {}_R\text{Mod}$ being f.g.p, then TFAE:

1. $[M] = [N]$.
2. M and N are stably isomorphic.

This equivalence strongly indicates that we need to build the idea of being stably isomorphic into the “idempotent picture” of K_0 . This leads us to the following generalization of Proposition 4.2.

Proposition 4.9

Let $R \in \text{Rng}^1$ and let p, q be idempotent matrices over R of possibly different sizes. TFAE:

1. $\text{ran}(p)$ and $\text{ran}(q)$ are isomorphic.
2. We can enlarge p and q to matrices over R of the same size n (by adding rows and columns of zeroes) so that the enlargements, denoted by $\hat{p}$ and $\hat{q}$, are conjugate under the group $\text{GL}(n, R)$.

Proof.

- (1) $\Rightarrow$ (2): We extend p (or q) to the same size by adding rows & columns of zeros; this produces two idempotent matrices whose ranges are $\text{ran}(p)$ and $\text{ran}(q)$ so they are isomorphic. Hence, they are algebraically equivalent. But statement (2) of Proposition 4.8 implies that they are similar after a further enlargement.
- (2) $\Rightarrow$ (1): Well, any such enlargement would preserve the ranges so they remain isomorphic and this means that we have idempotent matrices of the same size which are similar so they are algebraically equivalent. ■

We have a remark.

Remark 4.10. The result above is proved in [87, Lemma 1.2.1, p.7-8] and he effectively uses the proof of Proposition 4.8 in a direct way. However, we think our presentation is more conceptual & leads to you acquiring the actual techniques needed for working with K -theory. Also, having these results established separately paves the way for easier deductions of other characterizations of similarity/algebraic equivalence in the K -theory of operator algebras.

Let's introduce a notion of algebraic similarity for “infinite matrices”, inspired significantly by Proposition 4.9. To do this, we need a construction that makes sense of the notion of “infinite matrices” over a ring. Of course, this is just going to be a colimit construction but it is worth doing this explicitly so that we can establish notation.

³One can cook up a nice geometric picture which motivates this choice; we encourage you to do it.

Construction 4.11. Let $R \in \mathbf{Rng}^1$. For each $n \in \mathbb{N}$, we define:

$$\phi_n : M_{n \times n}(R) \rightarrow M_{(n+1) \times (n+1)}(R) , \quad a \mapsto \phi_n(a) \stackrel{\text{def}}{=} \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} ,$$

$$\psi_n : GL(n, R) \rightarrow GL(n+1, R) , \quad a \mapsto \psi_n(a) \stackrel{\text{def}}{=} \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} .$$

These maps give rise to inductive sequences in $\mathbf{Rng}$ and $\mathbf{Grp}$. Since these categories are cocomplete, these inductive sequences have colimits and we denote these by the following bit of notation:

$$\begin{aligned} \text{Mat}_\infty(R) &\stackrel{\text{def}}{=} \text{colim}(M_{n \times n}(R)), \\ GL_\infty(R) &\stackrel{\text{def}}{=} \text{colim}(GL(n, R)). \end{aligned}$$

It will be useful for us to view the elements of $\text{Mat}_\infty(R)$ as “infinite matrices” and, indeed, constructing this as a colimit really does allow us to use that intuitive picture. Here are two things to keep in mind:

1. We’ll let $\text{Idem}(\text{Mat}_\infty(R))$ denote the set of idempotent matrices in $\text{Mat}_\infty(R)$.
2. Note that $GL_\infty(R)$ acts on $\text{Mat}_\infty(R)$ by conjugation.

For the second point, the point is that $GL(n, R)$ acts on $M_{n \times n}(R)$ for each $n \in \mathbb{N}$. Since group actions can be described as functors, the action extends to $\text{Mat}_\infty(R)$ by universal property; it’s a good exercise to figure out why this works.

Here’s the main definition of interest. This is related to the notion of algebraic equivalence as defined in [Terminology 4.3](#).

Definition 4.12

Let $R \in \mathbf{Rng}^1$ and let $p, q \in \text{Mat}_\infty(R)$. p and q are algebraically similar iff:

$$\exists u \in GL_\infty(R) : p = uqu^{-1}.$$

That is, p and q are “similar” as infinite matrices.

So, the benefit of this definition is that it neatly packages the explicit thing that you do in [Proposition 4.9](#) of enlarging two matrices and then finding an invertible matrix of the appropriate size which makes those matrices similar. The inductive limit constructions above simply encode this enlargement process. With that said, here’s a simple exercise.

Claim 4.9. Algebraic similarity as described above is an equivalence relation. We denote this by writing:

$$p \sim_s q,$$

and let the equivalence classes be denoted by $[p]_s$.

With these results in hand, let us provide the idempotent construction of unital algebraic K_0 .

Construction 4.13. Let $R \in \mathbf{Rng}^1$ and let $p, q \in \text{Idem}(\text{Mat}_\infty)$. We write algebraic similarity a bit more cleanly as follows:

$$p \sim_s q \stackrel{\text{def}}{\iff} \exists u \in GL_\infty(R) : p = uqu^{-1}.$$

Now, let $S = \text{Idem}(\text{Mat}_\infty(R)) / \sim$ and let $p \in \text{Idem}(\text{Mat}_\infty(R))$. Then, the following notation:

$$[p]_s,$$

will refer to the equivalence class of p under $\sim$ and this will be called a conjugation orbit. With this, we define:

$$\forall p, q \in \text{Idem}(\text{Mat}_\infty(R)) : [p]_s + [q]_s \stackrel{\text{def}}{=} \left[\begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix} \right]_s .$$

So, this should make total sense; the creation of the matrix on the right-hand side by filling in the diagonals corresponds *exactly* to the direct sum in the case of f.g.p modules over R . Now, we are lead very naturally to the following result.

Claim. The operation above is well-defined and turns S into an abelian semigroup.

You’ll prove this in [Claim 4.10](#). So, the fact that this is an abelian semigroup means that we can apply the Grothendieck construction to it. What do we get when we do it? This leads us naturally to the main claim.

Claim. $\mathcal{G}(S) \equiv_{\mathbf{AbGrp}} K_0(R)$.

We start by defining a map:

$$f : S \rightarrow K_0(R) , \quad [p]_s \mapsto f([p]_s) \stackrel{\text{def}}{=} [\text{ran}(p)].$$

This is well-defined, by [Proposition 4.9](#) and additive^a. By universal property, there exists a group homomorphism:

$$\widehat{f} : \mathcal{G}(S) \rightarrow K_0(R) ,$$

and we need to check that this is a group isomorphism. Let’s check injectivity and surjectivity separately^b.

1. Let $[P] \in K_0(R)$, where P is a f.g.p module. Then, P is isomorphic to $\text{ran}(p)$ for some idempotent matrix p . So, we see that $[P] = [\text{ran}(p)] = f([p]_s)$, which implies the surjectivity of $\widehat{f}$.
2. Let’s check injectivity; let $p, q \in \text{Idem}(\text{Mat}_\infty(R))$ and let $\gamma_S : S \rightarrow \mathcal{G}(S)$ be the Grothendieck map so that:

$$\widehat{f}(\langle [p]_s, [q]_s \rangle) = 0.$$

This means that $f([p]_s) = f([q]_s)$. That is:

$$[\text{ran}(p)] = [\text{ran}(q)].$$

So, $\text{ran}(p)$ and $\text{ran}(q)$ are stably isomorphic and this means that p and q are conjugate^c.

This proves the claim and completes our alternative description of K_0 on unital rings.

^aWhy? Argue for this to be the case!

^bSo, this is the part which is, on a technical level, actually quite easy. However, keeping the notation and identifications straight in your head is an exercise in concentration, so please flip back & forth to continually remind yourself of notation & terms.

^cSee Proposition 4.9.

Here's the associated exercise.

| **Claim 4.10.** Prove the first claim of [Construction 4.13](#).

Here's some notation.

Notation 4.14. Let $R \in \text{Rng}^1$ and let $p \in \text{Idem}(\text{Mat}_\infty(R))$. Then, we denote by:

$$[p]_s,$$

the K_0 -element $[\text{ran}(p)]$.

This leads us to the following standard idempotent portrait for unital algebraic K_0 .

Proposition 4.15. The Standard Idempotent Portrait of Unital Algebraic K_0

Let $R \in \text{Rng}^1$. Then:

$$K_0(R) = \{[p]_s - [q]_s : p, q \in \text{Idem}(\text{Mat}_\infty(R))\} = \{[p]_s - [1_n]_s : n \in \mathbb{N} \wedge p \in \text{Idem}(\text{Mat}_\infty(R))\}.$$

If $p, q \in \text{Idem}(\text{Mat}_\infty(R))$, then TFAE:

1. $[p]_s = [q]_s$.
2. $\exists n \in \mathbb{N} : p \oplus 1_n \sim_s q \oplus 1_n$.

That is, p and q are stably similar.

Proof. There is nothing to prove here. We simply use the fact that f.g.p modules over a unital ring have a description in terms of idempotents and the fact that there is already a standard portrait of unital algebraic K_0 . ■

The above only gives a description of algebraic K_0 on the objects of Rng^1 ; we still need a description of what happens on the morphisms. Before we give such a description, we need some notation.

Notation 4.16. Let $R \in \text{Rng}$ and let $n \in \mathbb{N}$.

1. If R is unital, then 1_n will denote the identity of $M_{n \times n}(R)$ viewed as a subring of $\text{Mat}_\infty(R)$.
2. If $S \in \text{Rng}$ and $\phi \in \text{Rng}(R, S)$, then we define:

$$\phi^{(n)} : M_{n \times n}(R) \rightarrow M_{n \times n}(S) , (a_{ij}) \mapsto \phi^{(n)}((a_{ij})) = (\phi(a_{ij})).$$

This is known as the n -th matrix amplification of ϕ .

It will actually be convenient for us to denote the n -th matrix amplification by ϕ just to avoid cramming in a lot of notation into small spaces. Of course, we also do the same thing with idempotents in matrix algebras.

We have a remark about this piece of notation.

Remark 4.17. The notation above, incidentally, also makes it obvious that for $n \in \mathbb{N}$, we have a functor $\text{Mat}_n(-) : \text{Rng}^1 \rightarrow \text{Rng}^1$ which is reminiscent of a functor that we've already discussed for C^* -algebras in [Operator Algebraic Structures I](#). The properties of this functor will be relevant in later work within the K -theory of C^* -algebras.

We present the next construction as a continuation of [Construction 4.13](#).

Construction 4.18. Let $R, S \in \text{Rng}^1$, let $f \in \text{Rng}^1(R, S)$ and let $p \in \text{Idem}(\text{Mat}_\infty)$. Then, we see that:

$$f(p) \in \text{Idem}(\text{Mat}_\infty),$$

where we have used the appropriate matrix amplification of f . Now, it is the case that f preserves similarity of idempotents, so we can extend f to a function of abelian semigroups:

$$\widehat{f} : \text{Idem}(\text{Mat}_\infty(R)) / \sim \rightarrow \text{Idem}(\text{Mat}_\infty(S)) / \sim .$$

By the definition of the addition on these abelian semigroups, it follows that $\widehat{f}$ is a morphism of abelian semigroups. So, the universal property of the Grothendieck construction lifts $\widehat{f}$ to a morphism:

$$K_0(f)^1 : K_0(R) \rightarrow K_0(S) .$$

These observations prompt the following claim.

| **Claim.** $\mathbb{V}(f) = \widehat{f}$.

We leave this as a very useful exercise which uses all of the results we've proved in this section.

Here's a remark.

Remark 4.19. In other rings with more structure, it turns out that one can drastically simplify the standard portrait of algebraic K_0 . For example, if H is a $\mathbb{C}$ -Hilbert space, then one can work with projections as opposed to just idempotents and this allows one to conclude that $K_0(\mathcal{B}(H)) \simeq 0$ when H is infinite-dimensional^a.

^aWhen H is finite-dimensional, then $\mathcal{B}(H)$ is isomorphic to $M_{n \times n}(\mathbb{C})$ for some $n \in \mathbb{N}$ and one can calculate the K_0 -group of this ring to be $\mathbb{Z}$. Indeed, it's the same as $K_0(\mathbb{C})$ and this is not a specific coincidence.

Now, for the purposes that we have in mind, sticking to the unital situation is very restrictive. Indeed, we eventually wish to apply (at least some of) the machinery of algebraic K-theory in studying C^* -algebras and most of the interesting ones are certainly not unital. So, let us extend K_0 to Rng .

4.2 Extending Algebraic K_0 To All Rings

We'll start that endeavour with a bit of terminology which is a cultural accident.

Terminology 4.20. A ring extension is a diagram in Rng :

$$I \longrightarrow R \longrightarrow S$$

such that the following sequence in AbGrp :

$$0 \longrightarrow I \longrightarrow R \longrightarrow S \longrightarrow 0$$

is exact.

We have a remark about this bit of terminology.

Remark 4.21. Obviously, we're really just talking about an "exact sequence of rings" and for all intents & purposes, we will be very happy to use both pieces of terminology. People who care about homological algebra dislike this because they only like to speak of exact sequences in additive categories^a. So far, we have not found any need to talk about additive categories because it is simply unnecessary for our work at the moment.

^aIn general, the so-called Quillen exact categories minimally require the structure of an additive category in the background.

Here's a familiar construction.

Construction 4.22. Suppose we have a ring extension:

$$I \xrightarrow{i} R \xrightarrow{p} S$$

Then, we know that:

$$\text{ran}(i) = \ker(p).$$

So, we can draw the following commutative diagram:

$$\begin{array}{ccccc} I & \xrightarrow{i} & R & \xrightarrow{p} & S \\ \text{Id}_I \downarrow & & \downarrow \text{Id}_R & & \uparrow \hat{p} \\ I & \xrightarrow{i} & R & \xrightarrow{\pi} & R/\ker(p) \simeq R/\text{ran}(i) \end{array}$$

where the vertical maps are all isomorphisms and π is the canonical projection. So, we see that a ring extension is completely determined by I (and the map i). None of this crazy, it's mostly just standard homological algebra.

So, if you would like to view this construction "commutatively", then it's analogous to an excision-type construction. Indeed, "modding out an ideal" should be viewed as the noncommutative version of "quotienting out a subspace" of a topological space. Indeed, we will see that such an excision-type result for topological K^0 holds as a consequence of excision in algebraic K-theory. With that said, the construction provides another piece of intuition.

Intuition 4.23. So, the idea is that we want a functor:

$$K_0 : \text{Rng} \rightarrow \text{AbGrp},$$

such that for any ring extension:

$$I \xrightarrow{i} R \xrightarrow{p} S$$

we obtain an exact sequence:

$$0 \longrightarrow K_0(I) \xrightarrow{K_0(i)} K_0(R) \xrightarrow{K_0(p)} K_0(S) \longrightarrow 0$$

In other words, we want algebraic K_0 to be half-exact and therein lies the problem with the more obvious constructions we could do in order to extend algebraic K_0 ; these constructions do not yield half-exactness and we also lose other properties, making them practically useless. So, the way that we achieve this goal of extending algebraic K_0 is by considering a special ring extension that comes forth from the *unitization of rings*. Again, we expect that you might've seen this in previous studies^a but we go through the construction in detail^b for the sake of establishing notation.

^aYou've certainly seen it for $\mathbb{C}$ -algebras.

^bLeaving the main details as exercises.

Here's the construction.

Construction 4.24. Let $R \in \text{Rng}$. We define:

$$R^\sim \stackrel{\text{def}}{=} R \times \mathbb{Z},$$

equipped with the following operations:

$$\forall (x, t), (y, s) \in R^\sim : (x, t) + (y, s) \stackrel{\text{def}}{=} (x + y, t + s),$$

$$\forall (x, t), (y, s) \in R^\sim : (x, t) \cdot (y, s) \stackrel{\text{def}}{=} (xy + ty + sx, ts).$$

Here's our first claim.

Claim. With the given operations, we see that $R^\sim$ is a unital ring.

You'll check this in **Claim 4.11**. Of course, this ring comes equipped with an embedding:

$$\iota_R : R \rightarrow R^\sim, \quad x \mapsto \iota_R(x) \stackrel{\text{def}}{=} (x, 0).$$

The pair $(R^\sim, \iota_R)$ satisfies a universal property.

Claim. Let (S, f) be a pair consisting of the following data:

1. A unital ring S .
2. A ring homomorphism $f : R \rightarrow S$.

Then, there exists a unique unital ring homomorphism $f^\sim : R^\sim \rightarrow S$ such that the following diagram:

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow \iota_R & \nearrow f^\sim & \\ R^\sim & & \end{array}$$

commutes.

You'll check this in **Claim 4.12**. It is also easy to cook up a functor based on the results above.

Claim. There exists a functor:

$$\text{Rng} \rightarrow \text{Rng}^1,$$

with the following specifications:

1. If $R \in \text{Rng}$, then the functor takes it to $R^\sim$.
2. If $R, S \in \text{Rng}$ and $f : R \rightarrow S$ is a ring homomorphism, then we extract a unique unital ring homomorphism:

$$f^\sim : R^\sim \rightarrow S^\sim,$$

such that the following diagram:

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow \iota_R & & \downarrow \iota_S \\ R^\sim & \xrightarrow{f^\sim} & S^\sim \end{array}$$

commutes.

This will be known as the unitization functor.

You'll check this in **Claim 4.13**. From these results, we can create a ring extension. Let us first define a map:

$$\varepsilon : R^\sim \rightarrow \mathbb{Z}, \quad (x, t) \mapsto \varepsilon(x, t) \stackrel{\text{def}}{=} t.$$

Okay, so this fits in the following ring extension:

$$0 \longrightarrow R \xrightarrow{\iota_R} R^\sim \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

which is what we need to extend K_0 . Now, we have the following fact.

Claim. The exact sequence above is split-exact.

We define:

$$\lambda : \mathbb{Z} \rightarrow R^\sim, \quad t \mapsto \lambda(t) \stackrel{\text{def}}{=} (0, t).$$

Now, we see that:

$$\begin{aligned} \forall t \in \mathbb{Z} : (\varepsilon \circ \lambda)(t) &= \varepsilon(0, t) \\ &= t, \end{aligned}$$

which proves that λ is a section^a of ε , proving the claim.

Now, we define the so-called **scalar map**:

$$s : R^\sim \rightarrow R^\sim,$$

which is simply just the composition $s = \lambda \circ \varepsilon$.

^aOf course, one has to show that λ is a ring homomorphism but this is a consequence of a direct calculation.

Here are the exercises associated to the construction.

Claim 4.11. Prove claim (1) of [Construction 4.24](#).

Claim 4.12. Prove claim (2) of [Construction 4.24](#).

Claim 4.13. Prove claim (3) of [Construction 4.24](#).

The scalar mapping will be important as a technical tool in later work involving K_1 and we will always specify that we're referring to the scalar map whenever we do use it. So, just understand the technicalities of the above construction and forget about it till you need it. The more important point is that we can, now, define algebraic K_0 for rings. Before we do it, here's some nice intuition that motivates the definition in terms of the split-exact sequence above.

Intuition 4.25. Now, we would like to apply algebraic K_0 to this ring extension and obtain a 3-term exact sequence. Well, we need to be the case that “ $\text{ran}(K_0(\iota_R))$ ” equals $\ker(K_0(R))$. But since ι_R is an inclusion, we would really like for $K_0(\iota_R)$ to be viewed as the inclusion of the algebraic K_0 of R into the algebraic K_0 of $R^\sim$. So, in order to fulfill these desires, it seems most reasonable for us to make the following definition:

$$K_0(R) \stackrel{\text{def}}{=} \ker(K_0(\varepsilon)).$$

This will ensure, very artificially, that we have half-exactness in this special case.

Based on this intuition, we proceed with the following definition.

Definition 4.26

Let $R \in \text{Rng}$. Then, we define:

$$K_0^1(R) \stackrel{\text{def}}{=} \ker(K_0(\varepsilon)),$$

where ε is defined in [Construction 4.24](#).

So, we’ve used slightly different notation for this functor (we haven’t yet showed that it’s a functor but we will) and the reason is because one must consider the situation where R is already unital. In that case, it’s not completely clear that the definition above agrees with the one that we gave in the last lecture. To prove this, we need a few additional results.

Claim 4.14. Let $R \in \text{Rng}$. TFAE:

1. R is unital.
2. $R^\sim$ is isomorphic to the product ring $R \oplus \mathbb{Z}$.

Claim 4.15. Let $R \in \text{Rng}$. Then, the map $\varepsilon : R^\sim \rightarrow \mathbb{Z}$ is a split epi^a in Rng^1 .

^aThat is, it has a section in the given category.

Claim 4.16. Let $R \in \text{Rng}$. Then, the exact sequence below:

$$0 \longrightarrow R \xrightarrow{\iota_R} R^\sim \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

is split-exact. This will be known as the unitization sequence, as in the case with $\mathbb{C}$ -algebras.

We’re ready to show that our definition is consistent with the construction in the last lecture.

Proposition 4.27

Let $R \in \text{Rng}^1$. Then, $K_0(R) = K_0^1(R)$.

Proof. So, let’s take our sequence:

$$0 \longrightarrow R \xrightarrow{\iota_R} R^\sim \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

Since ε is split epi we see that:

$$K_0(\varepsilon) : K_0(R^\sim) \rightarrow K_0(\mathbb{Z}),$$

is split epi in AbGrp . But we know that $R^\sim$ is isomorphic to the product ring $R \times \mathbb{Z}$ so:

$$K_0(R^\sim) \simeq K_0(R) \oplus K_0(\mathbb{Z}).$$

We see that $\ker(K_0(\varepsilon))$ just picks out the first summand above, so it follows that:

$$K_0^1(R) = K_0(R),$$

as was desired. ■

At this stage, we have two remarks.

Remark 4.28. So, this construction (and discussion) explains why K_0 on Rng will not agree with K_0 on $C^*\text{Alg}$; the problem is that the unitization of a C^* -algebra is defined differently from the unitization of a ring so algebraic K_0 does not capture all of the required structure. Nevertheless, we do want to mention that it is still very interesting for one to study the algebraic K-theory of C^* -algebras and that the moral of the construction above will be applied when we extend operator K_0 from ${}^1C^*\text{Alg}$ to $C^*\text{Alg}$ in the coming lectures^a.

^aThe same idea will, of course, first be used in our discussion of Van Daele’s K-theory.

Remark 4.29. It is rather unfulfilling that the definition of algebraic K_0 seems to depend on the unitization of a ring, which depends on the commutative ring $\mathbb{Z}$. It would be nice if we could extend K_0 to any R -algebra, where $R \in \text{AbRng}^1$. It turns out that this is certainly possible and since we haven’t used any essential features of $\mathbb{Z}$ beyond its properties as a ring, it turns out that the **exact same** arguments as above will work. So, we can define the R -unitization of any R -algebra in the same way, we get a unitization split-exact sequence in the same way and so on.

In light of the result above, we can adopt the following notation.

Notation 4.30. Let $R \in \text{Rng}$. Then, we define:

$$K_0(R),$$

to be the algebraic K_0 of R in both the unital and non-unital cases.

Okay, let’s cook up a functor.

Construction 4.31. Let $R, S \in \mathbf{Rng}$ and let $f \in \mathbf{Rng}(R, S)$. Then, we have a diagram:

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow \iota_R & & \downarrow \iota_S \\ R^\sim & \xrightarrow{f^\sim} & S^\sim \end{array}$$

which commutes, where $f^\sim$ is a unital ring homomorphism. Well, we can extend this to the following diagram:

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow \iota_R & & \downarrow \iota_S \\ R^\sim & \xrightarrow{f^\sim} & S^\sim \\ \downarrow \varepsilon_R & & \downarrow \varepsilon_S \\ \mathbb{Z} & \xlongequal{\quad} & \mathbb{Z} \end{array}$$

Since the bottom square lives in $\mathbf{Rng}^1$, we may apply K_0 to it and we get:

$$\begin{array}{ccc} K_0(R^\sim) & \xrightarrow{K_0(f^\sim)} & K_0(S^\sim) \\ \downarrow K_0(\varepsilon_R) & & \downarrow K_0(\varepsilon_S) \\ K_0(\mathbb{Z}) & \xlongequal{\quad} & K_0(\mathbb{Z}) \end{array}$$

Since $K_0(R)$ and $K_0(S)$ are defined as the kernels of $K_0(\varepsilon_R)$ and $K_0(\varepsilon_S)$, we can just include them into $K_0(R^\sim)$ and $K_0(S^\sim)$. This allows us to extend the diagram above in the following way:

$$\begin{array}{ccc} K_0(R) & & K_0(S) \\ \downarrow & & \downarrow \\ K_0(R^\sim) & \xrightarrow{K_0(f^\sim)} & K_0(S^\sim) \\ \downarrow K_0(\varepsilon_R) & & \downarrow K_0(\varepsilon_S) \\ K_0(\mathbb{Z}) & \xlongequal{\quad} & K_0(\mathbb{Z}) \end{array}$$

Now, the exactness of the vertical sequences implies that:

$$\text{ran}(K_0(f^\sim)|_{K_0(R)}) \subseteq K_0(S).$$

So, we obtain a morphism $K_0(f)$ such that the diagram below:

$$\begin{array}{ccc} K_0(R) & \xrightarrow{\quad K_0(f) \quad} & K_0(S) \\ \downarrow & & \downarrow \\ K_0(R^\sim) & \xrightarrow{K_0(f^\sim)} & K_0(S^\sim) \\ \downarrow K_0(\varepsilon_R) & & \downarrow K_0(\varepsilon_S) \\ K_0(\mathbb{Z}) & \xlongequal{\quad} & K_0(\mathbb{Z}) \end{array}$$

commutes and this is done simply by restricting $K_0(f^\sim)$. So, we obtain a functor:

$$K_0 : \mathbf{Rng} \rightarrow \mathbf{AbGrp},$$

as was desired^a.

^aThis essentially follows by concatenation of diagrams and using the fact that this is functorial on $\mathbf{Rng}^1$.

Okay, let's try to aim for a more concrete description of algebraic K_0 in the same spirit as in the unital case.

Proposition 4.32. The Standard Idempotent Portrait of Algebraic K_0

Let $R \in \mathbf{Rng}$. Then:

$$K_0(R) = \{[p]_s - [1_n]_s : p \in \text{Idem}(\text{Mat}_\infty(R^\sim)) \wedge n \in \mathbb{N}_0 \wedge p - 1_n \in \text{Mat}_\infty(R)\}.$$

Of course, we only take $p - 1_n$ after we have increased both of them sufficiently in size for this to make sense.

Proof. Let $x \in K_0(R^\sim)$. By the standard portrait of unital algebraic K_0 , we see that:

$$x = [p]_s - [1_n]_s,$$

where $p \in \text{Idem}(\text{Mat}_\infty(R^\sim))$ and $n \in \mathbb{N}_0$. Using the notation of [Construction 4.24](#), we have a computation:

$$\begin{aligned} x \in K_0(R) &\Leftrightarrow K_0(\varepsilon)(x) = 0 \\ &\Leftrightarrow K_0(\varepsilon)([p]_s) = K_0(\varepsilon)([1_n]_s) \\ &\Leftrightarrow [\varepsilon(p)]_s = [1_n]_s, \end{aligned}$$

where the last equality holds in $K_0(\mathbb{Z})$. Now, we prove two inclusions.

$\supseteq$: The calculation above makes this clear, because if $p \in \text{Idem}(\text{Mat}_\infty(R^\sim))$ and $n \in \mathbb{N}_0$ such that:

$$p - 1_n \in \text{Mat}_\infty(R),$$

then we see that:

$$0 = \varepsilon(p - 1_n) = \varepsilon(p) - 1_n.$$

Therefore, we see that $[\varepsilon(p)]_s = [1_n]_s$ and this is enough for x to lie in $K_0(R)$.

$\subseteq$: Let $x \in K_0(R)$ and suppose that:

$$x = [p]_s - [1_n]_s,$$

where p and n are as given above. The calculation above reveals that:

$$\varepsilon(p) \sim_s 1_n.$$

The strategy is to change p and turn the above into an equality. We'll do this in two steps:

- (a) We'll replace stable isomorphism by algebraic similarity.
- (b) We'll replace similarity by equality.

Now, let $m \in \mathbb{N}$ such that $\varepsilon(p) \oplus 1_m \sim_s 1_n \oplus 1_m = 1_{m+n}$. But now, we note that:

$$1_m = \varepsilon(1_m),$$

so we let $f = p \oplus 1_m$ and we note that:

$$\varepsilon(f) \sim_s 1_k,$$

where $k = m + n$. The idea, then, is to note that $x = [f]_s - [1_k]_s$. But this description shows that $\varepsilon(f)$ is algebraically similar to 1_k . This is, again, not the end of the proof but the strategy, now, is to slightly change f . Since $\varepsilon(f)$ and 1_k are algebraically similar, we see that:

$$\exists v \in \text{GL}_\infty(\mathbb{Z}) : v\varepsilon(f)v^{-1} = 1_k.$$

Observe that $\text{GL}_\infty(\mathbb{Z})$ is embedded into $\text{GL}_\infty(R^\sim)$ so we view v as a matrix over $R^\sim$. Now, let $g = vfv^{-1}$. Then, g is algebraically similar to f by definition and this means that $x = [g]_s - [1_k]_s$. But we know that:

$$\varepsilon(g) = v\varepsilon(f)v^{-1} = 1_k,$$

which proves that $g - 1_k \in \text{Mat}_\infty(R)$, as was desired.

This completes the proof. ■

^aWe've also conflated 1_n in the last line with $\varepsilon(1_n)$ just to keep things clean.

We have a remark about the result above.

Remark 4.33. So, the moral of the proof is that whenever we have an idempotent matrix which is not convenient or does not satisfy the properties we would like it to satisfy, then we can try to find a way to replace that idempotent matrix with another one which satisfies our needs better.

Remark 4.34. Now, one major corollary of this standard portrait^a will be the split-exactness of K_0 , from which we'll deduce other properties of K_0 . We're not giving too many exercises here because, actually, coming to grips with the arguments and making sure you have a firm understanding of how we're using the standard portrait^b is already difficult enough. So, we strongly recommend that you work through these proofs thoroughly. However, don't lose the forest for the trees; for many computations in the future, the permanence properties are good enough.

^aWe will present this stuff in the next lecture.

^bIt's really important to understand this well in the unital case.

Let us end off our exposition here by briefly discussing the K -theory of group rings. Indeed, these are some of the most prominent examples of rings that are not necessarily commutative and which tend to pop up in applications to topology & geometry. Now, it turns out that the investigation of the algebraic K -theory of these group rings ties in very well with investigations of the K -theory of group operator algebras. This is a connection which we, unfortunately, cannot talk about in a whole load of detail here because it requires a whole text for itself. Nevertheless, we will try to provide as many snippets of these basic ideas as possible. In fact, this is a topic which is so nicely connected with many other aspects of mathematics so it actually takes effort for us to avoid it.

Conjecture 4.35 (K_0 -Conjecture). If $R \in \text{AbRng}^1$ is regular^a and $G \in \text{Grp}$ is torsion-free, then:

$$K_0(R[G]) / K_0(R) \simeq 0.$$

We refer to the left-hand side as the reduced K_0 -group of $R[G]$.

^aThis is a condition which comes naturally from geometric considerations. If you're not aware of the definition, then the examples to keep in mind are $\mathbb{Z}$ and (algebraically closed) fields; these cases are already very difficult.

Here's a remark.

| **Remark 4.36.** When R is a field of characteristic 0, the K_0 -Conjecture implies ??.

In fact, there is a similar conjecture for K_1 ; even though we haven't introduced this functor yet, the only thing you need to know is that it is supposed to be another algebraic invariant of rings. If you keep that in mind, then we can also state the analogous K_1 -Conjecture. The statement is only slightly different.

Conjecture 4.37 (K_1 -Conjecture). If $R \in \text{AbRng}^1$ is regular^a and $G \in \text{Grp}$ is torsion-free, then:

$$K_1(R[G]) / \langle K_1(R), K_0(R) \otimes R[G] \rangle \simeq 0.$$

We refer to the left-hand side as the Whithead group^b of G over R .

^aThis is a condition which comes naturally from geometric considerations. If you're not aware of the definition, then the examples to keep in mind are $\mathbb{Z}$ and (algebraically closed) fields; these cases are already very difficult.

^bFor this to make sense, we need to show that the quotient is well-defined. We will do this later.

Here is our remark about these conjectures.

Remark 4.38. The big difference is that the second conjecture is essentially far more geometric than the first one. We will discuss these conjectures in ??. By that time, we will have the basic theory of K_1 lined up for use. For now, just keep these conjectures in mind. One of our goals is to give a modern account of K -theory which will help you dive into the literature as needed if you wish to do it. For that to happen, it is crucial for us to introduce you to the basic conjectures of K -theory and the sort of things that people are interested in studying today.

At this stage, we have successfully given a concrete description of algebraic K_0 even in the non-unital situation and we have used that to establish its various properties. So, we're ready to start talking about topological K -theory on spaces and actually, we will be able to construct the K^0 -functor on spaces within the next lecture, also establishing its various interesting properties as a consequence of the properties of algebraic K_0 .

5 Properties of Algebraic K_0

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

Motivation.

5.1 Split Between Half-Exactness & Split-Exactness

Let's start with the following question.

Question 5.1. Let $R, S \in \text{Rng}^1$ and let $f \in \text{Rng}^1(R, S)$. Then, we know that:

$$f(R^\times) \subseteq S^\times.$$

If f is surjective, then is $f|_{R^\times}$ a surjection onto $S^\times$?

In general, the answer is a resounding no. Here are two examples.

Example 5.2. Let $p \geq 5$ be prime. Then, we consider the projection:

$$\mathbb{Z} \rightarrow \mathbb{Z}/p\mathbb{Z}.$$

This is a surjective unital ring homomorphism but one notes that:

$$|\mathbb{Z}^\times| = 2,$$

$$|(\mathbb{Z}/p\mathbb{Z})^\times| = p - 1,$$

and this obviously means that $\mathbb{Z}^\times$ cannot be surjected onto $(\mathbb{Z}/p\mathbb{Z})^\times$.

Example 5.3. Let $\mathbb{D}$ be the closed unit disk centered at 0 on $\mathbb{C}$ and consider the map:

$$\mathcal{C}(\mathbb{D}) \rightarrow \mathcal{C}(\mathbb{T}), \quad f \mapsto f|_{\mathbb{T}}.$$

This is a surjective unital ring homomorphism, by the Tietze Extension Theorem. Now, suppose that its restriction to $\mathcal{C}(\mathbb{D})^\times$ is surjective onto $\mathcal{C}(\mathbb{T})^\times$. Well, let's take the following inclusion map:

$$f : \mathbb{T} \rightarrow \mathbb{C} \setminus \{0\}, \quad x \mapsto f(x) \stackrel{\text{def}}{=} x.$$

Now, if the restriction above was surjective, then we could lift f to a continuous map $g : \mathbb{D} \rightarrow \mathbb{C} \setminus \{0\}$. But this implies, by a standard fact^a from [REF], that f is nullhomotopic. That is, $\text{Id}_{\mathbb{T}}$ is nullhomotopic and this means that $\mathbb{T}$ is contractible, which is false. So, this forms a neat counterexample.

^aIf $n \in \mathbb{N}_0$, $X \in \text{Top}$ and $f \in \mathcal{C}(S^n, X)$, then TFAE:

1. f is nullhomotopic.
2. There is a continuous map $g : D^{n+1} \rightarrow X$ with $g|_{S^n} = f$.
3. If $p \in S^n$, then $f \simeq \varepsilon_{f(p)}$, where the latter is the constant function on S^n with value $f(p)$.

This fact is not so difficult to prove.

Now, a lemma of Whitehead's actually ensures that we have a "lifting".

Proposition 5.4. Algebraic Whitehead Lemma (K_0)

Let $R, S \in \text{Rng}^1$, let $\phi \in \text{Rng}^1(R, S)$ be surjective, let $n \in \mathbb{N}$ and let $a \in \text{GL}_n(S)$. Then:

$$\exists u \in \text{GL}_{2n}(R) : \phi(u) = a \oplus a^{-1},$$

where $a \oplus a^{-1} = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$ as usual^a.

^aNote that we are using matrix amplifications here without specifying them. Also, see Notation 4.6.

We have a remark about this result.

Remark 5.5. This result has a bit of history behind it and has many corollaries in algebraic K-theory. In various K-theories on various categories of interest, there are variations of the lemma above and all of them are referred to as "Whitehead's Lemma" for the crucial role they play in computations. So, "Whitehead's Lemma" really refers to a collection of results which are related but not necessarily implying one another.

The proof of Whitehead's Lemma is exceedingly simple.

Proof of Proposition 5.4. The proof relies on an observation:

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -a^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

One may see this by explicitly computing the right-hand side. Note that:

$$\phi \left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and since this matrix is invertible, it follows that this has an invertible "lift". Now, ϕ is surjective so:

$$\exists b, c \in M_{n \times n}(R) : \phi(b) = a \wedge \phi(c) = a^{-1}.$$

But this means that:

$$\begin{aligned} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & \phi(b) \\ 0 & 1 \end{pmatrix} \\ &= \phi \left(\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \right), \end{aligned}$$

where $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$ is invertible in $M_{2n \times 2n}(R)$. A similar computation also yields an invertible lift for the second term of our decomposition derived at the start. Multiplying these lifts produces an invertible lift for $a \oplus a^{-1}$. ■

With this, we can prove the half-exactness of algebraic K_0 .

Proposition 5.6. Half-Exactness of Algebraic K_0

K_0 is a half-exact functor.

Proof. Suppose that we have a ring extension:

$$0 \longrightarrow I \xrightarrow{i} R \xrightarrow{p} S \longrightarrow 0$$

Now, we want to show that the sequence:

$$K_0(I) \longrightarrow K_0(R) \longrightarrow K_0(S)$$

is exact at $K_0(R)$. Since $p \circ i = 0$, we see that:

$$\begin{aligned} 0 &= K_0(0) \\ &= K_0(p \circ i) \\ &= K_0(p) \circ K_0(i), \end{aligned}$$

by functoriality. This implies that $\text{ran}(K_0(i)) \subseteq \ker(K_0(p))$. Therefore, it remains for us to prove the opposite inclusion. So, let $x \in \ker(K_0(p))$ and let's use the idempotent portrait of algebraic K_0 . So, we write:

$$x = [e]_s - [1_n]_s,$$

where $e \in \text{Idem}(\text{Mat}_\infty(R^\sim))$ and $n \in \mathbb{N}_0$ and $e - 1_n \in \text{Mat}_\infty(R)$. Then, we have that:

$$0 = K_0(p)(x) = K_0(p)([e]_s) - K_0(p)([1_n]_s),$$

which implies, by our explicit description previously given, is that:

$$[p^\sim(e)]_s = [1_n]_s,$$

where this equality holds in $K_0(S^\sim)$. Now, we apply the same trick as in our proof of the algebraic Whitehead Lemma and we just assume that $p^\sim(e) \sim_{\text{similar}} 1_n$ in $\text{Mat}_\infty(S^\sim)$ after replacing, if necessary, e by $e \oplus 1_m$ and 1_n by 1_{n+m} . So, there exists a k such that $v \in \text{GL}_k(S^\sim)$ such that:

$$vp^\sim(e)v^{-1} = 1_n.$$

$p^\sim$ is surjective & unital, so we apply Whitehead's Lemma and find a $u \in \text{GL}_{2k}(S^\sim)$ so that:

$$p^\sim(u) = v \oplus v^{-1}.$$

This allows us to define:

$$f \stackrel{\text{def}}{=} u(e \oplus 0)u^{-1}.$$

This is clearly idempotent and it is similar to e . Now, we have a calculation:

$$\begin{aligned} p^\sim(f) &= (v \oplus v^{-1})(p^\sim(e) \oplus 0)(v^{-1} \oplus v) \\ &= 1_n \oplus 0 \\ &= 1_n, \end{aligned}$$

which means that $f - 1_n \in \ker(p^\sim)$. Now, observe that:

$$0 \rightarrow M_{m \times m}(I) \rightarrow M_{m \times m}(R^\sim) \xrightarrow{p^\sim} M_{m \times m}(S^\sim) \rightarrow 0,$$

is exact for every $m \in \mathbb{N}$, which means that:

$$f - 1_n \in M_{2k \times 2k}(I).$$

This means that $f \in M_{2k \times 2k}(I^\sim)$. So, we can form an element:

$$y = [f]_s - [1_n]_s,$$

where $y \in K_0(I)$. Now, we need to check that:

$$K_0(i)(y) = x.$$

If this is true, then the proof will be complete, but it is true by construction. ■

Next, let us deal with split-exactness of algebraic K_0 .

Proposition 5.7. Split-Exactness of Algebraic K_0

K_0 is a split-exact functor.

Proof. Suppose that we have the following split-exact ring extension:

$$0 \longrightarrow I \xrightarrow{i} R \xrightarrow{p} R/I \longrightarrow 0$$

We want to show that the following sequence:

$$0 \longrightarrow K_0(I) \xrightarrow{K_0(i)} K_0(R) \xrightarrow{K_0(p)} K_0(R/I) \longrightarrow 0$$

is split-exact. Well, we get half-exactness of this sequence. Moreover, we see that the morphism $K_0(p)$ splits due to the functoriality of $K_0(p)$, which means that we definitely have an exact sequence:

$$K_0(I) \xrightarrow{K_0(i)} K_0(R) \xrightarrow{K_0(p)} K_0(R/I) \longrightarrow 0$$

It remains to show the injectivity of $K_0(i)$. We take inspiration from the proof of the half-exactness of algebraic K_0 and prove this using the idempotent portrait of algebraic K_0 . Let $x \in \ker(K_0(i))$ and note that we can write $x = [e]_s - [1_n]_s$, where $e \in \text{Idem}(\text{Mat}_\infty(I^\sim))$ and $n \in \mathbb{N}_0$ and $e - 1_n \in \text{Mat}_\infty(I)$. Then, we see that:

$$0 = K_0(i)(x) = K_0(i)([e]_s) - K_0(i)([1_n]_s),$$

which implies, by our explicit description previously given, that:

$$[i^\sim(e)]_s = [1_n]_s,$$

where this equality holds in $K_0(R^\sim)$. Now, we may assume that $i^\sim(e) \sim_{\text{similar}} 1_n$ in $\text{Mat}_\infty(R^\sim)$ after replacing, if necessary, e by $e \oplus 1_m$ and 1_n by 1_{n+m} . So, there exists a k such that $v \in \text{GL}_k(R^\sim)$ such that:

$$vi^\sim(e)v^{-1} = 1_n.$$

Now, let j be a section of p and define:

$$u \stackrel{\text{def}}{=} (j^\sim \circ p^\sim)(v^{-1})v.$$

Then, we see that u is invertible in $M_{n \times n}(R^\sim)$ and $p^\sim(u) = 1_k$. So, we see that $u - 1 \in \ker(p^\sim)$ but this means that $u - 1_k \in M_{k \times k}(I)$. So, $u \in \text{GL}_k(I^\sim)$. By calculating ueu^{-1} explicitly, we see that e is similar to 1_n so $x = 0$. ■

With split-exactness, the following corollary becomes easily accessible.

Corollary 5.8

Let $R \in \text{Rng}$. TFSH:

1. If $n \in \mathbb{N}$, then the map:

$$\phi : R \rightarrow M_{n \times n}(R) , a \mapsto \phi_n(a) \stackrel{\text{def}}{=} \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix},$$

induces an isomorphism on K_0 .

2. K_0 is inductively continuous.
3. $K_0(0) = 0$, where 0 on the left-hand side refers to the zero ring.
4. If $R, S \in \text{Rng}$ and $0 : R \rightarrow S$ is the zero map, then $K_0(0)$ is the zero map.

The proof of this is long and it's best we break it down. The major points are that we use some neat homological algebra to reduce everything to the unital case. Once we're there, we use our concrete picture of K_0 to deduce the results we desire.

Proof of Corollary 5.8. Let's get the minutiae out of the way, shall we?

| **Claim 5.1.** Prove statement (3).

| **Claim 5.2.** Prove statement (4).

Now, the proofs of statements (1) and (2) will be reduced to the unital cases first. Then, we'll use the unital idempotent portrait to finish off the argument. Let's start with the reduction of statement (1) to the unital case.

| **Claim.** Statement (1) reduces to the unital case.

So, one observes that we have a sequence:

$$0 \rightarrow R \rightarrow R^\sim \rightarrow \mathbb{Z} \rightarrow 0,$$

and matrix amplifying this, we get the following exact sequence^a:

$$0 \rightarrow M_{n \times n}(R) \rightarrow M_{n \times n}(R^\sim) \rightarrow M_{n \times n}(\mathbb{Z}) \rightarrow 0.$$

We note that both of these extensions are split; we know this for the unitization extension of R and functoriality of $M_{n \times n}(-)$ gives us the split-exactness of the amplified extension. Since K_0 is split-exact, we obtain:

$$\begin{array}{ccccccc} 0 & \longrightarrow & K_0(R) & \longrightarrow & K_0(R^\sim) & \longrightarrow & K_0(\mathbb{Z}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & K_0(M_{n \times n}(R)) & \longrightarrow & K_0(M_{n \times n}(R^\sim)) & \longrightarrow & K_0(M_{n \times n}(\mathbb{Z})) \longrightarrow 0 \end{array}$$

a commutative diagram at the level of K_0 . Note that if the result does hold for unital rings, then the two vertical maps on the right will be isomorphisms. By the Five Lemma, it follows that the vertical map on the left is an isomorphism as well. So, it remains to show this when R is assumed to be unital.

| **Claim.** Statement (1) holds when R is unital.

We'll prove this by constructing an explicit inverse for $K_0(\phi)$ and then showing that it works; this takes 3 steps.

1. Let's construct the inverse; let $\gamma_{n,k} : M_{k \times k}(M_{n \times n}(R)) \rightarrow M_{kn \times kn}(R)$ be the map, for each k , which “forgets parentheses”; this is a ring isomorphism. Now, we define the following map:

$$\gamma_n : \text{Idem}(\text{Mat}_\infty(M_{n \times n}(R))) \rightarrow K_0(R), \quad p \mapsto \gamma_n(p) \stackrel{\text{def}}{=} [\gamma_{n,k}(p)],$$

where $p \in \text{Idem}(M_{k \times k}(M_{n \times n}(R)))$. This is, in fact, an additive map between abelian semigroups so the universal property of the Grothendieck group tells us that we can lift this to a group homomorphism:

$$\alpha : K_0(M_{n \times n}(R)) \rightarrow K_0(R).$$

Now, we need to show that α is the inverse of $K_0(\phi)$.

2. Let $[p]_s \in K_0(M_{n \times n}(R))$. Then:

$$(K_0(\phi) \circ \alpha)([p]_s) = K_0(\phi)(\alpha([p]_s)) = K_0(\phi)([\gamma_{n,k}(p)]_s) = [\phi(\gamma_{n,k}(p))]_s.$$

So, we want to show that $\phi(\gamma_{n,k}(p))$ and p are stably similar. But this is clear, because both of these give rise to idempotents with isomorphic ranges. So, they must represent the same element in K-theory.

3. Let $[p]_s \in K_0(R)$. Then:

$$(\alpha \circ K_0(\phi))([p]_s) = \alpha([\phi(p)]_s) = [\gamma_{n,k}(\phi(p))]_s.$$

Again, this is just $[p]_s$ because of exactly the same argument as above.

This proves the result. Now, let's deal with statement (2) in, very roughly, the same way.

Let $F : I \rightarrow \text{Rng}$ be an inductive system. For clarity, if $i \rightarrow j \in \text{Mor}(I)$, then we define:

$$\phi_{ij} \stackrel{\text{def}}{=} F(i \rightarrow j).$$

Before we state the main claims, let's set some notation and also construct the natural map which we need to study^b.

1. $\forall i \in I : R_i \stackrel{\text{def}}{=} F(i)$.
2. Let R be the inductive limit of F , with cocone maps $(\psi_i^1 : R_i \rightarrow R)_{i \in I}$.
3. Let S be the inductive limit of $K_0 \circ F$, with cocone maps $(\psi_i^2 : K_0(R_i) \rightarrow S)_{i \in I}$.
4. Applying K_0 to each ψ_i^1 , we obtain a family of morphisms:

$$(K_0(\psi_i^1) : K_0(R_i) \rightarrow K_0(R)).$$

Observe that $\psi_j^1 \circ \phi_{ij} = \psi_i^1$. Applying K_0 on both sides, we see that $(K_0(\psi_i^1))_{i \in I}$ forms a family of cocone maps for $K_0 \circ F$. So, this induces, by universal property, the following morphism:

$$\Phi : S \rightarrow K_0(R),$$

and we wish to prove that this is an isomorphism.

Now, let's consider the situation where this is an isomorphism when R_i is unital, R is unital and all morphisms above are unital; this will, henceforth, be known as the “unital situation”. In this case, we have the following claim.

| Claim. Statement (2) reduces to the unital situation.

We're finally ready to finish the proof of this corollary.

| Claim. Statement (2) holds in the unital case.

This completes the proof. ■

^aThis relies on the previously known algebraic fact that matrix amplifications of rings form exact functors.

^bWe've already done this but let's do it again for your convenience.

Proof. We'll prove each of these one-by-one.

- 1.
 2. We'll prove this by showing surjectivity and injectivity, relying on ??.
- (a) Let's show surjectivity; let $y \in K_0(R)$. Then, we can write $y = [p]_s - [1_n]_s$ where $p \in \text{Idem}(\text{Mat}_\infty(R^\sim))$ and $p - 1_n \in \text{Mat}_\infty(R)$. We also know that:

$$R = \bigcup_{i \in I} \text{ran}(\psi_i^1).$$

Now, $p - 1_n$ is a matrix with finitely many non-zero entries and we can fit them into finitely many ranges that appear in the union. So, these entries can be built out of R_i for finitely many i ; let's select an index greater than the indices of all such R_i and called it j . Then, $p - 1_n$ is the image of some matrix over R_j and this means that $[p]_s - [1_n]_s$ is in the image of $K_0(\psi_j^1)$. Hence, it's in the image of Φ .

- (b) Let's show injectivity: let $x \in S$ so that $\Phi(x) = 0$. By ??, we know that:

$$S = \bigcup_{i \in I} \text{ran}(\psi_i^2),$$

so, $x \in \text{ran}(\psi_i^2)$ for some $i \in I$. So, there exists an idempotent p such that:

$$\psi_i^2([p]_s - [1_n]_s) = x,$$

where $p - 1_n \in \text{Mat}_\infty(R)$. Applying Φ on both sides, we get that:

$$(\Phi \circ \psi_i^2)([p]_s) = (\Phi \circ \psi_i^2)([1_n]_s),$$

which implies that the images of p and 1_n in $K_0(R)$ are equal, so they are stably isomorphic. But this

implies that they are conjugate under R and therefore, we see that:

$$[p]_s = [1_n]_s,$$

which means that $x = 0$.

This completes the proof.

- 3.
- 4.

■

We have a remark about this result.

Remark 5.9. These properties also happen to hold for operator K-theory and we will see that the proofs are pretty much exactly the same, with a few minor technical caveats due to the analytic structure of C^* -algebras. This is also why we've presented K-theory first in the algebraic setting, just so that you can get used to the structure of these arguments. Of course, we will also need to consider some analytic subtleties.

With the explicit description of algebraic K_0 available to us along with some properties, let us carry out one calculation which uses just about everything we've done so far.

Example 5.10. Let $R \in \text{Rng}^1$ and consider the following map:

$$\phi : R \rightarrow M_{n \times n}(R) , \quad a \mapsto \phi(a) \stackrel{\text{def}}{=} \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}.$$

Here's a claim.

Claim. $K_0(\phi)$ is an isomorphism.

For the next two claims, we will concern ourselves with rings which are C^* -algebras. In this case, we can replace the idempotents that come up in the idempotent portraits above with projections instead. This is the only simplification we use which we haven't explicitly proved. On the other hand, there is nothing to worry about it as we establish this in Proposition 7.3. In any event, our purpose in presenting this example is to illustrate that we could use the results above to compute the K_0 -groups of various special rings OR we could use natural maps associated to them. This process, when applied to basic situations, highlights the importance of *traces*.

Claim. Consider the map:

$$\text{Tr} : M_{n \times n}(\mathbb{C}) \rightarrow \mathbb{C} , \quad p \mapsto \text{Tr}(p).$$

Then, $K_0(\text{Tr})$ is an isomorphism.

Let's show that this is injective and surjective.

1. Let $g \in K_0(M_{n \times n}(\mathbb{C}))$ so that $K_0(\text{Tr})(g) = 0$. Well, we can write:

$$g = [p]_s - [q]_s,$$

where p, q are projections in $M_{k \times k}(M_{n \times n}(\mathbb{C}))$. So, it follows that:

$$\begin{aligned} 0 &= K_0(\text{Tr})(g) \\ &= [\text{Tr}(p)]_s - [\text{Tr}(q)]_s \\ &= \dim(\text{ran}(p)) - \dim(\text{ran}(q)). \end{aligned}$$

So, $\text{ran}(p)$ and $\text{ran}(q)$ have the same dimension, which means that these subspaces are isomorphic. Hence, we see that $[p]_s = [q]_s$ and that $g = 0$, which proves injectivity.

2. As for surjectivity, we have that $\text{ran}(K_0(\text{Tr}))$ is a subgroup of $\mathbb{Z}$ and it is $\mathbb{Z}$ iff it contains 1. So, we use any rank-one projection $e \in M_{n \times n}(\mathbb{C})$ and note that its trace is going to be 1.

This completes the proof of the claim.

Claim. If H is a separable, infinite-dimensional $\mathbb{C}$ -Hilbert space, then $K_0(\mathcal{K}(H)) \simeq \mathbb{Z}$.

In fact, there are two ways of doing this and both of them are very interesting.

1. The first way is by using the unbounded trace on $\mathcal{B}(H)$. In the same spirit as above, one can show that applying K-theory to the trace map produces an isomorphism.
2. A more appealing route would be to use the fact that $\mathcal{K}(H)$ is actually the inductive limit of full matrix algebras. Indeed, one may recall that we can form an inductive sequence of the following type:

$$\mathbb{C} \xrightarrow{\phi_0} M_{2 \times 2}(\mathbb{C}) \xrightarrow{\phi_1} M_{2^2 \times 2^2}(\mathbb{C}) \xrightarrow{\phi_2} \dots$$

where we define each ϕ_n as in the start of this example. Hitting this sequence with K_0 , we obtain that the above is an inductive limit of a sequence consisting only of the integers. So, we end up with the desired result.

This completes the proof of the claim and our exposition via this example.

5.2 The (Algebraic) Excision Theorem

Let's state it first.

Theorem 5.11. The Algebraic Excision Theorem

Let $R \in \text{Rng}^1$ and let $I \subseteq R$ be an ideal. Then, $K_0(R, I) \simeq K_0(I)$.

There is just one undefined term and we'll take care of its definition before we give the proof. One should note that the

theorem itself is quite geometric; it is supposed to be the algebraic modelling of excision of spaces.

Terminology 5.12. Let $R \in \mathbf{Rng}$ and let $I \subseteq R$ be an ideal. We define:

$$D(R, I) \stackrel{\text{def}}{=} \{(x, y) \in R \times R : x - y \in I\}.$$

This is known as the double of R along I .

Here's a nice, concrete instantiation of a double ring over an ideal.

Example 5.13. Let $X \in \mathbf{CHaus}$ and let $A \subseteq X$ be a closed set. Define:

$$I \stackrel{\text{def}}{=} \{f \in \mathcal{C}(X) : f|_A = 0\}.$$

As you already know, this is clearly an ideal of $\mathcal{C}(X)$. More importantly:

$$D(\mathcal{C}(X), I) = \{(f, g) \in \mathcal{C}(X) \times \mathcal{C}(X) : f|_A = g|_A\}.$$

Indeed, this just follows from the definition of the double ring along I .

Here are a few notable facts that will allow us to define the relative K-theory of a ring with respect to an ideal.

Claim 5.3. Let $R \in \mathbf{Rng}$ and let $I \subseteq R$ be an ideal. Then, $D(R, I)$ is a subring of $R \times R$.

Claim 5.4. Let $R \in \mathbf{Rng}$ and let $I \subseteq R$ be an ideal. Then, we have the following exact sequence:

$$0 \rightarrow I \rightarrow D(R, I) \xrightarrow{p_1} R \rightarrow 0,$$

of rings which splits^a.

^aThe first map is the embedding of I into the second coordinate and p_1 is the projection onto the first coordinate.

Claim 5.4 allows us to easily define the relative K_0 -groups.

Definition 5.14

Let $R \in \mathbf{Rng}^1$ and let $I \subseteq R$ be an ideal. We define:

$$K_0(R, I) \stackrel{\text{def}}{=} \ker(K_0(p_1)),$$

where p_1 is the projection in **Claim 5.4**.

At this stage, we are ready for the proof of the Excision Theorem.

Proof of Theorem 5.11. We construct a candidate group homomorphism from $K_0(I)$ to $K_0(R, I)$. Then, we'll use the idempotent picture to prove that it is injective and surjective. In other words, the proof has three steps:

Construction: The key is to use the split-exact sequence in **Claim 5.4**. Define a unital ring homomorphism:

$$\gamma : I^\sim \rightarrow D(R, I), \quad (x, n) \mapsto \gamma(x, n) \stackrel{\text{def}}{=} (n \cdot 1, n \cdot 1 + x).$$

Then, we can draw the following diagram:

$$\begin{array}{ccc} I^\sim & \xrightarrow{\gamma} & D(R, I) \\ \downarrow \rho & & \downarrow \iota \\ \mathbb{Z} & \xrightarrow{p_1} & R \end{array}$$

One may observe by direct calculation that this diagram commutes. So, we see that $K_0(\gamma) : K_0(I^\sim) \rightarrow K_0(D(R, I))$ sends $\ker(K_0(\rho))$ to $\ker(K_0(p_1))$. But the former is just $K_0(I)$ and the latter is $K_0(R, I)$, so $K_0(\gamma)$ restricted to $K_0(I)$ gives us our desired map from $K_0(I) \rightarrow K_0(R, I)$.

Injectivity: This follows by applying the idempotent portrait; see [87, Theorem 1.5.9, p.30-31].

Surjectivity: This follows by applying the idempotent portrait; see [87, Theorem 1.5.9, p.30-31].

This completes the proof. ■

6 Homotopical Structures on Categories of Operator Algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| Motivation.

6.1 Standard Homotopy-Theoretic Constructions

Since we want to build up the language of homotopy theory for equivariant C^* -algebras and Banach algebras, it is reasonable to start with the definition of an equivariant C^* -algebra/Banach algebra. To facilitate a clean presentation of the definitions/results, we will often mainly state the results for G - C^* -algebras and in remarks afterwards, we will say something about how those definitions/results can be adapted to the Banach algebra situation. **All algebras we work with are over $\mathbb{C}$.** With that said, let us define equivariant C^* -algebras.

Definition 6.1

Let $G \in \text{Grp}(\text{Top})$. A G - C^* -algebra is a pair (A, α) consisting of:

1. A C^* -algebra A .
2. A continuous group action $\alpha : G \times A \rightarrow A$ such that for each $g \in G$, the map:

$$\alpha(g, \cdot) : A \rightarrow A,$$

is a $\star$ -automorphism. We refer to this as a continuous action by G on A via automorphisms.

If $(A, \alpha), (B, \beta)$ are G - C^* -algebras, then a morphism:

$$\phi : (A, \alpha) \rightarrow (B, \beta),$$

is a $\star$ -homomorphism $\phi : A \rightarrow B$ such that the following diagram:

$$\begin{array}{ccc} G \times A & \xrightarrow{\alpha} & A \\ \text{Id}_G \times \phi \downarrow & & \downarrow \phi \\ G \times B & \xrightarrow{\beta} & B \end{array}$$

commutes. ϕ is, then, said to be a G -equivariant $\star$ -homomorphism.

We have a remark about this definition.

Remark 6.2. Definition 6.1 also makes sense when A is a $\mathbb{C}$ -Banach algebra and when the action is by automorphisms of A in the category BanAlg . The morphisms are required to satisfy the same condition as given. For the pair (A, α) where $A \in \text{BanAlg}$ and α being an action with the required properties, we will call this a G -Banach algebra.

Let us form a category out of this data.

Example 6.3. Let $G \in \text{Grp}(\text{Top})$. We form a structure G - $C^*\text{Alg}$ as follows:

1. The objects are G - C^* -algebras.
2. The morphisms are G -equivariant $\star$ -homomorphisms, with the usual composition.

We have the following claim.

| **Claim.** This structure forms a category.

The identity map on any G - C^* -algebra is an equivariant $\star$ -homomorphism. Moreover, equivariance is preserved under composition and since $\star$ -homomorphisms compose to form $\star$ -homomorphisms, it follows that the composition of two equivariant $\star$ -homomorphisms is an equivariant $\star$ -homomorphism.

This structure forms a category; it is the category of G - C^* -algebras. We can also define the equivariant categories:

$$G\text{-BanAlg}, G\text{-BanAlg}_{\leq 1},$$

of G -Banach algebras, with the main difference between them being the morphisms.

At this point, we will present a number of standard constructions that will be crucial for a systematic treatment of homotopy theory on these categories of equivariant operator algebras. It is also a chance to carry out some explicit verifications of facts which are used in the literature but which are not given proofs.

Construction 6.4. Let $G \in \text{Grp}(\text{LCH})$ and let $A \in G\text{-}C^*\text{Alg}$. Define:

$$\text{Cyl}_G(A) \stackrel{\text{def}}{=} \mathcal{C}([0, 1], A),$$

$$\text{Cone}_G(A) \stackrel{\text{def}}{=} \{\phi \in \text{Cyl}_G(A) : \phi(0) = 0\},$$

$$\text{Sus}_G(A) \stackrel{\text{def}}{=} \{\phi \in \text{Cyl}_G(A) : \phi(1) = \phi(0) = 0\}.$$

These constructions also make sense when $A \in G\text{-BanAlg}$. In all of these cases, the various operations of interest are induced point-wise on elements of each of these sets. Moreover, we define:

$$\forall g \in G : \forall f \in \text{Cyl}_G(A) : \forall t \in [0, 1] : (g \cdot f)(t) \stackrel{\text{def}}{=} gf(t).$$

Let us check that this goes into $\text{Cyl}_G(A)$.

| **Claim 6.1.** If $g \in G$ and $f \in \text{Cyl}_G(A)$, then $gf \in \text{Cyl}_G(A)$.

This action is also naturally induced on Cone_G and Sus_G . We name these constructs as follows:

1. $\text{Cyl}_G(A)$ is the (equivariant) cylinder of A .
2. $\text{Cone}_G(A)$ is the (equivariant) cone of A .
3. $\text{Sus}_G(A)$ is the (equivariant) suspension of A .

We have the following claim.

Claim 6.2. Let $A \in G\text{-C}^*\text{Alg}$. Then:

$$\text{Cyl}_G(A), \text{Cone}_G(A), \text{Sus}_G(A),$$

are $G\text{-C}^*\text{-algebras}$. The same proof goes through when $A \in G\text{-BanAlg}$.

This is proved on Page 272. Moreover, we define for each $t \in [0, 1]$ a map:

$$p_t^A : \text{Cyl}_G(A) \rightarrow A, \quad \phi \mapsto p_t^A(\phi) \stackrel{\text{def}}{=} \phi(t).$$

This is known as the t -th canonical inclusion^a or the evaluation map at^b t .

Claim 6.3. For each $t \in [0, 1]$ and $A \in G\text{-C}^*\text{Alg}$, the map p_t^A is an equivariant $\star$ -homomorphism. When $A \in G\text{-BanAlg}$, then this is an equivariant continuous algebra homomorphism.

This is proved on Page 272. The evaluation map is useful specifically in providing an alternative characterization of two equivariant $\star$ -homomorphisms being homotopic; see Proposition 6.18.

^aWe'll assume that the cylinder of A is always equipped with this family of maps.

^bThis terminology makes sense when $A \in G\text{-BanAlg}$ without any change.

In the next few examples, we will construct functors associated with the cylinder, cone & suspension.

Example 6.5. Let $G \in \text{Grp}(\text{LCH})$. Define a map of categories:

$$\text{Cyl}_G : G\text{-C}^*\text{Alg} \rightarrow G\text{-C}^*\text{Alg},$$

as follows:

1. If $A \in G\text{-C}^*\text{Alg}$, then $\text{Cyl}_G(A)$ is just the cylinder of A as defined above.
2. If $A, B \in G\text{-C}^*\text{Alg}$ and $f : A \rightarrow B$ is an equivariant $\star$ -homomorphism, then we define:

$$\text{Cyl}_G(f) : \text{Cyl}_G(A) \rightarrow \text{Cyl}_G(B), \quad \phi \mapsto \text{Cyl}_G(f)(\phi) \stackrel{\text{def}}{=} f \circ \phi.$$

It's easily checked that this is well-defined.

Let us show that this is a functor:

1. Let $A \in G\text{-C}^*\text{Alg}$. Then:

$$\begin{aligned} \forall \phi \in \text{Cyl}_G(A) : \text{Cyl}_G(\text{Id}_A)(\phi) &= \text{Id}_A \circ \phi \\ &= \phi \\ &= \text{Id}_{\text{Cyl}_G(A)}(\phi), \end{aligned}$$

$$\text{so } \text{Cyl}_G(\text{Id}_A) = \text{Id}_{\text{Cyl}_G(A)}.$$

2. Let $A, B, C \in G\text{-C}^*\text{Alg}$, let $f \in G\text{-C}^*\text{Alg}(A, B)$ and $g \in G\text{-C}^*\text{Alg}(B, C)$. Then:

$$\begin{aligned} \forall \phi \in \text{Cyl}_G(A) : \text{Cyl}_G(g \circ f)(\phi) &= (g \circ f) \circ \phi \\ &= g \circ (f \circ \phi) \\ &= g \circ \text{Cyl}_G(f)(\phi) \\ &= \text{Cyl}_G(g)(\text{Cyl}_G(f)(\phi)) \\ &= (\text{Cyl}_G(g) \circ \text{Cyl}_G(f))(\phi), \end{aligned}$$

from which we conclude that $\text{Cyl}_G(g \circ f) = \text{Cyl}_G(g) \circ \text{Cyl}_G(f)$.

This proves that this is a functor; we will refer to it as the cylinder functor.

Example 6.6. Let $G \in \text{Grp}(\text{LCH})$. Define a map of categories:

$$\text{Cone}_G : G\text{-C}^*\text{Alg} \rightarrow G\text{-C}^*\text{Alg},$$

as follows:

1. If $A \in G\text{-C}^*\text{Alg}$, then $\text{Cone}_G(A)$ is the cone of A as defined above.
2. If $A, B \in G\text{-C}^*\text{Alg}$ and $f \in G\text{-C}^*\text{Alg}(A, B)$, then we define:

$$\text{Cone}_G(f) \stackrel{\text{def}}{=} \text{Cyl}_G(f)|_{\text{Cone}_G(A)}.$$

Now, we need to check that $\text{Cone}_G(f)$ actually maps into $\text{Cone}_G(B)$; let $\phi \in \text{Cone}_G(A)$. Then:

$$\begin{aligned} \text{Cone}_G(f)(\phi)(0) &= (f \circ \phi)(0) \\ &= f(\phi(0)) \\ &= f(0) \\ &= 0, \end{aligned}$$

as was desired.

Since this is just a restriction of the cylinder functor, it is also a functor; we refer to it as the cone (functor).

Example 6.7. Let $G \in \text{Grp}(\text{LCH})$. Define a map of categories:

$$\text{Sus}_G : G\text{-C}^*\text{Alg} \rightarrow G\text{-C}^*\text{Alg},$$

as follows:

1. If $A \in G\text{-C}^*\text{Alg}$, then $\text{Sus}_G(A)$ is the suspension of A as defined above.
2. If $A, B \in G\text{-C}^*\text{Alg}$ and $f : A \rightarrow B$ is an equivariant $\star$ -homomorphism, then we define:

$$\text{Sus}_G(f) \stackrel{\text{def}}{=} \text{Cyl}_G(f)|_{\text{Sus}_G(A)}.$$

We need to show that $\text{Sus}_G(f)$ maps suspensions into suspensions and that is similar to [Example 6.6](#).

Since this is just a restriction of the cylinder functor, it is also a functor; we refer to it as the suspension (functor).

With the functors above, we can show that the time evaluation maps in [Construction 6.4](#) assemble into a natural transformation. In fact, there are a few technical results which are useful for us to establish right now.

Claim 6.4. Let $G \in \text{Grp}(\text{LCH})$ and for each $t \in [0, 1]$, define a map of functors:

$$\alpha_t : \text{Cyl}_G \Rightarrow \text{Id}, \quad \alpha_t^A \stackrel{\text{def}}{=} p_t^A,$$

on $G\text{-C}^*\text{Alg} / G\text{-BanAlg}$. This map of functors is a natural transformation. This is proved on [Page 272](#).

It is also a notable result that the cylinder and suspension “commute” naturally.

Claim 6.5. Let $G \in \text{Grp}(\text{LCH})$. There exists a natural isomorphism^a $\alpha : \text{Sus}_G \circ \text{Cyl}_G \Rightarrow \text{Cyl}_G \circ \text{Sus}_G$.

^aThis is proved on [Page 272](#).

Claim 6.6. Let $G \in \text{Grp}(\text{LCH})$ and let $A \in G\text{-C}^*\text{Alg}$ or $A \in G\text{-BanAlg}$. Then:

$$\forall t \in [0, 1] : p_t^{\text{Sus}_G(A)} \circ \alpha_A = \text{Sus}_G(p_t^A),$$

where α is the natural isomorphism of [Claim 6.5](#). This is proved on [Page 272](#).

In the next construction, we will place the cone and suspension into exact sequences.

Construction 6.8. Let $G \in \text{Grp}(\text{LCH})$ and let $A \in G\text{-C}^*\text{Alg}$. In the notation of [Construction 6.4](#), we have:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Cone}_G(A) & \xrightarrow{\iota} & \text{Cyl}_G(A) & \xrightarrow{p_1^A} & A \longrightarrow 0 \\ & & & & & & \\ 0 & \longrightarrow & \text{Sus}_G(A) & \xrightarrow{\iota} & \text{Cone}_G(A) & \xrightarrow{p_0^A} & A \longrightarrow 0 \end{array}$$

where the maps ι are two different inclusion maps which simply include an element of the cone/suspension into the cylinder. These sequences are defined even when A is a G -Banach algebra. We have the following notable result.

Claim 6.7. The sequences above are exact, even when $A \in G\text{-BanAlg}$.

This is proved on [Page 272](#). These facts are useful to have on standby.

As we mentioned in the introduction, our interest in the homotopy theory of operator algebras partly exists because it is useful for understanding various (co)homology theories on admissible subcategories of $G\text{-C}^*\text{Alg}$ and $G\text{-BanAlg}$. Regarding the former, this is a legitimate “problem” that we describe formally as a question.

Question 6.9. Is there an abstract framework which describes the homotopy theory of the following categories:

1. C^*Alg ,
2. $\omega\text{C}^*\text{Alg}$, the full subcategory of separable C^* -algebras,
3. $\omega\text{NucC}^*\text{Alg}$, the full subcategory of separable nuclear C^* -algebras,

all at once?

We have a remark about this question.

Remark 6.10. This question was implicitly posed as a remark in [\[89, p.408\]](#). In the same paper, it is suggested that one should use the closed model categories of Quillen to define the notion of an admissible subcategory and even provides some explicit conditions that one could impose on a full subcategory to make it admissible in [\[89, Remark 2.12, p.420-421\]](#). Nevertheless, the matter is essentially left unresolved. In ??, we will propose a working definition of an admissible subcategory of C^*Alg which addresses [Question 6.9](#) in the non-equivariant case.

In spite of the fact that we do not yet have the definition of an admissible subcategory ([Definition 9.1](#)), we will still use it for the purposes of explaining the notions that will be needed to motivate our work. This is not detrimental to our exposition since we will only use it as motivation, for now. In that case, let us introduce some notation.

Notation 6.11. Let $\mathcal{A}$ be an abelian category. We use the notation:

$$F : \text{C}^*\text{Alg} \rightarrow \mathcal{A},$$

to denote a functor which is defined on some admissible subcategory of C^*Alg . We will usually not need to have separate notation for such a category, because we can always use $\text{dom}(F)$ to specify it.

With that said, let us introduce some standard properties of functors from an admissible category of C^* -algebras.

Terminology 6.12. Let $\mathcal{A}$ be an abelian category & let $F : C^* \text{Alg} \rightarrow \mathcal{A}$ be a functor.

1. F is exact iff for any exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in the domain, the following sequence:

$$0 \longrightarrow F(A) \xrightarrow{F(\phi)} F(B) \xrightarrow{F(\psi)} F(C) \longrightarrow 0$$

is exact in $\mathcal{A}$.

2. F is split-exact iff for any split-exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightleftharpoons[\alpha]{\psi} C \longrightarrow 0$$

in the domain, the following sequence:

$$0 \longrightarrow F(A) \xrightarrow{F(\phi)} F(B) \xrightleftharpoons[F(\alpha)]{F(\psi)} F(C) \longrightarrow 0$$

is split exact in $\mathcal{A}$.

F is half-exact iff for any exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in the domain, the following sequence:

$$F(A) \xrightarrow{F(\phi)} F(B) \xrightarrow{F(\psi)} F(C)$$

is exact in $\mathcal{A}$.

We have a remark about Terminology 6.12.

Remark 6.13. The given terminology makes sense when $\mathcal{A}$ is replaced with any category for which we have a reasonable notion of *exact sequence*; there are many such categories which are not abelian (or even pre-additive) for which this is the case. It also makes sense when $C^* \text{Alg}$ is replaced by^a BanAlg . Finally, the given terminology also makes sense in the equivariant case, without any modification other than the morphisms we consider.

^aWe will, however, not discuss this situation explicitly in this thesis.

An important (and independently interesting) property of the suspension functor is that it is exact. This is sometimes used in technical arguments, so we will include a proof of this fact. To do this, we need some notation.

Notation 6.14. Let $X \in \text{LCH}$ and let $A \in \text{Ban}_{\mathbb{C}}$ with $a \in A$ and $f \in \mathcal{C}_0(X)$. Define:

$$fa : X \rightarrow A, \quad x \mapsto (fa)(x) \stackrel{\text{def}}{=} f(x)a.$$

These are known as the simple tensors in $\mathcal{C}_0(X, A)$; it is clear that they belong to this algebra.

The following technical lemma tells us that the span above generates $\mathcal{C}_0(X, A)$ with the given assumptions on X and A .

Lemma 6.15. [86, Lemma 10.1.1, p.175-176] Let $X \in \text{LCH}$ and let $A \in \text{Ban}_{\mathbb{C}}$. Then:

$$\text{Span}\{fa : f \in \mathcal{C}_0(X) \wedge a \in A\},$$

is dense in $\mathcal{C}_0(X, A)$.

The proof of the lemma in the reference is for the case where A is a C^* -algebra but the result holds in the stated generality by the same proof. The difficulty in showing exactness is proving the surjectivity of the last map in the image sequence. The lemma above allows us to obtain an approximate solution which is upgraded to a full solution.

Proposition 6.16

Let $G \in \text{Grp}(\text{LCH})$. Then, Sus_G is an exact functor.

Proof. Consider the following exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in $G\text{-}C^* \text{Alg}$. Then, we obtain the following sequence:

$$0 \longrightarrow \text{Sus}_G(A) \xrightarrow{\text{Sus}_G(\phi)} \text{Sus}_G(B) \xrightarrow{\text{Sus}_G(\psi)} \text{Sus}_G(C) \longrightarrow 0$$

whose exactness needs to be checked. We'll do this in three separate steps.

1. Let $f \in \text{Sus}_G(A)$ so that $\text{Sus}_G(\phi)(f) = 0$. This means that $\phi \circ f = 0$ so $\text{ran}(f) \subseteq \ker(\phi) = 0$. That is, $f = 0$.
2. We need to show that $\text{Sus}_G(\psi)$ is surjective. Note that the following set:

$$\text{Span}\{fc : f \in \mathcal{C}_0(0, 1) \wedge c \in C\},$$

approximates every element in $\text{Sus}_G(C)$ so it remains to show that the elements in the dense subspace above belong to $\text{ran}(\text{Sus}_G(\psi))$. Indeed, let $c \in C$. Since ψ is surjective, there is a $b \in B$ such that:

$$\psi(b) = c.$$

So, we see that if $f \in \mathcal{C}_0(0, 1)$, then:

$$\begin{aligned}\text{Sus}_G(\psi)(fb) &= \psi \circ fb \\ &= f\psi(b) \\ &= fc,\end{aligned}$$

which proves that $\text{Sus}_G(\psi)$ is surjective.

3. Let's show that the second sequence is exact at $\text{Sus}_G(B)$. By functoriality, we have that:

$$\begin{aligned}0 &= \text{Sus}_G(0) \\ &= \text{Sus}_G(\psi \circ \phi) \\ &= \text{Sus}_G(\psi) \circ \text{Sus}_G(\phi),\end{aligned}$$

which implies that $\text{ran}(\text{Sus}_G(\phi)) \subseteq \ker(\text{Sus}_G(\psi))$.

This completes the proof. ■

One property that functors of interest should have is *homotopy invariance*. For this to be defined, we need the notion of homotopy and luckily, the two categories of interest have enough structure to accommodate such a definition.

Terminology 6.17. Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f, g \in G\text{-}C^*\text{Alg}(A, B)$.

1. A homotopy from f to g is a family $(\phi_t : A \rightarrow B)_{t \in [0,1]}$ in $G\text{-}C^*\text{Alg}(A, B)$ such that:

$$H_a : [0, 1] \rightarrow B, t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a),$$

is continuous for each $a \in A$. Moreover, $\phi_0 = f$ and $\phi_1 = g$.

2. f is homotopic to g iff there exists a homotopy from f to g . We denote this by writing $f \simeq_h g$.

3. A and B are homotopy equivalent iff there exist $f \in G\text{-}C^*\text{Alg}(A, B)$ and $g \in G\text{-}C^*\text{Alg}(B, A)$ so that:

$$g \circ f \simeq_h \text{Id}_A,$$

$$f \circ g \simeq_h \text{Id}_B.$$

In this case, we refer to g as the homotopy inverse of f and f as the homotopy inverse of g .

Finally, A and B are homotopically equivalent iff there is an equivariant $\star$ -homomorphism:

$$f : A \rightarrow B,$$

which is also a homotopy equivalence. The same definition makes sense in $G\text{-BanAlg}$.

There is an alternative, useful characterization of homotopy that we present in the next result.

Proposition 6.18

Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f, g \in G\text{-}C^*\text{Alg}(A, B)$. TFAE:

1. f and g are homotopic.

2. There is an equivariant $\star$ -homomorphism $h : A \rightarrow \text{Cyl}_G(B)$ such that $p_0^B \circ h = f$ and $p_1^B \circ h = g$.

The morphism h in (2) is also called a homotopy from f to g . The result also holds when $A, B \in G\text{-BanAlg}$.

Proof. Let's prove this for equivariant C^* -algebras first.

(1) $\Rightarrow$ (2): By definition, for each $t \in [0, 1]$, we have a $\star$ -homomorphism $\phi_t : A \rightarrow B$ such that:

$$H_a : [0, 1] \rightarrow B, t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a),$$

is continuous for each $a \in A$. Moreover, we have that $\phi_0 = f$ and $\phi_1 = g$. Now, we define:

$$h : A \rightarrow \text{Cyl}_G(B), a \mapsto h(a) \stackrel{\text{def}}{=} H_a.$$

Let's show that h is an equivariant $\star$ -homomorphism.

| **Claim 6.8.** h is an equivariant $\star$ -homomorphism.

This is proved on Page 50. Next, we will show that $p_0^B \circ h = f$. This is a calculation:

$$\begin{aligned}\forall a \in A : (p_0^B \circ h)(a) &= p_0^B(h(a)) \\ &= p_0^B(H_a) \\ &= H_a(0) \\ &= \phi_0 \\ &= f.\end{aligned}$$

A similar calculation with p_1^B yields the implication.

(2) $\Rightarrow$ (1): Let $t \in [0, 1]$. Define:

$$\phi_t : A \rightarrow B, a \mapsto \phi_t(a) \stackrel{\text{def}}{=} h(a)(t).$$

Let's check that $\phi_0 = f$. We have the following computation:

$$\begin{aligned}\forall a \in A : \phi_0(a) &= h(a)(0) \\ &= (p_0^B \circ h)(a) \\ &= f(a),\end{aligned}$$

as was desired. A similar calculation gives us that $\phi_1 = g$. Finally, for each $a \in A$, the following map:

$$H_a : [0, 1] \rightarrow B, \quad t \mapsto H_a(t) \stackrel{\text{def}}{=} \phi_t(a),$$

is continuous because h maps into the cylinder of B .

For equivariant Banach algebras, the only technical point lies in $(1) \Rightarrow (2)$. In that case, we would've had to explicitly prove that h is continuous since that would not follow from its algebraic structure. For that, we note that $(\phi_t)_{t \in [0,1]}$ is a family of continuous linear maps which is pointwise bounded since each H_a is a continuous map on a compact space. By the Uniform Boundedness Principle, we see that there exists a $c > 0$ such that:

$$\forall t \in [0, 1] : \|\phi_t\| \leq c.$$

So, this means that:

$$\begin{aligned} \forall a \in A : \|H_a\| &= \sup_{t \in [0,1]} \|H_a(t)\| \\ &= \sup_{t \in [0,1]} \|\phi_t(a)\| \\ &\leq c\|a\|, \end{aligned}$$

which is exactly the statement that h is bounded. ■

It is slightly better to have a proof of the fact that homotopy is an equivalence relation using the second characterization of homotopy as presented in Proposition 6.18. We will also show that isomorphisms are homotopy equivalences.

Claim 6.9. Let $G \in \text{Grp}(\text{LCH})$. Then, $\simeq_h$ is an equivalence relation in $G\text{-C}^*\text{Alg} / G\text{-BanAlg}$.

^aWe prove this on Page 272.

Claim 6.10. If $G \in \text{Grp}(\text{LCH})$, then isomorphisms in $G\text{-C}^*\text{Alg} / G\text{-BanAlg}$ are homotopy equivalences^a.

^aWe prove this on Page 272.

We can also produce a neat and simple example of a homotopy equivalence.

Example 6.19. Let $A \in G\text{-C}^*\text{Alg}$ and define a map:

$$s : A \rightarrow \text{Cyl}_G(A), \quad a \mapsto s(a),$$

where $s(a) : [0, 1] \rightarrow A$ is the constant function with value a . We have the following claim.

Claim 6.11. s is a homotopy equivalence and p_0^A is a homotopy inverse. We prove this on Page 272.

The result also holds when $A \in G\text{-BanAlg}$, simply because every map we define remains continuous.

Equipped with the notion of homotopy, we can also define what it means for a functor to be homotopy-invariant. We will be concerned with the basic study of homotopy-invariant functors later in ??.

Terminology 6.20. Let $\mathcal{C}$ be a full subcategory of $G\text{-C}^*\text{Alg}$, let $\mathcal{D}$ be a category and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$. F is homotopy-invariant iff for two parallel morphisms $\phi, \psi \in \mathcal{C}$, we have that:

$$\phi \sim_h \psi \Rightarrow F(\phi) = F(\psi).$$

This also works for $G\text{-BanAlg}$.

This is a reasonable place to introduce the definition of contractibility of a C^* -algebra.

Terminology 6.21. Let $A \in G\text{-C}^*\text{Alg} / G\text{-BanAlg}$. A is contractible iff Id_A is homotopic to 0.

In the next claim, we will establish the contractibility of the cone over any equivariant operator algebra.

Claim 6.12. Let $A \in G\text{-C}^*\text{Alg} / G\text{-BanAlg}$. Then, $\text{Cone}_G(A)$ is contractible. We prove this on Page 272.

One should note that contractibility of Banach algebras/ C^* -algebras is not the same as contractibility of spaces.

Example 6.22. Let $A \in {}^1\text{BanAlg} / {}^1\text{C}^*\text{Alg}$.

Claim. A is not contractible.

If it was contractible, then $K_0(A) \simeq 0$ since K_0 is homotopy-invariant. In particular:

$$[1_A]_{K_0} = [0]_{K_0},$$

which implies that $1_A = 0$ and this is impossible.

In particular, $\mathbb{C}$ is **not** contractible as a Banach algebra.

The main result about homotopies between morphisms of equivariant operator algebras is given by the next proposition. There are two conceptual reasons why this result will be significant for us:

1. From this result, we can deduce all formal properties of homotopy in the two categories of interest.
2. This result allows us to define homotopy categories.

We will see, later in the thesis, that the following result also has explicit influence on how we equip categories of fibrant objects with sufficient structure to ensure that their internal notion of homotopy is compatible with compositions.

Proposition 6.23

Let $A, B, C \in G\text{-C}^*\text{Alg}$, let $f_1, f_2 \in G\text{-C}^*\text{Alg}(A, B)$ and let $g_1, g_2 \in G\text{-C}^*\text{Alg}(B, C)$ so that:

$$f_1 \simeq_h f_2,$$

$$g_1 \simeq_h g_2.$$

Then, $g_1 \circ f_1 \simeq_h g_2 \circ f_2$. This result also holds in $G\text{-BanAlg}$.

Proof. We will proceed in three steps.

1. Show that the result holds when $f_1 = f_2$.
2. Show that the result holds when $g_1 = g_2$.
3. Deduce the general case from the two specific situations above.

With this plan, we proceed with the proof.

1. Let $f_1 = f_2 = f$. Then, there is a homotopy $h : B \rightarrow \text{Cyl}_G(C)$ such that:

$$g_1 = p_0^C \circ h,$$

$$g_2 = p_1^C \circ h.$$

Now, we define:

$$H \stackrel{\text{def}}{=} h \circ f.$$

This is an equivariant $\star$ -homomorphism $A \rightarrow \text{Cyl}_G(C)$. It is clear that this is a

$$\begin{aligned} \forall i \in \{0, 1\} : p_i^C \circ H &= p_i^C \circ (h \circ f) \\ &= (p_i^C \circ h) \circ f \\ &= g_{i+1} \circ f, \end{aligned}$$

which means that H is a homotopy from $g_1 \circ f$ to $g_2 \circ f$.

2. Let $g_1 = g_2 = g$. Then, there is a homotopy $h : A \rightarrow \text{Cyl}_G(B)$ such that:

$$f_1 = p_0^B \circ h,$$

$$f_2 = p_1^B \circ h.$$

Now, we define:

$$H \stackrel{\text{def}}{=} \text{Cyl}_G(g) \circ h.$$

This is an equivariant $\star$ -homomorphism. Let's show that this is a homotopy from $g \circ f_1$ to $g \circ f_2$:

$$\begin{aligned} p_0^C \circ H &= p_0^C \circ (\text{Cyl}_G(g) \circ h) \\ &= (p_0^C \circ \text{Cyl}_G(g)) \circ h \\ &= (g \circ p_0^B) \circ h \\ &= g \circ (p_0^B \circ h) \\ &= g \circ f_1, \end{aligned}$$

where we used the naturality of the time evaluation maps as proved in **Claim 6.3**. Showing that:

$$p_1^C \circ H = g \circ f_2,$$

is a similar computation.

3. We have that:

$$\begin{aligned} g_1 \circ f_1 &\simeq_h g_2 \circ f_1 \\ &\simeq_h g_2 \circ f_2, \end{aligned}$$

as was desired.

This completes the proof. ■

As mentioned above, we can establish three other formal properties of homotopy on categories of operator algebras.

Claim 6.13. Let $G \in \text{Grp}(\text{LCH})$ and let $\mathcal{C}$ be either $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$. TFSH:

1. If $f, g_1, g_2 \in \text{Mor}(\mathcal{C})$ with g_1, g_2 being homotopy inverses of f , then $g_1 \simeq_h g_2$.
2. Let $f, g_1, g_2 \in \text{Mor}(\mathcal{C})$ where g_1 is a left homotopy inverse of f and g_2 is a right homotopy inverse of f . Then, we have that $g_1 \simeq_h g_2$. In particular, f is a homotopy equivalence.
3. Let $f, g \in \text{Mor}(\mathcal{C})$ such that any two of $f, g, g \circ f$ are homotopy equivalences, then so is the third.

We prove this on Page 272.

In light of the facts above, we can now introduce homotopy categories.

Construction 6.24. We define:

$$\mathbf{Ho}(G\text{-C}^*\text{Alg}) \stackrel{\text{def}}{=} G\text{-C}^*\text{Alg} / \sim,$$

where $\sim$ is the equivalence relation above on morphism classes. This is known as the homotopy category of $G\text{-C}^*\text{Alg}$.

The same construction can be adapted for $G\text{-BanAlg}$ without change.

In classical homotopy theory, it is useful to introduce the notion of (co)fibration alongside the different notions of “weak equivalence” available (typically, homotopy equivalence and weak homotopy equivalence). The reasons for introducing these objects is variable but for our purposes, they will be tools which will assist us in “controlling” the homotopy

categories associated to the categories above. In this case, we will focus primarily on a notion of *fibration* which seems to have been first introduced by (Claude) Schochet in [89]. Actually, he referred to these as “cofibrations” due to the fact that they actually correspond, in the commutative case, to cofibrations in topology.

Definition 6.25

Let $G \in \text{Grp}(\text{LCH})$, let $A, B \in G\text{-C*Alg}$ and let $p \in G\text{-C*Alg}(A, B)$. p is a Schochet fibration iff for any $D \in G\text{-C*Alg}$, for any equivariant homotopy $(h_t : D \rightarrow B)_{t \in [0,1]}$ and for any $f \in G\text{-C*Alg}(D, A)$ such that the diagram:

$$\begin{array}{ccc} & & A \\ & \nearrow f & \downarrow i \\ D & \xrightarrow{h_t} & B \end{array}$$

commutes when $t = 0$, then there exists an equivariant homotopy $(H_t : D \rightarrow A)_{t \in [0,1]}$ such that the following diagram:

A commutative diagram with three nodes: D at the bottom left, A at the top right, and B at the bottom right. A dashed arrow labeled H_t points from D to A . A solid arrow labeled f points from D to A . A solid arrow labeled h_t points from D to B . A solid arrow labeled i points from A to B .

commutes for each $t \in [0, 1]$ and $H_0 = f$. This definition also makes sense in G -BanAlg.

Let us mention an important example of a Schochet fibration.

Example 6.26. Let $G \in \text{Grp}(\text{LCH})$ and let $A \in G\text{-C}^*\text{Alg}$. Here's the main claim.

Claim. For each $t \in [0, 1]$, the canonical inclusion:

$$p_t : \text{Cyl}_G(A) \rightarrow A,$$

as defined in [Construction 6.4](#), is a Schochet fibration. This result also holds when $A \in G\text{-BanAlg}$.

This is proved in the non-equivariant case in [89, Lemma 1.3, p.407]. To apply that to the equivariant case, one needs to check that the calculated lift is equivariant and it is, because the morphisms we start with are equivariant.

Our eventual goal is to use the notion above to show that $G\text{-}C^*\text{Alg}$ and $G\text{-BanAlg}$ are categories of fibrant objects; see Proposition 23.34. For this to happen, we need to establish some key permanence properties of Schochet fibrations. These were actually “proved” in the non-equivariant case in [89], though full proofs were not always supplied so we will fill in those gaps. One novelty in our approach is that we will consistently use a characterization of Schochet fibrations which was mentioned in [89, Definition 1.2, p.406] for $\star$ -homomorphisms between C^* -algebras but was not explicitly established.

Claim 6.14. Let $G \in \text{Grp}(\text{LCH})$, $A, B \in G\text{-C}^*\text{Alg}$ and let $p \in G\text{-C}^*\text{Alg}(A, B)$. TFAE:

1. p is a Schochet fibration.
2. If $D \in G\text{-C}^*\text{Alg}$, $f : D \rightarrow A$ is a $\star$ -homomorphism and $h : D \rightarrow \text{Cyl}_G(B)$ is a homotopy of the composite $p \circ f$, then there exists a homotopy $H : D \rightarrow \text{Cyl}_G(A)$ of f such that the following diagram:

$$\begin{array}{ccc}
& & A \\
& \nearrow f & \uparrow p_0 \\
D & \xrightarrow{H} & \text{Cyl}_G(A) \\
& \searrow h & \downarrow \text{Cyl}_G(p) \\
& & \text{Cyl}_G(B)
\end{array}$$

commutes.

The same result holds in G -BanAlg. The proof is given on Page 272.

We have two remarks at this stage.

Remark 6.27. In statement (2) of Proposition 6.18, we characterized homotopies between morphisms of algebras completely via morphisms and compositions. Statement (2) of **Claim 6.14** does the same thing for Schochet fibrations.

Remark 6.28. There are modern reasons as to why it would not make sense to refer to Schochet fibrations as “cofibrations”. One reason for this is the duality between spaces and operator algebras. Indeed, cofibrations over spaces should correspond to fibrations & fibrations over spaces should correspond to cofibrations. Moreover, cofibrations are typically thought of as morphisms which are closed under pushout-type constructions, even in model category theory. As we will see through Proposition 6.29, this is not the case for Schochet fibrations.

With this said, let us deal with our required permanence properties.

Proposition 6.29

Let $G \in \text{Grp}(\text{LCH})$. TFSH:

1. Schochet fibrations in $G\text{-C}^*\text{Alg}$ are closed under composition.
2. Let $A \in G\text{-C}^*\text{Alg}$ and let $f \in G\text{-C}^*\text{Alg}(A, 0)$. Then, f is a Schochet fibration.
3. Let $A, B \in G\text{-C}^*\text{Alg}$ and let $f \in G\text{-C}^*\text{Alg}(A, B)$ be an isomorphism. f is a Schochet fibration.
4. The cylinder functor preserves pullbacks.
5. Schochet fibrations are preserved by base extensions.

These results also hold in $G\text{-BanAlg}$.

Let us recall the meaning of some property being preserved by base extensions.

Terminology 6.30. Let $\mathcal{C}$ be a category and let P be a property of morphisms in $\mathcal{C}$. We say that P is preserved by base extensions iff for any pullback diagram in $\mathcal{C}$ of the following form:

$$\begin{array}{ccc} p & \longrightarrow & x \\ \downarrow & & \downarrow \\ z & \longrightarrow & y \end{array}$$

if $x \rightarrow y$ has property P , then so does $p \rightarrow z$.

Now, we can prove our result.

Proof of Proposition 6.29. We will prove each statement one-by-one, mainly using **Claim 6.14**.

1. Let $A, B, C \in G\text{-C}^*\text{Alg}$, let $p \in G\text{-C}^*\text{Alg}(A, B)$ and $q \in G\text{-C}^*\text{Alg}(B, C)$ be Schochet fibrations. This part is just a two-step lifting and we approach by starting with the following diagram which we will explain afterwards:

$$\begin{array}{ccccc} & & & A & \\ & & & \uparrow p_0^A & \\ & & & \text{Cyl}_G(A) & \\ & f \nearrow & & \downarrow \text{Cyl}_G(p) & \\ D & \xrightarrow{H} & & \text{Cyl}_G(B) & \\ & \nwarrow \tilde{H} & & \downarrow \text{Cyl}_G(q) & \\ & & & \text{Cyl}_G(C) & \\ & h \searrow & & & \end{array}$$

Let us ignore the dotted arrows for the moment. We note that h is a homotopy of $(q \circ p) \circ f$ and that this is the same diagram as in statement (2) of **Claim 6.14**, except that we have used the functoriality of Cyl_G to separate p and q . Now, we want to fill in the diagram by the dotted arrows and let us explain why this is possible. So, h is, in particular, a homotopy of $q \circ (p \circ f)$. Since q is a Schochet fibration, we can find a homotopy lift $\tilde{H}$ to $\text{Cyl}_G(B)$ and this is a homotopy of $p \circ f$. Since p is a Schochet fibration, we can find a homotopy lift H to $\text{Cyl}_G(A)$ and this is a homotopy of f . But this was the desired lift for the original lifting problem.

2. Let $D \in G\text{-C}^*\text{Alg}$ and suppose we have a commutative diagram:

$$\begin{array}{ccc} & A & \\ p \nearrow & \uparrow p_0^A & \\ D & \text{Cyl}_G(A) & \\ h \searrow & \downarrow \text{Cyl}_G(f) & \\ & \text{Cyl}_G(0) & \end{array}$$

where h is a homotopy of $f \circ p$. We define:

$$s : D \rightarrow \text{Cyl}_G(D), \quad d \mapsto s(d),$$

where $s(d)$ is the constant function from $[0, 1]$ into D with value d . Now, we define:

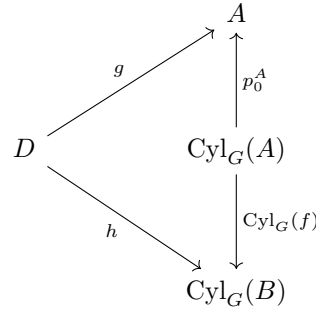
$$H \stackrel{\text{def}}{=} \text{Cyl}_G(p) \circ s.$$

Let us check that this works.

| **Claim 6.15.** H is a homotopy of p and it is the required homotopy lift. This is proved on Page 272.

This proves the result.

3. Consider the following homotopy lifting problem:



where h is a homotopy of $f \circ g$. Since f is an isomorphism, we simply define:

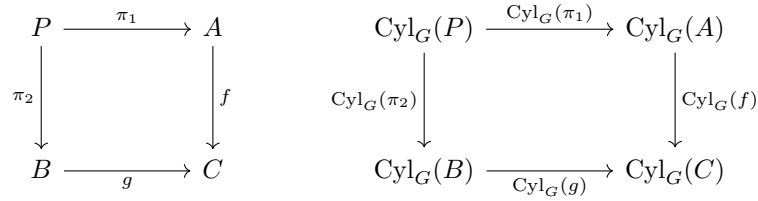
$$H \stackrel{\text{def}}{=} \text{Cyl}_G(f^{-1}) \circ h.$$

Let us check that this works.

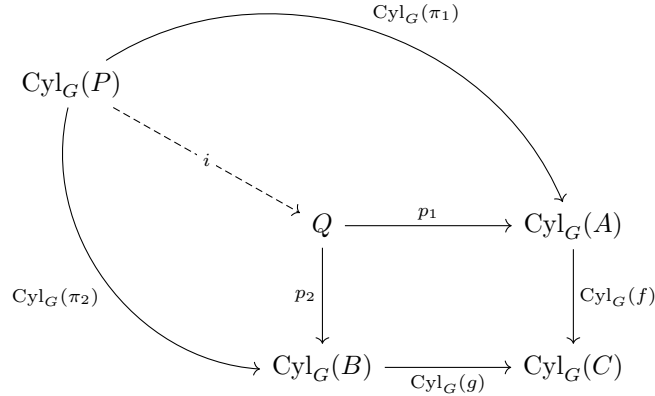
| **Claim 6.16.** H is a homotopy of g and is the desired homotopy lift. This is proved on Page 272.

This proves the result.

4. Consider the following diagrams:



where the left diagram is a pullback in $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$. We want to show that the diagram on the right is a pullback diagram. Let Q denote the actual pullback of the diagram on the right and note that the universal property of the pullback gives us a unique morphism $i : \text{Cyl}_G(P) \rightarrow Q$ such that the diagram:



commutes. Since we have an explicit form for P and for Q , we do have an explicit form for i . Namely:

$$\forall \phi \in \text{Cyl}_G(P) : i(\phi) = (\pi_1 \circ \phi, \pi_2 \circ \phi).$$

Now, we wish to define a map $j : Q \rightarrow \text{Cyl}_G(P)$, where we note that:

$$Q = \{(\phi, \psi) \in \text{Cyl}_G(A) \times \text{Cyl}_G(B) : f \circ \phi = g \circ \psi\}.$$

For $(\phi, \psi) \in Q$, we define:

$$j(\phi, \psi) : [0, 1] \rightarrow A \times B, \quad t \mapsto j(\phi, \psi)(t) \stackrel{\text{def}}{=} (\phi(t), \psi(t)).$$

Let us check that this is valued in P .

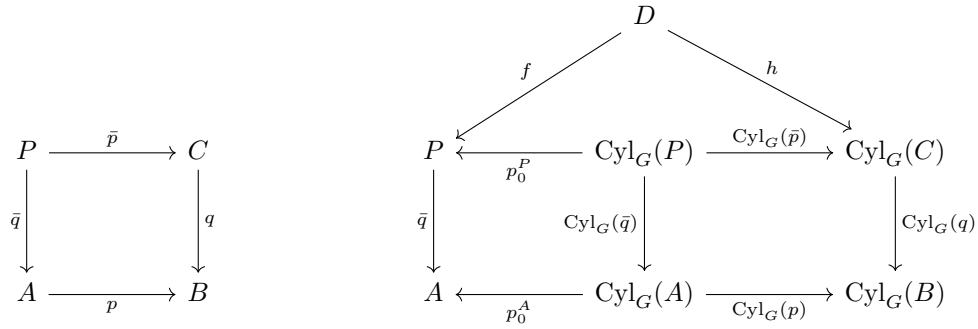
| **Claim 6.17.** $j(\phi, \psi)$ is valued in P . This is proved on Page 272.

$j(\phi, \psi)$ is clearly continuous since its components are continuous, so j is a well-defined function into $\text{Cyl}_G(P)$. It is clear that j is a homomorphism. Finally, we check that i and j are inverse to each other.

| **Claim 6.18.** i and j are inverse to each other. This is proved on Page 272.

So, i and j are inverse to each other. Equivariance follows from the fact that all morphisms involved are equivariant. In the case of $G\text{-BanAlg}$, the continuity of j follows from the continuity of its components.

5. Consider the following diagrams in $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$:



The left diagram is a pullback which we start with, where p is a Schochet fibration. To obtain the diagram on the right, we will first apply the functor Cyl_G to the left diagram. Then, we will extend the resulting diagram from the top by copying and pasting the diagram in (2) of **Claim 6.14**. Now, the proof proceeds as follows:

- Compute $\text{Cyl}_G(q) \circ h$ as a homotopy of the composite $p \circ (\bar{q} \circ f)$.
- Since p is a Schochet fibration, this induces a morphism from D to $\text{Cyl}_G(A)$ which is equal to $\text{Cyl}_G(q) \circ h$.
- The above creates a pullback cone for the pullback diagram of cylinders and since the pullback of this diagram is $\text{Cyl}_G(P)$, this induces a morphism from $D \rightarrow \text{Cyl}_G(P)$ which is the morphism we wanted.

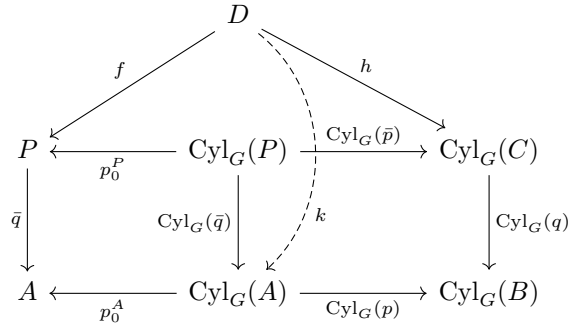
With this outline, let us proceed with filling in details.

- We have the following computation:

$$\begin{aligned}
 p_0^B \circ \text{Cyl}_G(q) \circ h &= q \circ p_0^C \circ h \\
 &= q \circ (\bar{p} \circ f) \\
 &= (q \circ \bar{p}) \circ f \\
 &= (p \circ \bar{q}) \circ f \\
 &= p \circ (\bar{q} \circ f),
 \end{aligned}$$

as was desired. So, $\text{Cyl}_G(q) \circ h$ is a homotopy of $p \circ (\bar{q} \circ f)$.

- As we mentioned above, p being a Schochet fibration gives us a diagram:



with $k : D \rightarrow \text{Cyl}_G(A)$ being a homotopy of $\bar{q} \circ f$. Moreover, we have that:

$$k \circ \text{Cyl}_G(p) = \text{Cyl}_G(q) \circ h.$$

by statement (2) of our characterization of Schochet fibrations.

- There is a pullback cone formed from $D \rightarrow \text{Cyl}_G(A) \rightarrow \text{Cyl}_G(B)$ and $D \rightarrow \text{Cyl}_G(C) \rightarrow \text{Cyl}_G(B)$. By the universal property of the pullback, there is a unique morphism from $D \rightarrow \text{Cyl}_G(P)$ such that all sub-diagrams formed commute. So, this morphism is a homotopy for f and $\bar{p}$ is a Schochet fibration.

This completes the proof. ■

Here is one more notable result which allows us to conclude that a $\star$ -homomorphism has a split from knowing that it is a Schochet fibration. We do not believe that this result is actually useful in practice, as we will eventually see that checking that a given morphism is a Schochet fibration can be a rather difficult problem. In fact, we are barely able to do it in an elementary way for even the Calkin projection on an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space; we will discuss this extensively in ?? and also when we discuss open problems in ??.

Proposition 6.31. A. Vats, 2026

Let $G \in \text{Grp}(\text{LCH})$, let $\mathcal{C}$ be either $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$, let $A, B \in \mathcal{C}$ and let $f \in \mathcal{C}(A, B)$ be a Schochet fibration such that there is a $g \in \mathcal{C}(B, A)$ which is a right homotopy inverse of f . Then, f is split epi in $G\text{-C}^*\text{Alg}$ and any left homotopy inverse of f is homotopic to its section.

Proof. Let us start with the following lifting problem:

$$\begin{array}{ccc}
 & & A \\
 & \nearrow g & \uparrow p_0^A \\
 B & \xrightarrow{H} & \text{Cyl}_G(A) \\
 & \searrow h & \downarrow \text{Cyl}_G(f) \\
 & & \text{Cyl}_G(B)
 \end{array}$$

where h is a homotopy from $f \circ g$ to Id_B . Since f is a Schochet fibration, we have a homotopy lift H and this is a homotopy of g . Now, we have the following computation:

$$\begin{aligned}
 f \circ (p_1^A \circ H) &= (f \circ p_1^A) \circ H \\
 &= (p_1^B \circ \text{Cyl}_G(f)) \circ H \\
 &= p_1^B \circ (\text{Cyl}_G(f) \circ H) \\
 &= p_1^B \circ h \\
 &= \text{Id}_B
 \end{aligned}$$

so f is split surjective. Now, let p be a left homotopy inverse of f and let g be the section of f . Then:

$$\begin{aligned}
 p &= p \circ \text{Id}_B \\
 &\simeq_h p \circ (f \circ g) \\
 &= (p \circ f) \circ g \\
 &\simeq_h g,
 \end{aligned}$$

as was desired. ■

This result is interesting and we have a remark about it that we will provide momentarily. Before that, we wish to introduce another class of morphisms which can be used as the “category of fibrations” over the category of C^* -algebras.

Terminology 6.32. Let $G \in \text{Grp}(\text{LCH})$, let $A, B \in G\text{-}C^*\text{Alg}$ and let $\phi \in G\text{-}C^*\text{Alg}(A, B)$. ϕ is a semisplit surjection iff it is a surjection which has an equivariant contractive & completely positive (ccp) linear section.

Here is a corollary of the proposition above.

Corollary 6.33. A. Vats, 2026

Let $G \in \text{Grp}(\text{LCH})$, let $A, B \in G\text{-}C^*\text{Alg}$ and let $f \in G\text{-}C^*\text{Alg}(A, B)$ be a Schochet fibration which admits a right homotopy inverse. Then, f is a semisplit surjection.

Proof. Indeed, the proposition above implies that f has a right inverse in $G\text{-}C^*\text{Alg}$ so we’re done. ■

Checking that some morphism is a Schochet fibration is difficult and one would like to construct homotopy equivalences without having to construct such fibrations. So, we introduce good deformation retracts to facilitate this process.

Definition 6.34

Let $G \in \text{Grp}(\text{LCH})$, let $\mathcal{C}$ be $G\text{-}C^*\text{Alg}$ or $G\text{-}Ban\text{Alg}$, let $A, B \in \mathcal{C}$ and let $f \in \mathcal{C}(A, B)$ be surjective. f is a good deformation retract iff there exists a homotopy $F : A \rightarrow \text{Cyl}_G(A)$ such that:

1. $p_0^A \circ F = \text{Id}_A$.
2. $\forall t \in [0, 1] : p_t^B \circ \text{Cyl}_G(f) \circ F = f$.
3. $(p_1^A \circ F)(\ker(f)) = 0$.

We will abbreviate good deformation retracts as GDRs.

The next result tells us that surjective GDRs give us homotopy equivalences. It is important to note that the next result **does not** necessarily hold for equivariant Banach algebras; see [Remark 6.36](#) for elaboration.

Proposition 6.35

Let $G \in \text{Grp}(\text{LCH})$. A surjective GDR in $G\text{-}C^*\text{Alg}$ is a homotopy equivalence.

Proof. Let $A, B \in G\text{-}C^*\text{Alg}$ and let $f \in G\text{-}C^*\text{Alg}(A, B)$ be a surjective GDR. There is a homotopy:

$$F : A \rightarrow \text{Cyl}_G(A),$$

such that the conditions of [Definition 6.34](#) hold. Let us factorize $p_1^A \circ F$ so that it is the composition $r \circ f$. To do this, we need to define r on B so let $b \in B$. Since f is surjective, we find $a \in A$ so $f(a) = b$. We define:

$$r(f(a)) \stackrel{\text{def}}{=} (p_1^A \circ F)(a).$$

Indeed, this is well-defined because if $a' \in A$ so that $b = f(a')$, then $a - a' \in \ker(f)$ and we know that $p_1^A \circ F$ vanishes on $\ker(f)$. The fact that r is a $\star$ -homomorphism follows by direct calculation with its definition and, in particular, the fact that f, p_1^A and F are $\star$ -homomorphisms. The rest of the proof involves establishing the next claim.

Claim. $r \circ f \simeq_h \text{Id}_A$ and $f \circ r = \text{Id}_B$.

The first statement holds because the map F satisfies the following relations:

$$\begin{aligned} p_0^A \circ F &= \text{Id}_A, \\ p_1^A \circ F &= r \circ f. \end{aligned}$$

For the second relation, let $b \in B$. Then, we find $a \in A$ so that $b = f(a)$. This leads us to a computation:

$$\begin{aligned} (f \circ r)(b) &= (f \circ r \circ f)(a) \\ &= (f \circ p_1^A \circ F)(a) \\ &= (p_1^B \circ \text{Cyl}_G(f) \circ F)(a) \\ &= f(a) \\ &= b, \end{aligned}$$

as was desired. ■

We have a remark about this result.

Remark 6.36. The proposition above is interesting for two reasons:

1. So far, most of the definitions and results we've presented work in the category of equivariant Banach algebras. It is not clear, however, that the next result holds in that category; the problem is that we construct a candidate homotopy inverse in the proof and there is no reason why this needs to be continuous.
2. In [89, p.412, Proposition 1.13a], it is claimed that if we have a Schochet fibration, then that morphism is a homotopy equivalence iff it is a GDR. We were not able to establish this directly ourselves and we do not think that it is true. However, we were not able to construct an explicit counterexample.

We will somewhat return to the issue in (2) within Remark 6.37.

The non-equivariant version of the next result is in [89, Proposition 1.13b, p.412]. However, a proof is not provided and a reference is given to the proof of an analogous result on spaces. That reference proves a more general result which does not seem to have an analogue in our setting. For this reason, we provide a self-contained full proof with details.

Claim 6.19. GDRs are preserved by base extensions.

At this point, we come to another result which we wanted to show but could not find a suitable argument to do it.

Remark 6.37. Let us state this as a claim.

Claim. Consider the following pullback diagram in $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$:

$$\begin{array}{ccc} P & \xrightarrow{\bar{p}} & C \\ \bar{q} \downarrow & & \downarrow q \\ A & \xrightarrow{p} & B \end{array}$$

If p is a homotopy equivalence and a Schochet fibration, then $\bar{p}$ is a homotopy equivalence and a Schochet fibration.

For the non-equivariant case and in the context of C^*Alg specifically, this result has been proved in [96, Theorem 2.19, p.10]. The proof uses a topological characterization of homotopy equivalences and Schochet fibrations:

1. the homotopy equivalences are characterized in terms of weak homotopy equivalences of mapping spaces and;
2. the Schochet fibrations are characterized in terms of Serre fibrations between the mapping spaces.

These characterizations exist because C^*Alg is enriched in the category of compactly-generated weakly Hausdorff spaces; we believe that these results extend to BanAlg . However, the proof of the claim above in these cases uses the fact that the category of compactly-generated weakly Hausdorff spaces equipped with the weak homotopy equivalences and Serre fibrations must be a *category of fibrant objects*. On the other hand, the presence of an action complicates matters for us; at the moment, it is unclear to us if the characterizations above extend to $G\text{-C}^*\text{Alg}$ or $G\text{-BanAlg}$. If it were the case that a Schochet fibration were a homotopy equivalence iff it was a GDR, then we would be able to provide a very easy proof of the claim. Since this is meant to emphasize a gap in the literature which does not seem to have an obvious resolution, let us present this alternative argument now.

Alternative Argument for Claim. The proof proceeds in just three steps:

1. Since p is a Schochet fibration, we see that $\bar{p}$ is a Schochet fibration.
2. To show that $\bar{p}$ is a homotopy equivalence, it's enough to show that it's a GDR.
3. Since p is a Schochet fibration and a homotopy equivalence, it also a GDR.
4. Since GDRs are preserved by base extensions, we see that $\bar{p}$ is a GDR.

This completes the proof.

This alternative is simple but it does not immediately work in $G\text{-BanAlg}$ since Proposition 6.35 may not hold. Now, we attempted to concoct a noncommutative argument to prove the claim above but there is an obstruction towards the end which we could not overcome. It effectively has to do with keeping a homotopy within a pullback.

Proof Attempt. We will prove this in the following concrete model for the pullback:

$$P = \{(a, c) \in A \times C : p(a) = q(c)\}.$$

Since p is a Schochet fibration, we see that $\bar{p}$ is a Schochet fibration by statement (5) of Proposition 6.29. Let $r : B \rightarrow A$ be a homotopy inverse of p ; by Proposition 6.31, r can be chosen so that it is a section of p . Define a map:

$$i_C : C \rightarrow P, \quad c \mapsto i_C(c) \stackrel{\text{def}}{=} ((r \circ q)(c), c).$$

Since r is a section of p , we see that i_C does, indeed, map into P and we also note that it is an equivariant $\star$ -homomorphism, since all maps involved in defining it are equivariant $\star$ -homomorphisms.

Subclaim. i_C is a homotopy inverse of $\bar{p}$.

Indeed, we have the following computation:

$$\begin{aligned} \forall c \in C : (\bar{p} \circ i_C)(c) &= \bar{p}((r \circ q)(c), c) \\ &= c \\ &= \text{Id}_C(c), \end{aligned}$$

which shows that i_C is a right inverse of $\bar{p}$. Moreover:

$$\begin{aligned} \forall (a, c) \in P : (i_C \circ \bar{p})(a, c) &= i_C(\bar{p}(a, c)) \\ &= i_C(c) \\ &= ((r \circ q)(c), c) \\ &= (r(q(c)), c) \\ &= (r(p(a)), c) \\ &= ((r \circ p) \times \text{Id}_C)|_P(a, c). \end{aligned}$$

So, it is sufficient for us to show that:

$$((r \circ p) \times \text{Id}_C)|_P \simeq_h \text{Id}_P.$$

Let $h : A \rightarrow \text{Cyl}_G(A)$ be a homotopy from Id_A to $r \circ p$. We could define:

$$H : P \rightarrow A \times B, \quad (a, c) \mapsto H(a, c) \stackrel{\text{def}}{=} (h_t(a), c),$$

but this does not lie in P for each $t \in [0, 1]$. We had the idea to define a two-parameter family:

$$(\Phi_{(s,t)} : P \rightarrow A \times C)_{(s,t) \times [0,1]^2},$$

where $\Phi_{(0,t)} = H_t$ for each $t \in [0, 1]$ and $\Phi_{(1,t)}$ lies in P for each $t \in [0, 1]$. We define:

$$\forall (a, c) \in P : \forall (s, t) \in [0, 1]^2 : \Phi_{(s,t)}(a, c) \stackrel{\text{def}}{=} \begin{cases} H_t(a, c) & \text{if } s = 0, \\ (a, c) & \text{if } t = 0, \\ H_{1-s}((r \circ p)(a), c) & \text{if } t = 1, \\ H_{(1-s)t}(a, c) & \text{otherwise} \end{cases}.$$

This was inspired by [98, Theorem 5.9, p.25]. However, $(\Phi_{(1,t)})_{t \in [0,1]}$ does not seem to be our desired homotopy. Indeed, for $t < 1$, this is just $H_0 = \text{Id}_P$. On the other hand, at $t = 1$, this is just $((r \circ p) \times \text{Id}_C)|_P$. The point, then, is that we cannot have point-norm continuity of this family of maps and the failure occurs at $t = 1$.

In spite of the obstruction laid out above, we firmly believe that the claim does hold. We believe this to be the case because the two categories in question do satisfy all other axioms of a category of fibrant objects & their structure is not so far from $C^*\text{Alg}$ that we expect this claim to fail.

At this stage, we have successfully developed the basics of equivariant homotopy theory on categories of operator algebras. Let us turn our attention towards a concrete conjecture which seems to have some unexpected applications.

6.2 Lifting Problems, Schochet C^* -Algebras & Semiprojectivity

Let us start by setting some notation.

Notation 6.38. Unless otherwise stated, H is an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space. Moreover:

$$\mathcal{Q}(H) \stackrel{\text{def}}{=} \mathcal{B}(H)/\mathcal{K}(H).$$

Finally, $\pi : \mathcal{B}(H) \rightarrow \mathcal{Q}(H)$ denotes the Calkin projection unless otherwise stated.

Our first goal is to provide a solution to the following conjecture due to Schochet.

Conjecture 6.39 (C. Schochet, 1984, [89, p.407]). The Calkin projection $\pi : \mathcal{B}(H) \rightarrow \mathcal{Q}(H)$ is not a Schochet fibration.

We will provide further motivation for interest in this conjecture momentarily. Here is an important remark.

Remark 6.40. Strictly speaking, Conjecture 6.39 has actually been resolved (positively) in [40, Example 6.3, p. 153]. In that paper, Loring and Exel introduce the notion of a spherical filtration of the K-theory of a unital C^* -algebra and go on to prove that if the surjective $\star$ -homomorphism in a given exact sequence is a Schochet fibration, then the associated K-theory boundary map respects the given spherical filtration (see Theorem 5.6 in the same paper, p.151-152). In the case of the Calkin extension, they show explicitly that the boundary map does not respect the given spherical filtration, so the Calkin projection cannot be a Schochet fibration. Nevertheless, a rather strong downside to their result is that an explicit C^* -algebra for which the fibration property of the Calkin projection fails is not provided. This leads us into the novelty of our approach, of which there are three main points worth noting:

1. Conjecture 6.39 is actually intimately related to an older question of Schochet's and (Jerome) Kaminker,

2. we will give an explicit example of a C^* -algebra for which the fibration property fails and;
3. our approach serves as a possible template for constructing counterexamples for homotopy-lifting in general.

Beyond these facts, we will also use our methods, which revolve around basic Fredholm theory, to resolve [Conjecture 6.39](#) in the case where H is non-separable. We will see that this has some natural implications for extensions of this problem which one can consider; we will discuss these in ??.

We will start by providing an extended discussion of the context behind the conjecture and its relationship with older problems in operator theory. Let us record down a useful piece of terminology that will fix language.

Terminology 6.41. Let P be a property that is satisfied by elements of a C^* -algebra and let $T \in \mathcal{B}(H)$. Then, T is essentially P iff $\pi(T)$ is P in $\mathcal{Q}(H)$. Moreover, the essential spectrum of T is defined by $\sigma_{\text{ess}}(T) \stackrel{\text{def}}{=} \sigma_{\mathcal{Q}(H)}(\pi(T))$.

We will mainly be concerned with essentially normal operators⁴. It is a stunning result of Brown-Douglas-Fillmore theory that there is a classification of such operators. To state this theorem formally, we will need a construction.

Construction 6.42. Let H be a $\mathbb{C}$ -Hilbert space and let $T \in \mathcal{B}(H)$ be Fredholm. We define:

$$\phi_T : \mathbb{C} \setminus \sigma_{\text{ess}}(T) \rightarrow \mathbb{Z}, \lambda \mapsto \phi_T(\lambda) \stackrel{\text{def}}{=} \text{Index}(T - \lambda \text{Id}_H).$$

This is known as the index function of T . This makes sense, because $\lambda \notin \sigma_{\text{ess}}(T)$ so $\pi(T) - \lambda 1_{\mathcal{Q}(H)}$ is invertible. Hence, $T - \lambda \text{Id}_H$ is a Fredholm operator and the index above is well-defined.

Now, we can state our result.

Theorem 6.43. The Brown-Douglas-Fillmore Theorem, [41, Corollary 11.2, p.118], [43]

Let H be an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space and let $T \in \mathcal{B}(H)$ be essentially normal. TFAE:

1. $T \in N_{\mathcal{B}(H)} + \mathcal{K}(H)$.
2. ϕ_T is trivial.

The key tool used to prove the theorem above (and related theorems classifying such operators) are the so-called Brown-Douglas-Fillmore (BDF) Ext groups. We will, now, provide some intuition for how these groups arise naturally; our exposition is heavily inspired by the introduction of [18]. If $T \in \mathcal{B}(H)$ is essentially normal, then T (along with $\mathcal{K}(H)$ and Id_H) generates a C^* -subalgebra A of $\mathcal{B}(H)$. With this data, we can construct a morphism $A \rightarrow C^*(\pi(T), \text{Id}_{\mathcal{Q}(H)})$, whose kernel is $\mathcal{K}(H)$. On the other hand, the C^* -algebra on the right-hand side of the arrow is a commutative unital C^* -algebra. By Gelfand Duality and some more concrete results about functional calculus, we see that:

$$C^*(\pi(T), \text{Id}_{\mathcal{Q}(H)}) \simeq \mathcal{C}(\sigma_{\text{ess}}(T)).$$

So, this data fits into an exact sequence:

$$0 \rightarrow \mathcal{K}(H) \rightarrow A \rightarrow \mathcal{C}(\sigma_{\text{ess}}(T)) \rightarrow 0.$$

In other words, essentially normal operators naturally give rise to extensions of the form given above and now, it becomes reasonable to replace $\sigma_{\text{ess}}(T)$ by some compact metrizable space X and study extensions of the following form:

$$0 \rightarrow \mathcal{K}(H) \rightarrow A \rightarrow \mathcal{C}(X).$$

This leads us to the following definition.

Definition 6.44

Let X be a compact, metrizable space.

1. A extension of $\mathcal{K}(H)$ by $\mathcal{C}(X)$ is a pair:

$$(A, \phi),$$

where A is a C^* -subalgebra of $\mathcal{B}(H)$ containing $\mathcal{K}(H)$ and Id_H & $\phi \in C^*\text{Alg}(A, \mathcal{C}(X))$ with $\ker(\phi) = \mathcal{K}(H)$.

2. If $(A_1, \phi_1), (A_2, \phi_2)$ are extensions of $\mathcal{K}(H)$ by $\mathcal{C}(X)$, then these extensions are strongly equivalent^a iff there exists a $\star$ -isomorphism $\psi : A_2 \rightarrow A_1$ such that $\phi_2 = \phi_1 \circ \psi$.

The set of equivalence classes of extensions of $\mathcal{K}(H)$ by $\mathcal{C}(X)$ is denoted by $\text{Ext}(X)$.

^aThis can be checked directly to be an equivalence relation.

We will not spend any time discussing the proofs of various results in BDF theory since that would take up too much space. The main result that needs to be known is that $\text{Ext}(X)$ can be made into a group and that Ext is functorial.

Proposition 6.45. [18, Corollary 1.18, Theorem 1.23, p. 272-273, p. 275]

We have a functor $\text{Ext}(-)$ from the category of compact metrizable spaces to AbGrp .

In fact, this functor happens to be homotopy-invariant. The proof of homotopy-invariance as discussed in [18, Section 2, p.275-286] is somewhat tricky and an alternative, simpler proof was desired. To that end, there was another functor constructed by Schochet & Kaminker which homotopy-invariance is “obvious”.

Construction 6.46. Let X be a compact metrizable space. We define:

$$\text{inj}^h(\mathcal{C}(X), \mathcal{Q}(H)),$$

⁴For an extended discussion of essentially normal operators & their history, as well as recent developments, see [33].

to be the set of all homotopy classes of injective $\star$ -homomorphisms $\mathcal{C}(X) \rightarrow \mathcal{Q}(H)$. Using the methods of [57], this can be shown to be an abelian group. This is also a functor^a, denoted by inj^h .

^aFrom the category of compact metrizable spaces to AbGrp .

These functors coincide on a very special class of spaces. To define them, we will need some terminology.

Terminology 6.47. Let $A, B \in C^*\text{Alg}$ and let $f, g \in C^*\text{Alg}(A, B)$. f and g are $\star$ -homotopic iff there is a continuous map^a $\phi : A \times [0, 1] \rightarrow B$ such that $\phi(\cdot, t)$ is a $\star$ -homomorphism for each $t \in [0, 1]$, with $\phi(\cdot, 0) = f$ and $\phi(\cdot, 1) = g$.

^aUnsurprisingly, we call this a $\star$ -homotopy.

This notion was introduced in [29] and [30]. Let us show that it is not different from the usual notion of homotopy. While this has not been shown explicitly in the literature, the computation used to establish it is standard.

Claim 6.20. Let $A, B \in C^*\text{Alg}$ and let $f, g \in C^*\text{Alg}(A, B)$. TFAE:

1. f and g are homotopic.
2. f and g are $\star$ -homotopic.

We also get explicit ways of defining homotopies from $\star$ -homotopies and vice versa. This is proved on Page 272.

Now, we can introduce the C^* -Covering Homotopy Property.

Definition 6.48. J. Conway, [29]

Let $A \in C^*\text{Alg}$. A has the C^* -Covering Homotopy Property (or satisfies the C^* -CHP or is C^* -CHP) iff for any $\star$ -homotopy $\phi : A \times [0, 1] \rightarrow \mathcal{Q}(H)$ and any $f \in C^*\text{Alg}(A, \mathcal{B}(H))$ such that:

$$\pi \circ f = \phi(\cdot, 0),$$

there is a $\star$ -homotopy $\Phi : A \times [0, 1] \rightarrow \mathcal{B}(H)$ such that $\pi \circ \Phi = \phi$ and $\Phi(\cdot, 0) = f$.

The following result explains the relationship between the BDF Ext groups and the functor in Construction 6.46.

Proposition 6.49. C. Schochet, J. Kaminker, [57, p.139]

If X is a compact metrizable space which is C^* -CHP, then:

$$\text{Ext}(X) \simeq \text{inj}^h(\mathcal{C}(X), \mathcal{Q}(H)),$$

and the isomorphism is natural on the full subcategory of compact metrizable C^* -CHP spaces.

In light of the fact that inj^h is homotopy-invariant, Schochet and Kaminker asked the following question.

Question 6.50 ([57, p.139]). Is every compact metrizable space C^* -CHP?

In the aforementioned papers [29] and [30], (John) Conway actually showed that there are a number of interesting examples of C^* -algebras which satisfy the C^* -CHP. In particular, he showed that AF algebras are C^* -CHP, $\mathcal{C}(S^1)$ is C^* -CHP and $\mathcal{C}([0, 1])$ is C^* -CHP. He also posed the following question in less formal form.

Question 6.51 (J. Conway, [29, p.136]). Are the following spaces:

1. $S^1 \vee S^1$,
2. $S^1 \vee [1, 2]$,
3. $[-1, 0] \cup \{2^{-n} : n \in \mathbb{N}_0\}$,

C^* -CHP?

It seems difficult to construct many examples of C^* -algebras which are C^* -CHP. It is likely that the spaces in Question 6.51 were chosen since they were just out of reach using the methods of the aforementioned papers. This question will be addressed later, when we have introduced semiprojectivity into our exposition. Now, we have to relate the discussion above to Conjecture 6.39. Let us introduce the first main class of C^* -algebras whose properties will concern us.

Definition 6.52. A. Vats, 2026

Let $A \in C^*\text{Alg}$. A is Schochet iff for any infinite-dimensional, separable $\mathbb{C}$ -Hilbert space H and for any solid diagram:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \end{array}$$

(A dashed curved arrow labeled $(H_t)_{t \in [0, 1]}$ points from A to $\mathcal{B}(H)$.)

which commutes for $t = 0$, there exists a homotopy $(H_t)_{t \in [0, 1]}$ such that:

$$H_0 = f,$$

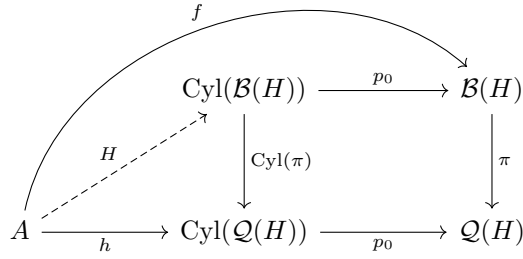
$$\forall t \in [0, 1] : \pi \circ H_t = h_t.$$

Let $X \in \text{LCH}$ and $G \in \text{Grp}(\text{LCH})$. X is Schochet iff $\mathcal{C}_0(X)$ is Schochet. G is Schochet iff $C_{\text{red}}^*(G)$ is Schochet.

Let us characterize the Schochet property for C^* -algebras in a way that will be useful.

Claim 6.21. Let $A \in C^*\text{Alg}$. TFAE:

1. A is Schochet.
2. The following lifting problem (indicated by the solid diagram):



admits a solution (indicated by the dotted arrows).

This is proved on Page 272.

In the next result, we will clarify the relationship between Schochet C^* -algebras and C^* -CHP C^* -algebras.

Claim 6.22. Let $A \in C^*\text{Alg}$. TFAE:

1. A is Schochet.
2. A is C^* -CHP.

Having given all of this background and having posed some interesting questions, it is time that we started working towards answering them and also developing this theory a bit more systematically. Let us, before doing anything else, mention how [Conjecture 6.39](#) can be understood in the context of the material introduced previously.

Proposition 6.53. A. Vats, 2026

The Calkin projection is not a GDR.

Proof. Let us suppose that it's a GDR. By Proposition 6.35, the Calkin projection must be a homotopy equivalence so it follows that $\mathcal{B}(H)$ and $\mathcal{Q}(H)$ have isomorphic K_1 -groups. However, this is clearly not the case as:

$$K_1(\mathcal{B}(H)) \simeq 0,$$

and $K_1(\mathcal{Q}(H)) \simeq \mathbb{Z}$. ■

We have two remarks about this proof.

Remark 6.54. In fact, we can use a similar argument to show that the canonical projection cannot even have a right homotopy inverse. So, there does not seem to be any hope in utilizing Proposition 6.31 to obtain a contradiction.

Remark 6.55. The argument used above can be generalized to show that other morphisms of C^* -algebras are not GDRs. Indeed, we can do this for morphisms where the domain and codomain do not have isomorphic K -groups.

Let us give our positive solution to [Conjecture 6.39](#).

Theorem 6.56. A. Vats, 2026

The following universal C^* -algebra:

$$A \stackrel{\text{def}}{=} C^*(x | xx^* = x^*x \wedge \|x\| \leq 1).$$

is not Schochet. That is, [Conjecture 6.39](#) holds.

Proof. Let $H = l^2(\mathbb{N})$, let S be the unilateral shift in $\mathcal{B}(H)$ and let s be the image of S in $\mathcal{Q}(H)$ under the Calkin projection. Then, the C^* -algebra generated by S in $\mathcal{B}(H)$ is the Toeplitz C^* -algebra $\mathcal{T}$ and it contains $\mathcal{K}(H)$. Moreover, it is a known fact that $\mathcal{T}/\mathcal{K}(H) \simeq \mathcal{C}(S^1)$. Now, we will need two subclaims.

Subclaim. There is no normal element $a \in \mathcal{B}(H)$ so that $\pi(a) = s$.

This simple fact is stated without proof in [17, p.1]; we will establish it at the end of this proof, after we have dealt with the main result. Now, the second subclaim is much more standard so we will point to a reference.

Subclaim. Let $E = \{x_i : i \in I\}$ be a set of generators, let R be relations between these generators such that the universal C^* -algebra $C^*(E|R)$ exists and let $B \in C^*\text{Alg}$ contain a set $E' = \{y_i : i \in I\}$. If the elements of E' satisfy the relations R , there is a unique $\star$ -homomorphism $\phi : C^*(E|R) \rightarrow B$ sending x_i to y_i for each $i \in I$.

For the proof and further systematic discussion about universal C^* -algebras, see [61, Proposition 6.7, p.71]. By subclaim (1), s does not have a unitary preimage in $\mathcal{B}(H)$ via the Calkin projection. Now, we define a map

$$\phi(x) : [0, 1] \rightarrow \mathcal{Q}(H), \quad t \mapsto \phi(x)(t) \stackrel{\text{def}}{=} ts.$$

This is continuous since:

$$\begin{aligned}\forall t, r \in [0, 1] : \|\phi(x)(t) - \phi(x)(r)\| &= \|ts - rs\| \\ &= |t - r| \cdot \|s\| \\ &\leq |t - s|.\end{aligned}$$

So, $\phi(x) \in \text{Cyl}(\mathcal{Q}(H))$. Moreover, $\phi(x)$ actually maps into $\text{Cyl}(\mathcal{T}/\mathcal{K}(H))$ and this is commutative so it follows that $\phi(x)$ is normal in $\mathcal{Q}(H)$. Now, we also have the following estimate:

$$\begin{aligned}\|\phi(x)\| &= \sup_{t \in [0, 1]} \|\phi(x)(t)\| \\ &= \sup_{t \in [0, 1]} \|ts\| \\ &\leq 1.\end{aligned}$$

So, $\phi(x)$ is a normal contraction in $\text{Cyl}(\mathcal{Q}(H))$. By the universal property of universal C^* -algebras (subclaim (2)), this induces a $\star$ -homomorphism $\phi : A \rightarrow \text{Cyl}(\mathcal{Q}(H))$. By construction, this is a homotopy from ϕ_0 to ϕ_1 . Now, the key point is that ϕ_0 lifts to $\mathcal{B}(H)$ while ϕ_1 does not lift to $\mathcal{B}(H)$ due to subclaim (1). Indeed, if ϕ_1 did have a lift $\psi : A \rightarrow \mathcal{B}(H)$, then $\pi(\psi(x)) = s$ with $\psi(x)$ being a normal contraction because ψ is contractive as a $\star$ -homomorphism. So, we have constructed a homotopy from ϕ_0 to ϕ_1 with ϕ_0 being liftable and ϕ_1 not being liftable, so we're done.

Now, let us deal with the proof of subclaim (1). A very classical and direct index theory calculation reveals that:

$$\text{Index}(S) = -1.$$

If $T \in \mathcal{B}(H)$ is normal, then we know that:

$$\text{coker}(T) \simeq \ker(T^*).$$

So, we see that:

$$\begin{aligned}\text{Index}(T) &= \dim(\ker(T)) - \dim(\text{coker}(T)) \\ &= \dim(\ker(T)) - \dim(\ker(T^*)) \\ &= 0,\end{aligned}$$

where the last statement holds, again, because normality is equivalent to the assertion that:

$$\forall x \in H : \|T(x)\| = \|T^*(x)\|.$$

On the other hand, let $T \in \mathcal{B}(H)$ so that:

$$\pi(T) = s = \pi(S),$$

which implies that $T - S \in \mathcal{K}(H)$. But this means that their Fredholm indices coincide but this is impossible. ■

As a consequence, we also resolve [Question 6.50](#).

Corollary 6.57. A. Vats, 2026

There exists a compact metrizable space which is not C^* -CHP. So, [Question 6.50](#) has a negative answer.

Proof. Let us take the compact unit disk $\mathbb{D}$. Then:

$$\mathcal{C}(\mathbb{D}) \simeq C^*(x, 1 | xx^* = x^*x \wedge \|x\| \leq 1).$$

This C^* -algebra is actually the unitization of the one used in [Theorem 6.56](#). In that case, we simply construct a homotopy from the $\star$ -homomorphism which takes x to 0 and 1 to $\text{Id}_{\mathcal{Q}(H)}$ to the $\star$ -homomorphism which takes x to s (as defined in the proof of the aforementioned theorem) and 1 to $\text{Id}_{\mathcal{Q}(H)}$. The first $\star$ -homomorphism is clearly liftable while the latter is not liftable, but both are homotopic. So, the given homotopy cannot lift. ■

Let us describe the inspiration behind the construction above.

Remark 6.58. Our argument in [Theorem 6.56](#) was inspired by the proof of (Bruce) Blackadar that the C^* -algebra constructed in the theorem is not l -closed. We will introduce this notion in [Definition 6.64](#) and explain why it made sense to look at different examples of l -closed C^* -algebras; see [[14](#), Example 6.5].

As we promised in [Remark 6.40](#), let us show that the Calkin projection is not a fibration for **any** infinite-dimensional $\mathbb{C}$ -Hilbert space. The proof requires just a little more input from Fredholm theory.

Proposition 6.59. A. Vats, 2026

Let H be any infinite-dimensional $\mathbb{C}$ -Hilbert space. Then, the Calkin projection over H is not a Schochet fibration.

Proof. We will adapt the argument of [Theorem 6.56](#) to this situation. In particular, define:

$$A \stackrel{\text{def}}{=} C^*(x, 1 | x^*x = xx^* \wedge \|x\| \leq 1).$$

We want to show that the Calkin projection fails the fibration property for this C^* -algebra. We have already shown this in the separable case, so assume that H is non-separable. We will construct a $T \in \mathcal{B}(H)$ such that:

1. T is Fredholm;
2. $\text{Index}(T) \neq 0$ and;
3. $\pi(T)$ is a normal contraction.

If we can do this, then exactly the same argument as in the separable case will work. We need a general claim.

Claim. Let $X, X', Y, Y' \in \text{Ban}_{\mathbb{C}}$, let $T \in \text{Ban}_{\mathbb{C}}(X, Y)$ and let $S \in \text{Ban}_{\mathbb{C}}(X', Y')$.

1. The linear map $T \oplus S$ is bounded, with $\|T \oplus S\| = \max\{\|T\|, \|S\|\}$.
2. If T and S are Fredholm operators, then $T \oplus S$ is a Fredholm operator.
3. If (2) holds, then $\text{Index}(T \oplus S) = \text{Index}(T) + \text{Index}(S)$.

We refer to $T \oplus S$ as the direct sum of T and S .

This can be found in [71, Exercise 19, p.33]; the proof involves relating $\ker(T \oplus S)$ to $\ker(T)$ and $\ker(S)$, while also relating $\text{ran}(T \oplus S)$ to $\text{ran}(T)$ and $\text{ran}(S)$. Now, let $M \subseteq H$ be a separable, closed subspace and write $H = M \oplus N$, where this is a topological direct sum; this is possible because closed subspaces of $\mathbb{C}$ -Hilbert spaces are complemented. Next, let S be the unilateral shift on M with respect to some orthonormal basis of M and define:

$$T \stackrel{\text{def}}{=} S \oplus \text{Id}_N.$$

Since S and Id_N are Fredholm operators, the claim implies that T is a Fredholm operator. Moreover:

$$\text{Index}(T) = \text{Index}(S) + \text{Index}(\text{Id}_N) = -1.$$

Let us also note that:

$$\|T\| = \|S \oplus \text{Id}_N\| = \max\{\|S\|, \|\text{Id}_N\|\} = 1.$$

Next, we have the following claim.

Claim. $\pi(T)$ is a normal contraction.

The fact that this is a contraction is easy:

$$\|\pi(T)\| \leq \|T\| = 1.$$

As for normality, we have the following computation:

$$\begin{aligned} TT^* - T^*T &= (S \oplus \text{Id}_N)(S^* \oplus \text{Id}_N) - (S^* \oplus \text{Id}_N)(S \oplus \text{Id}_N) \\ &= SS^* \oplus \text{Id}_N - \text{Id}_M \oplus \text{Id}_N \\ &= (SS^* - \text{Id}_M) \oplus 0, \end{aligned}$$

where $SS^* - \text{Id}_M$ is a compact operator. So, the above is a finite-rank operator and, therefore, compact. Hence, its image in the Calkin algebra vanishes and this implies that $\pi(T)$ is normal. So, the claim holds.

So, we have constructed our operator with the desired properties and we are done. ■

We have a remark about the result above.

Remark 6.60. We have written out the argument above in excess detail just to make sure that each step is convincing. It will also lead rather nicely into our discussion of a more general version of this problem in ??.

It is interesting to note that we do not even know if $\mathcal{C}(\mathbb{D} \setminus \{(0,0)\})$ is l -closed. To explain what this means, we need a bit of preparation. Let us introduce two pieces of convenient terminology and a construction.

Terminology 6.61. Let $A \in C^*\text{Alg}$ and let $I \subseteq A$ be a closed ideal. Then, the pair:

$$(A, I),$$

is known as a marked C^* -algebra.

Terminology 6.62. Let $A \in C^*\text{Alg}$, let (B, I) be a marked C^* -algebra, let $\pi_I : B \rightarrow B/I$ be the canonical projection and let $f \in C^*\text{Alg}(A, B/I)$. f is liftable iff there exists a $\star$ -homomorphism $\bar{f}$ such that the following diagram:

$$\begin{array}{ccc} & & B \\ & \nearrow \bar{f} & \downarrow \pi_I \\ A & \xrightarrow{f} & B/I \end{array}$$

commutes.

Here is our construction.

Construction 6.63. Let $A, B \in C^*\text{Alg}$. For each $a \in A$, we define a semi-metric:

$$p_a : \text{Ban}_{\mathbb{C}}(A, B) \times \text{Ban}_{\mathbb{C}}(A, B) \rightarrow [0, \infty), (f, g) \mapsto p_a(f, g) \stackrel{\text{def}}{=} \|f(a) - g(a)\|.$$

This gives us a family of seminorms on $\text{Ban}_{\mathbb{C}}(A, B)$ and this generates a locally convex topology on $\text{Ban}_{\mathbb{C}}(A, B)$; we will refer to this as the point-norm topology. Now, we take the subspace topology on $C^*\text{Alg}(A, B)$ and this set equipped with this topology will be denoted by $\text{Hom}(A, B)$. If A and B are unital, then we denote the (topological) subspace of unital $\star$ -homomorphisms by $\text{Hom}_1(A, B)$. Finally, if $I \subseteq B$ is a closed ideal, then we define:

$$\text{Hom}(A, B, I) \stackrel{\text{def}}{=} \pi_I \circ \text{Hom}(A, B),$$

as the topological subspace of liftable morphisms in $\text{Hom}(A, B/I)$.

Now, we can define l -openness and closedness.

Definition 6.64

Let $A \in C^*\text{Alg}$ and let (B, I) be a marked C^* -algebra.

1. A is (B, I) - l -open iff $\text{Hom}(A, B, I)$ is open in $\text{Hom}(A, B/I)$.
2. A is (B, I) - l -closed iff $\text{Hom}(A, B, I)$ is closed in $\text{Hom}(A, B/I)$.

Finally, A is l -open (closed) iff it is (B, I) - l -open (closed) for every marked C^* -algebra^a (B, I) .

^aIf we are working with the marked C^* -algebra $(\mathcal{B}(H), \mathcal{K}(H))$, then we will say that A is either H - l -open or H - l -closed.

Let us state our desired question more formally.

Question 6.65 (B. Blackadar, 2016, [14, Example 6.5, p.8-9]). Is $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ l -closed^a?

^a $\mathbb{D}$ is the compact unit disk in $\mathbb{C}$ centered at the origin.

We were not able to answer this question fully but we have managed to “explain” why one may expect it to be a difficult problem. Before we get to that, let us review what is known in the literature, starting with some terminology.

Terminology 6.66. Let (A, I) be a marked C^* -algebra and let $b \in A/I$ satisfy some property P which is satisfied by elements of C^* -algebras. b is P -liftable iff there exists an $a \in A$ such that $\pi(a) = b$ and a satisfies P .

We will also need some notation.

Notation 6.67. Let $A \in C^*\text{Alg}$. Then, the set of normal elements in A is denoted by N_A .

The following result characterizes when Question 6.65 has a positive answer.

Proposition 6.68. B. Blackadar (2016), H. Thiel (2020), [14, Example 6.5, p.8-9], [94]

TFAE:

1. $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is l -closed.
2. If (A, I) is a marked C^* -algebra and let $(x_n)_{n \in \mathbb{N}}$ be a sequence of normal elements in A/I which converges to some $x \in A/I$. If x_n is normal-liftable for each $n \in \mathbb{N}$, then x is also normal-liftable.
3. If (A, I) is a marked C^* -algebra, then $\pi(N_A)$ is closed in B/I , where $\pi : A \rightarrow A/I$ is the canonical projection.

Now, we state our result.

Theorem 6.69. A. Vats, 2026

Let (A, I) be a marked C^* -algebra. TFAE:

1. The sum $N_A + I$ is closed in A .
2. Any convergent sequence in $N_{A/I}$ whose terms are normal-liftable must have a normal-liftable limit.

The proof of this result requires the C^* -Bartle-Graves Theorem. The Banach-space version of this result is an old theorem proved in [69, p.364], though the version that we will provide now is a much newer result due to (Tatiana) Shulman which we hope to use later as a way of simplifying some proofs involving the Schochet property.

Theorem 6.70. The C^* -Bartle-Graves Theorem, T. Shulman, [91, Theorem 5, p.7-8]

Let $A, B \in C^*\text{Alg}$ and let $\pi \in C^*\text{Alg}(A, B)$ be surjective. Then, there exists a $s \in \mathcal{C}(B, A)$ so that:

1. s is contractive and;
2. s is unital (if A and B are unital, with π also being unital) and;
3. $\forall b \in B : s(b^*) = s(b)^*$ and;
4. $\forall b \in B^+ : s(b) \in A^+$.

Moreover, if $a_0 \in A$ and $b_0 \in B$ with $\pi(a_0) = b_0$, then s can be chosen so that:

$$s(b_0) = a_0,$$

and so that it still satisfies the above properties. We refer to s as the C^* -section of π .

This leads us to the proof of our result.

Proof of Theorem 6.69.

(1) $\Rightarrow$ (2): We will use the C^* -Bartle-Graves Theorem as stated in Theorem 6.70; since π is a surjective $\star$ -homomorphism, it has a C^* -section $s : A/I \rightarrow A$. Now, let $(x_n + I)_{n \in \mathbb{N}}$ be a convergent sequence in A/I with each x_n being normal in A and let $x + I$ be the limit. We define:

$$\begin{aligned} \forall n \in \mathbb{N} : y_n &\stackrel{\text{def}}{=} s(x_n + I), \\ y &\stackrel{\text{def}}{=} s(x + I). \end{aligned}$$

Since s is continuous, we see that $y_n \rightarrow y$. Now, observe that:

$$\begin{aligned}\forall n \in \mathbb{N} : y_n + I &= \pi(y_n) \\ &= \pi(s(x_n + I)) \\ &= x_n + I,\end{aligned}$$

which means that $y_n - x_n \in I$ for each $n \in \mathbb{N}$. In other words, each $y_n \in N_A + I$; we use the normality of each x_n over here. Since $(y_n)_{n \in \mathbb{N}}$ is convergent and $N_A + I$ is closed, we see that $y \in N_A + I$. So, we write:

$$y = a + k,$$

for some normal a and $k \in I$. But now, we see that:

$$\begin{aligned}x + I &= (\pi \circ s)(x + I) \\ &= \pi(s(x + I)) \\ &= \pi(y) \\ &= \pi(a + k) \\ &= \pi(a) + \pi(k) \\ &= \pi(a),\end{aligned}$$

which proves the result.

(2) $\Rightarrow$ (1): Let $(a_n + k_n)_{n \in \mathbb{N}}$ be a sequence in $N_A + I$ which converges to some $x \in A$, where each a_n is normal and each $k_n \in I$. Then, we have that $(\pi(a_n + k_n))_{n \in \mathbb{N}}$ is a sequence in A/I which converges to $\pi(x)$. Moreover, this sequence in A/I consists of terms which are normal-liftable, so its limit is normal-liftable. That is:

$$\exists a \in N_A : \pi(a) = \pi(x).$$

This implies that $x - a \in I$ so $x \in N_A + I$. ■

The following is an immediate corollary.

Corollary 6.71. A. Vats, 2026

Let (A, I) be a marked C^* -algebra. TFAE:

1. $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is (A, I) - l -closed.
2. $N_A + I$, as defined in [Theorem 6.69](#), is closed in A .

So, $\mathcal{C}(\mathbb{D} \setminus \{(0, 0)\})$ is l -closed iff for every marked C^* -algebra (A, I) , we have that $N_A + I$ is closed in A .

Proof. This follows by combining Proposition 6.68 and [Theorem 6.69](#). ■

This leads us to introduce the following piece of terminology.

Terminology 6.72. Let $A \in C^* \text{Alg}$. A is normally closed iff for every closed ideal $I \subseteq A$, we have that:

$$N_A + I,$$

is closed in A .

In fact, it turns out that there are many examples of normally closed C^* -algebras.

Example 6.73. Let $A \in \text{Ab}C^* \text{Alg}$. Then, A is clearly normally closed, since the normal elements of A are all elements of A . So, if $I \subseteq A$ is a closed ideal, then $N_A + I = A + I$ and this is closed.

Example 6.74. Let $A \in C^* \text{Alg}$ be simple. Then, A is clearly normally closed. Indeed, let $I \subseteq A$ be a closed ideal and note that $I = 0$ or $I = A$. If I is trivial, then we see that $N_A + I = N_A$ and this is closed. If $I = A$, then $N_A + A$ is also closed; the limit x of any convergent sequence in $N_A + A$ can be written just as $0 + x$.

Example 6.75. Let H be an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space. Then, $\mathcal{B}(H)$ is normally closed. Of course, the main point is to simply prove that $N_{\mathcal{B}(H)} + \mathcal{K}(H)$ is closed in $\mathcal{B}(H)$. Indeed, that is true but the classical proof uses Brown-Douglas-Fillmore theory; see [\[41, Corollary 11.4, p.119\]](#).

Example 6.76. Let H be an infinite-dimensional, nonseparable $\mathbb{C}$ -Hilbert space and let $I \subseteq \mathcal{B}(H)$ be a closed ideal which is not the compact operators in $\mathcal{B}(H)$. The results of [\[100\]](#) tell us that every normal element in $\mathcal{B}(H)/I$ is normal-liftable through the canonical projection. So, $N_{\mathcal{B}(H)} + I$ is closed in $\mathcal{B}(H)$. The problem for the compact operators is still open and the proof in the aforementioned paper does not seem to generalize to other C^* -algebras.

Normally closed C^* -algebras actually arise naturally in the context of lifting problems. In [\[94\]](#), Thiel introduces the notion of a universally liftable normal element in a C^* -algebra and presents some reasonable questions about this notion. Let us refine his definition just a bit and examine it more closely.

Definition 6.77

Let $\mathcal{C}$ be a full subcategory of $C^* \text{Alg}$, let $A \in C^* \text{Alg}$ and let $a \in A$ be normal.

1. a is universally liftable for $\mathcal{C}$ iff for every $B \in \mathcal{C}$ and every surjective $\star$ -homomorphism:

$$\pi : B \rightarrow A,$$

there exists a normal $b \in B$ such that $\pi(b) = a$.

2. a is universally liftable iff it is universally liftable for $C^* \text{Alg}$.

The set of normal elements in A which are universally liftable for $\mathcal{C}$ is denoted by^a $\text{UnivLift}_{\mathcal{C}}(A)$.

^aWe omit the $\mathcal{C}$ when $\mathcal{C}$ is just $C^*\text{Alg}$.

There are two main questions of interest regarding universally liftable elements.

| **Question 6.78** (H. Thiel, 2020, [94]). Let $A \in C^*\text{Alg}$. Is $\text{UnivLift}(A)$ closed?

| **Question 6.79** (H. Thiel, 2020, [94]). Let $A \in C^*\text{Alg}$. Is $\text{UnivLift}(A)$ open?

The next result indicates progress on Question 6.78.

Proposition 6.80. A. Vats, 2026

Let $A \in C^*\text{Alg}$. The set of universally liftable elements for normally closed C^* -algebras is closed.

Proof. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of universally liftable normal elements^a in A which converges to some normal $a \in A$. Let $B \in C^*\text{Alg}$ be normally closed and let $\pi : B \rightarrow A$ be a surjective $\star$ -homomorphism. For each $n \in \mathbb{N}$, there is a normal $b_n \in B$ so that $\pi(b_n) = a_n$. Now, we identify A with $B/\ker(\pi)$. Since B is normally closed, we can simply use Theorem 6.69 to conclude that a is normal-liftable so we're done. ■

^aFor normally closed C^* -algebras.

Due to the availability of stronger theorems and more palatable descriptions of certain C^* -algebras, it is actually possible for us to establish that the examples highlighted by Conway in [29] and [30] are actually Schochet C^* -algebras. Indeed, his methods were rather technical and we are in a position where we can give some simplified proofs. To illustrate that it can be completely non-trivial to verify the Schochet property, let us do it in the case of $\mathbb{C}$.

Example 6.81. We want to show that $\mathbb{C}$ is a Schochet C^* -algebra. We will rely on the following subclaim.

Subclaim. Let $A, B \in {}^1C^*\text{Alg}$, let $\pi \in C^*\text{Alg}(A, B)$ be surjective and let $(p_t)_{t \in [0,1]}$ be a path of projections in B such that p_0 lifts to a projection $P_0 \in A$ through π . The entire path lifts to a path $(P_t)_{t \in [0,1]}$ of projections in A .

The proof of this is somewhat technical, so we defer it to a reference [38, Exercise 8.1.28, p. 311], which contains essentially the whole proof. It is useful for us to simply sketch out the main points of the proof:

1. We will start by noticing that π is unital, since it is surjective so $\pi(1_A)$ is an identity for B .
2. Paths of projections in a unital C^* can be given by unitary conjugation. To be precise, there exists a path $(u_t)_{t \in [0,1]}$ of unitaries in B such that $p_t = u_t p_1 u_t^*$. This follows essentially from the fact that two projections which are sufficiently close must be unitarily equivalent.
3. So, the problem, now, reduces to lifting paths of unitaries. This can be done essentially by writing $u_t = \exp(ix_t)$ for some path $(x_t)_{t \in [0,1]}$ of self-adjoint elements (using the logarithm) and reducing the problem to lifting this path of self-adjoint elements to a path of self-adjoint elements in A , which is easy.

With this in mind, we start with the following lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ \mathbb{C} & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

Now, $(h_t(1))_{t \in [0,1]}$ is a path of projections in $\mathcal{Q}(H)$ so the subclaim tells us that there is a path $(S_t)_{t \in [0,1]}$ of projections in $\mathcal{B}(H)$ which is a lift for the given path in $\mathcal{Q}(H)$, with $S_0 = f(1)$. Now, we define the following map for each $t \in [0,1]$:

$$H_t : \mathbb{C} \rightarrow \mathcal{B}(H), \quad z \mapsto H_t(z) \stackrel{\text{def}}{=} zS_t.$$

We observe that this is a $\star$ -homomorphism^a. In fact, the family $(H_t)_{t \in [0,1]}$ forms a homotopy because $(S_t)_{t \in [0,1]}$ is a path and scalar multiplication is continuous. Now, let us check that this works:

$$\begin{aligned} \forall z \in \mathbb{C} : H_0(z) &= zS_0 \\ &= z f(1) \\ &= f(z), \end{aligned}$$

which means that $H_0 = f$. Moreover:

$$\begin{aligned} \forall t \in [0,1] : \forall z \in \mathbb{C} : (\pi \circ H_t)(z) &= \pi(H_t(z)) \\ &= \pi(zS_t) \\ &= z\pi(S_t) \\ &= zh_t(1) \\ &= h_t(z), \end{aligned}$$

which shows that $\pi \circ H_t = h_t$ so we are done.

So, even with a C^* -algebra as simple as $\mathbb{C}$, it does take some effort to prove the Schochet property directly.

^aLinearity is clear simply by the properties of scalar multiplication of operators. The fact that this preserves products follows from the fact that S_t is idempotent. Finally, the fact that this preserves the involution follows from the self-adjointness of S_t .

The next example generalizes the preceding one and it was one of those shown by Conway which we will not reprove here, as there is a simpler argument which already exists. So, we will point to that reference.

Example 6.82. Let A be an AF algebra.

| **Claim.** A is Schochet.

While this was proved by Conway in [30], the proof seems to not be so amenable to generalization. There is another proof in [73, Theorem 2.2.9] and it seems more reasonable to attempt to generalize this proof. The main obstruction to this is Lemma 2.2.7 in the same reference. We could not find the cited paper which proves that lemma, so we point to the proof of [76, Lemma 1.3, p.958-959] instead. Now, we also have the following claim.

| **Claim.** All totally disconnected compact Hausdorff spaces are Schochet spaces.

Indeed, continuous functions over such spaces are unital commutative AF algebras so we're done.

In fact, all commutative von Neumann algebras are Schochet C^* -algebras.

Example 6.83. Let $A \subseteq \mathcal{B}(H)$ be a commutative von Neumann algebra.

| **Claim.** A is a Schochet C^* -algebra.

Indeed, the Gelfand spectrum of any commutative von Neumann algebra is extremally disconnected and Hausdorff, so it is totally disconnected. Hence, it is a Schochet space and this implies the desired claim.

This argument works for any commutative C^* -algebra with extremally disconnected spectrum.

In the next example, we will show that $[0, 1]$ is a Schochet space.

Example 6.84. Let $A = \mathcal{C}([0, 1])$.

| **Claim.** $\mathcal{C}([0, 1])$ is a Schochet C^* -algebra. That is, $[0, 1]$ is a Schochet space.

Observe that we can write $\mathcal{C}([0, 1]) = C^*(u, 1 | 0 \leq u \leq 1)$. Next, we consider the following lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ \mathcal{C}([0, 1]) & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \end{array}$$

Observe that $(h_t(u))_{t \in [0, 1]}$ is a path of positive contractions in $\mathcal{Q}(H)$ with an initial lift given by $f(u)$ and so, the problem reduces to lifting paths of positive contractions. We will accomplish this by using a continuous section of π . By the C^* -Bartle-Graves Theorem, there is a unital C^* -section $s : \mathcal{Q}(H) \rightarrow \mathcal{B}(H)$ such that:

$$s(h_0(u)) = f(u).$$

So, now, we consider the path $(s(h_t(u)))_{t \in [0, 1]}$ and let us show that this is a path of positive contractions in $\mathcal{B}(H)$. Indeed, the positivity of s implies that this is a path of positive elements. So, it remains to show that:

$$\forall t \in [0, 1] : s(h_t(u)) \leq 1.$$

But this is equivalent to the claim that^a:

$$\forall t \in [0, 1] : \|s(h_t(u))\| \leq 1.$$

This holds because s is a contraction and each h_t is a contraction. So, we define:

$$H(u) : [0, 1] \rightarrow \mathcal{B}(H), \quad t \mapsto H(u)(t) \stackrel{\text{def}}{=} s(h_t(u)).$$

This is a continuous map because it is a composition of continuous maps. Moreover, our work above shows that it is a positive contraction. On the other hand, we have a path $(h_t(1))_{t \in [0, 1]}$ of projections with initial lift $f(1)$ & using the subclaim of Example 6.81, we can lift this to a path $(S_t)_{t \in [0, 1]}$ of projections in $\mathcal{B}(H)$ with:

$$S_0 = f(1),$$

$$\forall t \in [0, 1] : \pi(S_t) = h_t(1).$$

This allows us to define a $\star$ -homomorphism:

$$H : \mathcal{C}([0, 1]) \rightarrow \text{Cyl}(\mathcal{B}(H)),$$

by the universal property of $\mathcal{C}([0, 1])$. This sends u to $H(u)$ as defined above and sends 1 to the map $t \mapsto S_t$. Let us check that this is the desired homotopy lift. We will do this in the following claim.

| **Claim 6.23.** H is the desired homotopy lift. This is proved on Page 272.

This completes the proof.

Again, the proof here is only simplified by the fact that we are using a much stronger form of the Bartle-Graves Theorem which tells us so much more about the continuous section of our surjective $\star$ -homomorphism.

^aThis is a general fact. Let $A \in {}^1C^*\text{Alg}$ and let $x \in A^+$. Let's prove that $x \leq 1_A$ iff $\|x\| \leq 1$. Indeed, let B be the unital commutative C^* -algebra generated by x and 1_A . By the Gelfand-Naimark Theorem, this is $\star$ -isomorphic to $\mathcal{C}(X)$ for some compact Hausdorff space X . Now, observe that x corresponds to some map f in $\mathcal{C}(X)$ and we note that this result is clearly true for f . So, it is also true for x .

Since $[0, 1]$ is an example of a Schochet space, we would like to conclude that $(0, 1]$ is also Schochet. Since $[0, 1]$ is just the one-point compactification of $(0, 1]$, we will need to demonstrate some compatibility between the Schochet property and one-point compactifications. At the level of C^* -algebras, this amounts to showing some compatibility between the Schochet property and unitizations. To start discussing this compatibility, we will need to introduce unital homotopies into the picture. This also leads us to introducing a unital version of the Schochet property.

Terminology 6.85. Let $A, B \in {}^1C^*\text{Alg}$.

1. If $(\phi_t : A \rightarrow B)_{t \in [0, 1]}$ is a homotopy, we say that it is unital iff ϕ_t is unital for each $t \in [0, 1]$.
2. A is 1-Schochet iff for the following lifting problem^a:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \end{array}$$

(H_t)_{t ∈ [0, 1]}

where $(h_t)_{t \in [0, 1]}$ is unital and f is unital, we always have a homotopy lift satisfying the usual conditions. Note that the notion of being 1-Schochet only applies to unital C^* -algebras.

^aFor convenience, we will refer to this as a unital lifting problem.

We have a remark about this terminology.

Remark 6.86. The lift above does not have to be unital but it's not far away from it. Indeed:

$$\forall t \in [0, 1] : H_t(1_A) + \mathcal{K}(H) = \pi(H_t(1_A)) = h_t(1_A) = \text{Id}_H + \mathcal{K}(H),$$

which implies that $H_t(1_A) - \text{Id}_H$ is compact for each $t \in [0, 1]$.

Even though this is very easy to check, we still need to show that the unitization of a homotopy is a homotopy.

Claim 6.24. Let $A, B \in C^*\text{Alg}$ and let $(\phi_t : A \rightarrow B)_{t \in [0, 1]}$ be a homotopy. Then, the unitization:

$$(\phi_t^\sim : A^\sim \rightarrow B^\sim)_{t \in [0, 1]},$$

is a homotopy.

Now, we should also discuss some permanence properties since this will give us access to some examples.

Proposition 6.87. A. Vats, 2026

TFSH:

1. If $A, B \in C^*\text{Alg}$ are Schochet, then $A \star B$ is Schochet.
2. If $A, B \in C^*\text{Alg}$ with A being Schochet and a split epi $\phi \in C^*\text{Alg}(A, B)$, then B is Schochet.
3. If $A, B \in C^*\text{Alg}$ with A and B being $\star$ -isomorphic and A being Schochet, then B is Schochet.
4. If $A, B \in C^*\text{Alg}$ with A being Schochet and a Schochet fibration $f \in C^*\text{Alg}(A, B)$ so that:
$$\exists g \in C^*\text{Alg}(B, A) : f \circ g \simeq_h \text{Id}_B,$$
then B is Schochet.
5. If $A, B \in C^*\text{Alg}$ with $A \oplus B$ being Schochet, then A and B are Schochet.
6. If $A \in C^*\text{Alg}$ and $M_{n \times n}(A)$ is Schochet for some $n \in \mathbb{N}$, then A is Schochet.
7. If $A \in C^*\text{Alg}$ and $\text{Cyl}(A)$ is Schochet, then A is Schochet.
8. If $A, B \in {}^1C^*\text{Alg}$ with A being 1-Schochet and a split epi $\phi \in {}^1C^*\text{Alg}(A, B)$, then B is 1-Schochet.
9. If $A, B \in {}^1C^*\text{Alg}$ with A and B being $\star$ -isomorphic and A being 1-Schochet, then B is 1-Schochet.

Proof. We'll prove each statement one-by-one. In each case, we will start with the following diagram:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ D & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \end{array} \tag{1}$$

where $(h_t)_{t \in [0, 1]}$ is a homotopy such that the diagram above commutes when $t = 0$.

1. Let $D = A \star B$ and let ι_A, ι_B be the respective inclusion maps. In the case of A , we have a diagram:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f \circ \iota_A & \downarrow \pi \\ A & \xrightarrow{(h_t \circ \iota_A)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

which commutes at $t = 0$ because the one in (1) commutes when $t = 0$; a similar diagram exists in the case of B . Since A is Schochet, there is a homotopy lift $(H_t^A)_{t \in [0,1]}$ such that $H_0^A = f \circ \iota_A$ and we have that:

$$\forall t \in [0, 1] : \pi \circ H_t^A = h_t \circ \iota_A.$$

Similarly, we have a homotopy lift $(H_t^B)_{t \in [0,1]}$ for the corresponding diagram for B . By the universal property of the free product, we obtain a unique homotopy $(H_t : A \star B \rightarrow \mathcal{B}(H))_{t \in [0,1]}$ such that each H_t restricts to H_t^A and H_t^B . Now, we need to check that this works. Indeed, we need to check that $H_0 = f$. For this, we note that:

$$H_0 \circ \iota_A = H_0^A = f \circ \iota_A,$$

$$H_0 \circ \iota_B = H_0^B = f \circ \iota_B.$$

By the uniqueness part of the universal property for the free product, we see that $H_0 = f$. Let's check that:

$$\forall t \in [0, 1] : \pi \circ H_t = h_t.$$

So, let $t \in [0, 1]$. Then:

$$(\pi \circ H_t) \circ \iota_A = \pi \circ (H_t \circ \iota_A) = \pi \circ H_t^A = h_t \circ \iota_A,$$

$$(\pi \circ H_t) \circ \iota_B = \pi \circ (H_t \circ \iota_B) = \pi \circ H_t^B = h_t \circ \iota_B,$$

which implies, by uniqueness again, that our desired equality holds.

2. Let $D = B$ and consider the following diagram:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f \circ \phi & \downarrow \pi \\ A & \xrightarrow{(h_t \circ \phi)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

Since A is Schochet, we obtain a homotopy lift $(F_t : A \rightarrow \mathcal{B}(H))_{t \in [0,1]}$ such that:

$$F_0 = f \circ \phi,$$

$$\forall t \in [0, 1] : \pi \circ F_t = h_t \circ \phi.$$

Now, let ψ be a split for ϕ . Then, we define:

$$\forall t \in [0, 1] : H_t \stackrel{\text{def}}{=} F_t \circ \psi.$$

In that case, we see that:

$$H_0 = F_0 \circ \psi = (f \circ \phi) \circ \psi = f,$$

and we also have that:

$$\forall t \in [0, 1] : \pi \circ H_t = \pi \circ F_t \circ \psi = h_t \circ \phi \circ \psi = h_t,$$

which proves the result.

3. This follows from the previous statement, since $\star$ -isomorphisms are split epis.
4. We see that f is split epi by Proposition 6.31 so we're done.
5. Let us consider the projection $\pi_A : A \oplus B \rightarrow A$ and define:

$$\iota_A : A \rightarrow A \oplus B, \quad a \mapsto \iota_A(a) \stackrel{\text{def}}{=} (a, 0).$$

Then, we see that:

$$\begin{aligned} \forall a \in A : (\pi_A \circ \iota_A)(a) &= \pi_A(a, 0) \\ &= a, \end{aligned}$$

which means that π_A is a split-epi since ι_A is a $\star$ -homomorphism. So, A is Schochet and similarly, so is B .

6. We consider the following homotopy lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

where $h_0 = \pi \circ f$. Now, we apply the functor $M_{n \times n}(-)$ to this diagram so we obtain:

$$\begin{array}{ccc} & & M_{n \times n}(\mathcal{B}(H)) \\ & \nearrow^{M_{n \times n}(f)} & \downarrow^{M_{n \times n}(\pi)} \\ M_{n \times n}(A) & \xrightarrow{(M_{n \times n}(h_t))_{t \in [0,1]}} & M_{n \times n}(\mathcal{Q}(H)) \end{array}$$

Next, we will make some identifications as follows:

- (a) We will identify $M_{n \times n}(\mathcal{B}(H))$ with $\mathcal{B}(H^{\oplus n})$, where we note that $H^{\oplus n}$ is still an infinite-dimensional, separable $\mathbb{C}$ -Hilbert space so we are still in the same set-up.
- (b) We will identify $M_{n \times n}(\mathcal{Q}(H))$ with $\mathcal{Q}(H^{\oplus n})$.
- (c) We will identify $M_{n \times n}(\pi)$ with the Calkin projection on $\mathcal{B}(H^{\oplus n})$.
- (d) If we view the homotopy $(h_t)_{t \in [0,1]}$ as a map from A into $\text{Cyl}(\mathcal{Q}(H))$, then we can view $(M_{n \times n}(h_t))_{t \in [0,1]}$ as a homotopy from $M_{n \times n}(A)$ into $\text{Cyl}(M_{n \times n}(\mathcal{Q}(H)))$ and due to our identification above, this is just a homotopy from $M_{n \times n}(A)$ into $\text{Cyl}(\mathcal{Q}(H^{\oplus n}))$.

With these identifications, we see that we just have a lifting problem for $M_{n \times n}(A)$ and since this is Schochet, we find a homotopy lift $(F_t : M_{n \times n}(A) \rightarrow \mathcal{B}(H^{\oplus n}))_{t \in [0,1]}$. Now, we precompose each F_t with the left-upper corner inclusion of A into $M_{n \times n}(A)$ and postcompose it with the canonical isomorphism between $M_{n \times n}(\mathcal{B}(H))$ and $\mathcal{B}(H^{\oplus n})$. This is the homotopy lift for the lifting problem associated to A .

7. Let's consider the $\star$ -homomorphism $p_0^A : \text{Cyl}(A) \rightarrow A$ and define:

$$s : A \rightarrow \text{Cyl}(A), \quad a \mapsto s(a),$$

where $s(a)$ is the constant function on $[0,1]$ with value a . In [Example 6.19](#), we actually showed that s is a section for p_0^A . But this implies that A is Schochet.

Finally, statements 8 and 9 follow in the same way as statements (2) and (3). ■

The next corollary is due to Conway but we prove it using the proposition.

Corollary 6.88

Let $X, Y \in \text{LCH}$ be homeomorphic, with X being Schochet. Then, Y is Schochet.

Proof. Since X and Y are homeomorphic, we see that $\mathcal{C}_0(X)$ and $\mathcal{C}_0(Y)$ are $\star$ -isomorphic and $\mathcal{C}_0(X)$ is Schochet. Hence, it follows that $\mathcal{C}_0(Y)$ is Schochet and this means that Y is Schochet. ■

The next one gives us more examples of Schochet spaces.

Corollary 6.89. A. Vats, 2026

Let $X \in \text{CHaus}$. TFAE:

1. X is Schochet.
2. Every clopen subset Y of X is Schochet.

Proof. (2) $\Rightarrow$ (1) is clear, since X is clopen in itself. So, it remains to prove (1) $\Rightarrow$ (2). We define:

$$\phi : \mathcal{C}(X) \rightarrow \mathcal{C}(Y), \quad f \mapsto \phi(f) \stackrel{\text{def}}{=} f|_Y.$$

Note that this is a $\star$ -homomorphism. Now, we define:

$$\psi : \mathcal{C}(Y) \rightarrow \mathcal{C}(X), \quad f \mapsto \psi(f),$$

where we simply take a function and extend it to X through continuation by 0; this extension will be continuous^a and the map above is a $\star$ -homomorphism. Now, we observe that ψ is a split for ϕ and since X is Schochet, it follows from [Proposition 6.87](#) that Y is Schochet. ■

^aWe illustrate the argument for points in $X \setminus Y$ and note that the argument for continuity at points in Y will rely on it being an open set. Let $p \in X \setminus Y$ and let $(p_\delta)_{\delta \in D}$ be a net that converges to p . Since Y is closed, we see that $X \setminus Y$ is open so the net eventually lies in $X \setminus Y$. But this means that $f(p_\delta) = 0$ for each $\delta \in D$ and this implies that $f(p) = 0$.

In order to relate the Schochet property to the 1-Schochet property, we will need the following theorem.

Theorem 6.90. A. Vats, 2026

Let $A, B \in {}^1\text{C}\star\text{Alg}$ be 1-Schochet. Then, $A \oplus B$ is 1-Schochet.

To prove this, we will need two technical lemmas about paths of projections.

Lemma 6.91. J. Conway, [29, Lemma 5, p.138-139] Let $(Q_t)_{t \in [0,1]}$ be a path of projections in $\mathcal{B}(H)$ and let $(p_t)_{t \in [0,1]}$ be a path of projections in $\mathcal{Q}(H)$ satisfying the following conditions:

- $\forall t \in [0,1] : \pi(Q_t) \geq p_t$ and;

- there is a projection $P_0 \in \mathcal{B}(H)$ such that $P_0 \leq Q_0$ and $\pi(P_0) = p_0$.

Then, there is a path $(P_t)_{t \in [0,1]}$ of projections in $\mathcal{B}(H)$ so that:

1. $\forall t \in [0, 1] : \pi(P_t) = p_t$,
2. $\forall t \in [0, 1] : P_t \leq Q_t$.

In other words, paths of projections which are order-bounded by a liftable path of projections are liftable.

Actually, it turns out that [Lemma 6.91](#) is an essential step in Conway's proof that totally disconnected compact Hausdorff spaces are C^* -CHP. So, we could replace the use of the lemma by that result instead but we will just use the lemma as we have stated it. The next result lets us construct paths of partial isometries from a given path of projections.

Lemma 6.92. J. Conway, [29, Lemma 7, 140–141] Let $(p_t)_{t \in [0,1]}$ be a path of projections in $\mathcal{B}(H)$, then there exist paths of partial isometries $(u_t)_{t \in [0,1]}, (v_t)_{t \in [0,1]}$ such that the following relations are satisfied:

$$\begin{aligned} u_0 &= p_0 \\ v_0 &= 1 - p_0 \\ \forall t \in [0, 1] : u_t^* u_t &= p_0 \\ \forall t \in [0, 1] : v_t^* v_t &= 1 - p_0 \\ \forall t \in [0, 1] : u_t u_t^* &= p_t \\ \forall t \in [0, 1] : v_t v_t^* &= 1 - p_t, \end{aligned}$$

where 1 is used in place of Id_H above.

Our proof of [Theorem 6.90](#) is modelled after the proof of the analogous fact in [29, Theorem 8, p.141]. There is another proof of a similar result due to (Dolapo) Oyetunbi in [73, Theorem 2.2.10, p.40] but we will not proceed by that approach.

Proof of Theorem 6.90. Let us consider the following unital lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ A \oplus B & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

Let $i_A : A \rightarrow A \oplus B$ and $i_B : B \rightarrow A \oplus B$ the canonical inclusions; these are not unital $\star$ -homomorphisms. We will view the homotopy as a morphism into $\text{Cyl}(\mathcal{Q}(H))$. Since $(1_A, 0)$ and $(0, 1_B)$ are orthogonal projections in $A \oplus B$, we can define the following orthogonal projections in $\text{Cyl}(\mathcal{Q}(H))$:

$$\begin{aligned} Q_1 &\stackrel{\text{def}}{=} h(1_A, 0), \\ Q_2 &\stackrel{\text{def}}{=} h(0, 1_B), \end{aligned}$$

such that $Q_1 + Q_2 = 1$, since $(1_A, 0) + (0, 1_B) = (1_A, 1_B)$. These are paths of projections which are order-bounded by a liftable (constant) path. Moreover, both of these paths lift at $t = 0$ in a way which satisfies the conditions of [Lemma 6.91](#). So, we can find a path $(P_t^1)_{t \in [0,1]}$ of projections in $\mathcal{B}(H)$ such that:

1. $\forall t \in [0, 1] : \pi(P_t^1) = Q_1(t)$,
2. $\forall t \in [0, 1] : P_t^1 \leq 1$,
3. $P_0^1 = f(1_A, 0)$.

If we define:

$$\forall t \in [0, 1] : P_t^2 \stackrel{\text{def}}{=} 1 - P_t^1,$$

then we see that $(P_t^2)_{t \in [0,1]}$ is also a path of projections such that the following facts hold:

1. $\forall t \in [0, 1] : \pi(P_t^2) = \pi(1) - \pi(P_t^1) = 1 - Q_1(t) = Q_2(t)$.
2. $\forall t \in [0, 1] : P_t^2 \leq 1$,
3. $P_0^2 = 1 - f(1_A, 0) = f(1_A, 1_B) - f(1_A, 0) = f(0, 1_B)$.

For $i \in \{1, 2\}$, we may now define the following objects:

$$\begin{aligned} P^i &\stackrel{\text{def}}{=} P_0^i \\ H_i &\stackrel{\text{def}}{=} P^i(H). \end{aligned}$$

In particular, we can identify $\mathcal{B}(H_i)$ with $P^i \mathcal{B}(H) P^i$ and we can identify $\mathcal{B}(H_i)/\mathcal{K}(H_i)$ with $\pi(P^i) \mathcal{Q}(H) \pi(P^i)$; we will denote the latter by $\mathcal{Q}(H_i)$. Now, we have two paths of projections in $\mathcal{B}(H)$ so we can use [Lemma 6.92](#) to find two paths of partial isometries $(U_i(t))_{t \in [0,1]}$ for $i \in \{1, 2\}$ such that:

1. $U_i(0) = P^i$,
2. $\forall t \in [0, 1] : U_i(t)^* U_i(t) = P^i$,
3. $\forall t \in [0, 1] : U_i(t) U_i(t)^* = P_t^i$.

Next, let $t \in [0, 1]$ be fixed. We will define four maps:

$$\Phi_t^A : A \rightarrow \mathcal{Q}(H_1), \quad a \mapsto \Phi_t^A(a) \stackrel{\text{def}}{=} \pi(U_1(t))^* h_t(a, 0) \pi(U_1(t)),$$

$$\Phi_t^B : A \rightarrow \mathcal{Q}(H_2), \quad b \mapsto \Phi_t^B(b) \stackrel{\text{def}}{=} \pi(U_2(t))^* h_t(0, b) \pi(U_2(t)),$$

$$\psi^A : A \rightarrow \mathcal{B}(H_1), \quad a \mapsto \psi^A(a) \stackrel{\text{def}}{=} P^1 f(a, 0) P^1,$$

$$\psi^B : B \rightarrow \mathcal{B}(H_2), \quad b \mapsto \psi^B(b) \stackrel{\text{def}}{=} P^2 f(0, b) P^2.$$

Next, we have the following claim.

Claim 6.25. The maps above are $\star$ -homomorphisms and the families:

$$(\Phi_t^A)_{t \in [0, 1]}, \quad (\Phi_t^B)_{t \in [0, 1]},$$

are homotopies. This is proved on Page 272.

Now, we want to fit these into lifting problems whose solutions can be glued together to form a solution for the original lifting problem. In particular, we have the following candidate diagrams under consideration:

$$\begin{array}{ccc} & \mathcal{B}(H_1) & \\ \psi^A \nearrow & \downarrow \pi & \\ A & \xrightarrow{(\Phi_t^A)_{t \in [0, 1]}} & \mathcal{Q}(H_1) \end{array} \quad \begin{array}{ccc} & \mathcal{B}(H_2) & \\ \psi^B \nearrow & \downarrow \pi & \\ B & \xrightarrow{(\Phi_t^B)_{t \in [0, 1]}} & \mathcal{Q}(H_2) \end{array}$$

Next, we will verify the following claim.

Claim 6.26. The two diagrams given commute when $t = 0$. This is proved on Page 272.

So, this shows that we have two lifting problems for A and B respectively. Since A and B are Schochet, these two lifting problems have solutions $(\Psi_t^A)_{t \in [0, 1]}$ and $(\Psi_t^B)_{t \in [0, 1]}$. For each $t \in [0, 1]$, we now define:

$$\Psi_t : A \oplus B \rightarrow \mathcal{B}(H), \quad (a, b) \mapsto \Psi_t(a, b) \stackrel{\text{def}}{=} U_1(t) \Psi_t^A(a) U_1(t)^* + U_2(t) \Psi_t^B(b) U_2(t)^*.$$

A similar computation as in **Claim 6.25** shows that Ψ_t is a $\star$ -homomorphism. In fact, it is a homotopy, as seen by a similar argument in the aforementioned claim. Finally, we have our main claim.

Claim 6.27. $(\Psi_t)_{t \in [0, 1]}$ is a solution to the original lifting problem. This is proved on Page 272.

This claim finishes the proof of this implication. ■

We can use this theorem to deduce the relationship between the 1-Schochet property & Schochet property.

Corollary 6.93. A. Vats, 2026

Let $A \in {}^1\mathbf{C}^\star\mathbf{Alg}$. TFAE:

1. A is Schochet.
2. A is 1-Schochet.

Proof.

(1) $\Rightarrow$ (2): This follows clearly from the definition of being Schochet.

(2) $\Rightarrow$ (1): We consider the following diagram:

$$\begin{array}{ccc} & \mathcal{B}(H) & \\ f \sim \nearrow & \downarrow \pi & \\ A & \xrightarrow{(h_t)_{t \in [0, 1]}} & \mathcal{Q}(H) \\ i \nwarrow & \uparrow (h_t^\sim)_{t \in [0, 1]} & \\ A^\sim & & \end{array}$$

where $i : A \rightarrow A^\sim$ is the canonical inclusion, the triangle is a lifting problem for A and the outer arrows are the universal extensions of f and each h_t to the unitization through i . Since A is unital, we see that:

$$A^\sim \cong {}^1\mathbf{C}^\star\mathbf{Alg} \ A \oplus \mathbb{C},$$

with the right-hand side being 1-Schochet so $A^\sim$ is 1-Schochet^a. In that case, we can find a homotopy lift $(F_t : A^\sim \rightarrow \mathcal{B}(H))_{t \in [0, 1]}$ which solves the unital lifting problem. Now, we define:

$$\forall t \in [0, 1] : H_t \stackrel{\text{def}}{=} F_t \circ i.$$

Let us check that this works:

$$\begin{aligned} H_0 &= F_0 \circ i = f^\sim \circ i = f, \\ \forall t \in [0, 1] : \pi \circ H_t &= \pi \circ (F_t \circ i) = h_t^\sim \circ i = h_t, \end{aligned}$$

as was desired. ■

^aThis is where we use [Theorem 6.90](#).

As an immediate corollary, we obtain an even stronger result.

Corollary 6.94

Let $A, B \in {}^1\mathbf{C}^*\mathbf{Alg}$ be Schochet. Then, $A \oplus B$ is Schochet.

Proof. Since A, B are unital and Schochet, it is the case that they are 1-Schochet. Hence, $A \oplus B$ is 1-Schochet and since it is unital, it follows that it is Schochet. Over here, we simply apply [Corollary 6.93](#) multiple times. ■

The corollary above is rather pivotal in proving the next result.

Corollary 6.95. A. Vats, 2026

Let $A \in \mathbf{C}^*\mathbf{Alg}$. TFAE:

1. A is Schochet.
2. $A^\sim$ is Schochet.

Proof.

(1) $\Rightarrow$ (2): Since $A^\sim$ is unital, we can just show that it is 1-Schochet. Consider the following diagrams:

$$\begin{array}{ccc} & \mathcal{B}(H) & \\ & \downarrow \pi & \\ A^\sim & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array} \quad \begin{array}{ccc} & \mathcal{B}(H) & \\ & \downarrow \pi & \\ A & \xrightarrow{(h_t \circ i)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

(Left diagram: $A^\sim \xrightarrow{f} \mathcal{B}(H)$ and $A^\sim \xrightarrow{(h_t)_{t \in [0,1]}} \mathcal{Q}(H)$ with a vertical arrow $\pi: \mathcal{B}(H) \rightarrow \mathcal{Q}(H)$. Right diagram: $A \xrightarrow{f \circ i} \mathcal{B}(H)$ and $A \xrightarrow{(h_t \circ i)_{t \in [0,1]}} \mathcal{Q}(H)$ with a vertical arrow $\pi: \mathcal{B}(H) \rightarrow \mathcal{Q}(H)$. A dashed curved arrow labeled $(F_t)_{t \in [0,1]}$ points from A to $A^\sim$.)

where the diagram on the left is a unital lifting problem and the one on the right is obtained by composing the relevant morphisms with the canonical inclusion $i: A \rightarrow A^\sim$. In particular, the one on the right is a lifting problem for A . Since A is Schochet, we have a homotopy lift $(F_t)_{t \in [0,1]}$. Now, we define:

$$\forall t \in [0,1] : H_t \stackrel{\text{def}}{=} F_t^\sim.$$

By [Claim 6.24](#), this is a homotopy so let's check that it works. If $t = 0$, then:

$$H_0 = F_0^\sim = (f \circ i)^\sim = f,$$

because f is already unital and defined on $A^\sim$. Next, let $t \in [0,1]$ and consider the following computation:

$$\begin{aligned} \forall a + \lambda 1_{A^\sim} \in A^\sim : (\pi \circ H_t)(a + \lambda 1_{A^\sim}) &= \pi(H_t(a)) + \lambda \pi(H_t(1_{A^\sim})) \\ &= h_t(a) + \lambda \pi(\text{Id}_H) \\ &= h_t(a) + \lambda (\text{Id}_H + \mathcal{K}(H)) \\ &= h_t(a) + \lambda h_t(1_{A^\sim}) \\ &= h_t(a + \lambda 1_{A^\sim}), \end{aligned}$$

which proves that we have the required lift for the original problem.

(2) $\Rightarrow$ (1): Consider the canonical projection:

$$p: A^\sim \rightarrow A, (a, \lambda) \mapsto p(a, \lambda) \stackrel{\text{def}}{=} a.$$

One observes that:

$$\forall a \in A : (p \circ i)(a) = p(a, 0) = a,$$

which proves that p is a split epi. Since $A^\sim$ is Schochet, we see that A is Schochet. ■

Here is a corollary of the result above.

Corollary 6.96. A. Vats, 2026

Let $X \in \mathbf{LCH}$. TFAE:

1. X is Schochet.
2. X^+ , its one-point compactification, is Schochet.

Proof. We know that:

$$\mathcal{C}_0(X)^\sim \simeq \mathcal{C}(X^+).$$

Now, X is Schochet iff $\mathcal{C}_0(X)$ is Schochet iff $\mathcal{C}_0(X)^\sim$ is Schochet iff $\mathcal{C}(X^+)$ is Schochet iff X^+ is Schochet. ■

With this corollary, we can show that the half-open unit interval is a Schochet space.

Example 6.97. We have the following main claim.

Claim. $(0, 1]$ is a Schochet space.

We note that $(0, 1]$ is Schochet iff $(0, 1]^+ \simeq [0, 1]$ is Schochet. But the latter statement is true.

At this stage, we have developed most of the tools needed to show that S^1 is a Schochet space; this is one of the major examples of Conway and we give a slightly simplified proof. The point is that $\mathcal{C}(S^1)$ is a unital C^* -algebra so it is sufficient to show that it is 1-Schochet. This simplification will allow us to use lifting results for paths of unitaries.

Example 6.98. We have the following claim.

Claim. S^1 is a Schochet space.

Observe that we can write:

$$\mathcal{C}(S^1) = C^*(u, 1 | uu^* = u^*u = 1).$$

It is sufficient for us to show that $\mathcal{C}(S^1)$ is 1-Schochet. With the universal description of $\mathcal{C}(S^1)$, we see that the proof reduces to lifting paths of unitaries. In fact, this already been “used” in [Example 6.81](#).

Subclaim. Let $A, B \in {}^1C^*\text{Alg}$ and let $\pi \in C^*\text{Alg}(A, B)$ be surjective, with $(u_t)_{t \in [0,1]}$ being a path of unitaries in B so that there exists a unitary $U_0 \in A$ such that $\pi(U_0) = u_0$. Then, there exists a path $(U_t)_{t \in [0,1]}$ of unitaries in A such that $\pi(U_t) = u_t$ for each^a $t \in [0, 1]$. This is proved in [[52](#), Proposition 4.3.14a, p.95-96].

^aThe statement can be phrased a bit more succinctly by saying that $\pi|_{\text{Unitary}(A)}$ has the path-lifting property.

We consider the following unital lifting problem:

$$\begin{array}{ccc} & & \mathcal{B}(H) \\ & \nearrow f & \downarrow \pi \\ \mathcal{C}(S^1) & \xrightarrow{(h_t)_{t \in [0,1]}} & \mathcal{Q}(H) \end{array}$$

Observe that $\mathcal{C}(S^1)$ is the universal C^* -algebra generated by a unitary; let u denote a generator. Since each h_t is unital, we see that $(h_t(u))_{t \in [0,1]}$ is a path of Calkin unitaries. Using the lemma, we obtain a unitary lift $(v_t)_{t \in [0,1]}$ through the Calkin projection, with $v_0 = f(u)$. Now, we just define for each $t \in [0, 1]$:

$$H_t(u) \stackrel{\text{def}}{=} v_t,$$

and this defines, by universal property, a $\star$ -homomorphism $H_t : \mathcal{C}(S^1) \rightarrow \text{Cyl}(\mathcal{B}(H))$ with the desired property.

Ideally, we want to extend [Example 6.97](#) to $(0, 1)$ as well and we can do it.

Example 6.99. $(0, 1)$ is a Schochet space because $(0, 1)^+ \simeq S^1$ is a Schochet space.

We have discussed the Schochet property relative to spaces. One can ask what the basic situation looks like for groups and we have some results in that direction. In the next example, we see that all locally finite groups are Schochet.

Example 6.100. Let $G \in \text{Grp}$ be locally finite (that is, every finitely-generated subgroup of G is finite). Then, we note that $C_{\text{red}}^*(G)$ is an AF algebra and this is a Schochet C^* -algebra. So, we see that G is a Schochet group.

Which can also characterize Schochet groups among the LCH abelian groups.

Proposition 6.101. A. Vats, 2026

Let $G \in \text{Ab}(\text{LCH})$. TFSH:

1. TFAE:
 - (a) G is a Schochet group.
 - (b) $\widehat{G}$, the Pontryagin dual of G , is a Schochet space.
 In particular, $\widehat{G}$ is a Schochet group iff G is a Schochet space.
2. If G is compact, then G is Schochet.

Proof. We will prove each statement one-by-one.

1. Since G is abelian, we have the following chain of isomorphisms in $C^*\text{Alg}$:

$$C_{\text{red}}^*(G) \simeq C_{\text{max}}^*(G) \simeq \mathcal{C}_0(\widehat{G}),$$

where the first one holds because G is amenable and the second one holds as a general result of harmonic analysis. Now, G is Schochet iff $C_{\text{red}}^*(G)$ is Schochet iff $\mathcal{C}_0(\widehat{G})$ is Schochet iff $\widehat{G}$ is Schochet. The last statement follows via Pontryagin duality, as well as the fact that the Schochet property for groups is preserved by isomorphisms.

2. This follows from statement (1), since $\widehat{G}$ is discrete^a so it is Schochet. ■

^aThey are, in particular, locally compact Hausdorff totally disconnected spaces.

With this corollary, we can provide more interesting examples.

Example 6.102. We have that $\mathbb{T}$ is a Schochet group since it is a compact Hausdorff abelian group.

Example 6.103. Let $G = \mathbb{Z}$, the additive group of the integers. Then, we know that:

$$\widehat{G} \simeq S^1,$$

and the latter is a Schochet space. So, we see that G is a Schochet group.

Example 6.104. Let $G = \mathbb{R}$, the additive group of $\mathbb{R}$. Then, we know that:

$$\widehat{G} \simeq \mathbb{R},$$

and the latter is a Schochet space because $(0, 1)$ is a Schochet space. So, G is a Schochet group.

It would, of course, be very desirable for us to have more examples (or non-examples) of Schochet groups. We will discuss a possible way forward in ???. For now, we want to turn towards partially resolving Question 6.51. Actually, it turns out that the theory we have developed so far is not sufficient in tackling this problem. For that purpose, we will need to introduce the notion of *semiprojectivity*. This notion has its origins in the shape theory of C^* -algebras initiated by (Bruce) Blackadar in [13] (see Remark 6.109, however). We have a more practical reason to be interested in this notion. To explain this, we need some terminology which generalizes what we have done with the Schochet property.

Terminology 6.105. A pair (A, π) consisting of:

1. an $A \in C^*\text{Alg}$ and;
2. a surjective $\pi \in C^*\text{Alg}(E, B)$;

satisfies the homotopy lifting property iff the following lifting problem^a:

$$\begin{array}{ccc} & & E \\ & \nearrow f & \downarrow \pi \\ A & \xrightarrow{(h_t)_{t \in [0,1]}} & B \end{array}$$

(H_t)_{t ∈ [0,1]}

always admits a solution in $C^*\text{Alg}$. We also say that (A, π) is HLP in this case.

^aThis is defined in exactly the same sense in which lifting problems are defined for the Schochet property.

Our main reason for being interested in semiprojectivity is the following theorem.

Theorem 6.106. Blackadar's Homotopy Lifting Theorem, [14, Theorem 5.1, p.297-298]

Let $A \in C^*\text{Alg}$ be semiprojective. Then, every pair:

$$(A, \pi : E \rightarrow B),$$

where π is a surjective $\star$ -homomorphism, is HLP.

Before we explain (approximate) semiprojectivity, we have a remark regarding our solution to Conjecture 6.39.

Remark 6.107. So, in principle, Theorem 6.106 provides us a way to show that certain C^* -algebras are not semiprojective; we just have to show that they fail the HLP. For instance, this provides a proof that:

$$A = C^*(x | xx^* = x^*x \wedge \|x\| \leq),$$

is not semiprojective (this can be shown directly). We note that showing the failure of the HLP is difficult.

Let us explain semiprojectivity and also introduce approximate semiprojectivity simultaneously. These notions are mainly interesting for separable C^* -algebras, so this is the realm in which we'll consider them.

Definition 6.108

Let $A \in {}_\omega C^*\text{Alg}$.

1. A is semiprojective iff for any $B \in C^*\text{Alg}$, any increasing sequence $(J_n)_{n \in \mathbb{N}}$ of closed ideals in B with J being the closed ideal generated by the sequence and any $\star$ -homomorphism $\phi : A \rightarrow B/J$, we have that this can be lifted to a $\star$ -homomorphism $\psi : A \rightarrow B/J_n$ for sufficiently large n .
2. A is approximately semiprojective iff it is isomorphic to a sequential inductive limit of semiprojective C^* -algebras.

In the literature, an approximately semiprojective C^* -algebra is often also said to have a strong shape system.

In the next remark, we will provide some context for these notions.

Remark 6.109. In [13], it is claimed that a paper, "Homotopy Continuity & Shape Theory for C^* -Algebras", by (Edward) Effros and Kaminker was responsible for developing a noncommutative version of shape theory. This theory was originally introduced in the topological setting to study certain spaces by focusing on homotopy types of the spaces that are used to construct them; for more details on these matters and history, see [45]. We could not locate the aforementioned paper but it is referenced in a paper by Blackadar ([13]) which seems to be the first one which introduced the notion of semiprojectivity which is in use within modern^a research.

^aSee Definition 2.10, p.256 of the aforementioned paper.

With this background, we can finally (partially) address Question 6.51 through the following theorem.

Theorem 6.110. A. Vats, 2026

The spaces in statements (1) and (2) of Question 6.51 are Schochet spaces.

The proof relies on a difficult permanence property of semiprojectivity that we will not prove.

Proposition 6.111. D. Enders, 2014, [39, Corollary 3.4, p.8]

Suppose we have the following pullback square:

$$\begin{array}{ccc} A \oplus_F B & \longrightarrow & B \\ \downarrow & & \downarrow \\ A & \longrightarrow & F \end{array}$$

in $C^*\text{Alg}$, where F is finite-dimensional. If A and B are semiprojective, then $A \oplus_F B$ is semiprojective.

Here is the proof of our theorem.

Proof of Theorem 6.110. We will deal with each of these spaces individually.

1. Observe that:

$$\mathcal{C}(S^1 \vee S^1) \simeq \mathcal{C}(S^1) \oplus_{\mathbb{C}} \mathcal{C}(S^1).$$

It is known that $\mathcal{C}(S^1)$ is semiprojective. So, the above is a pullback of two semiprojective C^* -algebras over a finite-dimensional C^* -algebra so Proposition 6.111 tells us that it must be semiprojective and, hence, Schochet.

2. Observe that:

$$\mathcal{C}(S^1 \vee [0, 1]) \simeq \mathcal{C}(S^1) \oplus_{\mathbb{C}} \mathcal{C}([0, 1]).$$

By the same argument as in the proof of statement (1), we can conclude that this pullback is Schochet. ■

Let us mention a recent result which improves our intrinsic understanding of the homotopy lifting property. In fact, it was partly this result which prompted us to look into l -closed C^* -algebras when it came to our resolution of Conjecture 6.39.

Theorem 6.112. D. Oyetunbi, A. Tikuisis, [72, Theorem 3.8, p.7]

Let $A \in C^*\text{Alg}$. TFAE:

1. A is l -open.
2. The system of maps:

$$\text{Hom}(A, B) \rightarrow \text{Hom}(A, B/J),$$

for all marked C^* -algebras (B, J) , is uniformly open^a.

3. If (B, I) is a marked C^* -algebra, then $(A, \pi : B \rightarrow B/I)$ is HLP and $\text{Hom}(A, B)$ is locally path-connected.

^aWe will not define this term as it will not be needed; one can find it in the reference.

In the last result of this section, we will collect some facts that are derived from our work and that of others.

Proposition 6.113. A. Vats, 2026

Let $A \in C^*\text{Alg}$. TFSH:

1. (A, Id_A) is HLP.
2. A is Schochet iff (A, π) is HLP, with π being the Calkin projection.
3. A , as defined in Theorem 6.56, is approximately semiprojective.
4. The Schochet property is not closed under sequential inductive limits.
5. The HLP is not closed under sequential inductive limits.
6. If A is l -open, then it is Schochet.
7. There exists a Schochet C^* -algebra which is not l -open.
8. There exists a Schochet C^* -algebra which is not semiprojective.
9. The full subcategories of ${}_{\omega}C^*\text{Alg}$:
 - (a) the category of Schochet C^* -algebras and;
 - (b) the category of approximately semiprojective C^* -algebras;
 do not coincide.
10. If (b) is a subcategory of (a) , then Conjecture 6.114 is false.
11. Consider the full subcategories of ${}_{\omega}C^*\text{Alg}$:
 - (a) the category of approximately Schochet C^* -algebras and;
 - (b) the category of approximately semiprojective C^* -algebras.
 Then, (b) is a subcategory of (a) .

Proof. We will prove each statement one-by-one.

1. This is clear; we just need to take the lift to be the original homotopy.
2. This is the definition of the Schochet property.
3. This was a difficult result but it has been recently shown to be the case; see [62, p. 723].
4. Since semiprojective C^* -algebras are Schochet, the C^* -algebra of Theorem 6.56 is exactly an example of a sequential inductive limit of Schochet C^* -algebras which is not Schochet.
5. Since there is a non-Schochet C^* -algebra, this C^* -algebra also cannot be HLP and one can construct such an

example which is approximately semiprojective, so it is an inductive limit of HLP C^* -algebras^a.

6. This follows by [Theorem 6.112](#).
7. In [\[72, Example 3.9, p.7\]](#), a unital AF algebra is given which is not l -open.
8. We simply need a Schochet C^* -algebra which is not l -open and we are done^b.
9. Since there exists a non-Schochet C^* -algebra which is approximately semiprojective, we see that the category of Schochet C^* -algebras is not closed under sequential inductive limits. On the other hand, the category of approximately semiprojective C^* -algebras is closed under sequential inductive limits; this was shown in [\[93, Theorem 4.9, p.13\]](#). So, it follows that these two categories cannot coincide.
10. Indeed, if every approximately semiprojective C^* -algebra is Schochet, then we can just take a separable non-Schochet C^* -algebra and this cannot be approximately semiprojective so we're done.
11. This follows from the fact that semiprojective C^* -algebras are Schochet. ■

^aThis follows by Blackadar's Homotopy Lifting Theorem.

^bThis is because semiprojectivity implies l -openness.

Before we end this section, let us briefly explain why [Proposition 6.113](#) is significant. The answer has to do with a long-standing conjecture on semiprojectivity due to Blackadar. We have all the language necessary to state this conjecture.

| **Conjecture 6.114** (B. Blackadar, 1985, [\[13, 4.4, p.266\]](#)). Every separable C^* -algebra is approximately semiprojective.

Through our work with Schochet C^* -algebras, we have shown that the approximate Schochet property serves as a possible obstruction to [Conjecture 6.114](#). That is, if we can prove that there exists an approximately Schochet separable C^* -algebra which is not approximately semiprojective, then this would resolve the conjecture in the negative. In [??](#), we will discuss our reasons for believing that this could be a useful obstruction and this will involve relating it to other obstructions in the literature. This brings us to the end of our exposition on the concrete aspects of the homotopy theory on categories of operator algebras. We will, now, begin to explore abstract homotopy theory, so the tone of the thesis will change. Nevertheless, we are guided by much of the work done here and will refer back to it at key points.

7 Constructing Operator K_0

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| Motivation.

7.1 Two Pictures of K_0

7.1.1 The Projection Picture

With this in mind, let's start with some terminology.

Terminology 7.1. If $A \in {}^1\text{Alg}$ and let $p, q \in A$, then p and q are unitarily equivalent iff:

$$\exists u \in \text{Unitary}(A) : upu^* = q.$$

The unitary u is known as a change-of-basis between p and q .

As we've seen before, two idempotents that are similar must be algebraically equivalent. In a unital C^* -algebra, it's clear that unitary equivalence implies algebraic similarity. In fact, the converse holds for projections⁵.

Proposition 7.2

Let $A \in {}^1C^*\text{Alg}$ and let $p, q \in A$ be projections. TFAE:

1. p and q are algebraically similar.
2. p and q are unitarily equivalent.

Proof. The trick is to use the polar decomposition of invertibles in a unital C^* -algebra.

(1) $\Rightarrow$ (2): Let $a \in A^\times$ such that:

$$ap = qa.$$

Now, a has a polar decomposition $a = u|a|$ with u being a unitary. We want to show that:

$$up = qu.$$

Well, let's compute from the left-hand side and pretend that things work. Once we've done that, we'll just identify the thing we need for the calculation to go through and we'll deal with that separately.

$$\begin{aligned} up &= a|a|^{-1}p \\ &= ap|a|^{-1} \\ &= qa|a|^{-1} \\ &= qu. \end{aligned}$$

So, what we need is for p to commute with $|a|^{-1}$. Well, we really just need p to commute with $|a|$. By the properties of functional calculus^a, we just need to ensure that p commutes with a^*a . So, we have:

$$\begin{aligned} p(a^*a) &= (pa^*)a \\ &= (ap^*)^*a \\ &= (ap)^*a \\ &= (qa)^*a \\ &= a^*q^*a \\ &= a^*qa \\ &= a^*(ap) \\ &= (a^*a)p, \end{aligned}$$

as was desired.

(2) $\Rightarrow$ (1): This is clear. ■

^aSo, in particular, $|a|$ is just the limit of a sequence of polynomials in a^*a . Therefore, if p commutes with a^*a , then it must commute with any such polynomial and since multiplication is continuous, it follows that p commutes with $|a|$.

So, it would be very nice if, in a unital C^* -algebra, it was true that an idempotent is similar to a projection. In that case, we could just "replace" every idempotent by a projection and this would mean that we can consider unitary changes-of-basis rather than simply invertible ones. This luxury is afforded to us by the next result.

Proposition 7.3

Let $A \in {}^1C^*\text{Alg}$. TFSH:

1. Every idempotent in A is algebraically similar to a projection.
2. Let p, q be algebraically equivalent projections in $M_{n \times n}(A)$ such that:

$$\begin{aligned} p &= \alpha\beta, \\ q &= \beta\alpha, \end{aligned}$$

⁵Such a reduction is similar, in principle, to the one that we do for studying representations (of topological groups or C^* -algebras).

where $\alpha, \beta \in M_{n \times n}(A)$. There exists a partial isometry $u \in M_{n \times m}(A)$ so that:

$$\alpha = uu^*,$$

$$\beta = u^*u.$$

In particular, $p = u^*u$ and $q = uu^*$.

We'll prove this together.

Proof of Proposition 7.3. Let's deal with each statement one-by-one.

| **Claim.** Statement (1) holds.

Let A be isometrically embedded into some $\mathcal{B}(H)$ for a $\mathbb{C}$ -Hilbert space and let $e \in A$ be an idempotent. Then, $\text{ran}(e)$ is a closed subspace of H and we know that H decomposes as:

$$H = \text{ran}(e) \oplus \text{ran}(e)^\perp.$$

So, e has the following block matrix representation:

$$e = \begin{pmatrix} 1 & T \\ 0 & 0 \end{pmatrix},$$

for some operator T . Now, we let p be the operator with matrix representation $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and let s be the operator with matrix representation $\begin{pmatrix} 1 & T \\ 0 & 1 \end{pmatrix}$. This leads us towards the following computation:

$$\begin{aligned} se &= \begin{pmatrix} 1 & T \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & T \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & T \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & T \\ 0 & 1 \end{pmatrix} \\ &= ps. \end{aligned}$$

Moreover, s is invertible so it remains to be shown that p and s both lie in A . Note that:

$$s = 1 + e - p,$$

where 1 represents the identity matrix. So, the idea is to show that p is in the C^* -algebra generated by 1 and e . To see this, we compute with the matrices given above and find that:

$$p(1 + ee^* + e^*e - (e + e^*)) = ee^*.$$

We also see that:

$$1 + ee^* + e^*e - (e + e^*) = 1 + (e - e^*)(e - e^*)^*,$$

and the right-hand side is an element of the form $1 + xx^*$. Such an element is invertible in A so:

$$p = ee^*(1 + ee^* + e^*e - (e + e^*))^{-1},$$

which means that $p \in A$.

| **Claim.** Statement (2) holds.

Suppose that we have $\alpha', \beta' \in M_{n \times n}(A)$ such that:

$$p = \alpha'\beta',$$

$$q = \beta'\alpha'.$$

Now, let $\alpha = p\alpha'q$ and $\beta = q\beta'p$. Here's the first computation.

| **Claim 7.1.** We have that $\alpha\beta = p$ and $\beta\alpha = q$.

So, we've shown that p and q are algebraically equivalent and that this is observed by different elements. In this case, however, we have that these satisfy the additional equalities:

$$p\alpha q = \alpha,$$

$$q\beta p = \beta.$$

Now, we have the following result.

| **Claim 7.2.** We have that:

$$p \leq \|\alpha\|^2 \beta^* \beta,$$

$$q \leq \|\beta\|^2 \alpha^* \alpha.$$

In particular, $\alpha^* \alpha$ and $\beta^* \beta$ are invertible in qAq and pAp respectively.

Now, we define:

$$u \stackrel{\text{def}}{=} \beta|\beta|^{-1},$$

where u is an isometry in pAp . Now, observe that:

$$u^*u = |\beta|^{-1}\beta^*\beta|\beta|^{-1} = |\beta|^{-1}|\beta|^2|\beta|^{-1} = p,$$

where we compute the above in the C^* -algebra pAp , where the identity is p . Now, we have another claim.

| **Claim.** $uu^* = q$.

Since u is a partial isometry in A , we see that uu^* is a projection. Since⁶ $|\beta| \in pAp$, it follows that:

$$p|\beta|^{-2} = |\beta|^{-2}.$$

This leads us to a computation:

$$\begin{aligned} quu^* &= q\beta|\beta|^{-2}\beta^* \\ &= q\beta p|\beta|^{-2}\beta^* \\ &= \beta|\beta|^{-2}\beta^* \\ &= uu^*, \end{aligned}$$

where we're using the fact that $q\beta p = \beta$ as derived above. On the other hand, we have that:

$$q = qq^* = \beta\alpha\alpha^*\beta^* \leq \|\alpha\|^2\beta\beta^*.$$

We also know that $\beta\beta^* = u\beta^*\beta u^*$, so we get that:

$$q \leq \|\alpha\|^2 u\beta^*\beta u^* \leq \|\alpha\|^2 \|\beta\|^2 uu^*.$$

So, the claim now follows and the proof is complete.

This result leads us to the notion of *generalized Murray-von Neumann equivalence*.

Definition 7.4

Let $A \in {}^*\text{Alg}$ and let p, q be projections in matrix algebras over A . We say that p and q are generalized Murray-von Neumann equivalent iff there exists a matrix u such that the following equalities hold:

$$\begin{aligned} p &= u^*u, \\ q &= uu^* \end{aligned}$$

We denote this by writing $p \sim_{K_0} q$.

We have two remarks about this definition.

Remark 7.5. We are deliberately being vague about the specific matrix algebras in which p and q , as given in the definition, live. The problem with some texts is that they get too caught up in specifying such details when, in reality, such specifics will not matter for doing the crucial computations that are at the core of many proofs. Nevertheless, let's just be explicit about this *once*, just to show that it can be done. We'll do this in the next remark.

Remark 7.6. Let $m, n \in \mathbb{N}$ and let $p \in M_{n \times n}(A)$, $q \in M_{m \times m}(A)$ be projections. Then, $p \sim_{K_0} q$ iff there exists a matrix u such that $p = u^*u$ and $q = uu^*$. By looking at the sizes of p and q , it's clear that u will have to be a $m \times n$ -matrix. Once again, there is usually no need to fuss about this in computations since it is irrelevant.

Let's show that $\sim_{K_0}$ is an equivalence relation through the next exercise.

| **Claim 7.3.** Let $A \in {}^*\text{Alg}$. Then, $\sim_{K_0}$, given in Definition 7.4, is an equivalence relation.

Now, let us carry out a construction of K_0 for unital C^* -algebras in a similar spirit as Construction 4.13. Of course, this does not have to be the same as algebraic K_0 for unital C^* -algebras but we will see that they, indeed, do form the same invariant; this will follow from the results established earlier in this section.

Construction 7.7. Let $A \in {}^1C^*\text{Alg}$. For each $n \in \mathbb{N}$, we define:

$$\begin{aligned} \phi_n : M_{n \times n}(A) &\rightarrow M_{(n+1) \times (n+1)}(A), \quad a \mapsto \phi_n(a) \stackrel{\text{def}}{=} \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}, \\ \psi_n : \text{Unitary}(n, A) &\rightarrow \text{Unitary}(n+1, A), \quad a \mapsto \psi_n(a) \stackrel{\text{def}}{=} \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

These maps give rise to inductive sequences in $C^*\text{Alg}$ and Grp . Since these categories are cocomplete, these inductive sequences have colimits and we denote these by the following bit of notation:

$$\begin{aligned} \text{Mat}_\infty(A) &\stackrel{\text{def}}{=} \text{colim}(M_{n \times n}(A)), \\ \text{Unitary}_\infty(A) &\stackrel{\text{def}}{=} \text{colim}(\text{Unitary}(n, A)). \end{aligned}$$

We will view the elements of $\text{Mat}_\infty(A)$ as “infinite matrices” over A and, indeed, constructing this as a colimit really does allow us to use that intuitive picture. Here are two things to keep in mind:

1. We'll let $\text{Proj}(\text{Mat}_\infty(A))$ denote the set of projections in $\text{Mat}_\infty(A)$.
2. Note that $\text{Unitary}_\infty(A)$ acts on $\text{Mat}_\infty(A)$ by conjugation.

⁶Why? Give an argument!

Now, we let $p, q \in \text{Proj}(\text{Mat}_\infty(A))$. We write unitary similarity a bit more cleanly as follows:

$$p \sim_s q \stackrel{\text{def}}{\iff} \exists u \in \text{Unitary}_\infty(A) : p = uqu^*.$$

Let $S = \text{Proj}(\text{Mat}_\infty(A)) / \sim_s$ and let $p \in \text{Proj}(\text{Mat}_\infty(A))$. Then, the following notation:

$$[p]_s,$$

will refer to the equivalence class of p under $\sim$ and this is the unitary conjugation orbit. With this, we define:

$$\forall p, q \in \text{Proj}(\text{Mat}_\infty(A)) : [p]_s + [q]_s \stackrel{\text{def}}{=} \left[\begin{pmatrix} p & 0 \\ 0 & q \end{pmatrix} \right]_s.$$

So, this should make total sense; the creation of the matrix on the right-hand side by filling in the diagonals corresponds *exactly* to the direct sum in the case of f.g.p modules over R . Now, we are lead very naturally to the following result.

| **Claim.** The operation above is well-defined and turns S into an abelian semigroup.

You'll prove this in **Claim 7.4**. So, the fact that this is an abelian semigroup means that we can apply the Grothendieck construction to it. What do we get when we do it? Well, we define:

$$K_0(A) \stackrel{\text{def}}{=} \mathcal{G}(S),$$

as the operator K-theory of A .

Here's the associated exercise.

| **Claim 7.4.** Prove the first claim of [Construction 7.7](#).

7.1.2 The Homotopical Picture

The fact that such a description might exist is not so surprising if you've worked through ??; we do not assume that you have done so, which is why we will repeat some of the definitions, terminology & examples from that section over here. The bigger point is that the additional topological structure available on a C^* -algebra should allow us to obtain a geometric description of the K_0 -groups (and later, the K_1 -groups).

What you can imagine is that if we have two projections p, q in a C^* -algebra which are "close enough", then one should be able to connect them by a path of projections and this should allow us to say something about their ranges. But, well, their ranges being isomorphic is exactly what we need to "build" K_0 -groups. The goal of this section is to formalize that intuition and we start with the following piece of terminology.

| **Terminology 7.8.** Let $X \in \text{Top}$ and let $x, y \in X$. x is homotopic to y iff there exists a path from x to y .

This bit of terminology is a little strange, at first glance, because we are somehow conflating the usual use of the word "homotopy" with "path-connectedness". The next example explains the logic behind our choice of terminology here.

Example 7.9. Let $Y \in \text{CHaus}$, let $E \in \text{Nor}_\mathbb{C}$ and let $Z \subseteq E$. Now, we topologize:

$$X = \mathcal{C}(Y, Z),$$

by the topology induced from the $\mathbb{C}$ -normed space $\mathcal{C}(Y, E)$. This leads us to the following claim:

| **Claim.** If $f, g \in X$, then TFAE:

1. f and g are homotopic in X .
2. f and g are homotopic as continuous maps.

We'll prove each implication separately.

(1) $\Rightarrow$ (2):

(2) $\Rightarrow$ (1):

This proves the claim.

Now, we are interested in the behavior of homotopic points when X , as in [Terminology 7.8](#), is the set of all projections in a C^* -algebra. For the sake of keeping things neat and tidy, let's introduce a bit of temporary notation.

| **Notation 7.10.** Let $A \in C^*\text{Alg}$. Then, $\text{Proj}(A)$ is the set of all projections in A .

As we mentioned before, it will be technically convenient to introduce a further equivalence relation on this set (of idempotents) which has a more topological character.

Terminology 7.11. Let $A \in {}^1C^*\text{Alg}$ and let $p, q \in \text{Proj}(A)$. p and q are simeotopic iff there exists a path:

$$u : [0, 1] \rightarrow A^\times,$$

such that:

1. $u_0 = 1_A$.
2. $u_1 p u_1^{-1} = q$.

This relation is denoted by writing $p \sim_{\text{simeotopic}} q$.

We have a remark about this bit of terminology.

| **Remark 7.12.** Of course, we could generalize this bit of terminology to arbitrary C^* -algebras by working in the unitization. Indeed, we could say that two projections in a C^* -algebra are simeotopic iff they are simeotopic as

idempotents in $A^\sim$. In that case, one would have to prove that two projections in a unital C^* -algebra are simeotopic iff they are simeotopic in the unitization; this would ensure that we don't always have to explicitly deal with the unital and non-unital cases separately. We'll return to some of these technicalities a little later; they mattered much less in the setting of $\mathbb{C}$ -Banach algebras since we only used that as motivation for the algebraic K-theorist.

Here are two simple exercises.

Claim 7.5. If $A \in {}^1C^*Alg$, then simeotopy is an equivalence relation on $Proj(A)$.

Claim 7.6. Let $A \in {}^1C^*Alg$. Then, simeotopic projections in A are similar^a and homotopic in $Idem(A)$.

^aIn the sense of Terminology 4.3.

We have a remark about the preceding exercise.

Remark 7.13. At the moment, the strength of **Claim 7.7** is the same as the strength of ???. We'll fix that when we give an alternative description of simeotopies of projections in C^* -algebras.

Since we are working in C^* -algebras, one can significantly simplify the definition of simeotopy.

Proposition 7.14

Let $A \in {}^1C^*Alg$ and let $x, y \in A_{s.a.}$ be similar. Then, they are unitarily equivalent.

We've written out the proposition in a form that's easy to remember but the proof shows more. Indeed, you learn how to construct the unitary from the invertible element and as you might've guessed, this is done via the polar decomposition.

Proof of Proposition 7.14. The proof involves using the polar decomposition for invertible elements in a unital C^* -algebra. Indeed, let $u \in A^\times$ so that $x = uyu^{-1}$. Then, we must have that:

$$xu = uy,$$

which implies, by taking adjoints on both sides and using the self-adjointness of x and y , that:

$$u^*x = yu^*.$$

So, we see that:

$$|u|^2y = (u^*u)y = u^*xu = yu^*u = y|u|^2.$$

In other words, $|u|^2$ commutes with y and by approximation, y commutes with all elements in $C^*(1, |u|^2)$. In particular, y commutes with $|u|^{-1}$. This is crucial, because we have the polar decomposition of u :

$$u = a|u|,$$

where $a \in \text{Unitary}(A)$. Therefore, we see that:

$$\begin{aligned} aya^* &= a|u||u|^{-1}ya^* \\ &= uy|u|^{-1}a^* \\ &= yu^{-1} \\ &= x, \end{aligned}$$

as was desired. ■

This leads us to the following corollary.

Corollary 7.15

Let $A \in {}^1C^*Alg$ and let $p, q \in Proj(A)$. TFAE:

1. $p \sim_{\text{simeotopic}} q$.
2. There exists a path:

$$u : [0, 1] \rightarrow \text{Unitary}(A),$$

such that $u_0 = 1_A$ and $u_1pu_1^* = q$.

Proof. (2) $\Rightarrow$ (1) is clear. Now, let $v : [0, 1] \rightarrow A^\times$ be a simeotopy from p to q . For each $t \in [0, 1]$, we have:

$$v_t = a_t|v_t|,$$

where a_t is a unitary. Now, we define a map:

$$u : [0, 1] \rightarrow \text{Unitary}(A), \quad t \mapsto u_t \stackrel{\text{def}}{=} a_t.$$

This is continuous, because it is essentially obtained by point-wise multiplication of two continuous maps. Moreover, it is a path of unitaries by the polar decomposition in unital C^* -algebras. Finally, the proof of Proposition 7.14 tells us specifically that this choice of path works. ■

Using these results, we can strengthen **Claim 7.7**.

Claim 7.7. Let $A \in {}^1C^*Alg$. Then, simeotopic projections in A are similar^a and homotopic in $Proj(A)$.

^aIn the sense of Terminology 4.3.

If we have projections which are close enough in a unital C^* -algebra, then they are automatically simeotopic.

Proposition 7.16

Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$ and let $p, q \in \text{Proj}(A)$ so that:

$$\|p - q\| \leq \frac{1}{\|2p - 1\|}.$$

Then, $p \sim_{\text{simeotopic}} q$.

The proof of this result is exactly the same as the corresponding result for idempotents in a unital $\mathbb{C}$ -Banach algebra as proved in ???. The technical hypothesis above just makes sure that the proof goes through but the idea is to effectively show that p and q are similar.

Proof of Proposition 7.16. Observe that:

$$(p - q)(2p - 1) = 2p^2 - 2pq - p + q = p + q - 2pq.$$

Now, the technical condition implies (via the Neumann Series Theorem) that:

$$\|p + q - 2pq\| < 1,$$

which means that:

$$u = 1 - p - q + 2pq,$$

is invertible. Now, observe that:

$$pu = p - p^2 - pq + 2p^2q = pq,$$

$$uq = q - pq - q^2 + 2pq^2 = pq.$$

This implies that p and q are similar. But the important point is that $u \in B_A(1, 1)$ and since this open ball consists only of invertible elements, it follows that we can simply join u and 1 by a straight line since the open ball is convex. In that case, we see that this straight line gives us a simeotopy between p and q . ■

The usefulness of this result lies in the way it gives us a conceptual understanding of the set of idempotents in a unital $\mathbb{C}$ -Banach algebra as a topological space. Before we explain what we mean, we need some terminology.

Terminology 7.17. Let $X \in \text{Top}$. X is locally path-connected iff for every $x \in X$ and for every nbhd U of x , there exists a path-connected nbhd $V \subseteq U$ of x . This has the same template as all other “local” definitions in topology^a.

^aSee [REF].

Now, the desired corollary of our result above can be presented rather easily.

Corollary 7.18

Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$. Then, $\text{Proj}(A)$ is locally path-connected.

Proof. Let $p \in \text{Proj}(A)$ and let U be a nbhd of p in $\text{Proj}(A)$. Then, any projection in a small enough ball around p , by the proposition, is simeotopic to p . Since simeotopic projections are homotopic, we obtain the result. ■

The significance of this corollary is explained by the following purely topological result.

Claim 7.8. Let $X \in \text{Top}$ be locally path-connected, let $x \in X$ and let $S \subseteq X$. TFAE:

1. S is a path-connected component of x .
2. S is a locally path-connected component of x .

Now, it turns out that the notion of simeotopy was somewhat irrelevant for idempotents.

Proposition 7.19

Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$ and let $p, q \in \text{Proj}(A)$. TFAE:

1. p is homotopic to q in $\text{Proj}(A)$.
2. p is simeotopic to q .

Proof.

(1) $\Rightarrow$ (2): Let $\gamma : [0, 1] \rightarrow \text{Proj}(A)$ be a path from p to q . By subdividing $[0, 1]$ into small enough intervals and using the uniform continuity of γ , we can arrange for p and q to be connected by a finite sequence of idempotents which are simeotopic. It follows that p and q are simeotopic.

(2) $\Rightarrow$ (1): This is proved in ???. ■

In general, it is not the case that two idempotents being similar implies that they are homotopic; we will provide a counterexample after we have discussed the Bott Periodicity Theorem in the next lecture. Nevertheless, the relationship between these notions is about the same as similarity & algebraic equivalence. In particular, we have the following result.

Proposition 7.20

Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$ and let $p, q \in \text{Proj}(A)$ be similar. Then, $p \oplus 0$ and $q \oplus 0$ are homotopic in $\text{Proj}(M_{2 \times 2}(A))$.

To prove this result, we will need a topological version of Whitehead’s Lemma.

Lemma 7.21. Whitehead's Lemma

Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$ and let $x, y \in \text{Unitary}(A)$. Then, the following matrices:

1. $xy \oplus 1$,
2. $x \oplus y$,
3. $y \oplus x$,
4. $yx \oplus 1$,
5. $1 \oplus xy$,

are homotopic in $\text{Unitary}(\mathbf{M}_{2 \times 2}(A))$.

We'll carry out the proof together.

The proof proceeds by a rotation trick. We define:

$$\forall t \in [0, 1] : u_t \stackrel{\text{def}}{=} \begin{pmatrix} \cos\left(\frac{\pi t}{2}\right) & -\sin\left(\frac{\pi t}{2}\right) \\ \sin\left(\frac{\pi t}{2}\right) & \cos\left(\frac{\pi t}{2}\right) \end{pmatrix}.$$

For each $t \in [0, 1]$, we see that this is invertible in $\mathbf{M}_{2 \times 2}(\mathbb{C})$. Now, we define a map:

$$t \mapsto (x \oplus 1)u_t(y \oplus 1)u_t^{-1}.$$

This gives us a path from $xy \oplus 1$ to $x \oplus y$. A similar construction gives us that:

Claim 7.9. $x \oplus y$ and $y \oplus x$ are homotopic^a in $\text{Unitary}(\mathbf{M}_{2 \times 2}(A))$.

^aUse that $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are homotopic in $\text{Unitary}(\mathbf{M}_{2 \times 2}(A))$.

Okay, so we have shown that (1), (2) and (3) are homotopic. But since (1) is homotopic to (2), we see that $y \oplus x$ is homotopic to $yx \oplus 1$. Since $x \oplus y$ is homotopic to $y \oplus x$, it follows that (1) is homotopic to (2). But this means that all 5 matrices are homotopic to each other and this completes the proof of the result.

This leads us to the proof of our desired proposition.

Proof of Proposition 7.20. Let $u \in A^\times$ so that:

$$p = uqu^{-1}.$$

Now, we have the following calculation:

$$\begin{aligned} p \oplus 0 &\sim_h (uqu^{-1}) \oplus 0 \\ &\sim_h (uq \oplus 0) \\ &\sim_h q \oplus 0, \end{aligned}$$

as was desired^a. ■

^aWe use $\sim_h$ to mean homotopy in our sense, as described above. This will not be permanent notation.

So, what happens when we pass to infinite matrices? Well, if $A \in {}^1\mathbf{C}^*\mathbf{Alg}$, then the ring $\text{Mat}_\infty(A)$ formed in [Construction 4.13](#) is not necessarily a $\mathbb{C}$ -Banach algebra since the category ${}^1\mathbf{C}^*\mathbf{Alg}$ is not cocomplete (or at least, inductively complete). Nevertheless, we can still make sense of the notions introduced above by restricting attention in specific matrix subalgebras of $\text{Mat}_\infty(A)$. Let us be more precise with our next definition of homotopic/simeotopic idempotents.

Definition 7.22

Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$ and let $p, q \in \text{Proj}(\text{Mat}_\infty(A))$.

1. p and q are homotopic iff they have this property in $\mathbf{M}_{n \times n}(A)$ for some $n \in \mathbb{N}$.
2. p and q are simeotopic iff they have this property in $\mathbf{M}_{n \times n}(A)$ for some $n \in \mathbb{N}$.

Of course, both these items hold only after appropriately stabilizing p and q to the same size.

As a corollary, we obtain the following alternative description of K_0 .

Corollary 7.23

Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$. TFSH:

1. Algebraic equivalence, algebraic similarity, homotopy and simeotopy are the same on $\text{Proj}(\text{Mat}_\infty(A))$.
2. $K_0(A) \simeq G(\text{Idem}(\text{Mat}_\infty(A))/\sim)$, where $\sim$ is any of the equivalence relations in (1).

Proof. We'll prove each statement one-by-one.

1. This follows by a combination of all versions of Whitehead's Lemma proved so far.
2. We simply replace algebraic similarity with any of the others in the idempotent construction of K_0 . ■

We have a remark about this corollary.

Remark 7.24. So, this corollary is massively important because we can, now, use it as motivation for the construction of our desired functor. Indeed, all we need to do is to define an algebra of infinite invertible matrices over some (unital) $\mathbb{C}$ -Banach algebra, mod out an appropriate equivalence relation and then take the Grothendieck group of the

| resulting quotient. This will give us a functor which will, hopefully, be naturally isomorphic to K_1 .

7.2 The Properties of Operator K_0

In this section, we want to establish some properties of operator K_0 . This will be a rather long section as well and the proofs will be a little tedious. However, we'll try to make the ideas very clear and we'll split our discussion of the various properties as neatly as possible. This section will be divided into three main parts.

1. The homotopy-invariance & half-exactness of K_0 .
2. The continuity of K_0 .
3. The stability of K_0 .

We would recommend going through (1) and (2) together, while leaving (3) and (4) to separate lectures each. Fortunately, once we've done all of this grunt work over here and once we've established some general facts about homology theories on C^* -algebras, we will find it incredibly easy to prove most of these results for operator⁷ K_1 . Without further ado, let us get started by discussing the basic properties of K_0 . Let's start with the homotopy-invariance of operator K_0 .

Proposition 7.25. Homotopy Invariance of Operator K_0

Operator K_0 is homotopy-invariant.

The proof is slightly technical but it is a good idea to see “why” the result should be true before you look at the proof.

Proof of Proposition 7.25.

Let's deal with half-exactness.

Proposition 7.26. Half-Exactness of Operator K_0

Operator K_0 is half-exact.

Proof. The proof is given along the same lines as the usual proof for

Now, we'll deal with the inductive continuity of operator K_0 .

Proposition 7.27. Inductive Continuity of Operator K_0

K_0 is inductively continuous.

In the interest of keeping our presentation clean, we will establish notation separately.

Notation 7.28. Let $F : I \rightarrow C^* \text{Alg}$ be an inductive system. For clarity, if $i \rightarrow j \in \text{Mor}(I)$, then we define:

$$\phi_{ij} \stackrel{\text{def}}{=} F(i \rightarrow j).$$

Now, let's set some notation and also construct the natural map which we need to study^a.

1. $\forall i \in I : A_i \stackrel{\text{def}}{=} F(i)$.
2. Let A be the inductive limit of F , with cocone maps $(\psi_i^1 : A_i \rightarrow A)_{i \in I}$.
3. Let B be the inductive limit of $K_0 \circ F$, with cocone maps $(\psi_i^2 : K_0(A_i) \rightarrow B)_{i \in I}$.

Applying K_0 to each ψ_i^1 , we obtain a family of morphisms:

$$(K_0(\psi_i^1) : K_0(A_i) \rightarrow K_0(A)).$$

Observe that $\psi_j^1 \circ \phi_{ij} = \psi_i^1$. Applying K_0 on both sides, we see that $(K_0(\psi_i^1))_{i \in I}$ forms a family of cocone maps for $K_0 \circ F$. So, this induces, by universal property, the following morphism:

$$\Phi : B \rightarrow K_0(A),$$

and this is the map we wish to study.

^aOf course, this is a part of the definition of inductive continuity. However, seeing as we gave that definition in Operator Algebraic Structures I, we do want to remind you of the conventions set over there, even if you'll have to flip back to that text for the actual results.

The first step in this entire process is to reduce to the unital situation.

Proof of Proposition 7.27.

We'll prove this by showing surjectivity and injectivity, relying on ??.

1. Let's show surjectivity.
2. Let's show injectivity.

This completes the proof.

Finally, let's show that K_0 is C^* -stable.

⁷This is because we will, as in the case with $\mathbb{Z}_2$ -graded Banach algebras, define K_1 in this situation as the composition $K_0 \circ \text{Sus}$ and the properties of Sus alongside the properties of K_0 will yield the nice features of K_1 .

Proposition 7.29. Stability of Operator K_0

Operator K_0 is C^* -stable.

Proof.

■

We're ready to proceed towards the last major property of K_0 .

8 Classifying Unital AF Algebras

Summary

In this lecture, we will discuss the following topics:

1. (Partially) Ordered abelian groups.
2. The order structure of K_0 .

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

Motivation. In the construction of $\mathbb{Z}$ from $\mathbb{N}$, we made use of the fact that $\mathbb{N}$ comes naturally equipped with a total order to induce a total order on $\mathbb{Z}$. We did this by identifying the “positive cone” of $\mathbb{Z}$, denoted by $\mathbb{Z}^+$, and using that to define the order. Of course, this is the same procedure that one uses for C^* -algebras as well and since $A \in C^*\text{Alg}$ is equipped with an order, we expect that $K_0(A)$ will also be equipped with a natural order. Our expectations are justified under certain assumptions and the goal of this lecture is to demonstrate that operator $K_0(A)$ does have a nice order structure when A is a unital, **stably finite** C^* -algebra.

8.1 Bringing Order $\rightarrow K_0$

Let us be a bit more precise and state the main result of this lecture.

Proposition 8.1

Let $A \in C^*\text{Alg}$. We define the positive cone of $K_0(A)$ as follows:

$$K_0(A)^+ \stackrel{\text{def}}{=} \{[p]_{K_0} : p \in \text{Idem}_{s.a.\infty}(A)\} \subseteq K_0(A).$$

The following statements hold:

1. $\forall x, y \in K_0(A)^+ : x + y \in K_0(A)^+$.
2. If A is unital, then $K_0(A)^+ - K_0(A)^+ = K_0(A)$.
3. If A is stably finite, then $K_0(A)^+ \cap (-K_0(A)^+) = \{0\}$.

Finally, if A is unital and stably finite, then $(K_0(A), K_0(A)^+)$ is an ordered abelian group.

We have a preliminary remark about this result.

Remark 8.2. The standard portrait *potentially* provides us with a way to equip $K_0(A)$ with the structure of an ordered abelian group. Indeed, the idea is to select the following set as the positive cone:

$$K_0(A)^+ \stackrel{\text{def}}{=} \{[p]_{K_0} : p \in \text{Idem}_{s.a.\infty}(A^\sim)\}.$$

It isn't the case that this gives us a partially ordered abelian group for ALL C^* -algebras. However, there are interesting classes of C^* -algebras for which we do obtain such an order structure. This leads us to the following bit of terminology.

As always, the underlined terms need to be defined. This result could reasonably be seen as an extension of the standard portrait of operator K_0 for unital, stably finite C^* -algebras. We will deal with matters concerning (stable) finiteness in the next section, while this section will be dedicated towards some order theory. Let's start with the following definition.

Definition 8.3

A partially ordered abelian group is a pair (G, G^+) such that G is an abelian group and $G^+ \subseteq G$ such that:

1. $\forall x, y \in G^+ : x + y \in G^+$.
2. $G^+ \cap (-G^+) = \{0\}$.

Moreover, we define:

$$\forall x, y \in G : x \preceq y \stackrel{\text{def}}{\iff} y - x \in G^+.$$

The set G^+ is known as the positive cone^a of G .

^aIts elements are the positive elements of G .

In light of this definition, we have the following easy exercise.

Claim 8.1. Let (G, G^+) be a partially ordered abelian group. Then, $\preceq$ is a partial order on G such that:

$$\forall x, y, z \in G : x \preceq y \Rightarrow x + z \preceq y + z.$$

In other words, the positive cone induces a partial order that is compatible with the group structure.

The next exercise deals with something of a converse.

Claim 8.2. Let $G \in \text{AbGrp}$ and let $\preceq$ be a partial order on G such that:

$$\forall x, y, z \in G : x \preceq y \Rightarrow x + z \preceq y + z.$$

Prove that a set $G^+ \subseteq G$ can be defined so that:

1. (G, G^+) satisfies properties listed in **Definition 8.3**.
2. The partial order obtained from (G, G^+) as in the definition is just $\preceq$.

So, one can choose to start with a positive cone or with a partial order that is compatible with the group structure.

We have two remarks about the discussion so far.

Remark 8.4. Of course, you should be able to think of plenty of examples of ordered abelian groups. Indeed, any partially ordered vector space is certainly a partially ordered abelian group and that gives us a plethora of basic examples. Even though the definition above is purely algebraic, it is rather surprising that these objects are closely connected operator-theoretic structures. Indeed, we'll certainly see that some of their structure is closely related to the K-theory of AF algebras. We'll also give further references to the literature later for those of you who'd like to have an overview of other possible directions of interest.

Remark 8.5. From now on, whenever we have a partially ordered abelian group (G, G^+) and when we have $x \in G$, writing $0 \preceq x$ is synonymous with writing $x \in G^+$. Of course, the former also makes sense because the structure of a partially ordered abelian group induces a compatible partial order on G , as we have seen in the preceding exercise. In any case, we won't specify anything specific about $\preceq$ and we'll just assume that it's the induced order.

Here's a rather neat example of a partially ordered abelian group.

Example 8.6. Let $G = c_0(\mathbb{N}, \mathbb{Z})$ be the set of all sequences in $\mathbb{Z}$ which are eventually zero. This has the structure of an abelian group under point-wise addition. If we let G^+ be the set of all such sequences which are non-negative in every term, then (G, G^+) is a partially ordered abelian group.

Let's try to cook up a category with these as the objects. To do this, we need to specify the morphisms.

Terminology 8.7. Let $(G, G^+), (H, H^+)$ be partially ordered abelian groups and let $\phi : G \rightarrow H$ be a map. ϕ is a positive group homomorphism iff it is a group homomorphism which preserves positive cones. A positive group homomorphism is also known as an (partially) ordered group homomorphism.

Here are a few basic exercises for you.

Claim 8.3. Using the notation of Terminology 8.7, TFAE:

1. ϕ is a positive group homomorphism.
 2. ϕ is a group homomorphism such that for all $x \in G$ with $0 \preceq x$, we have that $0 \preceq \phi(x)$.
- (2) can be rephrased as follows; the positive elements of G are preserved by positive group homomorphisms.

Claim 8.4. Let (G, G^+) be a partially ordered abelian group. Then, Id_G is a positive group homomorphism.

Claim 8.5. The composition of positive group homomorphisms is a positive group homomorphism.

These exercises allow us to define a new category.

Example 8.8. Let $\text{OrdAbGrp}^{\text{partially}}$ be a structure consisting of the following data:

1. The objects in $\text{OrdAbGrp}^{\text{partially}}$ are the partially ordered abelian groups, of course.
2. The morphisms are just positive group homomorphisms between partially ordered abelian groups.

This gives us a category; this is the so-called category of partially ordered abelian groups.

Now, this discussion is supposed to serve the need of dealing with AF algebras and since C^* -algebras have the wonderful property that self-adjoint elements can be decomposed as a difference of positive elements⁸, we expect that $K_0(A)$ would also have that property too. So, let us generalize that and introduce some terminology. At the same time, we'll introduce another property that partially ordered abelian groups tend to have & we certainly want access to this property.

Terminology 8.9. Let (G, G^+) be a partially ordered abelian group.

1. (G, G^+) is an ordered abelian group iff $G = G^+ - G^+$; this is known as the Jordan decomposition property^a.
2. Let $u \in G^+$. This is an order unit iff:

$$\forall g \in G : \exists n \in \mathbb{N} : -nu \preceq g \preceq nu.$$

The triple (G, G^+, u) is known as an ordered abelian group with distinguished order unit.

Finally, an ordered abelian group (G, G^+) is simple iff every non-zero $u \in G^+$ is an order unit.

^aThis is not its formal name in the literature but it seems fitting.

Let's come back to order units in just a moment. The fact that ordered abelian groups are no different from partially ordered abelian groups *other than in the decomposition property* means we can define a full subcategory.

Example 8.10. Let OrdAbGrp be the full subcategory of $\text{OrdAbGrp}^{\text{partially}}$ consisting of ordered abelian groups.

The most difficult fact about OrdAbGrp that we will need is the existence of inductive limits. Technically, since we are only interested in using K_0 to classify separable AF-algebras, it might be reasonable for us to remain consistent with that goal & simply prove our preparatory results for sequential inductive limits. This leads us into the next proposition.

Proposition 8.11

Consider the following inductive sequence:

$$G_1 \xrightarrow{\beta_1} G_2 \xrightarrow{\beta_2} G_3 \xrightarrow{\beta_3} \dots$$

in OrdAbGrp and let $(G, (\alpha_n : G_n \rightarrow G)_{n \in \mathbb{N}})$ be its inductive limit^a in AbGrp . Define:

$$G^+ \stackrel{\text{def}}{=} \bigcup_{n \in \mathbb{N}} \alpha_n(G_n^+).$$

TFSH:

1. (G, G^+) is an ordered abelian group.
2. α_n is a positive group homomorphism for every $n \in \mathbb{N}$.
3. $((G, G^+), (\alpha_n : G_n \rightarrow G)_{n \in \mathbb{N}})$ is an inductive limit of the sequence above in OrdAbGrp .

⁸See ??.

In other words, OrdAbGrp is closed under sequential inductive limits.

^aThis exists by the construction from Operator Algebraic Structures I.

Proof. We'll prove each statement one-by-one.

1. Let's prove that (G, G^+) satisfies the required properties.
 - (a) Let $x, y \in G^+$. Then, there exist $u \in G_n^+$ and $v \in G_m^+$ so that:

$$x = \alpha_n(u),$$

$$y = \alpha_m(v).$$

If $n = m$, then we may just see that:

$$x + y = \alpha_n(u + v),$$

and since $u + v \in G_n^+$, we see that $x + y \in G^+$. Otherwise, we may assume that $n < m$. In that case, we apply $\beta_{m-1} \circ \dots \circ \beta_n$ to u ; this gives us a positive element in G_m .

This proves statement (1).

2. Observe that each α_n is a group homomorphism and we note that:

$$\forall n \in \mathbb{N} : \alpha_n(G_n^+) \subseteq G^+,$$

so it follows that each α_n is positive.

3. Let $(H, (\alpha_n : G_n \rightarrow H)_{n \in \mathbb{N}})$ be another inductive cocone of the given inductive sequence. By the universal property of inductive limits of abelian groups, we obtain a unique group homomorphism $f : G \rightarrow H$ between the inductive cocones. Now, we want to show that this is positive.

This completes the proof. ■

For the moment, this is all that interests us with regards to the basic theory of ordered abelian groups. Let's turn our attention to the relationships with K_0 , particularly the ones that started this entire discussion in the first place.

Terminology 8.12. Let $A \in C^*\text{Alg}$ and let $p \in A$.

1. If $q \in A$ satisfies $q \preceq p$ and is a projection, then q is known as a subprojection.
2. A projection $q \in A$ is a proper subprojection of p iff it is a subprojection of p and $q \neq p$ (so $q \prec p$).
3. p is infinite iff there is a proper subprojection q such that $p \sim_{M \vee N} q$.
4. p is finite iff it is not infinite.
5. If A is unital, then it is finite iff e is a finite projection.
6. If A is unital, then it is infinite iff it is not finite.
7. If A is unital, then it is stably finite iff $M_{n \times n}(A)$ is finite for all $n \in \mathbb{N}$.

If A is non-unital, then it is finite/stably finite/infinite iff its unitization is finite/stably finite/infinite.

We are, for the moment, looking towards a convenient characterization of finiteness of a C^* -algebra. Here is a simple exercise concerning a characterization of when a projection is finite.

Claim 8.6. Let $A \in C^*\text{Alg}$ and let $p \in A$. Prove that TFAE:

1. p is finite.
2. pAp is finite.

In order to characterize the finiteness of some unital C^* -algebra, we'll need a lemma involving invertibility.

Lemma 8.13 ▮ Let $A \in {}^1C^*\text{Alg}$ and let $a \in A$. TFAE:

1. a is left-invertible.
2. a^*a is invertible.

Proof.

(1) $\Rightarrow$ (2):

(2) $\Rightarrow$ (1): So, there is a $b \in A$ such that $ba^*a = e$. Well, ba^* is a left inverse for a so we're done. ■

Similarly, one can also establish a characterization of right invertibility, which will also be useful.

Claim 8.7. Let $A \in {}^1C^*\text{Alg}$ and let $a \in A$. TFAE:

1. a is right-invertible^a.
2. aa^* is invertible.

^aAgain, this just means that there is some $b \in A$ with $ab = e$.

Okay, here's our characterization of finite (unital) C^* -algebras.

Proposition 8.14

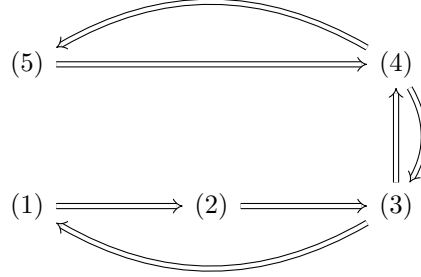
Let $A \in {}^1\mathbf{C}^*\mathbf{Alg}$. TFAE:

1. A is finite.
2. All isometries in A are unitary.
3. All projections in A are finite.
4. Every left-invertible element in A is invertible.
5. Every right-invertible element in A is invertible.

This result was in^a [86, Lemma 5.1.2].

^aThe proof over here is longer than the one in the cited text because we filled in all details.

Proof. The following arrow diagram indicates the implications we'll establish.



Okay, let's proceed with the bottom row.

(1) $\Rightarrow$ (2): Let $s \in A$ be an isometry. Then, $s^*s = e$. Now, observe that:

$$\begin{aligned}
 (ss^*)^2 &= ss^*ss^* \\
 &= ses^* \\
 &= ss^*,
 \end{aligned}$$

which means that ss^* is a projection. By ??, we see that $ss^* \sim s^*s = e$. If we can show that $ss^* \preceq e$, then we'd be able to conclude that $ss^* = e$ and that s is a unitary. By ??, we know that $\|ss^*\|e - ss^*$ is positive. Now, we have the following short computation:

$$\begin{aligned}
 \|ss^*\| &= \|s^*\|^2 \\
 &= \|s\|^2 \\
 &= \|s^*s\| \\
 &= 1,
 \end{aligned}$$

which proves the required inequality.

(2) $\Rightarrow$ (3): Let $p \in A$ be a projection and let $q \in A$ be a subprojection of p such that:

$$p \sim_{\text{MvN}} q.$$

By definition, we must have that:

$$\exists v \in A : p = v^*v \wedge q = vv^*.$$

Now, we define the following element:

$$s \stackrel{\text{def}}{=} v + (e - p).$$

Let us compute ss^* and s^*s .

$$\begin{aligned}
 ss^* &= (v + (e - p))(v^* + (e - p)) \\
 &= vv^* + (e - p)v^* + v(e - p) + (e - p)^2 \\
 &= q + (e - p)v^* + v(e - p) + (e - p) \\
 &= e + (q - p) + (e - p)v^* + v(e - p).
 \end{aligned}$$

Similarly, we also have that:

$$\begin{aligned}
 s^*s &= (v^* + (e - p))(v + (e - p)) \\
 &= v^*v + (e - p)v + v^*(e - p) + (e - p)^2 \\
 &= e + (e - p)v + v^*(e - p).
 \end{aligned}$$

In both computations above, $(e - p)v$ and $v^*(e - p)$ appear. We note that they are dual to each other and we're interested in computing them individually. This leads us to the following claim.

Claim. $(e - p)v = 0$.

We'll use the identities in ??. We have that:

$$\begin{aligned}
 (e - p)v &= v - pv \\
 &= v - v \\
 &= 0,
 \end{aligned}$$

as was desired.

Since e and p are projections, the claim above implies (by taking adjoints) that:

$$v^*(e - p) = 0.$$

This allows us to simplify our expressions for s^*s and ss^* to the following:

$$ss^* = e + (q - p),$$

$$s^*s = e.$$

Well, these show that s is an isometry so it must be unitary. Therefore:

$$e = ss^* = e + (q - p),$$

which implies that $p = q$.

(3) $\Rightarrow$ (1): This is obvious.

Now, we need to do the top row of our diagram above.

(4) $\Rightarrow$ (5): Let $a \in A$ be right-invertible. Then, a^* is left-invertible so it is invertible. Hence, a is invertible.

(5) $\Rightarrow$ (4): Let $a \in A$ be left-invertible. Then, a^* is right-invertible so it is invertible. Hence, a is invertible.

Finally, we must establish a bridge between the two rows.

(2) $\Rightarrow$ (5): Let $x \in A$ be right-invertible. By **Claim 8.7**, we see that xx^* is invertible. That is:

$$\exists y \in A : yxx^* = e = xx^*y.$$

Well, this demonstrates easily that x is right-invertible so we're done.

(5) $\Rightarrow$ (2): Let $s \in A$ be an isometry. Then, $s^*s = e$. So, s^* is right invertible and, therefore, invertible. But then it's inverse must just be s . This implies that s is unitary.

This completes the proof. ■

Here's a nice exercise related to this proposition.

Claim 8.8. Give a direct proof of (2) $\Rightarrow$ (1) above.

We do have one (loooong) remark about the non-unital situation.

Remark 8.15. Let $A \in C^*\text{Alg}$. We have the following claim.

Claim. If A is finite, then every projection in A is finite.

If A is unital, then this follows from the proposition. If A is not unital, then A being finite means that A_e is finite. Now, let $p \in A$ be a projection and let q be any subprojection of p such that $p \sim_{\text{MvN}} q$. Then, it is true that $(p, 0) \sim_{\text{MvN}} (q, 0)$ and that $(q, 0) \preceq (p, 0)$. Well, $(p, 0)$ is a projection in A_e and these must be finite, by the proposition above. So, it follows that $(q, 0) = (p, 0)$ and this means that $p = q$.

There are, however, infinite non-unital C^* -algebras for which all projections are finite. Let us provide an explicit example in this regard [GO OVER THE COMPUTATION]

Claim.

Moreover, there are finite C^* -algebras which are not stably finite. Let's also provide an example in this regard.

The next step is to give examples of finite C^* -algebras. For the moment, we will focus our attention on two examples.

Example 8.16. Let $V \in C^*\text{Alg}$ be finite-dimensional. Then, we know that V is unital & that every left-invertible element in V is invertible, so the proposition implies that V is finite. Of course, this "makes sense" from the colloquial viewpoint. In fact, we have the following easy claim.

Claim. V is stably finite.

For each $n \in \mathbb{N}$, we know that $M_{n \times n}(V)$ is a finite-dimensional C^* -algebra so it must be finite.

Again, if finite-dimensional C^* -algebras weren't finite in this sense, then we would sort of be screwed.

Example 8.17. Let $V \in C^*\text{Alg}$ be an AF-algebra. We have the following claim.

Claim. V is stably finite.

Let $n \in \mathbb{N}$ and note that $M_{n \times n}(V)$ can be embedded within $M_{n \times n}(V_e)$, where the latter is also an AF algebra. In other words, it's enough to prove that every unital AF algebra is stably finite, so let V be unital as well. For the proof, we will be using the local property of AF algebras.

Now, we have everything we need to equip $K_0(A)$ with the structure(ish) of an ordered abelian group. Actually, we won't quite be able to do it for **all** C^* -algebras but we'll be able to do it for a reasonably large class of them.

Proof Of Proposition 8.1. We'll prove each statement one-by-one, of course.

1. We have that:

$$\forall p, q \in \text{Idem}_{\text{s.a.}\infty}(A) : [p]_{K_0} + [q]_{K_0} = [p \oplus q]_{K_0},$$

by statement (1) of

2. This follows from the standard portrait of K_0 for unital C^* -algebras^a, every $x \in K_0(A)$ can be written as a difference $[p]_{K_0} - [q]_{K_0}$ with $p, q \in \text{Idem}_{\text{s.a.}\infty}(A)$. But $[p]_{K_0}, [q]_{K_0} \in K_0(A)^+$ so we're done.

3. Let $x \in K_0(A)^+ \cap (-K_0(A)^+)$. Then, we must have that:

$$\exists p, q \in \text{Idem}_{s.a.\infty}(A) : [p]_{K_0} = -[q]_{K_0}.$$

This implies that $[p \oplus q]_{K_0} = 0$ in $K_0(A_e)$. This means that $p \oplus q$ is stable equivalent^b to 0 so there is an $r \in \text{Idem}_{s.a.\infty}(A_e)$ such that $p \oplus q \oplus r \sim_{K_0} r$.

Finally, if A is unital and stably finite, then $K_0(A)^+$ satisfies all required conditions of a positive cone in an abelian group so the given pair must be an ordered abelian group by ??.

^aSee ??.

^bThis follows from the standard portrait of K_0 .

The last result we want to prove has to do with the “continuity” of K_0 . This will be absolutely essential in the proof of Elliott’s Classification Theorem but it is also of independent interest on its own as a property of the K_0 -functor. To establish that it is continuous in this sense, we will need a technical lemma.

Lemma 8.18 Let $A \in C^* \text{Alg}$.

1. If $a \in A$ is self-adjoint and $\delta = \|a - a^2\| < \frac{1}{4}$, then:

$$\exists p \in \mathcal{P}(A) : \|a - p\| \leq 2\delta.$$

That is, self-adjoint elements are “closely” approximated by projections.

2. Let $p, q \in \mathcal{P}(A)$ and let $x \in A$ such that:

$$\begin{aligned} \|x^*x - p\| &< \frac{1}{2}, \\ \|x^*x - q\| &< \frac{1}{2}. \end{aligned}$$

Then, $p \sim_{\text{MvN}} q$.

Proof. See [86, Lemma 6.3.1, p.98].

Now, we’re ready for the continuity of K_0 . Again, this is a property of K_0 which is very easy to understand but its proof is just extremely annoying because there are very many details to check.

Theorem 8.19. Continuity of K_0

Suppose we have an inductive sequence:

$$A_1 \xrightarrow{\phi_1} A_2 \xrightarrow{\phi_2} A_3 \xrightarrow{\phi_3} \dots$$

in $C^* \text{Alg}$ with inductive limit $(A, (\mu_n : A_n \rightarrow A)_{n \in \mathbb{N}})$. This induces an inductive sequence:

$$K_0(A_1) \xrightarrow{K_0(\phi_1)} K_0(A_2) \xrightarrow{K_0(\phi_2)} K_0(A_3) \xrightarrow{K_0(\phi_3)} \dots$$

in AbGrp with inductive limit $(G, (\beta_n : K_0(A_n) \rightarrow G)_{n \in \mathbb{N}})$. The following statements hold:

1. $K_0(A) \equiv_{\text{AbGrp}} G$.
2. If $(K_0(A_n), K_0(A_n)^+) \in \text{OrdAbGrp}$ for each $n \in \mathbb{N}$, then $K_0(A) \equiv_{\text{OrdAbGrp}} G$.
3. $K_0(A) = \bigcup_{n \in \mathbb{N}} K_0(\mu_n)(K_0(A_n))$.
4. $K_0(A)^+ = \bigcup_{n \in \mathbb{N}} K_0(\mu_n)(K_0(A_n)^+)$.
5. $\forall n \in \mathbb{N} : \ker(K_0(\mu_n)) = \bigcup_{m=n+1}^{\infty} \ker(K_0(\phi_{m,n}))$.

Informally, we write that $K_0(\text{colim}(A_n)) = \text{colim}(K_0(A_n))$.

Proof. See [86, Theorem 6.3.2, p.98-101]

8.2 The Supernaturality of Elliott’s Theorem

Notation 8.20. Let $A \in {}^1C^* \text{Alg}$ and let $u \in A$ be unitary. We define:

$$\text{Ad}(u) : A \rightarrow A, x \mapsto \text{Ad}(u)(x) \stackrel{\text{def}}{=} uxu^*.$$

We have the following claim.

Claim. $\text{Ad}(u)$ is a $\star$ -isomorphism & $K_0(\text{Ad}(u)) = \text{Id}_{K_0(A)}$.

It remains for us to show that $K_0(\text{Ad}(u)) = \text{Id}_{K_0(A)}$. Indeed, we have the following calculation:

$$\begin{aligned} \forall p \in \mathcal{P}_{\infty}(A) : K_0(\text{Ad}(u))([p]_{K_0}) &= [\text{Ad}(u)(p)]_{K_0} \\ &= [upu^*]_{K_0} \\ &= [p]_{K_0}, \end{aligned}$$

where we’ve used the fact that unitary equivalence implies Murray-von Neumann equivalence so we’re done. It is easy to show that this is a $\star$ -isomorphism and we leave that as a simple exercise.

The first lemma we’ll prove tells us something about the properties of $\star$ -homomorphisms from a finite-dimensional C^* -algebra to a unital C^* -algebra having the cancellation property. Obviously, our interest mainly lies in the AF algebras themselves but it’s easier to just prove the general statement. Again, we emphasize that the lemmas we’ll establish now

Lemma 8.21 Let $A, B \in C^*\text{Alg}$ with A being finite-dimensional & B being unital and cancellative.

1. If $\alpha : K_0(A) \rightarrow K_0(B)$ is a positive group homomorphism such that:

$$\alpha([1_A]_{K_0}) \leq [1_B]_{K_0},$$

then there is a $\star$ -homomorphism $\phi : A \rightarrow B$ such that $K_0(\phi) = \alpha$.

2. If $\alpha([1_A]_{K_0}) = [1_B]_{K_0}$ in (1), then ϕ is unital.
3. Let $\phi, \psi : A \rightarrow B$ be $\star$ -homomorphisms. TFAE:
 - (a) $K_0(\phi) = K_0(\psi)$.
 - (b) There is a unitary $u \in B$ such that $\psi = \text{Ad}(u) \circ \phi$.

For convenience, one says that ψ and ϕ are “unitarily equivalent”.

Proof. We’ll prove each statement one-by-one.

- 1.
- 2.
3. Let’s prove each direction separately.
 - (a) $\Rightarrow$ (b):
 - (b) $\Rightarrow$ (a): We have the following calculation:

$$\begin{aligned} K_0(\psi) &= K_0(\text{Ad}(u) \circ \phi) \\ &= K_0(\text{Ad}(u)) \circ K_0(\phi) \\ &= K_0(\phi), \end{aligned}$$

as was desired.

This completes the proof of this equivalence. ■

You see what we mean? The proof doesn’t really teach you anything about the lemma that you couldn’t have gleaned just from reading its statement. So, while we’ve tried to make the proof very readable and clear, **don’t** worry about working through it in detail for the moment. The next lemma is basically the same in this regard.

Lemma 8.22 Consider the following sequence:

$$A_1 \xrightarrow{\phi_1} A_2 \xrightarrow{\phi_2} A_3 \xrightarrow{\phi_3} \dots$$

of finite-dimensional C^* -algebras with inductive limit $(A, (\mu_m)_{m \in \mathbb{N}})$, let $B \in C^*\text{Alg}$ be finite-dimensional and suppose that there are positive group homomorphisms $\alpha : K_0(A_1) \rightarrow K_0(B)$, $\gamma : K_0(B) \rightarrow K_0(A)$ such that $\gamma \circ \alpha = K_0(\mu_1)$. Then, there is an $n \in \mathbb{N}$ and there is a positive group homomorphism $\beta : K_0(B) \rightarrow K_0(A_n)$ such that the diagram:

$$\begin{array}{ccccc} K_0(A_1) & \xrightarrow{K_0(\phi_{n,1})} & K_0(A_n) & \xrightarrow{K_0(\mu_n)} & K_0(A) \\ & \searrow \alpha & \uparrow \beta & \nearrow \gamma & \\ & & K_0(B) & & \end{array}$$

commutes. If each ϕ_k is unital and α is unital, then β is also unital.

Proof. ■

We have two remarks about the lemma above.

Remark 8.23. Again, you can clearly see that the lemma itself is entirely understandable without the proof. The funny thing is that the lemma above will just be used once in the proof of the classification theorem & that will be considered a “technical” step that will end as quickly as it begins. So, don’t worry too much about working through the details of the proof right now. Nevertheless, perhaps the lemma itself does deserve a bit more explanation.

Remark 8.24. Well, we already drew a diagram in the statement of [Lemma 8.22](#) but it might be useful to draw another one that explains the set-up. Okay, so our inductive sequence induces the following diagram in K -theory:

$$\begin{array}{ccccccc} K_0(A_1) & \longrightarrow & K_0(A_2) & \longrightarrow & \dots & \longrightarrow & K_0(A_n) & \longrightarrow & \dots \\ & \searrow \alpha & & & & & \searrow K_0(\mu_n) & & \\ & & K_0(B) & \xrightarrow{\gamma} & & & K_0(A) & & \end{array}$$

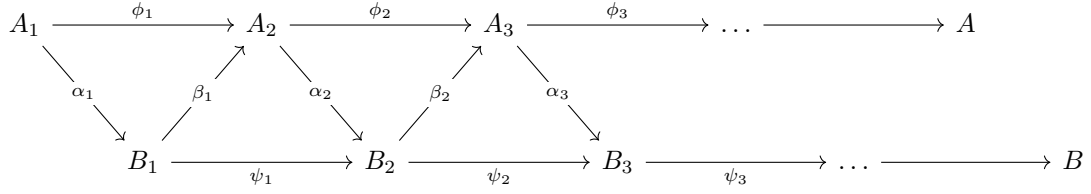
Now, what you have to imagine is that in the second row above, we don’t just have ONE finite-dimensional C^* -algebra but an entire sequence of them. So, the idea here is that the lemma allows us to “intertwine” both sequences at the level of K -theory. Now, that’s not really what’s happening; we’re really just intertwining parts of both sequences but this doesn’t affect anything due to previously known results that we’ll explicitly reference later.

Once we’ve intertwined both sequences at the level of K -theory, the hope is that one can “sink” that intertwining to the level of C^* -algebras. Indeed, such an intertwining will give us a $\star$ -isomorphism between the inductive limits and

the hope is that this isomorphism “gives us” the group isomorphism via K_0 . Indeed, this is **exactly** the strategy for the proof of the classification theorem and we hope that we’ve successfully explained how the second lemma above will be used. Of course, we will provide an even more detailed explanation in the proof itself.

Finally, there is a useful intertwining result about inductive sequences which will, really, form a highlight of Elliott’s classification argument for AF algebras. You can simply just view this as an interesting property of inductive sequences in the category of C^* -algebras. It says that if two inductive sequences of C^* -algebras are appropriately “intertwined”, then their inductive limits are isomorphic.

Claim 8.9. Consider the following commutative diagram:



in $C^*\text{Alg}$, with $(A, (\mu_n)_{n \in \mathbb{N}})$ being the inductive limit of the sequence in the first row and $(B, (\lambda_n)_{n \in \mathbb{N}})$ being the inductive limit of the sequence in the second row. Then, there exist $\star$ -isomorphisms $\alpha : A \rightarrow B$ and $\beta : B \rightarrow A$ such that the diagram above commutes with those arrows included. Moreover, $\alpha = \beta^{-1}$.

We have just one remark regarding this exercise.

Remark 8.25. So, Elliott’s Classification Theorem^a makes use of an “intertwining” argument, where we constructs an intertwining diagram at the level of K -theory & reduces it to an intertwining diagram of the form given above. In this way, the argument allows us to construct a particular $\star$ -isomorphism between two inductive limits. Even if it isn’t historically true that the above result is where (George) Elliott may have gotten inspired to produce this kind of proof, I like to think that this is a main motivation for the argument. Certainly, you’d agree that it makes for a nice narrative & that’s what I’ll push when we actually get to that lecture.

^aThis is [Theorem 8.26](#).

With all of that, we are ready to give the proof of the classification theorem.

Theorem 8.26. Elliott’s Classification Theorem (Unital AF Algebras)

Let A, B be unital AF algebras. TFAE:

1. A and B are isomorphic in ${}^1C^*\text{Alg}$.
2. The triples $(K_0(A), K_0(A)^+, [1_A]_{K_0})$ and $(K_0(B), K_0(B)^+, [1_B]_{K_0})$ are isomorphic in OrdAbGrp_u .

In particular, for any such unital positive group isomorphism α , there is a $\star$ -isomorphism ϕ such that $K_0(\phi) = \alpha$.

Proof.

(1) $\Rightarrow$ (2): This follows by functoriality.

(2) $\Rightarrow$ (1): We’ll begin with two inductive sequences:

$$A_1 \xrightarrow{f_1} A_2 \xrightarrow{f_2} A_3 \xrightarrow{f_3} \dots \longrightarrow A$$

$$B_1 \xrightarrow{g_1} B_2 \xrightarrow{g_2} B_3 \xrightarrow{g_3} \dots \longrightarrow B$$

with inductive limits $(A, (\mu_n)_{n \in \mathbb{N}})$ and $(B, (\lambda_n)_{n \in \mathbb{N}})$ respectively. Moreover, we also have a unital positive group isomorphism α between the two stated triples. In light of ??, we may assume that the connecting morphisms in each inductive sequence are unital. Both of the inductive sequences induce inductive sequences at the level of K -theory: the first goal is to obtain an “intertwining” of these induced diagrams. This will be done inductively, though we’ll do the first step in detail. Define:

$$B_0 \stackrel{\text{def}}{=} \mathbb{C}.$$

Now, there exist unique unital $\star$ -homomorphisms $g_0 : B_0 \rightarrow B_1$ and $\psi_0 : B_0 \rightarrow A_1$, so we define:

$$\beta_0 \stackrel{\text{def}}{=} K_0(\psi_0).$$

With all of this data, we have the following diagram in K -theory:

$$\begin{array}{ccccccc}
 & & K_0(A_1) & \xrightarrow{K_0(f_1)} & K_0(A_2) & \xrightarrow{K_0(f_2)} & K_0(A_3) & \xrightarrow{K_0(f_3)} & \dots \\
 & & & & & & & & \searrow \\
 & & & & & & & & K_0(A) \\
 & & & & & & & & \downarrow \alpha \\
 & & & & & & & & K_0(B) \\
 & \nearrow \beta_0 & & & \nearrow K_0(\mu_1) & & \nearrow \alpha \circ K_0(\mu_1) & & \\
 K_0(B_0) & \xrightarrow{K_0(g_0)} & K_0(B_1) & \xrightarrow{K_0(g_1)} & K_0(B_2) & \xrightarrow{K_0(g_2)} & K_0(B_3) & \xrightarrow{K_0(g_3)} & \dots
 \end{array}$$

What we want is to build a group homomorphism from $K_0(A_1)$ to some $K_0(B_{m_1})$ and the key to doing this is via [Lemma 8.22](#). However, applying the lemma requires that the following hold:

$$\alpha \circ K_0(\mu_1) \circ \beta_0 = K_0(\lambda_0),$$

where $\lambda_0 : B_0 \rightarrow B$ is a $\star$ -homomorphism which extends the second inductive sequence of C^* -algebras above. Note that both the group homomorphisms above are defined on $\mathbb{Z}$, so we just need to check that they coincide on 1. But this is obviously true^a, so we can apply the aforementioned lemma and find some $m_1 \in \mathbb{N}$ along with a group homomorphism $\alpha_1 : K_0(A_1) \rightarrow K_0(B_{m_1})$ such that the following diagram:

$$\begin{array}{ccc}
 K_0(A_1) & \xrightarrow{\alpha \circ K_0(\mu_1)} & K_0(B) \\
 \beta_0 \nearrow & & \nwarrow \alpha_1 \\
 K_0(B_0) & \xrightarrow{K_0(g_{m_1,0})} & K_0(B_{m_1}) \\
 & & \nearrow K_0(\lambda_{m_1})
 \end{array}$$

commutes. Now, the idea is to “ping-pong” back to the sequence $(K_0(A_n))_{n \in \mathbb{N}}$ by using the exact same construction as above, except that we use α^{-1} instead of α . This produces an intertwining diagram:

$$\begin{array}{ccccccc}
 K_0(A_1) & \xrightarrow{\quad} & K_0(A_{n_2}) & \xrightarrow{\quad} & \dots & \xrightarrow{\quad} & K_0(A) \\
 \beta_0 \nearrow & & \nwarrow \alpha_1 & & \nearrow \beta_1 & & \nwarrow \alpha_2 & & \nearrow \beta_2 & & \nwarrow \alpha^{-1} \\
 K_0(B_0) & \xrightarrow{\quad} & K_0(B_{m_1}) & \xrightarrow{\quad} & K_0(B_{m_2}) & \xrightarrow{\quad} & \dots & \xrightarrow{\quad} & K_0(B)
 \end{array}$$

which commutes **by construction**. This gives us two inductive sequences:

$$A_{n_1} \xrightarrow{f_{n_2,n_1}} A_{n_2} \xrightarrow{f_{n_3,n_2}} A_{n_3} \xrightarrow{f_{n_4,n_3}} \dots \longrightarrow A$$

$$B_{m_1} \xrightarrow{g_{m_2,m_3}} B_{m_2} \xrightarrow{g_{m_3,m_2}} B_{m_3} \xrightarrow{g_{m_4,m_3}} \dots \longrightarrow B$$

where $n_1 = 1$, of course. Both of these sequences still have A and B respectively as their inductive limit C^* -algebras (with the “same” maps from each C^* -algebra to the inductive limit)^b. Now, we have our intertwining diagram at the level of K -theory and we want to bring it down to the level of C^* -algebras. That’s where [Lemma 8.21](#) comes into the picture. For each $j \in \mathbb{N}$, we use statement (1) & (2) of the aforementioned lemma to find unital $\star$ -homomorphisms $\phi'_j : A_{n_j} \rightarrow B_{n_j}$ and $\psi'_j : B_{m_j} \rightarrow A_{n_{j+1}}$ such that:

$$\begin{aligned}
 K_0(f_{n_{j+1},n_j}) &= \beta_j \circ \alpha_j = K_0(\psi'_j \circ \phi'_j), \\
 K_0(g_{n_{j+1},n_j}) &= \alpha_{j+1} \circ \beta_j = K_0(\phi'_{j+1} \circ \psi'_j),
 \end{aligned}$$

where we’ve used the functoriality of K_0 and the fact that the intertwining diagram above commutes. Of course, in the second case, we set $\psi'_0 \stackrel{\text{def}}{=} \psi_0$ so that the second equation above makes sense even when $j = 0$. Now, we apply statement (3) of [Lemma 8.21](#) to construct two sequences of unitaries as follows:

(a) Let $j = 1$. Then, we know that:

$$\begin{aligned}
 K_0(f_{n_2,n_1}) &= K_0(\psi'_1 \circ \phi'_1), \\
 K_0(g_{n_2,n_1}) &= K_0(\phi'_2 \circ \psi'_1).
 \end{aligned}$$

By the aforementioned lemma, we find a unitary $u_1 \in A_{n_2}$ such that:

$$f_{n_2,n_1} = \text{Ad}(u_1) \circ \psi'_1 \circ \phi'_1.$$

We further define:

$$\begin{aligned}
 v_1 &\stackrel{\text{def}}{=} 1_B, \\
 \phi_1 &\stackrel{\text{def}}{=} \text{Ad}(1_B) \circ \phi'_1, \\
 \psi_1 &\stackrel{\text{def}}{=} \text{Ad}(u_1) \circ \psi'_1.
 \end{aligned}$$

We also get a unitary $v_2 \in B_2$ such that:

$$g_{n_2, n_1} = \text{Ad}(v_2) \circ \phi'_2 \circ \text{Ad}(v_1) \circ \psi'_1.$$

Now, we set $\phi_2 \stackrel{\text{def}}{=} \text{Ad}(v_2) \circ \phi'_2$. With all of these definitions, we arrive at the following equations:

$$f_{n_2, n_1} = \psi_1 \circ \phi_1,$$

$$g_{n_2, n_1} = \phi_2 \circ \psi_1.$$

which is the end result of the first step.

(b) Suppose we have two finite sequences $(u_j)_{j=1}^n$ and $(v_j)_{j=1}^n$ of unitaries such that:

$$f_j = \psi_j \circ \phi_j,$$

$$g_j = \phi_{j+1} \circ \psi_j,$$

where $\phi_j = \text{Ad}(v_j) \circ \phi'_j$ and $\psi_j = \text{Ad}(u_j) \circ \psi'_j$. By proceeding **exactly** as we did in the first step, we can extend both sequences while still maintaining the relations above in the case where $j = n + 1$.

Now, this construction gives us a commutative intertwining diagram at the level of C^* -algebras:

$$\begin{array}{ccccccc} & & A_1 & \xrightarrow{f_1} & A_2 & \xrightarrow{f_2} & A_3 \xrightarrow{f_3} \dots \\ & \nearrow \psi_0 & \searrow \phi_1 & & \nearrow \psi_1 & \searrow \phi_2 & \nearrow \psi_2 \\ B_0 & \xrightarrow{g_0} & B_1 & \xrightarrow{g_1} & B_2 & \xrightarrow{g_2} & \dots \end{array}$$

By **Claim 8.9**, we get a $\star$ -isomorphism $\phi : A \rightarrow B$. Let's show that:

$$K_0(\phi) = \alpha.$$

To prove this, we'll show that $K_0(\phi)$ and α agree on the range of $K_0(\mu_j)$ for each $j \in \mathbb{N}$ and then use the continuity of K_0 to conclude that they must be equal since $K_0(A)$ is covered by $(\text{ran}(K_0(\mu_j)))_{j \in \mathbb{N}}$. Indeed, this is just statement (3) of **Theorem 8.19**. $\blacksquare$

^aThis is where we use the assumption that α is a unital positive group isomorphism.

^bSee ??.

So, we have finally proved this result in its entirety and it is a huge milestone because it represents the start of the classification programme for C^* -algebras. Let us say some words about that in the next few remarks.

Remark 8.27. One can take the classification result above even further and classify UHF-algebras based on the so-called supernatural number. Let us state the result here as a claim just to illustrate the point.

Claim. Let A, B be UHF-algebras with associated supernatural numbers m and n respectively. TFAE:

1. A and B are isomorphic.
2. $m = n$.
3. $(K_0(A), [1]_{K_0})$ and $(K_0(B), [1]_{K_0})$ are isomorphic.
4. $(K_0(A), K_0(A)^+, [1]_{K_0})$ and $(K_0(B), K_0(B)^+, [1]_{K_0})$ are isomorphic.

In fact, the main result says quite a bit more and we refer the reader to [86, Theorem 7.4.5, p.128-129] for the proof of this result (and a bit more), as well as an elementary (but not so conceptually neat) development of the theory of UHF algebras. The reason why this classification result is very notable is due to the fact that it was originally proved via technical and analytic methods by Glimm in^a [46]. The fact that a short proof is available via Elliott's Classification Theorem which, itself, has a relatively short proof is pretty insane.

^aA nice little fun fact is that this is the first paper in which I've seen something close to an $\frac{\epsilon}{75}$ -argument, a fact that I found hilarious at the time because of its sheer absurdity.

Remark 8.28.

In the next lecture, we complete our discussion of K_0 by extending it to arbitrary C^* -algebras. This extension is not the same as K_0 on Rng but the method of extension is essentially the same. Once we've extended K_0 to arbitrary C^* -algebras, we'll indicate how one can modify Elliott's Classification Theorem to arbitrary AF algebras.

8.3 Appendix: The Structure of Dimension Groups

We know that:

$$\forall n \in \mathbb{N} : K_0(M_{n \times n}(\mathbb{C})) \simeq K_0(\mathbb{C}) \simeq \mathbb{Z}.$$

In particular, we've shown before that for every finite-dimensional $A \in C^*\text{Alg}$, we have that:

$$\exists k \in \mathbb{N}_0 : K_0(A) \simeq \mathbb{Z}^k.$$

Okay, so consider an AF algebra A and the following inductive sequence of which it is the inductive limit:

$$A_1 \xrightarrow{\phi_1} A_2 \xrightarrow{\phi_2} A_3 \xrightarrow{\phi_3} \dots$$

At the level of K -theory, we obtain the following inductive sequence:

$$K_0(A_1) \xrightarrow{K_0(\phi_1)} K_0(A_2) \xrightarrow{K_0(\phi_2)} K_0(A_3) \xrightarrow{K_0(\phi_3)} \dots$$

and since K_0 is inductively continuous, the inductive limit of this sequence is $K_0(A)$. So, this is just stuff that we've seen before and we're just repeating it step-by-step because we want to take it further. So, in particular, $K_0(A)$ is an ordered

abelian group which is the inductive limit of the following sequence:

$$\mathbb{Z}^{n_1} \xrightarrow{\alpha_1} \mathbb{Z}^{n_2} \xrightarrow{\alpha_2} \mathbb{Z}^{n_3} \xrightarrow{\alpha_3} \dots$$

Of course, one can consider inductive sequences of the type given above independently and study the ordered abelian groups that arise as their inductive limits. These are the so-called dimension groups and we've just shown that every unital AF algebra gives rise to a dimension group. This prompts some immediate questions:

1. Is there an intrinsic characterization of dimension groups?
2. Does every dimension group arise as the K_0 -group of some AF algebra?

We shall spend some time on these questions and, hopefully, address more in the process. This leads us straight to the definition of a dimension group.

Definition 8.29

Let $G \in \text{OrdAbGrp}$. It is a dimension group iff it is (isomorphic to) the inductive limit of the inductive sequence:

$$\mathbb{Z}^{n_1} \xrightarrow{\alpha_1} \mathbb{Z}^{n_2} \xrightarrow{\alpha_2} \mathbb{Z}^{n_3} \xrightarrow{\alpha_3} \dots$$

of ordered abelian groups, where each $n_k \in \mathbb{N}$.

We have a remark about this definition.

Remark 8.30. With this definition, we can specify a full subcategory of OrdAbGrp which we'll denote by DimGrp ; this is the category consisting of the dimension groups as objects. So, the K_0 functor takes the category of AF-algebras into the category of dimension groups^a. With this observation, one can easily see that the second question we asked in our prior motivation amounts to asking if this functor is essentially surjective.

^aWe've already argued for this in our motivation above but it's a useful exercise for you to reference the specific results which make that argument work properly at each step.

The question raised in the preceding remark has a positive answer.

Proposition 8.31

The K_0 -functor, when viewed as a functor from the category of AF algebras to DimGrp , is essentially surjective.

Proof. Let (G, G^+) be a dimension group. So, it is the inductive limit of some inductive sequence:

$$\mathbb{Z}^{n_1} \xrightarrow{\alpha_1} \mathbb{Z}^{n_2} \xrightarrow{\alpha_2} \mathbb{Z}^{n_3} \xrightarrow{\alpha_3} \dots$$

For each $j \in \mathbb{N}$, choose order units:

$$u_j \stackrel{\text{def}}{=} (k_1^{(j)}, k_2^{(j)}, \dots, k_{n_j}^{(j)}) \in (\mathbb{Z}^{n_j})^+,$$

such that $\alpha_j(u_j) \leq u_{j+1}$. Since each u_j is an order unit, we see that:

$$\forall j \in \mathbb{N} : \forall i \in \{1, \dots, n_j\} : k_i^{(j)} \geq 1.$$

So, now we define:

$$\forall j \in \mathbb{N} : A_j \stackrel{\text{def}}{=} \bigoplus_{l=1}^{n_j} M_{n_l^{(j)} \times n_l^{(j)}}(\mathbb{C}),$$

and let $\gamma_j : \mathbb{Z}^{n_j} \rightarrow K_0(A_j)$ be the canonical order isomorphism satisfying $\gamma_j(u_j) = [1_{A_j}]_{K_0}$. ■

Conceptually, Definition 8.38 is a very nice definition but one would like to find examples of dimension groups and the definition above does not seem particularly amenable to this desire. So, our next goal will be to obtain an axiomatic characterization of dimension groups which will allow us to construct examples & non-examples. Now, we may just restrict ourselves to dealing with countable groups, because dimension groups are necessarily countable. The trick is to consider two additional properties that certain “nice” ordered abelian groups satisfy.

Definition 8.32

Let $G \in \text{OrdAbGrp}$.

1. G is unperforated iff for every $x \in G$, if there is an $n \in \mathbb{N}$ such that $nx \geq 0$, then $x \geq 0$.
2. G satisfies the Riesz interpolation property iff for every $x_1, x_2, y_1, y_2 \in G$ where $x_i \leq y_j$ for $i, j \in \{1, 2\}$, there exists a $z \in G$ such that $x_i \leq z \leq y_j$ for $i, j \in \{1, 2\}$.

If G is unperforated, then it is called an unperforated OAG. If it satisfies the Riesz interpolation property, then it is called a Riesz interpolated OAG. Finally, it is a Riesz group iff it satisfies both properties.

Okay, so we have a few remarks about this definition.

Remark 8.33. So, the word “perforated” means “pierced with a hole or holes”. Therefore, “unperforated” fits nicely in the definition above. We're just saying that if we positively scale an element of G and it is positive, then the un-scaled element is also positive. So, there are no “holes” in the positive elements of G .

Remark 8.34. The name for condition (2) comes from the Riesz-Thorin Interpolation Theorem, a very nice result of analysis that we haven't covered in our texts but you have the background to learn about it, in any event. The idea here is

Remark 8.35. We will show that a countable ordered abelian group is a Riesz group iff it is a dimension group. Actually, it's been expressed in several papers^a that one can remove the assumption of countability by considering arbitrary filtered colimits instead. While we believe that this might be possible, it's notable that we haven't actually seen this be done explicitly in any resource we've encountered.

^aFor instance, in [36] but also in the original paper, [90], by Shen on these groups.

Theorem 8.36. The Effros-Handelman-Shen Theorem

Let $G \in \text{OrdAbGrp}$ be countable. TFAE:

1. G is a dimension group.
2. G is a Riesz group.

Proof.

(1) $\Rightarrow$ (2):

(2) $\Rightarrow$ (1):

■

We have a remark regarding this theorem.

Remark 8.37. We won't provide a proof of the theorem here since it would require going through some theory that's a bit irrelevant for us. Nevertheless, it is an interesting bit of research in group theory with reasonably nice connections back to C^* -algebra theory; we'll discuss these later in this section. For the proof of this theorem, we recommend working through [36] since it is short (and quite sweet).

This leads us straight to the definition of a dimension group.

Definition 8.38

Let $G \in \text{OrdAbGrp}$. It is a dimension group iff it is (isomorphic to) the inductive limit of the inductive sequence:

$$\mathbb{Z}^{n_1} \xrightarrow{\alpha_1} \mathbb{Z}^{n_2} \xrightarrow{\alpha_2} \mathbb{Z}^{n_3} \xrightarrow{\alpha_3} \dots$$

of ordered abelian groups, where each $n_k \in \mathbb{N}$.

We have a remark about this definition.

Remark 8.39. With this definition, we can specify a full subcategory of OrdAbGrp which we'll denote by DimGrp ; this is the category consisting of the dimension groups as objects. So, the K_0 functor takes the category of AF-algebras into the category of dimension groups^a. With this observation, one can easily see that the second question we asked in our prior motivation amounts to asking if this functor is essentially surjective.

^aWe've already argued for this in our motivation above but it's a useful exercise for you to reference the specific results which make that argument work properly at each step.

The question raised in the preceding remark has a positive answer.

Proposition 8.40

The K_0 -functor, when viewed as a functor from the category of AF algebras to DimGrp , is essentially surjective.

Proof. See [86, Proposition 7.2.8, p.117-118]

■

A similar result isn't true for K_1 ; here's a counterexample.

Example 8.41.

9 Axiomatic (Co)Homology I: The Bott Periodicity Theorem

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

Motivation.

9.1 Basic Properties of C^* -Homology Theories

At this stage, we are ready to define what it means for a given full subcategory of C^*Alg to be admissible.

Definition 9.1. A. Vats, 2026

Let $\mathcal{C}$ be a full subcategory of C^*Alg . We say that $\mathcal{C}$ is admissible iff:

1. $\mathcal{C}$ is closed under the cylinder.
2. $\mathcal{C}$ is closed under tensoring by $M_{n \times n}(\mathbb{C})$ for each $n \in \mathbb{N}$.
3. $\mathcal{C}$ is closed under sequential inductive limits.
4. $\mathcal{C}$ is closed under ideals, quotients and extensions.

Moreover, we require that $\mathcal{C}$ also contains 0 and $\mathbb{C}$.

In fact, we can show that any admissible subcategory of C^*Alg will be closed under a selection of constructions. Before we prove this, let us introduce another construction which will be relevant for later work in this subsection.

Construction 9.2. Let $A, B \in C^*Alg$ and let $f \in C^*Alg(A, B)$. We define:

$$\text{cone}(f),$$

to be a C^* -algebra that fits into the following pullback:

$$\begin{array}{ccc} \text{cone}(f) & \longrightarrow & \text{Cone}(B) \\ \downarrow & & \downarrow p_0^B \\ A & \xrightarrow{f} & B \end{array}$$

We refer to this as the mapping cone of f .

One of the main reasons why admissible categories are crucial for us is because we can deduce that they contain many interesting C^* -algebras. In particular, they should be closed under “homotopical constructions” and should contain sufficiently many C^* -algebras such that the classical results of axiomatic (co)homology as deduced by Schochet on C^*Alg actually apply to those subcategories as well. The next proposition partially handles this for us.

Proposition 9.3. A. Vats, 2026

Let $\mathcal{A}$ be an admissible subcategory of C^*Alg .

1. $\mathcal{A}$ is closed under pullbacks.
2. $\mathcal{A}$ is closed under the cone and suspensions.
3. $\mathcal{A}$ is closed under the mapping cone and mapping cylinder.
4. $\mathcal{A}$ is closed under the minimal unitization.
5. If $A, B \in C^*Alg$ with $A \in \mathcal{A}$ and if $f \in C^*Alg(A, B)$, then $\text{ran}(f), \ker(f) \in \mathcal{A}$.
6. $\mathcal{C}(S^1) \in \mathcal{A}$.
7. $\mathcal{K} \in \mathcal{A}$.
8. $\mathcal{T} \in \mathcal{A}$; this is the Toeplitz C^* -algebra.

Moreover, $\mathcal{A}$ is a fibrant subcategory of C^*Alg .

Proof. Let's prove each statement one-by-one.

1. This is [74, Proposition 3.4, p.259]. It is only implied by Axiom 4 of Definition 9.1.
2. Let $A \in \mathcal{A}$. Then, we have the following pullback diagrams:

$$\begin{array}{ccc} \text{Cone}(A) & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ \text{Cyl}(A) & \xrightarrow{p_1^A} & A \end{array} \quad \begin{array}{ccc} \text{Sus}(A) & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ \text{Cyl}(A) & \xrightarrow{(p_0^A, p_1^A)} & A \oplus A \end{array}$$

Since these pullback diagrams live in $\mathcal{A}$, it follows that $\text{Cone}(A)$ and $\text{Sus}(A) \in \mathcal{A}$.

3. These are defined by pullback diagrams whose vertices are in $\mathcal{A}$, so they must lie in $\mathcal{A}$.
4. If $A \in \mathcal{A}$, then we have the following extension:

$$0 \longrightarrow A \xrightarrow{i} A^\sim \xrightarrow{p} \mathbb{C} \longrightarrow 0$$

and since $\mathbb{C} \in \mathcal{A}$, it follows that $A^\sim \in \mathcal{A}$.

5. This follows from the fact that $\ker(f)$ and $\text{ran}(f)$ are an ideal and quotient of A , respectively.

6. We consider the following commutative diagram:

$$\begin{array}{ccc} \mathcal{C}(S^1) & \longrightarrow & \mathcal{C}([0, 1]) \\ \downarrow & & \downarrow \\ \mathbb{C} & \xrightarrow{\Delta_{\mathbb{C}}} & \mathbb{C} \oplus \mathbb{C} \end{array}$$

where $\Delta_{\mathbb{C}}$ is the diagonal map and the vertical map takes a function $f \in \mathcal{C}([0, 1])$ & maps it to $(f(0), f(1))$. Since $\mathbb{C} \in \mathcal{A}$ and since $\mathcal{A}$ is closed under cylinders, we know that $\mathcal{C}([0, 1]) = \text{Cyl}(\mathbb{C}) \in \mathcal{A}$. By closure under pullbacks, we see that $\mathbb{C} \oplus \mathbb{C} \in \mathcal{A}$. This implies, by closure under pullbacks, that $\mathcal{C}(S^1) \in \mathcal{A}$.

7. $\mathcal{K}$ can be written as an inductive limit of full matrix algebras and these belong to $\mathcal{A}$, so $\mathcal{K}$ belongs to $\mathcal{A}$.

8. The Topelitz C^* -algebra fits into the following exact sequence:

$$0 \rightarrow \mathcal{K} \rightarrow \mathcal{T} \rightarrow \mathcal{C}(S^1) \rightarrow 0,$$

and since $\mathcal{K}, \mathcal{C}(S^1)$ lie in $\mathcal{A}$, we see that $\mathcal{T} \in \mathcal{A}$.

Finally, the last statement holds because $\mathcal{A}$ is closed under pullbacks. ■

The next result is a neat observation which is clear.

Claim 9.1. Let $\mathcal{C}, \mathcal{D}$ be admissible subcategories of $C^*\text{Alg}$. Then, $\mathcal{C} \cap \mathcal{D}$ is also admissible.

Of course, our definition also captures the usual examples.

Example 9.4. $C^*\text{Alg}$ is certainly an admissible subcategory of $C^*\text{Alg}$.

Example 9.5. ${}_{\omega}C^*\text{Alg}$ is an admissible subcategory. We'll have to verify each of the required conditions one-by-one.

1. Clearly, 0 and $\mathbb{C}$ lie in ${}_{\omega}C^*\text{Alg}$.
2. ${}_{\omega}C^*\text{Alg}$ is closed under the cylinder because the tensor product of two separable C^* -algebras is separable^a.
3. ${}_{\omega}C^*\text{Alg}$ is closed under tensoring by $M_{n \times n}(\mathbb{C})$ for the same reasons as in point^b (2).
4. Let $A \in C^*\text{Alg}$ be the inductive limit of a sequence of separable C^* -algebras. Then, A is the closure of the directed countable union of a sequence of separable C^* -subalgebras.
5. ${}_{\omega}C^*\text{Alg}$ is clearly closed under ideals, quotients and extensions.

This proves the result.

^aOver here, we don't have to fuss over which tensor product is used since $\mathcal{C}[0, 1]$ is commutative and, therefore, nuclear.

^bAgain, no need to fuss over the tensor product used because $M_{n \times n}(\mathbb{C})$ is nuclear.

Example 9.6. $\text{Nuc}C^*\text{Alg}$ is an admissible subcategory^a of $C^*\text{Alg}$. Let us verify each property one-by-one.

1. 0 and $\mathbb{C}$ are nuclear C^* -algebras.
2. Let $A \in \text{Nuc}C^*\text{Alg}$. Then, its tensor product with any other nuclear C^* -algebra will be nuclear; see [19, Proposition 10.1.7, p.302]. So, we see that $\text{Cyl}(A) \in \text{Nuc}C^*\text{Alg}$.
3. Since $M_{2 \times 2}(A) \simeq M_{n \times n}(\mathbb{C}) \otimes A$ for any $A \in C^*\text{Alg}$, the result follows by the statement of the previous point.
4. Let's show that $\text{Nuc}C^*\text{Alg}$ is closed under ideals, quotients and extensions. This is the most difficult bit because it is a consequence of deep theorems. For closure under ideals, see [19, Exercise 2.3.4, p.36]. For closure under extensions, see [19, Exercise 3.8.1, p.105]. For closure under quotients, see [19, Corollary 9.4.4, p. 297].

This proves the result.

Using **Claim 9.1** and **Example 9.5**, give us that the category of separable nuclear C^* -algebras is also admissible.

^aThis is the full subcategory of $C^*\text{Alg}$ consisting of nuclear C^* -algebras.

In particular, we see that there are three other examples of categories of fibrant objects added to the list of such examples in the literature. At this stage, we are ready to move on to a more explicit set of applications of the ideas introduced above. In particular, we want to demonstrate that it is possible to do axiomatic (co)homology on fibrant subcategories of $C^*\text{Alg}$. This provides an enhancement of some of the basic results found in [89].

Since that paper is quite lengthy, our treatment will not be complete. Instead, we wish to simply convince the reader that this should be possible by showing that a number of basic results follow without trouble & that even the Bott Periodicity Theorem follows without any trouble. We will mainly focus on homology; since we are dealing with functors valued in abelian categories, these will simply encode the same basic results for cohomology theories by duality. Before we describe these C^* -homology theories, let us recall Notation 6.11 and also introduce another piece of notation.

Notation 9.7. Unless stated otherwise, $\mathcal{A}$ will be an abelian category.

Abelian categories will be the codomains of the functors we consider in this subsection. This is mainly a formality so that the results of this subsection also apply to cohomology theories on $C^*\text{Alg}$ by a duality argument. By the Freyd-Mitchell Embedding Theorem, we will always pretend that our functors are valued (at least, locally) in some module category. Now, we shall give the definition of an equivariant C^* -homology theory.

Definition 9.8

Let $(h_n : C^*Alg \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a sequence of functors^a. It is a C^* -homology theory iff:

1. Each h_n is a homotopy-invariant functor.
2. For any exact sequence:

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightarrow{\psi} C \longrightarrow 0$$

in $\text{dom}(h_n)$, there exists a sequence $(\partial : h_n(C) \rightarrow h_{n-1}(A))_{n \in \mathbb{Z}}$ of morphisms and a long-exact sequence:

$$\dots \longrightarrow h_n(A) \xrightarrow{h_n(\phi)} h_n(B) \xrightarrow{h_n(\psi)} h_n(C) \xrightarrow{\partial} h_{n-1}(A) \longrightarrow \dots$$

such that ∂ is natural with respect to morphisms of exact sequences.

The maps ∂ are known as boundary homomorphisms.

^aThe domain of each is the same, so it does not vary with n ! Moreover, we also allow for one-sided C^* -homology theories.

We have a remark about this definition.

Remark 9.9. Let's be more precise about what it means for ∂ to be natural with respect to morphisms of exact sequences. Indeed, suppose we have the following commutative diagram in C^*Alg with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \end{array}$$

then, the resulting squares:

$$\begin{array}{ccc} h_n(C) & \longrightarrow & h_{n-1}(A) \\ \downarrow & & \downarrow \\ h_n(C') & \longrightarrow & h_{n-1}(A') \end{array}$$

commute for every $n \in \mathbb{Z}$.

Sometimes, one assumes that a homology theory is, at least, finitely-additive. Let us introduce this notion somewhat generally. One could generalize this to countable additivity or arbitrary additivity but we will not need that here, so we will not work with that level of generality. We will need the following basic construction.

Construction 9.10. Let $(A_i)_{i \in I}$ be a finite family in an admissible subcategory of C^*Alg . Then, we know that:

$$A = \bigoplus_{i \in I} A_i \in C^*Alg,$$

and that it lies in the given admissible subcategory. Now, we have inclusions:

$$\iota_j : A_j \rightarrow A.$$

If $(h_n : C^*Alg \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ is a sequence of functors^a, then we get maps:

$$h_n(\iota_j) : h_n(A_j) \rightarrow h_n(A).$$

Now, the universal property of the coproduct in $\mathcal{A}$ gives a morphism:

$$\bigoplus_{j \in I} h_n(\iota_j) : \bigoplus_{j \in I} h_n(A_j) \rightarrow h_n(A),$$

for each $n \in \mathbb{Z}$.

^aThe domains are all the same.

This construction leads us to the following bit of terminology.

Terminology 9.11. Let $(h_n : C^*Alg \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a sequence of functors (the domains are all assumed to be the same) and let $(A_i)_{i \in I}$ be a finite family in the common domain of the given sequence of functors. The given sequence is finitely-additive iff the canonical morphism in [Construction 9.10](#) is an isomorphism for each $n \in \mathbb{Z}$.

Here are some basic properties that we can already derive.

Proposition 9.12

Let $(h_n : C^*Alg \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a C^* -homology theory.

1. The boundary homomorphisms obtained from a split exact sequence are zero.
2. The given sequence is finitely-additive.
3. Each h_n preserves finite coproducts.
4. $h_n(0) = 0$.
5. If $n \in \mathbb{Z}$, $A \in C^*Alg$ and $h_n(\mathbb{C}) \simeq 0$, then $h_n(\iota)$ is an isomorphism^a.

^a $\iota : A \rightarrow A^\sim$ is the canonical inclusion.

Proof. We'll prove each statement one-by-one.

1. Consider a split-exact sequence in $C^*\text{Alg}$.

$$0 \longrightarrow A \xrightarrow{\phi} B \begin{array}{c} \xrightarrow{\psi} \\ \xleftarrow{\alpha} \end{array} C \longrightarrow 0$$

This implies that $\psi \circ \alpha = \text{Id}_C$. Now, let $n \in \mathbb{Z}$ be fixed. By functoriality, we conclude that $h_n(\psi)$ is surjective. But this means that its range is just $h_n(C)$. On the other hand, the split-exact sequence above gives us a long-exact sequence with boundary homomorphisms $(\partial_n : h_n(C) \rightarrow h_{n-1}(A))_{n \in \mathbb{Z}}$. Since ∂_n connects to $h_n(\psi)$ and the long-exact sequence is exact at $h_n(C)$, it follows that $\ker(\partial_n) = h_n(C)$. That is, $\partial_n = 0$.

2. Consider a split-exact sequence in $C^*\text{Alg}$.

$$0 \longrightarrow A \xrightarrow{\phi} B \begin{array}{c} \xrightarrow{\psi} \\ \xleftarrow{\alpha} \end{array} C \longrightarrow 0$$

This induces a long-exact sequence:

$$\dots \longrightarrow h_n(A) \xrightarrow{h_n(\phi)} h_n(B) \xrightarrow{h_n(\psi)} h_n(C) \xrightarrow{\partial} h_{n-1}(A) \longrightarrow \dots$$

By (1), we know that $\partial = 0$ and we know that $h_n(\psi)$ is a split epimorphism by functoriality. So, the homology theory takes the split-exact sequence above to a split-exact one.

3. Let $A, B \in C^*\text{Alg}$ and let us be given the following maps:

$$\begin{aligned} \pi_A : A \oplus B &\rightarrow A, (a, b) \mapsto \pi_A(a, b) \stackrel{\text{def}}{=} a, \\ \pi_B : A \oplus B &\rightarrow B, (a, b) \mapsto \pi_B(a, b) \stackrel{\text{def}}{=} b, \\ \iota_A : A &\rightarrow A \oplus B, a \mapsto \iota_A(a) \stackrel{\text{def}}{=} (a, 0), \\ \iota_B : B &\rightarrow A \oplus B, b \mapsto \iota_B(b) \stackrel{\text{def}}{=} (0, b). \end{aligned}$$

With these maps, we have two split-exact sequences:

$$0 \longrightarrow A \xrightarrow{\iota_A} A \oplus B \xrightarrow{\pi_B} B \longrightarrow 0$$

$$0 \longrightarrow B \xrightarrow{\iota_B} A \oplus B \xrightarrow{\pi_A} A \longrightarrow 0$$

Both of these induce split-exact sequences in $\mathcal{A}$ and this implies the result by the splitting lemma.

4. By statement (3), we see that $h_n(0 \oplus 0) \simeq h_n(0) \oplus h_n(0) \simeq h_n(0)$ so we're done.
5. Consider the split-exact sequence:

$$0 \longrightarrow A \xrightarrow{\iota} A^\sim \xrightarrow{p} \mathbb{C} \longrightarrow 0$$

associated to the unitization. For each $n \in \mathbb{Z}$, we obtain a sequence:

$$0 \longrightarrow h_n(A) \xrightarrow{h_n(\iota)} h_n(A^\sim) \xrightarrow{h_n(p)} h_n(\mathbb{C}) \longrightarrow 0$$

which is split-exact. But note that $h_n(\mathbb{C}) \simeq 0$ so we see that $h_n(\iota)$ is injective & surjective ■

It is also reasonable for us to establish that C^* -homology theories are additive on morphisms. For this to make sense, we need to introduce the notion of orthogonality of $\star$ -homomorphisms. This is known and quite standard.

Terminology 9.13. Let $A, A', B \in C^*\text{Alg}$, let $f \in C^*\text{Alg}(A, B)$ and $g \in C^*\text{Alg}(A', B)$. f and g are orthogonal iff:

$$f(A)g(A') = 0.$$

In this case, we write that $f \perp g$.

We just said that we wish to establish that C^* -homology theories are additive on morphisms but the next result, which does do what we said, does not deal with C^* -homology theories. We will see, in Proposition 9.23, that homotopy-invariant and half-exact functors are the “correct” way to form (one-sided) C^* -homology theories.

Proposition 9.14

Let $A, B \in C^*\text{Alg}$ and let $f, g \in C^*\text{Alg}(A, B)$ so that $f \perp g$.

1. The pointwise sum $f + g$ is a $\star$ -homomorphism.
2. If $F : C^*\text{Alg} \rightarrow \mathcal{A}$ is homotopy-invariant and half-exact, then it is also split-exact.
3. If $F : C^*\text{Alg} \rightarrow \mathcal{A}$ is homotopy-invariant and half-exact and $A, B \in \text{dom}(F)$, then:

$$F(f + g) = F(f) + F(g).$$

That is, split-exact functors are additive on orthogonal morphisms.

Proof. We will prove each statement one-by-one.

1. The sum of bounded linear maps is known to be bounded. Next, we observe that:

$$\begin{aligned}\forall a \in A : (f + g)(a^*) &= f(a^*) + g(a^*) \\ &= f(a)^* + g(a)^* \\ &= (f(a) + g(a))^* \\ &= (f + g)(a)^*,\end{aligned}$$

so $f + g$ preserves the involution. Finally, we show that it preserves the product:

$$\begin{aligned}\forall a, b \in A : (f + g)(ab) &= f(ab) + g(ab) \\ &= f(a)f(b) + g(a)g(b) \\ &= f(a)f(b) + f(a)g(b) + g(a)f(b) + g(a)g(b) \\ &= f(a)(f(b) + g(b)) + g(a)(f(b) + g(b)) \\ &= (f(a) + g(a))(f(b) + g(b)) \\ &= (f + g)(a)(f + g)(b),\end{aligned}$$

where we are allowed to add the mixed terms since they are just zero as $f \perp g$.

2. We will imitate the argument in Proposition 9.12. Consider a split-exact sequence in $C^*\text{Alg}$.

$$0 \longrightarrow A \xrightarrow{\phi} B \xrightleftharpoons[\alpha]{\psi} C \longrightarrow 0$$

This implies that $\psi \circ \alpha = \text{Id}_C$. Now, let $n \in \mathbb{Z}$ be fixed. By functoriality, we conclude that $F(\psi)$ is surjective. But this means that its range is just $F(C)$. On the other hand, the split-exact sequence above gives us a long-exact sequence with boundary homomorphisms $(\partial_n : F_n(C) \rightarrow F_{n-1}(A))_{n \geq 1}$. Since ∂_n connects to $F_n(\psi)$ and the long-exact sequence is exact at $F_n(C)$ for $n \geq 1$, it follows that $\ker(\partial_n) = F_n(C)$ for $n \geq 1$. That is, $\partial_n = 0$ for $n \geq 1$. The boundary homomorphisms in the resulting long-exact sequence are zero, so we obtain:

$$0 \longrightarrow F(A) \xrightarrow{F(\phi)} F(B) \xrightleftharpoons[F(\alpha)]{F(\psi)} F(C) \longrightarrow 0$$

where we do have surjectivity of $F(\psi)$ due to the argument we gave above.

3. We observe that:

$$\text{ran}(f + g) \subseteq \text{ran}(f) \oplus \text{ran}(g),$$

where the internal sum in B on the right-hand side is a direct sum because $\text{ran}(f) \cap \text{ran}(g)$ is trivial. Indeed, if $y \in \text{ran}(f) \cap \text{ran}(g)$, then $y = f(a)$ and $y = g(b)$ for $a, b \in A$. So, we have the following computation:

$$\begin{aligned}\|y\|^2 &= \|yy^*\| \\ &= \|f(a)g(b^*)\| \\ &= 0,\end{aligned}$$

from which we conclude that $y = 0$. Since $\text{ran}(f), \text{ran}(g) \in \text{dom}(F)$, we see that their direct sum $\text{ran}(f) \oplus \text{ran}(g) \in \text{dom}(F)$. So, we may replace B by $\text{ran}(f) \oplus \text{ran}(g)$ and note that since F is split-exact:

$$F(B) \simeq F(\text{ran}(f)) \oplus F(\text{ran}(g)).$$

Now, $F(f)$ is a map into $F(\text{ran}(f))$ and $F(g)$ is a map into $F(\text{ran}(g))$. Now, we use the universal property of the product to induce a unique morphism $\beta : F(A) \rightarrow F(\text{ran}(f)) \oplus F(\text{ran}(g))$ such that the obvious diagram commutes. But now, we see that $F(f + g)$ and $F(f) + F(g)$ would make the same diagram commute. ■

Our next goal is to establish that homology theories are “essentially” formed by just one functor and the suspension. In other words, if we have h_n , then we can calculate it using h_{n-1} . Before we state this precisely, here’s a construction.

Construction 9.15. Let $(h_n)_{n \in \mathbb{Z}}$ be a C^* -homology theory, let $A \in C^*\text{Alg}$ & consider the sequence:

$$0 \longrightarrow \text{Sus}(A) \longrightarrow \text{Cone}(A) \longrightarrow A \longrightarrow 0$$

as introduced in Construction 6.8. This gives us the following data:

1. A sequence $(\sigma_A : h_n(A) \rightarrow h_{n-1}(\text{Sus}(A)))_{n \in \mathbb{Z}}$ of morphisms and;
2. A long-exact sequence:

$$\dots \longrightarrow h_n(\text{Sus}(A)) \longrightarrow h_n(\text{Cone}(A)) \longrightarrow h_n(A) \xrightarrow{\sigma_A} h_{n-1}(\text{Sus}(A)) \longrightarrow \dots$$

such that the boundary homomorphisms are natural with respect to morphisms of exact sequences.

We refer to each σ_A as a suspension map.

With this construction at hand, we have the following key result about the structure of C^* -homology theories.

Proposition 9.16

Let $(h_n : C^*\text{Alg} \rightarrow \mathcal{A})_{n \in \mathbb{Z}}$ be a C^* -homology theory & let $A \in \text{dom}(h_0)$.

1. If $B \in \text{dom}(h_0)$ and $f \in C^*\text{Alg}(A, B)$ so that $f = 0$, then $h_n(f) = 0$ for each $n \in \mathbb{Z}$.
2. $\forall n \in \mathbb{Z} : h_n(\text{Cone}(A)) = 0$.

In particular, the suspension map σ_A is an isomorphism.

Proof. We'll prove each of these statements one-by-one.

1. Observe that we can factorize f as follows:

$$A \rightarrow 0 \rightarrow B.$$

Applying each h_n to f , we get that:

$$h_n(f) = h_n(0 \rightarrow B) \circ h_n(A \rightarrow 0),$$

and since $h_n(0) = 0$, it follows that $h_n(f) = 0$.

2. $\text{Cone}(A)$ is contractible by **Claim 6.12**, so $\text{Id}_{\text{Cone}(A)}$ is homotopic to 0. This means that:

$$\begin{aligned} 0 &= h_n(0) \\ &= h_n(\text{Id}_{\text{Cone}(A)}) \\ &= \text{Id}_{h_n(\text{Cone}(A))}, \end{aligned}$$

and this only happens when $h_n(\text{Cone}(A)) = 0$.

Now, let $n \in \mathbb{N}$ and consider the suspension map:

$$\sigma_A : h_n(A) \rightarrow h_{n-1}(\text{Sus}(A)),$$

within the following long-exact sequence:

$$\dots \longrightarrow h_n(\text{Sus}(A)) \longrightarrow h_n(\text{Cone}(A)) \longrightarrow h_n(A) \xrightarrow{\sigma_A} h_{n-1}(\text{Sus}(A)) \longrightarrow \dots$$

Now, we'll show that σ_A is an isomorphism by showing that it is bijective.

1. We see that $h_n(\text{Cone}(A)) = 0$, so σ_A is injective since the sequence above is exact.
2. We see that $h_{n-1}(\text{Cone}(A)) = 0$, so σ_A is surjective since the sequence above is exact.

It follows that σ_A is bijective so we're done. ■

So, the result of Proposition 9.16 motivates the following bit of notation.

Notation 9.17. Let $F : C^*\text{Alg} \rightarrow \mathcal{A}$ be a functor and define:

$$\forall n \in \mathbb{N}_0 : F_n \stackrel{\text{def}}{=} F \circ \text{Sus}^n.$$

This is known as the n -th iterant of F . We refer to this as the suspension spectrum of F .

We have a remark about Notation 9.17.

Remark 9.18. For this to make sense, we need to ensure that Sus lands inside the domain of F . This is easy to ensure by restricting Sus to the domain of F , since admissible subcategories are closed under suspension^a.

^aSee statement (2) of Proposition 9.3.

We can already prove some basic facts about the suspension spectrum of some functor. Before we do this, let us also introduce the notion of C^* -stability. We will need this when discussing the Bott Periodicity Theorem later.

Terminology 9.19. Let $F : C^*\text{Alg} \rightarrow \mathcal{A}$ be a functor, let $A \in \text{dom}(F)$ and let $e \in \mathcal{K}$ be a rank-one projection. Define:

$$p_A : A \rightarrow A \otimes \mathcal{K}, \quad a \mapsto p_A(a) \stackrel{\text{def}}{=} a \otimes e.$$

F is C^* -stable iff $F(p_A)$ is an isomorphism^a for each $A \in \text{dom}(F)$.

^aThis makes sense because $A \otimes \mathcal{K} \in \text{dom}(F)$ by Proposition 9.3.

The next result, which extends what is in the literature, gives us permanence properties of the suspension spectrum.

Proposition 9.20

Let $F : C^*\text{Alg} \rightarrow \mathcal{A}$ be a functor.

1. If F is homotopy-invariant, then its suspension spectrum consists of homotopy-invariant functors.
2. If F is half-exact, then its suspension spectrum consists of half-exact functors.
3. If F is split-exact, then its suspension spectrum consists of split-exact functors.
4. If F is C^* -stable, then its suspension spectrum consists of C^* -stable functors.

Proof. Let $\text{dom}(F)$ be denoted by $\mathcal{C}$ throughout the proofs of these statements. Now, it is sufficient to prove that $F \circ \text{Sus}$ satisfies these properties, since this can be extended to the rest of the spectrum by induction.

1. Let $A, B \in \mathcal{C}$ and let $f, g \in \mathcal{C}(A, B)$ be homotopic, with $h : A \rightarrow \text{Cyl}(B)$ being a homotopy between them. We apply Sus to this morphism and let $\alpha : \text{Sus} \circ \text{Cyl} \Rightarrow \text{Cyl} \circ \text{Sus}$ be the natural isomorphism of **Claim 6.5**. Now, we wish to define a homotopy from $\text{Sus}(f)$ to $\text{Sus}(g)$ and we do this as follows:

$$H \stackrel{\text{def}}{=} \alpha_B \circ \text{Sus}(h).$$

Let us show that this is a homotopy from $\text{Sus}(f)$ to $\text{Sus}(g)$. If this is the case, then we can conclude by the homotopy-invariance of F that $F(\text{Sus}(f)) = F(\text{Sus}(g))$ and this implies that $F \circ \text{Sus}$ is, of course, homotopy-invariant. Indeed, let $i \in \{0, 1\}$ and consider the evaluation map $p_i^{\text{Sus}(B)} : \text{Cyl}(\text{Sus}(B)) \rightarrow \text{Sus}(B)$. We need to determine the exact relationship between $p_i^{\text{Sus}(B)}$ and $\text{Sus}(p_i^B)$. In fact, we have the following claim.

Claim. $p_i^{\text{Sus}(B)} \circ \alpha_B = \text{Sus}(p_i^B)$.

This was proved in **Claim 6.6**. Now, we have the following computation:

$$\begin{aligned}
p_i^{\text{Sus}(B)} \circ H &= p_i^{\text{Sus}(B)} \circ (\alpha_B \circ \text{Sus}(h)) \\
&= (p_i^{\text{Sus}(B)} \circ \alpha_B) \circ \text{Sus}(h) \\
&= \text{Sus}(p_i^B) \circ \text{Sus}(h) \\
&= \text{Sus}(p_i^B \circ \text{Sus}(h)),
\end{aligned}$$

which evaluates to $\text{Sus}(f)$ or $\text{Sus}(g)$ depending on whether i is chosen to be 0 or 1.

2. Since Sus is an exact functor, it takes an exact sequence to an exact sequence, so F takes that exact sequence to a half-exact sequence. In other words, $F \circ \text{Sus}$ is half-exact.
3. Since Sus is an exact functor, it takes an split-sequence to a split-exact sequence, so F takes that split-exact sequence to a split-exact sequence. In other words, $F \circ \text{Sus}$ is split-exact.
4. Let $G \stackrel{\text{def}}{=} - \otimes \mathcal{K}$ be defined on $\mathcal{C}$. We start with the following claim.

| Claim. The maps p_A form the components of a natural transformation from $\text{Id}_{\mathcal{C}}$ to G .

Again, this makes sense because the codomain of G lands in $\mathcal{C}$ by the strength of Proposition 9.3. Now, let $A, B \in \mathcal{C}$ and let $f \in \mathcal{C}(A, B)$. Then, we need to show that the following diagram:

$$\begin{array}{ccc}
A & \xrightarrow{p_A} & A \otimes \mathcal{K} \\
f \downarrow & & \downarrow f \otimes \text{Id}_{\mathcal{K}} \\
B & \xrightarrow{p_B} & B \otimes \mathcal{K}
\end{array}$$

commutes. This is a calculation as follows:

$$\begin{aligned}
\forall a \in A : (f \otimes \text{Id}_{\mathcal{K}} \circ p_A)(a) &= f \otimes \text{Id}_{\mathcal{K}}(p_A(a)) \\
&= f \otimes \text{Id}_{\mathcal{K}}(a \otimes e) \\
&= f(a) \otimes e \\
&= p_B(f(a)) \\
&= (p_B \circ f)(a),
\end{aligned}$$

as was desired. This proves the claim.

If the natural transformation of the preceding claim is denoted by $p : \text{Id}_{\mathcal{C}} \Rightarrow G$, then let us consider the left-whiskered natural transformation $q \stackrel{\text{def}}{=} p \circ \text{Sus} : \text{Sus} \Rightarrow G \circ \text{Sus}$. That is, if $A \in \mathcal{C}$, then this is defined as:

$$q_A \stackrel{\text{def}}{=} p_{\text{Sus}(A)}.$$

What we want to show is that $F(\text{Sus}(p_A))$ is an isomorphism and if it was the case that $\text{Sus}(p_A)$ could be “replaced” by q_A , then we would be done. Let us observe that if $A \in \mathcal{C}$, then we have a computation:

$$\begin{aligned}
\text{Sus}(A \otimes \mathcal{K}) &\simeq \mathcal{C}_0(\mathbb{R}) \otimes (A \otimes \mathcal{K}) \\
&\simeq (\mathcal{C}_0(\mathbb{R}) \otimes A) \otimes \mathcal{K} \\
&\simeq \text{Sus}(A) \otimes \mathcal{K}.
\end{aligned}$$

This simply uses the associativity of the tensor product and since this is natural, we see that the above establishes a natural isomorphism $\alpha : \text{Sus} \circ G \Rightarrow G \circ \text{Sus}$. Now, we are naturally led to define the right-whiskering:

$$r \stackrel{\text{def}}{=} \text{Sus} \circ p : \text{Sus} \Rightarrow \text{Sus} \circ G,$$

of p . This leads us to the following claim.

| Claim. $\alpha \circ r = q$.

If this claim is true, then we note that $r_A = \text{Sus}(p_A)$ for each $A \in \mathcal{C}$. So:

$$\begin{aligned}
F(q_A) &= F(\alpha_A \circ r_A) \\
&= F(\alpha_A) \circ F(r_A) \\
&= F(\alpha_A) \circ F(\text{Sus}(p_A)).
\end{aligned}$$

Since $F(q_A) = F(p_{\text{Sus}(A)})$ is an isomorphism (due to the C^* -stability of F) and $F(\alpha_A)$ is an isomorphism, it follows that $F(\text{Sus}(p_A))$ is an isomorphism and this shows that $F \circ \text{Sus}$ is C^* -stable. It remains to show the claim. To prove this, we need to carefully compute the left-hand side. So, let $A \in \mathcal{C}$ and let $f \in \text{Sus}(A)$:

$$\begin{aligned}
(\alpha \circ r)_A(f) &= \alpha_A(r_A(f)) \\
&= \alpha_A(\text{Sus}(p_A)(f)) \\
&= \alpha_A((\text{Id}_{\text{Sus}(\mathcal{C})} \otimes p_A)(f))
\end{aligned}$$

We cannot progress further without making further assumptions on f . In fact, it suffices to show that q_A and $(\alpha \circ r)_A$ coincide on a dense subset of $\text{Sus}(A)$ and we choose the dense subset in Lemma 6.15. Indeed, let $f = ga$

for some $g \in \mathcal{C}_0(\mathbb{R}) = \text{Sus}(\mathbb{C})$ and $a \in A$. Continuing the computation of $(\alpha \circ r)_A(f)$ above, we have that:

$$\begin{aligned} (\alpha \circ r)_A(f) &= \alpha_A((\text{Id}_{\text{Sus}(\mathbb{C})} \otimes p_A)(f)) \\ &= \alpha_A((\text{Id}_{\text{Sus}(\mathbb{C})} \otimes p_A)(ga)) \\ &= \alpha_A(g \otimes p_A(a)) \\ &= \alpha_A(g \otimes (a \otimes e)) \\ &= (g \otimes a) \otimes e \\ &= ga \otimes e \\ &= e \otimes_f \\ &= q_A(f) \end{aligned}$$

which proves the claim. ■

We should, now, address the following natural question.

| **Question 9.21.** Let $F : C^* \text{Alg} \rightarrow \mathcal{A}$ be a functor. When is its suspension spectrum a C^* -homology theory?

We will give an answer to this through the proof of Proposition 9.23. That result was known for functors defined on $C^* \text{Alg}$ and we extend it to functors defined on admissible subcategories. To prove it, we will need a technical lemma.

Lemma 9.22. [19, Corollary 3.7.4, p.93] Consider the following exact sequence of C^* -algebras:

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0$$

and let $B \in C^* \text{Alg}$. If A/I is nuclear or B is nuclear, then the tensored sequence:

$$0 \longrightarrow I \otimes_{\min} B \longrightarrow A \otimes_{\min} B \longrightarrow (A/I) \otimes_{\min} B \longrightarrow 0$$

is exact.

We are ready to discuss the desired proposition.

Proposition 9.23. The Barratt-Puppe Sequence

Let $F : C^* \text{Alg} \rightarrow \mathcal{A}$ be homotopy-invariant & half-exact. Its suspension spectrum is a one-sided C^* -homology theory.

Proof. Suppose we have our extension of C^* -algebras in $\text{dom}(F)$:

$$0 \longrightarrow I \xrightarrow{\iota_1} A \xrightarrow{\pi_1} A/I \longrightarrow 0$$

The proof is predicated around the following claim.

Claim. There exists an extension:

$$0 \longrightarrow \text{Sus}(A/I) \xrightarrow{\iota_2} \text{cone}(\pi_1) \xrightarrow{\pi_2} A \longrightarrow 0$$

with the following properties:

1. There exists a $\star$ -homomorphism $\alpha_1 : I \rightarrow \text{cone}(\pi_1)$ which induces an isomorphism on F_n for all $n \in \mathbb{N}$.
2. The composite $\pi_2 \circ \alpha_1$ induces the same map on F_n as the inclusion ι_1 for each $n \in \mathbb{N}$.

Let's start by noting that this claim makes sense in $\text{dom}(F)$. Indeed, since $A/I \in \text{dom}(F)$, we see that $\text{Sus}(A/I) \in \text{dom}(F)$. So, if the extension in the claim exists, then $\text{cone}(\pi_1) \in \text{dom}(F)$ since $\text{dom}(F)$ is extension-closed. So, the properties also make sense. At this point, the proof splits into two parts. In the first part, we'll check that we actually obtain the given long-exact sequence provided that the claim holds. In the second part, we'll prove the claim.

1. We obtain a sequence $(G_n)_{n \in \mathbb{N}}$ of exact sequences by the following specification:

$$G_1 : \quad 0 \longrightarrow I \xrightarrow{\iota_1} A \xrightarrow{\pi_1} A/I \longrightarrow 0$$

$$G_2 : \quad 0 \longrightarrow \text{Sus}(A/I) \xrightarrow{\iota_2} \text{cone}(\pi_1) \xrightarrow{\pi_2} A \longrightarrow 0$$

$$G_3 : \quad 0 \longrightarrow \text{Sus}(A) \xrightarrow{\iota_3} \text{cone}(\pi_2) \xrightarrow{\pi_3} \text{cone}(\pi_1) \longrightarrow 0$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

That is, G_1 is the extension we started with and for $n \in \mathbb{N}$, we have that the extension G_{n+1} is obtained from the extension G_n by the mapping cone construction in the claim. Now, we apply F to each of the sequences

above and utilize the fact that F is a half-exact functor to obtain the following exact sequences:

$$\begin{array}{ccccccc}
F(G_1) : & & F(I) & \xrightarrow{F(\iota_1)} & F(A) & \xrightarrow{F(\pi_1)} & F(A/I) \\
F(G_2) : & & F(\text{Sus}(A/I)) & \xrightarrow{F(\iota_2)} & F(\text{cone}(\pi_1)) & \xrightarrow{F(\pi_2)} & F(A) \\
F(G_3) : & & F(\text{Sus}(A)) & \xrightarrow{F(\iota_3)} & F(\text{cone}(\pi_2)) & \xrightarrow{F(\pi_3)} & F(\text{cone}(\pi_1)) \\
& & \vdots & & \vdots & & \vdots
\end{array}$$

Now, the idea is to just do the following “replacements”:

$$F(\text{cone}(\pi_1)) \rightsquigarrow F(I)$$

$$F(\text{cone}(\pi_2)) \rightsquigarrow F(\text{Sus}(A/I)) = F_1(A/I)$$

and glue the resulting sequences once we’ve done the replacements. This will actually yield the first part of the given long-exact sequence and continuing the process as we have done above will yield the rest of it. So, let’s illustrate why we can carry out the replacements above. Indeed, the claim gives us a commutative diagram:

$$\begin{array}{ccccccc}
F(\text{Sus}(A/I)) & \xrightarrow{F(\iota_2)} & F(\text{cone}(\pi_1)) & \xrightarrow{F(\pi_2)} & F(A) & \xrightarrow{F(\pi_1)} & F(A/I) \\
\parallel & & \downarrow F(\alpha_1)^{-1} & & \parallel & & \parallel \\
F(\text{Sus}(A/I)) & \xrightarrow{F(\alpha_1)^{-1} \circ F(\iota_2)} & F(I) & \xrightarrow{F(\iota_1)} & F(A) & \xrightarrow{F(\pi_1)} & F(A/I)
\end{array}$$

where the sequence on top is exact, so the sequence on the bottom is also exact. Now, we need to play a similar game with $F(G_3)$. Indeed, we have the following commutative diagram:

$$\begin{array}{ccccccc}
F(\text{Sus}(A)) & \xrightarrow{F(\iota_3)} & F(\text{cone}(\pi_2)) & \xrightarrow{F(\pi_3)} & F(\text{cone}(\pi_1)) \\
\parallel & & \downarrow F(\alpha_2)^{-1} & & \downarrow F(\alpha_1)^{-1} \\
F_1(A) & \xrightarrow{F(\alpha_2)^{-1} \circ F(\iota_3)} & F_1(A/I) & \xrightarrow{F(\alpha_1)^{-1} \circ F(\pi_3) \circ F(\alpha_2)} & F(I)
\end{array}$$

The top sequence is just $F(G_3)$ and this is exact, so the sequence on the bottom is exact. On the other hand, we can only sew this to the sequence we obtained from $F(G_1)$ and $F(G_2)$ if both morphisms from $F_1(A/I)$ to $F(I)$ coincide. These coinciding means precisely that the following equality is satisfied:

$$F(\alpha_1)^{-1} \circ F(\pi_3) \circ F(\alpha_2) = F(\alpha_1)^{-1} \circ F(\iota_2),$$

which is equivalent to the statement that:

$$F(\pi_3) \circ F(\alpha_2) = F(\iota_2).$$

By functoriality, the left-hand side is just $F(\pi_3 \circ \alpha_2)$ and this translates to property (2) of the claim above. So, we can sew this short exact sequence to the one obtained from $F(G_1)$ and $F(G_2)$.

- For this, we use the explicit form of the mapping cone defined in [Construction 9.2](#). We have that:

$$\text{cone}(\pi_1) = \{(f, a) \in \text{Cone}(A/I) \times A : f(0) = 0 \wedge f(1) = \pi_1(a)\}.$$

So, we define the three maps as follows:

$$\iota_2 : \text{Sus}(A/I) \rightarrow \text{cone}(\pi_1), \quad f \mapsto \iota_2(f) \stackrel{\text{def}}{=} (f, 0),$$

$$\pi_2 : \text{cone}(\pi_1) \rightarrow A, \quad (f, a) \mapsto \pi_2(f, a) \stackrel{\text{def}}{=} a,$$

$$\alpha_1 : I \rightarrow \text{cone}(\pi_1), \quad i \mapsto \alpha_1(i) \stackrel{\text{def}}{=} (0, i).$$

The fact that $\pi_2 \circ \alpha_1$ induces the same map as the inclusion is clear. Now, we also have an extension:

$$0 \longrightarrow I \xrightarrow{\alpha_1} \text{cone}(\pi_1) \longrightarrow \text{Cone}(A/I) \longrightarrow 0$$

with the second map being the projection to $\text{Cone}(A/I)$. By [Claim 6.12](#), we know that $\text{Cone}(A/I)$ is contractible. This means that $F(\alpha_1)$ is surjective. To show injectivity, we will replace I with a C^* -algebra Q so that I and Q are homotopy-equivalent and Q fits into an exact sequence with $\text{Cone}(I)$ on the left. So, we define:

$$Q \stackrel{\text{def}}{=} \{f \in \text{Cone}(A) : f(0) \in I\}.$$

We have the following claim.

| **Claim 9.2.** Q and I are homotopy equivalent. This is proved on Page 273.

Let us consider the following exact sequence:

$$0 \longrightarrow \text{Cone}(I) \longrightarrow Q \longrightarrow \text{cone}(\pi_1) \longrightarrow 0$$

where the first map is just an inclusion of functions and the second map is defined as:

$$k : f \mapsto (\pi_1 \circ f, f(1)).$$

The fact that these are $\star$ -homomorphisms will follow by an explicit computation and it also uses the fact that π_1 is a $\star$ -homomorphism. Since $I \in \text{dom}(F)$, we see that $\text{Cone}(I) \in \text{dom}(F)$ by Proposition 9.3. By our argument earlier in the proof, we know that $\text{cone}(\pi_1) \in \text{dom}(F)$. Since Q fits into the extension above, we see that $Q \in \text{dom}(F)$. Now, we use the contractibility of $\text{Cone}(I)$ to conclude that $F(k)$ is injective. Let us establish a formula which connects the map k to α_1 . If we define, as in the proof of **Claim 9.2**, the map:

$$\psi : I \rightarrow Q, \quad j \mapsto \psi(j),$$

where $\psi(j)$ is the constant function j on $[0, 1]$. Composing this with k , we see that:

$$\begin{aligned} \forall j \in I : (k \circ \psi)(i) &= k(\psi(i)) \\ &= (\pi_1 \circ \psi(i), \psi(i)(1)) \\ &= (0, i) \\ &= \alpha_1(i). \end{aligned}$$

So, we conclude that $k \circ \psi = \alpha_1$ and this means that:

$$\begin{aligned} F(\alpha_1) &= F(k \circ \psi) \\ &= F(k) \circ F(\psi). \end{aligned}$$

Since $F(k)$ is injective and $F(\psi)$ is an isomorphism, we conclude that $F(\alpha_1)$ is injective.

3. Of course, the claim requires that the actual conclusion be true for each F_n and not just for F_0 . Using the fact that the suspension is an exact functor, we see that iterated suspension of each exact sequence that appears above produces an exact sequence. By the argument in statement (1) of Proposition 9.20, iterated suspensions also preserve homotopies, in the sense that if two morphisms are homotopic, then their suspensions are also homotopic. In particular, iterated suspensions of Q and I are also homotopy-equivalent. Finally, we note that each F_n in the suspension spectrum is also a homotopy-invariant and half-exact functor.
4. Finally, we have to argue that the boundary maps obtained are natural with respect to morphisms of exact sequences. One notes that the boundary maps are formed from the exact sequence of the claim and also the exact sequence for Q . Suppose, then, that we have the following morphism of exact sequences in $\mathbf{C}^*\text{Alg}$:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \xrightarrow{\iota_1} & A & \xrightarrow{\pi_1} & A/I & \longrightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\ 0 & \longrightarrow & J & \xrightarrow{\iota_2} & B & \xrightarrow{\pi_2} & B/J & \longrightarrow & 0 \end{array}$$

Now, consider the following diagram:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \text{Sus}(A/I) & \xrightarrow{i_1} & \text{cone}(\pi_1) & \xrightarrow{p_1} & A & \longrightarrow & 0 \\ & & \downarrow \text{Sus}(\gamma) & & \downarrow & & \downarrow \beta & & \\ 0 & \longrightarrow & \text{Sus}(B/J) & \xrightarrow{i_2} & \text{cone}(\pi_2) & \xrightarrow{p_2} & B & \longrightarrow & 0 \end{array}$$

So, what's happening here is that we start with two exact sequences which have a morphism between them. Each of these exact sequences form another exact sequence which is the one from the claim made earlier in the proof of this result. Now, we want to form a morphism between those exact sequences. We define:

$$\hat{\pi} : \text{cone}(\pi_1) \rightarrow \text{cone}(\pi_2), \quad (f, a) \mapsto \hat{\pi}(f, a) \stackrel{\text{def}}{=} (\text{Cone}(\gamma)(f), \beta(a)).$$

Let us make sure that this makes sense first:

$$\begin{aligned} \forall (f, a) \in \text{cone}(\pi_1) : \text{Cone}(\gamma)(f)(1) &= (\gamma \circ f)(1) \\ &= \gamma(f(1)) \\ &= \gamma(\pi_1(a)) \\ &= (\gamma \circ \pi_1)(a) \\ &= (\pi_2 \circ \beta)(a) \\ &= \pi_2(\beta(a)), \end{aligned}$$

which shows, alongside the fact that $\text{Cone}(\gamma)(f)(0) = 0$, that $\hat{\pi}$ makes sense. Of course, it is also clearly a $\star$ -homomorphism so it remains to be checked that it gives us a commutative diagram above. We see that:

$$\begin{aligned} \forall f \in \text{Sus}(A/I) : (\hat{\pi} \circ i_1)(f) &= \hat{\pi}(f, 0) \\ &= (\text{Cone}(\gamma)(f), 0) \\ &= i_2(\text{Cone}(\gamma)(f)) \\ &= i_2(\text{Sus}(\gamma)(f)), \end{aligned}$$

where this works because the suspension “pinches” at both ends of $[0, 1]$ while the cone just “pinches” at one

end of $[0, 1]$. The commutativity of the right square is actually evident, based on the definition of $\widehat{\pi}$. Similarly, we want to form the following diagram and show that it commutes:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \text{Cone}(I) & \longrightarrow & Q_A & \longrightarrow & \text{cone}(\pi_1) \longrightarrow 0 \\
& & \downarrow \text{Cone}(\alpha) & & \downarrow & & \downarrow \widehat{\pi} \\
0 & \longrightarrow & \text{Cone}(J) & \longrightarrow & Q_B & \longrightarrow & \text{cone}(\pi_2) \longrightarrow 0
\end{array}$$

Again, from the two exact sequences we started with in this part of the proof, we form the exact sequences involving the auxiliary C^* -algebra Q (we have labeled this for each exact sequence) and the purpose is to relate these resulting exact sequences. Indeed, the map in the middle is defined as follows:

$$\widehat{q} \stackrel{\text{def}}{=} \text{Cone}(\beta)|_{Q_A}.$$

We need to check that this maps into Q_B . Indeed:

$$\begin{aligned}
\forall f \in Q_A : \widehat{q}(f)(0) &= \text{Cone}(\beta)|_{Q_A}(f)(0) \\
&= (\beta \circ f)(0) \\
&= \beta(f(0)) \\
&= \beta(\iota_1(f(0))) \\
&= \iota_2(\alpha(f(0))),
\end{aligned}$$

where the last term belongs to J . In the middle, we use the fact that if $f \in Q_A$, then $f(0) \in J$. Clearly, $\widehat{q}$ is a $\star$ -homomorphism and it can be used to ensure that our diagram just above commutes. However, this is sufficient to establish naturality of the boundary morphisms. After all, the boundary morphisms are formed precisely using the morphisms in these exact sequences and what we have established above is that the two boundary morphisms formed from the original exact sequence must fit into some commuting diagram. This commuting diagram will be precisely the required naturality square.

This completes the proof. ■

We have a remark about this result.

Remark 9.24. For functors defined on all of $C^*\text{Alg}$, this result can be found in [85, Proposition 8.4, p.34-35] with a reasonably detailed proof. We provide more details and explicitly show that this result can be adapted to our requirements. This requires us to take care of many more details than if we were just working on $C^*\text{Alg}$. Also, we are aware that we are being slightly loose with our argument for naturality. However, the last part just requires one to concatenate appropriate diagrams and identify the relevant boundary morphisms. Once these identifications are made, all of the maps that we defined earlier in the proof can be used to construct the desired naturality square.

9.2 An Abstract Proof of the Bott Periodicity Theorem

Having formulated these basic results, we now come to the main test result; the Bott Periodicity Theorem.

Theorem 9.25. The Bott Periodicity Theorem

Let $F : C^*\text{Alg} \rightarrow \mathcal{A}$ be a homotopy-invariant, half-exact and C^* -stable functor^a. Then, $F \simeq F \circ \text{Sus}^2$.

^aFrom now on, Notation 9.17 is in effect.

Here is some context for this theorem.

Remark 9.26. The Bott Periodicity Theorem, of course, has an extensive history that we will not discuss here since it is not relevant for us; one could also discuss the operator-algebraic version and the topological version(s) separately, alongside attempts to obtain periodicity for algebraic K-theory. Nevertheless, we mention that the rather general theorem above was proved by Cuntz in [9, Volume 1046, Theorem 4.4, p.63] for functors satisfying the listed properties and defined on all of $C^*\text{Alg}$. A more extensive proof with details is provided also in (Nigel) Higson's PhD thesis [50, Theorem 2.1.14, p.28]. The main contribution here is to go through Cuntz's proof & keep track of the auxiliary algebras introduced along the way, making sure that they are always contained in the domain of the functor F .

Let us start working towards Cuntz's proof of this theorem. A major step in that proof is to work with the *reduced* Toeplitz extension. In the next construction, we will study the Toeplitz extension and its reduced version. The main content of the construction lies in showing that the auxiliary C^* -algebras are in admissible subcategories of $C^*\text{Alg}$.

Construction 9.27. Let us consider the standard Toeplitz extension as follows:

$$0 \longrightarrow \mathcal{K} \xrightarrow{i_1} \mathcal{T} \xrightarrow{p_1} \mathcal{C}(S^1) \longrightarrow 0$$

For a detailed discussion of this extension and the proof of its existence, we strongly recommend the treatment of [61, Chapter 6.8, p.76-79]. We will briefly recall the essential points for your convenience. The key is to view $\mathcal{T}$ as the universal C^* -algebra generated by an isometry and $\mathcal{C}(S^1)$ as the universal C^* -algebra generated by a unitary. In that case, we can see that the map p_1 is induced by mapping our isometry of $\mathcal{T}$ (the generator) to the unitary of $\mathcal{C}(S^1)$ (the generator). If $v \in \mathcal{T}$ is the generator, then it will become the case that:

$$\mathcal{K} \simeq \langle 1_{\mathcal{T}} - vv^* \rangle,$$

with the right-hand side being an ideal in $\mathcal{T}$. The “reason” for this isomorphism is explained in the aforementioned notes very well, so we will not dwell on it. What we will do is define:

$$e_{00} \stackrel{\text{def}}{=} 1_{\mathcal{T}} - vv^*.$$

This will be used in our proof of the Bott Periodicity Theorem and we will recall it when necessary; it will be a very useful formula. Now, from the Toeplitz extension, we can construct the following diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_1} & \mathcal{T} & \xrightarrow{p_1} & \mathcal{C}(S^1) \longrightarrow 0 \\ & & & & \downarrow \varepsilon = \text{ev}_1 \circ p_1 & & \downarrow \text{ev}_1 \\ & & & & \mathbb{C} & \xlongequal{\quad} & \mathbb{C} \end{array}$$

where ev_1 is the evaluation map at $1 \in S^1$. Now, we put in $\ker(\varepsilon)$ into the diagram:

$$\begin{array}{ccccccc} & & & & 0 & & \\ & & & & \downarrow & & \\ & & & & \mathcal{T}_0 \stackrel{\text{def}}{=} \ker(\varepsilon) & & \\ & & & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_1} & \mathcal{T} & \xrightarrow{p_1} & \mathcal{C}(S^1) \longrightarrow 0 \\ & & & & \downarrow \varepsilon = \text{ev}_1 \circ p_1 & & \downarrow \text{ev}_1 \\ & & & & \mathbb{C} & \xlongequal{\quad} & \mathbb{C} \end{array}$$

Now, we have the following claim.

| Claim. A copy of $\mathcal{K}$ is contained in $\mathcal{T}_0$, which we will call the reduced Toeplitz C^* -algebra. This is nuclear.

Let $T \in \mathcal{K}$. Then, we need to show that $i_1(T) \in \mathcal{T}_0 = \ker(\varepsilon)$. For this to work, we need to show that:

$$\varepsilon(i_1(T)) = 0.$$

But now, we see that:

$$\text{ev}_1(p_1(i_1(T))) = 0,$$

which proves the result. Finally, nuclearity follows from the fact that it sits between two C^* -algebras which are known to be nuclear; $\mathcal{K}$ is an AF algebra so it is nuclear^a and $\mathcal{C}_0(S^1 \setminus \{0\})$ is commutative so it is nuclear^b.

In light of this result, we can place the reduced Toeplitz C^* -algebra in an extension as follows:

$$\begin{array}{ccccccc} & & & & 0 & & \\ & & & & \downarrow & & \\ 0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_2} & \mathcal{T}_0 & \xrightarrow{p_2} & \mathcal{C}_0(S^1 \setminus \{1\}) \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{K} & \xrightarrow{i_1} & \mathcal{T} & \xrightarrow{p_1} & \mathcal{C}(S^1) \longrightarrow 0 \\ & & & & \downarrow \varepsilon & & \downarrow \text{ev}_1 \\ & & & & \mathbb{C} & \xlongequal{\quad} & \mathbb{C} \end{array}$$

where the morphisms are chosen so that the given squares commute. The extension obtained involving $\mathcal{T}_0$ is known as the reduced Toeplitz extension. There are three facts worth noting:

1. i_2 is just i_1 .
2. p_2 is just p_1 restricted to $\mathcal{T}_0$.
3. $\mathcal{C}(S^1 \setminus \{1\})$ is just $\mathcal{C}_0((0, 1))$.

Now, the key point lies in the following results.

| Claim. Let $\mathcal{C}$ be an admissible subcategory of $C^*\text{Alg}$. Then, every C^* -algebra in the diagram above lies in $\mathcal{C}$.

We just need to check this for $\mathcal{T}_0$ and $\mathcal{C}(S^1 \setminus \{1\})$, since every other C^* -algebra above has been shown to lie in admissible subcategories. We note that $\mathcal{C}(S^1 \setminus \{1\}) \simeq \mathcal{C}_0(\mathbb{R}) = \text{Sus}(\mathbb{C})$ so this certainly lies in $\mathcal{C}$. Since $\mathcal{T}_0$ fits into the middle of an exact sequence where the other algebras belong to $\mathcal{C}$, we see that $\mathcal{T}_0 \in \mathcal{C}$.

We also need the following result.

Claim. Let $\mathcal{C}$ be an admissible subcategory of $C^*\text{Alg}$, let $A \in \mathcal{C}$ and suppose that B is any of the C^* -algebras appearing in the diagram above. Then, $A \otimes B \in \mathcal{C}$; every B above is nuclear so the notation is unambiguous.

We need to check this result for all of the algebras individually.

1. Let $B = \mathbb{C}$. Then, $A \otimes \mathbb{C} \simeq A$ so $A \otimes B \in \mathcal{C}$.
2. Let $B = \mathcal{K}$. We know that $M_{n \times n}(A) \in \mathcal{C}$ for each $n \in \mathbb{N}$. Now, we can write $A \otimes \mathcal{K}$ as an inductive limit of full matrix algebras over A and since $\mathcal{C}$ is closed under inductive limits, it follows that $A \otimes \mathcal{K} \in \mathcal{C}$.
3. Let $B = \mathcal{C}(S^1)$. Then, $A \otimes B \simeq \mathcal{C}(S^1, A)$ and we can fit this into a pullback diagram:

$$\begin{array}{ccc} \mathcal{C}(S^1, A) & \longrightarrow & \text{Cyl}(A) \\ \downarrow & & \downarrow (\text{ev}_0, \text{ev}_1) \\ A & \xrightarrow{\Delta_A} & A \oplus A \end{array}$$

Now, $A \oplus A \in \mathcal{C}$ and $\text{Cyl}(A) \in \mathcal{C}$, so we conclude that $\mathcal{C}(S^1, A) \in \mathcal{C}$.

4. Let $B = \mathcal{T}$. Let us tensor the Toeplitz exact sequence on the right by A and note that by Lemma 9.22, the resulting sequence is exact since $\mathcal{C}(S^1)$ is nuclear. Now, we know that $\mathcal{K} \otimes A \in \mathcal{C}$ and $\mathcal{C}(S^1, A) \in \mathcal{C}$. So, we use the extension property of nuclearity^c to conclude that $\mathcal{T} \otimes A \in \mathcal{C}$.
5. Let $B = \mathcal{C}_0(S^1 \setminus \{1\})$. This is just $\text{Sus}(\mathbb{C})$, so tensoring it with A produces $\text{Sus}(A)$ and this lies in $\mathcal{C}$.
6. Let $B = \mathcal{T}_0$. The same argument as we used for $\mathcal{T}$ will work.

This proves the claim.

This completes our desired construction.

^aSee [19, Proposition 2.4.1, p. 39, Theorem 3.8.7, p.104]. The first result stated in the citation shows nuclearity of $\mathcal{K}$ through a different definition. The second result shows that this definition is equivalent to the characterization we use for the claim.

^bSee [19, Proposition 2.4.2, p. 38-39, Theorem 3.8.7, p.104].

^cThis was discussed (and referenced) in Example 9.6.

Next, we need two technical lemmas; the first is about tensor products.

Lemma 9.28 Let $A_i, B_i \in C^*\text{Alg}$ and $\phi_i \in C^*\text{Alg}(A_i, B_i)$ for $i \in \{1, 2\}$. TFSH:

1. If each ϕ_i is injective, then so is $\phi_1 \otimes_{\min} \phi_2$.
2. If each ϕ_i is surjective, then so is $\phi_1 \otimes_{\min} \phi_2$.

Proof. We will prove each statement one-by-one.

1. Let $B_1 \subseteq \mathcal{B}(H_1)$ and $B_2 \subseteq \mathcal{B}(H_2)$ be faithfully represented on $\mathbb{C}$ -Hilbert spaces by the Gelfand-Naimark-Segal Theorem. Composing these faithful $\star$ -representations with each ϕ_i , we see that each ϕ_i forms a faithful $\star$ -representation of each A_i . Now, let $x \in A_1 \otimes_{\min} A_2$ be a mixed tensor, represented as follows:

$$x = \sum_{i=1}^m a_i \otimes b_i.$$

Next, we have the following computation:

$$\begin{aligned} \|(\phi_1 \otimes_{\min} \phi_2)(x)\| &= \left\| \sum_{i=1}^m \phi_1(a_i) \otimes \phi_2(b_i) \right\| \\ &= \|x\|, \end{aligned}$$

where we have used the definition of the minimal C^* -norm for the second equality above. This computation implies that $\phi_1 \otimes_{\min} \phi_2$ is isometric on the mixed tensors and since every element of $A_1 \otimes_{\min} A_2$ is a limit of mixed tensors, it follows that $\phi_1 \otimes_{\min} \phi_2$ is isometric on $A_1 \otimes_{\min} A_2$.

2. We use the fact that $\phi_1 \otimes_{\mathbb{C}} \phi_2$ is surjective^a. This implies that $\text{ran}(\phi_1 \otimes_{\min} \phi_2)$ is dense in $B_1 \otimes_{\min} B_2$. But this is also closed in $B_1 \otimes_{\min} B_2$, since $\star$ -homomorphisms have closed ranges. So, $\phi_1 \otimes_{\min} \phi_2$ is surjective. ■

^aTake an element $y \in B_1 \otimes_{\mathbb{C}} B_2$ and write it as a linear combination of simple tensors. Each simple tensor certainly has a preimage in $A_1 \otimes_{\mathbb{C}} A_2$ since ϕ_1 and ϕ_2 are surjective, so we can use that to construct a preimage for y .

Next, we also need a lemma about paths of self-adjoint unitaries.

Lemma 9.29 Let $A \in {}^1C^*\text{Alg}$ and let $u \in A$ be a self-adjoint unitary. There is a path of unitaries from u to 1_A .

Proof. The proof just involves explicitly constructing a path. Indeed, we define:

$$\phi : [0, 1] \rightarrow A, \quad t \mapsto \phi(t) \stackrel{\text{def}}{=} \frac{1}{2}(1_A + u) + \frac{1}{2}\exp(\pi it)(u - 1_A).$$

Let us check that this works.

| **Claim 9.3.** ϕ is a path of unitaries from u to 1_A .

At this stage, we are ready to prove the Bott Periodicity Theorem.

Proof of Theorem 9.25. We tensor the reduced Toeplitz extension by A so we get an extension^a:

$$0 \longrightarrow \mathcal{K} \otimes A \longrightarrow \mathcal{T}_0 \otimes A \xrightarrow{p_2 \otimes \text{Id}_A} \text{Sus}(A) \longrightarrow 0$$

Applying the Barratt-Puppe sequence, we obtain a long-exact sequence:

$$\dots \longrightarrow F_1(\text{Sus}(A)) \longrightarrow F_0(\mathcal{K} \otimes A) \longrightarrow F_0(\mathcal{T}_0 \otimes A) \longrightarrow F_0(\text{Sus}(A))$$

where F_n is defined by Notation 9.17 for each $n \in \mathbb{N}_0$. Now, we have the boundary map:

$$\partial : F_1(\text{Sus}(A)) \rightarrow F_0(\mathcal{K} \otimes A)$$

squeezed in-between $F_0(\mathcal{T}_0 \otimes A)$ and $F_1(\mathcal{T}_0 \otimes A)$. On the other hand, we also notice that:

$$\begin{aligned} F_1(\text{Sus}(A)) &= F_0(\text{Sus}^2(A)) = F_2(A), \\ F_0(\mathcal{K} \otimes A) &\simeq F_0(A), \end{aligned}$$

where the second isomorphism is induced from the C^* -stability. In fact, the proof of statement (4) of Proposition 9.20 showed that this second isomorphism is induced from a natural transformation p from the identity to right-tensoring by $\mathcal{K}$, so let α be the right-whiskering of F_0 with p . Now, we consider the composition $\alpha \circ \partial$. This is a natural transformation and it would be a natural isomorphism provided that for each $A \in \text{dom}(F)$, we have that:

$$F_0(\mathcal{T}_0 \otimes A) = F_1(\mathcal{T}_0 \otimes A) \simeq 0.$$

So, our next goal is to prove the following claim.

Claim. $\forall i \in \{0, 1\} : F_i(\mathcal{T}_0 \otimes A) \simeq 0$.

The rest of the proof is quite complicated and technical. We will essentially show this by a series of subclaims which act as reductions of the claim. Let us start by introducing the following auxiliary C^* -algebra:

$$\widehat{\mathcal{T}} \stackrel{\text{def}}{=} \mathcal{K} \otimes \mathcal{T}_0 + \mathcal{T} \otimes \mathbb{C}1_{\mathcal{T}} \subseteq \mathcal{T} \otimes \mathcal{T}.$$

In fact, we need to prove that this is a C^* -algebra.

Subclaim. $\widehat{\mathcal{T}}$ is a C^* -algebra.

This will follow from three facts:

1. $\mathcal{K} \otimes \mathcal{T}_0$ is a closed ideal in $\mathcal{T} \otimes \mathcal{T}$. This follows from the fact that both $\mathcal{K}$ and $\mathcal{T}_0$ include into $\mathcal{T}$ as ideals, so we just apply Lemma 9.28 to those inclusions to obtain an inclusion of $\mathcal{K} \otimes \mathcal{T}_0$ into $\mathcal{T} \otimes \mathcal{T}$. This is an ideal simply because $\mathcal{K}$ and $\mathcal{T}_0$ are both ideals in $\mathcal{T}$; refer back to the (reduced) Toeplitz extension.
2. $\mathcal{T} \otimes \mathbb{C}1_{\mathcal{T}}$ is a C^* -subalgebra of $\mathcal{T} \otimes \mathcal{T}$. This is clear.
3. The sum of a C^* -subalgebra and a closed ideal is a C^* -subalgebra. Let $A \in C^*\text{Alg}$, let $B \subseteq A$ be a C^* -subalgebra and let $I \subseteq A$ be a closed ideal. If $\pi : A \rightarrow A/I$ is the canonical projection, then $\pi(B)$ is closed in A/I since B is a C^* -algebra and π is a $\star$ -homomorphism. On the other hand, we have that:

$$B + I = \pi^{\leftarrow}(\pi(B)),$$

and so, we conclude that $B + I$ is closed.

This proves the subclaim.

Now, we wish to fit $\widehat{\mathcal{T}}$ into an extension. By construction, $\mathcal{K} \otimes \mathcal{T}_0$ is an ideal in $\widehat{\mathcal{T}}$. We note that $\mathcal{T} \otimes \mathbb{C}1_{\mathcal{T}} \cap \mathcal{K} \otimes \mathcal{T}_0 = \{0\}$ since the former contains the identity and $\mathcal{T}_0$ does not contain the identity since it is a codimension 1 closed ideal in $\mathcal{T}$. So, we see that $\widehat{\mathcal{T}}/(\mathcal{K} \otimes \mathcal{T}_0) \simeq \mathcal{T}$. Therefore, $\widehat{\mathcal{T}}$ fits into the following extension:

$$0 \longrightarrow \mathcal{K} \otimes \mathcal{T}_0 \longrightarrow \widehat{\mathcal{T}} \longrightarrow \mathcal{T} \longrightarrow 0$$

This leads us to another subclaim.

Subclaim. $\widehat{\mathcal{T}}$ is nuclear, lies in $\text{dom}(F)$ and $\widehat{\mathcal{T}} \otimes B \in \text{dom}(F)$ when $B \in \text{dom}(F)$.

Since both $\mathcal{T}_0$ and $\mathcal{K}$ are nuclear, we see that $\mathcal{K} \otimes \mathcal{T}_0$ is nuclear. Since $\mathcal{T}$ is nuclear, it follows that $\widehat{\mathcal{T}}$ is nuclear. Now, using the same extension, the fact that $\mathcal{K} \otimes \mathcal{T}_0 \in \text{dom}(F)$ by the last claim of Construction 9.27 and the fact that $\mathcal{T} \in \text{dom}(F)$, we conclude that $\widehat{\mathcal{T}} \in \text{dom}(F)$. Finally, let $B \in \text{dom}(F)$. Using the nuclearity of $\mathcal{T}$, we can tensor the extension above by B . Applying Lemma 9.22, we obtain an extension:

$$0 \longrightarrow \mathcal{K} \otimes \mathcal{T}_0 \otimes B \longrightarrow \widehat{\mathcal{T}} \otimes B \longrightarrow \mathcal{T} \otimes B \longrightarrow 0$$

Using the last claim of Construction 9.27, we see that $\mathcal{K} \otimes \mathcal{T}_0 \otimes B \in \text{dom}(F)$ and $\mathcal{T} \otimes B \in \text{dom}(F)$. So, it follows that $\widehat{\mathcal{T}} \otimes B \in \text{dom}(F)$ and we are done.

Since $\mathcal{K} \otimes \mathcal{T}_0 \in \text{dom}(F)$ and $\mathcal{T} \in \text{dom}(F)$, we conclude that $\widehat{\mathcal{T}} \in \text{dom}(F)$. Moreover, $\mathcal{T}$ is nuclear so we can tensor the extension above by A and by applying Lemma 9.22, we obtain an extension:

$$0 \longrightarrow \mathcal{K} \otimes \mathcal{T}_0 \otimes A \longrightarrow \widehat{\mathcal{T}} \otimes A \longrightarrow \mathcal{T} \otimes A \longrightarrow 0$$

We want to note that all three algebras here lie in $\text{dom}(F)$; the first and third one lie in $\text{dom}(F)$ due to the third claim of Construction 9.27 and the middle one lies in $\text{dom}(F)$ since $\text{dom}(F)$ is extension-closed. Now, we note that

this sequence actually splits. Indeed, we can define the following $\star$ -homomorphism:

$$k : \mathcal{T} \otimes A \rightarrow \widehat{\mathcal{T}} \otimes A, \quad b \otimes a \mapsto k(b \otimes a) \stackrel{\text{def}}{=} b \otimes 1_{\mathcal{T}} \otimes a.$$

This is clearly a section for the map $\widehat{\mathcal{T}} \otimes A \rightarrow \mathcal{T} \otimes A$. If we put this arrow into the diagram above and also include $\mathcal{T}_0 \otimes A$ into its stabilization as usual, then we obtain the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{K} \otimes \mathcal{T}_0 \otimes A & \xrightarrow{i} & \widehat{\mathcal{T}} \otimes A & \xleftarrow{k} & \mathcal{T} \otimes A \longrightarrow 0 \\ & & \uparrow p & & \nearrow j & & \\ & & \mathcal{T}_0 \otimes A & & & & \end{array}$$

where we explicitly note that p is defined by the formula $a \otimes b \mapsto e_{00} \otimes ab$, with e_{00} having been defined at the start of [Construction 9.27](#). This leads us to our second subclaim.

Subclaim. For $t \in \{0, 1\}$, we have that $F_t(j)$ is injective.

Since $F = F_0$ is homotopy-invariant, half-exact and C^* -stable, we can deduce that the same is true for F_1 by the strength of [Proposition 9.20](#). We note that homotopy-invariance & half-exactness implies, by statement (2) of [Proposition 9.14](#), that F_0 is split-exact and this means that F_1 is also split-exact. Now, let $t \in \{0, 1\}$ and note that $F_t(p)$ is an isomorphism since F_t is C^* -stable, while $F_t(i)$ is injective because F_t is split-exact and our sequence above is split-exact. So, it follows that $F_t(j)$ is injective.

So, if we want to show our main claim, it suffices to prove that:

$$\forall i \in \{0, 1\} : F_i(j) = 0.$$

To prove this, we will construct another map from $\mathcal{T}_0 \otimes A \rightarrow \widehat{\mathcal{T}} \otimes A$. Now, we need a general result.

Claim 9.4. Let $B \in {}^1C^*\text{Alg}$ and let $u \in B$ be an isometry. Then, the map:

$$\text{Ad}(u) : B \rightarrow B, \quad x \mapsto \text{Ad}(u)(x) \stackrel{\text{def}}{=} uxu^*,$$

is a $\star$ -homomorphism. This is proved on [Page 273](#).

If v is the isometry generating $\mathcal{T}$, then we define:

$$\psi : \widehat{\mathcal{T}} \otimes A^\sim \rightarrow \widehat{\mathcal{T}} \otimes A^\sim,$$

as $\text{Ad}(v \otimes 1_{\mathcal{T}} \otimes 1_{A^\sim})$. The input into $\text{Ad}(-)$ is an isometry since v is an isometry, so ψ is a $\star$ -homomorphism by [Claim 9.4](#). Now, we observe that $\psi(\widehat{\mathcal{T}} \otimes A) \subseteq \widehat{\mathcal{T}} \otimes A$; this is clear by directly using the definition of ψ and noting that the conjugating isometry has $1_{A^\sim}$ as its last entry. Now, we have our next subclaim.

Subclaim. $\psi \perp j$. That is, ψ and j are orthogonal^a.

^aRecall that this notion was introduced in [Terminology 9.13](#).

We observe that, explicitly, we have that:

$$\forall x \in \mathcal{T}_0 \otimes A : j(x)\psi(x) = (e_{00} \otimes x)(v \otimes 1_{\mathcal{T}} \otimes 1_{A^\sim})x(v^* \otimes 1_{\mathcal{T}} \otimes 1_{A^\sim}),$$

where e_{00} is a rank-one projection in $\mathcal{K}$ and is used to define p explicitly. Now, if we multiply out the first two terms, then we see that we need to evaluate $e_{00}(v \otimes 1_{\mathcal{T}})$. This is the same as taking a unilateral shift on $l^2(\mathbb{N})$ and projecting down to the first coordinate; in that case, we get that this must be 0 and this proves the subclaim.

Now, we construct the map:

$$\psi_1 \stackrel{\text{def}}{=} \psi \circ k|_{\mathcal{T}_0 \otimes A},$$

and this acts from $\mathcal{T}_0 \otimes A$ to $\widehat{\mathcal{T}} \otimes A$. By the subclaim above, we deduce that:

$$\psi_1 \perp j.$$

Let $\psi_0 \stackrel{\text{def}}{=} j + \psi_1$; this is a $\star$ -homomorphism, by statement (1) of [Proposition 9.14](#). Since F_0 and F_1 are both homotopy-invariant and half-exact, we can apply statement (4) of the aforementioned result to conclude that:

$$\forall i \in \{0, 1\} : F_i(\psi_0) = F_i(j) + F_i(\psi_1).$$

This equation leads us to our next reduction.

Subclaim. It suffices to prove that ψ_0 and ψ_1 are homotopic $\star$ -homomorphisms.

If we do this, then the homotopy-invariance of each F_i implies that:

$$F_i(\psi_0) = F_i(\psi_1),$$

which implies that $F_i(j) = 0$ for $i \in \{0, 1\}$, which is exactly what we wanted to show.

Now, the rest of the proof involves constructing a homotopy from ψ_0 to ψ_1 . Firstly, let us show that we can reduce to the case where $A = \mathbb{C}$. To prove this, we need the result of the following claim.

Claim 9.5. Let $A, B \in \text{Nuc}C^*\text{Alg}$, let $C \in C^*\text{Alg}$ and let $f, g \in \text{Nuc}C^*\text{Alg}(A, B)$ be homotopic. Then:

$$f \otimes \text{Id}_C \simeq_h g \otimes \text{Id}_C.$$

That is, tensoring by C preserves homotopies. This is proved on [Page 273](#).

Now, ψ_0 and ψ_1 both depend on A , so we may reflect this dependence by writing them as ψ_0^A and ψ_1^A . If we, then, run through the explicit construction of these maps, we will see that:

$$\forall i \in \{0, 1\} : \psi_i^A = \psi_i^{\mathbb{C}} \otimes \text{Id}_{\mathbb{C}}.$$

So, if we can show that $\psi_0^{\mathbb{C}}$ and $\psi_1^{\mathbb{C}}$ are homotopic, then **Claim 9.5** will give us a homotopy between ψ_0^A and ψ_1^A . Therefore, we assume that $A = \mathbb{C}$. If that is the case, then ψ_0 and ψ_1 are maps $\mathcal{T}_0 \rightarrow \widehat{\mathcal{T}}$. Since $\mathcal{T}_0$ fits into a split-exact sequence with $\mathcal{T}$ as a codimension 1 closed ideal, it is the case that:

$$\mathcal{T}_0^{\sim} \simeq \mathcal{T}.$$

So, we can extend ψ_0 and ψ_1 uniquely to $\mathcal{T}$; we denote these extensions by π_0 and π_1 . We do this so that we can construct our homotopy through universal property by simply specifying it on the generator v of $\mathcal{T}$. We define:

$$v_0 \stackrel{\text{def}}{=} \pi_0(v),$$

$$v_1 \stackrel{\text{def}}{=} \pi_1(v).$$

Therefore, constructing a homotopy from π_0 to π_1 is the same as specifying a path of isometries from v_0 to v_1 . Since π_0 and π_1 are unital, let us observe that v_0 and v_1 are isometries. Now, we have the final subclaim of this proof.

Subclaim. We can find unitaries u_0, u_1 such that:

$$\forall i \in \{0, 1\} : v_i = u_i(v \otimes 1_{\mathcal{T}}).$$

Moreover, these unitaries are self-adjoint.

Let us momentarily postpone the proof of this subclaim and, instead, construct our path of isometries from v_0 to v_1 . If the subclaim holds, then **Lemma 9.29** gives a path of unitaries $(u_t)_{t \in [0,1]}$ from u_0 to u_1 . Now, we simply define:

$$\forall t \in [0, 1] : v_t \stackrel{\text{def}}{=} u_t(v \otimes 1_{\mathcal{T}}).$$

It is readily checked by an explicit computation that each v_t is an isometry. Moreover, this is a path because $(u_t)_{t \in [0,1]}$ is a path and multiplication is a continuous map. Finally, this is clearly a path from v_0 to v_1 , given the content of the subclaim. So, modulo this subclaim, this completes the proof of the Bott Periodicity Theorem.

Let us prove the subclaim. For this, we need to simply come up with explicit formulas for u_0 and u_1 . Motivating these formulas requires a fairly extended discussion where we embed $\mathcal{T} \otimes \mathcal{T}$ into $\mathcal{B}(l^2(\mathbb{N}_0 \times \mathbb{N}_0))$ and study the action of each v_i on the canonical orthonormal basis. This would provide motivation for the formulas that we will present:

$$u_0 \stackrel{\text{def}}{=} v^2(v^*)^2 \otimes 1_{\mathcal{T}} + e_{00}v^* \otimes v + ve_{00} \otimes v^* + e_{00} \otimes e_{00}$$

$$u_1 \stackrel{\text{def}}{=} v^2(v^*)^2 \otimes 1_{\mathcal{T}} + e_{00}v^* \otimes 1_{\mathcal{T}} + ve_{00} \otimes 1_{\mathcal{T}},$$

where $e_{00} = 1 - vv^*$. Now, there are several statements that need to be shown and all of them require, unfortunately, very detailed computations. We will carry them out in the next few claims just for u_0 and v_0 . However, we stress that the philosophy is exactly the same with u_1 and v_1 . For the sake of brevity, we will write 1 whenever we mean $1_{\mathcal{T}}$ and this will not cause us issues. With that said, let us proceed towards our desired results.

| **Claim 9.6.** $\pi_0(v) = v^2v^* \otimes 1 + e_{00} \otimes v$.

| **Claim 9.7.** $\pi_0(v) = u_0(v \otimes 1)$

| **Claim 9.8.** u_0 is a self-adjoint unitary.

Again, similar calculations go through for u_1 and v_1 , so we're done. ■

^aThis is an application of **Lemma 9.22**, where the lemma applies since $\mathcal{C}_0((0, 1))$ is nuclear.

Through our work above, we hope that we have demonstrated that many results involving axiomatic (co)homology on C^* -algebras can be deduced formally.

10 Axiomatic (Co)Homology II: (Co)Limits & The Mayer-Vietoris Theorem

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

10.1 Mayer-Vietoris Theorem & Further Aspects of Homotopy Theory

10.2 Inductive Continuity of (Co)Homology Theories on C^* -Algebras

11 Constructing Operator K_1

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| Motivation.

11.1 The Unitary Picture of K_1

We will start with a purely algebraic discussion which builds towards the definition of algebraic K_1 . Then, we will modify that definition slightly to produce operator K_1 .

11.2 The Homotopical Picture of K_1

With these two constructions in hand, we can start extending some of the previously defined topological notions to the situation with invertible matrices over an arbitrary $\mathbb{C}$ -Banach algebra. Again, the benefit of these constructions is that we have to fuss much less about doing things separately for the unital & non-unital cases.

Terminology 11.1. Let $A \in C^*Alg$ and let $x, y \in GL_\infty(A)$.

1. x and y are homotopic iff they are homotopic in $GL_n(A)$ for some $n \in \mathbb{N}$.
2. We further define:

$$GL_\infty(A)_0 \stackrel{\text{def}}{=} \{a \in GL_\infty(A) : a \sim_h 1\},$$

where $\sim_h$ is the symbol used to denote that two elements are homotopic in the sense described above.

Of course, the identity in (2) comes from the unitization.

Here are two simple results.

Claim 11.1. Let $A \in C^*Alg$. Then, $GL_\infty(A)_0$ is a normal subgroup of $GL_\infty(A)$.

Claim 11.2. Let $A \in C^*Alg$ and let $a, b \in GL_\infty(A)$. TFAE:

1. a and b are homotopic.
2. a and b lie in the same coset modulo $GL_\infty(A)_0$.

With these results in tow, we are finally ready to construct our desired functor. Let's start by defining it on the objects; extending it to morphisms will be a bit tricky.

Definition 11.2

Let $A \in C^*Alg$. We define:

$$F(A) \stackrel{\text{def}}{=} GL_\infty(A) / GL_\infty(A)_0.$$

That is, we take the infinite invertibles modulo those elements homotopic to the identity.

Here's a simple calculation.

Example 11.3. So, here's the claim.

| **Claim.** $F(\mathbb{C}) \sim 0$.

It's sufficient to show that every $a \in GL_\infty(\mathbb{C})$ is homotopic to the identity. But this reduces to showing that every $a \in GL_n(\mathbb{C})$ is homotopic to the identity. Now, this requires some holomorphic functional calculus:

$$\forall t \in [0, 1] : z_t \stackrel{\text{def}}{=} a^t,$$

where the right-hand side is defined using holomorphic functional calculus as applied to some specific branch of the logarithm^a. This forms a path in $GL_n(\mathbb{C})$ from a to 1_n so we're done^b.

^aWe actually established the holomorphic functional calculus with full details in Operator Algebraic Structures I.

^bIt is possible to give other clean, linear algebraic proofs that rely on simpler continuity facts about determinants.

An alternative approach to this definition is discussed in the next exercise. The idea is that since $\mathbb{C}$ -Banach algebras have a natural topological structure, their general linear groups also have that natural topological structure and this means that GL_∞ of a $\mathbb{C}$ -Banach algebra should have a natural topological structure. We will account for this structure.

Claim 11.3. Let $A \in C^*Alg$. We topologize $GL_\infty(A)$ by giving it the strongest topology so that all inclusions:

$$GL_n(A) \rightarrow GL_\infty(A),$$

are continuous. TFSH:

1. $GL_\infty(A)$ is a topological group.
2. $GL_\infty(A) \simeq \text{colim}(GL_n(A))$ in $\text{Grp}(\text{Top})$.
3. If $a, b \in GL_\infty(A)$, then a and b are homotopic iff they are homotopic w.r.t the canonical topology.
4. $GL_\infty(A)$ is locally path-connected.
5. $GL_\infty(A)$ is the path component of the identity.
6. If $G \in \text{Grp}(\text{Top})$, then $\pi_0(G)$ is naturally isomorphic to the quotient G/G_0 , where G_0 is the path-connected component of the identity. In particular, we see that G_0 is a normal subgroup of G .
7. $F(A) \simeq \pi_0(GL_\infty(A))$.

We have a remark about this exercise.

Remark 11.4. The exercise may look innocent but it is actually a small indication of how we should think about higher K-groups in algebraic K-theory. Indeed, we will see that we should think of them as the “higher homotopy groups of spaces”; what this exactly entails remains to be seen. However, one should note that some of this work will form **some** motivation for our later work in classical homotopy theory & our bid to fix the issues in classical homotopy theory will eventually require us to work in the realm of higher categories.

Now, let us establish that we actually have a functor.

Construction 11.5. Let $A, B \in \mathbf{C^*Alg}$ and let $f \in \mathbf{C^*Alg}(A, B)$. This map induces a group homomorphism:

$$\mathrm{GL}_n(f) : \mathrm{GL}_n(A) \rightarrow \mathrm{GL}_n(B),$$

and by universal property, we obtain a group homomorphism:

$$F(f) : F(A) \rightarrow F(B),$$

which gives us a functor $F : \mathbf{C^*Alg} \rightarrow \mathbf{AbGrp}$.

Now, our goal is to show that the special functor F “is just” K_1 on $\mathbf{C^*Alg}$. Actually, we’ll show the slightly weaker claim that K_1 and F are naturally isomorphic. However, on objects, this just means that they are literally isomorphic in $\mathbf{C^*Alg}$ and this entails that we have an automatic standard portrait for topological K_1 on $\mathbb{C}$ -Banach algebras. To do this, we will need the following non-unital generalization of Whitehead’s Lemma; see Proposition 5.4.

Proposition 11.6. Non-Unital Whitehead Lemma

Let $R, S \in \mathbf{Rng}$, let $\phi \in \mathbf{Rng}(R, S)$ be surjective and let $a \in \mathrm{GL}_n(S)$. Then, $a \oplus a^{-1}$ lifts to $\mathrm{GL}_{2n}(R)$.

Proof. The proof is a reduction to the unital case. Since ϕ is surjective, we see that $\phi^\sim$ is surjective so we apply Whitehead’s Lemma and conclude that $a \oplus a^{-1}$ lifts to $\mathrm{GL}_{2n}(R^\sim)$. That is, there exists a $v \in \mathrm{GL}_{2n}(R^\sim)$ such that:

$$\phi(v) = a \oplus a^{-1}.$$

Now, we define:

$$u \stackrel{\text{def}}{=} vs(v^{-1}),$$

where $s : R^\sim \rightarrow R^\sim$ is the scalar map. Now, we see that^a:

$$\begin{aligned} \phi(u) &= \phi(vs(v^{-1})) \\ &= \phi(v)\phi(s(v^{-1})) \\ &= \phi(v) \\ &= a \oplus a^{-1}, \end{aligned}$$

as was desired. ■

^aSee the last part of [Construction 4.24](#).

Okay, let’s state the result we wish to prove.

Theorem 11.7

F is naturally isomorphic to $K_1 = K_0 \circ \text{Sus}$ on $\mathbf{C^*Alg}$.

Our first step will be to construct the natural isomorphism and show naturality. We do this separately because we want to focus on showing invertibility in the actual proof of this result.

Construction 11.8. Let $A \in \mathbf{C^*Alg}$, let $u \in \mathrm{GL}_n(A^\sim)$ and consider the split-exact sequence:

$$0 \longrightarrow A \xrightarrow{\iota} A^\sim \begin{array}{c} \xrightarrow{p} \\ \xleftarrow{\varepsilon} \end{array} \mathbb{C} \longrightarrow 0$$

where $p \circ \varepsilon = \text{Id}_{\mathbb{C}}$. Now, we define the scalar map as:

$$s : A^\sim \rightarrow A^\sim, (x, t) \mapsto s(x, t) \stackrel{\text{def}}{=} (\varepsilon \circ p)(x, t) \stackrel{\text{def}}{=} t \cdot 1_{A^\sim}.$$

Of course, we defined this explicitly in our treatment of unitizations of $\mathbb{C}$ -Banach algebras in *Operator Algebraic Structures I*; we’ve also defined the analogous map for rings in [Construction 4.24](#). Since s is a unital $\star$ -homomorphism, it maps invertibles to invertibles. Now, we have the following claim.

Claim. We can find a $v \in \mathrm{GL}_\infty(A^\sim)$ homotopic to u such that $s(v) = 1_n$.

Indeed, we note that $s(u) = k1_{A^\sim}$ for some $k \neq 0$. Now, just let $v = \frac{1}{k}u$. By Whitehead’s Lemma, we see that u and v are homotopic in $\mathrm{GL}_\infty(A^\sim)$. and the claim is proved. Now, by the topological version of Whitehead’s Lemma^a, we know that there exists a path $(z_t)_{t \in [0,1]}$ from $u \oplus u^{-1}$ to 1_{2n} in $\mathrm{GL}_{2n}(A^\sim)$. Here’s our second claim.

Claim. We can arrange for the path to satisfy:

$$\forall t \in [0, 1] : s(z_t) = 1_{2n}.$$

In particular, the path is normalized under the scalar map.

Indeed, we define:

$$\forall t \in [0, 1] : z'(t) \stackrel{\text{def}}{=} s(z_t)^{-1} z_t.$$

Clearly, this function has the desired properties. Now, we define:

$$\forall t \in [0, 1] : p_t = z_t \begin{pmatrix} 1_n & 0 \\ 0 & 0 \end{pmatrix} z_t^{-1},$$

and we denote $p = (p_t)_{t \in [0, 1]}$. By direct calculation, we see that each p_t is an idempotent. Collectively, we have an idempotent in $\text{Mat}_\infty(\text{Sus}(A)^\sim)$. So, this gives us the idea to define our map $\theta_A : K_1(A) \rightarrow K_0(\text{Sus}(A))$ by:

$$\theta_A([u]) \stackrel{\text{def}}{=} [p]_s - [s(p)]_s.$$

where $[u]$ is the equivalence class of u in K_1 . Now, we need to check that this is well-defined and we can conjure up a natural transformation from this construction. Both of these are handled by the next claim.

Claim. θ_A is well-defined. Also, the family:

$$\theta_A : K_1(A) \rightarrow K_0(\text{Sus}(A)),$$

forms the components of a natural transformation $\theta : K_1 \Rightarrow K_0 \circ \text{Sus}$.

We'll deal with well-definedness and naturality separately.

Well-Definedness:

Naturality:

This completes the proof of the claim.

^aSee Lemma 7.21.

With this construction under our belts, we can show that our special functor is nothing but K_1 .

Proof of Theorem 11.7. We simply need to show the injectivity and surjectivity of θ_A .

Injectivity:

Surjectivity:

This completes the proof. ■

So, in particular, this result allows us to simply calculate K_1 on the objects by using the standard description of F . For instance, we have the following nice corollary.

12 Properties of Operator K-Theory & Some Computations

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| Motivation.

12.1 The Index Map of K-Theory

12.2 Further K-Theory Computations

So, how do we get higher operator K-theory? Well, we'll start by proving that K_0 , when composed with iterated suspensions, actually nearly produces a C^* -homology theory. As we hinted in ??, extending the long-exact sequence we obtain in the following corollary to the right involves having the Bott Periodicity Theorem in our hands.

Corollary 12.1

The family $(K_0 \circ \text{Sus}^n)_{n \in \mathbb{N}_0}$ is a C^* -homology theory.

Proof. Since K_0 is homotopy-invariant and half-exact on $C^*\text{Alg}$, we can use ?? to construct a long-exact sequence out of the functors formed by composing with the iterated suspension. Since $K_0 \circ \text{Sus}^n$ is homotopy-invariant for each $n \in \mathbb{N}_0$, we see that this sequence gives rise to a C^* -homology theory. ■

This motivates our definition of the higher K-functors.

Definition 12.2

For each $n \in \mathbb{N}$, we define:

$$K_n \stackrel{\text{def}}{=} K_0 \circ \text{Sus}^n,$$

with K_n being the n -th operator K-functor^a.

^aWe will often refer to these functors as the higher operator K-functors or higher operator K-theory.

Here's a rather neat result which is a consequence of our previous labour.

Proposition 12.3

Let $n \geq 1$. TFSH:

1. K_n is homotopy-invariant.
2. K_n is half-exact.
3. K_n is split-exact.
4. K_n is C^* -stable.

In particular, $(K_n)_{n \in \mathbb{N}_0}$ is Bott periodic, with periodicity 2.

Proof. We'll prove each statement one-by-one.

1. Since Sus is homotopy-invariant, it follows that Sus^n is homotopy-invariant for each $n \in \mathbb{N}$. This implies that K_n is homotopy-invariant since K_0 is homotopy-invariant.
2. Since Sus is an exact functor, we see that Sus^n is an exact functor. But the half-exactness of K_0 ensures that K_n remains half-exact.
3. Again, the exactness of Sus ensures that split-exact sequences get taken to split-exact sequences, which means that Sus^n is also a split-exact functor. Since K_0 is split-exact, it follows that K_n is split-exact for each $n \in \mathbb{N}$.
4. This follows from the associativity, up to isomorphism, of tensor products along with the C^* -stability of K_0 .

The last statement follows from the fact that any sequence of functors which are homotopy-invariant, split-exact and C^* -stable will be Bott periodic. ■

The fact that operator K-theory is Bott periodic (with periodicity 2) means that, for the purpose of calculating K-groups, we just need to always focus on operator K_0 and K_1 groups. In the next two examples, we discuss two facts about operator K_1 .

Example 12.4. Here's the main claim.

| **Claim.** $K_1(\mathbb{C}) \simeq 0$.

Indeed, we have that:

$$\begin{aligned} K_1(\mathbb{C}) &= K_0(\text{Sus}(\mathbb{C})) \\ &\simeq K_0(\mathcal{C}_0(0, 1)) \\ &\simeq K^0(0, 1). \end{aligned}$$

Since $(0, 1)$ is contractible, we see that the far right-hand side above is trivial and we're done.

Moreover, it is also the case that operator K_1 does not “see” unitization.

Example 12.5. Let $A \in C^*\text{Alg}$.

| **Claim.** Let $A \in C^*\text{Alg}$ and $\iota : A \rightarrow A^\sim$ is the canonical inclusion, then $K_1(\iota)$ is an isomorphism.

We consider the unitization split-exact sequence:

$$0 \longrightarrow A \xrightarrow{\iota} A^\sim \xrightarrow{p} \mathbb{C} \longrightarrow 0$$

Since K_1 is split-exact, we see that:

$$0 \longrightarrow K_1(A) \xrightarrow{K_1(\iota)} K_1(A^\sim) \xrightarrow{K_1(p)} K_1(\mathbb{C}) \longrightarrow 0$$

is split-exact. But since $K_1(\mathbb{C}) \simeq 0$, we have our result.

Here's a remark.

Remark 12.6. One might be tempted to think that operator K_1 is the same as algebraic K_1 . However, this is far from being the case but it is not obvious at all. We will show that this is the case by coming up with a more concrete characterization of K_1 . In fact, we will see that those characterizations make this obvious. The main difficulty is in showing that this alternative characterization is actually equivalent to the one presented above.

Even though we gave an abstract proof of Bott Periodicity before, we did not use it to compute the K-theory of interesting C^* -algebras, We will fix that in this section by providing a number of interesting examples. We strongly recommend working through them in detail.

Example 12.7. If $A \in C^*\text{Alg}$, then one notes that:

$$\text{Sus}(A) \simeq_{C^*\text{Alg}} \mathcal{C}_0(\mathbb{R}, A),$$

where the right-hand side is a C^* -algebra^a. Then, we have a claim.

Claim. We have that:

$$\forall n \in \mathbb{N}_0 : K_0(\mathcal{C}_0(\mathbb{R}^n)) \simeq \begin{cases} \mathbb{Z} & \text{if } n \text{ is even,} \\ 0 & \text{if } n \text{ is odd} \end{cases}.$$

From this, we also determine $K_1(\mathcal{C}_0(\mathbb{R}^n))$ for each $n \in \mathbb{N}_0$.

Observe that for each $n \in \mathbb{N}_0$, we have that:

$$K_0(\mathcal{C}_0(\mathbb{R}^n)) \simeq K_0(\text{Sus}^n(\mathbb{C})).$$

If n is even, then this reduces to computing:

$$K_0(\mathbb{C}) \simeq \mathbb{Z}.$$

If n is odd, then this reduces to computing:

$$K_1(\mathbb{C}) \simeq 0,$$

by the Bott Periodicity Theorem so we're done with the first part of the claim. For the second part, we note that:

$$K_1(\mathcal{C}_0(\mathbb{R}^n)) = K_0(\mathcal{C}_0(\mathbb{R}^{n+1})),$$

for each $n \in \mathbb{N}_0$. If n is even, then $n+1$ is odd so this group is just 0. If n is odd, then $n+1$ is even so this group is just $\mathbb{Z}$. In other words, we deduce the following fact about the K_1 -groups:

$$\forall n \in \mathbb{N}_0 : K_1(\mathcal{C}_0(\mathbb{R}^n)) = \begin{cases} \mathbb{Z} & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even} \end{cases},$$

as was desired.

^aWe have presented this explicitly as a construction in Operator Algebraic Structures I.

Example 12.8. Our goal, in this example, is to compute the K-groups of $\mathcal{C}(S^n)$ for all $n \in \mathbb{N}$. The main strategy that gets used in practice is to find, for each $n \in \mathbb{N}$, a suitable split-exact sequence where $\mathcal{C}(S^n)$ is in the middle and is surrounded by algebras whose K-theory is “easy to compute” or has been computed. So far, we haven't done many computations so we'll want a split-exact sequence of the following form:

$$0 \longrightarrow \mathcal{C}_0(\mathbb{R}^n) \longrightarrow \mathcal{C}(S^n) \longrightarrow \mathbb{C} \longrightarrow 0$$

In fact, this is precisely a unitization exact sequence; we know that S^n is the one-point compactification of $\mathbb{R}^n$ and since unitizations & compactifications correspond to one another, we obtain a split-exact structure on the sequence above by viewing it as a unitization sequence. So, this means that we get two split-exact sequences of groups:

$$0 \longrightarrow K_0(\mathcal{C}_0(\mathbb{R}^n)) \longrightarrow K_0(\mathcal{C}(S^n)) \longrightarrow K_0(\mathbb{C}) \longrightarrow 0$$

$$0 \longrightarrow K_1(\mathcal{C}_0(\mathbb{R}^n)) \longrightarrow K_1(\mathcal{C}(S^n)) \longrightarrow K_1(\mathbb{C}) \longrightarrow 0$$

Now, we have to deal with two cases, using the result of Example 12.7.

n is odd: If n is odd, then we see that:

$$K_0(\mathcal{C}_0(\mathbb{R}^n)) \simeq 0.$$

By the first sequence above, we see that:

$$K_0(\mathcal{C}(S^n)) \simeq K_0(\mathbb{C}) \simeq \mathbb{Z}.$$

On the other hand, we know that:

$$K_1(\mathbb{C}) \simeq 0,$$

so the second sequence above tells us that:

$$K_1(\mathcal{C}(S^n)) \simeq K_1(\mathcal{C}_0(\mathbb{R}^n)) \simeq \mathbb{Z},$$

as was desired.

n is even: If n is even, then we see that:

$$K_0(\mathcal{C}_0(\mathbb{R}^n)) \simeq \mathbb{Z},$$

which implies, by the first sequence above, that:

$$K_0(\mathcal{C}(S^n)) \simeq \mathbb{Z} \oplus \mathbb{Z}.$$

On the other hand, we know that:

$$K_1(\mathcal{C}_0(\mathbb{R}^n)) \simeq 0,$$

which means that $K_1(\mathcal{C}(S^n)) \simeq K_1(\mathbb{C}) \simeq 0$.

This gives us the following computation of the K -groups of spheres for each $n \in \mathbb{N}$:

$$K_0(\mathcal{C}(S^n)) \simeq \begin{cases} \mathbb{Z} & \text{if } n \text{ is odd,} \\ \mathbb{Z} \oplus \mathbb{Z} & \text{if } n \text{ is even,} \end{cases}$$

$$K_1(\mathcal{C}(S^n)) \simeq \begin{cases} \mathbb{Z} & \text{if } n \text{ is odd,} \\ 0 & \text{if } n \text{ is even,} \end{cases}$$

as was desired.

Example 12.9. With this example, our goal is to compute:

$$K_*(\mathcal{C}(\mathbb{T}^n)),$$

for each $n \in \mathbb{N}$. Let us first note that when $n = 1$, then $\mathbb{T}$ is homeomorphic to S^1 . So:

$$K_0(\mathcal{C}(\mathbb{T})) \simeq \mathbb{Z} \simeq K_1(\mathcal{C}(\mathbb{T})).$$

Now, the idea is to find a split-exact sequence:

$$0 \longrightarrow \text{Sus}(\mathcal{C}(\mathbb{T}^n)) \longrightarrow \mathcal{C}(\mathbb{T}^{n+1}) \longrightarrow \mathcal{C}(\mathbb{T}^n) \longrightarrow 0$$

If we take the existence of this split-exact sequence for granted, then we get two split-exact sequences:

$$0 \longrightarrow K_0(\text{Sus}(\mathcal{C}(\mathbb{T}^n))) \longrightarrow K_0(\mathcal{C}(\mathbb{T}^{n+1})) \longrightarrow K_0(\mathcal{C}(\mathbb{T}^n)) \longrightarrow 0$$

$$0 \longrightarrow K_1(\text{Sus}(\mathcal{C}(\mathbb{T}^n))) \longrightarrow K_1(\mathcal{C}(\mathbb{T}^{n+1})) \longrightarrow K_1(\mathcal{C}(\mathbb{T}^n)) \longrightarrow 0$$

These two split-exact sequences imply that:

$$K_0(\mathcal{C}(\mathbb{T}^{n+1})) \simeq K_0(\text{Sus}(\mathcal{C}(\mathbb{T}^n))) \oplus K_0(\mathcal{C}(\mathbb{T}^n)) \simeq K_1(\mathcal{C}(\mathbb{T}^n)) \oplus K_0(\mathcal{C}(\mathbb{T}^n)),$$

$$K_1(\mathcal{C}(\mathbb{T}^{n+1})) \simeq K_1(\mathcal{C}(\mathbb{T}^n)) \oplus K_1(\text{Sus}(\mathcal{C}(\mathbb{T}^n))) \simeq K_1(\mathcal{C}(\mathbb{T}^n)) \oplus K_0(\mathcal{C}(\mathbb{T}^n)),$$

where we have used the Bott Periodicity Theorem in the above simplification. From this, we see that^a:

$$\forall n \in \mathbb{N} : K_0(\mathcal{C}(\mathbb{T}^n)) \simeq K_1(\mathcal{C}(\mathbb{T}^n)),$$

so it is enough to focus our attention on the K_0 -groups of $\mathcal{C}(\mathbb{T}^n)$ for all $n \in \mathbb{N}$. In that case, we will compute things explicitly using the inductive formula above in the case of $n = 1$ and $n = 2$ so we can come up with an appropriate general claim. Then, we will prove that using induction. With that said, let us proceed with the cases.

$n = 1$: In this case, we see that:

$$K_0(\mathcal{C}(\mathbb{T}^2)) \simeq K_1(\mathcal{C}(\mathbb{T})) \oplus K_0(\mathcal{C}(\mathbb{T})) \simeq \mathbb{Z}^2,$$

as was desired.

$n = 2$: In this case, we see that:

$$K_0(\mathcal{C}(\mathbb{T}^3)) \simeq K_1(\mathcal{C}(\mathbb{T}^2)) \oplus K_0(\mathcal{C}(\mathbb{T}^2)) \simeq \mathbb{Z}^2 \oplus \mathbb{Z}^2 \simeq \mathbb{Z}^4,$$

as was desired.

So, the next claim reflects the natural guess.

Claim. $\forall n \in \mathbb{N} : K_0(\mathcal{C}(\mathbb{T}^n)) \simeq \mathbb{Z}^{2^{n-1}}$.

We'll proceed by induction. Again, previous considerations yield that this holds in the base case. Now, assume that it holds for some $n \geq 1$. Then, we have the following calculation:

$$\begin{aligned} K_0(\mathcal{C}(\mathbb{T}^{n+1})) &\simeq K_1(\mathcal{C}(\mathbb{T}^n)) \oplus K_0(\mathcal{C}(\mathbb{T}^n)) \\ &\simeq K_0(\mathcal{C}(\mathbb{T}^n)) \oplus K_0(\mathcal{C}(\mathbb{T}^n)) \\ &\simeq \mathbb{Z}^{2^{n-1}} \oplus \mathbb{Z}^{2^{n-1}} \\ &\simeq \mathbb{Z}^{2^n}, \end{aligned}$$

which completes the proof. Unfortunately, we aren't done yet. Throughout this (mostly very formal) calculation, we simply took the existence of our split-exact sequence for granted. But actually, we need to show that exists and that yields the final claim of this example.

Claim. Our desired split-exact sequence exists.

The second map is a projection by setting the $(n + 1)$ -th coordinate to 0. The first map is actually an inclusion via currying. Since the spaces in question are “nice enough”, these maps are actually continuous algebra homomorphisms so we're done.

^aOf course, this uses an inductive argument and the base case needs to be checked. But the base case holds by previous considerations

| (and more specifically, the calculation in the preceding example).

Now, there's one more piece of structure that we would like to discuss on operator K-theory, though this structure seems to only be available when K-theory is restricted to commutative C^* -algebras. This is the so-called cup product and its description is actually very simple. If $X, Y \in \text{CHaus}$, then what we want is a map:

$$K^0(X) \times K^0(Y) \rightarrow K^0(X \times Y).$$

We've actually already seen a similar type of construction with the algebraic K-theory of unital commutative rings and so, we dare to hope that the analogue of such a construction might hold in our setting. In fact, this is the case but it is a rather unfortunate fact that descriptions of this cup product rely on the vector bundle picture of topological K-theory. There is, in fact, a simpler C^* -algebraic picture that we present now.

Construction 12.10. Let $X, Y \in \text{CHaus}$. Then, we have two projection maps:

$$\pi_X : X \times Y \rightarrow X,$$

$$\pi_Y : X \times Y \rightarrow Y.$$

Now, we apply the continuous maps functor so we obtain morphisms:

$$\mathcal{C}(\pi_X) : \mathcal{C}(X) \rightarrow \mathcal{C}(X \times Y),$$

$$\mathcal{C}(\pi_Y) : \mathcal{C}(Y) \rightarrow \mathcal{C}(X \times Y).$$

Since these are $\star$ -homomorphisms, we can hit these morphisms with K-theory to obtain:

$$K_0(\mathcal{C}(\pi_X)) : K^0(X) \rightarrow K^0(X \times Y),$$

$$K_0(\mathcal{C}(\pi_Y)) : K^0(Y) \rightarrow K^0(X \times Y).$$

We can now form the cup-product map by the universal property of the direct sum^a:

$$\phi : K^0(X) \oplus K^0(Y) \rightarrow K^0(X \times Y).$$

as was desired. If $x \in K^0(X)$ and $y \in K^0(Y)$, then we define:

$$x \cup y \stackrel{\text{def}}{=} \phi(x, y),$$

and refer to this as the cup product of x & y .

Let us discuss the special case when $X = Y$ as spaces. Then, we have a diagonal map:

$$\Delta_X : X \rightarrow X \times X,$$

and hitting this with K-theory gives us:

$$K^0(\Delta_X) : K^0(X \times X) \rightarrow K^0(X).$$

Now, we define the product on $K^0(X)$ by the following composition:

$$K^0(\Delta_X) \circ \phi : K^0(X) \oplus K^0(X) \rightarrow K^0(X).$$

This map is known as the multiplicative structure on $K^0(X)$.

^aOf course, we could've hit the two projection maps with topological K^0 right from the start but the process above is a bit important to see explicitly at least once, especially since the downside of this is that we don't immediately have a formula for the cup product.

One needs to ensure that the cup product (and therefore, the multiplicative structure) satisfies reasonable properties, like associativity and commutativity. Well, we would like to phrase this for the general cup product and we will have to do this by constructing commuting diagrams. This will be done in the next two constructions.

Construction 12.11. Let $X, Y, Z \in \text{CHaus}$. We want to form the following commutative diagram:

$$\begin{array}{ccc} K^0(X) \oplus K^0(Y) \oplus K^0(Z) & \longrightarrow & K^0(X \times Y) \oplus K^0(Z) \\ \downarrow & & \downarrow \\ K^0(X) \oplus K^0(Y \times Z) & \longrightarrow & K^0(X \times Y \times Z) \end{array}$$

Intuitively, what's happening is that along the top part of the square above, we have the map:

$$(x, y, z) \mapsto (x \cup y, z) \mapsto (x \cup y) \cup z,$$

and along the bottom half, we have that:

$$(x, y, z) \mapsto (x, y \cup z) \mapsto x \cup (y \cup z),$$

and the commutativity just means that both paths yield the same result. Now, let us describe each map:

1. The top map is formed by gluing the cup product of X and Y with $\text{Id}_{K^0(Z)}$.
2. The bottom map is simply a cup product, where we implicitly use that $X \times (Y \times Z) \simeq (X \times Y) \times Z$.
3. The left map is formed by gluing the cup product of Y and Z with $\text{Id}_{K^0(X)}$.
4. The right map is simply a cup product, where we implicitly use that $X \times (Y \times Z) \simeq (X \times Y) \times Z$.

With these constructions in mind, we simply note that the commutativity of the direct sum ensures that the construction above is entirely symmetric along both paths in the diagram. So, it follows that the diagram must commute. So, the cup product and the multiplicative structure on $K^0(X)$, for any $X \in \text{CHaus}$, are both associative.

Construction 12.12. Let $X, Y \in \text{CHaus}$. We want to form the following commutative diagram:

$$\begin{array}{ccc} K^0(X) \oplus K^0(Y) & \longrightarrow & K^0(X \times Y) \\ \downarrow & & \downarrow \\ K^0(Y) \oplus K^0(X) & \longrightarrow & K^0(Y \times X) \end{array}$$

We leave it as a very instructive exercise, one which is similar to what we did in [Construction 12.11](#), to construct the maps in the diagram above and show that this diagram does, indeed, commute. So, the cup product and the multiplicative structure on $K^0(X)$, for any $X \in \text{CHaus}$, are both commutative.

We're well aware that we haven't done anything with the cup product on K-theory and this is mainly because there is nothing to do with it at this point. It is simply a useful piece of structure which plays the role of a tool and we will eventually use it to compute the K-theories of some well-known spaces; that part of the discussion will mostly be classical and normal. The cup product will be used in a more interesting way when we eventually start discussing C^* -superrigidity. For now, you should just bear in mind that this is an interesting piece of structure on K-theory, that we will use this as a computational tool and that we might possibly want to generalize this at some point in the distant future⁹.

⁹That last point will be elaborated upon when we've proved our superrigidity theorem.

13 Filtration of Operator K-Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

13.1 Constructing The Spherical Filtration of K-Theory

13.2 Computing Some Filtrations

14 The Pimsner-Voiculescu Sequence

Summary

In this lecture, we will cover the following topics:

1. The statement of the Pimsner-Voiculescu Theorem.
2. Two proofs of the Pimsner-Voiculescu Theorem.

Our main goal is to introduce a K-theory comp

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

We will provide motivation in Section 14.1, so we will not provide separate motivation here.

14.1 The Statement of the Pimsner-Voiculescu Theorem

Our main goal is to prove the following theorem.

Theorem 14.1. The Pimsner-Voiculescu Theorem

Let $A \in C^* \text{Alg}$, let $\alpha \in \text{Aut}(A)$ and let $\iota : A \rightarrow A \times_{\alpha} \mathbb{Z}$ be the canonical inclusion. Then, we have a diagram:

$$\begin{array}{ccccc} K_0(A) & \xrightarrow{\text{Id} - K_0(\alpha)} & K_0(A) & \xrightarrow{K_0(\iota)} & K_0(A \times_{\alpha} \mathbb{Z}) \\ \uparrow & & & & \downarrow \\ K_1(A \times_{\alpha} \mathbb{Z}) & \xleftarrow{K_1(\iota)} & K_1(A) & \xleftarrow{1 - K_1(\alpha)} & K_1(A) \end{array}$$

which is a cyclic 6-term exact sequence in K-theory. Moreover:

$$\forall i \in \{0, 1\} : K_i(A \times_{\alpha} \mathbb{Z}) \simeq K_{1-i}(M(A, \alpha)),$$

where $M(A, \alpha)$ is the mapping torus of α .

Well, the mapping torus is easy to define, so let us just define it.

Construction 14.2. Let $A \in C^* \text{Alg}$ and let $\alpha \in \text{Aut}(A)$. We define:

$$M(A, \alpha) \stackrel{\text{def}}{=} \{f \in \text{Cyl}(A) : f(1) = \alpha(f(0))\}.$$

This is known as the mapping torus of A with respect to α .

Here are a few remarks.

Remark 14.3. Let $A = \mathbb{C}$ and let $\alpha = \text{Id}_A$. Then:

$$M(A, \alpha) = \mathcal{C}(S^1),$$

which explains the name.

Remark 14.4. Well, one might wonder why we work with a single automorphism in the theorem above. The point is that an action $\mathbb{Z} \curvearrowright A$ by automorphisms is simply given by the data of a single automorphism; we simply evaluate the action at $n = 1$ and take iterated powers of the resulting automorphism to obtain the whole action. Of course, this has been explained in [REF] but it is reasonable to revisit this again.

Remark 14.5. The statement of [Theorem 14.1](#) makes it clear that the K-theory of the crossed product depends only on the homotopy class of the action. Moreover, that statement indicates that we should obtain a more conceptual understanding of the mapping torus and we will do that by the end of this section. Indeed, one thing we will need to explain is why the mapping torus even appears in the statement of the theorem.

Here are some basic exercises regarding the mapping torus.

Claim 14.1. Let $A \in C^* \text{Alg}$ and let $\alpha \in \text{Aut}(A)$. Then, we have that:

$$M(A, \alpha) \simeq \{f \in \mathcal{C}_b(\mathbb{R}, A) \mid \forall x \in \mathbb{R} : f(x+1) = \alpha(f(x))\},$$

where this isomorphism is canonical.

Claim 14.2. Let $A \in C^* \text{Alg}$ and let $\alpha \in \text{Aut}(A)$. We define $\alpha^{\sim} : \mathbb{R} \rightarrow \text{Aut}(\mathcal{C}_b(\mathbb{R}, A))$ as follows:

$$\forall x \in \mathbb{R} : \forall f \in \mathcal{C}_b(\mathbb{R}, A) : \forall t \in \mathbb{R} : \alpha^{\sim}(x)(f)(t) \stackrel{\text{def}}{=} f(t-x).$$

This is, in fact, an action on $\mathcal{C}_b(\mathbb{R}, A)$ and it restricts to an action by $\mathbb{R}$ on $M(A, \alpha)$.

Claim 14.3. Let $A \in C^* \text{Alg}$ and let $\alpha \in \text{Aut}(A)$. Then, the following sequence:

$$0 \rightarrow \text{Sus}(A) \rightarrow M(A, \alpha) \rightarrow A \rightarrow 0,$$

is a short-exact sequence of C^* -algebras.

Here is a remark.

Remark 14.6. So, [Claim 14.2](#) gives an explanation as to why the mapping torus might, at first sight, be relevant in the Pimsner-Voiculescu Theorem. Indeed, the point is that the data of [Theorem 14.1](#) naturally gives rise to an $\mathbb{R}$ - C^* -algebra in the form of the mapping torus. This increases the amount of evidence that the Pimsner-Voiculescu Theorem has a “reasonable” statement.

14.2 Two Proofs of the Pimsner-Voiculescu Theorem

14.2.1 Proof I: A Hands-On Approach

14.2.2 Proof II: An Approach Via The Connes-Thom Isomorphism

The second approach to the Pimsner-Voiculescu Theorem proceeds via the following theorem.

Theorem 14.7. The Connes-Thom Isomorphism, [28]

Let $A \in \mathbf{C}^*\mathbf{Alg}$ and let $\alpha \in \text{Aut}(A)$. Then, there exist natural isomorphisms:

$$K_i(A \times_{\alpha} \mathbb{R}) \simeq K_{1-i}(A),$$

for $i \in \{0, 1\}$.

We will prove this theorem within the next two lectures; it is a beautiful result which is an amalgamation of some neat pieces of algebra, geometry and analysis. Let us, nevertheless, provide some basic intuition for the theorem.

Intuition 14.8. Since $\mathbb{R}$ is contractible, any action of $\mathbb{R}$ is homotopic to the trivial action and their crossed products should have the same K-theory. In fact, this is not enough because homotopic actions do not produce homotopic crossed products. However, this intuition will play a role in the proof, as you will find out in the next two lectures.

Now, we are ready to provide the second proof of the Pimsner-Voiculescu Theorem.

Proof II of Theorem 14.1.

■

15 The Connes-Thom Isomorphism

Summary

In this lecture, we will discuss the following topics:

1. The uniqueness assertion.
2. The existence assertion.

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

| **Motivation.**

15.1 The Uniqueness Assertion

15.2 The Existence Assertion

16 Constructing Algebraic K_1

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

Motivation.

16.1 A Computational Description of Algebraic K_1

We will begin by establishing some notation that will be in effect throughout this lecture.

Notation 16.1. Let $R \in \text{Rng}^1$.

1. If $n \in \mathbb{N}$, then the set of invertible $n \times n$ -matrices over R is denoted by $\text{GL}(n, R)$.
2. We define, for each $n \in \mathbb{N}$, the following map:

$$\phi_n : \text{GL}(n, R) \rightarrow \text{GL}(n+1, R), \quad a \mapsto \phi_n(a) \stackrel{\text{def}}{=} \begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix},$$

which is known as the left upper-corner inclusion^a.

3. If $a \in R$ and $n \in \mathbb{N}$ with $1 \leq i, j \leq n$ and $i \neq j$, then we define an $n \times n$ matrix $e_{ij}(a)$ which is just 1 on the diagonal, a on the (i, j) -th entry and zero everywhere else. A matrix of this type is called primary^b.
4. The set of $n \times n$ -primary matrices over R is denoted by $P(n, R)$.

Finally, the inductive limit of the following sequences:

$$\text{GL}(1, R) \xrightarrow{\phi_1} \text{GL}(2, R) \rightarrow \dots,$$

$$P(1, R) \xrightarrow{\phi_1} P(2, R) \rightarrow \dots,$$

in Rng^1 are the rings $\text{GL}_\infty(R)$ and $P(R)$ respectively, the latter being known as the group of primary matrices.

^aOne can write an explicit inverse for $\phi_n(a)$ using the inverse for a .

^bIn [87, Definition 2.1.1, p.59], the word “elementary” is used. However, the notion of an elementary matrix is already known from linear algebra and that's not what is being described here, so we do not use that terminology here.

The next result summarizes a number of easy matrix identities.

Proposition 16.2

Let $R \in \text{Rng}^1$, let $a, b \in R$, let $n \in \mathbb{N}$ and let $1 \leq i, j, k \leq n$. TFSH:

1. $e_{ij}(a)e_{ij}(b) = e_{ij}(a+b)$. In particular, primary matrices are invertible.
2. $e_{ij}(a)e_{kl}(b) = e_{kl}(b)e_{ij}(a)$ when $j \neq k$ and $i \neq l$.
3. $e_{ij}(a)e_{jk}(b)e_{ij}(a)^{-1}e_{jk}(b)^{-1} = e_{ik}(ab)$ for i, j, k distinct.
4. $e_{ij}(a)e_{ki}(b)e_{ij}(a)^{-1}e_{ki}(b)^{-1} = e_{kj}(-ba)$ for i, j, k distinct.

Furthermore, any upper-triangular or lower triangular matrix with 1's on the diagonal belongs to $E(R)$.

Proof. We'll prove each statement one-by-one.

1. Now, let's show that primary matrices are invertible; let $1 \leq i, j \leq n$ and let $a \in R$. Then, we see that:

$$e_{ij}(a)e_{ij}(-a) = e_{ij}(a + (-a)) = e_{ij}(0) = \text{Id}_n.$$

Since $a + (-a) = 0 = (-a) + a$, it follows that $e_{ij}(a)$ is invertible. That is, primary matrices are invertible.

- 2.
- 3.
- 4.

Finally, let's deal with the last statement. ■

A corollary of the result above is that if we have many invertible matrices, then we can always produce many elements in the group of primary matrices. The proof involves a standard matrix identity that we have previously encountered.

Corollary 16.3

Let $R \in \text{Rng}^1$ and let $A \in \text{GL}(n, R)$. Then, $\begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix} \in P(2n, R)$.

Proof. Observe that we can write:

$$\begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix} = \begin{pmatrix} 1 & A \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -A^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & A \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The first three factors on the right-hand side are (upper or lower) triangular matrices, so Corollary 16.3 tells us that they belong to $P(R)$. As for the last factor, we can re-write it in the following way:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

and the matrices on the right-hand side are triangular matrices so they lie in $P(R)$ by the aforementioned result. ■

At this stage, we could easily define K_1 on unital rings but life is a little bit simpler if we automatically had a simpler description of it available. So, we'll prove a version of Whitehead's Lemma first.

Proposition 16.4. Algebraic Whitehead's Lemma (K_1)

Let $R \in \text{Rng}^1$. TFSH:

1. $[\text{GL}_\infty(R), \text{GL}_\infty(R)] = [P(R), P(R)] = P(R)$.
2. $P(R)$ is normal in $\text{GL}_\infty(R)$.
3. The quotient $\text{GL}_\infty(R)/P(R)$ is the abelianization $\text{GL}_\infty(R)_{\text{ab}}$.

Proof. Statements (2) and (3) follow immediately from statement (1), so we will just prove statement (1). At this point, the way we do this is by relying on the matrix identities in Proposition 16.2.

| **Claim.** $[P(R), P(R)] \subseteq [\text{GL}_\infty(R), \text{GL}_\infty(R)]$.

Indeed, since $P(R) \subseteq \text{GL}_\infty(R)$, the claim holds.

| **Claim.** $P[R] = [P(R), P(R)]$.

Let $a \in R$. Then, statement (3) of Proposition 16.2 tells us that:

$$e_{ij}(a)e_{jk}(1)e_{ij}(a)^{-1}e_{jk}(1)^{-1} = e_{ik}(a),$$

where $1 \leq i, j, k \leq n$ are distinct. So, the point is that the This proves the claim.

| **Claim.** $[\text{GL}_\infty(R), \text{GL}_\infty(R)] \subseteq P(R)$.

By statement (1) of Proposition 16.2, we see that: ■

With this result, we proceed with the definition of K_1 for unital rings.

Definition 16.5

Let $R \in \text{Rng}^1$. We define:

$$K_1(R) \stackrel{\text{def}}{=} \text{GL}_\infty(R)_{\text{ab}},$$

where the right-hand side is described also in terms of primary matrices as in Proposition 16.4.

Our first priority is to cook up a functor.

Construction 16.6. Let $R, S \in \text{Rng}^1$ and let $\phi \in \text{Rng}^1(R, S)$. Here's our first claim.

| **Claim.** For each $n \in \mathbb{N}$, the matrix amplification of ϕ :

$$\phi^{(n)} : M_{n \times n}(R) \rightarrow M_{n \times n}(S),$$

restricts to a group homomorphism $\text{GL}(n, R) \rightarrow \text{GL}(n, S)$.

Well, this is simple; the matrix amplification is just a unital ring homomorphism, so it preserves invertible elements.

Hence, our restriction above makes sense. Here's our second claim.

| **Claim.** Using the restrictions of the matrix amplifications, we can induce a morphism:

$$\text{GL}(\phi) : \text{GL}_\infty(R) \rightarrow \text{GL}_\infty(S),$$

which restricts to each $\phi^{(n)}$ for each $n \in \mathbb{N}$.

This is actually an entirely formal colimit argument.

Now, we hit $\text{GL}(\phi)$ with the abelianization. In other words:

$$K_1(\phi) \stackrel{\text{def}}{=} \text{GL}(\phi)_{\text{ab}}.$$

This leads us to the final claim.

| **Claim.** The construction above produces a functor.

This is completely clear, because we form the restriction maps by matrix amplification (which is functorial), we form $\text{GL}(\phi)$ by taking colimits (which is functorial) and we form $\text{GL}(\phi)_{\text{ab}}$ by hitting $\text{GL}(\phi)$ with the abelianization (which is functorial).

So, we have successfully defined algebraic K_1 for unital rings; we will eventually deal with the situation over arbitrary rings but let us milk as much as we can from this definition. We proceed towards some calculations.

Example 16.7. Let $R \in \text{AbRng}^1$. We have the following major claim.

| **Claim.** The determinant:

$$\det_n : \text{GL}(n, R) \rightarrow R^\times,$$

extends to a split epi $\text{GL}_\infty(R) \rightarrow R^\times$ and gives a split epi:

$$K_1(R) \rightarrow R^\times,$$

of groups.

This example leads us to some notation.

Notation 16.8. Let $R \in \text{AbRng}^1$. We recall that:

$$\forall n \in \mathbb{N} : \text{SL}(n, R) \stackrel{\text{def}}{=} \{A \in \text{GL}(n, R) : \det(A) = 1\},$$

and we define:

$$\text{SL}(R) \stackrel{\text{def}}{=} \{A \in \text{GL}(R) : \det(A) = 1\}.$$

Since $E(R) \subseteq \text{SL}(R)$, we finally define:

$$\text{SK}_1(R) \stackrel{\text{def}}{=} \text{SL}(R)/E(R),$$

and refer to this as the special K_1 -group of R .

Okay, here are further examples.

Example 16.9. Let $\mathbb{F}$ be a field.

Claim. $K_1(\mathbb{F}) \simeq \mathbb{F}^\times$.

Example 16.10. Let $R \in \text{AbRng}^1$ be local.

Claim. The map:

$$\det : K_1(R) \rightarrow R^\times,$$

is an isomorphism.

We have a remark about this example.

Remark 16.11. It is possible to deal with the situation where R is not commutative in the preceding example. Indeed, there are local rings (unital rings with exactly one left maximal ideal) which arise from operator theory, so it seems reasonable to consider.

At this point, we would like to extend our construction of algebraic K_1 to arbitrary rings. For this, we will utilize ?? and ??, since they create a version of the invertible group of matrices over non-unital rings. With these constructions in mind, the next definition makes sense.

Definition 16.12

Let $R \in \text{Rng}$. We define:

$$K_1(R)^+ \stackrel{\text{def}}{=} \text{GL}_\infty(R)/[\text{GL}_\infty(R), \text{GL}_\infty(R)],$$

where this is strictly not the same, a priori, as algebraic K_1 .

Construction 16.13. Let $R \in \text{Rng}^1$. For each $n \in \mathbb{N}$, we define:

$$\phi_n : \text{GL}_n(R) \rightarrow \text{GL}_{n+1}(R), \quad a \mapsto \phi_n(a) \stackrel{\text{def}}{=} \begin{pmatrix} a & 0 \\ 0 & 1_{A^\sim} \end{pmatrix}.$$

Here's a simple result.

Claim 16.1. For each $n \in \mathbb{N}$, we see that ϕ_n is a well-defined group homomorphism.

So, this gives us an inductive sequence of groups and we can, now, take the colimit:

$$\text{GL}_\infty(R) \stackrel{\text{def}}{=} \text{colim}(\text{GL}_n(R)).$$

Of course, we will find it very reasonable to identify elements of $\text{GL}_n(R)$ with their images in $\text{GL}_\infty(R)$, with the latter being viewed as the group of all "infinite invertible matrices over R ".

The next construction is also algebraic.

Construction 16.14. Let $R \in \text{Rng}$ and define the map:

$$\varepsilon : R^\sim \rightarrow \mathbb{Z}, \quad x + t1_{R^\sim} \mapsto \varepsilon(x + t1_{R^\sim}) \stackrel{\text{def}}{=} t.$$

Now, this induces a map:

$$\text{GL}_n(\varepsilon) : \text{GL}_n(R^\sim) \rightarrow \text{GL}_n(\mathbb{Z}),$$

in which case we define:

$$\text{GL}_n^\sim(R) \stackrel{\text{def}}{=} \ker(\text{GL}_n(\varepsilon)).$$

Here's an alternative description of this group.

Claim 16.2. We have that $\text{GL}_n^\sim(R) = \{1_n + a \mid a \in M_{n \times n}(R) \wedge 1_n + a \in \text{GL}_n(R^\sim)\}$.

Of course, the above makes it seem like the unitization plays a special role. It doesn't.

Claim 16.3. TFSH:

1. If we replace $R^\sim$ in the construction above with a $S \in \text{Rng}^1$ which contains R as an ideal and if we replace ε with the projection $R \rightarrow R/S$, then $\text{GL}_n^+(R) = \ker(\text{GL}_n(S) \rightarrow \text{GL}_n(S/R))$.
2. If R is a k -algebra for some unital commutative ring k , then $\text{GL}_n^+(R) \simeq \ker(\text{GL}_n(R^{k,\sim}) \rightarrow \text{GL}_n(k))$.
3. If R is unital, then $\text{GL}_n^+(R) = \text{GL}_n(R)$.

Due to the last statement, we will denote $\text{GL}_n^+(R)$ by $\text{GL}_n(R)$.

This completes the bulk of the construction.

16.2 Milnor Patching & The Bass-Milnor-Murthy Excision Exact Sequence

Okay, here's the definition of a Milnor square.

Definition 16.15

Suppose we have a commutative square:

$$\begin{array}{ccc} R & \xrightarrow{\phi_2} & R_2 \\ \phi_1 \downarrow & & \downarrow \psi_2 \\ R_1 & \xrightarrow{\psi_1} & S \end{array}$$

in Rng . This square is a Milnor square iff:

1. The square is a pullback.
2. One of the maps among ψ_1 and ψ_2 is surjective.

We also refer to this, sometimes, as a Milnor pullback.

We have a remark about this definition.

Remark 16.16. So, Milnor squares were introduced in [70, p.19]. It seems that his overarching goal was to establish statements like the Mayer-Vietoris Theorem or to cook up long-exact sequences in K-theory. To do this, the most difficult part is to cook up a boundary homomorphism and that requires being able to cook up (minimally) the elements of K_0 . But the elements of K_0 are “just finitely-generated projective modules”. So, ultimately, Milnor squares were introduced as tools for conveniently constructing finitely-generated projective modules.

Here's a useful characterization of Milnor squares.

Claim 16.4. Suppose we have a commutative square:

$$\begin{array}{ccc} R & \xrightarrow{\phi_2} & R_2 \\ \phi_1 \downarrow & & \downarrow \psi_2 \\ R_1 & \xrightarrow{\psi_1} & S \end{array}$$

in Rng . TFAE:

1. This is a Milnor square.
2. There is a sequence which is exact.
3. There is a sequence which is exact.

Our eventual goal is to establish the following result.

Theorem 16.17. The Bass-Milnor-Murthy Excision Exact Sequence

Suppose that we have a Milnor square:

$$\begin{array}{ccc} R & \xrightarrow{\phi_2} & R_2 \\ \phi_1 \downarrow & & \downarrow \psi_2 \\ R_1 & \xrightarrow{\psi_1} & S \end{array}$$

in Rng^1 . Then, there is an associated exact sequence:

$$\begin{array}{ccccccc} K_1(R) & \xrightarrow{(K_1(\phi_1), K_1(\phi_2))} & K_1(R_1) \oplus K_1(R_2) & \xrightarrow{K_1(\psi_1) \oplus K_1(\psi_2)} & K_1(S) & & \\ & & \downarrow \partial & & & & \\ & & K_0(R) & \longrightarrow & K_0(R_1) \oplus K_0(R_2) & \longrightarrow & K_0(S) \end{array}$$

in AbGrp .

The proof is not difficult but it does require us to collect a number of technical results about Milnor patching. Moreover, the actual value of the proof is that we get to go through Milnor patching, one of the key motivations for higher K-theory, in detail. This will lead us through some calculations that use the exact sequence above (and other facts we have previously derived). Okay, let's go ahead with some examples of Milnor squares.

Example 16.18. Let $R \in \text{AbRng}^1$ and consider the following diagram:

$$\begin{array}{ccc} & & R[x] \\ & & \downarrow \\ R[y] & \longrightarrow & R \end{array}$$

where both maps simply take the constant term of any polynomial coming out of $R[x]$ and $R[y]$; these maps are surjective so it remains for us to calculate the pullback. So, the easiest intuitive way of thinking about this is when R is a field. Now, what happens is that applying the spectrum functor, we have a pushout^a:

$$\begin{array}{ccc} \text{Spec}(R) & \longrightarrow & \text{Spec}(R[x]) \\ \downarrow & & \\ \text{Spec}(R[y]) & & \end{array}$$

Well, when R is a field, then $\text{Spec}(R)$ is just a point. On the other hand, we see that:

$$\text{Spec}(R[x]) = \text{Spec}(R[y]) = R \cup \{*\},$$

where $*$ represents the generic point. So, what we're really doing is gluing two lines [ENTER IMAGE]:

at the single marked point. But now, we can't glue them so that they are horizontal, because that would mean that we're identifying all points on both lines. Hence, we glue them into the following picture [ENTER IMAGE]:

But if we “deform” this picture, then we simply get the “ x -axis” intersecting with the “ y -axis”; the equation for this is given by $xy = 0$. So, our guess, when passing back to commutative rings, is that:

$$R[y] \times_R R[x] \simeq R[x, y]/(xy).$$

Now, here's our K-theoretic claim.

Claim. $K_0(R[x, y]/(xy)) \simeq \mathbb{Z}$ if R is a field.

^aI thank my friend, Fawzy Hegab, who managed to explain what's happening here geometrically.

Example 16.19. Let $R \in \text{AbRng}^1$ so that $2 \in R^\times$ and consider the diagram:

$$\begin{array}{ccc} & & R[t] \\ & & \downarrow (\text{ev}_1, \text{ev}_{-1}) \\ R & \xrightarrow{\Delta} & R \times R \end{array}$$

where Δ is the diagonal map and we also have the evaluation maps. Let's think about what this pullback should be intuitively. The only way^a to get some intuition is to do a calculation. So, suppose that we have a polynomial:

$$p(t) = a_0 + a_1t + a_2t^2 + \dots + a_nt^n.$$

Let us study what happens in the lower cases, by setting $p(1) = p(-1)$.

1. Let $n = 0$. In that case, we get nothing interesting.
2. Let $n = 1$. In that case, we see that:

$$a_0 + a_1 = a_0 - a_1,$$

which means that $2a_1 = 0$ so $a_1 = 0$.

3. Let $n = 2$. In that case, we see that:

$$a_0 + a_1 + a_2 = a_0 - a_1 + a_2,$$

which means that $a_1 = 0$.

4. Let $n = 3$. In that case, we see that:

$$a_0 + a_1 + a_2 + a_3 = a_0 - a_1 + a_2 - a_3,$$

which implies that:

$$a_1 + a_3 = 0.$$

Therefore, we obtain a simplification of our polynomial:

$$p(t) = a_0 + a_1t + a_2t^2 - a_1t^3 = a_0 + a_2t^2 + a_1(t - t^3),$$

which is interesting to note.

5. Let $n = 4$. Well, this reduces back to the case of $n = 3$. The pattern so far seems to indicate that it's sufficient to consider the odd cases only.
6. Let $n = 5$. By a similar calculation as above, we see that:

$$a_1 + a_3 + a_5 = 0,$$

which means that we obtain the following simplification:

$$\begin{aligned} p(t) &= a_0 + a_2t^2 + a_4t^4 + a_1t + a_3t^3 + (-a_1 - a_3)t^5 \\ &= a_0 + a_2t^2 + a_4t^4 + a_1t(1 - t^2)(1 + t^2) + a_3t^3(1 - t^2), \end{aligned}$$

as was desired.

So, the pattern seems to be that we get polynomials in the variables “ $t^2 - 1$ ” and “ $t - t^3$ ”. We define:

$$\begin{aligned} x &= t^2 - 1, \\ y &= t - t^3. \end{aligned}$$

Well, this parametrizes a curve in $R[x, y]$ and we can compute this curve as follows:

$$\begin{aligned} y^2 &= t^2(1 - t^2)^2 \\ &= (t^2 - 1 + 1)x^2 \\ &= (x + 1)x^2 \\ &= x^3 + x^2. \end{aligned}$$

So, the guess is that the pullback is just the ring:

$$R[x, y]/(y^2 - x^3 - x^2),$$

which represents the nodal curve.

^aThat I know of, at least! I would like to be proven wrong.

Example 16.20. Let $R \in \mathbf{AbRng}^1$ and consider the diagram:

$$\begin{array}{ccc} & & R[t] \\ & & \downarrow \pi \\ R & \longrightarrow & R[t]/(t^2) \end{array}$$

where π is the canonical projection.

Example 16.21.

17 Homotopical Structures on Rings

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

18 Constructing Negative K-Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

19 K-Theory: A Categorical Invariant

Motivation

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

Motivation.

19.1 Quillen-Exact Categories & Lower K-Theory

Indeed, Quillen-exact categories were introduced by Quillen in [77] for the purposes of higher algebraic K-theory and he certainly used standard homological algebra on abelian categories to motivate their definition. However, we think that they are intuitive enough that this isn't necessary and one can introduce abelian categories in a cleaner way.

The idea is that using enrichment as the road to abelian categories is the “wrong” thing to do, especially if we wish to reproduce the standard diagram lemmas that one usually encounters over module categories. Indeed, the way this is typically done is that one introduces abelian categories via the pathway laid out in the preceding section and, then, appeals to the Freyd-Mitchell Embedding Theorem to reduce problems over abelian categories to module categories. This would be fine, except that the embedding theorem is not an easy result and most people never see the proof of that result.

Moreover, this has the severe downside of us simply knowing the proofs of those diagram lemmas already over module categories, instead of obtaining new proofs that rely only on universal properties. The real problem is that the set-up of an abelian category is not ideal for proving those lemmas. What we really need is a category which encodes a calculus of exact sequences and this is given to us by Quillen-exact categories. Now, to define Quillen-exact categories in a clean way, we will need to construct a category which naturally helps us encode exact sequences.

Construction 19.1. Let $\mathcal{C}$ be a category. We define a structure:

$$\mathcal{C}^{\rightarrow\rightarrow},$$

as follows:

1. The objects are pairs:

$$\bullet \xrightarrow{i} \bullet \xrightarrow{p} \bullet$$

of composable morphisms.

2. Given two objects in the following diagram (indicated by solid arrows):

$$\begin{array}{ccccc} \bullet & \xrightarrow{i} & \bullet & \xrightarrow{p} & \bullet \\ \downarrow & & \downarrow & & \downarrow \\ \bullet & \xrightarrow{i'} & \bullet & \xrightarrow{p'} & \bullet \end{array}$$

a morphism of these two objects is the data of the three dashed arrows such that the diagram commutes.

This data forms a category^a. This is the category of composable morphisms in $\mathcal{C}$.

^aThis is checked by a direct calculation.

So, the point is that a Quillen-exact category will just be a category with a distinguished subcategory of its category of composable morphisms satisfying some axioms. Of course, this subcategory will consist of those composable morphisms which fit into a short exact sequence; it's worth defining what these are in the next bit of terminology.

Terminology 19.2. Let $\mathcal{C}$ be a pointed category and consider an object:

$$x' \xrightarrow{f} x \xrightarrow{g} x''$$

in $\mathcal{C}^{\rightarrow\rightarrow}$. This is a kernel-cokernel pair iff f is the kernel of g and g is the cokernel of f . Now, let $\mathcal{C}$ be additive and let $\text{ExSeq}(\mathcal{C})$ be a subcategory of $\mathcal{C}^{\rightarrow\rightarrow}$ consisting of kernel-cokernel pairs.

1. A deflation in $(\mathcal{C}, \text{ExSeq}(\mathcal{C}))$ is a $p \in \text{Mor}(\mathcal{C})$ such that there is an $i \in \text{Mor}(\mathcal{C})$ so that $(i, p) \in \text{ExSeq}(\mathcal{C})$.
2. An inflation in $(\mathcal{C}, \text{ExSeq}(\mathcal{C}))$ is a $i \in \text{Mor}(\mathcal{C})$ such that there is a $p \in \text{Mor}(\mathcal{C})$ so that $(i, p) \in \text{ExSeq}(\mathcal{C})$.
3. The objects of $\text{ExSeq}(\mathcal{C})$ are called conflations in $(\mathcal{C}, \text{ExSeq}(\mathcal{C}))$.

Conflations are also known as exact pairs.

We have a remark about this bit of terminology.

Remark 19.3. In other words, a pair of an inflation and a deflation make up a conflation. The reason for these bits of terminology becomes obvious if we think about what's happening on module categories; the inflation is an inclusion map so it “expands” its domain into its codomain, while the deflation is a surjection so it “deflates” its domain into its codomain. Again, this is not completely airtight but it's a good way of making sense of the terms.

Okay, so what are Quillen-exact categories?

Definition 19.4

Let $\mathcal{C}$ be an additive category and let $\text{ExSeq}(\mathcal{C})$ be a subcategory of $\mathcal{C}^{\rightarrow\rightarrow}$ so that:

1. $\text{ExSeq}(\mathcal{C})$ consists of kernel-cokernel pairs.
2. $\text{ExSeq}(\mathcal{C})$ is closed under isomorphisms.
3. The identity morphism of the zero object is a deflation.

4. A composition of two deflations is a deflation.
5. A composition of two inflations is an inflation.
6. Suppose we have a diagram:

$$\begin{array}{ccc} & & y' \\ & & \downarrow c \\ x & \xrightarrow{p} & y \end{array}$$

where p is a deflation, then it may be completed to a pullback square:

$$\begin{array}{ccc} y & \xrightarrow{p'} & y' \\ \downarrow b & & \downarrow c \\ x & \xrightarrow{p} & y \end{array}$$

where p' is a deflation.

7. Suppose we have a diagram:

$$\begin{array}{ccc} y & \xrightarrow{i} & y' \\ \downarrow a & & \\ x & & \end{array}$$

where i is an inflation, then it may be completed to a pushout square:

$$\begin{array}{ccc} y & \xrightarrow{i} & y' \\ \downarrow a & & \downarrow b \\ x & \xrightarrow{i'} & x' \end{array}$$

where i' is an inflation.

Then, the pair $(\mathcal{C}, \text{ExSeq}(\mathcal{C}))$ is known as a Quillen-exact category.

We have a few remarks about this definition.

Remark 19.5. In Quillen's original work and in other resources on exact categories, one refers to a deflation as an admissible epimorphism while an inflation is called an admissible monomorphism. The reasoning behind this terminology is very clear if we obviously just view every kernel-cokernel pair as an exact sequence. On the other hand, we don't use that terminology as it is a bit too verbose for our tastes. We think the replacement terminology makes sense. We do, however, allow for the additional bit of terminology provided above for conflations.

Remark 19.6. (Bernhard) Keller is the one who coined the terms inflation, deflation and conflation. Apparently, this stems from the work of Gabriel & Roiter on the representation theory of finite-dimensional algebras; see [58]. This seems, at least to me, an indication that Quillen-exact categories **might** very well be helpful in giving us a handle of the representation theory of C^* -algebras or other operator algebras. This could be something to investigate later.

Remark 19.7. Of course, we will typically omit explicit notation for the set of exact sequences because that is determined entirely by the notation for the category^a.

^aThis is how we set it up.

Here's a simple exercise with facts we'll use again and again.

Claim 19.1. Let $\mathcal{C}$ be a Quillen-exact category.

1. Every inflation is a monomorphism.
2. Every deflation is an epimorphism.
3. Every isomorphism in $\mathcal{C}$ is an inflation and a deflation.

Now, beyond the simple exercise above, you may be slightly discombobulated by the rather long definition above and it may not be so clear as to how you should reason with the axioms. The next proposition, while useful for later work, is meant to teach you how you might use the axioms in practice.

Proposition 19.8

Let $\mathcal{C}$ be a Quillen-exact category and suppose we have a pullback diagram:

$$\begin{array}{ccc} x' & \xrightarrow{p'} & y \\ i' \downarrow & & \downarrow i \\ x & \xrightarrow{p} & y' \end{array}$$

where p is a deflation and i is an inflation. Then, i' is an inflation.

Proof. See [20, Proposition 2.15, p.10]. ■

With exact categories in hand, we should say something about subcategories and functors which preserve the exact structure. This will also be helpful when we try to cook up examples.

Terminology 19.9. Let $\mathcal{C}, \mathcal{D}$ be Quillen-exact categories.

1. $\mathcal{D}$ is a fully exact subcategory of $\mathcal{C}$ iff it is a full additive subcategory of $\mathcal{C}$ which is closed under extensions. That is, if it contains the end terms of a conflation in $\mathcal{C}$, then it also contains the middle term.
2. If $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$, then F is exact iff it is an additive functor which preserves conflations.

Of course, we can also define left-and-right-exactness but this is not necessary.

Okay, it's best if we start discussing some examples of Quillen-exact categories. As you can see, these have quite a lot of structure and one really has to go through the painful process of verifying each axiom one-by-one. Obviously, there have been some simplifications to the axioms made in the literature but we've presented them in a way that makes them memorable. With that said, we'll make our work a little easier by starting with *constructions*, before we proceed towards specific examples. For the first construction, we'll need some terminology.

Terminology 19.10. Let $\mathcal{C}$ be an additive category. A sequence:

$$x \xrightarrow{f} y \xrightarrow{g} z,$$

in $\mathcal{C}$ is split-exact iff (f, g) is a kernel-cokernel pair and there is a commutative diagram:

$$\begin{array}{ccccc} x & \xrightarrow{f} & y & \xrightarrow{g} & z \\ \text{Id}_x \downarrow & & \downarrow \beta & & \downarrow \text{Id}_z \\ x & \xrightarrow{\iota_x} & x \oplus z & \xrightarrow{\pi_z} & z \end{array}$$

where β is an isomorphism.

Of course, one can characterize split-exactness in the usual way, where we ask for g to have a section or f to have a retraction. Rather than simply do it because we can, we'll do it when it is actually necessary for us to do it.

Construction 19.11. Let $\mathcal{A}$ be an additive category. We define:

$$\text{ExSeq}(\mathcal{A}) \stackrel{\text{def}}{=} \{\text{All split exact sequences of } \mathcal{A}\}.$$

Here's the claim.

Claim. With these exact sequences, $\mathcal{A}$ is a Quillen-exact category.

You'll prove this in **Claim 19.2**.

Here's the associated exercise.

Claim 19.2. Prove the claim of **Construction 19.11**.

The next construction gives us a way of incorporating duality arguments into the theory.

Construction 19.12. Let $\mathcal{C}$ be a Quillen-exact category.

Claim. $(\mathcal{C}^{\text{op}}, \text{ExSeq}(\mathcal{C})^{\text{op}})$ is a Quillen-exact category.

This is clear from the way that we've defined Quillen-exact categories.

Construction 19.13. Let $\mathcal{C}, \mathcal{D}$ be Quillen-exact categories. Then, we can equip the product:

$$\mathcal{C} \times \mathcal{D},$$

with the structure of a Quillen-exact category, with an exact sequence in $\mathcal{C} \times \mathcal{D}$ being a pair of exact sequences, with the first being from $\mathcal{C}$ and the second being from $\mathcal{D}$.

Okay, here are specific examples. One may note that it is a non-trivial exercise, in all of these cases, to actually verify that we have an exact category. This seems to not have been done systematically in the literature, presumably because it is "obvious". Nevertheless, we will do it explicitly for the small selection of examples here since there is something interesting to be learned from these computations.

Example 19.14. Let $R \in \mathbf{Rng}^1$. Here's our claim.

| **Claim.** ${}_R\mathbf{Mod}_{f.g.p.}$ is an exact category.

One observes that ${}_R\mathbf{Mod}_{f.g.p.}$ is an additive category where each short exact sequence splits, by the strength of Lemma 19.37 (this is proved later). It follows, by the preceding example, that we have an exact category.

Example 19.15. The category:

$$\mathbf{Ban}_{\mathbb{K}},$$

has the structure of a Quillen-exact category with all of its usual exact sequences as inherited from $\mathbf{Vec}_{\mathbb{K}}$.

Example 19.16. Let $G \in \mathbf{Grp}(\mathbf{Top})$. The category:

$$\mathbf{Rep}_G,$$

has the structure of a Quillen-exact category with exact sequences inherited from $\mathbf{Ban}_{\mathbb{K}}$.

Example 19.17. Let $X \in \mathbf{CHaus}$. The category:

$$\mathbf{Bund}_{\mathbb{K}}(X),$$

has the structure of a Quillen-exact category with exact sequences inherited from $\mathbf{Ban}_{\mathbb{K}}$.

We have a further remark about the axioms.

Remark 19.18. In my mind, the axioms for the chosen set of exact sequences in a Quillen-exact category don't seem to rely on the additive structure. Indeed, the definition makes complete sense if we just had a pointed category $\mathcal{C}$ and picked out a set $\mathbf{ExSeq}(\mathcal{C})$ of exact sequences which satisfied the axioms that have been presented. Indeed, there are interesting categories which can be equipped with such structures.

| **Claim.** $\mathbf{C}^*\mathbf{Alg}$, with its usual exact structure inherited from $\mathbf{Ban}_{\mathbb{C}}$, satisfies the required properties.

Okay, let us prove some basic results about exact categories from the axioms. Actually, there's quite a lot that one could say about these categories in general and we'll discuss the role they play in giving us a better understanding of diagram lemmas in ???. For now, here's a simple result which illustrates how we use these axioms.

Proposition 19.19

Let $\mathcal{C}$ be a Quillen-exact category.

1. A deflation in $\mathcal{C}$ which is a mono is an isomorphism.
2. An inflation in $\mathcal{C}$ which is an epi is an isomorphism.
3. If $x, y \in \mathcal{C}$, then the following sequence:

$$x \xrightarrow{\iota_x} x \oplus y \xrightarrow{\pi_y} y,$$

lies in $\mathbf{ExSeq}(\mathcal{C})$.

4. The direct sum of two short-exact sequences is short-exact.
5. Every split-exact sequence in $\mathcal{C}$ lies in $\mathbf{ExSeq}(\mathcal{C})$.

Proof. We'll prove each statement one-by-one.

1. Suppose that we have a deflation $f : y \rightarrow z$ which is a mono. Then, we have a short-exact sequence:

$$x \xrightarrow{g} y \xrightarrow{f} z,$$

in $\mathcal{C}$. Now, we apply ??. Since $f = \mathbf{CoKer}(g)$ and since f is mono, it follows that f is an isomorphism.

2. If we have an inflation in $\mathcal{C}$ which is an epi, then it is a deflation in $\mathcal{C}^{\text{op}}$ which is a mono. So, it is an isomorphism in $\mathcal{C}^{\text{op}}$ and, therefore, an isomorphism in $\mathcal{C}$.
3. The idea is that we can construct the direct sum as a pushout:

$$\begin{array}{ccc} 0 & \longrightarrow & x \\ \downarrow & & \downarrow \iota_x \\ y & \xrightarrow{\iota_y} & x \oplus y \end{array}$$

Now, the point is that $0 \rightarrow x$ is the kernel of Id_x and $0 \rightarrow y$ is the kernel of Id_y . Since identities are deflations, it follows that $0 \rightarrow x$ and $0 \rightarrow y$ are inflations. Since the above is a pushout square, we conclude that ι_x and ι_y are inflations. On the other hand, by the strength of ??, we know that:

$$(\iota_x, \pi_y),$$

forms a kernel-cokernel pair. So, it follows that the pair (ι_x, π_y) lies in $\mathbf{ExSeq}(\mathcal{C})$.

4. We consider the following two exact sequences:

$$x \longrightarrow x' \longrightarrow x''$$

$$y \longrightarrow y' \longrightarrow y''$$

in $\mathcal{C}$. We start with the following claim.

Claim. If $z \in \mathcal{C}$, then the following sequence:

$$x \oplus z \longrightarrow x' \oplus z \longrightarrow x''$$

is exact in $\mathcal{C}$.

Indeed, the second morphism is a deflation because we can form it by a composition:

$$x' \oplus z \rightarrow x' \rightarrow x'',$$

of deflations. The first morphism $x \oplus z \rightarrow x' \oplus z$ is the kernel of the second one so it is an inflation. Hence, the given sequence is exact in $\mathcal{C}$. Now, it follows from this claim that the morphism:

$$x \oplus y \rightarrow x' \oplus y',$$

is an inflation because we can write it as a composition:

$$x \oplus y \rightarrow x' \oplus y \rightarrow x' \oplus y',$$

of two inflations (they are inflations by the claim). But now, we see that our desired direct sum of the two exact sequences forms a kernel-cokernel pair, so this means that this sequence is exact in $\mathcal{C}$.

5. Every split-exact sequence in $\mathcal{C}$ is a kernel-cokernel pair which is isomorphic to an exact sequence of the form given in (3). Since that one lies in $\text{ExSeq}(\mathcal{C})$, it follows that a split-exact sequence lies in $\text{ExSeq}(\mathcal{C})$. ■

With our work on Quillen-exact categories done, we are ready to introduce abelian categories.

Terminology 19.20. Let $\mathcal{C}$ be a Quillen-exact category.

1. If $f : x \rightarrow y$ is a morphism in $\mathcal{C}$, then f is admissible iff there is a factorization:

$$f = i \circ p,$$

where i is an inflation and p is a deflation.

2. $\mathcal{C}$ is abelian iff every morphism in $\mathcal{C}$ is admissible.

Note that the pair (i, p) does not necessarily form a conflation, simply because $p \circ i$ may not be well-defined.

We have two remarks about this bit of terminology.

Remark 19.21. The idea is that an abelian category is trying to model categories of modules over a unital ring and their general properties. In particular, we know from basic algebra that the so-called factor theorem holds in ${}_R\text{Mod}$ when $R \in \text{Rng}^1$. So, we're saying that an abelian category is one in which the factor theorem holds.

Remark 19.22. Let's put it another way; if $\mathcal{C}$ is a Quillen-exact category and f is a morphism, then f is admissible iff there exists $(p, i) \in \mathcal{C}^{\rightarrow \rightarrow}$ such that $f = i \circ p$ and i is an inflation while p is a deflation.

Here's a simple fact about the exact structure of an abelian category.

Proposition 19.23

Let $\mathcal{C}$ be a Quillen-exact category.

1. The factorization of an admissible morphism is unique up to unique isomorphism.
2. If $\mathcal{C}$ is abelian, then $\text{ExSeq}(\mathcal{C})$ has all kernel-cokernel pairs in $\mathcal{C}$ as its objects.

Proof. We'll prove each statement one-by-one.

1. Suppose we have the following commutative diagram:

$$\begin{array}{ccc} x & \xrightarrow{p} & z \\ p' \downarrow & & \downarrow i \\ z' & \xrightarrow{i'} & y \end{array}$$

with p, p' being deflations and i, i' being inflations. Here's a claim.

Claim. Let's suppose that $p = p'$ and $i = i'$. Then, $z = z'$ and there is one and only one morphism:

$$z \rightarrow z,$$

which would make the diagram commute. It is, of course, Id_z .

Of course, Id_z does make the diagram commute. Now, let $f : z \rightarrow z$ be a morphism such that the diagram does commute. So, we would have that $i = i' \circ f$. But i is an inflation so it is a monomorphism and this means that $f = \text{Id}_z$, which proves the claim. Now, we go back to the general situation. Let k be the kernel of p . Then:

$$i \circ (p \circ k) = (i \circ p) \circ k = i' \circ (p' \circ k) = 0.$$

Since i' is a monomorphism, we see that:

$$p' \circ k = 0.$$

Now, the point is that p is a deflation so p is the cokernel of k . On the other hand, p' is a cokernel cocone for k , so there exists a unique cocone morphism $j : z \rightarrow z'$ such that:

$$j \circ p = p'.$$

On the other hand, we know that:

$$i \circ p = i' \circ p' = i' \circ j \circ p,$$

and since p is an epimorphism, it follows that $i = i' \circ j$. So, what we have done is constructed a morphism between two objects in $\mathcal{C}^{\rightarrow\rightarrow}$. Now, one can construct a morphism in the opposite direction using the same argument, except that we start with the kernel of p' . This will give us another morphism $j' : z' \rightarrow z$ and $j \circ j'$ and $j' \circ j$ give the identity via the claim.

2. Let (i, p) be a kernel-cokernel pair. Then, we have factorizations:

$$\begin{aligned} i &= m \circ e, \\ p &= m' \circ e', \end{aligned}$$

where m, m' are inflations and e, e' are deflations. Here's the claim.

Claim. e and m' are isomorphisms.

Our main tool for this will be statements (1) and (2) of Proposition 19.19. Since i is the kernel of p , it is a mono and since m is an inflation, it is a mono. Now, we'll show that e is a mono. Indeed, suppose that:

$$e \circ f = e \circ f'.$$

Then, we see that $i \circ f = i \circ f'$ by composing with m on the far left, which implies that $f = f'$ since i is a mono. So, e is an isomorphism. A similar argument yields that m' is an isomorphism. But now, we see that:

$$(i, p) \simeq (m, p) \simeq (m, e'),$$

where (m, e') is an exact pair. ■

Our first goal is to show that this agrees with the usual definition of an abelian category. To do this, we will need a construction in which we effectively take a morphism and “descend it via quotienting”. Indeed, this is the morphism which typically appears in the statement of the First Isomorphism Theorem.

Construction 19.24. Let $\mathcal{C}$ be an additive category and let $f : x \rightarrow y$ be a morphism. We define:

$$\text{ran}(f) \stackrel{\text{def}}{=} \ker(y \rightarrow \text{coker}(f)),$$

$$\text{coran}(f) \stackrel{\text{def}}{=} \text{coker}(\ker(f) \rightarrow x),$$

where $\text{ran}(f)$ is the range of f and $\text{coran}(f)$ is the corange^a of f . We have canonical morphisms:

$$\begin{aligned} \text{ran}(f) &\rightarrow y, \\ x &\rightarrow \text{coran}(f). \end{aligned}$$

Since the first is the kernel of some morphism, it is a monomorphism and since the second is the cokernel of some morphism, it is an epimorphism^b. With the additional data supplied above, we can draw a diagram:

$$\begin{array}{ccccccc} \ker(f) & \xrightarrow{\text{Ker}(f)} & x & \xrightarrow{f} & y & \xrightarrow{\text{CoKer}(f)} & \text{coker}(f) \\ & \searrow 0 & \downarrow & & \uparrow & \nearrow 0 & \\ & & \text{coran}(f) & & \text{ran}(f) & & \end{array}$$

Now, observe that:

$$\text{CoKer}(f) \circ f = 0.$$

So, f is an equalizer of $\text{CoKer}(f)$ and 0. By the universal property of the equalizer, we get a morphism from $x \rightarrow \text{ran}(f)$ such that the filled out diagram above commutes. Similarly, we use the universal property of the coequalizer to get a morphism from $\text{coran}(f) \rightarrow y$ such that the following diagram:

$$\begin{array}{ccccccc} \ker(f) & \xrightarrow{\text{Ker}(f)} & x & \xrightarrow{f} & y & \xrightarrow{\text{CoKer}(f)} & \text{coker}(f) \\ & \searrow 0 & \downarrow & \nearrow & \uparrow & \nearrow 0 & \\ & & \text{coran}(f) & & \text{ran}(f) & & \end{array}$$

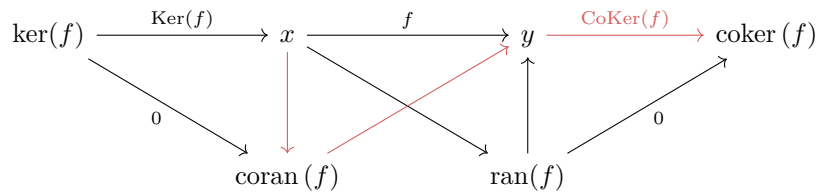
commutes. We have the following claim.

Claim. There exists a unique morphism $\hat{f} : \text{coran}(f) \rightarrow \text{ran}(f)$ such that the diagram above commutes.

The proof proceeds by a diagram chase. We want to show that the red path in the diagram below:

$$\begin{array}{ccccccc} \ker(f) & \xrightarrow{\text{Ker}(f)} & x & \xrightarrow{f} & y & \xrightarrow{\text{CoKer}(f)} & \text{coker}(f) \\ & \searrow 0 & \downarrow & \nearrow & \uparrow & \nearrow 0 & \\ & & \text{coran}(f) & & \text{ran}(f) & & \end{array}$$

is zero. Since $x \rightarrow \text{coran}(f)$ is an epimorphism, it's enough to show that the red path in the next diagram:



is zero, by ?? . But since the diagram above commutes, we see that going along the red path amounts to computing $\text{CoKer}(f) \circ f$ and this is certainly zero. By the universal property of $\text{ran}(f)$, we obtain our desired induced morphism. Of course, the fact that it is unique is guaranteed by the universal property.

^aIn the literature, one refers to the former as the image and the latter as the coimage.

^bThese statements just follow from the results above. However, the way you remember them is by noting that in a module category, you know exactly what these morphisms are supposed to represent and the analogues for injectivity/surjectivity are monicity or epicity.

Here's a remark about this construction.

Remark 19.25. Based on this construction, we can spell out exactly what we want from an abelian category. An abelian category should be a category in which (co)kernels exist **and** Noether's First Isomorphism Theorem holds *by definition*. This makes it easy for us to understand why categories in functional analysis are not going to be described by abelian categories; these are categories in which the First Isomorphism Theorem does not necessarily hold.

So, at least, we see the moral connection between the Terminology 19.20 and the usual way in which one would define abelian categories. Indeed, by the remark above, an abelian category is one in which Noether's First Isomorphism Theorem holds. On the other hand, the point is that this isomorphism theorem is typically derived from the factor theorem in most abelian categories that one concretely encounters in practice. The next result, then, tells us that asking for a factorization IS the same as asking for the First Isomorphism Theorem.

Proposition 19.26. The Factor Theorem (Quillen-Exact Categories)

Let $\mathcal{C}$ be a Quillen-exact category. TFAE:

1. $\mathcal{C}$ is abelian.
2. Every morphism in $\mathcal{C}$ has (co)kernels, every kernel-cokernel pair is in $\text{ExSeq}(\mathcal{C})$ and if $f \in \text{Mor}(\mathcal{C})$, then:

$$\hat{f} : \text{coran}(f) \rightarrow \text{ran}(f),$$

as defined in Construction 19.24, is an isomorphism.

The canonical morphism in (2) is known as the descended morphism.

As you work through the proof of this result, you will notice that we are using statements that have been proved but we aren't referring to those statements. This has been done on purpose; we want you to go back and identify those statements in the text and note them down over here. This ensures that you're actually engaging with this material.

Proof of Proposition 19.26.

(1) $\Rightarrow$ (2): Let $f \in \text{Mor}(\mathcal{C})$. Then, f admits a factorization $f = i \circ p$ with (i, p) being a conflation in $\mathcal{C}$. So, morally, the kernel of f should be the kernel of p and the cokernel of f should be the cokernel of i , with both the kernel of p and cokernel of i existing because these are deflations and inflations respectively.

(a) Let's show that:

$$(\ker(f), \text{Ker}(f)) = (\ker(p), \text{Ker}(p)).$$

Indeed, we observe that:

$$f \circ \text{Ker}(p) = (i \circ p) \circ \text{Ker}(p) = i \circ (p \circ \text{Ker}(p)) = 0.$$

Now, let q be any morphism such that $f \circ q = 0$. Then, we see that:

$$i \circ (p \circ q) = 0.$$

Since i is an inflation, so it is the kernel of some morphism in $\mathcal{C}$. But this means that it is a monomorphism so we may conclude that $p \circ q = 0$. But now, there is a unique morphism of cones from q to $\text{Ker}(p)$ and this proves the desired result.

(b) The idea is that:

$$(\text{coker}(f), \text{CoKer}(f)) = (\text{coker}(i), \text{CoKer}(i)).$$

To prove this, one uses exactly the same arguments as in (1), except applied to the morphism i .

(c) If $f : x \rightarrow y$ is our morphism, then:

$$\text{coran}(f) = \text{coker}(\text{Ker}(f)) = \text{coker}(\text{Ker}(p)),$$

and we also note that:

$$\text{ran}(f) = \ker(\text{CoKer}(f)) = \ker(\text{CoKer}(i)).$$

So, what we have is the following commutative diagram:

$$\begin{array}{ccccccc}
 \ker(f) & \xrightarrow{\text{Ker}(f)} & x & \xrightarrow{p} & z & \xrightarrow{i} & y \xrightarrow{\text{CoKer}(f)} \text{coker}(f) \\
 & \searrow 0 & \downarrow & \swarrow \simeq & \searrow \simeq & \uparrow & \nearrow 0 \\
 & & \text{coran}(f) & \xrightarrow{\hat{f}} & \text{ran}(f) & &
 \end{array}$$

which implies that $\hat{f}$ is an isomorphism.

This proves the implication.

(2) $\Rightarrow$ (1): We consider the following diagram:

$$\begin{array}{ccccccc}
 \ker(f) & \xrightarrow{\text{Ker}(f)} & x & \xrightarrow{f} & y & \xrightarrow{\text{CoKer}(f)} & \text{coker}(f) \\
 & \searrow 0 & \downarrow q_1 & & \uparrow q_2 & & \nearrow 0 \\
 & & \text{coran}(f) & \xrightarrow{\hat{f}} & \text{ran}(f) & &
 \end{array}$$

Well, this is a commutative diagram so we define:

$$\begin{aligned}
 p &\stackrel{\text{def}}{=} \hat{f} \circ q_1, \\
 i &\stackrel{\text{def}}{=} q_2.
 \end{aligned}$$

So, we have our factorization of f . Now, we have to show that i is an inflation and p is a deflation.

(a) Let's show that i is an inflation. The obvious sequence to consider here is:

$$\text{ran}(f) \xrightarrow{i} y \xrightarrow{\text{CoKer}(f)} \text{coker}(f).$$

We need to show that this is a kernel-cokernel pair; this will imply that it lies in $\text{ExSeq}(\mathcal{C})$.

- i. It is by definition that i is the kernel of $\text{CoKer}(f)$.
- ii. Clearly, $\text{CoKer}(f) \circ i = 0$. On the other hand, let $g : y \rightarrow z$ be a morphism such that $g \circ i = 0$. Then, we see that $g \circ f = 0$ so g is a cokernel cocone of f . This implies that there is a unique cocone morphism from $\text{CoKer}(f)$ to p and this proves that $\text{CoKer}(f)$ is the cokernel of i .

This proves that i is an inflation.

(b) Let's show that p is a deflation. The obvious sequence to consider here is:

$$\ker(f) \xrightarrow{\text{Ker}(f)} x \xrightarrow{p} \text{ran}(f).$$

So, we have to show that this is a kernel-cokernel pair; this will imply that it lies in $\text{ExSeq}(\mathcal{C})$.

- i. Well, by definition, we have that q_1 is the cokernel of $\text{Ker}(f)$ and since $\hat{f}$ is an isomorphism, it follows that p is also a cokernel of $\text{Ker}(f)$.
- ii. Observe that $p \circ \text{Ker}(f) = 0$. Now, let g be a morphism such that $p \circ g = 0$. Then:

$$f \circ g = 0,$$

so g is a kernel cone of f . Hence, there is a unique cone morphism from g to $\text{Ker}(f)$ and this proves that $\text{Ker}(f)$ is the kernel of p .

This proves that p is a deflation.

This proves the desired implication. ■

We have a remark about this result.

Remark 19.27. In [REF], we treated abelian categories directly via the characterization in statement (2). This is mainly because there was no need to introduce Quillen-exact categories. Nevertheless, we made an effort to actually emphasize a calculus of exact sequences as the approach for proving the diagram lemmas, as opposed to a direct computation with the universal properties. In other words, the philosophy of our exposition in that work is the same as the one here.

As we mentioned above, knowing too much about abelian categories won't do us much good at this point. The intuition is that these are supposed to behave like module categories and this is good enough for the moment. Nevertheless, here are a few elementary results which are useful to jot down now. The proofs rely on the diagrams drawn above.

Claim 19.3. Let $\mathcal{C}$ be an abelian category. Then, $\mathcal{C}^{\text{op}}$ is an abelian category.

Claim 19.4. Let $\mathcal{C}$ be an abelian category and let $f : x \rightarrow y$ be a morphism.

1. If f is a monomorphism, then the morphism $x \rightarrow \text{ran}(f)$ is an isomorphism.
2. If f is an epimorphism, then $\text{coran}(f) \rightarrow y$ is an isomorphism.
3. f is an isomorphism iff it is a monomorphism and an epimorphism.

This result clearly demonstrates that $\text{Ban}_{\mathbb{K}}$ is not an Abelian category. It does have (co)kernels but the epimorphisms are bounded linear maps which have dense range while monomorphisms are injective bounded linear maps. It's not too difficult to find examples of bounded linear injections with dense range which are not isomorphisms.

Claim 19.5. Let $\mathcal{C}$ be an abelian category. Then, $\mathcal{C}$ is finitely bicomplete.

Here's a remark about the axioms of an abelian category.

Remark 19.28. Note that the two conditions listed are not independent from each other. For the second condition to make sense, one needs to have (co)kernels available so that the (co)range can be defined. In other words, there isn't a clear intermediate structure between an additive and Abelian category that can be studied.

Now, there is one more result we wish to state regarding exact categories that we will utilize whenever convenient to us but we will not prove it. It establishes a relationship between exact categories and abelian categories that one might already notice for certain examples we've discussed. Indeed, one might note that most of the exact categories we've discussed are “naturally found in” some abelian category.

Theorem 19.29. The Gabriel-Quillen Embedding Theorem

Let $\mathcal{C}$ be a small exact category.

1. There is an abelian category $\mathcal{A}$ and a fully faithful functor $\iota : \mathcal{C} \rightarrow \mathcal{A}$ that reflects exactness.
2. $\mathcal{C}$ is closed under extensions in $\mathcal{A}$.
3. The category $\mathcal{A}$ may canonically be chosen to be $\text{Fun}^{\text{lex}}(\mathcal{C}^{\text{op}}, \text{AbGrp})$ and i to be the Yoneda embedding $i(A) = \mathcal{C}(-, A)$.
4. If $\mathcal{C}$ is weakly idempotent complete and $f \in \text{Mor}(\mathcal{C})$ so that $i(f)$ is an epimorphism in $\mathcal{A}$, then f is a deflation.

We have a few remarks about this theorem.

Remark 19.30. The theorem above is the analogue of the Freyd-Mitchell Embedding Theorem for exact categories, though it seems that the proof is a bit less involved than the one for exact categories. We will not go through a proof of this result as it is, like in the case of the Freyd-Mitchell Embedding Theorem, certainly besides the point of our exposition; you can find a very well-written proof in [20, Appendix A, p. 52-62]. Much of our exposition here has been inspired by that resource, so it is not the case that you need to worry about translating between what you've learned here & what you'll encounter in that document.

Remark 19.31. The theorem serves to explain why some authors who write about K-theory seem to automatically embed their exact categories into abelian categories. There are situations where this can be certainly helpful and indeed, we will see that it is necessary to do it for us to state the main theorems of algebraic K_0 and K_1 in the next section. Nevertheless, the abstract picture is also very useful and it will often be reasonable to view certain examples abstractly on their own. Such a perspective also happens to be useful when doing K-theory in the ∞ -categorical setting, something we will pursue in Towards NC Homotopy Theory II.

We're aware that there are some terms in the theorem that we haven't defined. We end off this lecture by defining them & giving some commentary about them that could make it easier for you to fit the theorem above into your head. Since statement (1) of the theorem is the most important one, let's deal with it first.

Statement (2) of the theorem deals with left-exact functors. Indeed, we haven't talked about functoriality between exact categories so the next definition deals with it. In fact, this just generalizes everything you already know about

Definition 19.32

Let $\mathcal{C}, \mathcal{D}$ be exact categories and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$.

1. A left exact sequence in $\mathcal{C}$ is a sequence:

$$\bullet \rightarrow \bullet \rightarrow \bullet,$$
 in $\mathcal{C}^{\rightarrow \rightarrow}$ such that the first arrow is an inflation.
2. A right-exact sequence in $\mathcal{C}$ is a sequence:

$$\bullet \rightarrow \bullet \rightarrow \bullet,$$
 in $\mathcal{C}^{\rightarrow \rightarrow}$ such that the second arrow is a deflation.
3. F is left-exact iff it takes an exact sequence in $\mathcal{C}$ to a left-exact sequence in $\mathcal{D}$.
4. F is right-exact iff it takes an exact sequence in $\mathcal{C}$ to a right-exact sequence in $\mathcal{D}$.

Finally, F is exact iff it takes an exact sequence in $\mathcal{C}$ to an exact sequence in $\mathcal{D}$.

Let's start with weak idempotent completeness.

Definition 19.33

Let $\mathcal{A}$ be an additive category. $\mathcal{A}$ is weakly idempotent complete iff every retraction in $\mathcal{A}$ has a kernel.

Here's our remark regarding the definition.

Remark 19.34. This notion is investigated further in [20, Section 7, p.23-25]; (Theo) Buehler attributes this notion to Thomason, who introduced it in [95, Appendix A.5.1]. In that reference, a proof of the Gabriel-Quillen Embedding Theorem is also given, though it seems to have a gap which is fixed by Buehler. It seems that the main reason this definition is interesting is because the property of being weakly idempotent complete is “satisfied by many exact categories”.

Having established a number of interesting properties of Quillen-exact categories, we are now ready to define their K-theory groups. We'll do this in the case of K_0 first and use that as our motivation for the definition of K_1 of a small¹⁰ exact category. This will, in turn, provide another piece of motivation for our concrete definition of K_1 of rings.

19.2 K_0 & K_1 of Exact Categories

Of course, we're choosing not to touch higher algebraic K-theory in this text. Indeed, a proper discussion of higher K-theory requires a fair amount of homotopy theory & that needs to be discussed systematically since the basic theory is extremely messy. Moreover, one would ideally like to show some significant applications of higher algebraic K-theory and this is impossible to do in this text. Indeed, the most important applications that are available in the literature are number-theoretic and algebro-geometric. We will cover all of that, including the background needed to understand these applications, in Towards NC Homotopy Theory I. With that said, here's K_0 on exact categories.

Definition 19.35

Let $\mathcal{C}$ be a small Quillen-exact category. We define:

$$K_0(\mathcal{C}),$$

to be the free abelian group on $\text{Obj}(\text{Skele}(\mathcal{C}))$ modulo the following relations:

1. $[x] = [x']$ if there is an isomorphism $x \rightarrow x'$ in $\mathcal{C}$.
2. $[x] = [x'] + [x'']$ iff there is an exact sequence:

$$0 \rightarrow x' \rightarrow x \rightarrow x'' \rightarrow 0,$$

in $\mathcal{C}$.

Over here, $[x]$ denotes the element of $K_0(\mathcal{C})$ corresponding to $x \in \text{Obj}(\text{Skele}(\mathcal{C}))$. Moreover, since every $x \in \text{Obj}(\mathcal{C})$ is isomorphic to an object in $\text{Skele}(\mathcal{C})$, the notation $[x]$ also makes sense for any object of $\mathcal{C}$.

The next theorem serves as our main sanity check for the validity of this definition.

Proposition 19.36

Let $R \in \text{Rng}^1$. Then, $K_0(R) \simeq K_0({}_R\text{Mod}_{\text{f.g.p.}})$.

The proof of this result relies on an algebraic fact.

Lemma 19.37

If $R \in \text{Rng}^1$, then every exact sequence in ${}_R\text{Mod}_{\text{f.g.p.}}$ splits.

Proof. Suppose we have an exact sequence:

$$0 \rightarrow M' \xrightarrow{i} M \xrightarrow{p} M'' \rightarrow 0,$$

in ${}_R\text{Mod}_{\text{f.g.p.}}$. Now, $p : M \rightarrow M''$ is a surjective R -linear map so it is split epi in ${}_R\text{Mod}$, by statement (3) of Proposition 2.2. So, the sequence above splits and this completes the proof. ■

With this, we can prove our desired result.

Proof of Proposition 19.36. Since both groups are defined by generators and relations, one simply needs to check that the generators, operations and relations coincide. This would be the standard way of doing this but there is a simpler way which relies on Construction 3.5. The point is that the following equality holds:

$$\text{Skele}({}_R\text{Mod}_{\text{f.g.p.}}) = \mathbb{V}(R),$$

as sets. Condition (2) of Definition 19.35 gives $\text{Skele}({}_R\text{Mod}_{\text{f.g.p.}})$ the structure of an abelian semigroup and ensures that it is isomorphic to $\mathbb{V}(R)$ in AbSemiGrp . Let's use the aforementioned construction and take the free abelian group of $\text{Skele}({}_R\text{Mod}_{\text{f.g.p.}})$. Since every exact sequence in ${}_R\text{Mod}_{\text{f.g.p.}}$ splits, dividing out by the relation in conditions (2) of the definition is the same as forming the Grothendieck group of $\text{Skele}({}_R\text{Mod}_{\text{f.g.p.}})$, which gives the result. ■

At this stage, we have two avenues through which we have motivated the definition of K_1 . We present the last avenue of motivation for the concrete definition as we'll present it in the next lecture.

Definition 19.38

Let $\mathcal{C}$ be a small Quillen-exact category. We define:

$$K_1(\mathcal{C}),$$

to be the free abelian group on pairs (x, α) , where $x \in \mathcal{C}$ and $\alpha \in \text{Aut}(x)$, modulo the following relations:

1. $[(x, \alpha)] + [(x, \beta)] = [(x, \alpha \circ \beta)]$.

¹⁰This can be circumvented by working in Grothendieck universes but we're not going to do it that way.

2. If there is a commutative diagram in $\mathcal{C}$ with exact rows:

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & x_1 & \xrightarrow{\iota} & x & \xrightarrow{\pi} & x_2 & \longrightarrow & 0 \\
 \parallel & & \downarrow \alpha_1 & & \downarrow \alpha & & \downarrow \alpha_2 & & \parallel \\
 0 & \longrightarrow & x_1 & \xrightarrow{\iota} & x & \xrightarrow{\pi} & x_2 & \longrightarrow & 0
 \end{array}$$

where $\alpha \in \text{Aut}(x)$, $\alpha_1 \in \text{Aut}(x_1)$ and $\alpha_2 \in \text{Aut}(x_2)$, then:

$$[(x, \alpha)] = [(x_1, \alpha_1)] + [(x_2, \alpha_2)],$$

where the left-hand side should be interpreted as a “direct sum” $x = x_1 \oplus x_2$ and $\alpha = \alpha_1 \oplus \alpha_2$.

One notes that we do not have a notion for splitting of sequences but we *treat* these sequences as though they do split.

So, now, we have three different ways in which we could think about K_1 ; we have motivation from topological K-theory, Milnor patching and K-theory on categories. From this, we obtain a concrete picture of algebraic K_1 and this will be used as the main motivation for how we should get things done in the non-unital setting. Once we have the definition of K_1 , we will prove a result of the form Proposition 19.36.

Remark 19.39. If $R \in \text{Rng}^1$, then we want to simply define:

$$K_1(R) \stackrel{\text{def}}{=} K_1({}_R\text{Mod}_{\text{f.g.p.}}).$$

So, let (M, α) be a pair of a fgp R -module and an automorphism $\alpha : M \rightarrow M$. Well, we know that M is a direct summand in R^n for some $n \in \mathbb{N}$. So, we find a $N \in {}_R\text{Mod}$ such that:

$$M \oplus N = R^n.$$

Now, what we have is a commutative diagram:

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & M & \xrightarrow{\iota} & R^n & \xrightarrow{\pi} & N & \longrightarrow & 0 \\
 \parallel & & \downarrow \alpha & & \downarrow \alpha \oplus 1_N & & \downarrow 1_N & & \parallel \\
 0 & \longrightarrow & M & \xrightarrow{\iota} & R^n & \xrightarrow{\pi} & N & \longrightarrow & 0
 \end{array}$$

which means that:

$$[(R^n, \alpha \oplus 1_N)] = [(M, \alpha)] + [(N, 1_N)].$$

So, the moral of this derivation, part of which will be used later to prove our result reminiscent of Proposition 19.36, is that all of the information about the pair (M, α) is contained in the pair $(R^n, \alpha \oplus 1_N)$. In particular, what we need to do is to consider invertible matrices over R of arbitrary sizes and simply increase sizes by adding 1's along the diagonal and 0's everywhere else. This is, very roughly, how the concrete construction of algebraic K_1 proceeds.

We need to do a sanity check to ensure that this definition makes sense.

Proposition 19.40

Let $R \in \text{Rng}^1$. Then, $K_1(R) \simeq K_1({}_R\text{Mod}_{\text{f.g.p.}})$.

Proof.

■

Okay, so this provides, everything accounted for, sufficient motivation for our concrete definition of K_1 to be given in the next lecture. Of course, our treatment of algebraic K_1 will mirror the way we've dealt with algebraic K_0 .

20 Higher Homotopy Groups & Topological (Co)Fibrations

Summary

In this lecture, we will cover the following topics:

- 1.
- 2.

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

Motivation. Our work with operator K-theory and algebraic K-theory in the past lectures strongly suggest that we need to define “higher” analogues of the fundamental group if we wish to gain access to the higher K-groups. Indeed, this lecture marks the start of our exploration of stable homotopy theory and so, operator-algebraic material becomes less dominant. Nevertheless, as we mentioned in Section 1, our main goal is to serve the needs of the operator-algebraist who might be interested in this intersection of homotopy theory & operator algebras. Indeed, we ultimately wish to discuss K-theory and we hope to do it in the most modern, general setting available while making sure that most of what we have here is self-contained. In the process, we will build up to many new results which should, hopefully, serve as motivation for further research.

20.1 Introducing The Higher Homotopy Groups

Let's start by setting up some notation.

Notation 20.1. Let $I = [0, 1]$, equipped with its standard topology. We define for $n \in \mathbb{N}$:

$$I^n \stackrel{\text{def}}{=} \prod_{k=1}^n I,$$

to be the compact n -cube equipped with the subspace topology. We define for $n \in \mathbb{N}_0$:

$$\partial I^n \stackrel{\text{def}}{=} \begin{cases} \partial_{\mathbb{R}^n} I & \text{if } n \in \mathbb{N}, \\ \emptyset & \text{if } n = 0. \end{cases}$$

This will be known as the n -cuboid.

Here are a few basic results.

Claim 20.1. Let $n, m \in \mathbb{N}_0$. Then^a, $\partial(I^n \times I^m) = (\partial I^n \times I^m) \cup (I^n \times \partial I^m)$.

^aThis is proved on Page 273.

Claim 20.2. Let $n \in \mathbb{N}$. Then:

$$\partial I^n = \{(x_1, \dots, x_n) \in I^n \mid \exists i \in \{1, \dots, n\} : x_i = 0\}.$$

So, this really is a cuboid and the boundary represents the faces of the n -cube^a.

^aThis is proved on Page 273.

Claim 20.3. Let $n \in \mathbb{N}$. Then^a, $I^n / \partial I^n$ is homeomorphic to S^n .

^aThis is proved on Page 273.

With these results in tow and with the motivation above, we can define the higher homotopy sets.

Definition 20.2

Let $n \geq 0$ and let $(X, x) \in \text{Top}_*$. Then, we define^a:

$$\pi_n(X, x) \stackrel{\text{def}}{=} \text{Top}_{\text{rel}}((I^n, \partial I^n), (X, x)) / \sim_h,$$

where the equivalence relation is given by relative homotopy.

^aOver here, we view the pointed space (X, x) as an object in Top_{rel} by viewing x as the set $\{x\}$.

One notes that when $n = 0$ and $n = 1$, our definition above agrees with the usual definitions of π_0 and π_1 covered in [REF]. In the next construction, we will equip these sets with further group-theoretic structure when $n \geq 2$.

Construction 20.3. Let $i \in \{1, \dots, n\}$ and define:

$$\tau_{1,i} : I^n \rightarrow I^n,$$

to be the map which interchanges the 1-st coordinate with the i -th coordinate. This is an automorphism which squares to the identity^a. Now, let $n \geq 1$, let $(X, x) \in \text{Top}_*$ and let $f, g \in \text{Top}_{\text{rel}}((I^n, \partial I^n), (X, x))$. Then, we define:

$$\forall (t_1, \dots, t_n) \in I^n : (f \star_1 g)(t_1, \dots, t_n) \stackrel{\text{def}}{=} \begin{cases} f(2t_1, t_2, \dots, t_n) & \text{if } t_1 \in [0, \frac{1}{2}], \\ g(2t_1 - 1, t_2, \dots, t_n) & \text{if } t_1 \in [\frac{1}{2}, 1] \end{cases}.$$

Using the automorphism $\tau_{1,i}$, we can further define^b:

$$f \star_i g \stackrel{\text{def}}{=} ((f \circ \tau_{1,i}) \star_1 (g \circ \tau_{1,i})) \circ \tau_{1,i}.$$

Now, we have the following claim.

Claim. Each of the operations above descend to $\pi_n(X, x)$.

This is not too difficult to see; each of the operations above is defined in a fixed way on each of the intervals $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$. Looking at that definition, we can build a homotopy of the product from homotopies on each of these fixed intervals. Of course, this can be done in such a way that we still have continuity at $\frac{1}{2}$.

In that case, $\pi_n(X, x)$ is known as the n -th homotopy set of (X, x) .

^aCheck this to be the case!

^bSo, the point is that in the definition above, we chose to just mess with the first coordinate but we could've done this with any of the other coordinates. In principle, working with each of these should yield different operations and that is exactly what we define here.

So, as we have higher homotopy sets which come with a bunch of relations. In fact, we have the following proposition.

Proposition 20.4

Let $n \geq 1$ and let $(X, x) \in \text{Top}_*$. TFSH:

1. For each $i \in \{1, \dots, n\}$, the operation $\star_i$ induces a group structure on $\pi_n(X, x)$ which does not depend on i .
2. If $n \geq 2$, then the group structure induced above is also commutative.

In particular, $\pi_n(X, x)$ for $n \geq 2$ is known as the sequence of higher homotopy groups of (X, x) .

The proof of this result relies on the so-called Eckmann-Hilton argument.

Lemma 20.5. Eckmann-Hilton Argument

Let M be a set with binary operations $\circ$ and $\star$ so that:

$$\forall v, w, x, y \in M : (v \circ w) \star (x \circ y) = (v \star x) \circ (w \star y).$$

Then, $\circ = \star$ and this operation is commutative.

Proof.

Now, we can prove our desired proposition.

Proof of Proposition 20.4.

To absolutely no one's surprise, we can turn these higher homotopy groups into functors.

Example 20.6. Let $n \in \mathbb{N}$. We define a map of categories:

$$\pi_n : \text{Top}_* \rightarrow \text{Grp},$$

as follows:

1. If $(X, x) \in \text{Top}_*$, then we get $\pi_n(X, x)$ as its image in Grp .
2. If $(X, x), (Y, y) \in \text{Top}_*$ and $f \in \text{Top}_*((X, x), (Y, y))$, then we define:

$$\pi_n(f) : \pi_n(X, x) \rightarrow \pi_n(Y, y), [\phi] \mapsto \pi_n(f)([\phi]) \stackrel{\text{def}}{=} [f \circ \phi].$$

Again, showing that this is well-defined can be done in the same way it is for $\pi_1(X, x)$; it involves knowing more general facts about relative homotopies and compositions as proved in Fundamentals of Topology & Logic.

This defines a functor^a; it is known as the n -th homotopy functor.

^aYou should mentally check this to be true!

Now, the higher homotopy groups are partly quite interesting because they admit an action by fundamental groups.

Construction 20.7. Let $(X, x) \in \text{Top}_*$ and let $n \in \mathbb{N}$. If $y \in \text{Top}$ and there is a path $f : [0, 1] \rightarrow X$ from x to y , then the idea is that this should induce an isomorphism between higher homotopy groups^a. Indeed, we define:

$$f \cdot : \pi_n(X, x) \rightarrow \pi_n(Y, y), [\phi] \mapsto f \cdot [\phi],$$

where the idea for defining this action is precisely by travelling from $y \rightarrow x$ by f^- (the reverse path), travelling from $x \rightarrow x$ by ϕ and then travelling from $x \rightarrow y$ by f . Of course, this is going to be invariant under homotopy, so the operation is well-defined and very clearly invertible^b. So, it follows that $f \cdot$ is an isomorphism of groups.

In particular, we have that:

$$\pi_1(X, x) \curvearrowright \pi_n(X, x),$$

via the action exactly described above^c.

^aThis is a generalization of exactly the same fact that you know to be true for the fundamental group.

^bIndeed, we can create a loop $x \rightarrow x$ from a loop $y \rightarrow y$ in exactly the analogous way.

^cSo, a loop in X based at x acts on the elements of $\pi_n(X, x)$ exactly as described above. But one can deform that loop into another one precisely because of the construction above, so we can have homotopy classes of loops acting on the elements of $\pi_n(X, x)$.

The higher homotopy groups actually introduce a second notion of “weak equivalence” on Top .

Definition 20.8

Let $X, Y \in \text{Top}$ and let $f \in \text{Top}(X, Y)$. f is a weak homotopy equivalence iff:

$$\pi_n(f) : \pi_n(X, x) \rightarrow \pi_n(Y, f(x)),$$

is an isomorphism for every $x \in X$ and every $n \in \mathbb{N}_0$.

Here is a remark about this definition.

Remark 20.9. As you may have noticed so far, we have now produced a number of examples of categories for which there naturally exist notions of “weak equivalence” between the objects. Indeed, we have (weak) homotopy

equivalence for $\mathbf{Top}$ and homotopy equivalence for equivariant C^* -algebras/Banach algebras. In all of these cases, we see that isomorphisms are definitely “weak equivalences” but there are “weak equivalences” which are not necessarily isomorphisms. This makes it reasonable for us to engage with the general study of pairs:

$$(\mathcal{C}, W),$$

consisting of a category $\mathcal{C}$ and some full subcategory $W \subseteq \mathbf{Arr}(\mathcal{C})$ of morphisms. The goal would be to artificially form a category:

$$\mathcal{C}[W^{-1}],$$

where each morphism in W becomes an isomorphism and the hope is to gain some amount of direct control in this category. We will formally introduce this problem and discuss it to some degree in Section 22, since it will play a significant role as partial motivation for many structures that will be introduced throughout this text. Somehow, it is precisely the problem of studying the category above that sits at the center of why we wish to study all of these intrinsically fascinating structures.

Here’s some further terminology related to higher homotopy groups.

Terminology 20.10. Let $X \in \mathbf{Top}$ and let $n \in \mathbb{N}_0$.

1. X is 0-connected iff it is path-connected.
2. X is 1-connected iff it is path-connected and simply connected.
3. X is n -connected iff it is path-connected and $\pi_j(X, x) = 0$ for each $j \leq n$ and for some $x \in X$.
4. If $(X, A) \in \mathbf{Top}_{\text{rel}}$, then this is relatively n -connected iff every map:

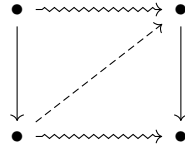
$$(B^j, S^{j-1}) \rightarrow (X, A),$$

with $j \leq n$ is homotopic relative to S^{j-1} to a map into A .

Finally, X is simple iff for any $x \in X$ and $n \in \mathbb{N}$, the action $\pi_1(X, x) \curvearrowright \pi_n(X, x)$ is trivial.

20.2 The Properties of (Co)Fibrations & Their Geometric Meaning

Now, we could just introduce topological (co)fibrations blindly and just start proving facts about them. However, it is best to understand these in the context of *lifting problems*. Indeed, the reader has certainly seen lifting problems before and we have already discussed a few instances of them in this text itself, particularly in relation to C^* -algebras. A general discussion will, at this stage of the game, do us a lot of good. Let us consider the following diagram:



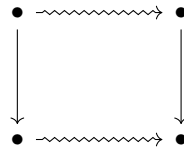
where bullets and arrows indicate a situation where it makes sense to consider such a diagram¹¹. The solid arrows represent given “morphisms”, the squiggly arrows represent any choice of “morphisms” and the dashed arrow represents a solution. So, a lifting problem is a commutative square for which there exists a solution. In order to introduce the definition of a lifting problem in the cleanest possible way, we introduce a somewhat uncommon construction.

Construction 20.11. Let $\mathcal{C}$ be a category. We define a structure:

$$\text{Arr}(\mathcal{C}),$$

as follows:

1. An object is an arrow^a in $\mathcal{C}$.
2. A morphism in this structure is a commutative diagram:



where the arrow on the left is the domain of the morphism, the arrow on the right is the codomain of the morphism and the two arrows on top & below form the data of the morphism. We define composition by the concatenation of commutative diagrams in the usual way through composition of morphisms in $\mathcal{C}$.

This data is easily shown to form a category; we refer to this as the arrow category of $\mathcal{C}$. In particular, if f, g are morphisms in $\mathcal{C}$ and $\alpha : f \Rightarrow g$ is a morphism in $\text{Arr}(\mathcal{C})$, then α is known as a lifting problem for f and g .

^aMore formally, it is a triple (x, y, f) with $x, y \in \mathcal{C}$ and $f \in \mathcal{C}(x, y)$ being an arrow.

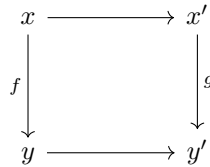
We have a remark.

Remark 20.12. Let $\mathcal{C}$ be a category. Sometimes, we will go back & forth between saying that $f \in \text{Mor}(\mathcal{C})$ or $f \in \text{Arr}(\mathcal{C})$. Both of these expressions “mean” the same thing in practice, namely that f is to be viewed as a morphism in $\mathcal{C}$. However, the latter obviously has an additional meaning as an object in the arrow category of $\mathcal{C}$ and that may sometimes carry additional implications. Nevertheless, what we really mean will be clear in all situations.

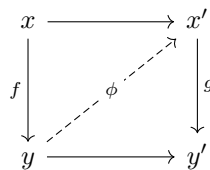
So, what are the solutions to lifting problems?

Definition 20.13

Let $\mathcal{C}$ be a category, let $x, y, x', y' \in \mathcal{C}$ and let $f : x \rightarrow y, g : x' \rightarrow y'$ be morphisms in $\mathcal{C}$. f has the left-lifting property (LLP) with respect to g (or g has the right-lifting property (RLP) with respect to f) iff any lifting problem:



has a solution $\phi : y \rightarrow x'$. That is, the following diagram:



commutes. We denote this by writing $f \square g$.

We have a remark about this definition.

Remark 20.14. The solution ϕ must work for all possible choices of morphisms from x to x' and y to y' . So, even though lifting problems do cover a wide range of phenomenon, the requirements of a solution are rather stringent.

We introduce two pieces of convenient terminology associated to the definition.

¹¹For instance, it would make sense to consider such a diagram in a category. It would also make sense for those arrows to be functors, as we will see in explicit examples later.

Terminology 20.15. Let $\mathcal{C}$ be a category and let f, g be morphisms such that:

$$f \sqsupseteq g.$$

Then, we say that f left-lifts g or g right-lifts f .

Terminology 20.16. Let $\mathcal{C}$ be a category and let $S, T \subseteq \text{Mor}(\mathcal{C})$. We say that S has the LLP with respect to T (or T has the RLP with respect to S) iff for every $f \in S$ and $g \in T$, we have that $f \sqsupseteq g$. We denote this by writing $S \sqsupseteq T$. Also, if $T = \{g\}$, then we'll often just write $S \sqsupseteq g$ instead of $S \sqsupseteq \{g\}$; the same is true when S is a singleton.

Before we prove anything with this definition, let us discuss some wonderful examples of lifting problems.

Example 20.17. Let $A, B \in \text{Set}$. We have the following claim.

Claim. If $f : A \rightarrow B$ is a map in Set , then TFAE:

1. f is surjective.
2. $(\emptyset \rightarrow *) \sqsupseteq f$.

We don't label the map from $\emptyset$ to $*$ because there's only one such map.

Before we prove this equivalence, here's the diagram lurking in the background.

$$\begin{array}{ccc} \emptyset & \longrightarrow & A \\ \downarrow & & \downarrow f \\ * & \longrightarrow & B \end{array}$$

Now, we prove each implication separately.

- (1) $\Rightarrow$ (2): Take the map from $*$ to B and let b be in its range. Then, there is an $a \in A$ such that $f(a) = b$. Now, define a map from $*$ to A whose value is a ; this is a solution to the lifting problem.
- (2) $\Rightarrow$ (1): Let $b \in B$. Then, we pick a map from $*$ to B with value b and we select the unique map from $\emptyset$ to A . This, of course, gives us a lifting problem which has a solution $\phi : * \rightarrow A$. The solution is literally the condition that $f(a) = b$, where the solution has value $a \in A$.

This completes the proof.

Example 20.18. Let $R \in \text{Rng}^1$ and let $M \in {}_R\text{Mod}$. The next claim characterizes projectivity of M via lifting problems.

Claim. TFAE:

1. M is projective.
2. $(0 \rightarrow M) \sqsupseteq \{\text{epimorphisms in } {}_R\text{Mod}\}$.

This follows essentially from the standard lifting result which characterizes projective modules; see Proposition 2.2.

Example 20.19. Let $\mathcal{C}, \mathcal{D}$ be categories and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$. We have the following claim.

Claim. TFAE:

1. F is surjective on objects.
2. F has the RLP with respect to $\emptyset \rightarrow [0]$.

We'll prove each implication separately.

(1) $\Rightarrow$ (2): Consider the following commutative diagram:

$$\begin{array}{ccc} \emptyset & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow F \\ [0] & \xrightarrow{d} & \mathcal{D} \end{array}$$

where d represents the constant functor which selects an object $d \in \mathcal{D}$. Since F is surjective on objects, there is a $c \in \mathcal{C}$ such that $F(c) = d$. Now, we define a lift $[0] \rightarrow \mathcal{C}$ which simply selects the object c in $\mathcal{C}$.

(2) $\Rightarrow$ (1): Let $d \in \mathcal{D}$. Then, the commutative diagram above has a lift; this lift just selects an object $c \in \mathcal{C}$. Since the lift causes the diagram to commute, we see that $F(c) = d$.

This completes the proof.

In some sense, we can extend Definition 20.13 to sets of morphisms, rather than limiting ourselves specifically to two morphisms. This is mainly just new notation, so we introduce it in that way.

Notation 20.20. Let $\mathcal{C}$ be a category and let $S \subseteq \text{Mor}(\mathcal{C})$. Define:

$$\begin{aligned} S \sqsupseteq^{\text{def}} & \{g \in \text{Mor}(\mathcal{C}) : S \sqsupseteq g\} \\ \sqsupseteq S^{\text{def}} & \{g \in \text{Mor}(\mathcal{C}) : g \sqsupseteq S\}. \end{aligned}$$

We refer to $S \sqsupseteq$ as the orthogonal lift of S and $\sqsupseteq S$ as the orthogonal prelift^a.

^aIn other words, $S^\square$ is the set of all morphisms in $\mathcal{C}$ that are left-lifted by S . ${}^\square S$ is the set of all morphisms in $\mathcal{C}$ that are right-lifted by S . This manner of stating how these objects are defined will be useful in making proofs intuitive.

Obviously, this terminology is inspired from the more elementary situation with (pre)-annihilators of subsets of Banach spaces and their dual spaces. In the next few result, we establish computational properties of the orthogonal (pre)-lifts.

Claim 20.4. Let $\mathcal{C}$ be a category, let $f, g \in \text{Mor}(\mathcal{C})$ so that:

$$f \square g,$$

in $\mathcal{C}$. Then, $g^{\text{op}} \square f^{\text{op}}$ in $\mathcal{C}^{\text{op}}$.

Claim 20.5. Let $\mathcal{C}$ be a category and let $S, T \subseteq \text{Mor}(\mathcal{C})$ so that:

$$S \square T.$$

Then^a, $T^{\text{op}} \square S^{\text{op}}$.

^aOf course, $S^{\text{op}} \stackrel{\text{def}}{=} \{f^{\text{op}} : f \in S\}$. This is a set of morphisms in $\mathcal{C}^{\text{op}}$.

Claim 20.6. Let $\mathcal{C}$ be a category and let $S, T \subseteq \text{Mor}(\mathcal{C})$. TFSH:

1. $S \subseteq {}^\square T \Leftrightarrow T \subseteq S^\square$.
2. If $S \subseteq T$, then $T^\square \subseteq S^\square$.
3. If $S \subseteq T$, then ${}^\square T \subseteq {}^\square S$.
4. $S \subseteq (S^\square)^\square$.
5. $S \subseteq ({}^\square S)^\square$.
6. $S^\square = \left({}^\square (S^\square) \right)^\square$.
7. ${}^\square S = \left(({}^\square S)^\square \right)^\square$.

We mostly found this in [25, Proposition 2.1.4, p.29-30], though a proof is not provided.

We have a remark about this result.

Remark 20.21. For a given set $S \subseteq \text{Mor}(\mathcal{C})$, one might hope to prove a “Double Annihilator Theorem”:

$$\bar{S} = (S^\square)^\square.$$

We’d also have to make sense of the left-hand side first. Actually, this is a bit tricky because we’d have to develop Quillen’s Small Object Argument first and it’s rather clear to us that the machinery for this is not necessary at all in the work we intend to do within this thesis. Nevertheless, there are two reasons why we’ll go through with it anyways:

1. It’s an important bit of mathematical culture related to this topic.
2. We’ll be developing all of the prerequisites to it anyways and once we do that, the proof of this fact falls out rather naturally from simple considerations.

Given this state of affairs, let us not worry too much about the prospects of proving such a “Double Annihilator Theorem” and, instead, proceed with developing the theory as normally as possible.

With that said, here’s some natural terminology.

Terminology 20.22. Let $\mathcal{C}$ be a category and let $x, y \in \mathcal{C}$. x is a retract of y iff there is a commutative diagram:

$$\begin{array}{ccccc} & & \text{Id}_x & & \\ & \nearrow & & \searrow & \\ x & \xrightarrow{f} & y & \xrightarrow{g} & x \end{array}$$

Also, if $f, g \in \text{Mor}(\mathcal{C})$, then f is a retract of g iff it is a retract of g in $\text{Arr}(\mathcal{C})$. Finally, if $S \subseteq \text{Mor}(\mathcal{C})$, then S is closed/stable under retracts iff every morphism in $\mathcal{C}$ which is a retract of some morphism in S must also lie in S .

In fact, an object being a retract of another object does not change under dualization.

Claim 20.7. Let $\mathcal{C}$ be a category and let $x, y \in \mathcal{C}$ with x being a retract of y . Then, x is a retract of y in $\mathcal{C}^{\text{op}}$.

The following result is easy to prove and it is described as a “trick” in Cisinski’s book. Indeed, the actual trick used in the proof is one that will be used a few times in this chapter.

Proposition 20.23. The Retract Lemma

Let $\mathcal{C}$ be a category, let $f : x \rightarrow y$ be a morphism and suppose we have a factorization:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ & \searrow i \quad \nearrow p & \\ & z & \end{array}$$

Then, TFSH:

1. If $i \dashv f$, then f is a retract of p .
2. If $f \dashv p$, then f is a retract of i .

Proof. We'll prove each statement one-by-one.

1. We need it to be the case that f is a retract of p in $\text{Arr}(\mathcal{C})$. Consider the diagram:

$$\begin{array}{ccccc} x & \xrightarrow{i} & z & \xrightarrow{\quad} & x \\ f \downarrow & & \downarrow p & & \downarrow f \\ y & \xrightarrow{\text{Id}_y} & y & \xrightarrow{\text{Id}_y} & y \end{array}$$

Basically, this is a morphism from f to p to f in the arrow category of $\mathcal{C}$ and we just need to find a morphism from z to x such that the two part composes to the identity. We start with the following lifting problem^a:

$$\begin{array}{ccc} x & \xrightarrow{\text{Id}_x} & x \\ i \downarrow & & \downarrow f \\ z & \xrightarrow{p} & y \end{array}$$

Note that this is actually just the commutative triangle given in the assumption and we've extended it into a square^b. Since $i \dashv f$, it follows that there is a solution $\phi : z \rightarrow x$. But this just means that $\phi \circ i = \text{Id}_x$.

2. If $f \dashv p$ in $\mathcal{C}$, then $p \dashv f$ in $\mathcal{C}^{\text{op}}$. So, f is a retract of i in $\mathcal{C}^{\text{op}}$ but that means that f is a retract of i in $\mathcal{C}$. ■

^aThe trick is to realize that triangles are secret squares.

^bThis is a common trick that one would've also encountered while studying basic category theory.

These general results concerning lifting problems are useful computational exercises for you to get your hands dirty with categorical reasoning about lifting problems; these will have individualized meaning in concrete settings and it will be important to have intrinsic intuition about those specific settings. With that said, we are ready to finally discuss (co)fibrations in this setting. For the sake of presenting a unified picture, let us define these all at once.

Definition 20.24

21 An Overview of the Q -Construction

Summary

In this lecture, we will cover the following topics:

- 1.
- 2.

This lecture was given by Abhijeet Vats, based partially on material covered by Bogdan Carasca during the K-theory seminar organized at HU Berlin in WiSe 2025/2026.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

Motivation.

21.1 The Basic Theory of Simplicial Sets

With the motivation above, we will begin by working towards the definition of simplicial sets. Actually, as a way of also making things a little easier for us much later, we are going to define simplicial categories and then specialize to simplicial sets. To do this, we'll need to set up some notation.

Notation 21.1. Let $n \in \mathbb{N}_0$. We define:

$$[n] \stackrel{\text{def}}{=} \{0 \leq 1 \leq \dots \leq n\},$$

where this inherits its partial order from $\mathbb{N}_0$. We define:

$$s^n : [n] \rightarrow 0, \quad i \mapsto s^n(i) \stackrel{\text{def}}{=} 0.$$

This map is known as the degeneracy; we do not distinguish all of the maps formed for each $n \in \mathbb{N}_0$ due to the fact that they just do “one thing”. On the other hand, let $n \in \mathbb{N}$ and let $0 \leq i \leq n$. Then, we define the following map:

$$d_i^n : [n-1] \rightarrow [n], \quad j \mapsto d_i^n(j) \stackrel{\text{def}}{=} \begin{cases} j & \text{if } j < i, \\ j+1 & \text{if } j \geq i. \end{cases}$$

These are known as the face maps^a. Also, we define:

$$s_i^n : [n] \rightarrow [n-1], \quad j \mapsto s_i^n(j) \stackrel{\text{def}}{=} \begin{cases} j & \text{if } j \leq i, \\ j-1 & \text{if } j \geq i+1. \end{cases}$$

These are the degeneracy maps^b. Since $[n]$ is a poset, it is a category; we refer to these as simplex categories.

^aOne way to think about these is that they're the unique injections which omit i .

^bOne way to think about these is that they're the unique surjections which hit i twice.

This notation will be very useful to have in hand for concrete calculations later. For now, we need to specify an example of a category since it will be used in the definition of a simplicial category.

Example 21.2. We define a structure Δ as follows:

1. The objects are finite non-empty totally ordered sets.
2. The morphisms are non-decreasing maps between these totally ordered sets, with the usual composition.

This data forms a category; we will refer to this as the simplicial conglomerate.

Here are a few remarks about the category Δ .

Remark 21.3. The extra name given to the category is my own invention; you won't find it in the literature. The problem is that, in this topic, we have words like “simplex” or “simplices” or “simplicial” flying around all over the place. What you need to bear in mind is how the category is defined, as well as the content of [Remark 21.4](#).

Remark 21.4. Every finite non-empty totally ordered set is isomorphic in Δ to $[n]$ for some $n \in \mathbb{N}_0$. So, we can simply **assume** that the objects in Δ are exactly of this form. This assumption is convenient for concrete calculations.

Remark 21.5. Note that we excluded the empty set from Δ . In [\[81\]](#), the morphisms of Δ are known as simplicial operators and that's because the objects of the simplex category are known also as the simplicial objects^a.

^aThey aren't quite the n -simplices; these will be defined differently later.

Remark 21.6. Here are two facts from your exposition to category theory that you might remember:

1. Posets can be viewed as categories.
2. Functors between posets viewed as categories are exactly the non-decreasing maps between them.

So, in particular, the objects of Δ can be viewed, themselves, as categories and the morphisms between them are precisely the functors between them. This is a rather simple fact but it will be useful at times.

Here are two pieces of notation associated to the category above.

Notation 21.7. Let $n, m \in \mathbb{N}_0$ and consider a morphism:

$$\delta : [n] \rightarrow [m].$$

It will be convenient, at times, to simply write this as a list of values:

$$\delta = \langle \delta_0 \delta_1 \dots \delta_n \rangle,$$

while stipulating that $\delta_i \leq \delta_{i+1}$ for all $i \in \{0, \dots, n-1\}$.

With the simplicial conglomerate in place, we can define (co)simplicial objects.

Definition 21.8

Let $\mathcal{C}$ be a category.

1. A cosimplicial object in $\mathcal{C}$ is a functor $X : \Delta \rightarrow \mathcal{C}$.
2. A simplicial object in $\mathcal{C}$ is a functor $X : \Delta^{\text{op}} \rightarrow \mathcal{C}$.

If $\mathcal{C} = \text{Set}$, then a simplicial object in $\mathcal{C}$ is called a simplicial set.

We have two remarks regarding this definition.

Remark 21.9. For this section, our interest will be directed towards proving statements about simplicial sets and simplicial spaces. We’ve provided the definition above in generality for the sake of completeness and convenience.

Remark 21.10. In Towards Noncommutative Homotopy Theory II, we will have reasons to turn towards the theory of ∞ -categories. As it turns out that there exists a special type of simplicial set, called a quasi-category, which provides a concrete model for the notion of an ∞ -category. Indeed, quasi-category theory is very similar to ordinary category theory. We will define quasi-categories later in this section but we’re not going to say much more about them beyond what is absolutely necessary for our work. Nevertheless, if you would like to have further geometric motivation for simplicial sets, then see [42].

Well, since the objects of Δ “are” just the sets $[n]$, we see that a simplicial set is a sequence of sets along with a collection of functions between those sets. The word “simplicial” is meant to indicate that these come with additional combinatorial data. This viewpoint is made more plausible by the next bit of terminology we introduce.

Terminology 21.11. Let $\mathcal{C}$ be a category and let X be a simplicial object in $\mathcal{C}$.

1. For each $n \in \mathbb{N}_0$, the set $X([n]) = X_n$ is known as an n -cell. Each X_n is known as a level of X .
2. X_0 is known as the set of vertices or points.
3. X_1 is known as the set of edges or paths.
4. If $m, n \in \mathbb{N}_0$ and $\delta : [m] \rightarrow [n]$ is a simplicial operator, then we define:

$$\forall a \in X_n : a\delta \stackrel{\text{def}}{=} X(\delta)(a).$$

This notation is used in Rezk’s notes^b.

When $\mathcal{C} = \text{Set}$, then the set of all n -cells will be the coproduct in Set of the family $(X_n)_{n \in \mathbb{N}_0}$.

^aOne may notice the similarities with CW complexes; this is not quite the right analogy but we will show that one can turn a simplicial set into a topological space via the geometric realization.

^bWe don’t use it much in this thesis but we introduce it as a matter of culture.

With this terminology in place, it’s easily seen that the data of a simplicial set consists of:

1. A sequence $(X_n)_{n \in \mathbb{N}_0}$ of sets.
2. A family of functions $X_n \rightarrow X_m$ given by simplicial operators $[m] \rightarrow [n]$.

Of course, we also require the given family of functions to satisfy the usual defining conditions for functoriality. There’s actually more notation that Rezk introduces which we will introduce only when it is necessary in context. With that said, we should explicitly define the category of simplicial sets; we do this a bit more generally.

Example 21.12. Let $\mathcal{C}$ be a category. We define a structure:

$$s\mathcal{C} \stackrel{\text{def}}{=} \text{Fun}(\Delta^{\text{op}}, \mathcal{C}).$$

This is a category because the structure on the right-hand side is a category. The morphisms in this category will be known as maps of simplicial objects in $\mathcal{C}$; these are just natural transformations^a.

^aNote also that there are some authors who use $\mathcal{C}_\Delta$ to denote the category of simplicial objects in $\mathcal{C}$.

Let us note the following fact right away.

Claim 21.1. If $\mathcal{C}$ is a category which is (co)complete, then $s\mathcal{C}$ is (co)complete.

Our first goal is to demonstrate that simplicial objects in certain categories have *some* of the key properties that we’d expect ∞ -categories to possess. We’ve already outlined some of these desirable properties in the last section and we want to “recover” some of them throughout our discussion. One of the reasons why we should focus on simplicial sets, in particular, is the fact that one can “embed” categories within them in an appropriate way via the next result.

Theorem 21.13

There exists a fully faithful functor $N : \text{Cat} \rightarrow s\text{Set}$.

This is one of the results that will lead us straight to the definition of a quasicategory. Once we have that definition, we’ll need to check that quasicategories actually do possess some of the base-line properties we’d want in a higher version of category theory. Before we deal with any of that, let us start simply by constructing the functor in **Theorem 21.13**.

Construction 21.14. Let $\mathcal{C}$ be a small category. We want to define a functor:

$$N[\mathcal{C}] : \Delta^{\text{op}} \rightarrow \text{Set}.$$

Well, we define it as follows:

1. If $k \in \mathbb{N}_0$, then we define:

$$(N[\mathcal{C}])_k \stackrel{\text{def}}{=} \text{Fun}([k], \mathcal{C}).$$

This is a set because $\mathcal{C}$ is small.

2. If $m, n \in \mathbb{N}_0$ and $\delta : [m] \rightarrow [n]$ is a simplicial operator, then we define:

$$N[\mathcal{C}](\delta) : \text{Fun}([n], \mathcal{C}) \rightarrow \text{Fun}([m], \mathcal{C}),$$

by pre-composition with δ .

It's obvious that this is a functor; we refer to $N[\mathcal{C}]$ as the nerve of $\mathcal{C}$. We further define:

$$N : \text{Cat} \rightarrow s\text{Set},$$

which acts as follows:

1. If $\mathcal{C}$ is a small category, then $N(\mathcal{C}) \stackrel{\text{def}}{=} N[\mathcal{C}]$.
2. If $\mathcal{C}, \mathcal{D}$ are small categories and $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$, then $N(F)$ is just a map between the nerves. But since the nerves are simplicial sets, the map $N(F)$ is just a natural transformation obtained in the obvious way^a.

Again, a computation will reveal that the nerve functor is actually a functor.

^aStart by specifying $N(F)$ on the components and then define its action by post-composition with F .

What's the intuition behind the construction above? Let's say something about it.

Intuition 21.15. Let $\mathcal{C}$ be a small category. Then, we note that:

1. $N[\mathcal{C}]_0 = \text{Fun}([0], \mathcal{C})$. Morally, these are just the objects of $\mathcal{C}$.
2. $N[\mathcal{C}]_1 = \text{Fun}([1], \mathcal{C})$. Morally, these are just the morphisms of $\mathcal{C}$.

The interpretations above are partly inspired by the discussion in the previous section. However, this would not be the right way to think about it; these interpretations are very natural in themselves and they provide motivation for our choices in the previous section. Nevertheless, this should make it more than plausible that the nerve of a category contains all of the information we'd want about the category. One also needs to check that the underlying identities and compositions are carried over to the nerve. This is where the "higher level of coherences" comes into play.

The next few miscellaneous results attempt to take the intuition presented above and build something concrete from it; these results are essentially summarized in [81, 4.3, p.13] and we provide the missing proofs.

Claim 21.2. Consider the simplicial operators:

$$\begin{aligned} \langle 0 \rangle &: [0] \rightarrow [1], \\ \langle 1 \rangle &: [0] \rightarrow [1], \end{aligned}$$

where we're using Notation 21.7. If $\mathcal{C}$ is a small category, then the following maps:

1. $N[\mathcal{C}](\langle 0 \rangle^{\text{op}})$,
2. $N[\mathcal{C}](\langle 1 \rangle^{\text{op}})$,

actually assign a morphism to its domain and codomain respectively.

Claim 21.3. Let $\mathcal{C}$ be a small category. Then, the map:

$$N[\mathcal{C}](\langle 00 \rangle) : N[\mathcal{C}]_0 \rightarrow N[\mathcal{C}]_1,$$

assigns an object to its identity morphism^a.

^aOver here, the morphism $\langle 00 \rangle$ goes from $[1] \rightarrow [0]$.

Claim 21.4. Let $\mathcal{C}$ be a small category. Then, $N[\mathcal{C}]_2$ is in bijection with sets of pairs of composable morphisms (f, g) . That is, the codomain of f is the same as the domain of g so that $g \circ f$ is well-defined.

So, we've formed a functor which takes any small category and constructs a simplicial set out of it. In fact, we'll show that this functor is fully faithful, thereby proving **Theorem 21.13**. For this result, we'll need some further preparatory discussion regarding the nerve of a small category. In particular, we need the following data:

1. The fact that any small category can be recovered from its nerve up to isomorphism.
2. The fact that we have an explicit description of the n -cells in the nerve of a small category.
3. The natural transformations to the nerve are determined by the edges.

In the same spirit as **Claim 21.4**, we have the following result.

Claim 21.5. Let $\mathcal{C}$ be a small category and let $k \in \mathbb{N}_0$.

1. There is a bijective correspondence:

$$(N[\mathcal{C}])_k \rightarrow \{(g_1, \dots, g_k) \in \text{Mor}(\mathcal{C})^k : \text{codom}(g_{i-1}) = \text{dom}(g_i)\},$$

which sends a functor $a : [k] \rightarrow \mathcal{C}$ to the sequence $(a \circ \langle 0, 1 \rangle, \dots, a \circ \langle n-1, n \rangle)$.

2. With respect to the correspondence in (1), if $\delta : [m] \rightarrow [n]$ is a simplicial operator in Δ , then the map:

$$N[\mathcal{C}](\delta^{\text{op}}) : N[\mathcal{C}]_n \rightarrow N[\mathcal{C}]_m,$$

induced by the simplicial set $N[\mathcal{C}]$ coincides with the function:

$$(g_1, \dots, g_n) \mapsto (h_1, \dots, h_m), \quad h_k = \begin{cases} \text{Id} & \text{if } \delta(k-1) = \delta(k), \\ g_j g_{j-1} \dots g_{i+1} & \text{if } \delta(k-1) = i < j = \delta(k). \end{cases}$$

It is also not too difficult to show that functors to the nerve of a category are determined precisely by the edges. This will also be used to prove that the nerve functor is fully faithful.

Claim 21.6. Let $\mathcal{C}$ be a category, let X be a simplicial set and let $f, g : X \rightarrow N[\mathcal{C}]$ be a natural transformation. Then, $f = g$ iff $f_1 = g_1$, where the latter two are maps from $X_1 \rightarrow (N[\mathcal{C}])_1$.

We're finally ready to prove that the nerve functor is fully faithful.

Proof of Theorem 21.13. Let $\mathcal{C}, \mathcal{D} \in \text{Cat}$. We want to prove that the following map:

$$N : \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Nat}(N[\mathcal{C}], N[\mathcal{D}]),$$

is a bijection. We'll show injectivity and surjectivity separately.

Injectivity: Let $f, g \in \text{Fun}(\mathcal{C}, \mathcal{D})$ so that $N(f) = N(g)$. Then, we need to show that f and g coincide on the objects and morphisms. Well, $N(f)$ and $N(g)$ coincide on $N[\mathcal{C}]_0$ and $N[\mathcal{C}]_1$ specifically; these are the objects and morphisms of $\mathcal{C}$, with $N(f), N(g)$ being “defined” as f and g when we restrict attention to the 0-cells & 1-cells. Hence, $f = g$.

Surjectivity: Let $\alpha : N[\mathcal{C}] \Rightarrow N[\mathcal{D}]$ be a natural transformation. Now, the point is that we can specify a functor $f : \mathcal{C} \rightarrow \mathcal{D}$ simply by defining it to be α_0 on the objects and α_1 on the morphisms. Now, we want to check that $N(f) = \alpha$. However, by **Claim 21.6**, we just need to check that these coincide on the 1-cells and they do, by construction.

This completes the proof. ■

So, we've successfully shown that 1-category theory can be embedded neatly into the theory of simplicial sets. If X is a simplicial set, then the intuition we've put forth through **Terminology 21.11** is that X_0 is the set of points in the simplicial set and X_1 is the set of paths between those points. So, we clearly have a structure that resembles a category but, in some sense, all of the extra X_n 's are meant to introduce higher levels of coherence and this will become somewhat apparent in our discussion of quasicategories. Speaking of that, we'll note that these are motivated in two steps:

1. One obtains a geometric characterization of simplicial sets which are isomorphic to the nerve of a category.
2. One weakens this geometric characterization to obtain quasicategories.

Actually, there are several interesting geometric characterizations of simplicial sets and we do want to take our time going through them in detail. Part of this will also entail some discussion of the connections to the homotopy theory of topological spaces. The next definition we'll give is motivated by a key property of the nerve proved in **Claim 21.5**. However, there is additional conceptual context behind this definition which we'd like to talk about it because it relates to issues we've brought up before in the synthetic approach. Our presentation here is inspired primarily by the contents of [79, p.7-8] and it starts by introducing the following bit of notation.

Notation 21.16. Let $\mathcal{C}$ be a small category. We define two maps:

$$\begin{aligned} s : \text{Mor}(\mathcal{C}) &\rightarrow \text{Obj}(\mathcal{C}) , \quad f \mapsto s(f) \stackrel{\text{def}}{=} \text{dom}(f), \\ t : \text{Mor}(\mathcal{C}) &\rightarrow \text{Obj}(\mathcal{C}) , \quad f \mapsto t(f) \stackrel{\text{def}}{=} \text{codom}(f). \end{aligned}$$

These are the source and target maps associated to the category.

Now, we can provide a very useful construction.

Construction 21.17. Let $\mathcal{C}$ be a small category. Consider the following diagram:

$$\begin{array}{ccc} & \text{Mor}(\mathcal{C}) & \\ & \downarrow s & \\ \text{Mor}(\mathcal{C}) & \xrightarrow{t} & \text{Obj}(\mathcal{C}) \end{array}$$

Now, we can calculate the pullback of this diagram in **Set**. So, we obtain:

$$\begin{array}{ccc} \text{Mor}(\mathcal{C}) \times_{\text{Obj}(\mathcal{C})} \text{Mor}(\mathcal{C}) & \xrightarrow{\pi_1} & \text{Mor}(\mathcal{C}) \\ \downarrow \pi_2 & & \downarrow s \\ \text{Mor}(\mathcal{C}) & \xrightarrow{t} & \text{Obj}(\mathcal{C}) \end{array}$$

Well, we know exactly what the pullback in **Set** is supposed to be; it consists of those pairs (f, g) with f, g being morphisms in $\mathcal{C}$ such that $t(f) = s(g)$. That is, the codomain of f is the same as the domain of g . In other words, the pullback above produces pairs of composable morphisms. In other words:

$$\text{Mor}(\mathcal{C}) \times_{\text{Obj}(\mathcal{C})} \text{Mor}(\mathcal{C}) = N[\mathcal{C}]_2.$$

In a similar way, we can construct the n -fold pullback:

$$\bigtimes_{\text{Obj}(\mathcal{C})}^n \text{Mor}(\mathcal{C}),$$

which is defined inductively in the usual way.

Now, we can try to play the same game with simplicial sets. To do this, we need to produce source and target maps from $X_1 \rightarrow X_0$, with X being a simplicial set. In this case, we can use **Claim 21.2** as motivation.

Notation 21.18. Let X be a simplicial set. Consider the simplicial operators:

$$\begin{aligned}\langle 0 \rangle &: [0] \rightarrow [1], \\ \langle 1 \rangle &: [0] \rightarrow [1]\end{aligned}$$

Now, we define the source and target maps as follows:

$$\begin{aligned}s &= X(\langle 0 \rangle^{\text{op}}), \\ t &= X(\langle 1 \rangle^{\text{op}}),\end{aligned}$$

where this makes sense because it produces maps from $X_1 \rightarrow X_0$.

With the source and target maps available, we can repeat [Construction 21.17](#) for simplicial sets. However, the situation is different simply because we do not have as explicit a description of the n -cells of an arbitrary simplicial set. For this reason, the construction below is quite a bit more complicated *but* that complication also leads us naturally to higher categorical notions of interest. We'll comment more on this topic as we proceed further.

Construction 21.19. Let X be a simplicial set. Consider the following diagram:

$$\begin{array}{ccc} & X_1 & \\ & \downarrow s & \\ X_1 & \xrightarrow{t} & X_0 \end{array}$$

This is a pullback diagram in Set , so we obtain the following commutative diagram:

$$\begin{array}{ccccc} X_1 \times_{X_0} X_1 & \xrightarrow{\pi_1} & X_1 & & \\ \pi_2 \downarrow & & \downarrow s & & \\ X_1 & \xrightarrow{t} & X_0 & & \end{array}$$

The point is that the fiber product, now, consists of pairs of edges (f, g) such that the source of g is the same as the target of f . That is, these edges are “composable”. In this situation, we can also construct the n -fold pullback:

$$\bigtimes_{X_0}^n X_1,$$

which is defined inductively in the usual way. Now, we want to induce a map:

$$X_2 \rightarrow X_1 \times_{X_0} X_1,$$

and the right way to do that is by employing the universal property of the pullback. For this to make sense, we need to construct maps from $X_2 \rightarrow X_1$ so that we obtain a pullback diagram relative to the maps s and t . This amounts to considering simplicial operators from $[1] \rightarrow [2]$. Since we care about composability, here are the ones we care about:

$$\begin{aligned}\langle 01 \rangle &: [1] \rightarrow [2], \\ \langle 12 \rangle &: [1] \rightarrow [2],\end{aligned}$$

So, $X(\langle 01 \rangle^{\text{op}})$ and $X(\langle 12 \rangle^{\text{op}})$ fit into a pullback diagram with s and t . This induces a unique map from X_2 to $X_1 \times_{X_0} X_1$ which is a morphism of pullback cones. One can repeat this for each $n \geq 2$ so that we get a map:

$$X_n \rightarrow \bigtimes_{X_0}^n X_1.$$

This is known as the n -th Segal map.

We have done quite a lot in the construction above but we've derived the existence of a family of maps whose properties will be used to deal with Segal spaces later. For now, we simply provide

Definition 21.20

A simplicial set X satisfies the Segal condition iff the n -th Segal map is a bijection for $n \geq 2$.

This definition leads us to our very first characterization of the nerve of a category.

Proposition 21.21

Let $X \in s\text{Set}$. TFAE:

1. X is naturally isomorphic to the nerve of some category.
2. X satisfies the Segal condition.

Proof.

(1) $\Rightarrow$ (2): This follows essentially from (1) of [Claim 21.5](#) along with [Construction 21.17](#).

(2) $\Rightarrow$ (1): We define a category $\mathcal{C}$ as follows:

- (a) The objects of $\mathcal{C}$ are the elements of X_0 .
- (b) The morphisms of $\mathcal{C}$ are the elements of X_1 .

If we assume that this is a category, then we have the following calculation:

$$\forall n \geq 2 : N[\mathcal{C}]_n \simeq \bigtimes_{N[\mathcal{C}]_0}^n N[\mathcal{C}]_1 \simeq \bigtimes_{X_0}^n X_1 \simeq X_n,$$

where each isomorphism above is natural, so it follows that $N[\mathcal{C}]$ is naturally isomorphic to X . This completes the part of the proof that the reader should care about; now we need to deal with the bit that no one cares about but needs to be addressed regardless. ■

We previously mentioned connections back to the homotopy theory of topological spaces. To facilitate discussion of these connections, let us introduce some familiar terminology.

Terminology 21.22. Let $n \in \mathbb{N}_0$. We define:

$$\Delta^n \stackrel{\text{def}}{=} \Delta(-, [n]).$$

This is known as the standard n -simplex.

Using the Yoneda Lemma and the standard simplices, we obtain a nice description of the levels of a simplicial set.

Claim 21.7. If $X \in s\text{Set}$ and $n \in \mathbb{N}_0$, then $\text{Nat}(\Delta^n, X)$ is in natural bijection with X_n .

The first thing we'll do is to show that one can construct topological spaces from finite ordered sets.

Example 21.23. We define a map of categories:

$$|-| : \Delta \rightarrow \text{Top},$$

as follows:

1. If $[n] \in \Delta$ for $n \in \mathbb{N}_0$, then we define:

$$|[n]| \stackrel{\text{def}}{=} \left\{ (t_i)_{i \in [n]} \in \mathbb{R}^{n+1} \mid (\forall i \in [n] : t_i \geq 0) \wedge \sum_{i \in [n]} t_i = 1 \right\},$$

equipped with the subspace topology.

2. If $[n], [m] \in \Delta$ and $f : [n] \rightarrow [m]$ is a morphism, then we define:

$$|f| : |[n]| \rightarrow |[m]|, (t_i)_{i \in [n]} \mapsto |f|((t_i)_{i \in [n]}) \stackrel{\text{def}}{=} \left(\sum_{i \in f^{-1}(\{j\})} t_i \right)_{j \in [m]}.$$

It is easily seen that this is continuous.

This data encodes a functor; it is known as the geometric realization functor.

Here's some notation related to the example.

Notation 21.24. Let $n \in \mathbb{N}_0$. Define:

$$|\Delta^n| \stackrel{\text{def}}{=} |[n]|,$$

where $[n] \in \Delta$ and the geometric realization on the right-hand side is as given above.

Using the geometric realization, we can construct simplicial sets from topological spaces.

Example 21.25. Let $X \in \text{Top}$. We define a map of categories:

$$\text{Sing}(X) : \Delta^{\text{op}} \rightarrow \text{Set},$$

as follows:

1. If $[n] \in \Delta$, then $\text{Sing}(X)_n \stackrel{\text{def}}{=} \mathcal{C}(|\Delta^n|, X)$.
2. If $[m], [n] \in \Delta$ and $f : [m] \rightarrow [n]$ is a morphism in Δ^{op} , then:

$$\text{Sing}(X)(f) : \mathcal{C}(|\Delta^m|, X) \rightarrow \mathcal{C}(|\Delta^n|, X), \phi \mapsto \text{Sing}(X)(f)(\phi) \stackrel{\text{def}}{=} \phi \circ |f^{\text{op}}|.$$

This feels confusing at first but just check that it's well-defined.

This is known as the singular simplicial set on X . We define a map of categories:

$$\text{Sing}(-) : \text{Top} \rightarrow s\text{Set},$$

as follows:

1. Let $X \in \text{Top}$. Then, $\text{Sing}(X)$ would just be the singular homology of X .
2. Let $X, Y \in \text{Top}$, let $f \in \mathcal{C}(X, Y)$ and let $n \in \mathbb{N}$. Then, we need to define a natural transformation:

$$\text{Sing}(f) : \text{Sing}(X) \Rightarrow \text{Sing}(Y).$$

Of course, it's enough to define this on the levels of the singular simplicial sets:

$$\text{Sing}(f)_n : \text{Sing}(X)_n \rightarrow \text{Sing}(Y)_n, \phi \mapsto \text{Sing}(f)_n(\phi) \stackrel{\text{def}}{=} f \circ \phi.$$

Checking that naturality holds comes down to a computation.

In fact, this map of categories is a functor as well and this is easy to check.

Now, here's a result.

Proposition 21.26

The functor $\text{Sing}(-)$ has a left adjoint; it is known as the geometric realization functor and denoted by $|\cdot|$.

Proof.

■

Are there other reasons why we should care about the standard simplices? Well, we do have the following notable result.

Claim 21.8. Δ^0 is a final object in $s\text{Set}$ and the empty simplicial set is an initial object in $s\text{Set}$.

The standard simplices are useful due to the fact that they can be used to build interesting simplicial sets. There's a relatively simple way to do this which requires some further terminology.

Terminology 21.27. Let $X, Y \in s\text{Set}$. X is a subsimplicial set of Y iff:

1. $\forall n \in \mathbb{N}_0 : X_n \subseteq Y_n$.
2. The inclusion map $\iota : X \rightarrow Y$ is a natural transformation.

We denote this by writing $X \subseteq Y$.

There are two ways of forming subsimplicial sets that are particularly interesting to us.

Example 21.28. Let $n \in \mathbb{N}_0$. We define:

$$\partial\Delta^n,$$

as follows:

$$\forall k \in \mathbb{N}_0 : (\partial\Delta^n)_k \stackrel{\text{def}}{=} \{f \in \Delta([k], [n]) : f \text{ is not surjective}\}.$$

This is known as the boundary of Δ^n

Example 21.29. Let $n \in \mathbb{N}$ and let $0 \leq j \leq n$. The horn Λ_j^n is the subsimplicial set of Δ^n defined by:

$$(\Lambda_j^n)_k \stackrel{\text{def}}{=} \{f \in \Delta([k], [n]) : [n] \setminus \{j\} \text{ is not a subset of } \text{ran}(f)\}.$$

Here's some further terminology:

1. When $0 < j < n$, then Λ_j^n is known as an inner horn.
2. When $j < n$, then Λ_j^n is a left horn.
3. When $0 < j$, then Λ_j^n is a right horn.
4. When $j \in \{0, n\}$, then Λ_j^n is an outer horn.

Of course, an outer horn is not inner and vice versa.

It's worth providing some visual intuition for these examples.

Intuition 21.30. Let us try to visualize these horns in the cases where n and j are sufficiently small.

1. Let $n = 1$ and let $j = 0$. For each $k \in \mathbb{N}_0$, we have that:

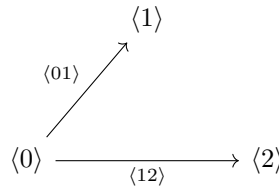
$$(\Lambda_0^1)_k = \{f \in \Delta([k], [1]) : [1] \setminus \{0\} \text{ is not a subset of } \text{ran}(f)\}.$$

Well, this means that for every f in the set above, we have that $f(i) = 1$ for each $i \in [k]$. In other words, each component of Λ_0^1 is a singleton and the same is also true for Λ_1^1 .

2. Let $n = 2$ and let $j = 0$. For each $k \in \mathbb{N}_0$, we have that:

$$(\Lambda_0^2)_k = \{f \in \Delta([k], [2]) : [2] \setminus \{0\} \text{ is not a subset of } \text{ran}(f)\}.$$

Let's draw this out in the case where $k = 0$ and $k = 1$. We obtain:



Of course, we don't include $\langle 12 \rangle$ because this would be a morphism $f : [1] \rightarrow [2]$ which contains $\{1, 2\}$ in its range and this is explicitly forbidden by the horn.

So, the intuition is that we obtain the j -th horn of Δ^n by deleting the face opposite the j -th vertex. Using this rule, it is easy to also draw corresponding diagrams for Λ_1^2 and Λ_2^2 .

Now, this is all that we really wanted to say regarding simplicial sets. At this stage, it is best for us to “get on with it” and start discussing classifying spaces.

21.2 Classifying Spaces & The Structure of the Q -Construction

In particular, we will need to restrict our attention to small categories, at least for the moment.

Definition 21.31

Let $\mathcal{C}$ be a small category. We define its classifying space:

$$B[\mathcal{C}],$$

to be the geometric realization of $N[\mathcal{C}]$.

Well, the classifying space construction is functorial.

Construction 21.32. Clearly, the classifying space construction is the following composition:

$$\text{Cat} \xrightarrow{N} s\text{Set} \xrightarrow{|\cdot|} \text{Top},$$

which implies that it is functorial.

We have a remark about this construction.

Remark 21.33. In Operator Algebraic Structures II, we gave a more topological (and far more painful) construction of the classifying space of a group and we even had to deal with classifying spaces of families. The above is meant to generalize the construction we gave and “clean it up” for future use.

The properties of the classifying space will play an absolutely indispensable role both in the study of the Q -construction and the $+$ -construction. So, we do recommend acquainting yourself with the basic properties quite well.

21.2.1 Properties of Classifying Spaces

We’ll start with a couple of crucial examples.

Example 21.34. Let X be a set and let $\mathcal{C}_X$ be the discrete category associated with^a X .

Claim. $B[\mathcal{C}_X]$ is X with the discrete topology.

For each $n \in \mathbb{N}_0$, we see that $N[\mathcal{C}_X]_n$ consists only of constant functors.

^aWe take the objects as the points of X and require that there are no arrows aside from the identity ones.

Example 21.35. Let $G \in \text{Grp}$ and let $\mathcal{C}_G$ be the associated groupoid category. Since this is a small category, we can attempt to compute its classifying space. In particular, we define the classifying space of a group:

$$B[G] \stackrel{\text{def}}{=} B[\mathcal{C}_G].$$

If G is abelian, then the group composition is a group homomorphism and we have a functor:

$$\mathcal{C}_G \times \mathcal{C}_G \rightarrow \mathcal{C}_G,$$

Since the classifying space construction takes a small category and turns it into a topological space, we can assign topological properties to small categories & functors between them simply by assigning those properties to the corresponding classifying constructs. For example, we do this in the next bit of terminology.

Terminology 21.36. Let P be a property of topological spaces, let $\mathcal{C} \in \text{Cat}$. $\mathcal{C}$ satisfies P iff $B[\mathcal{C}]$ satisfies P .

Terminology 21.37. Let P be a property of continuous maps between spaces, let $\mathcal{C}, \mathcal{D} \in \text{Cat}$ and let:

$$F : \mathcal{C} \rightarrow \mathcal{D},$$

be a functor. F satisfies Q iff $B[F]$ satisfies P .

Okay, so here’s an important result outlining two key properties of classifying spaces.

Proposition 21.38

Let $\mathcal{C}, \mathcal{D}$ be small categories.

1. If $F, G : \mathcal{C} \rightarrow \mathcal{D}$ are functors, a natural transformation:

$$\alpha : F \Rightarrow G,$$

induces a homotopy between $B[F]$ and $B[G]$.

2. An equivalence of categories gives rise to a homotopy equivalence of classifying spaces.

Proof. We’ll prove each statement one-by-one.

1. We view α as a functor:

$$\alpha : \mathcal{C} \times [1] \rightarrow \mathcal{D},$$

in the usual way. Now, we see that:

$$B[\alpha] : B[\mathcal{C} \times [1]] \simeq B[\mathcal{C}] \times B[1] \rightarrow B[\mathcal{D}],$$

is a continuous map. But $B[1] \simeq [0, 1]$, so we have a homotopy.

2. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an equivalence of categories, with quasi-inverse G . Then, we see that:

$$F \circ G \simeq \text{Id}_{\mathcal{D}},$$

$$G \circ F \simeq \text{Id}_{\mathcal{C}}.$$

Applying the classifying space to these equations, we see that:

$$B[F] \circ B[G] \simeq B[\text{Id}_{\mathcal{D}}] = \text{Id}_{B[\mathcal{D}]},$$

$$B[G] \circ B[F] \simeq B[\text{Id}_{\mathcal{C}}] = \text{Id}_{B[\mathcal{C}]},$$

which implies that $B[F]$ is a homotopy equivalence. ■

The two main corollaries of this result give us criteria for when classifying spaces are contractible.

Corollary 21.39

Let $\mathcal{C} \in \text{Cat}$. TFSH:

1. If $\mathcal{C}$ has an initial or final object, then $B[\mathcal{C}]$ is contractible.
2. If $\mathcal{C}$ has binary products, then $B[\mathcal{C}]$ is contractible.

Proof.

■

As we emphasized earlier, the reason we care for classifying spaces is because they are used explicitly in Quillen's definitions of higher algebraic K-theory (both in the $+$ and Q -constructions). This is why Quillen also spends some time giving the background on this topic in [77]. Of course, you may be wondering about his exact motivation for wanting to work with classifying spaces and it is essentially due to the following magnificent result.

Theorem 21.40

Let $\mathcal{C}$ be a small Quillen-exact category. Then, $\pi_1(B[Q[\mathcal{C}]], 0)$ is isomorphic to $K_0(\mathcal{C})$.

We'll explain $Q[\mathcal{C}]$ in [Construction 21.43](#) but let's deal with some remarks.

Remark 21.41. The importance of the result cannot be underestimated, even though the proof is really not so difficult. Indeed, it indicates that higher algebraic K-theory should be defined via a formula of the following type:

$$K_n(\mathcal{C}) \stackrel{\text{def}}{=} \pi_{n+1}(B[Q[\mathcal{C}]], 0).$$

The only way of interpreting the right-hand side is, somehow, to define “higher fundamental groups” (formally known as the higher homotopy groups) which extend the path-components functor and the fundamental group so we obtain a sequence of invariants. Well, as you might know by now, once we have a sequence of algebraic invariants, it is natural to want to look for some kind of long-exact sequence

Remark 21.42. The crux of Quillen's idea for defining higher algebraic K-theory, from an abstract viewpoint, is to define a *spectrum* for it. Very roughly speaking, this is a sequence of spaces to which we apply the higher homotopy groups to obtain a cohomology theory. Of course, this does not (and should not) make a lot of sense to you right now but we will discuss spectra in Towards NC Homotopy Theory I. We will see that spectra and cohomology theories are, really, one and the same thing. In particular, this will bring us much closer to the state-of-the-art in modern research, where the desire is usually to study the spectrum of K-theory rather than K-theory itself.

With that said, we need to clarify the Q as applied to our category in [Theorem 21.40](#). This will be done in the next construction and it's very important that you, for the moment, simply focus on making sure that you understand the technical details of the construction. Indeed, we actually had to fill all of the details in by hand since they don't seem to be available in any resource that we've seen. We don't claim, however, that the construction makes the intuition for it very obvious; this is something we address afterwards.

Construction 21.43. Let $\mathcal{C}$ be an small Quillen-exact category. We define a new structure:

$$Q[\mathcal{C}],$$

by the following data:

1. The objects of $Q[\mathcal{C}]$ are the same as those of $\mathcal{C}$.
2. Let $x, x' \in Q[\mathcal{C}]$. We consider all diagrams:

$$x \xleftarrow{j} y \xrightarrow{i} x'$$

in $\mathcal{C}$ where j is a deflation and i is an inflation; we refer to these as roofs as we can write the above as:

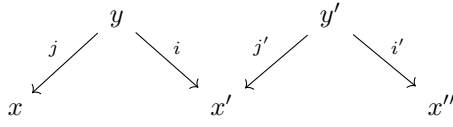
$$\begin{array}{ccc} & y & \\ j \swarrow & & \searrow i \\ x & & x' \end{array}$$

Now, suppose we have two such diagrams:

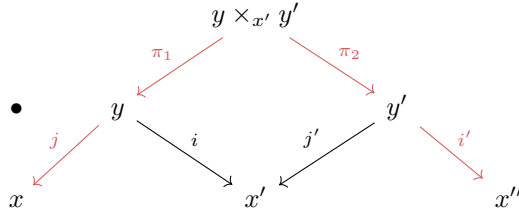
$$\begin{array}{ccccc} x & \xleftarrow{j} & y & \xrightarrow{i} & x' \\ \text{Id}_x \downarrow & & \downarrow \phi & & \downarrow \text{Id}_{x'} \\ x & \xleftarrow{j'} & y' & \xrightarrow{i'} & x' \end{array}$$

These two solid diagrams are equivalent iff there exists a dashed arrow ϕ such that ϕ is an isomorphism and so that the diagram above commutes. Now, we define $Q[\mathcal{C}](x, x')$ to be the set of isomorphism classes of all roof from x to x' . Now, let $x, x', x'' \in Q[\mathcal{C}]$ with morphisms $\phi : x \rightarrow x'$ and $\psi : x' \rightarrow x''$ being morphisms. Then, we

select representatives of ϕ & ψ and we will consider the following diagram:



To form the composite $\psi \circ \phi$, we take a pullback:



and take the equivalence class of the red roof. So, there are a few things going on here and we need to address them one-by-one so as to not confuse ourselves. Of course, we're not going to provide all the details but we'll provide enough that you should be able to fill in the rest on your own.

- (a) So, first of all, we can take the pullback above since j' is a deflation^a.
- (b) Since j' is a deflation, we see that π_1 is a deflation. So, $j \circ \pi_1$ is a deflation.
- (c) What we have is a pullback of an inflation along a deflation, so we can employ Proposition 19.8 and conclude that π_2 is an inflation. So, this means that $i' \circ \pi_2$ is an inflation. So, the red part is a roof.
- (d) One has to show that none of what we did above depends on the representatives. But this is quite easy, because the representatives are isomorphic to each other and pullbacks respect isomorphisms. So, changing representatives will not change the eventual equivalence class.

This completes our description of the composition.

This data forms a category^b; it is known as the Q -construction of $\mathcal{C}$. Here are a few claims.

Claim. $Q[\mathcal{C}]$ is small.

This is clear from the construction, following from the fact that $\mathcal{C}$ is small.

So, it makes sense to compute $B[Q[\mathcal{C}]]$. Furthermore, here's another claim.

Claim. $B[Q[\mathcal{C}]]$ has a canonical basepoint.

This is just given by the zero object (or rather, $B(0)$, with 0 being the zero object).

These are, of course, important points to keep in mind for later work.

^aThis follows by Axiom 6 of Definition 19.4.

^bOf course, one needs to check that this is the case but this can be done via the construction above.

After working through the details of this construction, one can ask a very obvious question.

Question 21.44. What in the world does the Q -construction mean?

Well, one explanation for this construction lies in Theorem 21.40; it is somehow a construction which allows that theorem to be true. Actually, this is what we want to focus on for the rest of this section; we want to make sense of why this result

Proof of Theorem 21.40.

■

21.2.2 Quillen's Theorems A & B: The Statements & Intuition

This leads us to the statement of Quillen's Theorem A.

Theorem 21.45. Quillen's Theorem A

Let $\mathcal{C}, \mathcal{D} \in \text{Cat}$, let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ so that for each $d \in \mathcal{D}$, the comma category d/F is contractible. Then:

$$B[F] : B[\mathcal{C}] \rightarrow B[\mathcal{D}],$$

is a homotopy equivalence.

This leads us to the statement of Quillen's Theorem B.

Theorem 21.46. Quillen's Theorem B

Let $\mathcal{C}, \mathcal{D} \in \text{Cat}$ and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ so that for each morphism $f : d \rightarrow d'$ in $\mathcal{D}$, the base change functor:

$$f : d'/F \rightarrow d/F,$$

is a homotopy equivalence. Then, for each $d \in \mathcal{D}$, the induced sequence of maps:

$$B[d/F] \xrightarrow{B[j]} B[\mathcal{C}] \xrightarrow{B[F]} B[\mathcal{D}],$$

of classifying spaces is a homotopy fibration sequence.

21.3 Appendix: Kan Extensions

Let's start with the definition of a Kan extension.

Definition 21.47

Let $\mathcal{C}, \mathcal{D}, \mathcal{E}$ be categories, let $F \in \text{Fun}(\mathcal{C}, \mathcal{E})$ and let $K \in \text{Fun}(\mathcal{C}, \mathcal{D})$.

1. A left Kan extension of F along K is a functor $\text{LKan}_K(F) : \mathcal{D} \rightarrow \mathcal{E}$ along with a natural transformation $\eta : F \Rightarrow (\text{LKan}_K(F) \circ K)$ with the following universal property: if (G, γ) is a pair consisting of a functor $G : \mathcal{D} \rightarrow \mathcal{E}$ and a natural transformation $\gamma : F \Rightarrow G$, then there is a unique natural transformation $\alpha : \text{LKan}_K(F) \Rightarrow G$ such that $\gamma = (\alpha K) \circ \eta$.
2. A right Kan extension of F along K is a functor $\text{RKan}_K(F) : \mathcal{D} \rightarrow \mathcal{E}$ along with a natural transformation $\epsilon : \text{RKan}_K(F) \Rightarrow F$ with the following universal property: if (G, δ) is a pair consisting of a functor $G : \mathcal{D} \rightarrow \mathcal{E}$ and a natural transformation $\delta : G \Rightarrow F$, then there exists a unique natural transformation $\beta : G \Rightarrow \text{RKan}_K(F)$ such that $\delta = \epsilon \circ (\beta K)$.

Note that we've used the whiskering process as described in ??.

We're aware that the definition is a lot to digest. Before we provide any kind of intuition and streamline the definition above into a more manageable form, let us discuss a rather neat situation in which Kan extensions arise naturally. Indeed, one can show that representable functors can be constructed by Kan extensions.

Construction 21.48. Let's start with representable functors; let $\mathcal{C}$ be a locally small category and let $x \in \mathcal{C}$.

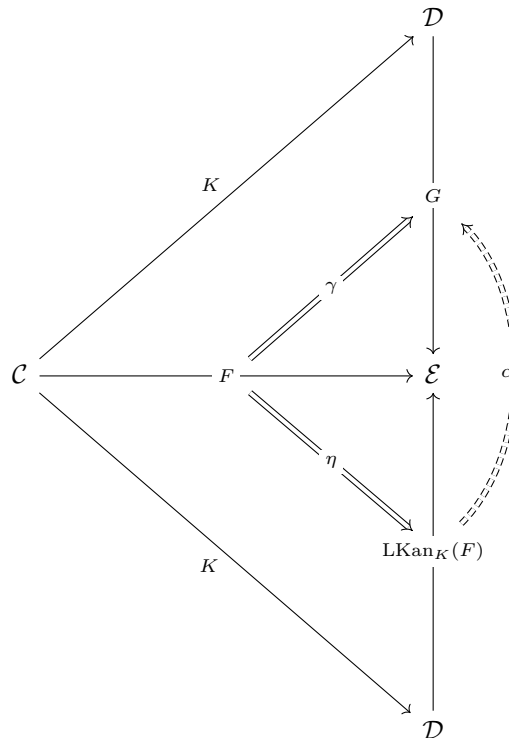
Claim. $\mathcal{C}(x, -)$ is the left Kan extension of some functor along another one.

We have a further remark about this example.

Remark 21.49. Actually, the fact that we can write representable functors as Kan extensions is a rather important fact. It means that there might be a Kan extension-theoretic proof of the Yoneda Lemma (which would be rather interesting just as a matter of unification) and indeed, there is certainly one. We'll discuss it in far more detail within ??, among other important aspects of the definition.

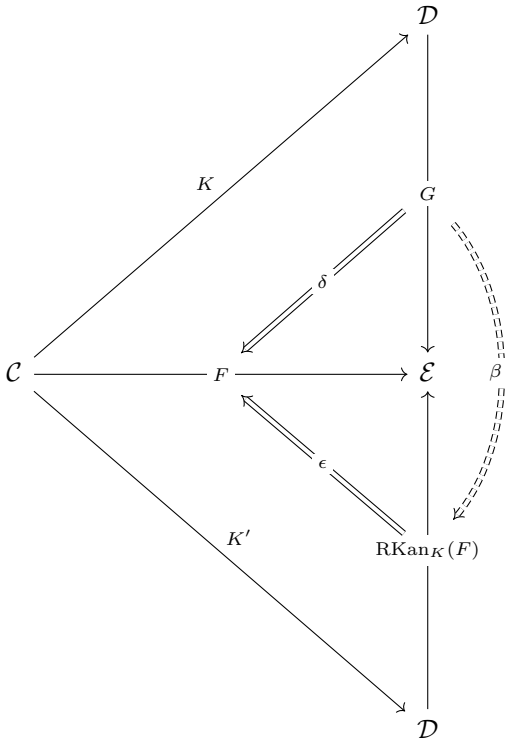
Okay, let's provide some intuition. We'll give you a diagram that encapsulates what's going on with both left and right Kan extensions; we'd suggest going back & forth between them for a little while before proceeding further.

Intuition 21.50. Let's consider left Kan extensions first. Using the data above, we draw the following diagram:



The diagram above actually introduces a great way to remember the universal property of a Kan extension; the diagram of natural transformations commutes up to whiskering. Nevertheless, it is somewhat misleading as well. For one thing, we've drawn a natural transformation from F to $\text{LKan}_K(F)$; this doesn't make sense because these functors are not defined on the same domain categories. Nevertheless, the mantra we've created above should help

negate any confusion. For right Kan extensions, we can draw another diagram of the same type:



Once again, the key point is that the universal property of a right Kan extension produces a diagram of natural transformations which commutes up to whiskering.

Okay, how about (co)limits?

Construction 21.51.

Finally, even adjunctions can be understood as Kan extensions.

Construction 21.52.

22 Localizations & Lifting Problems

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

Motivation.

22.1 Localizations of Categories

Let's start with some useful terminology.

Terminology 22.1. Let $\mathcal{C}, \mathcal{D}$ be categories, let $f \in \text{Mor}(\mathcal{C})$, let $W \subseteq \text{Mor}(\mathcal{C})$ and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$.

1. F inverts f iff $F(f)$ is an isomorphism.
2. F inverts W iff for every $f \in W$, we have that F inverts f .

We also define the following full subcategory of $\text{Fun}(\mathcal{C}, \mathcal{D})$ given by:

$$\text{Fun}_W(\mathcal{C}, \mathcal{D}),$$

which consists of functors $F : \mathcal{C} \rightarrow \mathcal{D}$ which invert W .

Okay, here's the definition of a weak/strict localization of categories.

Definition 22.2

Let $\mathcal{C}$ be a category and let $W \subseteq \text{Mor}(\mathcal{C})$. A weak localization of $\mathcal{C}$ with respect to W is a pair:

$$(\mathcal{C}_W, Q : \mathcal{C} \rightarrow \mathcal{C}_W),$$

consisting of a category $\mathcal{C}_W$ and the functor Q such that:

1. Q inverts W .
2. For any category $\mathcal{D}$, the following functor:

$$Q^* : \text{Fun}(\mathcal{C}_W, \mathcal{D}) \rightarrow \text{Fun}_W(\mathcal{C}, \mathcal{D}),$$

which is defined by pre-composition with Q is an equivalence.

The pair $(\mathcal{C}_W, Q)$ is a strict localization iff it satisfies (1) and the functor Q^* is an isomorphism^a.

^aIn both cases, we will often simply refer to $\mathcal{C}_W$ as a weak/strict localization.

Let's unpack what's being in both cases by property (2) within the definition.

Claim 22.1. Let $\mathcal{C}, \mathcal{D}$ be categories, let $W \subseteq \text{Mor}(\mathcal{C})$, let $F \in \text{Fun}_W(\mathcal{C}, \mathcal{D})$ and let $(\mathcal{C}_W, Q)$ be a weak localization of $\mathcal{C}$ with respect to W . Then, there exists a $\widehat{F} \in \text{Fun}(\mathcal{C}_W, \mathcal{D})$ so that $F \simeq \widehat{F} \circ Q$.

Claim 22.2. Let $\mathcal{C}, \mathcal{D}$ be categories, let $W \subseteq \text{Mor}(\mathcal{C})$, let $F \in \text{Fun}_W(\mathcal{C}, \mathcal{D})$ and let $(\mathcal{C}_W, Q)$ be a strict localization of $\mathcal{C}$ with respect to W . Then, there exists a $\widehat{F} \in \text{Fun}(\mathcal{C}_W, \mathcal{D})$ so that $F = \widehat{F} \circ Q$.

We have a remark about this result.

Remark 22.3. Using the notation given, the difference between the two is whether the following diagram:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ Q \downarrow & \nearrow \widehat{F} & \\ \mathcal{C}_W & & \end{array}$$

commutes exactly or up to natural isomorphism.

Since weak and strict localizations satisfy universal properties, they are both unique. However, due to the difference in how their respective universal properties manifest, they are unique in different ways.

Claim 22.3. Let $\mathcal{C}$ be a category and let $W \subseteq \text{Mor}(\mathcal{C})$.

1. If $(\mathcal{C}_1, Q_1), (\mathcal{C}_2, Q_2)$ are weak localizations of $\mathcal{C}$ with respect to W , then $\mathcal{C}_1$ and $\mathcal{C}_2$ are equivalent.
2. If $(\mathcal{C}_1, Q_1), (\mathcal{C}_2, Q_2)$ are strict localizations of $\mathcal{C}$ with respect to W , then $\mathcal{C}_1$ and $\mathcal{C}_2$ are isomorphic.

Clearly, any strict localization is a weak localization. So, if we want to know that weak localizations exist, then it suffices to simply construct strict localizations. This leads us to notation.

Notation 22.4. Let $\mathcal{C}$ be a category and let $W \subseteq \text{Mor}(\mathcal{C})$.

1. From now on, “localization” means strict localization.
2. The localization of $\mathcal{C}$ with respect to W is denoted by $\mathcal{C}[W^{-1}]$.

Other bits of notation for the localization include^a $\mathcal{C}_W$ or $W^{-1}\mathcal{C}$.

^aHowever, the latter will be used when we discuss calculi of fractions as a way of controlling localizations later.

It was shown in [44, Chapter 1.1, p.6-7] that it is possible to construct localizations and we will include a detailed construction. Before we get into it, it's worth spending some time on intuition.

Intuition 22.5. Let $\mathcal{C}$ be a category and let $W \subseteq \text{Mor}(\mathcal{C})$. Then, we want to form a set:

$$W^{-1} \stackrel{\text{def}}{=} \{f^{-1} : f \in W\},$$

of formal inverses so that we have form (finite) strings:

$$g_1 \circ f_1^{-1} \circ g_2 \circ f_2^{-1} \circ \dots$$

where each $f_i \in W$. Of course, this composition should make sense as a composition of morphisms in the localization of $\mathcal{C}$ at W . If we draw the expression above as a diagram, then we obtain the following diagram:

$$\bullet \xrightarrow{g_1} \bullet \xrightarrow{f_1^{-1}} \bullet \xrightarrow{g_2} \bullet \xrightarrow{f_2^{-1}} \dots$$

Again, we can't make sense of this diagram directly in $\mathcal{C}$ because the morphisms in W are not necessarily isomorphisms in $\mathcal{C}$. Nevertheless, we *can* make sense of the following zig-zag of morphisms in $\mathcal{C}$:

$$\bullet \xrightarrow{g_1} \bullet \xleftarrow{f_1} \bullet \xrightarrow{g_2} \bullet \xleftarrow{f_2} \dots$$

Of course, we can't compose the morphisms in this zig-zag within $\mathcal{C}$. However, the point is that we can construct the localization as a category of zig-zags where we identify two zig-zags appropriately by some relation. One should be able to see, intuitively, why the localization can get much larger than $\mathcal{C}$; it's mainly because the zig-zags above can be of arbitrary length and there's little we can do to control this directly. Here's our strategy:

1. Take $\mathcal{C}$ and do a "free" construction which gives the zig-zags above artificially.
2. Identify these zig-zags by an appropriate relation.

Once one has constructed the localization, one simply checks that it works.

The intuition we've laid out above motivates the following bit of terminology.

Terminology 22.6. A quiver is a triple $Q = (Q_0, Q_1, s, t)$ consisting of the following data:

1. A set of vertices.
2. A set of arrows between vertices^a.
3. Two maps $s, t : Q_1 \rightarrow Q_0$, known as the source and target maps.

If $a \in Q_1$, then we write it as $a : s(a) \rightarrow t(a)$.

^aThese can be explicitly modelled as pairs of vertices.

We'll start by constructing the path category of a quiver first.

Construction 22.7. Let $Q = (Q_0, Q_1, s, t)$ be a quiver. We define a structure $\text{Path}(Q)$ as follows:

1. The objects of this structure consist of the vertices of Q .
2. Let $a, b \in Q$. A morphism from a to b consists of a zig-zag:

$$(a_n = b, f_n, a_{n-1}, f_{n-1}, \dots, a_1, f_1, a_0 = a),$$

where each a_i is a vertex of Q and each f_i is an arrow from:

$$a_{i-1} \rightarrow a_i.$$

The composition is defined by the concatenation of two such lists to form a larger list^a.

It's easily seen that this is a category; if $a \in Q_0$, then the identity morphism is just (a, a) . Moreover, the composition in this category is clearly unital and associative, so we're done.

^aThe slight subtlety here is that in a composition $g \circ f$, the codomain of f will coincide with the domain of g . So, if we write these as lists and just blindly concatenate, then we do get that both the codomain of f and domain of g appear in the final list. So, we get rid of that possibility simply by not having the codomain of f and domain of g BOTH appear.

Construction 22.8. Let $\mathcal{C}$ be a category and let $W \subseteq \text{Mor}(\mathcal{C})$. We proceed via two steps.

1. We construct a quiver C from $\mathcal{C}$ and W .
2. We construct $\mathcal{C}[W^{-1}]$ as a quotient of $\text{Path}(C)$.

Okay, let's proceed with the first step. We define $C = (C_0, C_1, s, t)$ as follows:

1. C_0 is the set of objects of $\mathcal{C}$.
2. C_1 is the disjoint union of $\text{Mor}(\mathcal{C})$ with the following set:

$$W^{-1} \stackrel{\text{def}}{=} \{s^{-1} \mid s \in W\}.$$

Basically, W^{-1} IS just W but we've labelled each element of W differently.

Let the composition in $\text{Path}(C)$ be denoted by $\circ_P$. Consider the following relations:

1. If $f, g \in \text{Mor}(\mathcal{C})$ are composable, then $g \circ_P f = g \circ f$.
2. If $x \in \mathcal{C}$, then let Id_x^P be the identity of x in $\text{Path}(C)$. We demand that this is Id_x in $\mathcal{C}$.
3. If $s \in W$, then we have $s^{-1} \circ_P s = s \circ_P s^{-1} = \text{Id}_x^P$.

Let $\sim$ denote the equivalence relation on $\text{Mor}(C)$ generated by the relations above. Now, we define:

$$\mathcal{C}[W^{-1}] \stackrel{\text{def}}{=} \text{Path}(C) / \sim.$$

So, note that the objects of $\mathcal{C}[W^{-1}]$ are the objects of $\mathcal{C}$, while the morphisms from, say, x to y are just equivalence classes of zig-zags that will be given according to the following specification:

$$x \longrightarrow \bullet \xleftarrow{\in W} \bullet \longrightarrow \dots \xleftarrow{\in W} \bullet \longrightarrow y$$

We define a map of categories:

$$Q : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}],$$

as follows:

1. If $x \in \mathcal{C}$, then we let $Q(x) \stackrel{\text{def}}{=} x$.
2. On the morphisms, we define Q simply by the following composition:

$$\text{Mor}(\mathcal{C}) \longrightarrow \text{Mor}(\mathcal{C}) \amalg W^{-1} \longrightarrow \text{Mor}(\text{Path}(\mathcal{C})) \longrightarrow \text{Mor}(\mathcal{C}[W^{-1}])$$

One can “think of” $Q(f)$ as $\frac{f}{1}$ when f is a morphism in $\mathcal{C}$.

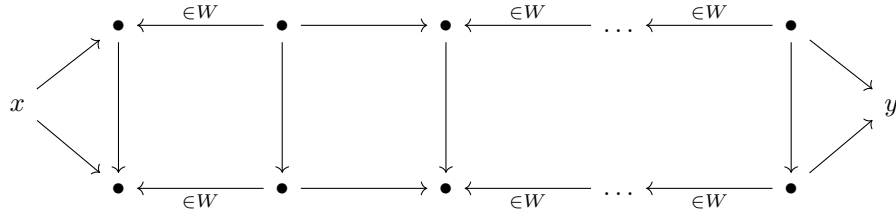
The equivalence relation $\sim$ defined above identifies two zig-zags by viewing a zig-zag as a “word” and “reducing” the word by using the explicit relations which generate $\sim$. Two zig-zags are equivalent iff they can be related^a:

1. by composing two consecutive arrows going in the same direction within $\mathcal{C}$ to form a single arrow;
2. by removing the identity morphisms by composition (see (1));
3. by removing instances of the following diagrams:

$$\begin{array}{c} \bullet \xrightarrow{f} \bullet \xleftarrow{f} \bullet \\ \bullet \xleftarrow{f} \bullet \xrightarrow{f} \bullet \end{array}$$

whenever $f \in W$ or;

4. through the existence of the following commutative diagram:



which resembles a hammock^b. Actually, it isn’t immediately obvious that two zig-zags should be identified if such a commutative diagram as the one above exists between them; we’ll explain why it works in [Intuition 22.9](#).

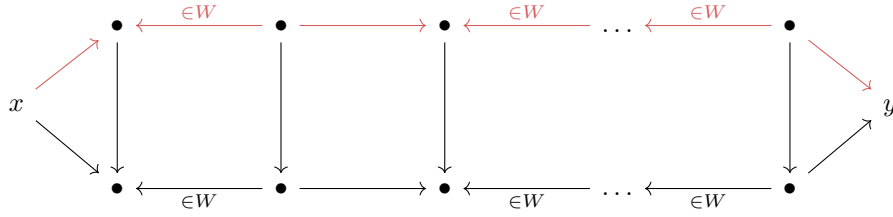
This construction gives us our desired pair $(\mathcal{C}[W^{-1}], Q)$.

^aThe author thanks his friend, Carl-Fredrik Lidgren, for providing this piece of insight.

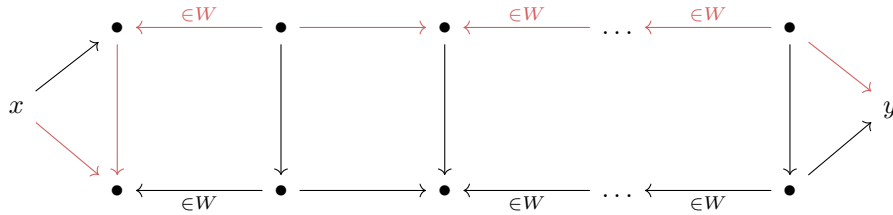
^bThis is why this localization is sometimes called the hammock localization.

Here’s some intuition regarding the “hammock” part of the construction above.

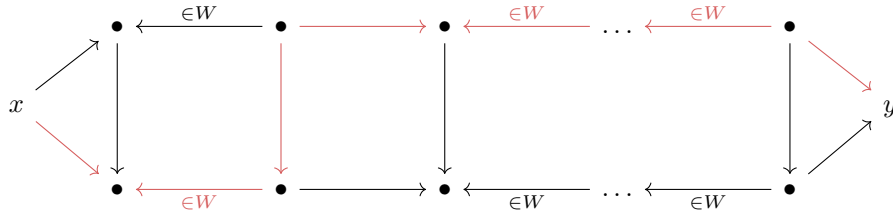
Intuition 22.9. So, the idea is that we start with the zig-zag at the top:



and note that since the left-most triangle is commutative, we can identify the given zig-zag with the following one:



and since the first square commutes, we can identify the given zig-zag with the following one:



and so on. That is, we proceed by identifying zig-zags in the commutative diagram through the relations described in the construction. Using these identifications, we can “morph” the top-most zig-zag into the bottom-most zig-zag. That is, the hammock relation is simply generated by the explicit relations we described in the construction.

To prove that localizations exist, we’ll need a lemma.

Lemma 22.10 Let $\mathcal{C}$ be a category, let $W \subseteq \text{Mor}(\mathcal{C})$ and suppose that we have a pair:

$$(\mathcal{C}_W, Q),$$

consisting of a category $\mathcal{C}_W$ and a functor $Q : \mathcal{C} \rightarrow \mathcal{C}_W$. TFAE:

1. The given pair is a strict localization of $\mathcal{C}$ with respect to W .
2. The functor^a Q^* is bijective on objects for any category $\mathcal{D}$.

^aSee Definition 22.2.

Proof.

(1) $\Rightarrow$ (2): This is clear.

(2) $\Rightarrow$ (1): It's enough to show that Q^* is fully faithful, so let $\mathcal{D}$ be a category and let $F, G \in \text{Fun}(\mathcal{C}_W, \mathcal{D})$. Then, we need to check that the following map:

$$Q^* : \text{Nat}(F, G) \rightarrow \text{Nat}(F \circ Q, G \circ Q),$$

is a bijection. Let $\alpha : F \circ Q \Rightarrow G \circ Q$ be a natural transformation and note that this corresponds uniquely to a functor $\alpha' : \mathcal{C} \rightarrow \text{Fun}(2, \mathcal{D})$, where 2 is the discrete category consisting of two objects. Now, we check that for each $f \in W$, it is the case that $\alpha'(f)$ is invertible; this will follow from the fact that F, G invert W and the naturality of α . Hence, there is a unique functor $\beta : \mathcal{C}_W \Rightarrow \text{Fun}(2, \mathcal{D})$ such that:

$$\alpha = \beta \circ Q = Q^*(\beta),$$

as was desired^a. ■

^aREWRITE THIS PROOF TO MAKE IT MORE INTUITIVE.

We're ready to show that localizations exist by verifying that Construction 22.8 works.

Theorem 22.11

Let $\mathcal{C}$ be a category and let $W \subseteq \text{Mor}(\mathcal{C})$. The localization of $\mathcal{C}$ with respect to W exists.

Proof. Of course, the pair that we wish to study is just $(\mathcal{C}[W^{-1}], Q)$, as provided in Construction 22.8. It remains for us to show that Q inverts W and also satisfies the required universal property. This leads us to the first claim.

Claim. Q inverts W .

Let $f \in W$. Then, we have that:

$$(\bullet \xrightarrow{f} \bullet) \quad \circ \quad (\bullet \xleftarrow{f} \bullet) \quad = \quad \bullet \xleftarrow{f} \bullet \xrightarrow{f} \bullet \quad = \quad \bullet \xrightarrow{\text{Id}} \bullet$$

with the other composition giving the same result. This proves the claim. The next step is to establish the universal property and in light of Lemma 22.10, it's enough for us to prove the following claim.

Claim. Let $\mathcal{D}$ be a category and let $F \in \text{Fun}_W(\mathcal{C}, \mathcal{D})$. Then, there exists a unique $\widehat{F} \in \text{Fun}(\mathcal{C}[W^{-1}], \mathcal{D})$ so that:

$$F = \widehat{F} \circ Q.$$

Since Q is the identity on objects and $\mathcal{C}$ has the same objects as its localization, it follows that $\widehat{F}$ is the identity on the objects. For morphisms, note that we can break up a zig-zag as the following composition:

$$\bullet \rightarrow \bullet \leftarrow \bullet \dots \bullet \leftarrow \bullet \rightarrow \bullet = (\bullet \rightarrow \bullet) \circ (\bullet \leftarrow \bullet) \circ \dots \circ (\bullet \leftarrow \bullet) \circ (\bullet \rightarrow \bullet).$$

This demonstrates that the value of $\widehat{F}$ on a given zig-zag depends on how it acts on $\bullet \rightarrow \bullet$ and $\bullet \leftarrow \bullet$. The former are precisely the morphisms in the range of Q , so we have that:

$$\widehat{F}(\bullet \rightarrow \bullet) = F(\bullet \rightarrow \bullet).$$

On the other hand, if $s : x \rightarrow y$ is a morphism in W , then we have that:

$$Q(s)^{-1} = (y \xleftarrow{s} x),$$

and every zig-zag of the form $\bullet \leftarrow \bullet$ arises in this way. Therefore, we must have that:

$$\widehat{F}(Q(s)^{-1}) = (F(Q(s)))^{-1},$$

which defines $\widehat{F}$ on the zig-zags and completes the proof^a. ■

^aThis particular rendition of the proof of existence owes itself to the lectures given to the author by Carl-Fredrik Lidgren.

We have two general remarks about localizations.

Remark 22.12. Morally, the proof of Theorem 22.11 is perfectly fine and standard. However, we've kindly swept size issues under the rug and actually, the fact that the zig-zags can be of arbitrary length means that the localization of a category with respect to some collection of morphisms can be very large compared to the starting category. Such issues can, of course, be resolved partly by working with Grothendieck universes but it gets even more complicated when one realizes that we **do not** have a reasonable way of working concretely with the localization.

Remark 22.13. Indeed, another problem with the localization is that the actual “homotopical meaning” of the construction is not completely clear. Indeed, the idea is that W is supposed to represent the set of “weak equivalences” that we pick out in a category; it's a weaker notion of equivalence which is, somehow, good enough for practical purposes. However, our construction of the localization does not shed light on the structure of these weak equivalences nor does it explain the relation to homotopy theory.

We will eventually deal with the problems above by introducing model categories relatively far down the road; one needs to do some classical homotopy theory in order to properly motivate those structures. We should stress that we will only

present the basics of the theory of model categories to convince you that there are frameworks for doing homotopy theory quite generally. However, we emphasize that the notion of a model category is not exactly the one which is relevant for the study of noncommutative homotopy theory; we will need to do much more work than that to have a general framework that works for categories of C^* -algebras, for example. With that said, let us speak about how localizations interact with adjoints.

Before we define a calculus of fractions, we need a bit of terminology.

Terminology 22.14. Let $\mathcal{C}$ be a category and let $S \subseteq \text{Mor}(\mathcal{C})$. S is a multiplicative system/set of morphisms in $\mathcal{C}$ iff it is closed under compositions and contains all identity morphisms.

Okay, here's the definition of a calculus of left fractions.

Definition 22.15

Let $\mathcal{C}$ be a category and let $W \subseteq \text{Mor}(\mathcal{C})$ be a multiplicative system. W admits a calculus of left fractions iff:

1. For any diagram of solid arrows:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ s \in W \downarrow & & \downarrow l \in W \\ x' & \xrightarrow{g} & y' \end{array}$$

we have a dashed arrow such that the square commutes^a.

2. For any diagram of solid arrows:

$$x' \xrightarrow{s \in W} x \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} y \xrightarrow{t \in W} y'$$

we have a dashed arrow such that the diagram commutes^b.

W admits a calculus of right fractions iff W^{op} admits a calculus of left fractions in $\mathcal{C}^{\text{op}}$.

^aNote that this also means that we have to specify y' .

^bNote that this also means we have to specify y' .

We have a remark about this definition.

Remark 22.16. The first condition allows us to replace an arrow of the form:

$$f \circ s^{-1},$$

with an arrow of the form $l^{-1} \circ g$. Of course, s and l are not necessarily invertible but they will be invertible in the localization with respect to S . The second condition tells that if we have a situation where:

$$f \circ s = g \circ s,$$

then we can, instead, turn that into a situation where:

$$t \circ f = t \circ g,$$

for some morphism $t \in W$.

As we tried to emphasize in our motivation above, the key is that calculus of fractions gives us a new way of describing the morphisms in the localization and gaining general control over the localization. Before we demonstrate the way in which this is the case, let us highlight a specific situation where these axioms are satisfied.

Claim 22.4. Let $\mathcal{C}, \mathcal{D}$ be categories and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ be a Bousfield localization. Define:

$$W \stackrel{\text{def}}{=} \{s \in \text{Mor}(\mathcal{C}) : F(s) \text{ is an isomorphism}\}.$$

Then, W is a multiplicative system and admits a calculus of left fractions.

22.2 Categorical Lifting Problems

In light of the motivation above, let us describe lifting problems in general. Before we provide a definition in an arbitrary category, let us sketch what such a problem should look like in general terms.

With the result above out of the way, let us proceed with explaining the notion of a weak factorization system. To motivate the definition, here's a nice miscellaneous result.

Claim 22.5. Let $\mathcal{L}$ denote the set of surjections in Set and let $\mathcal{R}$ denote the set of injections.

With retracts out of the way, we can specify weak factorization systems.

Terminology 22.17. Let $\mathcal{C}$ be a category and let $S, T \subseteq \text{Mor}(\mathcal{C})$ so that:

1. S and T are closed under retracts.
2. $S \square T$.
3. If $f \in \text{Mor}(\mathcal{C})$, then it has a factorization $f = p \circ i$ with $p \in T$ and $i \in S$.

The pair (S, T) is known as a weak factorization system (WFS).

With this, we can finally define model structures; this is different from the one given by Quillen in [78].

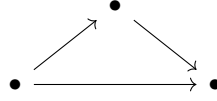
Definition 22.18

Let $\mathcal{C}$ be a category. A model structure on $\mathcal{C}$ consists of a triple:

$$(\text{Cof}, W, \text{Fib}),$$

where W, Fib are full subcategories of $\text{Arr}(\mathcal{C})$ such that:

1. W has the 2-out-of-3 property. Namely, if we have a commutative triangle:



with 2-out-of-3 morphisms in this triangle being in W , then the third is also in W .

2. The pairs $(\text{Cof}, W \cap \text{Fib})$ and $(W \cap \text{Cof}, \text{Fib})$ are WFSs.

A model category is the datum of a finitely bicomplete category $\mathcal{C}$ equipped with a fixed model structure.

We have some terminology that goes along well with the definition of a model category.

Terminology 22.19. Let $\mathcal{C}$ be a category with some model structure $(\text{Cof}, W, \text{Fib})$. Then:

1. Fib is the set of fibrations.
2. Cof is the set of cofibrations.
3. W is the set of weak equivalences.
4. $W \cap \text{Fib}$ is the set of all trivial fibrations.
5. $W \cap \text{Cof}$ is the set of all trivial cofibrations.

Moreover, (Fib, Cof) is known as the phantom structure on $\mathcal{C}$.

We have a remark about examples and our work so far.

Remark 22.20. In general, it is very difficult to provide examples of model categories because the verification that the appropriate datum on a category actually provides a model structure can be extremely difficult. There are three classes of examples that are most heavily prevalent in the literature, partly due to their classical nature but also because they get used in modern work in homotopy theory. These classes are listed as follows:

1. Topological spaces.
2. Simplicial sets.
3. Chain complexes.

Actually, [78] also gives other interesting examples of model categories, though complete verifications of such categories actually having a model structure are not provided. With regards to the three examples above, there are now many resources which give a systematic overview of various interesting model structures related to the objects listed above. Therefore, we will not repeat any of that over here, especially since we are barely going to need any of it^a. Instead, we refer the reader to the very detailed webpages [6], [7] and [5], which also contain more detailed references.

^aIn fact, we are strictly **not** going to need any of it for the explicit work that we do in this thesis. On the other hand, we do use the theory of ∞ -categories later in this thesis and while we use it as a “machine”, this machine does rely on several model structures on categories of simplicial sets which we do not discuss here, though we do mention some work done by (Denis-Charles) Cisinski on the construction of such model structures in ???. This is related to some of our work in the next subsection.

One may ask where operator algebras stand with respect to these developments. As we described in Remark 6.10, it was already noticed by Schochet that one might reasonably want to bring the homotopy theory of C^* -algebras into the framework of Quillen’s model categories. The point was to find a model structure on $C^*\text{Alg}$ such that the W would be the homotopy equivalences and Fib would consist of the Schochet fibrations. In fact, no such model structure exists.

Theorem 22.21

There is no model structure on $C^*\text{Alg}$ such that:

1. the weak equivalences are homotopy equivalences and;
2. the fibrations are Schochet fibrations.

Let us discuss this theorem and literature related to it in the next remark.

Remark 22.22. A variant of this theorem was proved by Andersen and Grodal in the unpublished preprint [2]; they showed that $C^*\text{Alg}$ can be equipped with the structure of a Baues fibration category and demonstrated the analogous impossibility result. The theorem above, however, was shown by (Otgonbayar) Uuye in [96, Theorem A.1, p.15-16]; he proves it by showing that the loop functor on the homotopy category of commutative C^* -algebras does not admit a left adjoint. Nevertheless, recent work by (Michael) Joachim and (Anupam) Datta suggests that it is possible to place a model structure on the category of locally multiplicative convex C^* -algebras; see [32]. Similar work of this type has been done in the past involving projective systems of C^* -algebras; see [10].

The benefit of Joachim and Datta’s work is that it addresses the homotopical nature of the so-called Cuntz q -construction; this is used in the algebraic construction of Kasparov’s KK-theory functor in [31]. Introducing this construction here would be a distraction but we will note that their (Joachim & Datta) work tells us that the Cuntz q should be viewed, up to stabilization, as a cofibrant replacement functor; see Corollary 8.10 on page 37 of their

paper. This seems to address a question of Meyer as stated in [67, p.19], so it is clear that a homotopical perspective elucidates known constructions and give us indications of the role they should play in the theory.

23 Introducing Categories of Fibrant Objects

Summary

In this lecture, we will cover the following topics:

- 1.
- 2.

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| Motivation.

23.1 Working Towards The Definition & Some Examples

Ultimately, we wish to study categories $\mathcal{C}$ which are equipped with two full subcategories:

$$W, \text{Fib},$$

of $\text{Arr}(\mathcal{C})$; the first will be a category of *weak equivalences* and the second will be a category of *fibrations*. The literature which deals with these categories systematically is somewhat sparse and we aim to extend some results about these categories while taking inspiration from our work on the homotopy theory of operator algebras as done in ???. With that general aim in mind, let us introduce some terminology that will only be relevant later in this subsection.

Terminology 23.1. Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$.

1. W satisfies the 2-out-of-3 property iff for any diagram of two composable arrows:

$$x \xrightarrow{f} y \xrightarrow{g} z$$

where any two of f, g and $g \circ f$ lies in W , then the third lies in W .

2. W satisfies the 2-out-of-6 property iff for a diagram of three composable arrows:

$$w \xrightarrow{f} x \xrightarrow{g} y \xrightarrow{h} z$$

where $g \circ f$ and $h \circ g$ lies in W , we have that f, g, h and $h \circ g \circ f$ lie in W .

If $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$ is the localization of $\mathcal{C}$ with respect to W , then we say that W is saturated in $\mathcal{C}$ iff for any morphism $f \in \text{Mor}(\mathcal{C})$ with $\gamma(f)$ being an isomorphism, we have that $f \in W$.

Here are some basic (but useful facts) which are strewn across the literature.

Claim 23.1. Let $\mathcal{C}$ be a category and let W be a full subcategory of $\text{Arr}(\mathcal{C})$.

1. The isomorphisms of $\mathcal{C}$, viewed as a full subcategory of $\text{Arr}(\mathcal{C})$, satisfy the 2-out-of-6 property.
2. If W satisfies the 2-out-of-6 property and contains the identity morphisms, then it is 2-out-of-3.
3. If W satisfies the 2-out-of-6 property and contains the identity morphisms, then it contains all isomorphisms.

This is proved on Page 274.

Claim 23.2. Let $\mathcal{C}, \mathcal{D}$ be categories, let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ and let W be a full subcategory of $\text{Arr}(\mathcal{C})$ consisting of all morphisms whose images under F become isomorphisms in $\mathcal{D}$. Then, W satisfies the 2-out-of-6 property^a.

^aThis is proved on Page 274.

Claim 23.3. Let $\mathcal{C}$ be a category and let W be a saturated full subcategory of $\text{Arr}(\mathcal{C})$. Then^a, W is 2-out-of-6.

^aThis is proved on Page 274.

We will work our way up to the definition of a category of fibrant objects, starting with the notion of a fibration category.

Definition 23.2

A fibration category is a pair:

$$(\mathcal{C}, \text{Fib}_{\mathcal{C}}),$$

of a pointed category $\mathcal{C}$ with a full subcategory $\text{Fib}_{\mathcal{C}}$ of $\text{Arr}(\mathcal{C})$ consisting of *fibrations* such that:

1. The isomorphisms in $\mathcal{C}$ are in $\text{Fib}_{\mathcal{C}}$.
2. The composition of any two fibrations is a fibration.
3. If $x \in \mathcal{C}$, then the unique morphism $x \rightarrow 0$ lies in $\text{Fib}_{\mathcal{C}}$.
4. Pullbacks by fibrations exist and fibrations are preserved by base extensions (recall Terminology 6.30).

A cofibration category is a category with cofibrations as defined on [97, p.320], being exactly dual to fibration categories.

Here are two simple results.

Claim 23.4. Let $\mathcal{C}$ be a category.

1. If $\mathcal{C}$ is a fibration category, then $\mathcal{C}^{\text{op}}$ is a cofibration category when paired with $\text{Fib}_{\mathcal{C}}^{\text{op}}$.
2. If $\mathcal{C}$ is a cofibration category, then $\mathcal{C}^{\text{op}}$ is a fibration category when paired with $\text{Cof}_{\mathcal{C}}^{\text{op}}$.

This is proved on Page 274.

Claim 23.5. Let $\mathcal{C}$ be a fibration category. Then, $\mathcal{C}$ admits all finite products^a.

^aThis is proved on Page 274.

Of course, one can also define structure-preserving functors, even though we will not use this notion.

Terminology 23.3. Let $\mathcal{C}, \mathcal{D}$ be fibration categories and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$. Then, F is exact iff it preserves the zero object, if it preserves fibrations and if it preserves base extensions along a fibration.

Here are two nice constructions of fibration categories.

Construction 23.4. Let $\mathcal{C}$ be a fibration category and let I be a small category. We define:

$$\text{Fib}_{\text{Fun}(I, \mathcal{C})},$$

as follows: if $F, G \in \text{Fun}(I, \mathcal{C})$ and $\alpha \in \text{Nat}(F, G)$, then α is a fibration iff its components are fibrations in $\mathcal{C}$.

| **Claim.** $\text{Fun}(I, \mathcal{C})$ becomes a fibration category with the data of $\text{Fib}_{\text{Fun}(I, \mathcal{C})}$.

We will show each axiom of a fibration category one-by-one.

1. If α is an isomorphism in the category $\text{Fun}(I, \mathcal{C})$, then its components are isomorphisms in $\mathcal{C}$ and these are certainly fibrations in $\mathcal{C}$. So, it follows that α is a fibration.
2. Let α and β be natural transformations so that $\beta \circ \alpha$ is well-defined and so that each of them is a fibration in $\text{Fun}(I, \mathcal{C})$. Now, let $x \in \mathcal{C}$ and note that $(\beta \circ \alpha)_x = \beta_x \circ \alpha_x$. Since β_x and α_x are fibrations, their composition is a fibration and this means that $\beta \circ \alpha$ is a fibration on its components so it is a fibration in $\text{Fun}(I, \mathcal{C})$.
3. Let $F \in \text{Fun}(I, \mathcal{C})$ and let $\alpha : F \Rightarrow 0$, where the zero functor in $\text{Fun}(I, \mathcal{C})$ is a zero object which takes every object in I to the zero object in $\mathcal{C}$. Now, the components of α are the unique morphisms from each $F(i)$ to the zero object of $\mathcal{C}$ so these are fibrations in $\mathcal{C}$. But this means that α is a fibration in $\mathcal{C}$.
4. Limits in $\text{Fun}(I, \mathcal{C})$ are computed point-wise, so fibrations are preserved by base extensions.

This completes the proof.

Construction 23.5. Let $\mathcal{C}$ be a pointed model category. We define a full subcategory:

$$\mathcal{C}_f,$$

consisting of the fibrant objects^a of $\mathcal{C}$.

| **Claim.** If we restrict the fibrations of $\mathcal{C}$ to fibrations between objects of $\mathcal{C}_f$, then $\mathcal{C}_f$ is a fibration category.

With the definition of a model category that we have given in [Definition 22.18](#), it seems non-trivial to verify that restricting to the fibrant objects as we have described above gives us a fibration category. Nevertheless, one can show that our definition is equivalent to the original one due to Quillen ([\[78, Definition 1.1.1, p. 14-15\]](#)) and that definition makes this result obvious. This equivalence is discussed to some degree in [\[25, Remark 2.2.3, p.33-34\]](#) and since we will not need this result beyond its philosophical significance, we choose to say nothing more about it.

^aAn object $x \in \mathcal{C}$ is fibrant iff the unique morphism $x \rightarrow 0$ is a fibration.

Finally, we have two examples coming from operator algebra theory.

Example 23.6. We have the following claim.

| **Claim.** TFSH:

1. $G\text{-C}^*\text{Alg}$ is a fibration category when equipped with the Schochet fibrations.
2. $G\text{-BanAlg}$ is a fibration category when equipped with the Schochet fibrations.

We'll verify each of the axioms as listed in [Definition 23.2](#) one-by-one.

1. Clearly, $G\text{-C}^*\text{Alg}$ and $G\text{-BanAlg}$ have zero objects; it's just $\{0\}$.
2. In both cases, axiom (1) is taken care of by statement (3) of [Proposition 6.29](#).
3. In both cases, axiom (2) is taken care of by statement (1) of [Proposition 6.29](#).
4. In both cases, axiom (3) is taken care of by statement (2) of [Proposition 6.29](#).
5. In both cases, axiom (4) is taken care of by statement (5) of [Proposition 6.29](#).

This proves the result.

For the next example, we will need the notion of a semisplit surjection as introduced in [Terminology 6.32](#)

Example 23.7. Let $G \in \text{Grp}(\text{LCH})$ and define:

$$\text{Fib} \stackrel{\text{def}}{=} \{\text{Semisplit surjections in } G\text{-C}^*\text{Alg}\}.$$

Here's the main claim.

| **Claim.** The pair $(G\text{-C}^*\text{Alg}, \text{Fib})$ forms a fibration category.

We'll verify each of the axioms as listed in [Definition 23.2](#) one-by-one.

1. Clearly, $G\text{-C}^*\text{Alg}$ and $G\text{-BanAlg}$ have zero objects; it's just $\{0\}$.
2. In both cases, axiom (1) is taken care of by statement (3) of [Proposition 6.29](#).
3. In both cases, axiom (2) is taken care of by statement (1) of [Proposition 6.29](#).
4. In both cases, axiom (3) is taken care of by statement (2) of [Proposition 6.29](#).
5. This is the most difficult one to verify and a complete verification has been done in the proof of [\[23, Proposition 2.10, p.26-27\]](#). That result is stated for discrete, countable groups but the proof works for LCH groups.

This proves the result.

Once we have a fibration category, we also would want to add an additional bit of data about *weak equivalences*; this is a second subcategory of the arrow category that we include in our data.

Definition 23.8

A weak fibration category is a triple:

$$(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}}),$$

such that:

1. $(\mathcal{C}, \text{Fib}_{\mathcal{C}})$ is a fibration category.
2. $W_{\mathcal{C}}$ is a full subcategory of $\text{Arr}(\mathcal{C})$; this is known as the subcategory of weak equivalences.
3. All isomorphisms in $\mathcal{C}$ are contained in $W_{\mathcal{C}}$.
4. Suppose we have a commutative diagram in $\mathcal{C}$:

$$\begin{array}{ccccc} b & \xrightarrow{\quad\quad} & a & \xleftarrow{\quad\quad} & c \\ \downarrow & & \downarrow & & \downarrow \\ b' & \xrightarrow{\quad\quad} & a' & \xleftarrow{\quad\quad} & c' \end{array}$$

where the red arrows are fibrations and the vertical arrows are weak equivalences. Then, the induced map:

$$b \times_a c \rightarrow b' \times_{a'} c',$$

is in $W_{\mathcal{C}}$. This requirement is known as the Gluing Lemma.

A weak cofibration category is a category with cofibrations and weak equivalences as defined in [97, p.326].

Since our notation for the two subcategories of a weak fibration category involve the notation for the category itself, we will find it reasonable to mostly omit explicit mention of the subcategories when referring to a weak fibration category. Here's some further terminology that will be repeated later for categories of fibrant objects.

Terminology 23.9. Let $\mathcal{C}$ be a category equipped with two full subcategories:

$$\text{Fib}_{\mathcal{C}}, W_{\mathcal{C}},$$

of $\text{Arr}(\mathcal{C})$. Then, the morphisms in the subcategory $\text{Fib}_{\mathcal{C}} \cap W_{\mathcal{C}}$ are known as trivial fibrations.

Before we give examples, let us develop some theory associated to this definition since it is rather interesting. In particular, the Gluing Lemma is ideally something we'd like to establish, rather than postulate. We have the following result, which is an amalgamation of facts that are scattered throughout the literature but are useful to have in a single place.

Proposition 23.10

Let $\mathcal{C}$ be a category and suppose that there are two full subcategories Fib, W of $\text{Arr}(\mathcal{C})$ so that:

- All isomorphisms of $\mathcal{C}$ are in W .
- W satisfies the 2-out-of-3 property.
- If $f \in \text{Mor}(\mathcal{C})$, then $f = p \circ i$, where $p \in \text{Fib}$ and $i \in W$.
- Fibrations are preserved by base extensions.

Then, the following statements hold:

1. If $f \in \text{Mor}(\mathcal{C})$ is a section of a morphism in W , then $f \in W$.
2. If $f \in \text{Mor}(\mathcal{C})$ with factorization:

$$f = p \circ i,$$

then $f \in W$ iff $p \in \text{Fib} \cap W$.

3. If $\mathcal{C}$ is closed under pullbacks, then TFAE:
 - (a) The Gluing Lemma holds.
 - (b) A pullback of a morphism in W along a morphism in Fib is in W .

Proof. We'll prove each statement one-by-one.

1. Let $g \in W$ so that $g \circ f = \text{Id}$. Since Id is an isomorphism, it is in W . On the other hand, $g \in W$ so the 2-out-of-3 property implies that $f \in W$.
2. We'll prove each implication separately.
 - $\Rightarrow$: If $f \in W$, then the 2-out-of-3 property implies that $p \in W$.
 - $\Leftarrow$: If $p \in W$, then the 2-out-of-3 property means that $f \in W$.

This completes the proof.

3. This is the dual of the result proved in [66]. ■

Here are some examples.

Example 23.11. Let $\mathcal{C}$ be a fibration category & let $W_{\mathcal{C}}$ be the full subcategory of $\text{Arr}(\mathcal{C})$ of all isomorphisms in $\mathcal{C}$.

Claim 23.6. The triple $(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}})$ is a weak fibration category.

This is proved on Page 274; it is known as the minimal weak fibration structure on a fibration category.

Example 23.12. Let $\mathcal{C}$ be a fibration category & let $W_{\mathcal{C}}$ just be $\text{Arr}(\mathcal{C})$.

Claim. The triple $(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}})$ is a weak fibration category.

This is completely clear; it is known as the maximal weak fibration structure on a fibration category.

Our interest is mainly in categories of fibrant objects; we will relate these to weak fibration categories later. For now, here is some terminology which will lead us towards categories of fibrant objects.

Notation 23.13. Let $\mathcal{C}$ be a category and suppose we have the following commutative diagram:

$$\begin{array}{ccc} x & \xrightarrow{\beta} & y \\ \alpha \downarrow & & \downarrow f \\ x' & \xrightarrow{g} & y' \end{array}$$

If the fibre product of the diagram above exists, then the commutative diagram above is a pullback cone. So, there exists a unique morphism $(\alpha, \beta) : x \rightarrow x' \times_{y'} y$ such that the resulting diagram formed is commutative. In particular, let $f : x \rightarrow y$ be a morphism in $\mathcal{C}$. Then, the diagonal induced by f is defined by:

$$\Delta_f \stackrel{\text{def}}{=} (\text{Id}_x, \text{Id}_x).$$

Diagram-wise, the one to imagine is the following:

$$\begin{array}{ccc} x & \xrightarrow{\text{Id}_x} & x \\ \text{Id}_x \downarrow & & \downarrow f \\ x & \xrightarrow{f} & y \end{array}$$

commutative diagram.

Terminology 23.14. Let $\mathcal{C}$ be a category with finite products, final object e and two full subcategories:

$$\text{Fib}, W \subseteq \text{Arr}(\mathcal{C}).$$

If $x \in \mathcal{C}$, then a path space of x is an object, usually called a path object, x^I together with morphisms:

$$x \xrightarrow{s} x^I \xrightarrow{(d_0, d_1)} x \times x$$

such that $s \in W$, $(d_0, d_1) \in \text{Fib}$ and the composite is the diagonal morphism^a.

^aThe notation (s, x^I, d_0, d_1) will be used for a given path space of x .

Here's some notation involving path spaces.

Notation 23.15. Let $\mathcal{C}$ be a category as in [Terminology 23.14](#) and let $x \in \mathcal{C}$. From now on, if (x^I, s, d_0, d_1) is a path space for x , then we will simply denote this by x^I unless it is absolutely necessary to specify the other morphisms. Moreover, if there is one path space being considered, then the notation as a quadruple above will be valid. So, we may say “Let x^I be the path space for x ” and later in a given proof, we might use d_0, d_1 whenever necessary.

With this preparation at hand, we can define categories of fibrant objects.

Definition 23.16

Let $\mathcal{C}$ be a pointed category and two full subcategories of $\text{Arr}(\mathcal{C})$:

$$\text{Fib}, W,$$

called *fibrations* and *weak equivalences* respectively. Consider the following axioms:

1. $(\mathcal{C}, \text{Fib})$ is a fibration category.
2. Any isomorphism in $\mathcal{C}$ is in W .
3. W satisfies the 2-out-of-3 property.
4. If $x \in \mathcal{C}$, there exists a path space (s, x^I, d_0, d_1) .
5. Morphisms in $\text{Fib} \cap W$ are preserved by base extensions.

The triple $(\mathcal{C}, \text{Fib}, W)$ is called a Brown category iff it satisfies axioms (1), (2), (3), (4). If such a category also satisfies axiom (5), then we refer to it as a category of fibrant objects.

We have a few remarks about this definition.

Remark 23.17. The main reason we isolate axiom (5) above is because it is usually the most difficult axiom to verify, with the path space axiom being comparable. Moreover, as we will see in [Proposition 23.34](#), for $G \in \text{Grp}(\text{LCH})$, we have only been able to prove that $G\text{-C}^*\text{Alg}$ and $G\text{-BanAlg}$ form Brown categories. As we specified towards the end of [Remark 6.37](#), we do expect this result to be true even for these equivariant categories.

Remark 23.18. In the original definition of Brown, the category $\mathcal{C}$ is not required to be pointed but simply has a final object. Also, the way we've stated the definition above is much less explicit than the one given by Brown. We have this assumption because it is a reasonable simplification given the main examples of interest to us.

In the next remark, we will discuss some of the history behind the notions we have introduced so far, though we will

refrain from speaking of the connections to operator algebras since that has been discussed in [Remark 22.22](#).

Remark 23.19. Categories of fibrant objects were first introduced by (Kenneth) Brown in [16] for the purpose of studying the homotopy theory of sheaves of simplicial sets and sheaves of spectra. It turns out that their homotopy theory is not satisfactorily described by the usual axioms of a model category and for the purpose of developing a cohomology theory for the aforementioned categories, Brown introduced categories of fibrant objects and proved results about them which are analogous to known statements about model categories. We also mention that weak fibration categories were essentially introduced in [97] by Waldhausen. Part of the underlying goal, at least of Chapter 1 of the paper, was to introduce an extension of Quillen’s definition of higher algebraic K-theory as described in [77]. In particular, Waldhausen demonstrated that the $S_\bullet$ -construction, as attributed to (Graeme) Segal, on categories with cofibrations and weak equivalences actually “recovers” the Q -construction on exact categories. We emphasize that Waldhausen’s approach to K-theory has been very influential on modern developments of algebraic K-theory, particularly those which involve its universal characterization. For more information in this direction, see [11] and [15].

Here is an immediate construction.

Construction 23.20. Let $\mathcal{C}$ be a pointed model category and consider the fibration category $\mathcal{C}_f$ of [Construction 23.5](#).

| **Claim.** With the weak equivalences of $\mathcal{C}$ restricted to the objects of $\mathcal{C}_f$, then $\mathcal{C}_f$ is a category of fibrant objects.

Again, our definition of a model category does not help us in verifying this claim. Nevertheless, the equivalent definition of a model category due to Quillen makes this claim obvious. In particular, this gives us a number of useful examples of categories of fibrant objects. Note that pointedness is not actually necessary; see [Remark 23.18](#).

Of course, one would hope that we can construct categories of fibrant objects by looking at special subcategories. This leads us to the notion of *fibrant subcategory* which we introduce as a definition. This seems to not be in the literature.

Definition 23.21. A. Vats, 2026

Let $\mathcal{C}$ be a Brown category and let $\mathcal{D}$ be a full subcategory of $\mathcal{C}$ so that:

1. $\mathcal{D}$ retains the zero object of $\mathcal{C}$.
2. $W_{\mathcal{D}}$ is a full subcategory of $W_{\mathcal{C}}$ with objects being morphisms in $\mathcal{D}$.
3. $\text{Fib}_{\mathcal{D}}$ is a full subcategory of $\text{Fib}_{\mathcal{C}}$ with objects being morphisms in $\mathcal{D}$.
4. $\mathcal{D}$ is closed under pullbacks along morphisms in $\text{Fib}_{\mathcal{D}}$ and the inclusion $\mathcal{D} \rightarrow \mathcal{C}$ preserves pullbacks.
5. Fibrations in $\mathcal{D}$ are preserved by base extensions.
6. If $x \in \mathcal{D}$, then there is a path space of x which lies in $\mathcal{D}$.

In that case, $(\mathcal{D}, W_{\mathcal{D}}, \text{Fib}_{\mathcal{D}})$ is known as a fibrant subcategory.

In fact, fibrant subcategories certainly do inherit appropriate structure.

Proposition 23.22. A. Vats, 2026

Let $\mathcal{C}$ be a Brown category.

1. Any fibrant subcategory of $\mathcal{C}$ is a Brown category.
2. If $\mathcal{C}$ is a category of fibrant objects, then any fibrant subcategory of $\mathcal{C}$ is a category of fibrant objects^a.

^aOf course, equipped with the specified structure in [Definition 23.21](#).

Proof. Let $\mathcal{D}$ be a fibrant subcategory of $\mathcal{C}$. In both cases above, we will need the following claim.

| **Claim.** $\mathcal{D}$ is a fibration category.

We note that $\mathcal{D}$ is a pointed category since it is a full subcategory of $\mathcal{C}$ which retains the zero object of $\mathcal{C}$. So, it remains for us to verify the axioms listed in [Definition 23.2](#).

1. Let f be any isomorphism in $\mathcal{D}$. Then, $f \in \text{Fib}_{\mathcal{C}}$ so $f \in \text{Fib}_{\mathcal{D}}$.
2. If $f, g \in \text{Fib}_{\mathcal{D}}$ such that $g \circ f$ exists, then $g \circ f \in \text{Fib}_{\mathcal{C}}$ so it lies in $\text{Fib}_{\mathcal{D}}$.
3. If $x \in \mathcal{D}$, then $x \rightarrow 0$ is a fibration in $\mathcal{C}$ and since $0 \in \mathcal{D}$, this is a fibration in $\mathcal{D}$.
4. Consider the following pullback diagram in $\mathcal{D}$:

$$\begin{array}{ccc} P & \longrightarrow & B \\ \bar{p} \downarrow & & \downarrow p \\ A & \longrightarrow & C \end{array}$$

where $p \in \text{Fib}_{\mathcal{D}}$. We know that this exists in $\mathcal{D}$ and we also note that this is also a pullback diagram in $\mathcal{C}$ (since the inclusion functor preserves pullbacks), so we see that $\bar{p} \in \text{Fib}_{\mathcal{C}}$. But this means that $\bar{p} \in \text{Fib}_{\mathcal{D}}$.

This proves the claim.

Now, assume that $\mathcal{C}$ is a Brown category. Let us verify each of the axioms listed in [Definition 23.16](#).

1. $(\mathcal{D}, \text{Fib}_{\mathcal{D}})$, as shown by the claim above, form a fibration category.
2. Let f be an isomorphism in $\mathcal{D}$. Then, it is an isomorphism in $\mathcal{C}$ and $f \in W_{\mathcal{C}}$, so $f \in W_{\mathcal{D}}$.
3. Let $f, g \in \text{Mor}(\mathcal{D})$ so that $g \circ f$ is defined and suppose that 2-of-3 of these are in $W_{\mathcal{D}}$. So, 2-of-3 of these are in $W_{\mathcal{C}}$, so we conclude that the third must be in $W_{\mathcal{C}}$. Hence, the third must also be in $W_{\mathcal{D}}$.
4. This is clear.

This proves statement (1).

Now, assume that $\mathcal{C}$ is a category of fibrant objects. Then, we have to verify axiom (5) of Definition 23.16, so let $f \in \text{Fib}_{\mathcal{D}} \cap W_{\mathcal{D}}$ and suppose that we take a pullback along f . Then, $f \in \text{Fib}_{\mathcal{C}} \cap W_{\mathcal{D}}$ and this pullback is also one in $\mathcal{C}$ (since the inclusion functor preserves pullbacks), so the morphism opposite to it is a trivial fibration in $\mathcal{C}$. But that morphism is one between objects in $\mathcal{D}$, so it is a trivial fibration in $\mathcal{D}$. This proves statement (2). ■

Whenever we have a functor between categories of fibrant objects and this functor preserves the weak equivalences of both categories, we can always induce a functor between the respective homotopy categories.

Construction 23.23. Let $\mathcal{C}, \mathcal{D}$ be categories of fibrant objects and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor such that:

$$F(W_{\mathcal{C}}) \subseteq W_{\mathcal{D}}.$$

Then, we note that $\gamma_{\mathcal{D}} \circ F : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{D})$ is a functor which inverts morphisms in $W_{\mathcal{C}}$. So, the universal property of the localization yields that there exists a unique functor $\mathbf{Ho}(\mathcal{C}) : \mathbf{Ho}(\mathcal{C}) \rightarrow \mathbf{Ho}(\mathcal{D})$ such that the following diagram:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{F} & \mathcal{D} \\ \gamma_{\mathcal{C}} \downarrow & & \downarrow \gamma_{\mathcal{D}} \\ \mathbf{Ho}(\mathcal{C}) & \xrightarrow{\mathbf{Ho}(F)} & \mathbf{Ho}(\mathcal{D}) \end{array}$$

commutes. This is known as the homotopy functor of F between homotopy categories. In fact, we can prove more.

Claim. For the construction above to go through, it is sufficient to assume that $F(W_{\mathcal{C}} \cap \text{Fib}_{\mathcal{C}}) \subseteq W_{\mathcal{D}}$.

This follows immediately from Corollary 23.30 and the fact that $W_{\mathcal{D}}$ satisfies the 2-out-of-3 property.

One may notice that categories of fibrant objects satisfy assumptions (1), (2) and (4) of Proposition 23.10. Our first goal will be to show that it satisfies (3) as well. Indeed, this result ends up being a standard tool of the theory.

Proposition 23.24. Brown's Factorization Lemma

Let $\mathcal{C}$ be a category of fibrant objects and let $f \in \text{Mor}(\mathcal{C})$. Then:

$$\exists p \in \text{Fib} : i \in W : f = p \circ i.$$

Such a factorization of f will be called a Brown factorization of f .

The proof of this result is not so difficult but it does require a bit of preparation. The first bit of preparation involves the following lemma, where we say a bit more about the evaluation maps in a path space.

Lemma 23.25 Let $\mathcal{C}$ be a category of fibrant objects, let $x \in \mathcal{C}$ and select a path space:

$$(s, x^I, d_0, d_1).$$

Then, $d_0, d_1 : x^I \rightarrow x$ are trivial fibrations.

This lemma was originally found in [78, Lemma 2, p. 6-7]; that result is about cofibrant objects which have cylinders in a model category. However, the structure of a category of fibrant objects is rich enough to yield the result.

Proof of Lemma 23.25. Let 0 be the zero object. Since $\mathcal{C}$ is a fibration category, we see that the unique morphism $x \rightarrow 0$ is a fibration. Now, we can fit the unique morphism $x \rightarrow 0$ in the following pullback diagram with $x \times_0 x$:

$$\begin{array}{ccc} x \times_0 x & \xrightarrow{\pi_2} & x \\ \pi_1 \downarrow & & \downarrow \\ x & \longrightarrow & 0 \end{array}$$

and we note that since fibrations are preserved by base extensions, it follows that π_1 and π_2 are fibrations.

Claim. d_0 is a fibration.

This follows from the fact that (d_0, d_1) is a fibration so composing it with π_1 produces a fibration. However, this composition is just d_0 so the claim follows, since fibrations compose to form fibrations.

Claim. d_0 is a weak equivalence.

The identity map is a trivial fibration, so diagonal maps must also be trivial fibrations. Now, $(d_0, d_1) \circ s$ is a weak equivalence because it is a diagonal map and since s is a weak equivalence, we conclude by Axiom 3 of Definition 23.16 that (d_0, d_1) is a weak equivalence. But this means that d_0 is a weak equivalence^a and we're done. ■

^aThe analogous proof for d_1 will go through as well.

A reasonably big component to the proof of the factorization lemma is the so-called mapping cylinder construction.

Construction 23.26. Let $\mathcal{C}$ be a category of fibrant objects, let $f \in \mathcal{C}(x, y)$ and let:

$$(s, y^I, d_0^y, d_1^y),$$

be a path space for y . We consider the following pullback^a:

$$\begin{array}{ccc} \text{cyl}(f) & \xrightarrow{\bar{f}} & y^I \\ \bar{d}_f \downarrow & & \downarrow d_0^y \\ x & \xrightarrow{f} & y \end{array}$$

$\text{cyl}(f)$ is called the mapping cylinder of f relative to the given path space.

^aThe pullback exists because d_0^y is a fibration.

We have a further remark about this construction, since we will need it later.

Remark 23.27. For some specific categories of fibrant objects, there might very well be a canonical choice of path space for any given object and in those cases, one computes the mapping cylinder with respect to that path space.

Here's another lemma where we record some basic properties of the construction.

Lemma 23.28 In the notation of [Construction 23.26](#), TFSH:

1. $\bar{d}_f$ is a trivial fibration.
2. If f is a fibration, then $d_i^y \circ \bar{f}$ is a fibration for $i \in \{0, 1\}$.
3. If f is arbitrary, then $d_1^y \circ \bar{f}$ is a fibration.

Proof. We will prove each statement one-by-one.

1. This follows from the fact that d_0^y is a trivial fibration and that trivial fibrations are preserved by base extensions.
2. If f is a fibration, then $\bar{f}$ is a fibration. Since each d_i^y is a fibration, the result holds.
3. We will rewrite $d_1^y \circ \bar{f}$ as a composition of two fibrations. That is, we write it as the composition:

$$\text{cyl}(f) \xrightarrow{g} x \times y \longrightarrow y$$

where the second morphism is a projection. The projection fits into a pullback diagram with the unique morphisms $x \rightarrow 0$ and $y \rightarrow 0$ as the legs. These are fibrations, so it follows that $x \times y \rightarrow y$ is a fibration^a. To deal with the first morphism, let us construct it by introducing a diagram that will be discussed throughout the proof. We will first draw the diagram and then explain its role in the proof:

$$\begin{array}{ccccc} \text{cyl}(f) & \longrightarrow & x \times y & \xrightarrow{\pi_x} & x \\ \downarrow & & \downarrow f \times \text{Id}_y & & \downarrow f \\ y^I & \xrightarrow{(d_0^y, d_1^y)} & y \times y & \xrightarrow{\pi_1} & y \end{array}$$

We will describe this diagram in a few steps:

- (a) We first take the pullback of (d_0^y, d_1^y) against $f \times \text{Id}_y$; this produces an object p in the upper left corner which we will identify with $\text{cyl}(f)$ later in the proof.
- (b) The right square is a part of the diagram used to construct $f \times \text{Id}_y$ and it is a pullback square.
- (c) Since both squares are pullbacks, the pasting law says that the outer rectangle is a pullback diagram. However, the outer rectangle is the pullback diagram for the mapping cylinder of f . In other words, there is an isomorphism from $\text{cyl}(f)$ to p . So, we will just replace p by $\text{cyl}(f)$ and we obtain the diagram above.

Let g be the morphism from $\text{cyl}(f)$ to $x \times y$. This is a fibration because the left square is a pullback and (d_0^y, d_1^y) is a fibration. We need to show that $\pi_y \circ g$ is $d_1^y \circ \bar{f}$. The commutativity of the resulting diagram gives us:

$$(f \times \text{Id}_y) \circ g = (d_0^y, d_1^y) \circ \bar{f}.$$

Next, we note that the construction of $f \times \text{Id}_y$ entails that:

$$\pi_2 \circ (f \times \text{Id}_y) = \text{Id}_y \circ \pi_y = \pi_y,$$

where π_2 and π_y are projection maps. This means that we have the following computation:

$$\begin{aligned} \pi_y \circ g &= \pi_2 \circ (f \times \text{Id}_y) \circ g \\ &= \pi_2 \circ (d_0^y, d_1^y) \circ \bar{f} \\ &= d_1^y \circ \bar{f}, \end{aligned}$$

as was desired. ■

^aWe have used an analogous argument in [Lemma 23.25](#).

We are ready to prove Brown's Factorization Lemma.

Proof of Proposition 23.24. The proof is conceptually simple. We will divide it into two parts: we'll first construct the decomposition of f and then show that it has the required properties. With that said, let us proceed.

1. Let $f : x \rightarrow y$ be a morphism and choose a path space:

$$(s, y^I, d_0^y, d_1^y).$$

Let us consider the mapping cylinder $\text{cyl}(f)$ relative to this path space. Consider the following diagram:

$$\begin{array}{ccccc} x & & \xrightarrow{s \circ f} & & y^I \\ & \searrow i & & \searrow \bar{f} & \\ & \text{cyl}(f) & \xrightarrow{\bar{f}} & & y^I \\ & \downarrow \bar{d}_f & & \downarrow d_0^y & \\ x & \xrightarrow{f} & & & y \\ & \uparrow \text{Id}_x & & & \end{array}$$

where the square is a pullback. We will define morphisms $i : x \rightarrow \text{cyl}(f)$ and $p : \text{cyl}(f) \rightarrow y$ so that:

$$f = p \circ i.$$

For p , this is easy to do:

$$p \stackrel{\text{def}}{=} d_1^y \circ \bar{f}.$$

Now, we want to show that the outer diagram above commutes:

$$\begin{aligned} d_0^y \circ (s \circ f) &= (d_0^y \circ s) \circ f \\ &= \text{Id}_y \circ f \\ &= f \\ &= f \circ \text{Id}_x. \end{aligned}$$

So, the whole diagram commutes and by the universal property of the pullback, there is a unique morphism $i : x \rightarrow \text{cyl}(f)$ so that the resulting diagram commutes. Finally, we check that this factorization works:

$$\begin{aligned} p \circ i &= (d_1^y \circ \bar{f}) \circ i \\ &= d_1^y \circ (\bar{f} \circ i) \\ &= d_1^y \circ (s \circ f) \\ &= (d_1^y \circ s) \circ f \\ &= \text{Id}_y \circ f \\ &= f, \end{aligned}$$

as was desired.

2. By statement (3) of Lemma 23.28, $p \in \text{Fib}_C$ so we should show that $i \in W$. We know that $\text{Id}_x = \bar{d}_f \circ i$. Since $\bar{d}_f$ is a trivial fibration, i is a section of a trivial fibration so statement (1) of Proposition 23.10 tells us that $i \in W$.

This completes the proof. ■

The next result is mentioned in some places in the literature but it is never proved, so we fill in the gap.

Proposition 23.29

Let $\mathcal{C}$ be a pointed category and suppose that there are two full subcategories:

$$\text{Fib}_C, W_C,$$

satisfying axioms (1) – (4) in Definition 23.16. TFAE:

1. This structure satisfies axiom (5).
2. If $f \in \text{Mor}(\mathcal{C})$, then there is a $p \in \text{Fib}_C$ and $i \in W$ such that $f = p \circ i$.

In fact, (2) $\Rightarrow$ (1) holds even when $\mathcal{C}$, with Fib_C , is just a fibration category.

Proof. (1) $\Rightarrow$ (2) is simply Brown's Factorization Lemma. For (2) $\Rightarrow$ (1), let $x \in \mathcal{C}$ and consider the diagonal map:

$$\Delta_x : x \rightarrow x \times x,$$

where this is well-defined because $\mathcal{C}$ is closed under finite products. Now, we factorize this as follows:

$$\Delta_x = p_x \circ i_x,$$

where $p_x \in \text{Fib}_C$ and $i_x \in W$. This leads us to the following claim which will finish the proof.

Claim. The quadruple $(i_x, \text{dom}(p_x), \pi_1 \circ p_x, \pi_2 \circ p_x)$ is a path space for x , where $\pi_1, \pi_2 : x \times x \rightarrow x$ are projections.

Indeed, we note that $s \stackrel{\text{def}}{=} i_x$ is a weak equivalence by construction. If we define:

$$d_0 \stackrel{\text{def}}{=} \pi_1 \circ p_x$$

$$d_1 \stackrel{\text{def}}{=} \pi_2 \circ p_x,$$

then $(d_0, d_1) = p_x$ and this is a fibration by construction so we're done. The last statement is clear. ■

Let us discuss some corollaries of the Brown Factorization Lemma. The first two results below appear in [16, Lemma 1, Lemma 2, p.422-423] with one-line indications towards proofs, so we fill in this gap. The second of these results will be used in the proof of Proposition 24.2, so it is best to work through them and gain a bit of intuition for how they work.

Claim 23.7. Let $\mathcal{C}$ be a category of fibrant objects. Then, any commutative diagram:

$$\begin{array}{ccc} x & \xrightarrow{\quad} & y \\ i \downarrow & & \downarrow p \\ x' & \xrightarrow{\quad} & y' \end{array}$$

with i being a weak equivalence and p being a fibration, can be embedded in a commutative diagram:

$$\begin{array}{ccccc} x & \xrightarrow{\quad} & z & \xrightarrow{\quad} & y \\ & \searrow i & \downarrow t & & \downarrow p \\ & & x' & \xrightarrow{\quad} & y' \end{array}$$

where t is a trivial fibration. This is proved on Page 274.

Claim 23.8. Let $\mathcal{C}$ be a category of fibrant objects, let $x, y \in \mathcal{C}$, let $u : x \rightarrow y$ be a morphism and suppose that we have two path spaces $(x^I, s, d_0, d_1), (y^I, s^y, d_0^y, d_1^y)$. Then, there is a path space (x^J, s^x, d_0^x, d_1^x) , a trivial fibration $t : x^J \rightarrow x^I$ and a morphism $\bar{u} : x^J \rightarrow y^I$ such that the following diagram:

$$\begin{array}{ccccc} & & & & y \\ & & & & \downarrow s^y \\ x & \xrightarrow{s^x} & x^J & \xrightarrow{\bar{u}} & y^I \\ \downarrow s & \swarrow t & \downarrow (d_0^x, d_1^x) & & \downarrow (d_0^y, d_1^y) \\ x^I & \xrightarrow{(d_0, d_1)} & x \times x & \xrightarrow{u \times u} & y \times y \end{array}$$

commutes. This is proved on Page 274.

The next result is a neat corollary of the factorization lemma.

Corollary 23.30. Ken Brown's Lemma

Let $\mathcal{C}$ be a category of fibrant objects, let $\mathcal{D}$ be a category with a set:

$$W \subseteq \text{Mor}(\mathcal{D}),$$

satisfying the 2-out-of-3 property and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$ so that $F(W_{\mathcal{C}} \cap \text{Fib}_{\mathcal{C}}) \subseteq W$. Then, $F(W_{\mathcal{C}}) \subseteq W$.

Proof. Let $f \in \text{Mor}(\mathcal{C})$ be a weak equivalence. We write:

$$f = p \circ i,$$

where p is a trivial fibration and i is a weak equivalence. So, $F(p) \in W$ and it would be sufficient to show that $F(i) \in W$. To do this, we note that i is right-inverse to a trivial fibration q so we must have that $q \circ i = \text{Id}$. Since Id is an isomorphism, it is a trivial fibration so we see that $F(q) \circ F(i) = F(\text{Id})$ with $F(q)$ and $F(\text{Id})$ being in W . It follows that $F(i) \in W$. But this means that $F(f) \in W$ and we're done. ■

The next result seems to have first been proved in the C^* -algebra context in [89, Proposition 1.13, p.412-413]. However, the proof actually generalizes so that the analogue of that result holds in any category of fibrant objects.

Corollary 23.31

Let $\mathcal{C}$ be a category of fibrant objects and suppose we have a pullback diagram in $\mathcal{C}$:

$$\begin{array}{ccc} u & \xrightarrow{g_1} & x \\ g_2 \downarrow & & \downarrow f_1 \\ y & \xrightarrow{f_2} & z \end{array}$$

where f_1 is a fibration and f_2 is a weak equivalence. Then, g_1 is a weak equivalence.

Proof. Let us proceed in two steps.

1. We'll prove this result assuming that f_2 is also a fibration.
2. We'll extend the result to f_2 just being a weak equivalence.

Again, we deal with these two steps separately:

1. If we assume that f_2 is also a fibration, then we have a pullback of a trivial fibration and this means that g_1 is a trivial fibration. In particular, g_1 is a weak equivalence.
2. We factorize f_2 as $j \circ r$, with j being a fibration and r being a weak equivalence. Now, we have a diagram:

$$\begin{array}{ccccc}
 u & \xrightarrow{g_1} & x & & \\
 \searrow \bar{r} & & \nearrow \bar{j} & & \\
 & m' & & & \\
 \downarrow g_2 & & \downarrow f_1 & & \\
 y & \xrightarrow{r} & m & \xrightarrow{j} & z
 \end{array}$$

Let us explain what's going on in the diagram:

- (a) m' is the pullback $x \times_z m$.
- (b) The maps g_1 and $r \circ g_2$ give rise to a commutative square and $\bar{r}$ is the morphism formed from the commutative square via the universal property of the pullback.

Having explained the diagram, we can now present it more cleanly:

$$\begin{array}{ccccc}
 u & \xrightarrow{\bar{r}} & m' & \xrightarrow{\bar{j}} & x \\
 \downarrow g_2 & & \downarrow & & \downarrow f_1 \\
 y & \xrightarrow{r} & m & \xrightarrow{j} & z
 \end{array}$$

Since f_2 is a weak equivalence and since r is a weak equivalence, we use the 2-out-of-3 property to conclude that j is a weak equivalence. So, j is a trivial fibration and we use (1) above to conclude that $\bar{j}$ is a weak equivalence.

But this means that $\bar{j} \circ \bar{r} = g_1$ is a weak equivalence.

This completes the proof. ■

As a consequence, we have the following corollary.

Corollary 23.32. A. Vats, 2026

Let $\mathcal{C}$ be a category of fibrant objects. Then, it is a weak fibration category.

Proof. Since a pullback of a morphism in $W_{\mathcal{C}}$ along a morphism in Fib is in W , we see that $W_{\mathcal{C}}$ satisfies the Gluing Lemma. It follows that our triple $(\mathcal{C}, \text{Fib}_{\mathcal{C}}, W_{\mathcal{C}})$ is a weak fibration subcategory. ■

The next result generalizes what has been proved for $C^* \text{Alg}$ in [89, Proposition 1.8, p.410].

Proposition 23.33. A. Vats, 2026

In the notation of [Construction 23.26](#), TFSH:

1. f is a weak equivalence iff $\bar{f}$ is a weak equivalence.
2. A Brown factorization of f produces a Brown factorization of $\bar{f}$.
3. If there is a morphism $\phi : x \rightarrow y^I$ such that $f = d_0^y \circ \phi$, then $\bar{d}_f$ is a split epi.
4. The morphism ϕ in statement (3) exists, so $\bar{d}_f$ is always split epi.

Proof. We will prove each statement one-by-one.

1. We'll prove each implication separately.

$\Rightarrow$: This is an application of Corollary 23.31.

$\Leftarrow$: We consider the pullback diagram defining the mapping cylinder and note that $d_0^y \circ \bar{f}$ is a weak equivalence since each factor is a weak equivalence. Hence, we see that $f \circ \bar{d}_f$ is a weak equivalence and since $\bar{d}_f$ is a weak equivalence, it follows by the 2-out-of-3 property that f is a weak equivalence.

This proves the result.

2. Let us factorize f as $j \circ r$, where $j : m \rightarrow y$ is a fibration and $r : x \rightarrow m$ is a weak equivalence. As in the proof of Corollary 23.31, we can put this in a large commutative diagram:

$$\begin{array}{ccccc}
 \text{cyl}(f) & \xrightarrow{\bar{r}} & m' & \xrightarrow{\bar{j}} & y^I \\
 \downarrow \bar{d}_f & & \downarrow & & \downarrow d_0^y \\
 x & \xrightarrow{r} & m & \xrightarrow{j} & y
 \end{array}$$

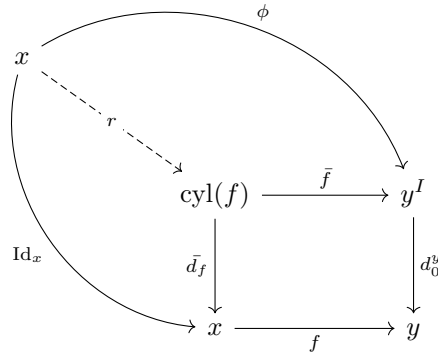
The key observation is that the right square above is a pullback, so the entire rectangle is a pullback iff the left square is a pullback^a. Since the entire rectangle is a pullback, we see that the left square is a pullback. Now:

$$\bar{f} = \bar{j} \circ \bar{r}.$$

Since j is a fibration, we see that $\bar{j}$ is a fibration. So, the morphism in the middle of the diagram is a fibration because d_0^y is a fibration. Since r is a weak equivalence, we can use Corollary 23.31 to conclude that $\bar{r}$ is a weak

equivalence. This produces a Brown factorization for $\bar{f}$ so we're done.

3. The point is to write the given information into the diagram of [Construction 23.26](#). To be precise, we have:



So, the given condition on ϕ ensures that the outer diagram above commutes. By the universal property of the pullback, we obtain a morphism $r : x \rightarrow \text{cyl}(f)$ such that the entire diagram above commutes. We see that:

$$\bar{d}_f \circ r = \text{Id}_x,$$

as was desired.

4. Let $s : y \rightarrow y^I$ be the morphism into the path object which comes as part of the data of the path space. Then:

$$d_0^y \circ s = \text{Id}_y,$$

so we see that:

$$d_0^y \circ s \circ f = \text{Id}_y \circ f = f.$$

So, we simply define $\phi \stackrel{\text{def}}{=} s \circ f$ and this works, so $\bar{d}_f$ is split epi. ■

^aWe are using the pasting law for pullbacks here; see [\[8\]](#).

We will start with the following proposition which is known in the non-equivariant case for C^* -algebras.

Proposition 23.34. A. Vats (2026)

Let $G \in \text{Grp}(\text{LCH})$ and let $\mathcal{C}$ be either $G\text{-}C^*\text{Alg}$ or $G\text{-}Ban\text{Alg}$ and define:

$$W \stackrel{\text{def}}{=} \{\text{homotopy equivalences in } \mathcal{C}\},$$

$$\text{Fib} \stackrel{\text{def}}{=} \{\text{Schochet fibrations in } \mathcal{C}\}.$$

Then, $(\mathcal{C}, W, \text{Fib})$ forms a Brown category.

Proof. We just need to verify the axioms one-by-one, in the order presented within [Definition 23.16](#).

1. $\mathcal{C}$ has a final object; it's just $\{0\}$.
2. $(\mathcal{C}, \text{Fib})$ is a fibration category, as established in [Example 23.6](#).
3. Every isomorphism in $\mathcal{C}$ is in W by [Claim 6.10](#).
4. W satisfies the 2-out-of-3 property by statement (3) of [Claim 6.13](#).
5. If $A \in \mathcal{C}$, then we can easily specify a quadruple:

$$(s, \text{Cyl}_G(A), p_0^A, p_1^A),$$

where $s : A \rightarrow \text{Cyl}_G(A)$ is the inclusion as constant functions & we could try to show that this is a path space for A . However, this seems to be difficult and, in fact, Uuye does not seem to give a complete argument for this as well; see [Remark 23.35](#). As a consequence, we will use (2) $\Rightarrow$ (1) of [Proposition 23.29](#) to deduce the existence of a path space for any $A \in \mathcal{C}$. That is, let us construct a factorization of any morphism in $\mathcal{C}$ and we do this by taking inspiration from the proof of the Brown Factorization Lemma. If $A, B \in \mathcal{C}$ and $f \in \mathcal{C}(A, B)$, then its mapping cylinder (with respect to Cyl_G) is still defined as in [Construction 23.26](#):

$$\begin{array}{ccc} \text{cyl}(f) & \xrightarrow{\bar{f}} & \text{Cyl}_G(B) \\ \bar{d}_f \downarrow & & \downarrow p_0^B \\ A & \xrightarrow{f} & B \end{array}$$

Now, we define:

$$p \stackrel{\text{def}}{=} p_1^B \circ \bar{f}.$$

This will be our Schochet fibration; we will show this later. To construct the homotopy equivalence $i : A \rightarrow \text{cyl}(f)$ such that $p \circ i = f$, we will use the universal property of the pullback. To be precise, we define a map:

$$s : B \rightarrow \text{Cyl}_G(B), \quad b \mapsto s(b),$$

where $s(b)$ is the constant function on $[0, 1]$ with value b . This fits into the following diagram:

$$\begin{array}{ccccc}
 & & & \text{sof} & \\
 & & & \curvearrowright & \\
 A & \xrightarrow{\quad i \quad} & \text{cyl}(f) & \xrightarrow{\quad \bar{f} \quad} & \text{Cyl}_G(B) \\
 & \searrow \text{Id}_A & \downarrow \bar{d}_f & & \downarrow p_0^B \\
 & & A & \xrightarrow{\quad f \quad} & B
 \end{array}$$

where the solid part of the diagram commutes; this follows from the definition of s . The universal property of the pullback induces a unique morphism $i : A \rightarrow \text{cyl}(f)$ such that the entire diagram commutes. So, we have:

$$\begin{aligned}
 p \circ i &= (p_1^B \circ \bar{f}) \circ i \\
 &= p_1^B \circ (\bar{f} \circ i) \\
 &= p_1^B \circ (s \circ f) \\
 &= (p_1^B \circ s) \circ f \\
 &= \text{Id}_B \circ f \\
 &= f,
 \end{aligned}$$

where the fact that $p_1^B \circ s$ can be deduced quickly from the explicit definitions of p_1^B and s . So, it remains for us to show that p is a Schochet fibration and i is a homotopy equivalence.

(a) Let us show that i is a homotopy equivalence. Observe that:

$$\bar{d}_f \circ i = \text{Id}_A,$$

since the diagram above commutes. To find a right homotopy inverse, let us make $\text{cyl}(f)$ more explicit:

$$\text{cyl}(f) = \{(\phi, a) \in \text{Cyl}_G(B) \times A : \phi(0) = f(a)\}.$$

Using this, we can deduce the exact formula for i from the commuting diagram as follows:

$$\forall a \in A : i(a) = (s(f(a)), a).$$

Let us consider $i \circ \bar{d}_f$ as follows:

$$\begin{aligned}
 \forall (\phi, a) \in \text{cyl}(f) : (i \circ \bar{d}_f)(\phi, a) &= i(\bar{d}_f(\phi, a)) \\
 &= i(a) \\
 &= (s(f(a)), a) \\
 &= (s(\phi(0)), a).
 \end{aligned}$$

This motivates us to define a homotopy $h : \text{cyl}(f) \rightarrow \text{Cyl}_G(\text{cyl}(f))$ as we will do now, so let $(\phi, a) \in \text{cyl}(f)$. For each $s \in [0, 1]$, we can define the following function:

$$\phi_s : [0, 1] \rightarrow A, \quad t \mapsto \phi_s(t) \stackrel{\text{def}}{=} \phi(st).$$

Since ϕ is continuous, we see that ϕ_s is continuous. Moreover:

$$\phi_s(0) = \phi(0) = f(a),$$

which means that $(\phi_s, a) \in \text{cyl}(f)$ for each $s \in [0, 1]$. Now, we consider the map:

$$h(\phi, a) : [0, 1] \rightarrow \text{cyl}(f), \quad s \mapsto h(\phi, a)(s) \stackrel{\text{def}}{=} (\phi_s, a).$$

We have to argue that this is continuous. If it is continuous, then we can form a homotopy:

$$h : \text{cyl}(f) \rightarrow \text{Cyl}_G(\text{cyl}(f)),$$

which will be a $\star$ -homomorphism by direct computation. More importantly, it does what we need it to do because if $s = 0$, then this gives us $i \circ \bar{d}_f$ and when $s = 1$, this gives us $\text{Id}_{\text{cyl}(f)}$. So, it remains to argue that $h(\phi, a)$ is continuous. This follows from the following estimate:

$$\begin{aligned}
 \forall s, s' \in [0, 1] : \|h(\phi, a)(s) - h(\phi, a)(s')\| &= \|(\phi_s, a) - (\phi_{s'}, a)\| \\
 &= \|(\phi_s - \phi_{s'}, 0)\| \\
 &= \sup_{t \in [0, 1]} \|\phi(st) - \phi(s't)\|.
 \end{aligned}$$

Next, we observe that $|st - s't| = |s - s'| \cdot |t| \leq |s - s'|$. As a consequence, if we have s and s' close enough, then st is close enough to $s't$ and this implies, by the continuity of ϕ , that $h(\phi, a)$ is continuous. We want to explicitly note that all morphisms formed above are equivariant because they are formed from equivariant morphisms. So, this proves that the map i is a homotopy equivalence.

(b) Let us prove that p is a Schochet fibration. Let us consider the following lifting problem:

$$\begin{array}{ccc}
 & & \text{cyl}(f) \\
 & \nearrow k & \uparrow p_0^{\text{cyl}(f)} \\
 D & & \text{Cyl}_G(\text{cyl}(f)) \\
 & \searrow h & \downarrow \text{Cyl}_G(p) \\
 & & \text{Cyl}_G(B)
 \end{array}$$

where h is a homotopy of $p \circ k$. Since $\text{cyl}(f) \subseteq \text{Cyl}_G(B) \times A$, we can project k down to each component and write it as a pair $k = (k', k'')$. Let us record that this pair satisfies the following property:

$$\forall d \in D : k'(d)(0) = f(k''(d)).$$

So, we will define a homotopy lift H into $\text{Cyl}_G(\text{Cyl}_G(B) \times A)$ and show that this is contained in $\text{Cyl}_G(\text{cyl}(f))$. Since Cyl_G preserves pullbacks, we will conveniently identify $\text{Cyl}_G(\text{Cyl}_G(B) \times A)$ with $\text{Cyl}_G(\text{Cyl}_G(B)) \times \text{Cyl}_G(A)$. For any function into this algebra, we simply recover the situation in $\text{Cyl}_G(B) \times A$ by evaluating that function additionally at a pair $(s, t) \in [0, 1] \times [0, 1]$; this will be illustrated below. We define:

$$\forall d \in D : \forall (s, t) \in [0, 1] \times [0, 1] : h'(d)(s, t) \stackrel{\text{def}}{=} \begin{cases} k'(d)\left(\frac{2s}{2-t}\right) & \text{if } 2s \leq 2-t, \\ h(d)(2s+t-2) & \text{if } 2s > 2-t \end{cases}.$$

If we fix d and s , then this defines a map $[0, 1] \rightarrow B$ and this is continuous everywhere except for possibly at $t = 2 - 2s$. At that point, the right-hand limit exists and it is equal to $k'(d)(1)$, while the left-hand limit is equal to $h(d)(0)$. But we see that $h(d)(0) = p(k(d))$ and we note that this is just $k'(d)(1)$. Since the two one-sided limits coincide, we see that the map $[0, 1] \rightarrow B$ is continuous. For similar reasons, we have continuity when we fix d so that $h'(d) \in \text{Cyl}_G(\text{Cyl}_G(B))$ and since k' & h are equivariant $\star$ -homomorphisms, we see that h' is an equivariant $\star$ -homomorphism. Next, we define a morphism from A to $\text{Cyl}_G(A)$:

$$h'' \stackrel{\text{def}}{=} s^A \circ k'',$$

where we now use the morphism $s^A : A \rightarrow \text{Cyl}_G(A)$ which takes an element of A to the constant function on $[0, 1]$ with the value a . We observe that h'' is an equivariant $\star$ -homomorphism. We define:

$$H \stackrel{\text{def}}{=} (h', h''),$$

and this is an equivariant $\star$ -homomorphism from D to $\text{Cyl}_G(\text{Cyl}_G(B)) \times \text{Cyl}_G(A)$. Again, we specify values in $\text{Cyl}_G(B) \times A$ by evaluating H as $H(-)(-, -)$, where it is the second variable in time which controls the homotopy. Now, we need to check that this lies in the mapping cylinder of f and that this is a homotopy of k . Let $d \in D$ and $s, t \in [0, 1]$. Then, we want to show that:

$$H(d)(s, t) = (h'(d)(s, t), h''(d)(t)) \in \text{cyl}(f).$$

Let $t = 0$. In that case, we see that:

$$H(d)(s, 0) = (h'(d)(s, 0), h''(d)(0)) = (k'(d)(s), k''(d)) = k(d)$$

So, H is a homotopy of k . Next, we let s, t be arbitrary. Then, we have the following computation:

$$\begin{aligned}
 f(h''(d)(t)) &= f(k''(d)) \\
 &= k'(d)(0) \\
 &= h'(d)(0, t),
 \end{aligned}$$

which proves that we are in the mapping cylinder throughout the homotopy. Finally, we will show that:

$$\text{Cyl}_G(p) \circ H = h.$$

So, let $d \in D$ and observe that:

$$\begin{aligned}
 \forall t \in [0, 1] : h(d)(t) &= h(d)(2 \cdot 1 + t - 2) \\
 &= h'(d)(1, t).
 \end{aligned}$$

On the other hand, we have that:

$$\begin{aligned}
 \forall t \in [0, 1] : (\text{Cyl}_G(p) \circ H)(d)(t) &= (p \circ H(d))(t) \\
 &= (p_1^B \circ \bar{d}_f \circ H(d))(t) \\
 &= h'(d)(1, -)(t) \\
 &= h'(d)(1, t),
 \end{aligned}$$

from which we conclude that $\text{Cyl}_G(p) \circ H = h$.

This completes our construction of the Brown factorization of f . Now, we are in a position to apply Proposition 23.29 to conclude that each object in $\mathcal{C}$ admits a path space.

This proves the claim. ■

^aNote that when $t = 2 - 2s$, $s \neq 0$ since that would imply that $t = 2$ and this cannot happen. So, $\frac{2s}{2-t} = 1$.

We have two remarks about this proof.

Remark 23.35. The result above was deduced in the non-equivariant set-up by Uuye in [96, Theorem 2.19, p.10]. However, we want to point out that his proof of the existence of path spaces (in particular, all of the necessary verifications) seems to be incomplete. This is actually related to a significant disadvantage of his approach which we will discuss in Remark 23.36. As the reader can see, our solution to this problem was to simply establish Brown factorizations for $\star$ -homomorphisms in the same way that was done in [89, Proposition 1.8, p.410] for the non-equivariant set-up, supplemented with many more details, and then utilize Proposition 23.29. This proof seems to have a good chance of generalizing to other categories of equivariant topological algebras.

Remark 23.36. There are two reasons why the approach used by Uuye does not generalize to the equivariant case:

1. The proof relies on purely topological characterizations of homotopy equivalences and Schochet fibrations in terms of weak homotopy equivalences and Serre fibrations between spaces. For the latter, a necessary component of the proof seems to be to use the fact that the maximal tensor product of C^* -algebras preserves pullbacks. This does not seem to be true for the projective tensor product of Banach algebras and there seems to be no easy fix for this defect in the setting of Banach algebras. One might consider using the injective tensor product of Banach algebras but it is not clear that the resulting tensor product is actually an algebra.
2. The proof does not account for equivariance. Indeed, it is not clear that these characterizations hold when one needs to account for equivariance. In fact, to work with the model structures available for categories of equivariant spaces, it is necessary that even stronger characterizations hold^a. It is not clear that this is the case.

Nevertheless, these (potential) hurdles with Uuye's proof seem to suggest that more work needs to be done to relate equivariant classical homotopy theory and equivariant homotopy theory on C^* -algebras^b.

^aSee [4].

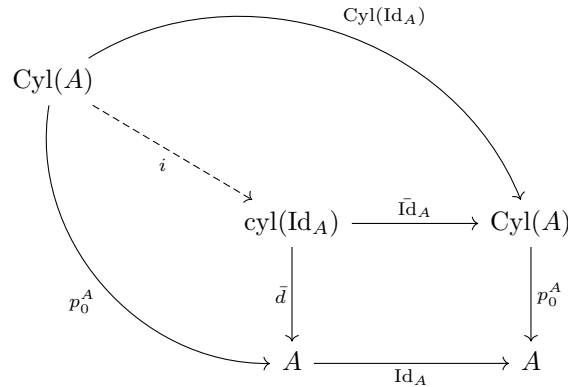
^bWe will discuss some questions in this direction in ??.

Given the role that the mapping cylinder plays in the construction of the factorization above, it is useful to have a computation of it in a very concrete example, so we will discuss that now. Indeed, the mapping cylinder of the diagonal morphism on a C^* -algebra can be used to explain why it has that name in the first place.

Example 23.37. Let $A \in C^*Alg$, for simplicity^a. We have the following claim.

| Claim. $Cyl(A)$ is isomorphic to $cyl(Id_A)$.

Let us consider the following diagram:



So, the main square is the pullback diagram for the cylinder, of course. The outer diagram is actually the naturality square for the time evaluation natural transformation $p_0 : Cyl \Rightarrow Id$, so it commutes. Therefore, the universal property of the pullback gives a natural morphism from $Cyl(A) \rightarrow cyl(Id_A)$. We do have an explicit description of $cyl(Id_A)$:

$$cyl(Id_A) = \{(a, \phi) \in A \times Cyl(A) : a = \phi(0)\}.$$

In particular, this allows us to have an explicit form for the map i :

$$i : Cyl(A) \rightarrow cyl(Id_A), \quad \phi \mapsto i(\phi) \stackrel{\text{def}}{=} (\phi(0), \phi).$$

This is clearly actually valued in $cyl(Id_A)$ and it is a $\star$ -homomorphism^b. Now, we define:

$$j : cyl(Id_A) \rightarrow Cyl(A), \quad (a, \phi) \mapsto j(a, \phi) \stackrel{\text{def}}{=} \phi,$$

where $\pi_1 : A \times A \rightarrow A$ is the projection onto the first factor. This is a $\star$ -homomorphism, as can be checked with a direct computation; as above, one will need that π_1 is a $\star$ -homomorphism. One can check that these are inverses.

| Claim 23.9. i and j are inverses.

This is proved on Page 274. So, this proves that $Cyl(A)$ and $cyl(Id_A)$ are $\star$ -isomorphic.

^aWe can also do this with equivariant C^* -algebras or Banach algebras but this keeps notation simple.

^bChecking this amounts to a direct computation with the formula.

At this point, we would like to bring actual “homotopies” into the picture. As motivation, recall that in Section 6, we established that homotopies and Schochet fibrations in the realm of equivariant operator algebras can be characterized entirely in terms of morphisms, compositions and certain functors & natural transformations. In other words, “homotopy” can be viewed as a categorical notion and we see that it is possible to define it in a category of fibrant objects.

23.2 The Homotopy Theory of Categories of Fibrant Objects

24 The Homotopy Theory of Categories of Fibrant Objects

Summary

In this lecture, we will cover the following topics:

- 1.
- 2.

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

Motivation.

24.1 A Purely Categorical Notion of Homotopy

This leads us to the following bit of terminology. Over here, we take a bit of liberty to define **two** notions of homotopy; as it turns out, only one of them is typically well-behaved with respect to compositions. The other one (the more intuitive one, actually) is only well-behaved under compositions under some further conditions that are the subject of some recent work¹². Nevertheless, we will see that the chosen notion of homotopy is the “right one”.

Terminology 24.1. Let $\mathcal{C}$ be a category of fibrant objects and let $f, g : x \rightarrow y$ be morphisms.

1. f is right-homotopic to g iff there exists a path space (y^I, s, d_0, d_1) and a morphism $h : x \rightarrow y^I$ such that:

$$f = d_0 \circ h,$$

$$g = d_1 \circ h,$$

We denote this by writing $f \simeq_{h,r} g$ and we say that h is a right-homotopy from f to g .

2. f is homotopic to g iff there exists a weak equivalence $t : z \rightarrow x$ such that:

$$f \circ t \simeq_{h,r} g \circ t.$$

We denote this by writing $f \simeq_h g$ and we say that f is homotopic to g .

Clearly, if two morphisms are right-homotopic, then they are homotopic.

The next result was essentially mentioned as a comment in [16, p.422] and it's a good exercise.

Claim 24.1. Let $\mathcal{C}$ be a category of fibrant objects, let $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$ be the localization and let $f, g \in \mathcal{C}(x, y)$ so:

$$f \simeq_{h,r} g,$$

in $\mathcal{C}$. Then, $\gamma(f) = \gamma(g)$. In particular, this is true when f and g are homotopic.

The next result is somewhat technical but it is the obvious precursor to defining “homotopy categories”.

Proposition 24.2. K. Brown, 1973, [16, Proposition 1, p.423-424]

Let $\mathcal{C}$ be a category of fibrant objects and let $f, g \in \mathcal{C}(x, y)$ so that:

$$f \simeq_{h,r} g.$$

TFSH:

1. If $u \in \mathcal{C}(z, x)$, then $f \circ u \simeq_{h,r} g \circ u$.
2. If $u \in \mathcal{C}(y, z)$, then there is a trivial fibration $t : x' \rightarrow x$ so that $u \circ f \circ t \simeq_{h,r} u \circ g \circ t$.

Proof. Let us prove each statement one-by-one.

1. We pick a path space y^I of y and consider a right-homotopy $h : x \rightarrow y^I$ so that:

$$f = d_0 \circ h,$$

$$g = d_1 \circ h.$$

We define $H \stackrel{\text{def}}{=} h \circ u$. In that case, we see that:

$$d_0 \circ H = d_0 \circ (h \circ u) = (d_0 \circ h) \circ u = f \circ u,$$

$$d_1 \circ H = d_1 \circ (h \circ u) = (d_1 \circ h) \circ u = g \circ u,$$

as was desired. So, $f \circ u$ and $g \circ u$ are right-homotopic.

2. Let y^I be a path space for y , let $h : x \rightarrow y^I$ be a homotopy from f to g , let z^I be a path space for z and use **Claim 23.8** to find a path space y^J , a trivial fibration $k : y^J \rightarrow y^I$ and a morphism $\bar{u} : y^J \rightarrow z^I$ so that:

$$\begin{array}{ccccc}
 & & & & z \\
 & & & & \downarrow s^z \\
 & & & & z^I \\
 & & & \swarrow \bar{u} & \downarrow (d_0^z, d_1^z) \\
 y & \xrightarrow{s^y} & y^J & \xrightarrow{\quad} & z^I \\
 \downarrow s & \swarrow k & \downarrow (d_0^y, d_1^y) & & \downarrow (d_0^z, d_1^z) \\
 y^I & \xrightarrow{(d_0, d_1)} & y \times y & \xrightarrow{u \times u} & z \times z
 \end{array}$$

¹²This was done in the author's master's thesis [REF].

commutes. Next, we consider the following pullback in $\mathcal{C}$:

$$\begin{array}{ccc} x' & \xrightarrow{t'} & y^J \\ \downarrow t & & \downarrow k \\ x & \xrightarrow{h} & y^I \end{array}$$

Since k is a trivial fibration, we see that t is also a trivial fibration. Next, we define:

$$H \stackrel{\text{def}}{=} \bar{u} \circ t'.$$

Now, we need to check that this is a homotopy from $u \circ f \circ t$ to $u \circ g \circ t$:

$$\begin{aligned} d_0^z \circ H &= d_0^z \circ (\bar{u} \circ t') \\ &= (d_0^z \circ \bar{u}) \circ t' \\ &= (u \circ d_0^y) \circ t' \\ &= u \circ (d_0^y \circ t') \\ &= u \circ (d_0 \circ k \circ t') \\ &= u \circ (d_0 \circ h \circ t) \\ &= u \circ f \circ t. \end{aligned}$$

A similar computation reveals that $d_0^z \circ H = u \circ g \circ t$. ■

Now, we want to prove the following proposition.

Theorem 24.3. K. Brown, 1973, [16]

Let $\mathcal{C}$ be a category of fibrant objects. TFSH:

1. $\simeq_h$ is an equivalence relations
2. $\simeq_h$ is compatible with compositions so that:

$$\pi(\mathcal{C}) \stackrel{\text{def}}{=} \mathcal{C} / \simeq_h,$$

is well-defined as a category. This category is equipped with a projection $\pi : \mathcal{C} \rightarrow \pi(\mathcal{C})$.

Yes, this is a whole-ass theorem because the proof of statement (1), particularly, does not proceed without showing that we actually have a calculus-of-fractions thing going on here. Now, one can work through a lot of theory regarding this and we will do so in an appendix to this lecture. However, it is a good idea to get some intuition for it by seeing it first developed in this concrete setting. To be precise, we have the following lemma which has a rather educational proof.

Lemma 24.4 Let $\mathcal{C}$ be a category of fibrant objects. TFSH:

1. Consider the diagram:

$$x \xrightarrow{u} z \xleftarrow{v} y$$

in $\mathcal{C}$. Then, the projection $x \times_z \text{cyl}(v) \rightarrow x$ is a fibration^a. It is trivial if v is a weak equivalence.

2. Given the following solid diagram in $\mathcal{C}$:

$$\begin{array}{ccc} x' & \xrightarrow{\bar{f}} & y \\ \downarrow \bar{t} & & \downarrow t \\ x & \xrightarrow{f} & z \end{array}$$

where t is a weak equivalence. Then, we can find dashed arrows above such that:

$$t \circ \bar{f} \simeq_h f \circ \bar{t},$$

and so that $\bar{t}$ is a weak equivalence. That is, this diagram homotopy-commutative.

3. Suppose that we have two morphisms $f, g : x \rightarrow y$ for which there is a weak equivalence $t : y \rightarrow z$ so that:

$$t \circ f \simeq_h t \circ g.$$

Then, there exists a weak equivalence $t' : x' \rightarrow x$ such that $f \circ t' \simeq_h g \circ t'$.

^aWe compute the mapping cylinder relative to some path space of z . We will specify this pullback later.

Proof. Let us prove the given statements individually.

1. Let us put everything into one diagram:

$$\begin{array}{ccccc} \text{cyl}(v) & \xrightarrow{\bar{v}} & z^I & & \\ \downarrow \bar{d}_v & & \downarrow d_0^z & & \\ y & \xrightarrow{v} & z & \xleftarrow{u} & x \end{array}$$

The projection in statement (1) is constructed via the following pullback diagram:

$$\begin{array}{ccc} x \times_z \text{cyl}(v) & \xrightarrow{\quad} & x \\ \downarrow & & \downarrow u \\ \text{cyl}(v) & \xrightarrow{d_1^z \circ \bar{v}} & z \end{array}$$

Observe that $d_1^z \circ \bar{v}$ is the fibration which appears in the Brown factorization of v . As a consequence, we see that the top morphism, which is our desired projection, is a fibration. If v is a weak equivalence, then we can use statement (2) of Proposition 23.10 to conclude that $d_1^z \circ \bar{v}$ is a trivial fibration. Since the above square is a pullback, we conclude that our desired projection is a trivial fibration in this case.

2. By statement (1), we may select $x' \stackrel{\text{def}}{=} x \times_z \text{cyl}(v)$ and choose $\bar{t}$ to be the projection of x' onto x . Since t is a weak equivalence, we can use statement (1) to conclude that $\bar{t}$ is a trivial fibration.
- 3.

■

Now, we can prove our desired theorem.

Proof of Theorem 24.3. We will prove each statement one-by-one.

1. Let us prove each property individually.

Reflexivity: Let $f : x \rightarrow y$ be a morphism and let y^I be a path space of y . Then, we define:

$$h \stackrel{\text{def}}{=} s^y \circ f.$$

Now, we observe that:

$$\begin{aligned} (d_0^y, d_1^y) \circ h &= (d_0^y, d_1^y) \circ (s^y \circ f) \\ &= ((d_0^y, d_1^y) \circ s^y) \circ f \\ &= \Delta_y \circ f, \end{aligned}$$

which proves that h is a homotopy from f to itself.

Symmetric: Let $f, g : x \rightarrow y$ so that $f \simeq_h g$. Then, there exists a weak equivalence t such that $f \circ t \simeq_{h,r} g \circ t$. So, it suffices to show that right-homotopies are symmetric. So, assume that $f \simeq_{h,r} g$ and let $h : x \rightarrow y^I$ be a homotopy so that:

$$(d_0, d_1) \circ h = (f, g).$$

Intuitively, we should use the same homotopy with path space (y^I, s^y, p_0, p_1) , where:

$$\forall i \in \{0, 1\} : p_i = d_{i+1},$$

where i is computed modulo 2. For this to work, we need to ensure that (d_1, d_0) is a fibration. For this to work, we need to ensure that for $x, y \in \mathcal{C}$, the interchange morphism $x \times y \rightarrow y \times x$ is a fibration. If this is the case, then we can compose it with (d_0, d_1) to form a fibration. But the interchange morphism fits into a pullback diagram where the legs are fibrations, so we're done.

Transitive: Let $f, g, h \in \mathcal{C}(x, y)$ so that:

$$\begin{aligned} f &\simeq_h g, \\ g &\simeq_h h. \end{aligned}$$

So, there exist weak equivalences t_1, t_2 such that:

$$\begin{aligned} f \circ t_1 &\simeq_{h,r} g \circ t_1, \\ g \circ t_2 &\simeq_{h,r} h \circ t_2. \end{aligned}$$

Let us note that the codomains of t_1 and t_2 are the same, so we may use statement (2) of Lemma 24.4 to find morphisms $\bar{t}_1$ and $\bar{t}_2$ (with $\bar{t}_2$ being a weak equivalence) so that:

$$t_1 \circ \bar{t}_2 \simeq_h t_2 \circ \bar{t}_1.$$

So, this leads us to the following calculation:

$$\begin{aligned} f \circ (t_1 \circ \bar{t}_2) &= (f \circ t_1) \circ \bar{t}_2 \\ &\simeq_h (g \circ t_1) \circ \bar{t}_2 \\ &= g \circ (t_1 \circ \bar{t}_2) \\ &\simeq_h g \circ (t_2 \circ \bar{t}_1) \\ &= (g \circ t_2) \circ \bar{t}_1 \\ &\simeq_h (h \circ t_2) \circ \bar{t}_1 \\ &= h \circ (t_2 \circ \bar{t}_1) \\ &\simeq_h h \circ (t_1 \circ \bar{t}_2), \end{aligned}$$

which proves that we have a transitive relation.

This proves statement (1).

2. Let $f, g \in \text{Mor}(\mathcal{C})$ so that $f \simeq_h g$ and let $t \in W$ so that:

$$f \circ t \simeq_{h,r} g \circ t.$$

Now, let $u \in \text{Mor}(\mathcal{C})$ so that $u \circ f$ and $u \circ g$ are defined. Then, we know that:

$$u \circ f \circ t \simeq_{h,r} u \circ g \circ t,$$

which means that $u \circ f \simeq_h u \circ g$. Next, let $v \in \text{Mor}(\mathcal{C})$ so that $f \circ v$ and $g \circ v$ are defined. So:

$$f \circ t \circ v \simeq_{h,r} g \circ t \circ v.$$

By statement (3) of Lemma 24.4, we can find a weak equivalence t' so that:

$$t \circ v \simeq_h v \circ t'.$$

So, we see that:

$$f \circ v \circ t' \simeq_{h,r} g \circ v \circ t',$$

which means that $f \circ v \simeq_h g \circ v$. ■

With the result above in hand, we can set some notation which also doubles up as terminology.

Notation 24.5. Let $\mathcal{C}$ be a category of fibrant objects and let $f \in \text{Mor}(\mathcal{C})$. f is a homotopy equivalence in $\mathcal{C}$ iff $\pi(f)$ is an isomorphism. The full subcategory of homotopy equivalences in $\text{Arr}(\mathcal{C})$ is denoted by $W_{\mathcal{C},h}$ or W_h .

Here is a notable result.

Claim 24.2. Let $\mathcal{C}$ be a category of fibrant objects. Then^a, $W_{\mathcal{C},h}$ is 2-out-of-6.

^aThis is proved on Page 275.

We have a remark about this simple result.

Remark 24.6. So, even though we do not necessarily assume that the weak equivalences in a category of fibrant objects is 2-out-of-6, it is the case that it produces two subcategories of morphisms which are 2-out-of-6.

One of the most important results about categories of fibrant objects which was proved by Brown has to do with a way to control the size of the homotopy category of a category of fibrant objects. This goes back to the reason why Section 1.4 exists in this thesis. What makes this notable is that such a result is true for the homotopy category of a model category and categories of fibrant objects can be quite far from being model categories.

Theorem 24.7. K. Brown, [16, Theorem 1, p.425]

Let $\mathcal{C}$ be a category of fibrant objects. TFSH:

1. $\pi(W) \subseteq \pi(\mathcal{C})$ admits a calculus of right fractions.
2. $\forall x, y \in \mathcal{C} : [x, y]_{\mathcal{C}} \simeq \text{colim}_{x' \xrightarrow{\sim} x} \pi(\mathcal{C})(x', y)$.

Moreover, if $\gamma : \mathcal{C} \rightarrow \mathbf{Ho}(\mathcal{C})$ is the localization functor, then TFSH:

1. If $x, y \in \mathbf{Ho}(\mathcal{C})$, then any morphism in $[x, y]_{\mathcal{C}}$ can be written as a right fraction:

$$x \xleftarrow{\gamma(t)} x' \xrightarrow{\gamma(f)} y$$

where $t \in W$ is a weak equivalence.

2. If $x, y \in \mathcal{C}$ and $f_0, f_1 \in \mathcal{C}(x, y)$, then $\gamma(f_0) = \gamma(f_1)$ iff f_0 and f_1 are homotopic.

With that said, let us return to the fact that Proposition 24.2 is rather unappealing. Indeed, it does not allow for compatibility of right-homotopy with compositions to the highest degree. Let us fix this by taking some inspiration from our work with homotopy theory on equivariant operator algebras.

Definition 24.8. A. Vats, 2026

Let $\mathcal{C}$ be a category of fibrant objects and suppose that we have further data of:

1. a functor $T : \mathcal{C} \rightarrow \mathcal{C}$ such that for each $x \in \mathcal{C}$, $T(x)$ gives us a path space (the canonical path space) and;
2. two natural transformations $\alpha_0, \alpha_1 : T \Rightarrow \text{Id}_{\mathcal{C}}$ such that for each $x \in \mathcal{C}$, the morphisms:

$$\alpha_0^x, \alpha_1^x : T(x) \rightarrow x,$$

are the evaluation morphisms in the path space of $T(x)$,

such that for any $x, y \in \mathcal{C}$ and $f, g \in \mathcal{C}(x, y)$, the following are equivalent:

1. $f \simeq_{h,r} g$.
2. There exists a morphism $h : x \rightarrow T(y)$ such that $\alpha_0^y \circ h = f$ and $\alpha_1^y \circ h = g$.

Then, the quadruple $(\mathcal{C}, T, \alpha_0, \alpha_1)$ forms a h-category of fibrant objects^a.

^aAll constructions involving path spaces will, then, occur with respect to the canonical path space.

In such a category, we see that right-homotopy is fully compatible with compositions.

Proposition 24.9. A. Vats, 2026

Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h-category of fibrant objects, let $x, y, z \in \mathcal{C}$, let $f_1, f_2 \in \mathcal{C}(x, y)$ and let $g_1, g_2 \in \mathcal{C}(y, z)$ so that:

$$f_1 \simeq_{h,r} f_2,$$

$$g_1 \simeq_{h,r} g_2.$$

Then, $g_1 \circ f_1 \simeq_{h,r} g_2 \circ f_2$.

Proof. The key is to proceed exactly as in the proof of Proposition 6.23. The first step of the argument is simply statement (1) of Proposition 24.2, while the last step is purely formal and proceeds as written. So, what we need to do is to clarify step (2). So, let $g_1 = g_2 = g$ and let $h : x \rightarrow T(y)$ be a homotopy from f_1 to f_2 . We define:

$$H \stackrel{\text{def}}{=} T(g) \circ h.$$

This is a morphism from x to $T(z)$. Now, let us show that this is a homotopy from $g \circ f_1$ to $g \circ f_2$ by a computation:

$$\begin{aligned} \alpha_i^z \circ H &= \alpha_i^z \circ T(g) \circ h \\ &= g \circ \alpha_i^y \circ h \\ &= g \circ f_{i+1}, \end{aligned}$$

for $i \in \{0, 1\}$, which completes the proof. ■

We have a remark about this result and **Definition 24.8**.

Remark 24.10. Looking at the proof above, we see that it is the functoriality of the path space and the naturality of the evaluation morphisms which allowed us to conclude that right-homotopies are compatible with post-composition.

It is the case that a h -category of fibrant objects admits an internal notion of homotopy which agrees with the notion of homotopy defined in concrete categories. It is also the case that such a category will admit an internal notion of fibration. To define this, we will utilize statement (2) of **Claim 6.14** as inspiration.

Terminology 24.11. Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects and let $p \in \mathcal{C}(x, y)$. p is an internal fibration (int-fibration) iff for any $z \in \mathcal{C}$, any $f \in \mathcal{C}(z, x)$ and for any $h \in \mathcal{C}(z, T(y))$ which is a homotopy of $p \circ f$:

$$\alpha_0^y \circ h = p \circ f,$$

there exists a $H \in \mathcal{C}(z, T(x))$ such that the following diagram:

$$\begin{array}{ccc} & & x \\ & \nearrow f & \uparrow \alpha_0^x \\ z & \text{---} H \text{---} & T(x) \\ & \searrow h & \downarrow T(p) \\ & & T(y) \end{array}$$

commutes.

With a h -category of fibrant objects, we can also characterize fibrations in a different way. The discussion here was inspired by [89, Proposition 1.10, p.410-411]. We will, first of all, require a simple construction.

Construction 24.12. Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects, let $f \in \mathcal{C}(x, y)$ and consider the diagram:

$$\begin{array}{ccccc} & & & \alpha_0^x & \\ & & & \curvearrowright & \\ T(x) & & & & x \\ & \searrow v & \xrightarrow{\bar{d}_f} & & \downarrow f \\ & \text{cyl}(f) & & & y \\ & \downarrow \bar{f} & & & \\ & T(y) & \xrightarrow{\alpha_0^y} & & \end{array}$$

So, the key point is that we have the pullback square for the mapping cylinder and the outer diagram commutes due to the naturality of α_0 . By the universal property of the pullback, we induce a unique morphism $v : T(x) \rightarrow \text{cyl}(f)$ such that the entire diagram commutes. We refer to this as the cyl-mapcyl morphism.

We should collect some properties of internal fibrations. Actually, these properties will show that a suitable h -category of fibrant objects actually admits **another** fibration category structure.

Proposition 24.13. A. Vats, 2026

Let $(\mathcal{C}, T, \alpha_0, \alpha_1)$ be a h -category of fibrant objects so that T preserves pullbacks & zero objects. TFSH:

1. The int-fibrations are closed under composition.
2. If $x \in \mathcal{C}$, then the unique map $x \rightarrow 0$ is an int-fibration.

3. Let $f \in \text{Mor}(\mathcal{C})$ be an isomorphism. Then, it is an int-fibration.

4. Int-fibrations are preserved by base extensions.

In particular, $\mathcal{C}$ is a fibration category when equipped with its int-fibrations.

Proof. Due to the work we have already put into making the proof of Proposition 6.29 as formal as possible, so the proof of this result is simply the proof of the aforementioned proposition with the cylinder replaced with T and with the time-evaluation maps replaced with the components of α_0 and α_1 . The additional requirement on T to preserve pullbacks ensures that the proof of statement (4) in the aforementioned result also carries over here. ■

Let us characterize internal fibrations in a h -category of fibrant objects among the morphisms.

Proposition 24.14. C. Schochet, [89, Proposition 1.10, p.410-412], A. Vats, 2026

In the notation of Construction 24.12 & assuming that T preserves pullbacks, TFAE:

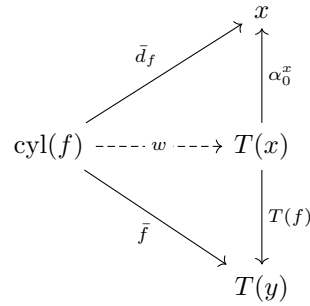
1. f is an int-fibration.
2. The cyl-map_{cyl} morphism v is a split epi.

Proof. We will use the notation of Construction 24.12 throughout the proof.

(1) $\Rightarrow$ (2): Observe that:

$$\alpha_0^y \circ \bar{f} = f \circ \bar{d}_f.$$

So, this means that $\bar{f}$ is a homotopy of $\bar{d} \circ f$. Now, we consider the following homotopy lifting problem:



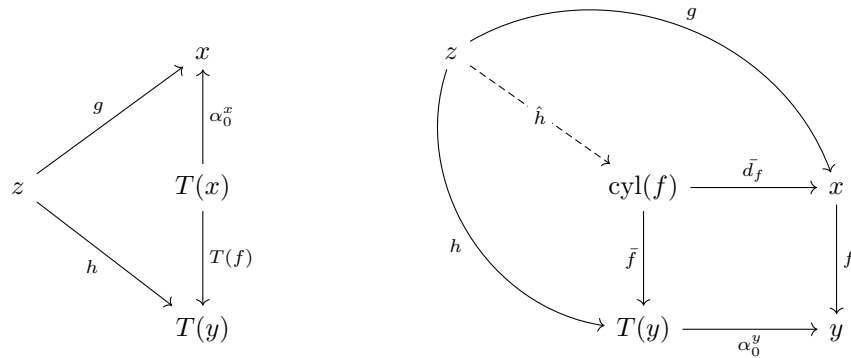
Since f is an internal fibration, we see there exists a homotopy lift $w : \text{cyl}(f) \rightarrow T(x)$. Let us check that this is, in fact, a right inverse to the map v . Observe that we have the following equations:

$$\bar{d}_f = \alpha_0^x \circ w = \bar{d}_f \circ (v \circ w),$$

$$\bar{f} = T(f) \circ w = \bar{f} \circ (v \circ w).$$

Now, we apply the universal property of the pullback to conclude that $v \circ w = \text{Id}_{\text{cyl}(f)}$.

(2) $\Rightarrow$ (1): Let $w : \text{cyl}(f) \rightarrow T(x)$ be a split for w and consider the following diagrams following lifting problem:



The diagram on the left is a lifting problem, where h is a homotopy of $f \circ g$. The diagram on the right is a pullback diagram for the mapping cylinder of f and the outer diagram commutes because h is a homotopy of $f \circ g$. By the universal property of the pullback, we have a morphism $\hat{h} : z \rightarrow \text{cyl}(f)$. We define:

$$k \stackrel{\text{def}}{=} w \circ \hat{h}.$$

This is a morphism from z to $T(x)$, so it is a candidate homotopy lift. Let's show that it works.

| **Claim 24.3.** k is the desired lift.

This is proved on Page 275. So, f is an internal fibration. ■

24.2 Hints of Homotopy-Induced Algebraic Structures

Let us turn towards a more general result about categories of fibrant objects which deserves some attention. Introducing this result (??) requires some preparation and in fact, we will not discuss all of the terminology used within the theorem within this subsection, seeing as it is too much to cover. We will mainly use the remainder of this subsection to work towards the statement of that theorem and provide some context around it. Let us start with some terminology.

Terminology 24.15. Let $\mathcal{C}$ be a pointed category and let $f \in \mathcal{C}(x, y)$. Then, we define:

$$\text{Fibre}(f) \stackrel{\text{def}}{=} \text{Eq}(f, 0),$$

the fibre of f , if this exists. The cofibre of f , denoted by $\text{Cofibre}(f)$, is its fibre in $\mathcal{C}^{\text{op}}$.

Now, we use this to construct the loop of an object in a category of fibrant objects.

Construction 24.16. Let $\mathcal{C}$ be a category of fibrant objects and let $x \in \mathcal{C}$. The loop-object of x is the fibre:

$$\Omega(x) \stackrel{\text{def}}{=} \text{Fibre}((d_0, d_1)),$$

where (s, x^I, d_0, d_1) is a path-space of x in $\mathcal{C}$.

In fact, this construction is not functorial on the underlying category of fibrant objects in general. The next result, however, tells us that the loop can be defined as a functor on the *homotopy category* of a category of fibrant objects.

Proposition 24.17. K. Brown, [16, Theorem 3, Proposition 3, p.430-432]

Let $\mathcal{C}$ be category of fibrant objects. Then, Ω defines a functor:

$$\Omega : \mathbf{Ho}(\mathcal{C}) \rightarrow \mathbf{Ho}(\mathcal{C}),$$

called the loop-object functor. This functor satisfies the following properties:

1. If $x \in \mathcal{C}$, then $\Omega(x)$ is a group object in $\mathbf{Ho}(\mathcal{C})$ and $\Omega^2(x)$ is an abelian group object in $\mathbf{Ho}(\mathcal{C})$.
2. If $p : x \rightarrow y$ is a fibration with fibre F , then there is a natural right action:

$$F \times \Omega(y) \rightarrow F,$$

in $\mathbf{Ho}(\mathcal{C})$. In particular, we have a natural morphism $\Omega(y) \rightarrow F$ in $\mathbf{Ho}(\mathcal{C})$.

Let us introduce (abelian) group objects in a category. The idea is that the properties of (abelian) groups can be represented by the commutativity of some diagrams and this can be mimicked in a fairly general way within a category. Our treatment of these notions is based on the excellent exposition in [35, Part I, Section 2, p.3-4].

Terminology 24.18. Let $\mathcal{C}$ be a pointed category^a with finite products. A group object in $\mathcal{C}$ is a quadruple:

$$(G, m, e, i),$$

consisting of $G \in \mathcal{C}$ and three morphisms:

$$m : G \times G \rightarrow G,$$

$$e : 1 \rightarrow G,$$

$$i : G \rightarrow G,$$

such that the following diagrams:

$$\begin{array}{ccc} G \times (G \times G) & \xrightarrow{\cong} & (G \times G) \times G \xrightarrow{m \times \text{Id}_G} G \times G \\ \text{Id}_G \times m \downarrow & & \downarrow m \\ G \times G & \xrightarrow{m} & G \end{array}$$

$$\begin{array}{ccccc} 1 \times G & \xrightarrow{e \times \text{Id}_G} & G \times G & \xleftarrow{\text{Id}_G \times e} & G \times 1 \\ & \searrow \cong & \downarrow m & \swarrow \cong & \\ & & G & & \end{array}$$

$$\begin{array}{ccc} G & \xrightarrow{(1, i)} & G \times G \\ \downarrow & & \downarrow m \\ 1 & \xrightarrow{e} & G \end{array}$$

commute. This group object is an abelian group object iff the following diagram:

$$\begin{array}{ccc} G \times G & \xrightarrow{T} & G \times G \\ & \searrow m & \downarrow m \\ & & G \end{array}$$

commutes, where T is the interchange map^b.

^aThe zero object is denoted by 1.

^bThis is constructed by the universal property of the product on the canonical projections.

Of course, the (abelian) group objects in Set are just (abelian) groups, because the diagrams above precisely encode the defining axioms of (abelian) groups. Now, we should also explain the notion of an action of a group object on another object in a category. Again, group actions of usual groups on sets are encoded by maps and equations, the latter of which are encoded by diagrams. So, we simply adapt those diagrams to a category.

Terminology 24.19. Let $\mathcal{C}$ be a category with finite products and a final object 1 , let G be a group object in $\mathcal{C}$ and let $x \in \mathcal{C}$. A (right) action of G on x is a map $\nu : x \times G \rightarrow x$ such that the following diagram:

$$\begin{array}{ccccc} (x \times G) \times G & \xrightarrow{\cong} & x \times (G \times G) & \xrightarrow{\text{Id}_x \times m} & x \times G & \xleftarrow{\text{Id}_x \times e} & x \times 1 \\ \downarrow \nu \times \text{Id}_G & & & & \downarrow \nu & \swarrow \cong & \\ x \times G & \xrightarrow{\quad \nu \quad} & & & x & & \end{array}$$

commutes.

Now, there is a classical result of model category theory which says that if $\mathcal{M}$ is a model category and $x, y \in \mathcal{M}$, then $[y, \Omega(x)]$ has the structure of a group and $[y, \Omega^2(x)]$ has the structure of an abelian group. In that setting, this result is difficult to prove. For instance, it is essentially shown within [54, Corollary 6.1.6, p.151] and makes use of the full structure available on model categories. However, it seems that a significant simplification of that proof is available and it just uses Proposition 24.17 and a formal characterization of (abelian) group objects. Let us sketch the latter first.

Proposition 24.20

Let $\mathcal{C}$ be a locally small category with a final object and have finite products & let $x \in \mathcal{C}$. TFAE:

1. x has a group structure in $\mathcal{C}$.
2. The functor $\mathcal{C}(-, x)$ has a group structure in $\text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$.

If either (1) or (2) holds, then x is an abelian group object iff $\mathcal{C}(-, x)$ is an abelian group object in $\text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$.

Proof Sketch. This is explained in [65, Proposition 1, p.75-76]. However, the basic idea is that we can translate between morphisms in $\mathcal{C}$ and morphisms between hom-sets by using the Yoneda Lemma. For example, let $m : x \times x \rightarrow x$ be the multiplication morphism if x is a group object and let $y \in \mathcal{C}$. Then, we have the following composition:

$$\mathcal{C}(y, x) \times \mathcal{C}(y, x) \rightarrow \mathcal{C}(y, x \times x) \rightarrow \mathcal{C}(y, x),$$

where the first map is a natural isomorphism because representable functors preserve products while the second map is obtained by applying $\mathcal{C}(y, -)$ to the morphism m . So, the above forms a “multiplication” map for the hom-set $\mathcal{C}(y, x)$ and one checks that this multiplication actually satisfies the relevant properties. Going backwards, suppose that $\mathcal{C}(y, x)$ is a group for each $y \in \mathcal{C}$. So, this has a multiplication map denoted as follows:

$$\bar{m} : \mathcal{C}(y, x) \times \mathcal{C}(y, x) \rightarrow \mathcal{C}(y, x).$$

We precompose this with the natural isomorphism $p : \mathcal{C}(y, x) \times \mathcal{C}(y, x) \simeq \mathcal{C}(y, x \times x)$ and we get:

$$\bar{m} \circ p : \mathcal{C}(y, x \times x) \rightarrow \mathcal{C}(y, x).$$

Due to the naturality of $\bar{m}$ and the naturality of p , we see that the above is a component of a natural transformation $\mathcal{C}(-, x \times x) \Rightarrow \mathcal{C}(-, x)$. Now, we apply the Yoneda Lemma and note that this natural transformation corresponds precisely to a morphism in $\mathcal{C}(x \times x, x)$ and this is our multiplication map. ■

Now, we can state and prove our result.

Theorem 24.21

Let $\mathcal{C}$ be a category of fibrant objects and let $x, y \in \mathcal{C}$.

1. $[y, \Omega(x)]$ is a group.
2. $[y, \Omega^2(x)]$ is an abelian group.

Proof. Observe that $\mathbf{Ho}(\mathcal{C})$ is a locally small category with a final object and admits finite products. Since $\Omega(x)$ is a group object in $\mathbf{Ho}(\mathcal{C})$, we see that the functor $\mathbf{Ho}(\mathcal{C})(-, \Omega(x))$ is a group object in $\text{Fun}(\mathbf{Ho}(\mathcal{C})^{\text{op}}, \text{Set})$. So, $\mathbf{Ho}(\mathcal{C})(y, \Omega(x)) = [y, \Omega(x)]$ is a group, which proves statement (1). For statement (2), the argument is precisely analogous, except that we use the fact that $\Omega^2(x)$ is an abelian group object in $\mathbf{Ho}(\mathcal{C})$. ■

As is the case with homotopy theory on spaces, there is a long-exact sequence obtained from a fibration.

Proposition 24.22. Barratt-Puppe Fibration Sequence, [78, Proposition 1.3.3-1.3.4, p.42-46]

Let $\mathcal{C}$ be a category of fibrant objects and let $p : x \rightarrow y$ be a fibration with fibre F . There is an exact sequence:

$$\dots \rightarrow [z, \Omega^2(y)]_{\mathcal{C}} \rightarrow [z, \Omega(F)]_{\mathcal{C}} \rightarrow [z, \Omega(x)]_{\mathcal{C}} \rightarrow [z, \Omega(y)]_{\mathcal{C}} \rightarrow [z, F]_{\mathcal{C}} \rightarrow [z, x]_{\mathcal{C}} \rightarrow [z, y]_{\mathcal{C}},$$

for any $z \in \mathcal{C}$.

We have a remark about this proposition.

Remark 24.23. We need to explain what exactness means in the context above, because the first three terms of the sequence are simply pointed sets. Let us define exactness here. Consider the following sequence of pointed sets:

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

The sequence is exact at Y iff:

$$\text{ran}(f) = \ker(g) \stackrel{\text{def}}{=} g^{\leftarrow}(\{z\}),$$

where $z \in Z$ is the basepoint. In the case of groups and abelian groups, these come equipped with a natural basepoint

(the identity) and one recovers the natural notion of exactness in those cases by using what we have defined above. Of course, we should also mention that for the latter maps which appear in the fibration sequence of Proposition 24.22, it is desirable to have them be group homomorphisms and, in fact, it turns out that they are group homomorphisms.

25 Some Standard Theorems of Classical Homotopy Theory

Summary

In this lecture, we will cover the following topics:

- 1.
- 2.

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

26 Introducing Model Structures

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

26.1 Two Approaches Towards Discussing Model Categories

26.2 Notions of Homotopy Internal to Model Categories

27 A Survey of Examples of Model Categories

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

28 Intrinsic Homotopy in Model Categories

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

29 Comparisons of Model Structures

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

30 The Basic Axioms of ∞ -Categories

Summary

In this chapter, we discuss the following topics:

1. The fundamentals of the synthetic view.
2. Pullbacks, functor categories & the big three axioms.

We strongly recommend reading this chapter linearly until the last section, where it is a technical view into the construction of the Yoneda embedding at the ∞ -categorical level.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

1-category theory - This refers to the ordinary category theory which is assumed background for readers.

Let's start with some motivation.

Motivation.

30.1 The Set-Up Of The Axiomatic Theory

The motivation above compels us to start by introducing an axiom which reflects our desire to deal with structures which have a notion of “isomorphisms up to an infinite level of coherence”. This will just be an analogue of [26, Axiom A.1, p.15-16]; our changes only improve upon the clarity of the statement.

Axiom 30.1. Existence of ∞ -Categories/Functors

We may speak of ∞ -categories which are denoted by the same symbols as usual categories. If $\mathcal{C}, \mathcal{D}$ are ∞ -categories, then there exist ∞ -functors^a between them which we denote by $f : \mathcal{C} \rightarrow \mathcal{D}$. Moreover:

1. If $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are ∞ -categories and $f : \mathcal{A} \rightarrow \mathcal{B}, g : \mathcal{B} \rightarrow \mathcal{C}$ are functors, then there exists a unique functor:

$$g \circ f : \mathcal{A} \rightarrow \mathcal{C},$$

which is known as the composition of g and f .

2. If $\mathcal{A}, \mathcal{B}$ are ∞ -categories and $f, g : \mathcal{A} \rightarrow \mathcal{B}$ are functors, then we may speak of natural isomorphisms between functors $\alpha : f \simeq g$. If $h : \mathcal{A} \rightarrow \mathcal{B}$ is a functor and we have two natural isomorphisms:

$$\alpha : f \simeq g,$$

$$\beta : g \simeq h,$$

then there exists a natural isomorphism $\beta \circ \alpha : f \simeq h$. Natural isomorphisms are also known as 2-(iso)morphisms.

3. If α, β are natural isomorphisms between functors, then we may speak of isomorphisms of natural isomorphisms $\alpha \simeq \beta$ with the following composition property. If $\alpha \simeq \beta$ and $\beta \simeq \gamma$ are isomorphisms of natural isomorphisms, then we may pre-compose the former with the latter to form an isomorphism of natural isomorphisms:

$$\alpha \simeq \gamma.$$

Such isomorphisms are known as 3-isomorphisms.

4. Proceeding inductively, we demand that for every $n \geq 4$, a notion of an n -isomorphism between two $(n-1)$ -isomorphisms satisfying the usual composition property as stipulated above.

Two functors g, f between ∞ -categories for which $g \circ f$ exists are said to be composable functors.

^aWe will simply refer to them as functors.

Let us introduce some new terminology associated to the axiom.

Terminology 30.2. A functor between ∞ -categories is also known as a 1-morphism. Moreover, when we refer to n -morphisms for $n \geq 2$, we implicitly assume that these are isomorphisms, keeping in line with Axiom 30.1.

Terminology 30.3. Let $n \in \mathbb{N}_0$ and suppose we have two ∞ -categories for which we can consider n -morphisms.

1. n is said to be the level of coherence of the morphism.
2. If we have two n -morphisms, then we say that these morphisms are at the n -th level of coherence.

If we say “consider two morphisms at some level of coherence”, then they are both at the same level of coherence.

Here's a further remark.

Remark 30.4. In [26], the term ∞ -category used above is replaced with “synthetic category”, ostensibly as a reference to the synthetic approach. However, Cnossen's notes on stable homotopy theory ([27]) simply refer to these as ∞ -categories. Since we are no longer interested in working in a given model, we'll just refer to these as ∞ -categories.

Now, the next axiom introduces unitality, inverses and associativity of the composition for higher morphisms.

Axiom 30.5. Properties of Compositions

Let $n \in \mathbb{N}$.

1. For each n -morphism f , there exists an $(n+1)$ -morphism $\text{Id}_f : f \simeq f$ such that for any n -morphism g and any $(n+1)$ -morphisms $\alpha : f \simeq g$ & $\beta : g \simeq f$, we have that $\alpha \circ \text{Id}_f \simeq \alpha$ and $\text{Id}_f \circ \beta \simeq \beta$.
2. If f, g, h are n -morphisms such that $g \circ h$ exists and $f \circ (g \circ h)$ exists, then:

$$f \circ (g \circ h) \simeq (f \circ g) \circ h.$$

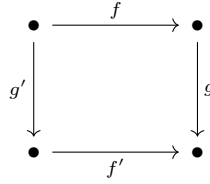
In other words, composition is associative up to isomorphism.

3. Let $n \geq 1$, let f, g be n -morphisms and let $\alpha : f \simeq g$ be a $(n+1)$ -morphism. Then, there exists a $(n+1)$ -morphism $\alpha^{-1} : g \simeq f$ such that $\alpha \circ \alpha^{-1} \simeq \text{Id}_g$ and $\alpha^{-1} \circ \alpha \simeq \text{Id}_f$.

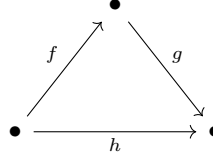
So, we replace all equalities of morphisms at a level of coherence by isomorphisms of coherence one step higher.

Now, we can introduce what it means for certain diagrams of morphisms at the same level of coherence to commute.

Terminology 30.6. Let $n \in \mathbb{N}$ and consider the following diagram of n -morphisms:



This is a commutative square of n -morphisms iff $g \circ f \simeq f' \circ g'$. That is, the square commutes iff we can provide a $(n+1)$ -morphism between these n -morphisms. Similarly, consider the following triangle of n -morphisms:



This is a commutative triangle of n -morphisms iff $g \circ f \simeq h$, precisely in the sense as stipulated above with squares^a.

^aIn both cases, we require the actual isomorphism to be part of the data of the commutative triangle. This is meant to be a nod towards the fundamental principle of homotopy theory, where we also want to remember *how* two things are “the same”.

With this basic language for dealing with ∞ -categories along with some manipulation rules for the symbols, let us define a notion familiar to us from 1-category theory and work with it.

Definition 30.7

Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories and let $f : \mathcal{C} \rightarrow \mathcal{D}$ be a functor.

1. f is an equivalence of ∞ -categories iff there exists a functor:

$$g : \mathcal{D} \rightarrow \mathcal{C},$$

along with distinguished natural isomorphisms $\text{Id}_{\mathcal{C}} \simeq g \circ f$ and $\text{Id}_{\mathcal{D}} \simeq f \circ g$.

2. $\mathcal{C}$ and $\mathcal{D}$ is an equivalence of ∞ -categories iff there exists a functor f as given above which is also an equivalence of ∞ -categories. We denote this situation by writing $\mathcal{C} \simeq \mathcal{D}$.

In (1), we refer to g as a quasi-inverse of f .

Here's a remark.

Remark 30.8. Let $\mathcal{C}, \mathcal{D}$ be 1-categories and let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Here's a nice characterization:

1. F is an equivalence of categories.
2. F is fully faithful and essentially surjective.

This is easy enough to prove in 1-category theory. In ∞ -category theory, one would need to define what it means for a functor to be fully faithful & essentially surjective. In fact, it turns out that this is quite hard to do even in a treatment of a concrete model. For example, this is proved only through a very, very long sequence of lemmas in [60, Theorem 2.3.20, p.130-132], with the proof of the result itself still being quite difficult even after all that preparation. In the formalization project, the analog of the equivalence above is only proved in [26, Theorem 6.3.3, p.187-188]. In other words, establishing the foundations of ∞ -category theory is still very difficult and requires a lot of work.

Here is a rather basic result.

Claim 30.1. Let $\mathcal{C}, \mathcal{D}, \mathcal{E}$ be ∞ -categories. TFSH:

1. $\mathcal{C} \simeq \mathcal{C}$.
2. If $\mathcal{C} \simeq \mathcal{D}$, then $\mathcal{D} \simeq \mathcal{C}$.
3. If $\mathcal{C} \simeq \mathcal{D}$ and $\mathcal{D} \simeq \mathcal{E}$, then $\mathcal{C} \simeq \mathcal{E}$.

Is it the case that the quasi-inverses of an equivalence are isomorphic?

Claim 30.2. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories and let $f : \mathcal{C} \rightarrow \mathcal{D}$ be an equivalence of categories with quasi-inverses:

$$h, h'.$$

Then, we have that h and h' are naturally isomorphic.

This prompts us to introduce some terminology.

Terminology 30.9. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories and let $f : \mathcal{C} \rightarrow \mathcal{D}$ be a functor.

1. A section of f is a functor $s : \mathcal{D} \rightarrow \mathcal{C}$ such that $f \circ s \simeq \text{Id}_{\mathcal{D}}$.
2. A retraction of f is a functor $r : \mathcal{D} \rightarrow \mathcal{C}$ such that $r \circ f \simeq \text{Id}_{\mathcal{C}}$.

We also refer to these as right & left quasi-inverses respectively.

To check that a functor is an equivalence, it's enough to check that it has a section and a retraction.

Claim 30.3. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories and let $f : \mathcal{C} \rightarrow \mathcal{D}$ be a functor. TFAE:

1. f is an equivalence of categories.
2. f admits both a section and a retraction.

If either of these conditions hold, then the section and retraction are naturally isomorphic.

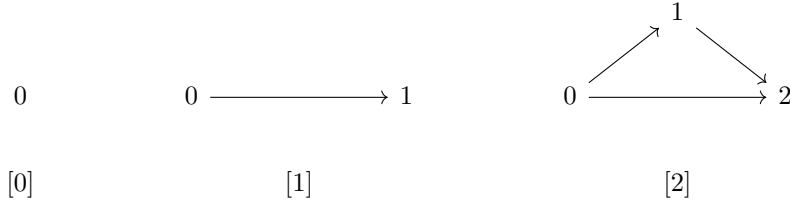
Now, equivalences of ∞ -categories also satisfy the 2-out-of-3 and the 2-out-of-6 properties.

Claim 30.4. Let $\mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{F}$ be ∞ -categories and let $f : \mathcal{C} \rightarrow \mathcal{D}, g : \mathcal{D} \rightarrow \mathcal{E}, h : \mathcal{E} \rightarrow \mathcal{F}$ be functors. TFSH:

1. If 2 out of $f, g, g \circ f$ are equivalences, then the third is also an equivalence.
2. If $g \circ f$ and $h \circ g$ are equivalences, then f, g, h and $h \circ g \circ f$ are equivalences.

Our motivation at the beginning also indicated a way for us to use functorial language to talk about the internal structure of a ∞ -category and the last chapter actually provides us with the tools to do this. Here's some intuition first.

Intuition 30.10. The categories in ?? can be drawn as follows:



So, these diagrams represent abstract arrows, with $[0]$ representing an abstract object. So, we can model an object in a ∞ -category $\mathcal{C}$ as a functor $x : [0] \rightarrow \mathcal{C}$. However, the issue is that $[0]$ is a category and $\mathcal{C}$ is a ∞ -category, so we don't know what it means for there to be a functor between them. Indeed, we haven't said anything at all about the relationship between ordinary categories and ∞ -categories.

This intuition leads us to an axiom which tells us that every ordinary category is certainly an ∞ -category and that every 1-morphism between ordinary categories is a functor in the usual sense.

Axiom 30.11. The Bridge To 1-Categories

Let $\mathcal{C}$ be a category.

1. $\mathcal{C}$ is also an ∞ -category.
2. If $\mathcal{D}$ is a category, then any $f \in \text{Fun}(\mathcal{C}, \mathcal{D})$ is a functor between ∞ -categories.
3. If $\mathcal{D}$ is a category and $f : \mathcal{C} \rightarrow \mathcal{D}$ is a 1-morphism, then it is a functor between ordinary categories.

Moreover, if f, g are functors between ordinary categories, then the natural isomorphisms between them are precisely the same one as those which arise between them as 1-morphisms.

We have two remarks about this axiom.

Remark 30.12. You will note that this axiom has been stated far less precisely than the two before it and the simple reason is that a fully axiomatic framework would simply not require it. Since we do not have the space for a fully axiomatic treatment and since the axiom above is plausible in light of our desire to see ∞ -category theory *generalize* ordinary category theory, we simply accept this state of affairs.

Remark 30.13. The imprecision mentioned in the preceding remark actually does cause a few problems later. For example, there are situations where we would like to know when an ∞ -category IS just a 1-category. Now, in a concrete model, this is very easy; we can say that a CSS is a 1-category iff it is naturally isomorphic to the nerve of a category. Over here, the situation seems much less clear; we suspect that this situation could be “fixed” if we had a universal characterization of the nerve functor. [INVESTIGATE FURTHER LATER?]

With this data, we can introduce terminology pertaining to the internal structure of a ∞ -category.

Terminology 30.14. Let $\mathcal{C}$ be an ∞ -category.

1. An object in $\mathcal{C}$ is a functor^a $f : [0] \rightarrow \mathcal{C}$.
2. A morphism of $\mathcal{C}$ is a functor $f : [1] \rightarrow \mathcal{C}$. We have additional terminology:
 - (a) The source/domain of f is denoted by $f(0)$.
 - (b) The target/codomain of f is denoted by $f(1)$.

These are defined explicitly by the following:

$$\begin{aligned} f(0) &\stackrel{\text{def}}{=} f \circ d_1, \\ f(1) &\stackrel{\text{def}}{=} f \circ d_0. \end{aligned}$$

We will also denote the morphism by the usual notation:

$$f : x \rightarrow y,$$

to indicate that f is accompanied by isomorphisms $x \simeq f(0)$ and $y \simeq f(1)$.

3. A commutative triangle in $\mathcal{C}$ is a functor $\sigma : [2] \rightarrow \mathcal{C}$.
4. A commutative square is a functor $\sigma : [1] \times [1] \rightarrow \mathcal{C}$.

With regards to the commutative triangle and square, we want to think of these in terms of the usual diagrams:

$$\begin{array}{ccc} x & \xrightarrow{h} & z \\ & \searrow f & \nearrow g \\ & y & \end{array} \qquad \begin{array}{ccc} x & \xrightarrow{g} & y \\ f \downarrow & & \downarrow h \\ z & \xrightarrow{r} & w \end{array}$$

where each morphism above is formed by composition of σ (in both cases) with some face map.

^aTypically, when we write “Let $x \in \mathcal{C}$ ” or “Let x be an object in $\mathcal{C}$ ”, this will just be short-hand for “Let $x : [0] \rightarrow \mathcal{C}$ be a functor”. The benefit is that many of the proofs we give for reasonable statements end up “looking like” proofs from ordinary category theory.

Let’s construct our first morphism in a ∞ -category.

Construction 30.15. Let $\mathcal{C}$ be an ∞ -category and let $x : [0] \rightarrow \mathcal{C}$ be an object in $\mathcal{C}$. Define:

$$\mathrm{Id}_x \stackrel{\mathrm{def}}{=} x \circ s_0^1.$$

This is a functor from $[1]$ to $\mathcal{C}$, so it is a morphism in $\mathcal{C}$.

| Claim. The source and target of Id_x are isomorphic to x .

Well, we note that:

$$\mathrm{Id}_x(0) = \mathrm{Id}_x \circ d_1^1 = x \circ s_0^1 \circ d_1^1.$$

$$\mathrm{Id}_x(1) = \mathrm{Id}_x \circ d_0^1 = x \circ s_0^1 \circ d_0^1.$$

Now, we have to compute $s_0^1 \circ d_1^1$ and $s_0^1 \circ d_0^1$. Well, both of these functors go from $[0]$ to $[0]$, so it follows that both equations above reduce to x on the right-hand side. That is, the source and target of Id_x is isomorphic to x .

We should expect the identity to have the “right property”. Namely, if $f : x \rightarrow y$ is a morphism, then:

$$f \circ \mathrm{Id}_x \simeq f,$$

$$\mathrm{Id}_y \circ f \simeq f.$$

Note that we haven’t defined what it means for two morphisms in a ∞ -category $\mathcal{C}$ to be composed. Actually, let us dedicate a whole remark to that business, because the solution is pretty stupid.

Remark 30.16. Let $\mathcal{C}$ be an ∞ -category and let $f : x \rightarrow y, g : y \rightarrow z$ be morphisms in $\mathcal{C}$. Then, we define the composite of g and f to be a morphism $h : x \rightarrow z$ equipped with a commutative triangle $\sigma : [2] \rightarrow \mathcal{C}$ given as follows:

$$\begin{array}{ccc} & y & \\ f \nearrow & & \searrow g \\ x & \xrightarrow{\quad h \quad} & z \end{array}$$

This is literally the (first part of) the content of [26, Definition 1.5.16, p.48]. This may seem reasonable, until one realizes that there’s no guarantee of uniqueness of this composite. To fix this issue, one needs to add another axiom into the picture; the Segal axiom and this is, of course, best motivated by the properties that one derives in a discussion of complete Segal spaces and we will spend some time discussing these issues later.

In spite of the issues put forth in the remark, we can be sure that the relations above hold in a ∞ -category.

Claim 30.5. Let $\mathcal{C}$ be a ∞ -category and let $f : x \rightarrow y$ be a morphism in $\mathcal{C}$. Then, there exist triangles:

$$\begin{array}{ccc} & x & \\ \mathrm{Id}_x \nearrow & & \searrow f \\ x & \xrightarrow{\quad f \quad} & y \end{array} \qquad \begin{array}{ccc} & y & \\ f \nearrow & & \searrow \mathrm{Id}_y \\ x & \xrightarrow{\quad f \quad} & y \end{array}$$

which commute.

We have a remark about the general behavior of objects and morphisms in a ∞ -category.

Remark 30.17. Our treatment of objects and morphisms primarily follows [27, Chapter 1.2.2, p.13-15]. There are multiple ways of generalizing the concept of an “object of a category” within higher structures. One typically takes inspiration from Homotopy Type Theory and defines the following structures within ∞ -categories:

1. Terms.
2. Contexts.
3. (Absolute) Objects.
4. (Absolute) Morphisms.

All of these are useful for very different reasons that we choose not to discuss here explicitly. More information about this type-theoretic approach to ∞ -categories can be found in [26].

Our next set of axioms concern several important constructions that we can do with ∞ -categories to produce new ones. Indeed, many notions of interest (such as hom-sets in a ∞ -category) are constructed using pullbacks & such, so it seems reasonable to have axioms that tell us how these constructions work. Let us summarize some of these constructions in the next remark; this is [27, Axiom B, p.27-28], modulo some notational modifications.

Remark 30.18. We demand the existence of the following ∞ -categories:

1. An initial ∞ -category $\emptyset$; if $\mathcal{C}$ is a ∞ -category, there is a functor $\iota_{\mathcal{C}} : \emptyset \rightarrow \mathcal{C}$ which is “unique”.

2. A final ∞ -category $*$; if $\mathcal{C}$ is a ∞ -category, there is a functor $p_{\mathcal{C}} : \mathcal{C} \rightarrow *$ which is “unique”.
3. The ∞ -category $\emptyset$ is strictly initial.
4. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories. Then, there exists a ∞ -category $\mathcal{C} \times \mathcal{D}$ and two functors:

$$\pi_{\mathcal{C}} : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C},$$

$$\pi_{\mathcal{D}} : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{D},$$

satisfying the universal property of the product (up to isomorphism).

5. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories. Then, there exists a ∞ -category $\mathcal{C} \amalg \mathcal{D}$ and two functors:

$$\iota_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C} \amalg \mathcal{D},$$

$$\iota_{\mathcal{D}} : \mathcal{D} \rightarrow \mathcal{C} \amalg \mathcal{D},$$

satisfying the universal property of the coproduct^a (up to isomorphism).

6. If $\mathcal{C}, \mathcal{D}, \mathcal{E}$ are ∞ -categories and we have the following diagram:

$$\begin{array}{ccc} & \mathcal{C} & \\ & \downarrow f & \\ \mathcal{D} & \xrightarrow{g} & \mathcal{E} \end{array}$$

Then, we require that this is a “pullback diagram up to homotopy”.

7. Finally, if $\mathcal{C}, \mathcal{D}$ are ∞ -categories, then there is a functor category:

$$\mathrm{Fun}(\mathcal{C}, \mathcal{D}) = \mathcal{D}^{\mathcal{C}}.$$

If $\mathcal{E}$ is another ∞ -category and $f : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D}$ is a functor, then there is a *curried* functor:

$$f_c : \mathcal{E} \rightarrow \mathrm{Fun}(\mathcal{C}, \mathcal{D}).$$

Conversely, every functor $g : \mathcal{E} \rightarrow \mathrm{Fun}(\mathcal{C}, \mathcal{D})$ gives rise to an *uncurried* functor:

$$g^u : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D},$$

satisfying the following requirements:

$$f \simeq (f_c)^u,$$

$$g \simeq (g^u)_c.$$

We may similarly curry/uncurry natural isomorphisms between functors.

In terms of content, none of these constructions should raise any eyebrows. On the other hand, their actual relevance in providing a working theory of ∞ -categories will be consistently questioned & we will attempt to provide reasonable background on their importance throughout our exposition.

^aWe also require that the coproduct is “universal”. This will be explained later.

These constructions are very intuitive and we will present them in their entirety. It’s natural to expect that some of them will require more discussion than others and for this reason, we’ll leave some of them to the next section rather than dealing with all of them over here. With this said, let’s proceed with the first axiom mentioned in the list of [Remark 30.18](#).

Axiom 30.19. Existence of Initial/Final ∞ -Categories

Let $\mathcal{C}$ be a ∞ -category.

1. The category $[0]$ is the final ∞ -category. Indeed, there exists a functor:

$$p_{\mathcal{C}} : \mathcal{C} \rightarrow [0],$$

and for any two functors $f, g : \mathcal{C} \rightarrow [0]$, there is a 2-morphism $f \simeq g$.

2. The empty category $\emptyset$ is the initial ∞ -category. There exists a functor:

$$\iota_{\mathcal{C}} : \emptyset \rightarrow \mathcal{C},$$

and for any two functors $f, g : \emptyset \rightarrow \mathcal{C}$, there is a 2-morphism $f \simeq g$.

Finally, if $f : \mathcal{C} \rightarrow \emptyset$ is a functor, then it is an equivalence^a.

^aSo, we encode this by stating that $\emptyset$ is strictly initial.

Let us introduce a piece of notation, followed by some terminology.

| **Notation 30.20.** When viewed as the final ∞ -category, $[0]$ will be denoted by $*$.

We also have a remark.

| **Remark 30.21.** If $\mathcal{C}$ is a ∞ -category, then we previously said that an object in $\mathcal{C}$ is a functor $f : [0] \rightarrow \mathcal{C}$. By the notation given above, it will also be sometimes convenient to write this as a functor $f : * \rightarrow \mathcal{C}$.

This axiom was standard fare and the only difference from 1-category theory is that we have to account for higher coherences. The next few axioms are far more complicated in appearance but they also have the following general form:

1. One introduces a ∞ -category $\mathcal{C}$, together with additional structure.
2. If $\mathcal{D}$ is another ∞ -category with the same structure, then there is a compatible functor $\mathcal{D} \rightarrow \mathcal{C}$.

3. If we have two functors that are both compatible as in (2), then they are naturally isomorphic¹³. One may quickly even see that Axiom 30.19 also has this general form. With that said, let us postulate the existence of products/coproducts of ∞ -categories. To keep these axioms at bite-sized lengths, we'll start with products.

Axiom 30.22. Existence of Products

Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories. There exists a ∞ -category $\mathcal{C} \times \mathcal{D}$ and two functors:

$$\pi_{\mathcal{C}} : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C},$$

$$\pi_{\mathcal{D}} : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{D},$$

known as the projection functors, satisfying the following universal property; if $f : \mathcal{E} \rightarrow \mathcal{C}$ and $g : \mathcal{E} \rightarrow \mathcal{D}$ are functors, then there exists a functor $(f, g) : \mathcal{E} \rightarrow \mathcal{C} \times \mathcal{D}$ such that the following isomorphisms hold:

$$f \simeq \pi_{\mathcal{C}} \circ (f, g),$$

$$g \simeq \pi_{\mathcal{D}} \circ (f, g).$$

Moreover, suppose we have two functors $h, k : \mathcal{E} \rightarrow \mathcal{C} \times \mathcal{D}$ such that:

$$\alpha : \pi_{\mathcal{C}} \circ h \simeq \pi_{\mathcal{C}} \circ k,$$

$$\beta : \pi_{\mathcal{D}} \circ h \simeq \pi_{\mathcal{D}} \circ k,$$

then there exists a natural isomorphism $(\alpha, \beta) : h \simeq k$

As a sanity check, we should ensure functoriality of taking products of categories through the next construction.

Construction 30.23. Let $\mathcal{C}, \mathcal{C}', \mathcal{D}, \mathcal{D}'$ be ∞ -categories and suppose we have functors:

$$f : \mathcal{C} \rightarrow \mathcal{C}',$$

$$g : \mathcal{D} \rightarrow \mathcal{D}'.$$

Then, we have the following diagram:

$$\begin{array}{ccccc} \mathcal{C} & \xleftarrow{\pi_{\mathcal{C}}} & \mathcal{C} \times \mathcal{D} & \xrightarrow{\pi_{\mathcal{D}}} & \mathcal{D} \\ f \downarrow & & & & \downarrow g \\ \mathcal{C}' & \xleftarrow{\pi_{\mathcal{C}'}} & \mathcal{C}' \times \mathcal{D}' & \xrightarrow{\pi_{\mathcal{D}'}} & \mathcal{D}' \end{array}$$

Observe that $f \circ \pi_{\mathcal{C}}$ and $g \circ \pi_{\mathcal{D}}$ are functors from $\mathcal{C} \times \mathcal{D}$ to $\mathcal{C}'$ and $\mathcal{D}'$ respectively. By the universal property of the product, this induces a functor $f \times g : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C}' \times \mathcal{D}'$ such that the following diagram:

$$\begin{array}{ccccc} \mathcal{C} & \xleftarrow{\pi_{\mathcal{C}}} & \mathcal{C} \times \mathcal{D} & \xrightarrow{\pi_{\mathcal{D}}} & \mathcal{D} \\ f \downarrow & & \downarrow f \times g & & \downarrow g \\ \mathcal{C}' & \xleftarrow{\pi_{\mathcal{C}'}} & \mathcal{C}' \times \mathcal{D}' & \xrightarrow{\pi_{\mathcal{D}'}} & \mathcal{D}' \end{array}$$

commutes up to isomorphism. In particular, the following relations hold:

$$f \circ \pi_{\mathcal{C}} \simeq \pi_{\mathcal{C}'} \circ (f \times g),$$

$$g \circ \pi_{\mathcal{D}} \simeq \pi_{\mathcal{D}'} \circ (f \times g).$$

Now, one needs to ensure that this construction is functorial.

Claim 30.6. The given construction is functorial. This involves showing the following statements:

1. $\text{Id}_{\mathcal{C} \times \mathcal{D}} \simeq \text{Id}_{\mathcal{C}} \times \text{Id}_{\mathcal{D}}$.
2. If $f' : \mathcal{C}' \rightarrow \mathcal{C}''$ and $g' : \mathcal{D}' \rightarrow \mathcal{D}''$ are 1-morphisms between ∞ -categories, then:

$$(f' \times g') \circ (f \times g) \simeq (f' \circ f) \times (g' \circ g).$$

This, of course, is the obvious compatibility with the composition.

Again, the proof is nothing crazy and it's effectively the same as the proof that one would give in 1-category theory, with the caveat that we also have to account for higher coherences. This can be annoying when filling in details but it also means that this general framework is quite conceptual.

Another sanity check would be to ensure that an object in the product ∞ -category is just a pair of objects from the component categories. Let's ensure that this is the case through the next construction.

Construction 30.24. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories, let $x \in \mathcal{C}$ and let $y \in \mathcal{D}$. Now, let $z \in \mathcal{C} \times \mathcal{D}$. Concretely, this means that z is a functor $[0] \rightarrow \mathcal{C} \times \mathcal{D}$. If $\pi_{\mathcal{C}}, \pi_{\mathcal{D}}$ are the projection functors, then we define:

$$x \stackrel{\text{def}}{=} \pi_{\mathcal{C}} \circ z,$$

$$y \stackrel{\text{def}}{=} \pi_{\mathcal{D}} \circ z.$$

Now, we have the following claim.

Claim. There is a 2-morphism $z \simeq (x, y)$.

¹³That is, there is a 2-morphism between them.

The functor $(x, y) : [0] \rightarrow \mathcal{C} \times \mathcal{D}$ is formed by universal property. By definition, we have that:

$$\pi_{\mathcal{C}} \circ z = x \simeq \pi_{\mathcal{C}} \circ (x, y),$$

$$\pi_{\mathcal{D}} \circ z = y \simeq \pi_{\mathcal{D}} \circ (x, y),$$

which implies that there is a 2-morphism $z \simeq (x, y)$.

We expect that products of ∞ -categories are unital, commutative and associative. In the next result, we will prove that this is the case. We will essentially be following the proof of [27, Lemma 1.3.7, Lemma 1.3.8, p.17], though we add more detail which is in line with the actual axiom.

Claim 30.7. Let $\mathcal{C}, \mathcal{D}, \mathcal{E}$ be ∞ -categories. TFSH:

1. $\mathcal{C} \times * \simeq \mathcal{C}$.
2. $\mathcal{C} \times \mathcal{D} \simeq \mathcal{D} \times \mathcal{C}$.
3. $\mathcal{C} \times (\mathcal{D} \times \mathcal{E}) \simeq (\mathcal{C} \times \mathcal{D}) \times \mathcal{E}$.

We have a general remark about these results that may help clarify why the situation is significantly more tricky.

Remark 30.25. So, the key problem is that all of the results above are trivial in 1-category theory because we can *explicitly construct equivalences* and we can, therefore, check that our specific choices work. Now, we have to employ the universal property as stipulated in Axiom 30.22. In fact, there's a worse problem and this is illustrated by statement (2) of **Claim 30.7**. Indeed, in addition to the ∞ -categories $\mathcal{C} \times \mathcal{D}$ and $\mathcal{D} \times \mathcal{C}$, we also have to account for 4 different projection functors & the relationships between these are not clear at all from the start, unlike the situation in 1-category theory. So, on the surface, these axioms *look* as if they just slightly strengthen different aspects of 1-category theory but actually, they are strictly more difficult to handle.

One can cook up a similar axiom for coproducts of ∞ -categories.

Axiom 30.26. Existence of Coproducts

Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories. There exists a ∞ -category $\mathcal{C} \amalg \mathcal{D}$ and two functors:

$$\iota_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C} \amalg \mathcal{D},$$

$$\iota_{\mathcal{D}} : \mathcal{D} \rightarrow \mathcal{C} \amalg \mathcal{D}$$

known as the inclusion functors, satisfying the following universal property; if $f : \mathcal{C} \rightarrow \mathcal{E}$ and $g : \mathcal{D} \rightarrow \mathcal{E}$ are functors, then there exists a functor $f \amalg g : \mathcal{C} \amalg \mathcal{D} \rightarrow \mathcal{E}$ such that the following isomorphisms hold:

$$f \simeq (f \amalg g) \circ \iota_{\mathcal{C}},$$

$$g \simeq (f \amalg g) \circ \iota_{\mathcal{D}}.$$

Moreover, suppose we have two functors $h, k : \mathcal{C} \amalg \mathcal{D} \rightarrow \mathcal{E}$ such that:

$$\alpha : h \circ \iota_{\mathcal{C}} \simeq k \circ \iota_{\mathcal{C}},$$

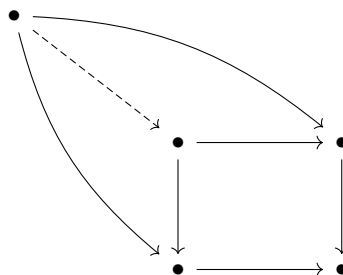
$$\beta : h \circ \iota_{\mathcal{D}} \simeq k \circ \iota_{\mathcal{D}},$$

then there exists a natural isomorphism $\alpha \amalg \beta : h \simeq k$.

We have a remark at this stage.

Remark 30.27. It's also possible to run through exactly the same kinds of arguments we did with products and obtain the “same” results. Morally, forming products and coproducts of categories should be dual to each other. However, you will note that we haven't even defined the opposite of a category, a key tool used to produce snappy proofs of dual statements in 1-category theory. Nevertheless, one can simply work informally with the overarching concept of duality and reverse the arrows in previous proofs to obtain similar results for coproducts of categories.

Okay, now we come to construction 6 as stipulated in **Remark 30.18**; this is the one involving pullbacks of ∞ -categories. We'll do what we need to do in order to state the axiom and, then, provide some intuition for why it happens to be an important construction. In [27, Axiom B.5, p.18-19], this axiom is certainly stated explicitly but some details are left out because they aren't needed later in the notes. However, having a precise formulation of the axiom will help drive home the fact that higher category theory does look a lot like 1-category theory. As a consequence, we'll follow the treatment as given in [26, Chapter 1.3, p.27-28]. The key is that the following diagram



can be made to make sense if the nodes are ∞ -categories and the arrows are functors. Now, the axiom itself will be easier to state if we introduce some terminology that summarizes parts of the diagram above.

Terminology 30.28. Let $\mathcal{C}, \mathcal{D}, \mathcal{E}$ be ∞ -categories, consider a diagram of functors:

$$\mathcal{C} \xrightarrow{f} \mathcal{E} \xleftarrow{g} \mathcal{D}$$

and let T be a ∞ -category. A pullback cone on T is a triple $c = (t_{\mathcal{C}}, t_{\mathcal{D}}, \tau)$ consisting of two functors:

$$\begin{aligned} t_{\mathcal{C}} &: T \rightarrow \mathcal{C}, \\ t_{\mathcal{D}} &: T \rightarrow \mathcal{D}, \end{aligned}$$

such that the following diagram:

$$\begin{array}{ccc} T & \xrightarrow{t_{\mathcal{C}}} & \mathcal{C} \\ t_{\mathcal{D}} \downarrow & & \downarrow f \\ \mathcal{D} & \xrightarrow{g} & \mathcal{E} \end{array}$$

commutes up to a natural isomorphism:

$$\tau : f \circ t_{\mathcal{C}} \simeq g \circ t_{\mathcal{D}}.$$

T is the zenith of the cone and $t_{\mathcal{C}}, t_{\mathcal{D}}$ are the cone threads.

1. If $c' = (t'_{\mathcal{C}}, t'_{\mathcal{D}}, \tau')$ is another cone diagram over T , then an isomorphism of cone diagrams $c \simeq c'$ is a triple:

$$\Phi_{c,c'} : (\phi_{\mathcal{C}}, \phi_{\mathcal{D}}, H),$$

with the following data:

- (a) A natural isomorphism $\phi_{\mathcal{C}} : t_{\mathcal{C}} \simeq t'_{\mathcal{C}}$.
- (b) A natural isomorphism $\phi_{\mathcal{D}} : t_{\mathcal{D}} \simeq t'_{\mathcal{D}}$.
- (c) A 3-morphism $H : \tau \simeq \tau'$.

We will refer to this as a cone isomorphism.

2. If $\Psi_{c,c'} = (\psi_{\mathcal{C}}, \psi_{\mathcal{D}}, K)$ is another isomorphism of cone diagrams, then it is isomorphic to $\Phi_{c,c'}$ iff there are isomorphisms $\phi_{\mathcal{C}} \simeq \psi_{\mathcal{C}}$ and $\phi_{\mathcal{D}} \simeq \psi_{\mathcal{D}}$ so that H becomes isomorphic to K .

When we use the words “natural isomorphism”, we obviously are just referring to morphisms of appropriate coherence.

We have a remark about this bit of terminology.

Remark 30.29. Suppose we have the following pullback cone:

$$\begin{array}{ccc} T & \xrightarrow{t_{\mathcal{C}}} & \mathcal{C} \\ t_{\mathcal{D}} \downarrow & & \downarrow f \\ \mathcal{D} & \xrightarrow{g} & \mathcal{E} \end{array}$$

of ∞ -categories and functors & let $\phi : S \rightarrow T$ be a functor between ∞ -categories. With the axioms we’ve introduced so far, we can form the following nice diagram:

$$\begin{array}{ccccc} & & & & \\ & & & & \\ & & & & \\ S & \xrightarrow{t_{\mathcal{C}} \circ \phi} & \mathcal{C} & & \\ & \searrow \phi & \downarrow f & & \\ & T & \xrightarrow{t_{\mathcal{C}}} & \mathcal{C} & \\ & \downarrow t_{\mathcal{D}} & & & \\ & \mathcal{D} & \xrightarrow{g} & \mathcal{E} & \\ & \nearrow t_{\mathcal{D}} \circ \phi & & & \end{array}$$

where the double arrows represent 2-morphisms. Now, we would like to say that the outer square gives us the data of a pullback cone. The problem comes about when we try to specify a 2-morphism as part of the data of a pullback cone. If $(t_{\mathcal{C}}, t_{\mathcal{D}}, \tau)$ is the pullback cone we started with, then it seems like a good candidate is the following 2-morphism:

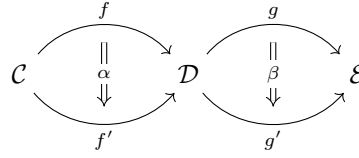
$$\tau \star \text{Id}_{\phi},$$

where this is to be interpreted as a “horizontal composition”. Well, such a notion does exist in 1-category theory and in order to have this construction work, we should introduce it as well. Moreover, we have to require that it is compatible with our previously defined notion of vertical composition.

So, in fact, before we can postulate the existence of pullbacks, we need axioms that deal with horizontal compositions.

Axiom 30.30. Existence of Vertical Compositions

Suppose we have the following diagram of ∞ -categories, 1-morphisms and 2-morphisms:



Then, there is a 2-morphism $\beta * \alpha : g \circ f \simeq g' \circ f'$. Moreover, the following properties hold:

1. We have that $\text{Id}_g * \text{Id}_f \simeq \text{Id}_{g \circ f}$.
2. Suppose we also have two functors $f'' : \mathcal{C} \rightarrow \mathcal{D}$ and $g'' : \mathcal{D} \rightarrow \mathcal{E}$ & two more 2-morphisms:

$$\begin{aligned}\alpha' : f' &\simeq f'', \\ \beta' : g' &\simeq g''.\end{aligned}$$

Then, we have a 3-morphism:

$$(\beta' \circ \beta) * (\alpha' \circ \alpha) \simeq (\beta' * \alpha') \circ (\beta * \alpha).$$

This is the compatibility condition between vertical compositions and horizontal compositions.

We refer to the 2-morphism $\beta * \alpha$ as the Godement product of these 2-morphisms. One also inductively defines Godement products of n -morphisms & imposes compatibility of higher coherences in the analogous way.

So, this axiom just postulates existence of such a composition & says something about its compatibility with horizontal compositions as introduced in Axiom 30.1. Such an axiom is inspired, obviously, by the properties that horizontal compositions of natural transformations have with respect to vertical compositions in 1-category theory. Next, let's introduce an axiom which tells us that Godement products are compatible with unitality & associativity of composites. This is supposed to ensure that this new product is compatible with the data introduced in Axiom 30.5.

Axiom 30.31. Unital-Associative Compatibility

Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories.

1. If $f, f' : \mathcal{C} \rightarrow \mathcal{D}$ are functors, then there exist the following diagrams:

$$\begin{array}{ccc} \text{Id}_{\mathcal{D}} \circ f & \xrightarrow{\text{Id}_{\text{Id}_{\mathcal{D}}} * \alpha} & \text{Id}_{\mathcal{D}} \circ f' \\ \downarrow \simeq & & \downarrow \simeq \\ f & \xrightarrow{\alpha} & f' \end{array} \quad \begin{array}{ccc} f \circ \text{Id}_{\mathcal{C}} & \xrightarrow{\alpha * \text{Id}_{\text{Id}_{\mathcal{C}}}} & f' \circ \text{Id}_{\mathcal{C}} \\ \downarrow \simeq & & \downarrow \simeq \\ f & \xrightarrow{\alpha} & f' \end{array}$$

which commute up to isomorphism.

2. If we have 2-morphisms $\alpha : f \simeq f', \beta : g \simeq g', \gamma : h \simeq h'$, then the following diagram:

$$\begin{array}{ccc} (h \circ g) \circ f & \xrightarrow{(\gamma * \beta) * \alpha} & (h' \circ g') \circ f' \\ \downarrow \simeq & & \downarrow \simeq \\ h \circ (g \circ f) & \xrightarrow{\gamma * (\beta * \alpha)} & h' \circ (g' \circ f') \end{array}$$

commutes up to isomorphism.

Similar conditions for higher coherences can be defined.

These axioms were taken from [26, Axiom A.1', Axiom A.2', p.18-19]; they were introduced much earlier in the text, alongside Axioms 30.1 & 30.5. One can view these axioms as extensions of the aforementioned axioms & they are presented in this way within the reference. We do not view this as a useful way of thinking about them for the simple reason that it is not clear, a priori, that the notion of horizontal composition of n -morphisms would be a useful one in higher category theory. Armed with all of this data, we can finally deal with the issue stated in Remark 30.29.

Claim 30.8. Suppose we have the following pullback cone:

$$\begin{array}{ccc} T & \xrightarrow{t_{\mathcal{C}}} & \mathcal{C} \\ t_{\mathcal{D}} \downarrow & & \downarrow f \\ \mathcal{D} & \xrightarrow{g} & \mathcal{E} \end{array}$$

of ∞ -categories and functors & let $\phi : S \rightarrow T$ be a functor between ∞ -categories. The triple:

$$\phi^*(c) \stackrel{\text{def}}{=} (t_{\mathcal{C}} \circ \phi, t_{\mathcal{D}} \circ \phi, \tau * \text{Id}_{\phi}),$$

forms a pullback cone for f and g , where $(t_{\mathcal{C}}, t_{\mathcal{D}}, \tau)$ is the pullback cone given.

So, one sees that even to state a reasonable axiom that deals with pullbacks of ∞ -categories, some more work is definitely required if one also wishes to have the nice properties of pullbacks. Since pullbacks are such an important facet of ∞ -categorical constructions, it only seems reasonable that we treat them with that level of importance and dedicate an

entire section which partly spends some time discussing them properly.

30.2 Pullbacks, Functor Categories & A Futuristic View

In any event, let us proceed with the axiom that postulates their existence.

Axiom 30.32. Existence of Pullbacks

Let $\mathcal{C}, \mathcal{D}, \mathcal{E}$ be ∞ -categories and consider a diagram of functors:

$$\mathcal{C} \xrightarrow{f} \mathcal{E} \xleftarrow{g} \mathcal{D}.$$

Then, the pullback of f and g is specified by the following data:

1. A ∞ -category $\mathcal{C} \times_{\mathcal{E}} \mathcal{D}$.
2. A cone diagram $c_{\text{univ}}(\pi_{\mathcal{C}}, \pi_{\mathcal{D}}, \tau_{\text{univ}})$:

$$\begin{array}{ccc} \mathcal{C} \times_{\mathcal{E}} \mathcal{D} & \xrightarrow{\pi_{\mathcal{C}}} & \mathcal{C} \\ \pi_{\mathcal{D}} \downarrow & & \downarrow f \\ \mathcal{D} & \xrightarrow{g} & \mathcal{E} \end{array}$$

which is universal.

The cone diagram above is universal in the following sense:

1. If T is a ∞ -category and $c = (t_{\mathcal{C}}, t_{\mathcal{D}}, T)$ is a cone diagram on T , then there exists a functor:

$$(t_{\mathcal{C}}, t_{\mathcal{D}}) : T \rightarrow \mathcal{C} \times_{\mathcal{E}} \mathcal{D},$$

together with an isomorphism $c \simeq (t_{\mathcal{C}}, t_{\mathcal{D}})^*(c_{\text{univ}})$.

2. Given two functors $s, t : T \rightarrow \mathcal{C} \times_{\mathcal{E}} \mathcal{D}$ equipped with an isomorphism of pullback cones:

$$s^*(c_{\text{univ}}) \simeq t^*(c_{\text{univ}}),$$

there exists a natural isomorphism $s \simeq t$ that induces this isomorphism of pullback cones.

Typically, one abuses notation and refers to $\mathcal{C} \times_{\mathcal{E}} \mathcal{D}$ as the pullback without reference to the two projection functors $\pi_{\mathcal{C}}$ and $\pi_{\mathcal{D}}$. Moreover, one often refers to the pullback as the fiber product.

We have a remark about this axiom.

Remark 30.33. There does not seem to be a direct analogous result which justifies this axiom in [64]. However, I believe that this specific definition essentially encodes the idea that we are working with homotopy pullbacks, as opposed to genuine pullbacks. In fact, Lurie makes a distinction between categorical pullbacks and homotopy pullbacks in Kerodon before giving several (useful) conditions which indicate when these are the same. This leads me to believe that the axiom, as formulated above, was meant to reflect the situations where these are the same. Unsurprisingly, these would be the ones of greatest interest.

Morally, the fiber product of two categories is “like” the product. With the pullback, we can define new ∞ -categories from given ones. For example, it becomes possible to define the fiber of a functor over an object.

Construction 30.34. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories, let $f : \mathcal{C} \rightarrow \mathcal{D}$ be a functor and let $y \in \mathcal{D}$. We define an ∞ -category:

$$\mathcal{C}_y,$$

by the following pullback diagram:

$$\begin{array}{ccc} \mathcal{C}_y & \longrightarrow & \mathcal{C} \\ \downarrow & & \downarrow f \\ * & \xrightarrow{x} & \mathcal{D} \end{array}$$

of ∞ -categories.

Of course, we can also define pullback squares of ∞ -categories.

Terminology 30.35. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories. A commutative square:

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{s} & \mathcal{C} \\ t \downarrow & & \downarrow f \\ \mathcal{B} & \xrightarrow{g} & \mathcal{D} \end{array}$$

of functors between ∞ -categories is a pullback square iff the induced functor:

$$s : \mathcal{A} \rightarrow \mathcal{B} \times_{\mathcal{D}} \mathcal{C},$$

is an equivalence of ∞ -categories.

We have a remark about this bit of terminology.

Remark 30.36. In the next lecture, we will define commutative squares within ∞ -categories; the definition will be straightforward and exactly what you might expect. The point is that we would like for commutative squares of functors between ∞ -categories to be an instantiation of the definition of a commutative square. To do this, we will need to invent an ∞ -category of ∞ -categories, in which case our definition would specialize^a.

^aThere is a slight non-trivial point of showing that mapping categories are anima but this will be proved later.

To demonstrate that we can do something with this theory, let us prove a non-trivial result using it. For the moment, this will just be a formal result of interest in the theory but this is the best we can do right now.

Proposition 30.37. Pasting Lemma (Pullback Squares)

Consider a commutative diagram:

$$\begin{array}{ccccc} C_1 & \xrightarrow{g_1} & C_2 & \xrightarrow{g_2} & C_3 \\ f_1 \downarrow & & f_2 \downarrow & & f_3 \downarrow \\ D_1 & \xrightarrow{h_1} & D_2 & \xrightarrow{h_2} & D_3 \end{array}$$

Then, TFSH:

1. There is a preferred equivalence $D_1 \times_{D_3} C_3 \rightarrow D_1 \times_{D_2} (D_2 \times_{D_3} C_3)$.
2. Suppose that the right-hand square is a pullback square. TFAE:
 - (a) The left-hand square is a pullback square.
 - (b) The outer rectangle is a pullback square.

In particular, we can paste together pullbacks to get pullbacks.

Proof. We'll prove each statement one-by-one.

1. Obviously, we define our candidate by projecting to D_1 first, then projecting to $D_2 \times_{D_3} C_3$ and then gluing the results. Consider the following diagram:

$$\begin{array}{ccccc} C_1 & \xrightarrow{g_1} & C_2 & \xrightarrow{g_2} & C_3 \\ & \searrow & \downarrow f_2 & \nearrow & \downarrow f_3 \\ & D_1 \times_{D_3} C_3 & & & \\ f_1 \downarrow & \nearrow & & & \\ D_1 & \xrightarrow{h_1} & D_2 & \xrightarrow{h_2} & D_3 \end{array}$$

The diagram shows a commutative diagram with a central node $D_1 \times_{D_3} C_3$. Arrows from C_1 and C_2 point to this node. A red arrow points from $D_1 \times_{D_3} C_3$ to D_1 . A green arrow points from $D_1 \times_{D_3} C_3$ to D_2 . The bottom row $D_1 \rightarrow D_2 \rightarrow D_3$ is also shown.

The green and red arrow to D_1 are the same arrows, we've just draw two copies for clarity. So, the morphism from $D_1 \times_{D_3} C_3 \rightarrow D_1$ is just the red arrow. On the other hand, the morphism from $D_1 \times_{D_3} C_3 \rightarrow D_2 \times_{D_3} C_3$ can be formed by gluing together the green arrows and the brown arrow. If we glue everything together, we get our preferred candidate and the goal is to construct an inverse. Consider the following diagram:

$$\begin{array}{ccccc} C_1 & \xrightarrow{g_1} & C_2 & \xrightarrow{g_2} & C_3 \\ f_1 \downarrow & & \downarrow f_2 & & \downarrow f_3 \\ D_1 & \xrightarrow{h_1} & D_2 & \xrightarrow{h_2} & D_3 \end{array}$$

The diagram shows a commutative diagram with a central node $D_1 \times_{D_2} \times (D_2 \times_{D_2} \times C_3)$. Arrows from C_1 and C_2 point to this node. A red arrow points from $D_1 \times_{D_2} \times (D_2 \times_{D_2} \times C_3)$ to $D_2 \times_{D_3} \times C_3$. A green arrow points from $D_1 \times_{D_2} \times (D_2 \times_{D_2} \times C_3)$ to D_1 . The bottom row $D_1 \rightarrow D_2 \rightarrow D_3$ is also shown.

Again, we form the candidate inverse by projecting to D_1 and C_3 . The diagram above clarifies exactly how we should define these projections. Now, checking that these work is a computation which uses the commutativity of these diagrams. But one sees, after unwrapping everything, that this computation actually leads us to a tautology.

2. We consider the following diagram:

$$\begin{array}{ccc} C_1 & \xrightarrow{\quad} & D_1 \times_{D_2} C_2 \\ \downarrow & & \downarrow \\ D_1 \times_{D_2} C_3 & \xrightarrow{\quad \simeq \quad} & D_1 \times_{D_2} (D_2 \times_{D_3} C_3) \end{array}$$

This commutative diagram is formed by writing out pullbacks in our two-square diagram above and drawing the projections from them to the vertices of the diagram. Since the right-hand square is a pullback, we see that the map $D_1 \times_{D_2} C_2 \rightarrow D_1 \times_{D_2} (D_2 \times_{D_3} C_3)$ is an equivalence since it is the gluing of Id_{D_1} with the equivalence

$\mathcal{C}_2 \rightarrow \mathcal{D}_2 \times_{\mathcal{D}_3} \mathcal{C}_3$. Now, we prove our desired equivalence:

- (1) $\Rightarrow$ (2): If the left-hand square is a pullback, then the functor $\mathcal{C}_1 \rightarrow \mathcal{D}_1 \times_{\mathcal{D}_2} \mathcal{C}_2$ is an equivalence. But this means, by the 2-out-of-3 property, that the functor $\mathcal{C}_1 \rightarrow \mathcal{D}_1 \times_{\mathcal{D}_3} \mathcal{C}_3$ is an equivalence. On the other hand, this means exactly that the whole rectangle is a pullback square.
- (2) $\Rightarrow$ (1): If the whole rectangle is a pullback square, then $\mathcal{C}_1 \rightarrow \mathcal{D}_1 \times_{\mathcal{D}_3} \mathcal{C}_3$ is an equivalence. So, by the 2-out-of-3 property, we see that $\mathcal{C}_1 \rightarrow \mathcal{D}_1 \times_{\mathcal{D}_2} \mathcal{C}_2$ is an equivalence. So, the left-hand square is a pullback.

This completes the proof. ■

At this point, we have a remark.

Remark 30.38. In [26], there is an additional axiom introduced which breaks the symmetry between products and coproducts of ∞ -categories; it is the so-called Universality Axiom for coproducts and it effectively tells us that gluings of two functors are compatible with pullbacks. However, this axiom seems to be mainly put in place to allow for technical arguments involve cocartesian fibrations to work; these are presented much later in that text. So, since we are not going to be doing anything with the axiom right now, we're not going to present it now.

So, many of the same results we established before about products also do hold; we leave it to the reader to formulate these and provide proofs¹⁴. In specific models discussed previously, we've seen that pullbacks are an important mechanism for constructing categorical notions of interest and this is no different in the synthetic world. This leads us to the following piece of intuition we provide that will be made precise in a bit.

Intuition 30.39. Let $\mathcal{C}$ be a ∞ -category and let $x, y \in \mathcal{C}$. Let's assume^a that there exists a ∞ -category $\text{Fun}([1], \mathcal{C})$ consisting of all functors from $[1]$ to $\mathcal{C}$. We want to define the hom-sets^b:

$$\mathcal{C}(x, y),$$

consisting of all morphisms from x to y . Consider the following diagram:

$$\begin{array}{ccc} x & \simeq & x' \\ f \downarrow & & \downarrow g \\ y & \simeq & y' \end{array}$$

So, even though g would be viewed as a different morphism in a 1-category, we want to shift away from that perspective and, instead, also view g as a morphism from x to y . The only way in which this makes sense is if $\mathcal{C}(x, y)$ is constructed as a category. In fact, it's constructed as pullback of the following diagram:

$$\begin{array}{ccc} \mathcal{C}(x, y) & \longrightarrow & \text{Fun}([1], \mathcal{C}) \\ \downarrow & & \downarrow (\text{ev}_0, \text{ev}_1) \\ * & \xrightarrow{(x, y)} & \mathcal{C} \times \mathcal{C} \end{array}$$

The functors ev_i are typically constructed alongside functor categories and since we haven't given an axiom dealing with them, you'll just have to be content with their intuitive meaning for now. However, it's easily seen that the above is the right definition. Indeed, the diagram above commutes up to isomorphism, so it's easy to see that we're restricting ourselves to morphisms in $\text{Fun}([1], \mathcal{C})$ whose domains are isomorphic to x and whose codomains are isomorphic to y . So, we see that the pullback of ∞ -categories gives us the right construct.

^aThis motivates us to introduce an axiom postulating the existence of functor categories; see Axiom 30.41.

^bIn many resources on ∞ -categories, these are called hom-*anima* for reasons that will be explained much later.

We have a general remark about the last part of the intuition above.

Remark 30.40. The intuition above captures part of the strategy of the synthetic approach. Since we're only identifying objects of interest in ∞ -category theory up to isomorphism, the key is to create a whole category in which they reside and then identify an object in that category in the exact way that we've been doing. So, for example, a morphism from $x \rightarrow y$ should, morally, be an object in $\mathcal{C}(x, y)$ and this is unique up to isomorphism because we've identified the objects in $\mathcal{C}(x, y)$ up to isomorphism by our work above. Of course, one might need to impose more axioms in order to identify various notions of isomorphism and this involves introducing more pullback diagrams. This is the benefit of the synthetic approach; it relies heavily on individual models for motivation for the axioms but once one has them, it is possible to work without any knowledge of the models themselves.

Intuition 30.39 gives us an actual reason to introduce an axiom which deals with functor categories.

Axiom 30.41. Existence of Functor Categories

Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories.

1. There exists an ∞ -category $\text{Fun}(\mathcal{C}, \mathcal{D})$ known as the functor category from $\mathcal{C}$ to $\mathcal{D}$.
2. $\text{Fun}(\mathcal{C}, \mathcal{D})$ is equipped with an evaluation functor $\text{ev} : \text{Fun}(\mathcal{C}, \mathcal{D}) \times \mathcal{C} \rightarrow \mathcal{D}$.

¹⁴This is explicitly done, however, on [26, p.29].

3. If $\mathcal{E}$ is a ∞ -category and $f : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D}$ is a functor, then there is a *curried* functor:

$$f_c : \mathcal{E} \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D}).$$

Moreover, it comes equipped with the following diagram:

$$\begin{array}{ccc} \mathcal{E} \times \mathcal{C} & \xrightarrow{f_c \times \text{Id}_{\mathcal{C}}} & \text{Fun}(\mathcal{C}, \mathcal{D}) \times \mathcal{C} \\ & \searrow f & \downarrow \text{ev} \\ & & \mathcal{D} \end{array}$$

which commutes.

Moreover, if we have two functors $g, h : \mathcal{E} \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$ and a 2-morphism:

$$\alpha : \text{ev} \circ (g \times \text{Id}_{\mathcal{C}}) \simeq \text{ev} \circ (h \times \text{Id}_{\mathcal{C}}),$$

we obtain a 2-morphism $\beta : g \simeq h$ such that $\alpha \simeq \text{ev} * (\beta \times \text{Id}_{\mathcal{C}})$.

Original formulations of this axiom included uncurrying as a part of the structure. However, this turns out to be unnecessary as we can construct uncurrying from the given data easily

Construction 30.42. Let $\mathcal{C}, \mathcal{E}, \mathcal{D}$ be ∞ -categories and suppose we have a functor:

$$g : \mathcal{E} \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D}).$$

Then, we define g^u by the following composition:

$$\mathcal{E} \times \mathcal{C} \xrightarrow{g \times \text{Id}_{\mathcal{C}}} \text{Fun}(\mathcal{C}, \mathcal{D}) \times \mathcal{C} \xrightarrow{\text{ev}} \mathcal{D}$$

This is known as the *uncurried* functor associated to g .

With uncurrying available to us, we can provide further intuition for Axiom 30.41.

Intuition 30.43. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories.

1. So, in particular, $\text{Fun}(\mathcal{C}, \mathcal{D})$ does have the functors as its objects because an object in it is just a functor $x : [0] \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$. By uncurrying, this is a functor $x : [0] \times \mathcal{C} \rightarrow \mathcal{D}$. Since we have that:

$$\mathcal{C} \simeq [0] \times \mathcal{C},$$

it follows that we just have a functor $x : \mathcal{C} \rightarrow \mathcal{D}$.

2. Morally, the morphisms in such a ∞ -category should be the “natural transformations” between functors. Well, these are just functors $f : [1] \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$. By uncurrying, we simply get functors $f : [1] \times \mathcal{C} \rightarrow \mathcal{D}$ and these are precisely just natural transformations in 1-category theory.

So, in other words, the axiom we’ve stipulated is perfectly reasonable.

It is not too difficult to prove that currying and uncurrying determine a one-to-one correspondence.

Proposition 30.44

Let $\mathcal{C}, \mathcal{D}, \mathcal{E}$ be ∞ -categories and let $f : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D}$ & $g : \mathcal{E} \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D})$ be functors.

1. There is a 2-morphism $f \simeq (f_c)^u$.
2. There is a 2-morphism $g \simeq (g^u)_c$.

Proof. We’ll prove each statement one-by-one.

1. Observe that:

$$(f_c)^u \simeq \text{ev} \circ (f_c \times \text{Id}_{\mathcal{C}}) \simeq f,$$

as was desired.

2. It suffices to construct a natural isomorphism after uncurrying. But then:

$$g^u \simeq ((g^u)_c)^u,$$

by statement (1) so we’re done. ■

As we have alluded to before, functor categories facilitate the construction of other categories of interest. For instance, we are now in a position to construct the arrow category of an ∞ -category, the hom-categories of morphisms between objects in an ∞ -category and natural transformations.

Construction 30.45. Let $\mathcal{C}$ be an ∞ -category. We define:

$$\text{Arr}(\mathcal{C}) \stackrel{\text{def}}{=} \text{Fun}([1], \mathcal{C}),$$

as the arrow category of $\mathcal{C}$. **Intuition 30.43** tells us that this does capture a category which contains the morphisms of $\mathcal{C}$ as objects, but it’s useful for us to also gain an understanding of the morphisms in the category. Well, if f is a morphism in $\text{Arr}(\mathcal{C})$, then f is a functor $[1] \rightarrow \text{Fun}([1], \mathcal{C})$. By currying, f is a functor from $[1] \times [1] \rightarrow \mathcal{C}$, which encodes the data of a certain commutative square, as is the case for the usual 1-categorical arrow category.

Construction 30.46. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories and let $x \in \mathcal{C}$. We define:

$$\text{ev}_x : \text{Fun}(\mathcal{C}, \mathcal{D}) \simeq \text{Fun}(\mathcal{C}, \mathcal{D}) \times * \xrightarrow{\text{Id} \times x} \text{Fun}(\mathcal{C}, \mathcal{D}) \times \mathcal{C} \xrightarrow{\text{ev}} \mathcal{D},$$

which makes sense.

Construction 30.47. Let $\mathcal{C}, \mathcal{D}, \mathcal{D}'$ be ∞ -categories and let $g \in \text{Fun}(\mathcal{D}, \mathcal{D}')$. We want to define a functor:

$$g \circ - : \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{D}').$$

The gut reaction is to try and define this in the obvious way that one would in 1-category theory. But one remembers that functors in ∞ -category theory are **not** specified by defining how they behave on objects & arrows. Indeed, the trick is to use the data of the axiom and the shift of perspective required involves us using the fact that specifying a functor above would be the same as specifying a functor:

$$* \rightarrow \text{Fun}(\text{Fun}(\mathcal{C}, \mathcal{D}) \times \mathcal{C}, \mathcal{D}').$$

In that case, there's only one conceivable option, namely given by the following diagram:

$$\text{Fun}(\mathcal{C}, \mathcal{D}) \times \mathcal{C} \xrightarrow{\text{ev}} \mathcal{D} \xrightarrow{g} \mathcal{D}'.$$

Of course, this simply specifies an object in a specific functor category so we simply need to curry it:

$$g \circ - \stackrel{\text{def}}{=} (g \circ \text{ev})_c.$$

In a similar way, we can define a functor:

$$- \circ g : \text{Fun}(\mathcal{D}', \mathcal{C}) \rightarrow \text{Fun}(\mathcal{D}, \mathcal{C}),$$

which models precomposition. Finally, it is possible to directly define a bifunctor:

$$- \circ - : \text{Fun}(\mathcal{D}, \mathcal{E}) \times \text{Fun}(\mathcal{C}, \mathcal{D}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{E}),$$

where $\mathcal{C}, \mathcal{E}, \mathcal{D}$ are ∞ -categories; it is interesting to consider how one can define this functor.

Incidentally, one can also use functor categories to define pushouts.

Construction 30.48. Consider a commutative square:

$$\begin{array}{ccc} \mathcal{C}' & \xrightarrow{u} & \mathcal{C} \\ f' \downarrow & & \downarrow f \\ \mathcal{D}' & \xrightarrow{v} & \mathcal{D} \end{array}$$

of ∞ -categories and functors. This is a pushout square^a iff for every ∞ -category $\mathcal{E}$, the induced square:

$$\begin{array}{ccc} \text{Fun}(\mathcal{D}, \mathcal{E}) & \xrightarrow{- \circ v} & \text{Fun}(\mathcal{D}', \mathcal{E}) \\ - \circ f \downarrow & & \downarrow - \circ f' \\ \text{Fun}(\mathcal{C}, \mathcal{E}) & \xrightarrow{- \circ u} & \text{Fun}(\mathcal{C}', \mathcal{E}) \end{array}$$

is a pullback of ∞ -categories.

^aOr a cocartesian square.

With this construction, we can state the axiom dealing with pushouts.

Axiom 30.49. Pushouts of ∞ -Categories

If $\mathcal{C}, \mathcal{D}, \mathcal{E}$ are ∞ -categories and we have a solid diagram of arrows:

$$\begin{array}{ccc} \mathcal{E} & \xrightarrow{f} & \mathcal{C} \\ g \downarrow & & \downarrow \iota_{\mathcal{C}} \\ \mathcal{D} & \dashrightarrow_{\iota_{\mathcal{D}}} & \mathcal{C} \amalg_{\mathcal{E}} \mathcal{D} \end{array}$$

between ∞ -categories, then there exists an ∞ -category $\mathcal{C} \amalg_{\mathcal{E}} \mathcal{D}$ and two inclusion functors $\iota_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C} \amalg_{\mathcal{E}} \mathcal{D}$ and $\iota_{\mathcal{D}} : \mathcal{D} \rightarrow \mathcal{C} \amalg_{\mathcal{E}} \mathcal{D}$ such that the square above is cocartesian. We refer to $\mathcal{C} \amalg_{\mathcal{E}} \mathcal{D}$ as the pushout of $\mathcal{C}$ and $\mathcal{D}$ along $\mathcal{E}$.

With the universal property of functor categories, we can show that functor categories are compatible with the initial/final ∞ -categories, (co)products of ∞ -categories and pullbacks. We summarize these properties below.

Proposition 30.50

Let I be a set, let $\mathcal{T}, \mathcal{C}, \mathcal{D}, \mathcal{E}$ and $(\mathcal{C}_i)_{i \in I}$ be ∞ -categories and let $f : \mathcal{C} \rightarrow \mathcal{E}, g : \mathcal{D} \rightarrow \mathcal{E}$ be functors. TFSH:

1. The functor $p_{\text{Fun}(\mathcal{T}, *)} : \text{Fun}(\mathcal{T}, *) \rightarrow *$ is an equivalence.
2. The functor:

$$(\pi_{\mathcal{C}} \circ -, \pi_{\mathcal{D}} \circ -) : \text{Fun}(\mathcal{T}, \mathcal{C} \times \mathcal{D}) \rightarrow \text{Fun}(\mathcal{T}, \mathcal{C}) \times \text{Fun}(\mathcal{T}, \mathcal{D}),$$

is an equivalence.

3. The functor:

$$(\pi_i \circ -)_i : \text{Fun} \left(\mathcal{T}, \prod_{i \in I} \mathcal{C}_i \right) \rightarrow \text{Fun}(\mathcal{T}, \mathcal{C}_i),$$

is an equivalence.

4. The functor:

$$(\pi_{\mathcal{C}} \circ -, \pi_{\mathcal{D}} \circ -) : \text{Fun}(\mathcal{T}, \mathcal{C} \times_{\mathcal{E}} \mathcal{D}) \rightarrow \text{Fun}(\mathcal{T}, \mathcal{C}) \times_{\text{Fun}(\mathcal{T}, \mathcal{E})} \text{Fun}(\mathcal{T}, \mathcal{D}),$$

induced by the isomorphism $(f \circ -) \circ (\pi_{\mathcal{C}} \circ -) \simeq (g \circ -) \circ (\pi_{\mathcal{D}} \circ -)$ is an equivalence.

5. The functor $\text{Fun}(\emptyset, \mathcal{T}) \rightarrow *$.

6. The functor:

$$(- \circ \iota_{\mathcal{C}}, - \circ \iota_{\mathcal{D}}) : \text{Fun}(\mathcal{C} \amalg \mathcal{D}, \mathcal{T}) \rightarrow \text{Fun}(\mathcal{C}, \mathcal{T}) \times \text{Fun}(\mathcal{D}, \mathcal{T}),$$

is an equivalence.

7. The functor:

$$(- \circ \iota_i)_i : \text{Fun}(\coprod_{i \in I} \mathcal{C}_i, \mathcal{T}) \rightarrow \prod_{i \in I} \text{Fun}(\mathcal{C}_i, \mathcal{T}),$$

is an equivalence.

8. The functor $\text{ev}_* : \text{Fun}(*, \mathcal{C}) \rightarrow \mathcal{C}$ is an equivalence.

9. The functor:

$$\text{Fun}(\mathcal{T}, \text{Fun}(\mathcal{C}, \mathcal{D})) \rightarrow \text{Fun}(\mathcal{T} \times \mathcal{C}, \mathcal{D}),$$

obtained by currying the composite:

$$\text{Fun}(\mathcal{T}, \text{Fun}(\mathcal{C}, \mathcal{D})) \times \mathcal{T} \times \mathcal{C} \xrightarrow{\text{ev} \times \text{Id}_{\mathcal{C}}} \text{Fun}(\mathcal{C}, \mathcal{D}) \times \mathcal{C} \xrightarrow{\text{ev}} \mathcal{D},$$

is an equivalence.

Each of these statements is a direct and tedious consequence of the universal properties of the constructions listed above. We will prove all of them in detail just to illustrate how one should use the universal property in a rigorous fashion. However, we only recommend specifically reading through the proof for statement (1), which is very short, and then moving on. Only come back to the proofs when you feel like you need to review how the reasoning works.

Proof of Proposition 30.50. We'll prove each statement one-by-one.

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.

9. We note a slightly technical thing in the construction of the given equivalence. So, in particular, we have a composite of more than what's been reflected in the actual statement. Namely, we have that:

$$\text{Fun}(\mathcal{T}, \text{Fun}(\mathcal{C}, \mathcal{D})) \rightarrow \text{Fun}(\mathcal{T} \times \mathcal{C}, \mathcal{D}),$$

gets uncurried into:

$$\text{Fun}(\mathcal{T}, \text{Fun}(\mathcal{C}, \mathcal{D})) \times (\mathcal{T} \times \mathcal{C}) \rightarrow \mathcal{D},$$

but this gets pre-composed with the equivalence:

$$(\text{Fun}(\mathcal{T}, \text{Fun}(\mathcal{C}, \mathcal{D})) \times \mathcal{T}) \times \mathcal{C} \simeq \text{Fun}(\mathcal{T}, \text{Fun}(\mathcal{C}, \mathcal{D})) \times (\mathcal{T} \times \mathcal{C}).$$

Indeed, the point is that the associativity of the product up to equivalences of ∞ -categories comes with a preferred equivalence which is constructed via the projection functors from the axiom of products of ∞ -categories. Since we do account for preferred functors when (un)currying, we must also account for these preferred equivalences and this means that the construction of our desired equivalence uses all of these preferred equivalences. Constructing a functor in the opposite direction also uses two instantiations of uncurrying and also the homotopy associativity, along with preferred equivalences. ■

Now, there's one more construction with functor categories, which goes back to our original motivation for introducing them, that we want to give before we end this lecture. The reason why we left it over here is because it directly forms the motivation for what is about to come next.

Construction 30.51. Let $\mathcal{C}$ be an ∞ -category and let $x, y \in \mathcal{C}$. We define^a:

$$\mathcal{C}(x, y),$$

the hom *anima* from x to y , by the following pullback diagram:

$$\begin{array}{ccc} \mathcal{C}(x, y) & \longrightarrow & \text{Arr}(\mathcal{C}) \\ \downarrow & & \downarrow (\text{ev}_0, \text{ev}_1) \\ * & \xrightarrow{(x, y)} & \mathcal{C} \times \mathcal{C} \end{array}$$

Here are a few things to note:

1. We will formally define the word “anima” a little later^b.

2. An object in $\mathcal{C}(x, y)$ is a functor $f : [0] \rightarrow \mathcal{C}(x, y)$.
3. A morphism in $\mathcal{C}(x, y)$ is a functor $f : [1] \rightarrow \mathcal{C}(x, y)$.

One notices that this construction mimics the one we did with complete Segal spaces. In fact, let $\mathcal{C}, \mathcal{D}$ be ∞ -categories and let $f, g : \mathcal{C} \rightarrow \mathcal{D}$ be functors. Then, we can now talk about natural transformations from f to g as follows:

$$\mathrm{Nat}(f, g) \stackrel{\mathrm{def}}{=} \mathrm{Fun}(\mathcal{C}, \mathcal{D})(f, g).$$

Over here, the objects in $\mathrm{Nat}(f, g)$ form the natural transformations from f to g .

^aWhen it is more convenient or more visually appealing, we will use the notation $\mathcal{C}(x, y)$.

^bThis is just analogous to the notion of a Segal groupoid, which is just an ∞ -category in which all morphisms are invertible.

The construction above raises two natural questions.

Question 30.52. et $\mathcal{C}, \mathcal{D}$ be ∞ -categories, let $x, y \in \mathcal{C}$ and let $f, g : \mathcal{C} \rightarrow \mathcal{D}$ be functors.

1. Is there a notion of isomorphism in $\mathcal{C}(x, y)$?
2. Do the 2-morphisms from f to g correspond *exactly* to the isomorphisms in $\mathrm{Nat}(f, g)$?

We will address this question in the next lecture. At this moment, it's a good idea for us to re-group a little bit and figure out what exactly we wish to achieve. In particular, what other axioms do we need to introduce and what are we trying to gun towards? Well, it would be very nice to have the Yoneda Lemma in this setting and to phrase it, we need a suitable replacement of Set . Indeed, this will allow us to define, for an ∞ -category $\mathcal{C}$, the Yoneda bifunctor:

$$\mathcal{C}(-, -) : \mathcal{C} \times \mathcal{C}^{\mathrm{op}} \rightarrow \text{Replacement of Set}.$$

Of course, to make sense of this, we also need to introduce the opposite of a category. So, already, we have to introduce at least one new axiom into the mix. In fact, we will need to introduce a separate axiom which deals with the replacement of Set and its general properties. That axiom is what we'll state now and it will be our goal.

Axiom 30.53. Fundamental Properties of Cat_{∞} and An

There exist ∞ -categories An and Cat_{∞} together with a fully faithful functor:

$$\mathrm{An} \rightarrow \mathrm{Cat}_{\infty},$$

satisfying the following properties:

1. There is a functor $\pi_{\mathrm{univ}} : (\mathrm{Cat}_{\infty})_{\bullet} \rightarrow \mathrm{Cat}_{\infty}$ called the universal cocartesian fibration. Every object:

$$c \in \mathrm{Cat}_{\infty},$$

defines an ∞ -category $\mathcal{C}$ via the following pullback square:

$$\begin{array}{ccc} \mathcal{C} & \longrightarrow & (\mathrm{Cat}_{\infty})_{\bullet} \\ \downarrow & & \downarrow \pi_{\mathrm{univ}} \\ * & \xrightarrow{c} & \mathrm{Cat}_{\infty} \end{array}$$

An ∞ -category which arises in this way is called a small ∞ -category. If c is contained in the full subcategory An , then $\mathcal{C}$ is an anima. Animae arising this way are called small animae.

2. For objects $c, d \in \mathrm{Cat}_{\infty}$, with corresponding ∞ -categories $\mathcal{C}$ and $\mathcal{D}$, there is an equivalence of animae:

$$\mathrm{Cat}_{\infty}(c, d) \rightarrow \mathrm{Maps}(\mathcal{C}, \mathcal{D}),$$

and in particular, morphisms in Cat_{∞} correspond to functors of ∞ -categories. Composition in Cat_{∞} corresponds to the usual composition of functors.

3. The ∞ -categories An and Cat_{∞} admit all small limits and colimits; here, the adjective “small” means that the indexing ∞ -category is required to be small. Moreover, the inclusion:

$$\mathrm{An} \rightarrow \mathrm{Cat}_{\infty},$$

is required to preserve all small (co)limits.

4. The initial and final ∞ -categories are small. Small ∞ -categories are closed under (co)products, pullbacks, functor categories, subcategories and localizations.
5. There is a fully faithful functor:

$$\Delta \rightarrow \mathrm{Cat}_{\infty},$$

given on objects by sending a linearly ordered (finite) set P to its associated ∞ -category.

6. There is a fully faithful functor:

$$\mathrm{Set} \rightarrow \mathrm{An},$$

given on objects by sending a set to its associated anima.

Sometimes, An and Cat_{∞} are referred to as the universe of animae and the universe of ∞ -categories.

We have a few remarks about this axiom.

Remark 30.54. We have underlined the terms that need to be defined in the next few lectures. Indeed, this axiom strongly indicates that we will need axioms which deal with the following ∞ -categorical notions:

1. (Full) Subcategories & embeddings.
2. Full faithfulness.

3. Cocartesian fibrations.
4. Localizations.
5. Anima.
6. (Co)limits.
7. Adjunctions.

Out of all of these, the *localization* is the construction which is most central to homotopy theory. As we have seen before, much of the homotopy theory we've developed, such as model categories, categories of (co)fibrant objects, calculus of fractions, was in a bid to control localizations. Localizations are how we're going to realize the ∞ -category of anima and a large part of ∞ -category theory is dedicated towards getting some control over them.

Remark 30.55. If we apply the formalism of Grothendieck universes to this axiom, then it is typical to also require the existence of two more universes $\widehat{\mathbf{An}}$ and $\widehat{\mathbf{Cat}_\infty}$ that satisfy the same conditions as $\mathbf{An}$ and $\mathbf{Cat}_\infty$. Also:

1. There are fully faithful inclusions $\mathbf{An} \rightarrow \widehat{\mathbf{An}}$ and $\mathbf{Cat}_\infty \rightarrow \widehat{\mathbf{Cat}_\infty}$.
2. The ∞ -categories $\mathbf{An}$ and $\mathbf{Cat}_\infty$ are small with respect to $\widehat{\mathbf{Cat}_\infty}$.

The objects of $\widehat{\mathbf{Cat}_\infty}$ are referred to as large ∞ -categories, while the objects of $\widehat{\mathbf{An}}$ are known as large anima.

31 Carrying Over 1-Categorical Constructs

Summary

In this lecture, we will cover the following topics:

1. Controlling isomorphisms in (and of) ∞ -categories.
2. Further constructions of ∞ -categories.

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

1-category theory - This refers to the ordinary category theory which is assumed background for readers.

Let's start with some motivation.

Motivation. In this lecture, we move one step closer to our goal of a better understanding of ∞ -category and, in particular, the content of Axiom 30.53.

31.1 Controlling Isomorphisms

In light of the motivation given above, we start with a discussion of isomorphisms and anima. This is mainly because all of this can be done in a way that “makes sense” without it being fully rigorous. Once the essence of the notions are introduced and one knows what to expect, it becomes reasonable to start introducing the more technical axioms. This discussion will take up all of this section and we will start establishing further results about ∞ -groupoids in Section 31.2, ending off with an exposition of Grothendieck’s Homotopy Hypothesis.

31.1.1 Isomorphisms, Anima & All That

We’ll start by introducing the notion of an isomorphism in an ∞ -category and the notion of an anima.

Definition 31.1

Let $\mathcal{C}$ be an ∞ -category and let $f : x \rightarrow y$ be a morphism. f is said to be invertible, or is said to be an isomorphism/equivalence, iff there exists a morphism $g : y \rightarrow x$ together with commutative triangles in $\mathcal{C}$ of the form:

$$\begin{array}{ccc} y & \xrightarrow{g} & x \\ & \searrow \text{Id}_y & \downarrow f \\ & & y \end{array} \qquad \begin{array}{ccc} x & \xrightarrow{f} & y \\ & \searrow \text{Id}_x & \downarrow g \\ & & x \end{array}$$

We say that $\mathcal{C}$ is an anima/ ∞ -groupoid iff every morphism in $\mathcal{C}$ is invertible.

We have a remark at this definition.

Remark 31.2. This is the first step towards the introduction of the ∞ -category $\mathbf{An}$, which consists of anima^a as objects. Incidentally, it’s also the first explicit step towards stable homotopy theory.

^aSometimes, “anima” is used as the plural for anima in the literature.

Now, we are ready to state the groupoid core axiom.

Axiom 31.3. The Groupoid Core Axiom

Let $\mathcal{C}$ be an ∞ -category. Then, there exists a pair:

$$(\mathcal{C}^\simeq, \gamma_{\mathcal{C}} : \mathcal{C}^\simeq \rightarrow \mathcal{C}),$$

consisting of:

1. An anima $\mathcal{C}^\simeq$,
2. A functor $\gamma_{\mathcal{C}} : \mathcal{C}^\simeq \rightarrow \mathcal{C}$.

Moreover, if X is an anima and for the following solid diagram of functors:

$$\begin{array}{ccc} X & \xrightarrow{F} & \mathcal{C} \\ & \searrow \widehat{F} & \nearrow \gamma_{\mathcal{C}} \\ & \mathcal{C}^\simeq & \end{array}$$

there exists a dashed arrow such that the diagram commutes. Moreover, suppose that $G, H : X \rightarrow \mathcal{C}^\simeq$ be functors such that there is a natural isomorphism $\gamma_{\mathcal{C}} \circ G \simeq \gamma_{\mathcal{C}} \circ H$. Then, there is a natural isomorphism $G \simeq H$. Furthermore, we require that the functor $\langle 0, 1 \rangle : * \amalg * \rightarrow [1]$ factors through an equivalence $* \amalg * \rightarrow [1]^\simeq$.

There’s an additional technical part of the axiom above.

Notation 31.4. Let $\mathcal{C}, \mathcal{D}$ be ∞ -categories. We define:

$$\text{Maps}(\mathcal{C}, \mathcal{D}) \stackrel{\text{def}}{=} \text{Fun}(\mathcal{C}, \mathcal{D})^\simeq,$$

as the mapping anima associated to the functor category $\text{Fun}(\mathcal{C}, \mathcal{D})$.

Here’s a result which illustrates how we use the axiom.

Proposition 31.5

Let X be an anima and let $\mathcal{C}$ be an ∞ -category. Then, the inclusion:

$$\gamma_{\mathcal{C}} : \mathcal{C}^\simeq \rightarrow \mathcal{C},$$

induces an equivalence:

$$\text{Maps}(X, \mathcal{C}^\simeq) \xrightarrow{\sim} \text{Maps}(X, \mathcal{C}),$$

of ∞ -categories.

Proof Sketch. We construct our functor in the forwards direction via the following diagram:

$$\begin{array}{ccc} \text{Fun}(X, \mathcal{C}^\simeq) & \xrightarrow{\gamma_{\mathcal{C}^\simeq}} & \text{Fun}(X, \mathcal{C}) \\ \downarrow & \searrow & \downarrow \\ \text{Maps}(X, \mathcal{C}^\simeq) & \dashrightarrow & \text{Maps}(X, \mathcal{C}) \end{array}$$

where we use the Groupoid Core Axiom to induce our desired functor. Now, constructing the inverse is a bit more tricky but it can be done using (un)currying. So, what we want to do is to construct a functor:

$$\text{Fun}(X, \mathcal{C}) \longrightarrow \text{Fun}(X, \mathcal{C}^\simeq)$$

Well, this is the same as constructing a functor:

$$* \rightarrow \text{Fun}(\text{Fun}(X, \mathcal{C}), \text{Fun}(X, \mathcal{C}^\simeq)),$$

which is the same as constructing a functor:

$$* \rightarrow \text{Fun}(\text{Fun}(X, \mathcal{C}) \times X, \mathcal{C}^\simeq),$$

Well, we don't have a functor like this one exactly but we do have the evaluation:

$$\text{ev} : \text{Fun}(X, \mathcal{C}) \times X \rightarrow \mathcal{C},$$

and by using the Groupoid Core Axiom, we obtain a functor:

$$\widehat{\text{ev}} : \text{Fun}(X, \mathcal{C}) \times X \rightarrow \mathcal{C}^\simeq,$$

which can, then, be used to construct our quasi-inverse. Once we have constructed the quasi-inverse, we must show that it is actually a quasi-inverse and this involves using all universal properties employed along with the commutativity of the diagrams that go along with them. ■

As we mentioned above, we skipped over a few preliminary technical axioms in favor of getting to the main point of this lecture, which was a discussion of anima and the groupoid core axiom. Nevertheless, the technical axioms need to be addressed and the set-up above gives us motivation for what we should put into the theory. Let us list a few problems:

1. The definition of an isomorphism clearly requires us to define the notion of a commutative triangle. Of course, the specific data of a commutative triangle must come with information about how we identify the vertices and edges.
2. We need a way of composing morphisms. Moreover, the data of a commutative triangle should be determined, up to homotopy, by the morphisms we wish to compose.
3. If $\mathcal{C}$ is an ∞ -category, then objects in $\mathcal{C}$ are just functors $x, y : * \rightarrow \mathcal{C}$. How do we know that a 2-morphism from x to y is the same as there being an isomorphism between them in $\mathcal{C}$?

The solutions to these problems are presented by the Commutative Square Axiom, The Segal Axiom and the Rezk Axiom. In fact, the Segal Axiom is called an axiom precisely because it is a condition placed on bisimplicial sets, as we have seen before, to allow us to do category theory on bisimplicial sets. However, the specific way in which we phrase the Commutative Square Axiom will ensure that the Segal Axiom becomes a result in the synthetic setting.

31.1.2 The Commutative Square Axiom & Some Simplicial Nonsense

Let's start with a few pieces of terminology concerning diagrams. None of this is going to look weird from the conceptual viewpoint; all of the diagrams we describe have been encountered by you in one way or another. The problem is understanding the necessity of dedicating a whole axiom towards them. Before we explain why we'll do that, let's just introduce the diagrams of interest. This is going to be a little ad-hoc but bear with us.

Terminology 31.6. Let $\mathcal{C}$ be an ∞ -category.

1. A commutative triangle is a functor $\sigma : [2] \rightarrow \mathcal{C}$.
2. A commutative square is a functor $\sigma : [1] \times [1] \rightarrow \mathcal{C}$.
3. Let $[\omega] = \mathbb{N}$ with its standard order, regarded as an ∞ -category formed from the underlying poset 1-category. Then, an infinite sequence of morphisms in $\mathcal{C}$ is a functor:

$$x_\bullet : [\omega] \rightarrow \mathcal{C},$$

which is typically displayed by the following notation:

$$x_1 \xrightarrow{f_1} x_2 \xrightarrow{f_2} \dots,$$

where x_n is the evaluation of $x_\bullet$ at $n \in [\omega]$ and f_n is obtained by composing:

$$[1] \simeq \{n \leq n+1\} \rightarrow [\omega] \rightarrow \mathcal{C},$$

which is completely reasonable.

4. Consider the subposet $\lceil \subseteq [1] \times [1]$ spanned by three objects $(0,0), (0,1), (1,1)$. A span in $\mathcal{C}$ is a functor $\lceil \rightarrow \mathcal{C}$ and it is typically denoted by the following image:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ \downarrow g & & \\ z & & \end{array}$$

Similarly, we may consider the subposet $\rfloor \subseteq [1] \times [1]$ spanned by three objects $(0,0), (1,0), (1,1)$. A cospan in

$\mathcal{C}$ is a functor $\mathcal{J} \rightarrow \mathcal{C}$ and it is typically denoted by the following image:

$$\begin{array}{ccc} & & y \\ & & \downarrow f \\ z & \xrightarrow{g} & x \end{array}$$

Note that every commutative square gives rise to both a span and cospan via restriction along the two inclusions. Finally, a span is also called a pushout and a cospan is also called a pullback.

We have two remarks about this bit of terminology.

Remark 31.7. Once we have Cat_∞ available, we will need to check that a commutative square of ∞ -categories in the sense defined previously is the same as a commutative square in the ∞ -category of ∞ -categories. From the way that we have phrased Axiom 30.53, it isn't completely clear that this bit of terminology specializes very well.

Remark 31.8. The interest in these specific diagrams is a partly necessary and partly technical. So, of course, the necessity comes from making sure that we can make sense of our definition of isomorphisms. But more importantly, this set-up will be necessary when we want to define specialized functors in special ∞ -categories later. For instance, when we have the structure of a stable ∞ -category, we will want to define a loop and suspension functor on that ∞ -category. Well, it will be easy to define loop and suspension objects via pullback and pushout diagrams. But once we have those objects, we'd like to conclude that we have an entire functor on our hands. The set-up we discuss here is a technically simple way of doing that which is somewhat different from the way it used to be done.

Let's state the commutative square axiom. For the statement of the axiom, we note that the reader should recall the notation for the face and degeneracy maps as discussed in Towards Noncommutative Homotopy Theory I.

Axiom 31.9. The Commutative Square Axiom

The following commutative squares:

$$\begin{array}{ccc} [0] & \xrightarrow{d_0} & [1] \\ d_1 \downarrow & & \downarrow d_2 \\ [1] & \xrightarrow{d_0} & [2] \end{array}$$

$$\begin{array}{ccc} [1] & \xrightarrow{d_1} & [2] \\ d_1 \downarrow & & \downarrow j_0 \\ [2] & \xrightarrow{j_1} & [1] \times [1] \end{array}$$

$$\begin{array}{ccc} [0] & \xrightarrow{0} & [1] \\ n-1 \downarrow & & \downarrow e_{n-1} \\ [n-1] & \xrightarrow{\quad} & [n] \end{array}$$

$$\begin{array}{ccc} [0] & \xrightarrow{d_0} & [1] \\ d_1 \downarrow & & \downarrow (0,0) \rightarrow (1,0) \\ [1] & \xrightarrow{(0,0) \rightarrow (0,1)} & \lceil \end{array}$$

$$\begin{array}{ccc} [0] & \xrightarrow{d_0} & [1] \\ d_0 \downarrow & & \downarrow (1,0) \rightarrow (1,1) \\ [1] & \xrightarrow{(0,1) \rightarrow (1,1)} & \rfloor \end{array}$$

$$\begin{array}{ccc} \coprod_{k \in \mathbb{N}} \{k+1\} & \xrightarrow{\quad} & \coprod_{k \in \mathbb{N}} \{2k \leq 2k+1\} \\ \downarrow & & \downarrow \\ \coprod_{k \in \mathbb{N}} \{2k+1 \leq 2k+2\} & \xrightarrow{\quad} & [\omega] \end{array}$$

are pushout squares of ∞ -categories.

We have a remark about this axiom.

Remark 31.10. So, the axiom may seem reasonable and the name for it may fit because we deal with a bunch of commutative squares. In reality, this axiom is somewhat unsatisfactory. Even though, as we will show below, the Segal Axiom becomes a result rather than an axiom in this framework, the reality is that the input of the axiom above leaves a bit to be desired. This is reflected in [26], where the Commutative Square Axiom actually deals with the

structure of a commutative square in an ∞ -category; it just states that such a square is the gluing of two commutative triangles. In that text, the Segal Axiom needs to be stated as an axiom and the proofs as we will do them below stay the same. So, whether you accept the axiom above or the Segal Axiom should just be a matter of taste.

In any event, here's the Segal Axiom proved as a result.

Proposition 31.11. The Segal Axiom

Let $\mathcal{C}$ be an ∞ -category. For $n \geq 2$, the restriction functor:

$$(d_0^*, d_2^*) : \text{Fun}([2], \mathcal{C}) \rightarrow \text{Fun}([1], \mathcal{C}) \times_{\mathcal{C}} \text{Fun}([1], \mathcal{C}),$$

is an equivalence of categories^a.

^aThe $\star$ -notation is another notation for the composition functors.

Proof. This follows by using Axiom 30.49 on the pushout square:

$$\begin{array}{ccc} [0] & \longrightarrow & [1] \\ \downarrow & & \downarrow \\ [1] & \longrightarrow & [2] \end{array}$$

where this gets turned into a pullback upon passing to functor categories. ■

The intuition for the Segal axiom is *precisely* the same intuition for the Segal condition that allows us to move from bisimplicial sets to Segal spaces. With this machine, we can (and will) construct the composition functor for morphisms in an ∞ -category. Since the construction of this functor in Segal spaces was formal,

Construction 31.12. Let $\mathcal{C}$ be an ∞ -category and consider the following diagram:

$$\text{Fun}([1], \mathcal{C}) \times_{\mathcal{C}} \text{Fun}([1], \mathcal{C}) \xleftarrow{(d_0^*, d_2^*)} \text{Fun}([2], \mathcal{C}) \xrightarrow{d_1^*} \text{Fun}([1], \mathcal{C})$$

By the Segal axiom, the first map is an equivalence of ∞ -categories so it admits a quasi-inverse and we obtain:

$$- \circ - : \text{Fun}([1], \mathcal{C}) \times_{\mathcal{C}} \text{Fun}([1], \mathcal{C}) \rightarrow \text{Fun}([2], \mathcal{C}) \xrightarrow{d_1^*} \text{Fun}([1], \mathcal{C}).$$

If we pass to the fibers, then we obtain precisely the composition of morphisms taken from two hom anima as usual. This is known as the ∞ -categorical composition functor.

Of course, we proved, in the setting of Segal spaces, that this composition functor respected identities and was associative. The argument, in that case, relied on a formal use of the Segal Axiom and the same argument goes through here.

Proposition 31.13

Let $\mathcal{C}$ be an ∞ -category, let $x, y, z \in \mathcal{C}$ and let $f \in \mathcal{C}(x, y), g \in \mathcal{C}(y, z), h \in \mathcal{C}(z, w)$. TFSH:

1. $f \circ \text{Id}_x \simeq f \simeq \text{Id}_y \circ f$.
2. $h \circ (g \circ f) \simeq (h \circ g) \circ f$.

Moreover, the isomorphisms above are natural.

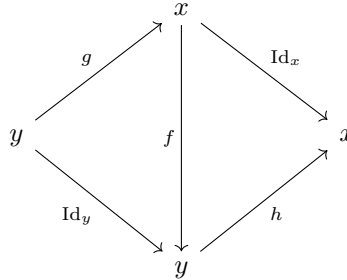
Proof. See [27, Lemma 1.4.8, Proposition 1.4.9, p.32-33]. ■

With this proposition in hand and with the material we've discussed so far, we can actually prove a number of facts about the hom anima in an ∞ -category. However, we still can't quite prove that the hom anima is an anima before we state the Rezk axiom and that's mainly a technical issue. Nevertheless, that is one of the most important results of the basic theory, so we do want to make sure that it is proved properly.

31.1.3 The Rezk Axiom

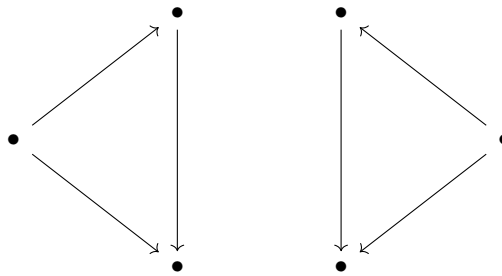
In order to state the Rezk axiom, we need a construction which is inspired by our work with complete Segal spaces. Indeed, the Rezk axiom is the completeness axiom in this setting and you will see that the formulation of this axiom is very similar to the way we derived the completeness condition. The only difference is that in that setting, there were concrete ∞ -categories that gave us several characterizations of the condition, whereas we'll only get one characterization here. Before we start with the statement of the axiom, it's best if we begin with a bit of intuition for what exactly we wish to accomplish. The intuition can, then, be turned into a proper construction.

Intuition 31.14. Let $\mathcal{C}$ be an ∞ -category. We want a category consisting of commutative diagrams in $\mathcal{C}$ of the form:

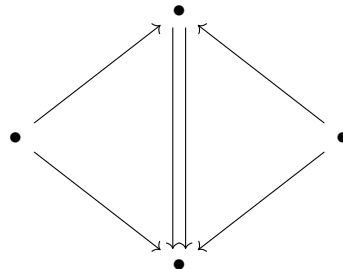


of morphisms. In other words, it's a category of commutative squares which encode isomorphisms. So, the steps for forming this category are very simple, even though we get an annoying pullback diagram in the end.

1. We start with two triangles:

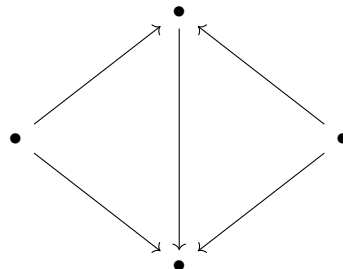


in $\mathcal{C}$ and what we want to do is to glue these together into a square without losing information about the morphism in the middle. So, we obtain something like the following diagram:



where the two morphisms down the middle are the same so we pretend that we have them identified.

2. Now, we would like to recognize the triangles in the following diagram:



as “isomorphism triangles”. In other words, we are trying to specify the left and right homotopy inverses of the morphism down the middle. So, all we need to do is to identify the appropriate edges as the appropriate identity morphism, depending on the source and target objects, of course.

Of course, each picture can, then, be turned into an appropriate commutative diagram of ∞ -categories and functors.

Okay, let's formalize this intuition.

Construction 31.15. Let $\mathcal{C}$ be an ∞ -category. Following [Intuition 31.14](#), we start with the following pullback:

$$\begin{array}{ccc} \mathrm{Fun}([2], \mathcal{C}) \times_{\mathrm{Fun}[1], \mathcal{C}} \mathrm{Fun}([2], \mathcal{C}) & \longrightarrow & \mathrm{Fun}([1], \mathcal{C}) \\ \downarrow & & \downarrow \Delta \\ \mathrm{Fun}([2], \mathcal{C}) \times \mathrm{Fun}([2], \mathcal{C}) & \longrightarrow & \mathrm{Fun}([1], \mathcal{C}) \times \mathrm{Fun}([1], \mathcal{C}) \end{array}$$

So, this represents the process of gluing two commutative triangles along an edge without discarding information about that edge. Of course, we have used an additional identification with the pullback but that just amounts to discarding the additional copy of the middle morphism. Now, we want to identify specific edges as identity morphisms:

$$\begin{array}{ccc}
 \text{Fun}([2], \mathcal{C}) \times_{\text{Fun}[1], \mathcal{C}} \text{Fun}([2], \mathcal{C}) & \xrightarrow{\quad\quad\quad} & \text{Fun}([1], \mathcal{C}) \\
 \downarrow & & \downarrow \Delta \\
 \text{Fun}([2], \mathcal{C}) \times \text{Fun}([2], \mathcal{C}) & \xrightarrow{\quad\quad\quad} & \text{Fun}([1], \mathcal{C}) \times \text{Fun}([1], \mathcal{C}) \\
 \downarrow & & \\
 \mathcal{C} \times \mathcal{C} & \xrightarrow{(x,y) \mapsto (\text{Id}_x, \text{Id}_y)} & \text{Fun}([1], \mathcal{C}) \times \text{Fun}([1], \mathcal{C})
 \end{array}$$

This is handled by the extension of our previous diagram to the one above. Now, we take the pullback of the arrows towards the left-hand side and this is how we obtain our category $\text{Iso}(\mathcal{C})$:

$$\begin{array}{ccccc}
 & & \pi_{\text{Iso}} & & \\
 & \text{Iso}(\mathcal{C}) & \xrightarrow{\quad\quad\quad} & \text{Fun}([2], \mathcal{C}) \times_{\text{Fun}[1], \mathcal{C}} \text{Fun}([2], \mathcal{C}) & \xrightarrow{\quad\quad\quad} & \text{Fun}([1], \mathcal{C}) \\
 \downarrow & & & \downarrow & & \downarrow \Delta \\
 & \text{Fun}([2], \mathcal{C}) \times \text{Fun}([2], \mathcal{C}) & \xrightarrow{\quad\quad\quad} & \text{Fun}([1], \mathcal{C}) \times \text{Fun}([1], \mathcal{C}) & & \\
 \downarrow & & & \downarrow & & \\
 \mathcal{C} \times \mathcal{C} & \xrightarrow{(x,y) \mapsto (\text{Id}_x, \text{Id}_y)} & \text{Fun}([1], \mathcal{C}) \times \text{Fun}([1], \mathcal{C}) & & &
 \end{array}$$

along with a functor:

$$\pi_{\text{Iso}} : \text{Iso}(\mathcal{C}) \rightarrow \text{Arr}(\mathcal{C}),$$

which is defined by the composition above. Now, the statement of the Rezk axiom involves another functor:

$$i : \mathcal{C} \rightarrow \text{Iso}(\mathcal{C}),$$

which, roughly speaking, takes an object $x \in \mathcal{C}$ and embeds it into $\text{Iso}(\mathcal{C})$ by the following square:

$$\begin{array}{ccccc}
 & & x & & \\
 & \nearrow \text{Id}_x & & \nwarrow \text{Id}_x & \\
 x & & & & x \\
 & \searrow \text{Id}_x & & \swarrow \text{Id}_x & \\
 & & x & &
 \end{array}$$

Let's do this in a few steps:

1. Consider the functor $p : [2] \rightarrow [0]$ between ∞ -categories. This induces a functor:

$$p^* : \text{Fun}([0], \mathcal{C}) \rightarrow \text{Fun}([2], \mathcal{C}),$$

which is just reduced to a functor:

$$p^* : \mathcal{C} \rightarrow \text{Fun}([2], \mathcal{C}),$$

by identifying $\text{Fun}([0], \mathcal{C})$ with $\mathcal{C}$.

2. Now, the functor p^* induces a functor:

$$(p^*, p^*) : \mathcal{C} \rightarrow \text{Fun}([2], \mathcal{C}) \times \text{Fun}([2], \mathcal{C}),$$

by the universal property of the product.

3. If we place this into the massive pullback diagram where we constructed $\text{Iso}(\mathcal{C})$, then we see that the universal property of the pullback implies the existence of a functor $i : \mathcal{C} \rightarrow \text{Iso}(\mathcal{C})$. Of course, one has to check explicitly that we have a pullback cone but this will follow from our construction above.

So, as a consequence, we obtain a way of shoving our ∞ -category via the functor:

$$i : \mathcal{C} \rightarrow \text{Iso}(\mathcal{C}),$$

and this is the one that will concern us in the statement of the Rezk axiom.

With this preparation, it becomes easy to state the Rezk axiom.

Axiom 31.16. The Rezk Axiom

Let $\mathcal{C}$ be an ∞ -category. Then, the functor $i : \mathcal{C} \rightarrow \text{Iso}(\mathcal{C})$ is an equivalence.

As a corollary of this axiom.

Corollary 31.17

Let $\mathcal{C}$ be an ∞ -category and let $x, y \in \mathcal{C}$ be isomorphic objects, with an invertible morphism:

$$f : x \rightarrow y.$$

Then, there exists a 2-morphism $x \simeq y$ as functors $\ast \rightarrow \mathcal{C}$.

Proof. Since the functor $i : \mathcal{C} \rightarrow \text{Iso}(\mathcal{C})$ is an equivalence of ∞ -categories, there exists a $z \in \mathcal{C}$ together with an isomorphism $i(z) \simeq f$ in $\text{Iso}(\mathcal{C})$. Through the source and target functors $s, t : \text{Iso}(\mathcal{C}) \rightarrow \mathcal{C}$, we see that the isomorphism in $\text{Iso}(\mathcal{C})$ produces isomorphisms $z \simeq x$ and $z \simeq y$ in $\mathcal{C}$. By inversion, we obtain that:

$$x \simeq z,$$

and by composition, we see that $x \simeq y$. ■

31.2 ∞ -Groupoids, Localizations & Subcategories: The Homotopy Hypothesis

We'll start by trying to develop the direct machinery needed to prove the following result.

Proposition 31.18

Let $\mathcal{C}$ be an ∞ -category and let $x, y \in \mathcal{C}$. Then, $\mathcal{C}(x, y)$ is an anima.

Proof Of Proposition 31.18.

■

32 (Co)Limits & Adjunctions of ∞ -Categories

Summary

In this lecture, we will cover the following topics:

- 1.
- 2.

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| Motivation.

32.1 A Synthetic View of the Yoneda Lemma

Our first goal is to work towards the statement of the Yoneda Lemma. To do this, we will rely on the following result.

Proposition 32.1

Let $\mathcal{C} \in \text{Cat}_\infty$. Then, there exists a functor:

$$\mathcal{C}(-, -) : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \text{An},$$

such that for $(x, y) \in \mathcal{C}^{\text{op}} \times \mathcal{C}$, we have that:

$$\mathcal{C}(-, -)(x, y) = \mathcal{C}(x, y).$$

This is known as the Hom functor.

We have a remark about this result.

Remark 32.2. Showing that such a functor exists for all ∞ -categories is completely non-trivial and requires the use of Lurie's (Un)Straightening Equivalence. In a specific model of ∞ -category theory, this is going to be a (difficult) result but we will treat it as an axiom. Once we have it (and all of the relevant terminology surrounding it), we will be able to construct the Hom functor above. This will be done in ??.

Now, we define the Yoneda embedding by a direct construction.

Construction 32.3. Let $\mathcal{C} \in \text{Cat}_\infty$. Then, we have an equivalence:

$$\text{Fun}(*, \text{Fun}(\mathcal{C}^{\text{op}} \times \mathcal{C}, \text{An})) \simeq \text{Fun}(\mathcal{C}, \text{Fun}(\mathcal{C}^{\text{op}}, \text{An})).$$

So, there exists a unique functor $\mathcal{Y} = \mathcal{Y}_{\mathcal{C}} : \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \text{An})$ which corresponds to the Hom functor from Proposition 32.1; we refer to this as the Yoneda embedding. For each $x \in \mathcal{C}$, we define:

$$\mathcal{C}(-, x) \stackrel{\text{def}}{=} \mathcal{Y}(x),$$

and refer to this as the presheaf associated to x .

This leads us to the statement of the Yoneda Lemma. Given the sheer importance of this result, we are very much justified in treating it as an axiom and as we said before, that's exactly our plan.

Axiom 32.4. The Yoneda Lemma

Let $\mathcal{C} \in \text{Cat}_\infty$. Then, the Yoneda embedding:

$$\mathcal{Y} : \mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \text{An}),$$

is fully faithful. Moreover, for any functor $f : \mathcal{C}^{\text{op}} \rightarrow \text{An}$, there is an equivalence:

$$\text{Nat}(\mathcal{C}(-, x), f) \rightarrow f(x),$$

for any $x \in \mathcal{C}$.

We have a remark regarding this axiom.

Remark 32.5. Well, using the Hom functor and the Yoneda Lemma, we can start doing actual category theory. In particular, many of our arguments, now, are going to be truly synthetic. Of course, there will be a few places in which we'll have to appeal to a specific result in a specific model but we'll be able to isolate those situations fairly well.

Our next goal is to give a conceptual discussion of adjunctions & (co)limits. Due to the nature of ∞ -category theory, there are some new features that we'll have to pay attention to because they're not automatically given to us. For instance, to define adjunctions in the usual way, we'll need to construct the two functors:

$$\mathcal{D}(F(-), -) \wedge \mathcal{C}(-, G(-)),$$

for ∞ -categories $\mathcal{C}, \mathcal{D}$ and antiparallel functors F and G . The way this is done in 1-category theory is fairly obvious but more thought is required to define them over here.

Construction 32.6. Let $\mathcal{C}, \mathcal{D} \in \text{Cat}_\infty$ and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$. We want a functor:

$$\mathcal{D}(F(-), -) : \mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \text{An},$$

such that:

$$\forall (x, y) \in \mathcal{C}^{\text{op}} \times \mathcal{D} : \mathcal{D}(F(-), -)(x, y) \stackrel{\text{def}}{=} \mathcal{D}(F(x), y).$$

Well, it's enough for us to specify a functor:

$$\mathcal{C}^{\text{op}} \rightarrow \text{Fun}(\mathcal{D}, \text{An}),$$

Well, we just define

Construction 32.7. Let $\mathcal{C}, \mathcal{D} \in \text{Cat}_\infty$ and let $G \in \text{Fun}(\mathcal{D}, \mathcal{C})$. We want a functor:

$$\mathcal{C}(-, G(-)) : \mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \text{An},$$

such that:

$$\forall (x, y) \in \mathcal{C}^{\text{op}} \times \mathcal{D} : \mathcal{C}(-, G(-))(x, y) \stackrel{\text{def}}{=} \mathcal{C}(x, G(y)).$$

Well, it's enough for us to specify a functor:

$$\mathcal{C}^{\text{op}} \rightarrow \text{Fun}(\mathcal{D}, \text{An}),$$

32.2 A Fibrational Approach Towards The Yoneda Lemma

A big part of the difficulty in working with quasicategories and complete Segal spaces is that they both seem to be lacking in the notion of a *representable functor*. This comes from the fact that compositions of morphisms are defined only up to contractible choice. This means that there is a significant lack in a suitable version of the Yoneda Lemma, which we state below.

Theorem 32.8. The Yoneda Lemma, [REF]

Let $\mathcal{C}$ be a locally small category, let $F \in \text{Fun}(\mathcal{C}, \text{Set})$ and let $x \in \mathcal{C}$. Then, the following map:

$$\Phi : \text{Nat}(\mathcal{C}(x, -), F) \rightarrow F(x), \quad \alpha \mapsto \Phi(\alpha) \stackrel{\text{def}}{=} \alpha_x(\text{Id}_x),$$

is a natural bijection.

Of course, we have proved this in our treatment of basic category theory within [REF]. Also, the meaning of “natural” is elucidated a bit more carefully in the aforementioned reference. In order to define (co)limits and adjunctions of CSSs, it would be very nice for us to have a version of the Yoneda Lemma for CSSs. However, the statement of the theorem above makes it clear that we need to have a notion of representability and this is not going to be easily manageable. This leads us to the following natural question.

Question 32.9. Is there an alternative version of representability which is suitable for our purposes?

The goal of this section is to demonstrate that there is an affirmative answer to this question. At the very end, we will present a fibrational version of the Yoneda Lemma which will be far more amenable to generalization in the setting of CSSs. Before we get started, let us give a brief historical overview of our approach.

Remark 32.10. Let $\mathcal{C}$ be a category. The idea is to look at a certain full subcategory of $\text{Cat}/\mathcal{C}$, which we will later refer to as the discrete Grothendieck opfibrations over $\mathcal{C}$, as an alternative characterization of $\text{Fun}(\mathcal{C}, \text{Set})$. The idea for this goes all the way back to [48, Chapter VI, p.119-152], though it has also come up in important works such as [47] and the notes [92]. This identification is possible by the so-called Grothendieck construction, something which is used in 1-category theory as a formal way of describing universal properties. For our purposes, we will be using it as the main tool in understanding these opfibrations and this will set us up for the next chapter.

With all of this motivation given, let us start by giving a characterization of representability of covariant functors.

Claim 32.1. Let $\mathcal{C}$ be a locally small category and let $F \in \text{Fun}(\mathcal{C}, \text{Set})$. TFAE:

1. F is representable.
2. There is a pair (c, x) such that:
 - (a) $c \in \mathcal{C}$.
 - (b) $x \in F(c)$.
 - (c) If (d, y) is a pair with $y \in F(d)$, then there is a unique morphism $f : c \rightarrow d$ such that $F(f)(x) = y$.

The pair (c, x) is known as the representing pair for the functor.

This result is [1, Proposition 1.7.7, p.50-51].

The result above suggests that studying a category consisting of pairs as described in statement (2) is the key to understanding representability of functors. This leads us to the following construction.

Construction 32.11. Let $\mathcal{C}$ be a locally small category and let $F \in \text{Fun}(\mathcal{C}, \text{Set})$. We define a structure:

$$\int_{\mathcal{C}} F,$$

as follows:

1. The objects are pairs (c, x) with $c \in \mathcal{C}$ and $x \in F(c)$.
2. If $(c, x), (d, y)$ are objects, then a morphism $f : (c, x) \rightarrow (d, y)$ is just a morphism $f : c \rightarrow d$ such that:

$$F(f)(x) = y.$$

The composition is defined as usual.

This structure forms a category; it is known as the category of elements of F . Define a map of categories:

$$\pi_F : \int_{\mathcal{C}} F \rightarrow \mathcal{C},$$

as follows:

1. If $(c, x) \in \int_{\mathcal{C}} F$, then $\pi_F(c, x) \stackrel{\text{def}}{=} c$.
2. If $f : (c, x) \rightarrow (d, y)$ is a morphism in $\int_{\mathcal{C}} F$, then $\pi_F(f) \stackrel{\text{def}}{=} f$.

This is clearly a functor. This is known as the projection associated to F .

An interesting fact about the projection associated to F is that it is conservative.

Terminology 32.12. Let $\mathcal{C}, \mathcal{D}$ be categories and let $F \in \text{Fun}(\mathcal{C}, \mathcal{D})$. F is conservative iff for any $f \in \text{Mor}(\mathcal{C})$ such that $F(f)$ is an isomorphism in $\mathcal{D}$, we have that f is an isomorphism.

Here's the miscellaneous result about the projection in the construction.

Claim 32.2. Using the notation of [Construction 32.11](#), we have that π_F is conservative.

We also have a category of elements for contravariant functors.

Construction 32.13. Let $\mathcal{C}$ be a locally small category and let $F \in \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$. We define a structure:

$$\int_{\mathcal{C}^{\text{op}}} F,$$

as follows:

1. The objects are pairs (c, x) with $c \in \mathcal{C}^{\text{op}}$ and $x \in F(c)$.
2. If $(c, x), (d, y)$ are objects, then a morphism $f : (c, x) \rightarrow (d, y)$ is a morphism $f : c \rightarrow d$ in $\mathcal{C}$ such that:

$$F(f)(y) = x.$$

The composition is defined as usual.

This structure forms a category; it is known as the category of coelements of F . In a similar way, one can define a functor $\pi_F : \int_{\mathcal{C}^{\text{op}}} F \rightarrow \mathcal{C}$ as we did in the preceding construction.

We have a general remark about these constructions.

Remark 32.14. Both of these constructions are grouped under the label “Grothendieck Construction”. In Riehl’s textbook ([83]), these constructions are used primarily as a mechanism for formalizing the notion of a universal property^a. The Grothendieck construction will allow us to translate between functors and fibrations.

^aThis seems to be a point of difference among certain texts in category theory. For example, the Grothendieck construction as presented in [82, Chapter 5, p.77-90] mainly seems to have been chunked in almost as an afterthought.

Using these constructions, we may simplify the statement of [Claim 32.1](#) and we do so in the next remark.

Remark 32.15. Let $\mathcal{C}$ be a locally small category and let $F \in \text{Fun}(\mathcal{C}, \text{Set})$. TFAE:

1. F is representable.
2. $\int_{\mathcal{C}} F$ has an initial object.

Of course, this statement is far more amenable to dualization.

Here's the dual assertion to [Claim 32.1](#).

Claim 32.3. Let $\mathcal{C}$ be a locally small category and let $F \in \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$. TFAE:

1. F is representable.
2. $\int_{\mathcal{C}^{\text{op}}} F$ has a final object.

The categories of elements of the representable functors are particularly interesting.

Example 32.16. Let $\mathcal{C}$ be a locally small category and let $x \in \mathcal{C}$. We want to calculate:

$$\int_{\mathcal{C}} \mathcal{C}(x, -).$$

This is the category given by the following data:

1. The objects are pairs (f, y) , where $y \in \mathcal{C}$ and $f \in \mathcal{C}(x, y)$.
2. The morphisms between objects $(f, y), (g, z)$ are maps $\phi : y \rightarrow z$ such that $\mathcal{C}(x, -)(\phi)(f) = g$.

Well, with this data, it's clear that $\int_{\mathcal{C}} \mathcal{C}(x, -) = x/\mathcal{C}$. Similarly, one can show that the category of coelements of the contravariant representable functor is just the slice category of the representing object over that object.

In the next construction, we will define the so-called Grothendieck functor.

Construction 32.17. Let $\mathcal{C}$ be a category. Define a map of categories^a:

$$\int_{\mathcal{C}} : \text{Fun}(\mathcal{C}, \text{Set}) \rightarrow \text{Cat} / \mathcal{C}.$$

as follows:

1. If $F \in \text{Fun}(\mathcal{C}, \text{Set})$, then this map just gives us $(\int_{\mathcal{C}} F, \pi_F)$.
2. If $F, G \in \text{Fun}(\mathcal{C}, \text{Set})$ and $\alpha : F \Rightarrow G$ is a natural transformation, then we define a map of categories:

$$\int_{\mathcal{C}} \alpha : \int_{\mathcal{C}} F \rightarrow \int_{\mathcal{C}} G,$$

as follows:

- (a) If $(c, x) \in \int_{\mathcal{C}} F$, then $\int_{\mathcal{C}} \alpha(c, x) \stackrel{\text{def}}{=} (c, \alpha_c(x))$.
- (b) If $(c, x), (c', x') \in \int_{\mathcal{C}} F$ and $f : c \rightarrow c'$ is a morphism of these objects in $\int_{\mathcal{C}} F$, then $\int_{\mathcal{C}} \alpha(f) = f$.

This is clearly a functor. We do need to check that it's a morphism in the slice category $\text{Cat} / \mathcal{C}$. To prove this, we need to demonstrate that $\pi_G \circ \int_{\mathcal{C}} \alpha = \pi_F$ and that amounts to checking it on objects and morphisms.

One may reasonably refer to this as the Grothendieck functor.

What are the properties of the functor obtained above? The next result collects some properties of interest.

Proposition 32.18

Let $\mathcal{C}$ be a category.

1. The Grothendieck functor on $\mathcal{C}$ is fully faithful.
2. The Grothendieck functor on $\mathcal{C}$ has a left and right adjoint.

In particular, the adjoints are Bousfield (co)localizations.

Partial Proof: We'll deal with each statement separately.

1. Let $F, G \in \text{Fun}(\mathcal{C}, \text{Set})$. Then, we want to show that the following map:

$$\Phi : \text{Nat}(F, G) \rightarrow (\text{Cat}/\mathcal{C}) \left(\left(\int_{\mathcal{C}} F, \pi_F \right), \left(\int_{\mathcal{C}} G, \pi_G \right) \right),$$

is a bijection. We will do this by constructing an explicit inverse. Let $f : \int_{\mathcal{C}} F \rightarrow \int_{\mathcal{C}} G$ be a morphism between the relevant objects in the slice category. That is, the following diagram of functors:

$$\begin{array}{ccc} \int_{\mathcal{C}} F & \xrightarrow{\pi_F} & \mathcal{C} \\ & \searrow f & \uparrow \pi_G \\ & & \int_{\mathcal{C}} G \end{array}$$

Now, let $(c, x) \in \int_{\mathcal{C}} F$. Then, we have that:

$$c = \pi_F(c, x) = \pi_G(f(c, x)).$$

We define a natural transformation $\Psi(f) : F \rightarrow G$ as follows:

$$\forall x \in \mathcal{C} : \forall u \in F(x) : \Psi(f)_x(u) \stackrel{\text{def}}{=} \pi_2(f(x, u)),$$

where π_2 denotes the projection onto the second factor.

| **Claim 32.4.** $\Psi(f)$ is actually a natural transformation.

Next, we have the following claim.

| **Claim 32.5.** The anti-parallel map (with respect to Φ) given by Ψ is the inverse of Φ .

Hence, this completes the proof of statement (1).

2. Giving a self-contained proof would require a few detours into concepts that are not particularly relevant for this thesis. A reasonably full proof is provided in [80, Definition 1.9, Proposition 1.10, p.11-12], though a number of computations are omitted and one would have to check *all* of them to ensure that the argument works. ■

Clearly, the Grothendieck functor is not essentially surjective in general; it maps to conservative functors by **Claim 32.2** and not every functor is conservative. Nevertheless, functors which do lie in the essential image of the Grothendieck functor form the starting point for the fibrational approach to the Yoneda Lemma.

Definition 32.19

Let $\mathcal{C}, \mathcal{D}$ be categories and let $F \in \text{Fun}(\mathcal{D}, \mathcal{C})$. F is a Grothendieck opfibration iff it is in the essential image of $\int_{\mathcal{C}}$.

We have a remark about terminology.

Remark 32.20. The paper by Rasekh also refers to Grothendieck opfibrations as “discrete Grothendieck opfibrations”. We're not too sure about the relevance of the word “discrete”. In any event, these refer to the same thing.

In the next result, we provide a concrete characterization of these objects. For the purposes of a more pleasant reading experience of the proof, we do advise re-visiting **Construction 32.11** again just to get notation straight.

Proposition 32.21

Let $\mathcal{C}, \mathcal{D}$ be categories and let $F \in \text{Fun}(\mathcal{D}, \mathcal{C})$. TFAE:

1. F is a Grothendieck opfibration.
2. If $f : c \rightarrow c'$ is a morphism in $\mathcal{C}$ and $d \in \mathcal{D}$ such that $F(d) = c$, then:

$$\exists! d' \in \mathcal{D} : \exists! g \in \mathcal{D}(d, d') : F(g) = f.$$

The morphism g is called a lift of f .

Proof.

(1) $\Rightarrow$ (2): There exists a $G \in \text{Fun}(\mathcal{C}, \text{Set})$ such that:

$$\left(\int_{\mathcal{C}} G, \pi_G \right) \simeq_{\text{Cat}/\mathcal{C}} F.$$

That is, there is an isomorphism^a $\alpha : \int_{\mathcal{C}} G \rightarrow \mathcal{D}$ such that the following diagram:

$$\begin{array}{ccc} \int_{\mathcal{C}} G & \xrightarrow{\pi_G} & \mathcal{C} \\ & \searrow \alpha & \uparrow F \\ & & \mathcal{D} \end{array}$$

commutes. Now, let $c, c' \in \mathcal{C}$, let $f \in \mathcal{C}(c, c')$ and let $d \in \mathcal{D}$ so that $F(d) = c$. Let's start with a claim.

Claim. $G(c) \neq \emptyset$.

Indeed, α being an isomorphism means that we can write $\alpha^{-1}(d) = (c'', x'')$. But we know that $\pi_G(c'', x'') = c''$ and this is just $(F \circ \alpha)(c'', x'') = F(d) = c$. In particular, this means that $c'' = c$ and that $x'' \in G(c)$, so $G(c) \neq \emptyset$. Next, let's show that the lift g exists and that it is unique.

Claim. The lift g exists and is unique.

Let $x \in G(c)$ and define:

$$x' \stackrel{\text{def}}{=} G(f)(x).$$

So, $(c', x') \in \int_{\mathcal{C}} G$ and $f : (c, x) \rightarrow (c', x')$ is a morphism in $\int_{\mathcal{C}} G$. Now, we define:

$$g \stackrel{\text{def}}{=} \alpha(f).$$

Of course, g is unique because α is an isomorphism. Hence, we have that:

$$F(g) = F(\alpha(f)) = (F \circ \alpha)(f) = \pi_G(f) = f,$$

as was desired.

(2) $\Rightarrow$ (1): For each $c \in \mathcal{C}$, we consider the fiber $F^{\leftarrow}(c)$ over c ; this is a discrete subcategory of $\mathcal{D}$ due to the uniqueness part of our lifting condition. This means that the fiber is actually a set. Now, we define a functor $G : \mathcal{C} \rightarrow \text{Set}$ as follows:

(a) If $c \in \mathcal{C}$, then $G(c) = F^{\leftarrow}(c)$.

(b) If $c, c' \in \mathcal{C}$ and $f : c \rightarrow c'$ is a morphism, then we want a morphism $G(f) : G(c) \rightarrow G(c')$. If $G(c) \neq \emptyset$, then there is a $d \in \mathcal{D}$ so that $F(d) = c$. Now, this means that there is a $d' \in \mathcal{D}$ and a unique $g \in \mathcal{D}(d, d')$ such that $F(g) = f$. In that case, we define $G(f)(d) = d'$.

It is, at this point, a straightforward check to show that this works. ■

^aThat is, an on-the-nose isomorphism, not an equivalence!

From the viewpoint of important constructions, we do have just about everything we need to prove the fibrational version of the Yoneda Lemma. However, it will be necessary to set up some notation that will make the result easier to state.

Notation 32.22. Let $\mathcal{C}$ be a category with final object $*$ and let $x, y \in \mathcal{C}$. Given a morphism $\phi_y : * \rightarrow y$, we will define a morphism $\{y\} : x \rightarrow y$ as the unique^a morphism which factors through ϕ_y .

^aIt is unique because $*$ is a final object.

The second bit of notation improves readability.

Notation 32.23. Let $\mathcal{C}$ be a category and let $x \in \mathcal{C}$. Then, we will just say that the objects of $\mathcal{C}/x$ are morphisms with codomain x . Applying this to the Grothendieck construction, we will say that if $F \in \text{Fun}(\mathcal{C}, \text{Set})$, then $\int_{\mathcal{C}}(F)$ is just π_F . Of course, we should be careful to distinguish between^a $\int_{\mathcal{C}}(F)$ and $\int_{\mathcal{C}} F$.

^aThe former is a functor, the latter is a category.

We are almost ready to establish the fibrational Yoneda Lemma; we need just one more bit of notation.

Notation 32.24. Let $\mathcal{C}, \mathcal{D}$ be categories, let $x \in \mathcal{C}$ and let $F \in \text{Fun}(\mathcal{D}, \mathcal{C})$. Consider the functor:

$$* \xrightarrow{x} \mathcal{C}$$

We will define a map:

$$\Phi_F : (\text{Cat} / \mathcal{C}) (\pi_{\mathcal{C}(x, -)}, F) \rightarrow (\text{Cat} / \mathcal{C}) (\{ \text{Id}_x \}, F).$$

Let $G \in (\text{Cat} / \mathcal{C}) (\pi_{\mathcal{C}(x, -)}, F)$. Then, we define:

$$\Phi_F(G) \stackrel{\text{def}}{=} G(\text{Id}_x).$$

This is known as the Yoneda function.

We are ready to prove the fibrational Yoneda Lemma.

Theorem 32.25. The Fibrational Yoneda Lemma

Let $\mathcal{C}, \mathcal{D}$ be categories, let $c \in \mathcal{C}$ and let $F \in \text{Fun}(\mathcal{D}, \mathcal{C})$ be a Grothendieck opfibration. Then, Φ_F is a bijection.

Proof. Let $G : \mathcal{C} \rightarrow \text{Set}$ be a functor such that:

$$\int_{\mathcal{C}} F \simeq \mathcal{D},$$

over $\mathcal{C}$. Consider the diagram:

$$\begin{array}{ccc} (\text{Cat}/\mathcal{C}) (\pi_{\mathcal{C}(c,-)}, F) & \longrightarrow & (\text{Cat}/\mathcal{C}) (\{\text{Id}_c\}, F) \\ \uparrow \int_c & & \uparrow \\ \text{Nat}(\mathcal{C}(c, -), G) & \xrightarrow{\text{Yon}} & G(c) \end{array}$$

The left side is a bijection due to the fact that the Grothendieck functor is fully faithful, the bottom is a natural bijection due to the usual Yoneda Lemma and the right side is a bijection by identification. ■

For all of the preparatory work we did, the actual proof was incredibly simple. From the viewpoint of 1-category theory, this version of the Yoneda Lemma is not particularly useful; working with slice categories is certainly less convenient than just working with natural transformations and so on. Nevertheless, the key take-away is that this fibrational version of the Yoneda Lemma can be proven relatively easily for complete Segal spaces.

Theorem 32.26

Let $L \rightarrow W$ be a left fibration. Then, the map induced by $F(0) \rightarrow x/W$ over W :

$$\text{Maps}_W(x/W, L) \rightarrow \text{Maps}_W(F(0), L) = \Delta^0 \times_{W_0} L_0,$$

is a trivial fibration of Kan complexes.

Proof Of Theorem 32.26. ■

33 Stability & Spectra I: The Fundamentals

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

1-category theory - This refers to the ordinary category theory which is assumed background for readers.

Let's start with some motivation.

Motivation. Stable ∞ -categories emerged from the search for a satisfactory homotopy-theoretic foundation for homological algebra and stable homotopy theory. Quillen's introduction of model categories in the 1960s provided an abstract framework for homotopy theory and made it possible to construct derived categories and stable homotopy categories as homotopy categories of suitable model categories. At the same time, differential graded (dg-)categories and, later, A_∞ -categories were developed as enhancements of triangulated categories, retaining information about complexes and higher homotopies that is lost upon passing to the homotopy category. These approaches revealed that triangulated categories, while indispensable in practice, do not adequately encode mapping spaces, homotopy limits and colimits, or coherent higher compositions. Motivated by these shortcomings, the theory of ∞ -categories, developed by Boardman and Vogt, Joyal, and others, provided a setting in which higher homotopical data could be retained intrinsically. Building on this foundation, Lurie introduced stable ∞ -categories as ∞ -categories in which finite limits and colimits exist and coincide, thereby unifying the perspectives of stable model categories, dg-categories, and triangulated categories. Their homotopy categories naturally inherit triangulated structures, while the ambient ∞ -categorical framework preserves the higher coherence necessary for modern developments in algebraic topology, algebraic geometry, and algebraic (K)-theory.

We are finally ready to start discussing stable ∞ -categories and the actual material needed to understand the enhancements of the important functors we have introduced throughout this thesis. In this chapter, we will introduce the language of spectra in the ∞ -categorical context and this will be useful for building the ∞ -categorical enhancement of KK-theory which we previously promised. In [27], the ∞ -category of (pointed) anima are discussed, along with generalized cohomology theories and the Brown Representability Theorem, in completely separate chapters and the material on stable ∞ -categories is discussed afterwards. For the sake of keeping the presentation clean and organized, we will simply combine our treatments, following the work of Lurie in [63].

However, before we can do this properly, we will need some background in the ordinary parts of ∞ -category theory. That is, we will have to introduce the basic language of (co)limits and adjunctions in the ∞ -categorical context. In fact, there is quite a lot to be said here and that is evidenced by the sheer size (and incompleteness!) of the formalization project. So, we will not be able to prove everything in detail. Nevertheless, we hope that the main ideas are communicated as well as possible, even if technical arguments are omitted.

33.1 Pointed ∞ -Categories & Suspensions/Loops

Before we can get started in discussing stable ∞ -categories and the language associated with them (alongside the results), we need to set up some preliminary language and this involves a further discussion of pointed ∞ -categories. Of course, we already introduced these in ???. The reader is familiar with 1-categorical examples of pointed categories and these are certainly pointed ∞ -categories. An example which isn't 1-categorical is presented here.

Example 33.1. Let $\mathbf{An}_*$ denote the ∞ -category of pointed anima^a. Clearly, this is a pointed ∞ -category.

^aThe contractible anima $*$ serves as the zero object.

Here's an easy criterion to determine if an ∞ -category is pointed or not.

Claim 33.1. Let $\mathcal{C}$ be an ∞ -category. TFAE:

1. $\mathcal{C}$ is pointed.
2. $\mathcal{C}$ has an initial object $\emptyset$, a final object 1 and a morphism $f : 1 \rightarrow \emptyset$.

As a consequence, we can introduce the following subcategory of $\mathbf{Cat}_\infty$.

Example 33.2. We denote by:

$$\mathbf{Cat}_\infty^{\text{fin}},$$

to be the smallest full subcategory of $\mathbf{Cat}_\infty$ such that:

1. It contains $\emptyset$, $*$ and $[1]$.
2. It is closed under pushouts.

The ∞ -categories in $\mathbf{Cat}_\infty^{\text{fin}}$ are known as finite categories.

We can also do the same thing with anima. In light of the example above, the corresponding notion for anima just becomes terminology.

Terminology 33.3. An anima is finite iff it is a finite ∞ -category. We denote by:

$$\mathbf{An}^{\text{fin}},$$

the full subcategory of $\mathbf{An}$ consisting of finite anima.

Of course, there's a more concrete description of the full subcategory of finite anima in the same spirit as Example 33.2.

Claim 33.2. $\mathbf{An}^{\text{fin}}$ is the smallest subcategory of $\mathbf{An}$ closed under finite colimits and containing the point.

When an ∞ -category has a final object, then there is a canonical way of turning it into a pointed ∞ -category.

Construction 33.4. Let $\mathcal{C}$ be an ∞ -category with final object $*$. Then, we define an ∞ -category:

$$\mathcal{C}_*,$$

of pointed objects by the following pullback square:

$$\begin{array}{ccc} \mathcal{C}_* & \longrightarrow & \text{Fun}([1], \mathcal{C}) \\ \downarrow & & \downarrow \text{ev}_0 \\ * & \xrightarrow{*} & \mathcal{C} \end{array}$$

where we obviously abuse notation in the bottom part of the diagram.

Here are a few basic results.

Claim 33.3. Let $\mathcal{C}$ be an ∞ -category with final object $*$. Then, $\mathcal{C}_*$ is pointed.

Claim 33.4. Let $\mathcal{C}$ be an ∞ -category with final object $*$. TFAE:

1. $\mathcal{C}$ is pointed.
2. The forgetful functor $\mathcal{C}_* \rightarrow \mathcal{C}$ is an equivalence.

Claim 33.5. Let $\mathcal{C}$ be an ∞ -category with final object $*$.

1. If $\mathcal{C}$ admits pullbacks, then $\mathcal{C}_\infty$ admits pullbacks.
2. The functor $\mathcal{C}_* \rightarrow \mathcal{C}$ preserves pullbacks.
3. If $\mathcal{C}$ admits pushouts, then $\mathcal{C}_\infty$ admits pushouts.
4. The functor $\mathcal{C}_* \rightarrow \mathcal{C}$ preserves pushouts.

In particular, An_* admits pushouts and pullbacks.

Of course, pointed ∞ -categories also have a notion of *zero morphism*, unsurprisingly.

Construction 33.5. Let $\mathcal{C}$ be a pointed ∞ -category and let $x, y \in \mathcal{C}$. Then, we have two unique morphisms:

$$\begin{aligned} f &: x \rightarrow 0, \\ g &: 0 \rightarrow y. \end{aligned}$$

Now, we define:

$$0_{x,y} \stackrel{\text{def}}{=} 0 \stackrel{\text{def}}{=} g \circ f.$$

That is, this is the unique morphism from x to y which factorizes through the universal morphisms above^a.

^aAs usual, we will often just denote this morphism by 0 , not specifying the (co)domain.

As we mentioned in our motivation, we were previously able to define homotopical constructs in various categories of interest. In fact, we can do the same on a pointed ∞ -category. In the next piece of terminology, we will introduce zero-sequences as well as (co)fiber sequences. This will set the stage for us to construct the loop/suspension functors.

Terminology 33.6. Let $\mathcal{C}$ be a pointed ∞ -category and consider the following commutative square:

$$\begin{array}{ccc} x & \xrightarrow{f} & y \\ \downarrow & & \downarrow g \\ 0 & \longrightarrow & z \end{array}$$

Such a square is known as a zero-sequence^a. Moreover, we have the following pieces of terminology.

1. The commutative square above is a fiber sequence iff it is a pullback square in $\mathcal{C}$. Sometimes, we write^b:

$$y/z,$$

for x .

2. The commutative square above is a cofiber sequence iff it is a pushout square in $\mathcal{C}$. Sometimes, we write:

$$y/x,$$

for z .

Moreover, we have the following:

1. $\mathcal{C}$ admits fibers iff pullbacks along the morphisms from the zero object exist in $\mathcal{C}$.
2. $\mathcal{C}$ admits cofibers iff pushouts along the morphisms to the zero object exist in $\mathcal{C}$.

We say that $\mathcal{C}$ is fiber-complete iff it admits cofibers and fibers.

^aThis is reasoned by simply looking at the top-half of the diagram and noting that $g \circ f \simeq 0$.

^bWe will almost never do this but this is common notation in the literature.

The next two constructions deal with the loop and suspension functors.

Construction 33.7. Let $\mathcal{C}$ be a pointed ∞ -category and let $x \in \mathcal{C}$.

1. Assume that $\mathcal{C}$ admits fibers. Then, the loop object of x in $\mathcal{C}$ is an object:

$$\Omega(x),$$

obtained by the following:

$$\begin{array}{ccc} \Omega(x) & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & x \end{array}$$

pullback diagram.

2. Assume that $\mathcal{C}$ admits cofibers. Then, the suspension object of x in $\mathcal{C}$ is an object:

$$\Sigma(x),$$

obtained by the following:

$$\begin{array}{ccc} x & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \Sigma(x) \end{array}$$

pushout diagram.

In both of these cases, we obtain two functors:

$$\begin{aligned} \Omega : \mathcal{C} &\rightarrow \mathcal{C}, \\ \Sigma : \mathcal{C} &\rightarrow \mathcal{C}, \end{aligned}$$

known as the loop functor & suspension functors.

This construction was actually taken from [27, Definition 4.1.7]. In fact, we have a further comment about the construction, because something doesn't quite sit right towards the end.

Remark 33.8. So, the jump that we've made in the last step of the construction is a significant one, seeing as there is nothing that we've introduced so far which suggests that we can cook up a functor between ∞ -categories from a pullback within that ∞ -category. Firstly, it is *believable* that such a construction would yield a functor, because pullbacks are limits and limits are functorial. Secondly, there are two ways of fixing this:

1. One can work in a specific model of ∞ -categories and construct the two functors above by hand. This is ostensibly done by Lurie in [63, p.7], though that reference does not provide all of the details. A reference with more details (but which works with ∞ -cosmoi instead) is [84].
2. One may employ the so-called "functoriality of universals" axiom which is given in [26, Section 5.10, p.159]. This axiom allows one to basically assemble a functor once we know how it is defined on objects and it is claimed to be one of the most important axioms due to its role in reducing statements in ∞ -category theory to the level of objects. As usual, we will leave these details to the aforementioned text as they are not relevant.

So, one need not worry about the functors above not existing; one way or another, they do exist.

As in ordinary, classical homotopy theory, the loop and suspension functors are adjoint to each other.

Proposition 33.9

Let $\mathcal{C}$ be a fiber-complete pointed ∞ -category. Then, Σ is left adjoint to Ω .

The proof of this result requires a bit of preparation. The problem is that we need to gain a better understanding of the structure of pullbacks and pushouts in ∞ -categories. Let's start with a simple construction.

Proof Of Proposition 33.9. For $x, y \in \mathcal{C}$, we have to show that:

$$\mathcal{C}(\Sigma(x), y) \simeq \mathcal{C}(x, \Omega(y)),$$

as ∞ -categories. ■

Now, how about functoriality? Well, with pointed ∞ -categories, it seems that the only additional, interesting property of functors appears to be preservation of the zero object.

Definition 33.10

Let $\mathcal{C}, \mathcal{D}$ be pointed ∞ -categories and let $f \in \text{Fun}(\mathcal{C}, \mathcal{D})$. f is a pointed functor iff f preserves the zero object.

We have a remark about this definition.

Remark 33.11. At the moment, this is about all that we can manage with the structure available. In the next section, we'll see that stable ∞ -categories admit enough structure that we can start talking about exactness of functors.

Speaking of the next section, let us start motivating the extra structure that will be involved in eventually defining exactness. In 1-category theory, it is clear that the most fundamental category is Set . If we're interested in "linear" categories, then the most fundamental category is AbGrp . For example, AbGrp naturally has the structure of an abelian category and it is to abelian categories what Set is to arbitrary categories. The most fundamental ∞ -category is, arguably, An and this is primarily because anima arise naturally as mapping spaces within ∞ -categories. Now, the point is that stable ∞ -categories are the analogue of abelian categories in the setting of higher structures.

However, before we start discussing stability, we do want to ground ourselves in a more concrete goal. Of course, for the reader with background in triangulated category theory or derived algebraic geometry, it will slowly become clear that one actually needs the set-up of a stable ∞ -category to model the sorts of examples that naturally crop up in practice. For instance, we'll see that stable ∞ -categories give us a better understanding of spectra and give us the most general version of the Brown Representability Theorem, a result which ultimately allows us to study (extraordinary) cohomology theories via spectra. Most of this will come in good time.

33.2 The Properties Of Stable ∞ -Categories

With our work in the preceding section, we are finally ready to discuss stable ∞ -categories. Let us define them.

Definition 33.12

Let $\mathcal{C}$ be an ∞ -category. $\mathcal{C}$ is stable iff:

1. $\mathcal{C}$ is pointed.
2. Pushouts and pullbacks exist in $\mathcal{C}$.
3. Every commutative square in $\mathcal{C}$ is a pullback square iff it is a pushout square.

The commutative squares in (3) are known as exact squares.

Here are two simple results.

| **Claim 33.6.** If $\mathcal{C}$ is a stable ∞ -category, then it is finitely bicomplete.

| **Claim 33.7.** If $\mathcal{C}$ is a stable ∞ -category, then $\mathcal{C}^{\text{op}}$ is also a stable ∞ -category.

We have a remark about the definition.

| **Remark 33.13.** [ADD HISTORICAL MOTIVATION FOR THE DEFINITION OF A STABLE INFTY CATEGORY]

Here's a useful bit of notation.

Notation 33.14. If $\mathcal{C}$ is a stable ∞ -category, then we'll denote that by writing^a $\mathcal{C} \in \text{Cat}_{\infty}^{\text{st}}$.

^aThis notation will become a formal part of the theory when we define the ∞ -category of small stable ∞ -categories later.

Of course, it is desirable to have criteria which guarantee that a given pointed ∞ -category is stable.

Proposition 33.15

Let $\mathcal{C}$ be a pointed ∞ -category. TFAE:

1. $\mathcal{C}$ is stable.
2. $\mathcal{C}$ admits both fibers and cofibers, with a sequence $x \rightarrow y \rightarrow z$ being a fiber sequence iff it is a cofiber sequence.
3. $\mathcal{C}$ admits fibers and the loop $\Omega : \mathcal{C} \rightarrow \mathcal{C}$ is an equivalence.
4. $\mathcal{C}$ admits cofibers and the suspension $\Sigma : \mathcal{C} \rightarrow \mathcal{C}$ is an equivalence.

Proof. See [27, Theorem 3.2.2, p.91-92].

■

Okay, what about functoriality? Indeed, it will be convenient for us to restrict to functors between stable ∞ -categories that possess the relevant compatibility properties relative to the structure on a stable ∞ -category.

Terminology 33.16. Let $\mathcal{C}, \mathcal{D} \in \text{Cat}_{\infty}^{\text{st}}$ and let $f \in \text{Fun}(\mathcal{C}, \mathcal{D})$. f is exact iff it preserves the zero object and sends exact sequences to exact sequences; we denote by $\text{Fun}^{\text{ex}}(\mathcal{C}, \mathcal{D})$ the full subcategory of $\text{Fun}(\mathcal{C}, \mathcal{D})$ spanned by the exact functors.

The following characterization of exact functors is rather useful and also easy.

Proposition 33.17

Let $\mathcal{C}, \mathcal{D} \in \text{Cat}_{\infty}^{\text{st}}$ and let $f \in \text{Fun}(\mathcal{C}, \mathcal{D})$. TFAE:

1. f is exact.
2. f preserves initial objects and pushouts.
3. f preserves final objects and pullbacks.

Proof.

(1) $\Rightarrow$ (2):

(2) $\Rightarrow$ (3):

(3) $\Rightarrow$ (1):

■

Now, our eventual goal is to “do” K-theory on ∞ -categories. For many modern purposes, a stable ∞ -category is perfectly enough to highlight what is going on

Nevertheless, we will introduce the more general setting of Waldhausen ∞ -categories as that is “sort of” necessary for the universal characaterization of higher K-theory due to Blumberg, Gepner and Tabuada. At the same time, this

leads us very well into further discussions concerning stable ∞ -categories and their algebraic structure. We'll start off by introducing semi-additive categories and giving a construction based off of them. Then, we'll introduce additive ∞ -categories and Waldhausen ∞ -categories.

Terminology 33.18. Let $\mathcal{C} \in \text{Cat}_\infty$. $\mathcal{C}$ is semi-additive iff for each $x, y \in \mathcal{C}$, the morphism:

$$\begin{pmatrix} \text{Id} & 0 \\ 0 & \text{Id} \end{pmatrix} : x \amalg y \rightarrow x \times y,$$

is an isomorphism. In this case, we call the biproducts in $\mathcal{C}$ direct sums and denote them by $x \oplus y$.

Okay, now we need a construction.

Construction 33.19. Let $\mathcal{C} \in \text{Cat}_\infty$ be semiadditive and let $x \in \mathcal{C}$. We construct a shear map:

$$\text{sh} \stackrel{\text{def}}{=} \begin{pmatrix} \text{Id} & 0 \\ \text{Id} & \text{Id} \end{pmatrix} : x \oplus x \rightarrow x \oplus x,$$

in two steps:

1. We construct the diagonal:

$$\Delta : x \rightarrow x \oplus x,$$

by gluing together two copies of Id_x .

2. So, now, we have the diagonal:

$$\Delta : x \rightarrow x \oplus x,$$

and the zero map $0 : x \rightarrow x \oplus x$. So, we glue these together to form the shear map.

Of course, we use the semi-additive structure above to ensure that we can work with biproducts.

Okay, here's more terminology.

Terminology 33.20. Let $\mathcal{C}$ be an ∞ -category.

1. $\mathcal{C}$ is additive iff it is semiadditive and the shear map is an isomorphism for all $x \in \mathcal{C}$.
2. Suppose that we have a pair:

$$(\mathcal{C}, \text{Cof}_\mathcal{C}),$$

of an ∞ -category $\mathcal{C}$ and a subcategory $\text{Cof}_\mathcal{C}$. This pair is known as a Waldhausen ∞ -category iff:

- (a) $\mathcal{C}$ is pointed.
- (b) If $x \in \mathcal{C}$, then $0 \rightarrow x$ lies in $\text{Cof}_\mathcal{C}$; that is, x is a “cofibrant object”.
- (c) $\text{Cof}_\mathcal{C}$ contains the groupoid core $\mathcal{C}^\simeq$.
- (d) Every span:

$$z \leftarrow x \rightarrow y,$$

with $x \rightarrow y$ in $\text{Cof}_\mathcal{C}$ admits a pushout. Moreover, in the following diagram:

$$\begin{array}{ccc} x & \longrightarrow & y \\ \downarrow & & \downarrow \\ z & \longrightarrow & y \cup_x z \end{array}$$

then the morphism $z \rightarrow y \cup_x z$ is a cofibration.

The morphisms in $\text{Cof}_\mathcal{C}$ are called cofibrations.

We have a remark about this bit of terminology.

Remark 33.21. We only introduce Waldhausen ∞ -categories due to their cultural significance in our approach to higher K-theory.

Okay, let's show that stable ∞ -categories are automatically additive.

Proposition 33.22

Let $\mathcal{C} \in \text{Cat}_\infty^{\text{st}}$. Then, $\mathcal{C}$ is additive.

Proof. Let's start by showing that $\mathcal{C}$ is semi-additive.

Claim. $\mathcal{C}$ is semi-additive.

Let $x, y \in \mathcal{C}$. We consider the following pullback diagrams:

$$\begin{array}{ccc} 0 & \longrightarrow & x \\ \downarrow & & \downarrow \text{Id}_x \\ 0 & \longrightarrow & x \end{array} \qquad \begin{array}{ccc} 0 & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ y & \xrightarrow{\text{Id}_y} & y \end{array}$$

Now, we can take their product so we get the following pullback diagram:

$$\begin{array}{ccc} 0 & \longrightarrow & x \\ \downarrow & & \downarrow \\ y & \longrightarrow & x \times y \end{array}$$

with the obvious arrows. Since $\mathcal{C}$ is stable, this diagram is also a pushout and that establishes that the natural morphism $x \amalg y \rightarrow x \times y$ is an isomorphism, so $\mathcal{C}$ is semi-additive. The next claim will finish the proof.

| **Claim.** If $x \in \mathcal{C}$, then the shear map of x is an isomorphism.

We can fit the shear map into the following diagram:

$$\begin{array}{ccccc} x & \xrightarrow{i_2} & x \oplus x & \xrightarrow{p_1} & x \\ \text{Id}_x \downarrow & & \downarrow \text{sh} & & \downarrow \text{Id}_x \\ x & \xrightarrow{i_2} & x \oplus x & \xrightarrow{p_1} & x \end{array}$$

where each square is a pullback, but this is sufficient in proving that the shear map is an isomorphism. ■

34 Stability & Spectra II: Stabilization & K-Theory

Summary

In this chapter, we will cover the following topics:

1. Stabilization and the ∞ -category of spectra.
2. The ∞ -category of spectra.

This lecture was written by Abhijeet Vats.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

1-category theory - This refers to the ordinary category theory which is assumed background for readers.

Let's start with some motivation.

Motivation. In this lecture, our main goal is to present a definition of K-theory on stable ∞ -categories. In fact, we're really getting ahead of ourselves here; what we really want to do is to define K-theory on exact ∞ -categories and Waldhausen ∞ -categories. However, we feel that giving you a taste of what higher K-theory actually entails & showing that, in fact, it does have a sane definition in this setting is a good idea. Indeed, it will allow us to already start putting the ∞ -category of spectra to good use; this category will be introduced in this lecture and its basic properties will be described. To a large extent, this material is a little bit technical but it is necessary to have it on standby because it gets used quite a lot.

34.1 The Process Of Stabilization

With that said, let us start by working towards the definition of the ∞ -category of spectra. In fact, we will define this as a special case of the more general procedure of *stabilization* that we introduce for ∞ -categories with finite limits.

Definition 34.1

Let $\mathcal{C} \in \text{Cat}_\infty$ be finitely complete. A stabilization of $\mathcal{C}$ is a pair:

$$(\text{Stab}(\mathcal{C}), \Omega^\infty),$$

consisting of:

1. an ∞ -category $\text{Stab}(\mathcal{C})$ and;
2. a left-exact functor $\Omega^\infty : \text{Stab}(\mathcal{C}) \rightarrow \mathcal{C}$,

such that for every $\mathcal{D} \in \text{Cat}_\infty^{\text{st}}$, the composition functor:

$$\Omega^\infty \circ - : \text{Fun}^{\text{lex}}(\mathcal{D}, \text{Stab}(\mathcal{C})) \rightarrow \text{Fun}^{\text{lex}}(\mathcal{D}, \mathcal{C}),$$

is an equivalence. We further define:

$$\text{Sp} \stackrel{\text{def}}{=} \text{Stab}(\text{An}),$$

as the ∞ -category of spectra, with its objects being known as spectra.

We will study Sp a lot more carefully in the next section. In this section, we need to provide:

1. a construction of the stabilization of a finitely complete ∞ -category and;
2. a list of nice properties satisfied by the stabilization.

We will provide two constructions of the stabilization in Section 34.1.1 and Section 34.1.2. Let us summarize them:

1. The excisive viewpoint, as discussed in Section 34.1.1, presents the stabilization as a category of functors $F : \text{An}^{\text{fin}}_\star \rightarrow \mathcal{C}$ which preserve the final object and send pushout squares to pullback squares; such functors are called *excisive*.
2. The classical viewpoint, as discussed in Section 34.1.2, presents the stabilization as a category of spectrum objects. This is, of course, the classical perspective on spectra and while it is very useful for proving some abstract properties of the stabilization, it is not the best perspective for thinking about how this ∞ -category behaves.

What are the main properties of the stabilization? Let $\mathcal{C} \in \text{Cat}_\infty$ be finitely complete. Let us list these as claims.

Claim. $\text{Stab}(\mathcal{C})$ admits finite limits.

This will be proved in.

Claim. If $\mathcal{C}$ is finitely cocomplete and the loop:

$$\Omega : \mathcal{C}_\star \rightarrow \mathcal{C}_\star,$$

preserves sequential colimits, then $\text{Stab}(\mathcal{C})$ admits colimits.

This will be proved in.

Claim. When $\text{Stab}(\mathcal{C})$ admits finite colimits, then Ω^∞ admits a left adjoint.

This will be proved in.

Most of these properties will be established in Section 34.1.3. If you are happy with taking these on faith for the moment, then we do recommend going through Section 34.1.1 and Section 34.1.2 very briefly before moving on to the next section. You can come back to the (rather technical) proofs of the claims above later.

34.1.1 The Excisive Perspective on Spectra

34.1.2 The Classical Perspective on Spectra

34.1.3 Some Technical Results

34.2 The ∞ -Category Of Spectra

As we defined [Definition 34.1](#), the ∞ -category Sp is the stabilization of An . In the previous section, we demonstrated that there are two concrete perspectives that one can take on the stabilization and one can use these to prove technical results about the stabilization. In the next corollary, we will list a number of relevant consequences of these technical results which apply to Sp . Once again, the point is that these properties do concretely hold and that one can just utilize them.

Corollary 34.2

TFSH:

1. Let $\mathcal{D}$ be a stable ∞ -category and let $F \in \mathrm{Fun}^{\mathrm{lex}}(\mathcal{D}, \mathrm{An})$. Then, there exists a unique exact functor $\widehat{F} : \mathcal{D} \rightarrow \mathrm{Sp}$ such that $\Omega^\infty \circ \widehat{F} = F$.
2. Sp is bicomplete.
3. The functors:

$$\Omega^\infty : \mathrm{Sp} \rightarrow \mathrm{An},$$

$$\Omega^\infty : \mathrm{Sp} \rightarrow \mathrm{An}_*,$$

admits left adjoints:

$$\Sigma^\infty : \mathrm{An}_* \rightarrow \mathrm{Sp},$$

$$\Sigma_+^\infty : \mathrm{An} \rightarrow \mathrm{Sp},$$

satisfying $\Sigma_+^\infty(X) \simeq \Sigma^\infty(X_+)$ for each $X \in \mathrm{An}$.

Proof. We'll prove each statement one-by-one.

- 1.
- 2.
- 3.

■

Here's a useful piece of terminology.

Terminology 34.3. If $X \in \mathrm{Sp}$, then we refer to:

$$\Omega^\infty(X),$$

as the underlying anima of the spectrum X .

With the corollary above, we can already give two constructions of spectra.

Construction 34.4. Let $X \in \mathrm{An}_*$. Then, we refer to the spectrum:

$$\Sigma^\infty(X),$$

as the reduced suspension spectrum of X .

Construction 34.5. Let $X \in \mathrm{An}$. Then, we refer to the spectrum:

$$\Sigma_+^\infty(X),$$

as the unreduced suspension spectrum of X . We further define:

$$\S \stackrel{\mathrm{def}}{=} \Sigma^\infty(S^0) \simeq \Sigma_+^\infty(*),$$

as the sphere spectrum.

What is the intuition behind the notion of spectra, beyond the two concrete ways in which we've provided in the previous section? Let us draw up a table of analogies between ordinary algebra and higher algebra.

Ordinary Algebra	Higher Algebra
Set	Anima
Abelian Group	Spectrum
Integers	Sphere Spectrum
Underlying Set	Underlying Anima
Free abelian group	Unreduced suspension spectrum

This should feel intuitive for you given everything we've discussed in the preceding section and in our more classical treatment of spectra in *Towards NC Homotopy Theory I*. In light of these analogies, here's some notation.

Notation 34.6. Let $X \in \mathrm{An}$. We define:

$$\mathbb{S}[X] \stackrel{\mathrm{def}}{=} \Sigma_+^\infty(X),$$

as the unreduced suspension spectrum of X .

In light of the analogies above, let us introduce some crucial intuition for suspension spectra.

Intuition 34.7. So, the point is that one should view a spectrum as an abelian group. Now, every abelian group can be written as the cokernel of a map of free abelian groups. But free abelian groups are unreduced suspension spectra. In other words, the intuition is that spectra should have a “presentation” in terms of unreduced suspension spectra. On the other hand, unreduced suspension spectra should be built, intuitively, from iterated suspensions of a pointed anima. We will make this precise within [Section 34.2.2](#).

The first usage of this new ∞ -category is to obtain a refinement of the Yoneda Lemma for stable ∞ -categories. To do this, we'll need a refinement of the hom functor for stable ∞ -categories. You should learn the proof of the next result very well because it shows us exactly how we can obtain similar refinements for functors that have $\mathbf{An}$ as their codomain.

Proposition 34.8

Let $\mathcal{C} \in \mathbf{Cat}_{\infty}^{\text{st}}$. There exists a unique functor:

$$\text{hom}_{\mathcal{C}}(-, -) : \mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathbf{Sp},$$

that is exact in both variables and fits in a diagram below:

$$\begin{array}{ccc} & & \mathbf{Sp} \\ & \nearrow \text{hom}_{\mathcal{C}}(-, -) & \downarrow \Omega^{\infty} \\ \mathcal{C}^{\text{op}} \times \mathcal{C} & \xrightarrow{\mathcal{C}(-, -)} & \mathbf{An} \end{array}$$

which commutes.

Proof.

Before we give the proof of this, which is very easy and synthetic, let's introduce relevant terminology.

Terminology 34.9. Let $\mathcal{C} \in \mathbf{Cat}_{\infty}^{\text{st}}$ and let $x, y \in \mathcal{C}$. Then:

$$\text{hom}_{\mathcal{C}}(x, y),$$

is known as the hom spectra or mapping spectra of $\mathcal{C}$ from x to y .

As a corollary, we obtain the spectral Yoneda Lemma.

Corollary 34.10. The Spectral Yoneda Lemma

Let $\mathcal{C} \in \mathbf{Cat}_{\infty}^{\text{st}}$. Then, the functor:

$$\mathcal{C} \rightarrow \mathbf{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Sp}), \quad y \mapsto \text{hom}_{\mathcal{C}}(-, y),$$

obtained by currying the functor in Proposition 34.8, is fully faithful.

Proof.

34.2.1 The Categorical Structure of $\mathbf{Sp}$

One of the complaints that we had with the usual 1-category of (topological) spectra is that it does not admit an obvious symmetric monoidal structure. Indeed, the point is that the smash product would be the most natural choice of a tensor product but there are many technical hurdles in turning this into a working theory. Moreover, this product has to be modified if we work in smaller categories (for instance, with CW complexes) so that we have something which is reasonably well-behaved. For these reasons, it is desirable to:

1. have a reasonable tensor product on $\mathbf{Sp}$ and;
2. have it be constructed from the internal data of $\mathbf{Sp}$.

The key to fulfilling these desires is to establish a separate universal property for $\mathbf{Sp}$ in terms of functors that go *out of it*; the one that we used to define $\mathbf{Sp}$ looked at the behavior of certain functors going into it. This universal property effectively says that colimit-preserving functors $\mathbf{Sp} \rightarrow \mathcal{C}$ into a cocomplete stable ∞ -category $\mathcal{C}$ are uniquely determined by their value on the sphere spectrum. Again, this is a reflection of the classical idea that understanding what happens on spheres effectively gives us everything on CW complexes. It's not surprising that this is reflected here, considering the fact that $\mathbf{An}$ is formed as a localization of the category of CW complexes.

Proposition 34.11

Let $\mathcal{C} \in \mathbf{Cat}_{\infty}^{\text{st}}$ be cocomplete. Then, there exists a unique functor:

$$- \otimes - : \mathbf{Sp} \times \mathcal{C} \rightarrow \mathcal{C},$$

equipped with a natural equivalence of spectra:

$$\text{hom}_{\mathcal{C}}(x \otimes y, z) \simeq \text{hom}_{\mathbf{Sp}}(x, \text{hom}_{\mathcal{C}}(y, z)),$$

for $x \in \mathbf{Sp}$ and $y, z \in \mathcal{C}$.

Proof.

34.2.2 The Standard Presentation of a Spectrum

Solutions

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| **Proof. 1.1.** THIS IS A TEST BOX.

Section 2

| **Proof. 2.1.**

| **Proof. 2.2.**

| **Proof. 2.3.**

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| **Proof. 2.5.**

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| **Proof. 2.7.**

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| **Proof. 2.9.**

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Section 11

| **Proof. 11.1.**

| **Proof. 11.2.**

| **Proof. 11.3.**

Section 14

| **Proof. 14.1.**

| **Proof. 14.2.**

| **Proof. 14.3.**

Section 16

| **Proof. 16.1.**

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| **Proof. 16.4.**

Section 19

| **Proof. 19.1.**

| **Proof. 19.2.**

| **Proof. 19.3.**

| **Proof. 19.4.**

| **Proof. 19.5.**

Section 20

| **Proof. 20.1.** This is just a general property of the boundary operator; see [REF].

| **Proof. 20.2.** This is clear when $n = 1$ and the general case can be obtained by induction, alongside **Claim 20.1**.

| **Proof. 20.3.** This follows from the fact that S^n is homeomorphic to the one-point compactification of $\mathbb{R}^n$. $\mathbb{R}^n$ is homeomorphic to the interior of I^n and $I/\partial I^n$ is homeomorphic to the one-point of compactification of $(0, 1)^n$. So, we get that S^n is homeomorphic to S^n .

| **Proof. 20.4.**

| **Proof. 20.5.**

| **Proof. 20.6.**

| **Proof. 20.7.**

Section 21

| **Proof. 21.1.** This follows from the fact that $s\mathcal{C}$ is a functor category, so (co)limits are evaluated point-wise.

| **Proof. 21.2.** Actually, this is obvious. So, let $f \in \text{Fun}([1], \mathcal{C})$. So, this functor gives us a morphism $f(0) \rightarrow f(1)$. Now, we compute $f \circ \langle 0 \rangle$; on the objects, this evaluates to $f(0)$ and on the morphisms, we simply get $\text{Id}_{f(0)}$. In other words, we recover the domain of $f() \rightarrow f(1)$. A similar statement can be made for the second map.

| **Proof. 21.3.** Indeed, let $x \in \text{Fun}([0], \mathcal{C})$. Then, the action of $N[\mathcal{C}](\langle 00 \rangle)$ on x simply forms $x \circ \langle 00 \rangle$. Well, this is a morphism whose domain is the same as its codomain. Moreover, the only morphism from 0 to 0 is Id_0 , so it follows that the morphism $x \circ \langle 00 \rangle$ is just Id_x .

| **Proof. 21.4.** THIS IS CLEAR.

| **Proof. 21.5.** We'll deal with each statement one-by-one.

1. One simply generalizes the proof of **Claim 21.4**;
2. If $a \in N[\mathcal{C}]_n$, then (1) implies that we can write:

$$a = (g_1, \dots, g_n),$$

where each g_i is a morphism in $\mathcal{C}$.

| **Proof. 21.6.** Clearly, $\Rightarrow$ holds so we want to prove that $f_1 = g_1$ implies that $f = g$. Now, let $n \in \mathbb{N}_0$ and let's show that $f_n = g_n$. Well, let $x \in X_n$ and note that $f_n(x), g_n(x) \in N[\mathcal{C}]_n$. But $N[\mathcal{C}]_n$ is in bijection with the set of n -tuples of composable morphisms. In other words, $f_n(x)$ and $g_n(x)$ are n -tuples of composable morphisms and since $f_1 = g_1$, it follows that these n -tuples are exactly the same. That is, $f_n(x) = g_n(x)$.

| **Proof. 21.7.** This follows immediately from the Co-Yoneda Lemma (this is just the contravariant form of the Yoneda Lemma).

| **Proof. 21.8.** The first statement is obvious, $[0]$ is a singleton so there is just one map that can be formed at all levels. The second statement is also obvious and follows from the fact that the empty set is an initial object in Set .

Section 22

| **Proof. 22.1.**

| **Proof. 22.2.**

| **Proof. 22.3.**

| **Proof. 22.4.**

| **Proof. 22.5.**

Section 23

| **Proof. 23.1.**

| **Proof. 23.2.**

| **Proof. 23.3.**

| **Proof. 23.4.**

| **Proof. 23.5.**

| **Proof. 23.6.**

| **Proof. 23.7.**

| **Proof. 23.8.**

| **Proof. 23.9.**

Section 24

| **Proof. 24.1.** If f, g are homotopic, then there is a weak equivalence t such that:

$$f \circ t \simeq_{h,r} g \circ t.$$

So, applying the first statement, we see that:

$$\begin{aligned} \gamma(f) \circ \gamma(t) &= \gamma(f \circ t) \\ &= \gamma(g \circ t) \\ &= \gamma(g) \circ \gamma(t), \end{aligned}$$

where we can cancel $\gamma(t)$ since it is an isomorphism. So, it is sufficient for us to prove the first statement. In that

case, let y^I be a path space and let $h : x \rightarrow y^I$ be a right-homotopy from f to g . Then, we see that:

$$\begin{aligned} f &= d_0 \circ h, \\ g &= d_1 \circ h. \end{aligned}$$

Now, we have a calculation^a:

$$\begin{aligned} \gamma(f) &= \gamma(d_0 \circ h) \\ &= \gamma(d_0) \circ \gamma(h) \\ &= \gamma(d_0) \circ \gamma(d_1)^{-1} \circ \gamma(d_1) \circ \gamma(h) \\ &= \gamma(d_0) \circ \gamma(d_1)^{-1} \circ \gamma(d_1 \circ h) \\ &= \gamma(d_0) \circ \gamma(d_1)^{-1} \circ \gamma(g). \end{aligned}$$

Now, we need to show that:

$$\gamma(d_0) = \gamma(d_1).$$

By definition of the path space, there is a morphism $s : x \rightarrow x^I$ as part of the data of a path space such that:

$$\begin{aligned} d_0 \circ s &= \text{Id}_x, \\ d_1 \circ s &= \text{Id}_x. \end{aligned}$$

Applying γ to both sides in both equations, we have that:

$$\begin{aligned} \gamma(d_0) \circ \gamma(s) &= \gamma(d_0 \circ s) \\ &= \gamma(\text{Id}_x) \\ &= \gamma(d_1) \circ \gamma(s). \end{aligned}$$

Since γ inverts weak equivalences and s is a weak equivalence, we can cancel $\gamma(s)$ and we're done.

^aThis is justified since d_1 is a trivial fibration so its image in $\mathbf{Ho}(\mathcal{C})$ is an isomorphism.

Proof. 24.2. We simply apply **Claim 23.2** to the quotient functors.

Proof. 24.3. We start with the following computation:

$$\begin{aligned} \alpha_0^x \circ k &= \alpha_0^x \circ (w \circ \hat{h}) \\ &= (\alpha_0^x \circ w) \circ \hat{h} \\ &= \bar{d}_f \circ \hat{h} \\ &= g. \end{aligned}$$

This shows that k is a homotopy of g . Next, we have the following computation:

$$\begin{aligned} T(f) \circ k &= T(f) \circ (w \circ \hat{h}) \\ &= (T(f) \circ w) \circ \hat{h} \\ &= \bar{f} \circ \hat{h} \\ &= h, \end{aligned}$$

as was desired.

Section 30

Proof. 30.1. We'll prove each statement one-by-one.

1. Let $\text{Id}_{\mathcal{C}}$ be the identity functor on $\mathcal{C}$. By Axiom 30.5, there is an identity natural isomorphism $\text{Id}_{\text{Id}_{\mathcal{C}}}$ from $\text{Id}_{\mathcal{C}}$ to $\text{Id}_{\mathcal{C}}$.
2. Let $f : \mathcal{C} \rightarrow \mathcal{D}$ be an equivalence of categories. Then, it has a quasi-inverse $g : \mathcal{D} \rightarrow \mathcal{C}$. By definition, g is an equivalence of categories, so $\mathcal{D} \simeq \mathcal{C}$.
3. Let $f_1 : \mathcal{C} \rightarrow \mathcal{D}$ and $f_2 : \mathcal{D} \rightarrow \mathcal{E}$ be equivalences of categories. Then, there exist quasi-inverses $g_1 : \mathcal{D} \rightarrow \mathcal{C}$ and $g_2 : \mathcal{E} \rightarrow \mathcal{D}$. Now, we claim that $f_2 \circ f_1$ is an equivalence of categories. Indeed, we have the following calculation:

$$\begin{aligned} (f_2 \circ f_1) \circ (g_1 \circ g_2) &\simeq f_2 \circ (f_1 \circ g_1) \circ g_2 \\ &\simeq f_2 \circ \text{Id}_{\mathcal{D}} \circ g_2 \\ &\simeq f_2 \circ g_2 \\ &\simeq \text{Id}_{\mathcal{E}}, \end{aligned}$$

as was desired.

Proof. 30.2. We have the following calculation:

$$\begin{aligned} h &\simeq h \circ \text{Id}_{\mathcal{D}} \\ &\simeq h \circ (f \circ h') \\ &\simeq (h \circ f) \circ h' \\ &\simeq \text{Id}_{\mathcal{C}} \circ h' \\ &\simeq h', \end{aligned}$$

as was desired.

Proof. 30.3.

(1) $\Rightarrow$ (2): This is clear.

(2) $\Rightarrow$ (1): Let s and r be a section & retraction of f respectively. It's enough to show that they are equal to each other; this will imply that s is a quasi-inverse of f so f must be an equivalence. In fact, this is just a small calculation:

$$\begin{aligned} s &\simeq \text{Id}_{\mathcal{C}} \circ s \\ &\simeq (r \circ f) \circ s \\ &\simeq r \circ (f \circ s) \\ &\simeq r \circ \text{Id}_{\mathcal{D}} \\ &\simeq r, \end{aligned}$$

as was desired.

Proof. 30.4. We'll prove each statement one-by-one.

1. We know that the result holds when f and g are equivalent; you effectively proved this in ???. So, assume that $g \circ f$ and f are equivalences of categories. Then, they have quasi-inverses h and f^{-1} respectively. By **Claim 30.3**, it's enough to show that g has a section and a retraction. Here's a short calculation:

$$\begin{aligned} \text{Id}_{\mathcal{E}} &\simeq (g \circ f) \circ h \\ &\simeq g \circ (f \circ h), \end{aligned}$$

which means that g has a section. Next, we also have that:

$$\begin{aligned} \text{Id}_{\mathcal{C}} &\simeq h \circ (g \circ f) \\ &\simeq (h \circ g) \circ f, \end{aligned}$$

which means that $h \circ g$ is a retraction of f so it is isomorphic to f^{-1} . This implies that $f \circ h$ is a retraction of g . Hence, it must be an equivalence of categories. In the case where $g \circ f$ and h are equivalences of categories, a similar argument will yield that f is also an equivalence so we're done.

2. If f, g, h are equivalences, then we know that $h \circ g \circ f$ are equivalences. If f is an equivalence, then we'll know that g is an equivalence by applying the proposition above and noting that $g \circ f$ is an equivalence. In that case, it will follow that h is an equivalence of categories, again by applying the proposition above and noting that $h \circ g$ is an equivalence. So, it remains to be shown that f is an equivalence. The point is that $g \circ f$ being an equivalence means that f has a retraction.

Proof. 30.5. Constructing two functors $\sigma, \eta : [2] \rightarrow \mathcal{C}$ with the relevant specifications is straightforward; we simply define these functors by composing them with the space maps and explicitly specifying the faces as f or Id_x .**Proof. 30.6.** Let's start with the first statement as listed in the construction after the claim. Let $f = \text{Id}_{\mathcal{C}}$ in the relations above and let $g = \text{Id}_{\mathcal{D}}$. In that case, we have that:

$$\begin{aligned} \pi_{\mathcal{C}} &\simeq \pi_{\mathcal{C}} \circ (\text{Id}_{\mathcal{C}} \times \text{Id}_{\mathcal{D}}) \\ \pi_{\mathcal{D}} &\simeq \pi_{\mathcal{D}} \circ (\text{Id}_{\mathcal{C}} \times \text{Id}_{\mathcal{D}}). \end{aligned}$$

which shows the first statement. For the second statement, it will be useful to have the following diagram as reference:

$$\begin{array}{ccccc} \mathcal{C} & \xleftarrow{\pi_{\mathcal{C}}} & \mathcal{C} \times \mathcal{D} & \xrightarrow{\pi_{\mathcal{D}}} & \mathcal{D} \\ \downarrow f & & \downarrow f \times g & & \downarrow g \\ \mathcal{C}' & \xleftarrow{\pi_{\mathcal{C}'}} & \mathcal{C}' \times \mathcal{D}' & \xrightarrow{\pi_{\mathcal{D}'}} & \mathcal{D}' \\ \downarrow f' & & \downarrow f' \times g' & & \downarrow g' \\ \mathcal{C}'' & \xleftarrow{\pi_{\mathcal{C}''}} & \mathcal{C}'' \times \mathcal{D}'' & \xrightarrow{\pi_{\mathcal{D}''}} & \mathcal{D}'' \end{array}$$

In the claim, the two functors in question have codomain $\mathcal{C}'' \times \mathcal{D}''$. Our strategy is to show that their compositions with the projection functors $\pi_{\mathcal{C}''}$ and $\pi_{\mathcal{D}''}$ produce the same results. Let's calculate as follows:

$$\begin{aligned} \pi_{\mathcal{C}''} \circ [(f' \times g') \circ (f \times g)] &\simeq [\pi_{\mathcal{C}''} \circ (f' \times g')] \circ (f \times g) \\ &\simeq (f' \circ \pi_{\mathcal{C}'}) \circ (f \times g) \\ &\simeq f' \circ (\pi_{\mathcal{C}'} \circ (f \times g)) \\ &\simeq f' \circ (f \circ \pi_{\mathcal{C}}) \\ &\simeq (f' \circ f) \circ \pi_{\mathcal{C}}. \end{aligned}$$

By construction, we know that:

$$\pi_{\mathcal{C}''} \circ [(f' \circ f) \times (g' \circ g)] \simeq f' \circ f.$$

So, it follows that:

$$\pi_{\mathcal{C}''} \circ [(f' \circ f) \times (g' \circ g)] \simeq \pi_{\mathcal{C}''} \circ [(f' \times g') \circ (f \times g)].$$

A similar result is obtained for $\pi_{\mathcal{D}''}$ and we're done.

Proof. 30.7. We'll prove each statement one-by-one.

1. Let $p_C : \mathcal{C} \rightarrow *$ be a functor. Note also that we have two projection functors:

$$\pi_1 : \mathcal{C} \times * \rightarrow \mathcal{C},$$

$$\pi_2 : \mathcal{C} \times * \rightarrow *.$$

Now, the goal is to construct a quasi-inverse for π_1 . Indeed, we also have the identity functor $\text{Id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$, so we can form a new functor $(\text{Id}_{\mathcal{C}}, p_C) : \mathcal{C} \rightarrow \mathcal{C} \times *$. Now, observe that:

$$\pi_1 \circ (\text{Id}_{\mathcal{C}}, p_C) \simeq \text{Id}_{\mathcal{C}}.$$

We have the following claim.

Claim. $(\text{Id}_{\mathcal{C}}, p_C) \circ \pi_1 \simeq \text{Id}_{\mathcal{C} \times *}$.

This is just checked by drawing a diagram of how the right-hand side is constructed and using the universal property of the product so we're done.

2. Consider the following three tier diagram:

$$\begin{array}{ccccc}
 \mathcal{C} & \xleftarrow{\pi'_C} & \mathcal{D} \times \mathcal{C} & \xrightarrow{\pi'_D} & \mathcal{D} \\
 \uparrow \text{Id}_{\mathcal{C}} & & \uparrow (\pi_D, \pi_C) & & \downarrow \text{Id}_{\mathcal{D}} \\
 \mathcal{C} & \xleftarrow{\pi_C} & \mathcal{C} \times \mathcal{D} & \xrightarrow{\pi_D} & \mathcal{D} \\
 \uparrow \text{Id}_{\mathcal{C}} & & \uparrow (\pi'_C, \pi'_D) & & \uparrow \text{Id}_{\mathcal{D}} \\
 \mathcal{C} & \xleftarrow{\pi'_C} & \mathcal{D} \times \mathcal{C} & \xrightarrow{\pi'_D} & \mathcal{D} \\
 \uparrow \text{Id}_{\mathcal{C}} & & \uparrow (\pi_D, \pi_C) & & \uparrow \text{Id}_{\mathcal{D}} \\
 \mathcal{C} & \xleftarrow{\pi_C} & \mathcal{C} \times \mathcal{D} & \xrightarrow{\pi_D} & \mathcal{D}
 \end{array}$$

where each individual square is homotopy commutative. Since homotopy commutative squares concatenate to form homotopy commutative squares, it follows that $(\pi_D, \pi_C) : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{D} \times \mathcal{C}$ is an equivalence of ∞ -categories.

3. We will construct a 1-morphism forwards and backwards; showing that it is an equivalence amounts to the same arguments we've been doing above via universal properties. So, suppose that we have the data indicated in the following diagram:

$$\begin{array}{ccccccc}
 (\mathcal{C} \times \mathcal{D}) \times \mathcal{E} & \xrightarrow{\pi^1} & \mathcal{C} \times \mathcal{D} & \xrightarrow{\pi_C} & \mathcal{C} & & \mathcal{E} \xleftarrow{\mu_{\mathcal{E}}} \mathcal{D} \times \mathcal{E} \xleftarrow{\mu^2} \mathcal{C} \times (\mathcal{D} \times \mathcal{E}) \\
 \downarrow \pi^2 & & \downarrow \pi_D & & & & \downarrow \mu_D \quad \downarrow \mu^1 \\
 \mathcal{E} & & \mathcal{D} & & & & \mathcal{D} \quad \mathcal{C}
 \end{array}$$

where all arrows are canonical projection functors. We will focus our attention on the left diagram. To define a functor from $(\mathcal{C} \times \mathcal{D}) \times \mathcal{E}$ into $\mathcal{C} \times (\mathcal{D} \times \mathcal{E})$, we need to define two functors; one with codomain $\mathcal{C}$ and the other with codomain $\mathcal{D} \times \mathcal{E}$. For the first, we simply take the composition $\pi_C \circ \pi^1$. As for the second, it's enough to define two further functors with codomain $\mathcal{D}$ and $\mathcal{E}$ respectively; we can just take $\pi_D \circ \pi^1$ and π^2 respectively. By gluing all of this together, we obtain a functor from $(\mathcal{C} \times \mathcal{D}) \times \mathcal{E} \rightarrow \mathcal{C} \times (\mathcal{D} \times \mathcal{E})$. Doing the same thing on the right diagram above yields a functor in the opposite direction and it is a tedious check by universal properties to show that these two functors are quasi-inverses of each other.

Proof. 30.8.

Section 32

Proof. 32.1.

Proof. 32.2.

Proof. 32.3.

Proof. 32.4.

Proof. 32.5.

Section 33

Proof. 33.1. (1) $\Rightarrow$ (2) is clear. For (2) $\Rightarrow$ (1), let's show that 1 is also initial by proving that f is an isomorphism. Indeed, there is a unique morphism $g : \emptyset \rightarrow 1$. So, observe that $f \circ g$ must be isomorphic to $\text{Id}_{\emptyset}$ since $\emptyset$ is an initial object and there is one & only one morphism $\emptyset \rightarrow \emptyset$. On the other hand, $g \circ f$ must be isomorphic to Id_1 , by a similar line of reasoning as in the preceding sentence so we're done.

Proof. 33.2. Clearly, An^{fin} is closed under finite colimits and contains the point. Now, let $\mathcal{C}$ be any other subcategory of An which is closed under finite colimits & contains the point.

Proof. 33.3. We need to show that $*$ is both initial and final in $\mathcal{C}_*$.

1. Let's prove that it is initial; note that the functor $\mathcal{C}(*, y) \rightarrow \mathcal{C}(*, y)$ given by pre-composing with Id_* is an equivalence.

2.

This completes the proof.

Proof. 33.4. $(2) \Rightarrow (1)$ is clear.

Proof. 33.5.

Proof. 33.6. This follows from the fact that $\mathcal{C}$ is pointed and that pullbacks/pushouts in $\mathcal{C}$ exist.

Proof. 33.7. This is clear from the definition.

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